



Thesis Project by Group CLS

THESIS 1-3

Thesis Project

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Section 1. Introduction

1.1 Problem Overview

Large-scale global optimization (LSGO) presents a critical challenge in computational intelligence, where the "curse of dimensionality" leads to an explosive increase in local optima and a high risk of premature convergence. While foundational algorithms like Differential Evolution (DE) are robust, their dependence on crossover and random mutation may restrict guided exploration — reducing particle diversity over time and leading to premature convergence.

This notebook implements and evaluates the proposed **Diversity-Preserving Level-Based Learning Differential Evolution (DP-LLDE)** algorithm, tested against the **CEC 2017 Large-Scale Global Optimization Benchmark** across 29 benchmark functions (F1-F30, excluding deprecated F2) at **10 dimensions**.

1.2 Research Objectives

1. Design and implement DP-LLDE, integrating sequential relative particle mutation and dynamic level-based learning into the Differential Evolution framework.
2. Evaluate DP-LLDE on the CEC 2017 benchmark suite (F1-F30) at 10 dimensions.
3. Examine convergence behavior, solution quality, and the effect of diversity-preserving strategies across unimodal, multimodal, hybrid,

and composition functions.

1.3 Algorithm Overview - DP-LLDE

DP-LLDE extends LBLDE (Qiao et al., 2022) with three core contributions:

1. **Sequential Level Assignment** — each individual is assigned a learning source level drawn randomly from $\{2, \dots, NL\}$ on the first generation, decaying toward the elite level (level 1) in subsequent generations.
2. **Diversity-Aware Exemplar Selection** — if two exemplars e_1, e_2 fall below a diversity threshold, one is replaced with the most distant candidate in that level, preventing exemplar collapse.
3. **Normalized Mutation Pressure (Eq. 10):**

$$V_i = e_1 + F_i \times (e_2 - X_i) \times \frac{NL - k}{NL}$$

The factor $(NL-k)/NL$ decays to zero as $k \rightarrow NL$, naturally reducing mutation pressure as learning quality improves.

Parameter	Value
Population Size (NP)	100
Number of Levels (NL)	4
Initial μ_{CR}	0.35
Max FES	$10,000 \times D$
Dimensions (D)	10
Seed	42

Section 2. Benchmark Function Description

2.1 Overview

The **CEC 2017 Large-Scale Global Optimization Benchmark** (Awad et al., 2017) is a standardized suite of 30 test functions designed to evaluate optimization algorithms on complex, high-dimensional search spaces. The benchmark tests an

algorithm's ability to avoid false local optima, convergence speed, and solution accuracy.

All functions are defined over the search space $[-100, 100]^D$ with their respective global optima offset by a function-specific bias value.

Library used: `cec2017` by Duncan Tilley — provides the official shift vectors, rotation matrices, and shuffle permutations from the CEC 2017 specification.

2.2 Unimodal Function Overview

Unimodal functions contain a **single global optimum and no local optima**. They are used to test the convergence speed and accuracy of an algorithm in a straightforward landscape.

No.	Function Name	Global Optimum
F1	Shifted and Rotated Bent Cigar	100
F2	Shifted and Rotated Sum of Different Power (<i>Deprecated</i>)	200
F3	Shifted and Rotated Zakharov	300

Note: F2 is officially deprecated from the CEC 2017 benchmark but is included here for completeness as it remains implemented in the `cec2017` library.

2.3 Multimodal Function Overview

Multimodal functions contain **multiple local optima**, making them significantly harder to solve. They are used to evaluate an algorithm's capacity to avoid premature convergence and escape false local optima.

No.	Function Name	Global Optimum
F4	Shifted and Rotated Rosenbrock's	400
F5	Shifted and Rotated Rastrigin's	500
F6	Shifted and Rotated Expanded Schaffer's F6	600
F7	Shifted and Rotated Lunacek Bi-Rastrigin	700
F8	Shifted and Rotated Non-Continuous Rastrigin's	800

No.	Function Name	Global Optimum
F9	Shifted and Rotated Levy	900
F10	Shifted and Rotated Schwefel's	1000

2.4 Hybrid Function Overview

Hybrid functions divide the search space into several regions, each governed by a **different basic function**. They assess an algorithm's adaptability to heterogeneous landscapes and mixed search behaviors.

No.	Component Functions	Global Optimum
F11	Zakharov; Rosenbrock; Rastrigin	1100
F12	Elliptic; Schwefel; Bent Cigar	1200
F13	Bent Cigar; Rosenbrock; Bi-Rastrigin	1300
F14	Elliptic; Ackley; Schaffer F7; Rastrigin	1400
F15	Bent Cigar; HGBat; Rastrigin; Rosenbrock	1500
F16	Schaffer F6; HGBat; Rosenbrock; Schwefel	1600
F17	Katsuura; Ackley; Griewank+Rosenbrock; Schwefel; Rastrigin	1700
F18	Elliptic; Ackley; Rastrigin; HGBat; Discus	1800
F19	Bent Cigar; Rastrigin; Griewank+Rosenbrock; Weierstrass; Schaffer F6	1900
F20	HappyCat; Katsuura; Ackley; Rastrigin; Schwefel; Schaffer F7	2000

2.5 Composite Function Overview

Composition functions **combine multiple shifted and rotated basic functions** into a single search space, creating highly complex and deceptive optimization landscapes. They are the most challenging category in the CEC 2017 suite and significantly test an algorithm's overall robustness and global search capability.

No.	Component Functions	Global Optimum
F21	Rosenbrock; Elliptic; Rastrigin	2100
F22	Rastrigin; Griewank; Schwefel	2200
F23	Rosenbrock; Ackley; Schwefel; Rastrigin	2300

No.	Component Functions	Global Optimum
F24	Ackley; Elliptic; Griewank; Rastrigin	2400
F25	Rastrigin; HappyCat; Ackley; Discus; Rosenbrock	2500
F26	Schaffer F6; Schwefel; Griewank; Rosenbrock; Rastrigin	2600
F27	HGBat; Rastrigin; Schwefel; Bent Cigar; Elliptic; Schaffer F6	2700
F28	Ackley; Griewank; Discus; Rosenbrock; HappyCat; Schaffer F6	2800
F29	F15; F16; F17	2900
F30	F15; F18; F19	3000

Section 3. List of Required Libraries

3.1 List of Libraries and Versions

The following libraries and modules were used for optimization, benchmarking, and visualization in this project:

Library / Module	Purpose	Version
<code>python</code>	Core programming language	3.10+
<code>numpy</code>	Numerical computing and array operations	2.0.2
<code>matplotlib</code>	Plotting and saving convergence graphs	3.10.0
<code>cec2017</code>	CEC 2017 benchmark functions (simple, hybrid, composition)	Latest (pip)
<code>cec2017.simple</code>	Simple benchmark functions F1-F10	—
<code>cec2017.hybrid</code>	Hybrid benchmark functions F11-F20	—
<code>cec2017.composition</code>	Composition benchmark functions F21-F30	—
<code>dplldc</code>	DP-LLDE optimization algorithm (uploaded locally)	Custom
<code>google.colab.files</code>	File upload utility for loading <code>dplldc.py</code> into the Colab environment	Built-in
<code>sys</code>	Clears cached module before re-importing <code>dplldc.py</code>	Built-in

Section 4. Environment Setup and Validation

4.1 Installation

Install `cec2017`

```
In [1]: # !pip install cec2017 --quiet
```

4.2 File Validation

Import `simple.py`, `hybrid.py`, and `composition.py` python files for benchmark functions

```
In [2]: from cec2017 import simple, hybrid, composition
print('cec2017 loaded!')
```

cec2017 loaded!

Import `sys` control over warning messages for verification that `lblde.py` exists

```
In [3]: from google.colab import files
import sys

uploaded = files.upload()

if 'lblde.py' in uploaded:
    if 'lblde' in sys.modules:
        del sys.modules['lblde']
    import lblde
    from lblde import LBLDE
    print('lblde.py uploaded and imported successfully!')
else:
    print('ERROR: Please upload a file named lblde.py')
```

Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

Saving `lblde.py` to `lblde.py`

lblde.py uploaded and imported successfully!

```
In [4]: import lblde
        from lblde import LBLDE
```

Section 5. Configuration and Helper Functions

5.1 Library Imports

Import libraries `numpy`, `matplotlib`, and `cec2017` for numerical computation, visualization, and CEC 2017 benchmark functions.

```
In [5]: import numpy as np
import matplotlib.pyplot as plt
from cec2017 import simple, hybrid, composition
```

5.2 Global Parameters

Define global parameters

```
In [6]: D = 30 # Change this when testing different dimensions
```

```
In [7]: bounds = np.array([[-100.0, 100.0]] * D)
MAX_FES = 10000 * D
NP, NL, MU_CR = 100, 4, 0.35
SEED = 42
```

Store all results for summary table under `all_results` dict

```
In [8]: all_results = {}
```

5.3 Function Definitions

Define `make_wrapper` to adapt CEC 2017 functions to accept a 1D input array from DP-LLDE, returning a large penalty value if the output is invalid (nan/inf).

```
In [9]: def make_wrapper(fn):
        """Wrap a cec2017 function (expects (1,D) input) to accept (D,) input."""
        def wrapper(x):
            val = fn(x.reshape(1, -1))
            if hasattr(val, '__len__'):
                val = float(val[0])
            else:
                val = float(val)
            return val if np.isfinite(val) else 1e18
        return wrapper
```

Define `run_and_plot` to execute a single DP-LLDE optimization run on a given benchmark function, print the results, store them for the summary table, and save the convergence graph.

```
In [10]: def run_and_plot(fn_name, fn_eval, f_global, fn_label):
    """Run LBLDE_s on a function and plot convergence + diversity graphs."""
    print('=' * 80)
    print(f'LBLDE_s on CEC 2017 {fn_name} – Single Run')
    print('=' * 80)
    print(f'Function      : {fn_label} ({D}D)')
    print(f'Bounds        : [-100, 100]')
    print(f'Global optimum : {f_global}')

    opt = LBLDE(
        objective_func=fn_eval,
        bounds=bounds,
        NP=NP, NL=NL, mu_CR_ini=MU_CR,
        max_fes=MAX_FES,
        seed=SEED
    )
    # ← Unpack 4 values now
    best_sol, best_fit, history, diversity_history = opt.optimize(verbose=True)

    error = abs(best_fit - f_global)
    print(f'\n{"="*80}')
    print('Results:')
    print(f'  Best Fitness      : {best_fit:.6e}')
    print(f'  Global Optimum    : {f_global:.6e}')
    print(f'  Error (|best-optimum|) : {error:.6e}')
    print(f'  Best Solution (5 dim) : {best_sol[:5]}')
    print(f'  D^(0) (init. diversity): {opt.D_0:.4e}')
    print(f'{"="*80}')

    # Store for summary
    all_results[fn_name] = {
        'label': fn_label,
        'best_fit': best_fit,
        'f_global': f_global,
        'error': error
    }

    fes_axis = np.linspace(0, MAX_FES, len(history))
    gen_axis = np.arange(len(diversity_history)) # Generations (0, 1, 2, ...)

    # — Two-panel figure —
    fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(10, 10))

    # Convergence curve (top) – FEs on x-axis
    ax1.plot(fes_axis, history, color='steelblue', linewidth=1.5)
    ax1.set_ylabel('Best Fitness', fontsize=12)
    ax1.set_xlabel('Function Evaluations (FEs)', fontsize=12)
    ax1.set_title(
        f'LBLDE – CEC 2017 {fn_name}: {fn_label} ({D}D)',
```



```

        fontsize=12
    )
    ax1.grid(True, which='both', linestyle='--', alpha=0.6)

    # Diversity curve (bottom) – Generations on x-axis
    ax2.plot(gen_axis, diversity_history, color='darkorange', linewidth=1.5)
    ax2.set_xlabel('Generation', fontsize=12)
    ax2.set_ylabel('Population Diversity\n(Mean Std Dev)', fontsize=12)
    ax2.set_title(f'Diversity Across Generations – {fn_name}', fontsize=12)
    ax2.grid(True, which='both', linestyle='--', alpha=0.6)

    plt.tight_layout()
    fname = f'convergence_{fn_name}.png'
    plt.savefig(fname, dpi=150)
    plt.show()
    print(f'Graph saved as {fname}\n')

```

Section 6. CEC2017 Benchmark Functions

6.1 Unimodal Functions (F1-F3)

```

In [11]: # =====
# F1 – Shifted and Rotated Bent Cigar
# =====
run_and_plot('F1', make_wrapper(simple.f1), 100.0, 'Shifted and Rotated Bent C

```

```

=====
=
LBLDE_s on CEC 2017 F1 – Single Run
=====
=
Function      : Shifted and Rotated Bent Cigar (30D)
Bounds       : [-100, 100]
Global optimum : 100.0
Initial D^(0) = 5.7976e+01
Generation 0: Best Fitness = 1.189140e+12 | Diversity = 5.7976e+01
Generation 10: Best = 2.519524e+11 FES = 1100 Diversity = 3.1837e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 1.016543e+11 FES = 2100 Diversity = 1.7915e+01 k_t
= [3, 3, 2, 4]
Generation 30: Best = 2.664432e+10 FES = 3100 Diversity = 1.0141e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 1.148318e+10 FES = 4100 Diversity = 6.0867e+00 k_t
= [3, 4, 3, 4]
Generation 50: Best = 4.103345e+09 FES = 5100 Diversity = 3.6563e+00 k_t
= [1, 1, 1, 1]
Generation 60: Best = 1.344250e+09 FES = 6100 Diversity = 2.1093e+00 k_t
= [2, 3, 4, 3]
Generation 70: Best = 4.448440e+08 FES = 7100 Diversity = 1.3033e+00 k_t
= [1, 1, 1, 1]
Generation 80: Best = 1.484474e+08 FES = 8100 Diversity = 8.0483e-01 k_t
= [4, 3, 4, 3]
Generation 90: Best = 6.357003e+07 FES = 9100 Diversity = 4.8927e-01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 2.396250e+07 FES = 10100 Diversity = 2.8213e-01 k_t
= [2, 4, 4, 3]
Generation 110: Best = 1.287650e+07 FES = 11100 Diversity = 1.7455e-01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 5.736231e+06 FES = 12100 Diversity = 1.1381e-01 k_t
= [3, 3, 3, 4]
Generation 130: Best = 2.312395e+06 FES = 13100 Diversity = 8.6685e-02 k_t
= [1, 1, 1, 1]
Generation 140: Best = 1.029870e+06 FES = 14100 Diversity = 6.6261e-02 k_t
= [2, 3, 3, 3]
Generation 150: Best = 5.265868e+05 FES = 15100 Diversity = 4.0268e-02 k_t
= [1, 1, 1, 1]
Generation 160: Best = 2.630268e+05 FES = 16100 Diversity = 2.7790e-02 k_t
= [4, 2, 3, 2]
Generation 170: Best = 1.107999e+05 FES = 17100 Diversity = 1.8886e-02 k_t
= [1, 1, 1, 1]
Generation 180: Best = 4.228909e+04 FES = 18100 Diversity = 1.2251e-02 k_t
= [2, 4, 3, 2]
Generation 190: Best = 2.057257e+04 FES = 19100 Diversity = 8.5333e-03 k_t
= [1, 1, 1, 1]
Generation 200: Best = 9.725983e+03 FES = 20100 Diversity = 5.2654e-03 k_t
= [4, 4, 4, 3]
Generation 210: Best = 5.947759e+03 FES = 21100 Diversity = 3.6041e-03 k_t
= [1, 1, 1, 1]
Generation 220: Best = 4.854109e+03 FES = 22100 Diversity = 2.3742e-03 k_t
= [2, 3, 3, 4]

```

Generation 230:	Best = 4.260000e+03	FES = 23100	Diversity = 1.6389e-03	k_t = [1, 1, 1, 1]
Generation 240:	Best = 4.081766e+03	FES = 24100	Diversity = 1.0914e-03	k_t = [4, 4, 4, 3]
Generation 250:	Best = 4.021762e+03	FES = 25100	Diversity = 7.9104e-04	k_t = [1, 1, 1, 1]
Generation 260:	Best = 3.975575e+03	FES = 26100	Diversity = 5.3362e-04	k_t = [4, 3, 4, 2]
Generation 270:	Best = 3.955125e+03	FES = 27100	Diversity = 3.9764e-04	k_t = [1, 1, 1, 1]
Generation 280:	Best = 3.949202e+03	FES = 28100	Diversity = 3.4211e-04	k_t = [4, 2, 2, 4]
Generation 290:	Best = 3.944851e+03	FES = 29100	Diversity = 2.5547e-04	k_t = [1, 1, 1, 1]
Generation 300:	Best = 3.943099e+03	FES = 30100	Diversity = 1.9669e-04	k_t = [3, 4, 3, 2]
Generation 310:	Best = 3.942111e+03	FES = 31100	Diversity = 1.7850e-04	k_t = [1, 1, 1, 1]
Generation 320:	Best = 3.941356e+03	FES = 32100	Diversity = 1.6473e-04	k_t = [4, 4, 4, 4]
Generation 330:	Best = 3.940763e+03	FES = 33100	Diversity = 1.1009e-04	k_t = [1, 1, 1, 1]
Generation 340:	Best = 3.940402e+03	FES = 34100	Diversity = 8.4391e-05	k_t = [3, 3, 2, 3]
Generation 350:	Best = 3.940165e+03	FES = 35100	Diversity = 7.6152e-05	k_t = [1, 1, 1, 1]
Generation 360:	Best = 3.939982e+03	FES = 36100	Diversity = 5.6116e-05	k_t = [4, 2, 2, 4]
Generation 370:	Best = 3.939754e+03	FES = 37100	Diversity = 3.7259e-05	k_t = [1, 1, 1, 1]
Generation 380:	Best = 3.939670e+03	FES = 38100	Diversity = 2.9679e-05	k_t = [2, 3, 3, 4]
Generation 390:	Best = 3.939535e+03	FES = 39100	Diversity = 2.5656e-05	k_t = [1, 1, 1, 1]
Generation 400:	Best = 3.939460e+03	FES = 40100	Diversity = 2.0750e-05	k_t = [4, 4, 2, 2]
Generation 410:	Best = 3.939394e+03	FES = 41100	Diversity = 2.0728e-05	k_t = [1, 1, 1, 1]
Generation 420:	Best = 3.939345e+03	FES = 42100	Diversity = 1.5742e-05	k_t = [3, 4, 2, 2]
Generation 430:	Best = 3.939301e+03	FES = 43100	Diversity = 1.5390e-05	k_t = [1, 1, 1, 1]
Generation 440:	Best = 3.939267e+03	FES = 44100	Diversity = 1.2193e-05	k_t = [3, 4, 4, 2]
Generation 450:	Best = 3.939211e+03	FES = 45100	Diversity = 1.0798e-05	k_t = [1, 1, 1, 1]
Generation 460:	Best = 3.939185e+03	FES = 46100	Diversity = 1.0279e-05	k_t = [2, 4, 3, 2]
Generation 470:	Best = 3.939160e+03	FES = 47100	Diversity = 8.6592e-06	k_t = [1, 1, 1, 1]
Generation 480:	Best = 3.939146e+03	FES = 48100	Diversity = 7.5116e-06	k_t = [4, 2, 3, 3]
Generation 490:	Best = 3.939128e+03	FES = 49100	Diversity = 6.1515e-06	k_t = [1, 1, 1, 1]

Generation 500:	Best = 3.939116e+03	FES = 50100	Diversity = 4.8492e-06	k_t
= [3, 2, 2, 4]				
Generation 510:	Best = 3.939106e+03	FES = 51100	Diversity = 4.5996e-06	k_t
= [1, 1, 1, 1]				
Generation 520:	Best = 3.939099e+03	FES = 52100	Diversity = 4.0841e-06	k_t
= [3, 3, 4, 3]				
Generation 530:	Best = 3.939086e+03	FES = 53100	Diversity = 4.5290e-06	k_t
= [1, 1, 1, 1]				
Generation 540:	Best = 3.939076e+03	FES = 54100	Diversity = 4.9628e-06	k_t
= [4, 2, 4, 2]				
Generation 550:	Best = 3.939062e+03	FES = 55100	Diversity = 4.4215e-06	k_t
= [1, 1, 1, 1]				
Generation 560:	Best = 3.939057e+03	FES = 56100	Diversity = 3.9603e-06	k_t
= [2, 4, 4, 4]				
Generation 570:	Best = 3.939046e+03	FES = 57100	Diversity = 3.6132e-06	k_t
= [1, 1, 1, 1]				
Generation 580:	Best = 3.939039e+03	FES = 58100	Diversity = 3.4153e-06	k_t
= [4, 2, 3, 4]				
Generation 590:	Best = 3.939030e+03	FES = 59100	Diversity = 3.8429e-06	k_t
= [1, 1, 1, 1]				
Generation 600:	Best = 3.939024e+03	FES = 60100	Diversity = 3.6872e-06	k_t
= [4, 2, 2, 3]				
Generation 610:	Best = 3.939018e+03	FES = 61100	Diversity = 2.9562e-06	k_t
= [1, 1, 1, 1]				
Generation 620:	Best = 3.939014e+03	FES = 62100	Diversity = 2.6078e-06	k_t
= [2, 3, 4, 2]				
Generation 630:	Best = 3.939007e+03	FES = 63100	Diversity = 2.9573e-06	k_t
= [1, 1, 1, 1]				
Generation 640:	Best = 3.938998e+03	FES = 64100	Diversity = 3.0719e-06	k_t
= [4, 4, 4, 4]				
Generation 650:	Best = 3.938993e+03	FES = 65100	Diversity = 2.9384e-06	k_t
= [1, 1, 1, 1]				
Generation 660:	Best = 3.938989e+03	FES = 66100	Diversity = 2.6222e-06	k_t
= [2, 2, 4, 4]				
Generation 670:	Best = 3.938982e+03	FES = 67100	Diversity = 2.9189e-06	k_t
= [1, 1, 1, 1]				
Generation 680:	Best = 3.938977e+03	FES = 68100	Diversity = 3.0108e-06	k_t
= [3, 4, 4, 4]				
Generation 690:	Best = 3.938970e+03	FES = 69100	Diversity = 3.0496e-06	k_t
= [1, 1, 1, 1]				
Generation 700:	Best = 3.938964e+03	FES = 70100	Diversity = 2.9427e-06	k_t
= [3, 4, 2, 4]				
Generation 710:	Best = 3.938961e+03	FES = 71100	Diversity = 2.6216e-06	k_t
= [1, 1, 1, 1]				
Generation 720:	Best = 3.938956e+03	FES = 72100	Diversity = 2.6044e-06	k_t
= [2, 3, 2, 3]				
Generation 730:	Best = 3.938953e+03	FES = 73100	Diversity = 2.2394e-06	k_t
= [1, 1, 1, 1]				
Generation 740:	Best = 3.938950e+03	FES = 74100	Diversity = 1.9972e-06	k_t
= [3, 4, 3, 2]				
Generation 750:	Best = 3.938947e+03	FES = 75100	Diversity = 1.9931e-06	k_t
= [1, 1, 1, 1]				
Generation 760:	Best = 3.938942e+03	FES = 76100	Diversity = 2.2458e-06	k_t
= [3, 2, 2, 4]				

Generation 770:	Best = 3.938937e+03	FES = 77100	Diversity = 2.8059e-06	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 3.938931e+03	FES = 78100	Diversity = 2.6606e-06	k_t
= [3, 4, 2, 4]				
Generation 790:	Best = 3.938927e+03	FES = 79100	Diversity = 2.4687e-06	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 3.938923e+03	FES = 80100	Diversity = 2.5151e-06	k_t
= [3, 2, 4, 2]				
Generation 810:	Best = 3.938915e+03	FES = 81100	Diversity = 3.1242e-06	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 3.938908e+03	FES = 82100	Diversity = 3.5270e-06	k_t
= [4, 3, 4, 2]				
Generation 830:	Best = 3.938897e+03	FES = 83100	Diversity = 4.2021e-06	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 3.938889e+03	FES = 84100	Diversity = 3.8404e-06	k_t
= [3, 4, 2, 3]				
Generation 850:	Best = 3.938879e+03	FES = 85100	Diversity = 3.9921e-06	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 3.938870e+03	FES = 86100	Diversity = 3.8203e-06	k_t
= [3, 2, 2, 4]				
Generation 870:	Best = 3.938861e+03	FES = 87100	Diversity = 3.7334e-06	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 3.938849e+03	FES = 88100	Diversity = 4.4531e-06	k_t
= [3, 3, 4, 2]				
Generation 890:	Best = 3.938842e+03	FES = 89100	Diversity = 4.1389e-06	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 3.938833e+03	FES = 90100	Diversity = 3.9810e-06	k_t
= [4, 3, 3, 3]				
Generation 910:	Best = 3.938826e+03	FES = 91100	Diversity = 3.3922e-06	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 3.938815e+03	FES = 92100	Diversity = 4.0848e-06	k_t
= [3, 4, 2, 2]				
Generation 930:	Best = 3.938807e+03	FES = 93100	Diversity = 3.6879e-06	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 3.938798e+03	FES = 94100	Diversity = 3.5749e-06	k_t
= [3, 3, 2, 2]				
Generation 950:	Best = 3.938792e+03	FES = 95100	Diversity = 3.5399e-06	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 3.938783e+03	FES = 96100	Diversity = 3.5612e-06	k_t
= [4, 3, 3, 4]				
Generation 970:	Best = 3.938770e+03	FES = 97100	Diversity = 4.1219e-06	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 3.938757e+03	FES = 98100	Diversity = 4.3368e-06	k_t
= [2, 2, 4, 4]				
Generation 990:	Best = 3.938748e+03	FES = 99100	Diversity = 4.1813e-06	k_t
= [1, 1, 1, 1]				
Generation 1000:	Best = 3.938735e+03	FES = 100100	Diversity = 4.5807e-06	k_t
= [3, 2, 4, 3]				
Generation 1010:	Best = 3.938726e+03	FES = 101100	Diversity = 4.2582e-06	k_t
= [1, 1, 1, 1]				
Generation 1020:	Best = 3.938719e+03	FES = 102100	Diversity = 3.8586e-06	k_t
= [2, 4, 3, 3]				
Generation 1030:	Best = 3.938711e+03	FES = 103100	Diversity = 3.1913e-06	k_t
= [1, 1, 1, 1]				

Generation 1040: Best = 3.938706e+03 = [4, 3, 2, 4]	FES = 104100	Diversity = 2.5479e-06	k_t
Generation 1050: Best = 3.938702e+03 = [1, 1, 1, 1]	FES = 105100	Diversity = 2.2765e-06	k_t
Generation 1060: Best = 3.938698e+03 = [3, 4, 2, 3]	FES = 106100	Diversity = 2.0098e-06	k_t
Generation 1070: Best = 3.938696e+03 = [1, 1, 1, 1]	FES = 107100	Diversity = 2.1210e-06	k_t
Generation 1080: Best = 3.938691e+03 = [3, 2, 4, 2]	FES = 108100	Diversity = 2.0806e-06	k_t
Generation 1090: Best = 3.938687e+03 = [1, 1, 1, 1]	FES = 109100	Diversity = 2.4097e-06	k_t
Generation 1100: Best = 3.938680e+03 = [2, 2, 3, 3]	FES = 110100	Diversity = 2.8620e-06	k_t
Generation 1110: Best = 3.938675e+03 = [1, 1, 1, 1]	FES = 111100	Diversity = 2.2501e-06	k_t
Generation 1120: Best = 3.938671e+03 = [4, 4, 3, 2]	FES = 112100	Diversity = 2.4465e-06	k_t
Generation 1130: Best = 3.938668e+03 = [1, 1, 1, 1]	FES = 113100	Diversity = 1.9921e-06	k_t
Generation 1140: Best = 3.938663e+03 = [2, 3, 2, 3]	FES = 114100	Diversity = 2.1291e-06	k_t
Generation 1150: Best = 3.938658e+03 = [1, 1, 1, 1]	FES = 115100	Diversity = 2.4501e-06	k_t
Generation 1160: Best = 3.938655e+03 = [3, 3, 3, 3]	FES = 116100	Diversity = 2.2401e-06	k_t
Generation 1170: Best = 3.938653e+03 = [1, 1, 1, 1]	FES = 117100	Diversity = 1.7117e-06	k_t
Generation 1180: Best = 3.938651e+03 = [2, 4, 4, 3]	FES = 118100	Diversity = 1.6725e-06	k_t
Generation 1190: Best = 3.938648e+03 = [1, 1, 1, 1]	FES = 119100	Diversity = 1.7632e-06	k_t
Generation 1200: Best = 3.938645e+03 = [3, 2, 2, 4]	FES = 120100	Diversity = 1.7778e-06	k_t
Generation 1210: Best = 3.938642e+03 = [1, 1, 1, 1]	FES = 121100	Diversity = 1.9440e-06	k_t
Generation 1220: Best = 3.938636e+03 = [2, 2, 2, 3]	FES = 122100	Diversity = 2.7301e-06	k_t
Generation 1230: Best = 3.938630e+03 = [1, 1, 1, 1]	FES = 123100	Diversity = 2.6671e-06	k_t
Generation 1240: Best = 3.938625e+03 = [3, 4, 3, 2]	FES = 124100	Diversity = 2.6632e-06	k_t
Generation 1250: Best = 3.938617e+03 = [1, 1, 1, 1]	FES = 125100	Diversity = 2.8489e-06	k_t
Generation 1260: Best = 3.938612e+03 = [4, 4, 4, 3]	FES = 126100	Diversity = 2.8664e-06	k_t
Generation 1270: Best = 3.938610e+03 = [1, 1, 1, 1]	FES = 127100	Diversity = 2.2268e-06	k_t
Generation 1280: Best = 3.938605e+03 = [3, 3, 2, 4]	FES = 128100	Diversity = 2.1497e-06	k_t
Generation 1290: Best = 3.938601e+03 = [1, 1, 1, 1]	FES = 129100	Diversity = 2.3239e-06	k_t
Generation 1300: Best = 3.938595e+03 = [4, 2, 3, 4]	FES = 130100	Diversity = 2.5910e-06	k_t

Generation 1310: Best = 3.938591e+03 = [1, 1, 1, 1]	FES = 131100	Diversity = 2.2179e-06	k_t
Generation 1320: Best = 3.938589e+03 = [2, 4, 4, 3]	FES = 132100	Diversity = 1.7698e-06	k_t
Generation 1330: Best = 3.938586e+03 = [1, 1, 1, 1]	FES = 133100	Diversity = 1.8653e-06	k_t
Generation 1340: Best = 3.938582e+03 = [2, 2, 3, 4]	FES = 134100	Diversity = 1.8514e-06	k_t
Generation 1350: Best = 3.938581e+03 = [1, 1, 1, 1]	FES = 135100	Diversity = 1.4264e-06	k_t
Generation 1360: Best = 3.938577e+03 = [3, 3, 3, 2]	FES = 136100	Diversity = 1.7832e-06	k_t
Generation 1370: Best = 3.938562e+03 = [1, 1, 1, 1]	FES = 137100	Diversity = 4.3604e-06	k_t
Generation 1380: Best = 3.938544e+03 = [2, 3, 3, 3]	FES = 138100	Diversity = 5.1221e-06	k_t
Generation 1390: Best = 3.938527e+03 = [1, 1, 1, 1]	FES = 139100	Diversity = 4.7704e-06	k_t
Generation 1400: Best = 3.938513e+03 = [2, 2, 3, 2]	FES = 140100	Diversity = 5.0330e-06	k_t
Generation 1410: Best = 3.938504e+03 = [1, 1, 1, 1]	FES = 141100	Diversity = 3.4589e-06	k_t
Generation 1420: Best = 3.938496e+03 = [2, 2, 3, 4]	FES = 142100	Diversity = 3.3943e-06	k_t
Generation 1430: Best = 3.938488e+03 = [1, 1, 1, 1]	FES = 143100	Diversity = 3.4030e-06	k_t
Generation 1440: Best = 3.938478e+03 = [2, 2, 2, 2]	FES = 144100	Diversity = 3.3380e-06	k_t
Generation 1450: Best = 3.938471e+03 = [1, 1, 1, 1]	FES = 145100	Diversity = 3.1193e-06	k_t
Generation 1460: Best = 3.938461e+03 = [3, 3, 4, 3]	FES = 146100	Diversity = 3.4804e-06	k_t
Generation 1470: Best = 3.938448e+03 = [1, 1, 1, 1]	FES = 147100	Diversity = 4.3848e-06	k_t
Generation 1480: Best = 3.938434e+03 = [2, 3, 4, 4]	FES = 148100	Diversity = 3.8745e-06	k_t
Generation 1490: Best = 3.938426e+03 = [1, 1, 1, 1]	FES = 149100	Diversity = 3.0499e-06	k_t
Generation 1500: Best = 3.938418e+03 = [2, 4, 2, 4]	FES = 150100	Diversity = 3.0996e-06	k_t
Generation 1510: Best = 3.938406e+03 = [1, 1, 1, 1]	FES = 151100	Diversity = 3.6733e-06	k_t
Generation 1520: Best = 3.938391e+03 = [2, 4, 2, 3]	FES = 152100	Diversity = 4.4949e-06	k_t
Generation 1530: Best = 3.938375e+03 = [1, 1, 1, 1]	FES = 153100	Diversity = 4.9858e-06	k_t
Generation 1540: Best = 3.938358e+03 = [4, 4, 4, 4]	FES = 154100	Diversity = 5.0420e-06	k_t
Generation 1550: Best = 3.938345e+03 = [1, 1, 1, 1]	FES = 155100	Diversity = 3.8868e-06	k_t
Generation 1560: Best = 3.938334e+03 = [4, 3, 3, 4]	FES = 156100	Diversity = 3.3967e-06	k_t
Generation 1570: Best = 3.938324e+03 = [1, 1, 1, 1]	FES = 157100	Diversity = 3.3425e-06	k_t

Generation 1580: Best = 3.938315e+03 = [3, 3, 2, 2]	FES = 158100	Diversity = 3.1117e-06	k_t
Generation 1590: Best = 3.938304e+03 = [1, 1, 1, 1]	FES = 159100	Diversity = 3.2581e-06	k_t
Generation 1600: Best = 3.938294e+03 = [2, 2, 2, 4]	FES = 160100	Diversity = 3.0788e-06	k_t
Generation 1610: Best = 3.938287e+03 = [1, 1, 1, 1]	FES = 161100	Diversity = 2.7543e-06	k_t
Generation 1620: Best = 3.938283e+03 = [4, 3, 3, 4]	FES = 162100	Diversity = 2.1710e-06	k_t
Generation 1630: Best = 3.938279e+03 = [1, 1, 1, 1]	FES = 163100	Diversity = 1.9892e-06	k_t
Generation 1640: Best = 3.938275e+03 = [3, 4, 4, 3]	FES = 164100	Diversity = 2.1917e-06	k_t
Generation 1650: Best = 3.938270e+03 = [1, 1, 1, 1]	FES = 165100	Diversity = 2.2843e-06	k_t
Generation 1660: Best = 3.938264e+03 = [3, 3, 2, 3]	FES = 166100	Diversity = 2.1885e-06	k_t
Generation 1670: Best = 3.938262e+03 = [1, 1, 1, 1]	FES = 167100	Diversity = 1.7957e-06	k_t
Generation 1680: Best = 3.938259e+03 = [3, 3, 3, 2]	FES = 168100	Diversity = 1.6687e-06	k_t
Generation 1690: Best = 3.938256e+03 = [1, 1, 1, 1]	FES = 169100	Diversity = 1.9652e-06	k_t
Generation 1700: Best = 3.938250e+03 = [2, 2, 4, 3]	FES = 170100	Diversity = 2.3535e-06	k_t
Generation 1710: Best = 3.938242e+03 = [1, 1, 1, 1]	FES = 171100	Diversity = 2.9110e-06	k_t
Generation 1720: Best = 3.938233e+03 = [2, 2, 3, 2]	FES = 172100	Diversity = 2.9112e-06	k_t
Generation 1730: Best = 3.938224e+03 = [1, 1, 1, 1]	FES = 173100	Diversity = 3.4251e-06	k_t
Generation 1740: Best = 3.938205e+03 = [3, 3, 3, 2]	FES = 174100	Diversity = 4.5561e-06	k_t
Generation 1750: Best = 3.938184e+03 = [1, 1, 1, 1]	FES = 175100	Diversity = 4.9048e-06	k_t
Generation 1760: Best = 3.938175e+03 = [4, 4, 2, 2]	FES = 176100	Diversity = 4.1787e-06	k_t
Generation 1770: Best = 3.938161e+03 = [1, 1, 1, 1]	FES = 177100	Diversity = 3.8962e-06	k_t
Generation 1780: Best = 3.938150e+03 = [2, 2, 4, 3]	FES = 178100	Diversity = 3.6599e-06	k_t
Generation 1790: Best = 3.938141e+03 = [1, 1, 1, 1]	FES = 179100	Diversity = 3.4168e-06	k_t
Generation 1800: Best = 3.938136e+03 = [3, 3, 2, 3]	FES = 180100	Diversity = 2.1830e-06	k_t
Generation 1810: Best = 3.938131e+03 = [1, 1, 1, 1]	FES = 181100	Diversity = 2.1312e-06	k_t
Generation 1820: Best = 3.938127e+03 = [4, 3, 3, 3]	FES = 182100	Diversity = 1.9591e-06	k_t
Generation 1830: Best = 3.938121e+03 = [1, 1, 1, 1]	FES = 183100	Diversity = 1.9013e-06	k_t
Generation 1840: Best = 3.938117e+03 = [4, 4, 4, 4]	FES = 184100	Diversity = 2.1399e-06	k_t

Generation 1850: Best = 3.938112e+03 = [1, 1, 1, 1]	FES = 185100	Diversity = 2.2592e-06	k_t
Generation 1860: Best = 3.938107e+03 = [2, 4, 2, 2]	FES = 186100	Diversity = 2.3482e-06	k_t
Generation 1870: Best = 3.938101e+03 = [1, 1, 1, 1]	FES = 187100	Diversity = 1.9388e-06	k_t
Generation 1880: Best = 3.938097e+03 = [3, 3, 2, 4]	FES = 188100	Diversity = 1.9790e-06	k_t
Generation 1890: Best = 3.938091e+03 = [1, 1, 1, 1]	FES = 189100	Diversity = 1.9794e-06	k_t
Generation 1900: Best = 3.938089e+03 = [3, 4, 4, 3]	FES = 190100	Diversity = 1.3930e-06	k_t
Generation 1910: Best = 3.938086e+03 = [1, 1, 1, 1]	FES = 191100	Diversity = 1.3421e-06	k_t
Generation 1920: Best = 3.938083e+03 = [4, 4, 3, 4]	FES = 192100	Diversity = 1.3910e-06	k_t
Generation 1930: Best = 3.938079e+03 = [1, 1, 1, 1]	FES = 193100	Diversity = 1.9955e-06	k_t
Generation 1940: Best = 3.938075e+03 = [2, 4, 2, 2]	FES = 194100	Diversity = 1.8530e-06	k_t
Generation 1950: Best = 3.938070e+03 = [1, 1, 1, 1]	FES = 195100	Diversity = 2.0289e-06	k_t
Generation 1960: Best = 3.938065e+03 = [3, 2, 2, 3]	FES = 196100	Diversity = 2.1339e-06	k_t
Generation 1970: Best = 3.938059e+03 = [1, 1, 1, 1]	FES = 197100	Diversity = 1.9031e-06	k_t
Generation 1980: Best = 3.938055e+03 = [4, 2, 2, 4]	FES = 198100	Diversity = 2.0390e-06	k_t
Generation 1990: Best = 3.938051e+03 = [1, 1, 1, 1]	FES = 199100	Diversity = 1.9242e-06	k_t
Generation 2000: Best = 3.938046e+03 = [2, 4, 3, 3]	FES = 200100	Diversity = 1.9676e-06	k_t
Generation 2010: Best = 3.938041e+03 = [1, 1, 1, 1]	FES = 201100	Diversity = 1.9239e-06	k_t
Generation 2020: Best = 3.938038e+03 = [2, 2, 3, 3]	FES = 202100	Diversity = 1.5132e-06	k_t
Generation 2030: Best = 3.938037e+03 = [1, 1, 1, 1]	FES = 203100	Diversity = 1.1397e-06	k_t
Generation 2040: Best = 3.938035e+03 = [3, 3, 4, 2]	FES = 204100	Diversity = 1.1539e-06	k_t
Generation 2050: Best = 3.938033e+03 = [1, 1, 1, 1]	FES = 205100	Diversity = 1.0799e-06	k_t
Generation 2060: Best = 3.938032e+03 = [4, 2, 2, 3]	FES = 206100	Diversity = 1.1663e-06	k_t
Generation 2070: Best = 3.938030e+03 = [1, 1, 1, 1]	FES = 207100	Diversity = 1.3376e-06	k_t
Generation 2080: Best = 3.938029e+03 = [4, 2, 4, 4]	FES = 208100	Diversity = 9.9854e-07	k_t
Generation 2090: Best = 3.938028e+03 = [1, 1, 1, 1]	FES = 209100	Diversity = 7.6895e-07	k_t
Generation 2100: Best = 3.938027e+03 = [4, 3, 2, 3]	FES = 210100	Diversity = 7.9002e-07	k_t
Generation 2110: Best = 3.938025e+03 = [1, 1, 1, 1]	FES = 211100	Diversity = 9.8375e-07	k_t

Generation 2120: Best = 3.938023e+03 = [3, 3, 2, 4]	FES = 212100	Diversity = 1.2378e-06	k_t
Generation 2130: Best = 3.938022e+03 = [1, 1, 1, 1]	FES = 213100	Diversity = 1.1651e-06	k_t
Generation 2140: Best = 3.938020e+03 = [2, 4, 4, 2]	FES = 214100	Diversity = 1.1279e-06	k_t
Generation 2150: Best = 3.938018e+03 = [1, 1, 1, 1]	FES = 215100	Diversity = 1.0713e-06	k_t
Generation 2160: Best = 3.938017e+03 = [3, 4, 2, 3]	FES = 216100	Diversity = 1.0896e-06	k_t
Generation 2170: Best = 3.938016e+03 = [1, 1, 1, 1]	FES = 217100	Diversity = 9.2459e-07	k_t
Generation 2180: Best = 3.938014e+03 = [3, 3, 4, 3]	FES = 218100	Diversity = 1.0928e-06	k_t
Generation 2190: Best = 3.938012e+03 = [1, 1, 1, 1]	FES = 219100	Diversity = 1.0886e-06	k_t
Generation 2200: Best = 3.938010e+03 = [2, 3, 2, 4]	FES = 220100	Diversity = 1.1229e-06	k_t
Generation 2210: Best = 3.938008e+03 = [1, 1, 1, 1]	FES = 221100	Diversity = 1.2205e-06	k_t
Generation 2220: Best = 3.938005e+03 = [3, 3, 4, 4]	FES = 222100	Diversity = 1.3679e-06	k_t
Generation 2230: Best = 3.938002e+03 = [1, 1, 1, 1]	FES = 223100	Diversity = 1.3393e-06	k_t
Generation 2240: Best = 3.938001e+03 = [2, 3, 2, 2]	FES = 224100	Diversity = 1.2722e-06	k_t
Generation 2250: Best = 3.937998e+03 = [1, 1, 1, 1]	FES = 225100	Diversity = 1.3053e-06	k_t
Generation 2260: Best = 3.937996e+03 = [3, 2, 4, 2]	FES = 226100	Diversity = 1.2710e-06	k_t
Generation 2270: Best = 3.937993e+03 = [1, 1, 1, 1]	FES = 227100	Diversity = 1.2800e-06	k_t
Generation 2280: Best = 3.937991e+03 = [4, 3, 3, 4]	FES = 228100	Diversity = 1.3340e-06	k_t
Generation 2290: Best = 3.937989e+03 = [1, 1, 1, 1]	FES = 229100	Diversity = 1.2596e-06	k_t
Generation 2300: Best = 3.937986e+03 = [3, 4, 2, 3]	FES = 230100	Diversity = 1.3141e-06	k_t
Generation 2310: Best = 3.937984e+03 = [1, 1, 1, 1]	FES = 231100	Diversity = 1.2911e-06	k_t
Generation 2320: Best = 3.937982e+03 = [2, 3, 4, 4]	FES = 232100	Diversity = 1.3198e-06	k_t
Generation 2330: Best = 3.937980e+03 = [1, 1, 1, 1]	FES = 233100	Diversity = 1.1955e-06	k_t
Generation 2340: Best = 3.937978e+03 = [4, 4, 3, 2]	FES = 234100	Diversity = 1.2441e-06	k_t
Generation 2350: Best = 3.937976e+03 = [1, 1, 1, 1]	FES = 235100	Diversity = 1.2093e-06	k_t
Generation 2360: Best = 3.937974e+03 = [3, 4, 3, 4]	FES = 236100	Diversity = 1.1644e-06	k_t
Generation 2370: Best = 3.937973e+03 = [1, 1, 1, 1]	FES = 237100	Diversity = 1.0187e-06	k_t
Generation 2380: Best = 3.937971e+03 = [2, 2, 3, 3]	FES = 238100	Diversity = 1.0792e-06	k_t

Generation 2390: Best = 3.937969e+03 = [1, 1, 1, 1]	FES = 239100	Diversity = 1.5257e-06	k_t
Generation 2400: Best = 3.937965e+03 = [2, 4, 2, 2]	FES = 240100	Diversity = 1.5838e-06	k_t
Generation 2410: Best = 3.937961e+03 = [1, 1, 1, 1]	FES = 241100	Diversity = 1.7601e-06	k_t
Generation 2420: Best = 3.937957e+03 = [2, 4, 4, 3]	FES = 242100	Diversity = 1.9793e-06	k_t
Generation 2430: Best = 3.937953e+03 = [1, 1, 1, 1]	FES = 243100	Diversity = 1.4629e-06	k_t
Generation 2440: Best = 3.937951e+03 = [4, 3, 3, 2]	FES = 244100	Diversity = 1.5883e-06	k_t
Generation 2450: Best = 3.937948e+03 = [1, 1, 1, 1]	FES = 245100	Diversity = 1.5285e-06	k_t
Generation 2460: Best = 3.937942e+03 = [4, 3, 4, 4]	FES = 246100	Diversity = 2.0926e-06	k_t
Generation 2470: Best = 3.937935e+03 = [1, 1, 1, 1]	FES = 247100	Diversity = 2.5662e-06	k_t
Generation 2480: Best = 3.937929e+03 = [4, 3, 3, 3]	FES = 248100	Diversity = 2.5373e-06	k_t
Generation 2490: Best = 3.937916e+03 = [1, 1, 1, 1]	FES = 249100	Diversity = 3.3857e-06	k_t
Generation 2500: Best = 3.937904e+03 = [3, 4, 2, 2]	FES = 250100	Diversity = 3.2077e-06	k_t
Generation 2510: Best = 3.937894e+03 = [1, 1, 1, 1]	FES = 251100	Diversity = 2.9427e-06	k_t
Generation 2520: Best = 3.937885e+03 = [3, 4, 3, 4]	FES = 252100	Diversity = 3.2064e-06	k_t
Generation 2530: Best = 3.937876e+03 = [1, 1, 1, 1]	FES = 253100	Diversity = 3.0704e-06	k_t
Generation 2540: Best = 3.937866e+03 = [4, 4, 3, 4]	FES = 254100	Diversity = 3.5739e-06	k_t
Generation 2550: Best = 3.937859e+03 = [1, 1, 1, 1]	FES = 255100	Diversity = 2.7448e-06	k_t
Generation 2560: Best = 3.937852e+03 = [4, 4, 2, 4]	FES = 256100	Diversity = 2.2177e-06	k_t
Generation 2570: Best = 3.937847e+03 = [1, 1, 1, 1]	FES = 257100	Diversity = 2.1368e-06	k_t
Generation 2580: Best = 3.937843e+03 = [2, 3, 3, 4]	FES = 258100	Diversity = 1.8515e-06	k_t
Generation 2590: Best = 3.937841e+03 = [1, 1, 1, 1]	FES = 259100	Diversity = 1.2491e-06	k_t
Generation 2600: Best = 3.937838e+03 = [4, 2, 3, 4]	FES = 260100	Diversity = 1.4141e-06	k_t
Generation 2610: Best = 3.937836e+03 = [1, 1, 1, 1]	FES = 261100	Diversity = 1.1769e-06	k_t
Generation 2620: Best = 3.937834e+03 = [4, 4, 3, 2]	FES = 262100	Diversity = 1.4640e-06	k_t
Generation 2630: Best = 3.937829e+03 = [1, 1, 1, 1]	FES = 263100	Diversity = 2.3708e-06	k_t
Generation 2640: Best = 3.937819e+03 = [3, 3, 2, 3]	FES = 264100	Diversity = 3.1748e-06	k_t
Generation 2650: Best = 3.937803e+03 = [1, 1, 1, 1]	FES = 265100	Diversity = 3.4886e-06	k_t

Generation 2660: Best = 3.937787e+03 = [3, 2, 4, 2]	FES = 266100	Diversity = 4.2683e-06	k_t
Generation 2670: Best = 3.937766e+03 = [1, 1, 1, 1]	FES = 267100	Diversity = 4.1858e-06	k_t
Generation 2680: Best = 3.937754e+03 = [3, 2, 4, 4]	FES = 268100	Diversity = 3.5446e-06	k_t
Generation 2690: Best = 3.937740e+03 = [1, 1, 1, 1]	FES = 269100	Diversity = 4.1481e-06	k_t
Generation 2700: Best = 3.937728e+03 = [3, 2, 2, 2]	FES = 270100	Diversity = 3.2127e-06	k_t
Generation 2710: Best = 3.937721e+03 = [1, 1, 1, 1]	FES = 271100	Diversity = 2.3918e-06	k_t
Generation 2720: Best = 3.937715e+03 = [3, 2, 4, 4]	FES = 272100	Diversity = 2.5796e-06	k_t
Generation 2730: Best = 3.937703e+03 = [1, 1, 1, 1]	FES = 273100	Diversity = 2.9316e-06	k_t
Generation 2740: Best = 3.937690e+03 = [3, 4, 4, 3]	FES = 274100	Diversity = 3.5398e-06	k_t
Generation 2750: Best = 3.937677e+03 = [1, 1, 1, 1]	FES = 275100	Diversity = 3.6404e-06	k_t
Generation 2760: Best = 3.937664e+03 = [4, 2, 2, 4]	FES = 276100	Diversity = 4.3828e-06	k_t
Generation 2770: Best = 3.937655e+03 = [1, 1, 1, 1]	FES = 277100	Diversity = 3.0074e-06	k_t
Generation 2780: Best = 3.937647e+03 = [3, 2, 4, 4]	FES = 278100	Diversity = 2.4829e-06	k_t
Generation 2790: Best = 3.937643e+03 = [1, 1, 1, 1]	FES = 279100	Diversity = 1.8986e-06	k_t
Generation 2800: Best = 3.937638e+03 = [3, 2, 2, 4]	FES = 280100	Diversity = 1.8403e-06	k_t
Generation 2810: Best = 3.937632e+03 = [1, 1, 1, 1]	FES = 281100	Diversity = 1.9149e-06	k_t
Generation 2820: Best = 3.937628e+03 = [3, 3, 2, 2]	FES = 282100	Diversity = 1.8330e-06	k_t
Generation 2830: Best = 3.937623e+03 = [1, 1, 1, 1]	FES = 283100	Diversity = 2.0881e-06	k_t
Generation 2840: Best = 3.937619e+03 = [4, 3, 2, 4]	FES = 284100	Diversity = 2.0074e-06	k_t
Generation 2850: Best = 3.937614e+03 = [1, 1, 1, 1]	FES = 285100	Diversity = 2.0464e-06	k_t
Generation 2860: Best = 3.937610e+03 = [3, 4, 3, 3]	FES = 286100	Diversity = 2.2732e-06	k_t
Generation 2870: Best = 3.937605e+03 = [1, 1, 1, 1]	FES = 287100	Diversity = 2.0881e-06	k_t
Generation 2880: Best = 3.937600e+03 = [4, 2, 4, 2]	FES = 288100	Diversity = 2.0964e-06	k_t
Generation 2890: Best = 3.937596e+03 = [1, 1, 1, 1]	FES = 289100	Diversity = 1.8825e-06	k_t
Generation 2900: Best = 3.937592e+03 = [2, 4, 3, 3]	FES = 290100	Diversity = 1.8063e-06	k_t
Generation 2910: Best = 3.937589e+03 = [1, 1, 1, 1]	FES = 291100	Diversity = 1.5146e-06	k_t
Generation 2920: Best = 3.937585e+03 = [3, 4, 3, 2]	FES = 292100	Diversity = 1.7984e-06	k_t

Generation 2930: Best = 3.937581e+03 FES = 293100 Diversity = 1.8735e-06 k_t
 = [1, 1, 1, 1]
 Generation 2940: Best = 3.937573e+03 FES = 294100 Diversity = 2.5648e-06 k_t
 = [4, 3, 2, 3]
 Generation 2950: Best = 3.937566e+03 FES = 295100 Diversity = 2.5975e-06 k_t
 = [1, 1, 1, 1]
 Generation 2960: Best = 3.937559e+03 FES = 296100 Diversity = 2.2228e-06 k_t
 = [2, 2, 2, 2]
 Generation 2970: Best = 3.937556e+03 FES = 297100 Diversity = 1.7622e-06 k_t
 = [1, 1, 1, 1]
 Generation 2980: Best = 3.937552e+03 FES = 298100 Diversity = 1.7006e-06 k_t
 = [4, 2, 3, 3]
 Generation 2990: Best = 3.937549e+03 FES = 299100 Diversity = 1.3717e-06 k_t
 = [1, 1, 1, 1]

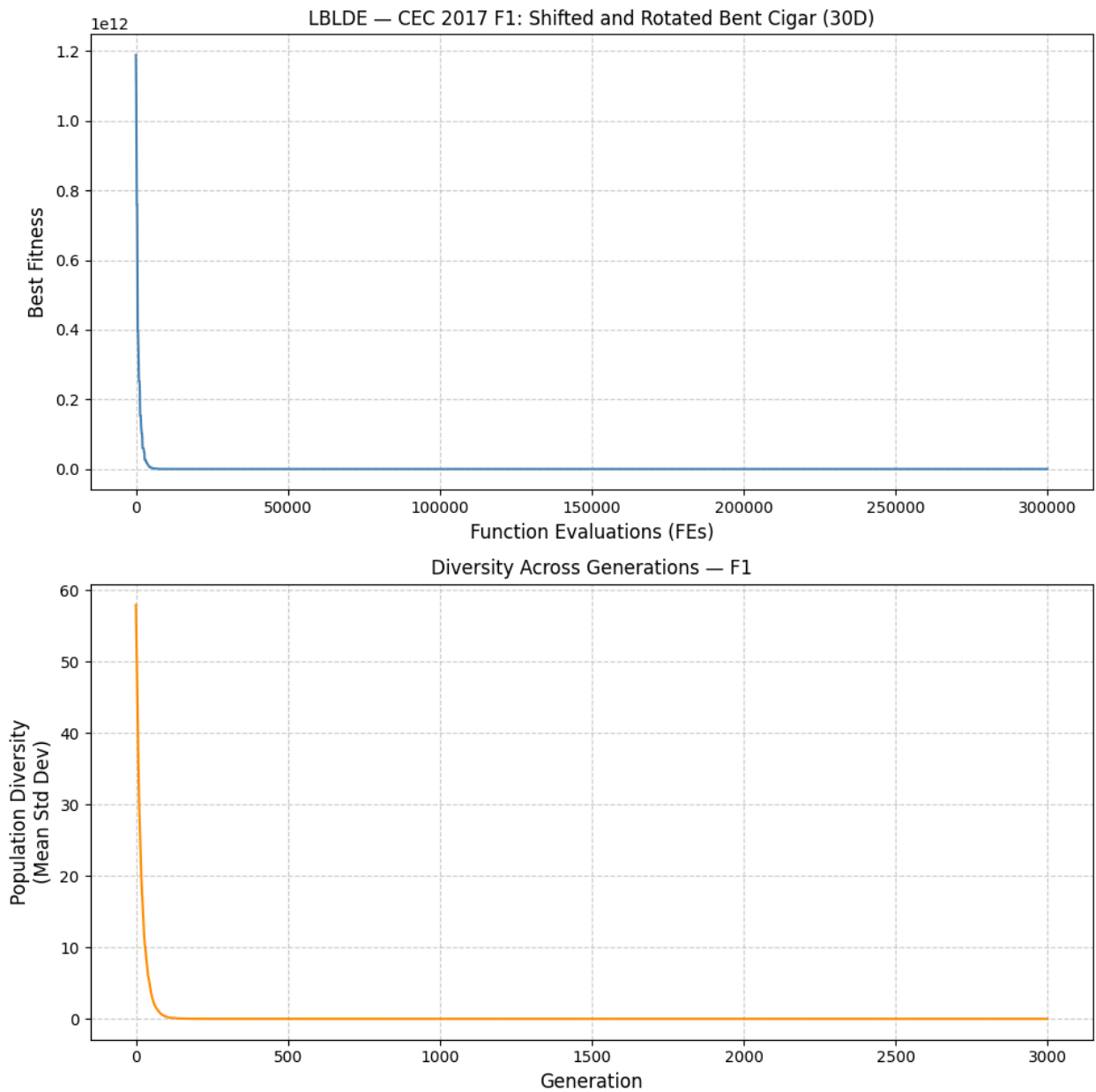
Optimization completed!

Final Best Fitness = 3.937546e+03

Total Function Evaluations = 300000

```

=====
=
Results:
  Best Fitness           : 3.937546e+03
  Global Optimum         : 1.000000e+02
  Error (|best-optimum|) : 3.837546e+03
  Best Solution (5 dim)  : [-31.94452231 -70.42956001 -29.6101798  -58.32676416
22.08960279]
  D^(0) (init. diversity): 5.7976e+01
=====
=
  
```



Graph saved as convergence_F1.png

```
In [12]: # =====
# F2 – Shifted and Rotated Sum of Different Power
# =====
run_and_plot('F2', make_wrapper(simple.f2), 200.0, 'Shifted and Rotated Sum of
```

```

=====
=
LBLDE_s on CEC 2017 F2 – Single Run
=====
=
Function      : Shifted and Rotated Sum of Different Power (Deprecated) (30D)
Bounds       : [-100, 100]
Global optimum : 200.0
WARNING: f2 has been deprecated from the CEC 2017 benchmark suite
Initial D^(0) = 5.7976e+01
Generation 0: Best Fitness = 3.044577e+48 | Diversity = 5.7976e+01
Generation 10: Best = 3.752381e+39 FES = 1100 Diversity = 4.5308e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 1.558348e+35 FES = 2100 Diversity = 3.7885e+01 k_t
= [3, 3, 2, 2]
Generation 30: Best = 6.164355e+32 FES = 3100 Diversity = 3.1059e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 1.331258e+32 FES = 4100 Diversity = 2.9076e+01 k_t
= [3, 4, 3, 4]
Generation 50: Best = 1.526693e+30 FES = 5100 Diversity = 2.4192e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 2.320369e+28 FES = 6100 Diversity = 2.0908e+01 k_t
= [4, 2, 2, 4]
Generation 70: Best = 2.330575e+27 FES = 7100 Diversity = 1.6516e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 2.716242e+26 FES = 8100 Diversity = 1.3227e+01 k_t
= [4, 4, 3, 3]
Generation 90: Best = 2.641820e+25 FES = 9100 Diversity = 1.1503e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 1.394246e+24 FES = 10100 Diversity = 1.0222e+01 k_t
= [3, 2, 4, 4]
Generation 110: Best = 4.867021e+22 FES = 11100 Diversity = 8.9581e+00 k_t
= [1, 1, 1, 1]
Generation 120: Best = 2.430425e+22 FES = 12100 Diversity = 8.2136e+00 k_t
= [3, 3, 2, 4]
Generation 130: Best = 2.482389e+21 FES = 13100 Diversity = 7.4146e+00 k_t
= [1, 1, 1, 1]
Generation 140: Best = 4.616599e+20 FES = 14100 Diversity = 6.5204e+00 k_t
= [2, 2, 3, 3]
Generation 150: Best = 1.173050e+20 FES = 15100 Diversity = 6.1633e+00 k_t
= [1, 1, 1, 1]
Generation 160: Best = 3.494328e+19 FES = 16100 Diversity = 5.9217e+00 k_t
= [2, 3, 2, 4]
Generation 170: Best = 1.490342e+19 FES = 17100 Diversity = 5.8497e+00 k_t
= [1, 1, 1, 1]
Generation 180: Best = 4.065215e+18 FES = 18100 Diversity = 5.5224e+00 k_t
= [3, 2, 4, 3]
Generation 190: Best = 3.080909e+18 FES = 19100 Diversity = 5.1701e+00 k_t
= [1, 1, 1, 1]
Generation 200: Best = 1.474483e+18 FES = 20100 Diversity = 5.0163e+00 k_t
= [3, 2, 2, 2]
Generation 210: Best = 8.498960e+17 FES = 21100 Diversity = 4.7855e+00 k_t
= [1, 1, 1, 1]
Generation 220: Best = 6.140659e+17 FES = 22100 Diversity = 4.7941e+00 k_t

```

```

= [2, 2, 4, 2]
Generation 230: Best = 3.336804e+17 FES = 23100 Diversity = 4.4777e+00 k_t
= [1, 1, 1, 1]
Generation 240: Best = 3.226037e+17 FES = 24100 Diversity = 4.3853e+00 k_t
= [4, 2, 3, 3]
Generation 250: Best = 2.312428e+17 FES = 25100 Diversity = 4.1330e+00 k_t
= [1, 1, 1, 1]
Generation 260: Best = 2.075889e+17 FES = 26100 Diversity = 4.2240e+00 k_t
= [3, 2, 4, 4]
Generation 270: Best = 1.885847e+17 FES = 27100 Diversity = 4.3310e+00 k_t
= [1, 1, 1, 1]
Generation 280: Best = 1.560441e+17 FES = 28100 Diversity = 4.3592e+00 k_t
= [3, 2, 3, 4]
Generation 290: Best = 8.194453e+16 FES = 29100 Diversity = 4.6122e+00 k_t
= [1, 1, 1, 1]
Generation 300: Best = 5.062228e+16 FES = 30100 Diversity = 4.2989e+00 k_t
= [3, 2, 2, 2]
Generation 310: Best = 3.603853e+16 FES = 31100 Diversity = 3.9779e+00 k_t
= [1, 1, 1, 1]
Generation 320: Best = 2.545331e+16 FES = 32100 Diversity = 4.0285e+00 k_t
= [3, 4, 2, 2]
Generation 330: Best = 1.748319e+16 FES = 33100 Diversity = 4.0400e+00 k_t
= [1, 1, 1, 1]
Generation 340: Best = 1.249398e+16 FES = 34100 Diversity = 3.9196e+00 k_t
= [2, 2, 2, 3]
Generation 350: Best = 9.506298e+15 FES = 35100 Diversity = 4.0372e+00 k_t
= [1, 1, 1, 1]
Generation 360: Best = 8.141985e+15 FES = 36100 Diversity = 3.6061e+00 k_t
= [2, 2, 3, 4]
Generation 370: Best = 5.474573e+15 FES = 37100 Diversity = 3.4556e+00 k_t
= [1, 1, 1, 1]
Generation 380: Best = 4.954827e+15 FES = 38100 Diversity = 3.6240e+00 k_t
= [2, 3, 3, 2]
Generation 390: Best = 4.075019e+15 FES = 39100 Diversity = 3.5421e+00 k_t
= [1, 1, 1, 1]
Generation 400: Best = 3.343795e+15 FES = 40100 Diversity = 3.4168e+00 k_t
= [3, 2, 2, 4]
Generation 410: Best = 2.388611e+15 FES = 41100 Diversity = 3.4723e+00 k_t
= [1, 1, 1, 1]
Generation 420: Best = 1.525470e+15 FES = 42100 Diversity = 3.8168e+00 k_t
= [4, 4, 4, 2]
Generation 430: Best = 1.306363e+15 FES = 43100 Diversity = 3.8409e+00 k_t
= [1, 1, 1, 1]
Generation 440: Best = 9.439105e+14 FES = 44100 Diversity = 3.8110e+00 k_t
= [2, 3, 3, 4]
Generation 450: Best = 3.556645e+14 FES = 45100 Diversity = 3.7628e+00 k_t
= [1, 1, 1, 1]
Generation 460: Best = 2.782746e+14 FES = 46100 Diversity = 3.3817e+00 k_t
= [3, 3, 4, 4]
Generation 470: Best = 2.087353e+14 FES = 47100 Diversity = 3.1637e+00 k_t
= [1, 1, 1, 1]
Generation 480: Best = 1.392265e+14 FES = 48100 Diversity = 2.9899e+00 k_t
= [3, 4, 2, 2]
Generation 490: Best = 7.650700e+13 FES = 49100 Diversity = 2.8192e+00 k_t

```



```

= [1, 1, 1, 1]
Generation 500: Best = 5.623301e+13 FES = 50100 Diversity = 2.6722e+00 k_t
= [2, 2, 3, 3]
Generation 510: Best = 4.568239e+13 FES = 51100 Diversity = 2.6442e+00 k_t
= [1, 1, 1, 1]
Generation 520: Best = 3.571982e+13 FES = 52100 Diversity = 2.8237e+00 k_t
= [3, 2, 3, 3]
Generation 530: Best = 2.824810e+13 FES = 53100 Diversity = 2.9012e+00 k_t
= [1, 1, 1, 1]
Generation 540: Best = 1.942245e+13 FES = 54100 Diversity = 2.7065e+00 k_t
= [3, 3, 3, 4]
Generation 550: Best = 1.264683e+13 FES = 55100 Diversity = 2.5655e+00 k_t
= [1, 1, 1, 1]
Generation 560: Best = 1.061196e+13 FES = 56100 Diversity = 2.5698e+00 k_t
= [2, 3, 2, 3]
Generation 570: Best = 8.615907e+12 FES = 57100 Diversity = 2.6032e+00 k_t
= [1, 1, 1, 1]
Generation 580: Best = 6.847369e+12 FES = 58100 Diversity = 2.4974e+00 k_t
= [2, 2, 2, 2]
Generation 590: Best = 5.505277e+12 FES = 59100 Diversity = 2.3422e+00 k_t
= [1, 1, 1, 1]
Generation 600: Best = 4.448961e+12 FES = 60100 Diversity = 2.3581e+00 k_t
= [4, 2, 3, 4]
Generation 610: Best = 3.944041e+12 FES = 61100 Diversity = 2.2033e+00 k_t
= [1, 1, 1, 1]
Generation 620: Best = 3.483550e+12 FES = 62100 Diversity = 2.1674e+00 k_t
= [2, 2, 4, 2]
Generation 630: Best = 3.178358e+12 FES = 63100 Diversity = 2.1488e+00 k_t
= [1, 1, 1, 1]
Generation 640: Best = 2.642331e+12 FES = 64100 Diversity = 2.1614e+00 k_t
= [3, 2, 3, 4]
Generation 650: Best = 2.237364e+12 FES = 65100 Diversity = 2.1416e+00 k_t
= [1, 1, 1, 1]
Generation 660: Best = 1.930650e+12 FES = 66100 Diversity = 2.0725e+00 k_t
= [3, 3, 3, 2]
Generation 670: Best = 1.588073e+12 FES = 67100 Diversity = 1.9348e+00 k_t
= [1, 1, 1, 1]
Generation 680: Best = 1.344368e+12 FES = 68100 Diversity = 1.8418e+00 k_t
= [3, 3, 3, 4]
Generation 690: Best = 1.131418e+12 FES = 69100 Diversity = 1.7449e+00 k_t
= [1, 1, 1, 1]
Generation 700: Best = 1.019717e+12 FES = 70100 Diversity = 1.6440e+00 k_t
= [4, 2, 4, 4]
Generation 710: Best = 9.408547e+11 FES = 71100 Diversity = 1.6071e+00 k_t
= [1, 1, 1, 1]
Generation 720: Best = 8.958948e+11 FES = 72100 Diversity = 1.5908e+00 k_t
= [4, 4, 4, 3]
Generation 730: Best = 7.737495e+11 FES = 73100 Diversity = 1.5170e+00 k_t
= [1, 1, 1, 1]
Generation 740: Best = 6.505841e+11 FES = 74100 Diversity = 1.4821e+00 k_t
= [2, 4, 2, 3]
Generation 750: Best = 6.204604e+11 FES = 75100 Diversity = 1.4012e+00 k_t
= [1, 1, 1, 1]
Generation 760: Best = 5.035978e+11 FES = 76100 Diversity = 1.4164e+00 k_t

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```

= [4, 2, 3, 3]
Generation 770: Best = 4.564043e+11  FES = 77100  Diversity = 1.3305e+00  k_t
= [1, 1, 1, 1]
Generation 780: Best = 3.078513e+11  FES = 78100  Diversity = 1.3426e+00  k_t
= [3, 4, 2, 2]
Generation 790: Best = 2.646942e+11  FES = 79100  Diversity = 1.4359e+00  k_t
= [1, 1, 1, 1]
Generation 800: Best = 2.333137e+11  FES = 80100  Diversity = 1.5147e+00  k_t
= [4, 2, 4, 4]
Generation 810: Best = 2.012939e+11  FES = 81100  Diversity = 1.3967e+00  k_t
= [1, 1, 1, 1]
Generation 820: Best = 1.853405e+11  FES = 82100  Diversity = 1.4016e+00  k_t
= [3, 4, 3, 2]
Generation 830: Best = 1.686560e+11  FES = 83100  Diversity = 1.3580e+00  k_t
= [1, 1, 1, 1]
Generation 840: Best = 1.624933e+11  FES = 84100  Diversity = 1.3482e+00  k_t
= [2, 4, 3, 2]
Generation 850: Best = 1.564216e+11  FES = 85100  Diversity = 1.2632e+00  k_t
= [1, 1, 1, 1]
Generation 860: Best = 1.518816e+11  FES = 86100  Diversity = 1.1668e+00  k_t
= [4, 3, 2, 3]
Generation 870: Best = 1.474918e+11  FES = 87100  Diversity = 1.0924e+00  k_t
= [1, 1, 1, 1]
Generation 880: Best = 1.449774e+11  FES = 88100  Diversity = 1.0986e+00  k_t
= [4, 4, 3, 2]
Generation 890: Best = 1.415604e+11  FES = 89100  Diversity = 1.0974e+00  k_t
= [1, 1, 1, 1]
Generation 900: Best = 1.391035e+11  FES = 90100  Diversity = 1.1270e+00  k_t
= [2, 3, 3, 4]
Generation 910: Best = 1.368156e+11  FES = 91100  Diversity = 1.1018e+00  k_t
= [1, 1, 1, 1]
Generation 920: Best = 1.345890e+11  FES = 92100  Diversity = 1.1444e+00  k_t
= [2, 4, 2, 4]
Generation 930: Best = 1.325787e+11  FES = 93100  Diversity = 1.0981e+00  k_t
= [1, 1, 1, 1]
Generation 940: Best = 1.320236e+11  FES = 94100  Diversity = 1.1885e+00  k_t
= [2, 2, 2, 4]
Generation 950: Best = 1.314807e+11  FES = 95100  Diversity = 1.1400e+00  k_t
= [1, 1, 1, 1]
Generation 960: Best = 1.312569e+11  FES = 96100  Diversity = 1.0391e+00  k_t
= [3, 3, 4, 3]
Generation 970: Best = 1.308399e+11  FES = 97100  Diversity = 1.0438e+00  k_t
= [1, 1, 1, 1]
Generation 980: Best = 1.307449e+11  FES = 98100  Diversity = 1.1087e+00  k_t
= [3, 2, 2, 4]
Generation 990: Best = 1.305056e+11  FES = 99100  Diversity = 1.0446e+00  k_t
= [1, 1, 1, 1]
Generation 1000: Best = 1.303030e+11  FES = 100100  Diversity = 1.0958e+00  k_t
= [4, 4, 4, 4]
Generation 1010: Best = 1.301685e+11  FES = 101100  Diversity = 1.0986e+00  k_t
= [1, 1, 1, 1]
Generation 1020: Best = 1.300793e+11  FES = 102100  Diversity = 1.0542e+00  k_t
= [4, 2, 3, 3]
Generation 1030: Best = 1.299678e+11  FES = 103100  Diversity = 1.0075e+00  k_t

```

```

= [1, 1, 1, 1]
Generation 1040: Best = 1.298888e+11  FES = 104100  Diversity = 9.9940e-01  k_t
= [3, 3, 4, 3]
Generation 1050: Best = 1.298334e+11  FES = 105100  Diversity = 9.6980e-01  k_t
= [1, 1, 1, 1]
Generation 1060: Best = 1.297726e+11  FES = 106100  Diversity = 9.7054e-01  k_t
= [3, 3, 2, 2]
Generation 1070: Best = 1.296985e+11  FES = 107100  Diversity = 9.6835e-01  k_t
= [1, 1, 1, 1]
Generation 1080: Best = 1.296272e+11  FES = 108100  Diversity = 1.0093e+00  k_t
= [2, 3, 4, 3]
Generation 1090: Best = 1.295851e+11  FES = 109100  Diversity = 1.0044e+00  k_t
= [1, 1, 1, 1]
Generation 1100: Best = 1.293668e+11  FES = 110100  Diversity = 1.0808e+00  k_t
= [3, 2, 3, 3]
Generation 1110: Best = 1.292765e+11  FES = 111100  Diversity = 1.1033e+00  k_t
= [1, 1, 1, 1]
Generation 1120: Best = 1.291824e+11  FES = 112100  Diversity = 1.0110e+00  k_t
= [2, 3, 3, 3]
Generation 1130: Best = 1.290821e+11  FES = 113100  Diversity = 9.7536e-01  k_t
= [1, 1, 1, 1]
Generation 1140: Best = 1.290154e+11  FES = 114100  Diversity = 9.8142e-01  k_t
= [3, 3, 4, 4]
Generation 1150: Best = 1.289498e+11  FES = 115100  Diversity = 1.0071e+00  k_t
= [1, 1, 1, 1]
Generation 1160: Best = 1.288290e+11  FES = 116100  Diversity = 1.0139e+00  k_t
= [3, 3, 2, 3]
Generation 1170: Best = 1.286829e+11  FES = 117100  Diversity = 1.0137e+00  k_t
= [1, 1, 1, 1]
Generation 1180: Best = 1.286188e+11  FES = 118100  Diversity = 1.0136e+00  k_t
= [4, 4, 2, 3]
Generation 1190: Best = 1.285143e+11  FES = 119100  Diversity = 1.0209e+00  k_t
= [1, 1, 1, 1]
Generation 1200: Best = 1.284052e+11  FES = 120100  Diversity = 9.9007e-01  k_t
= [2, 2, 4, 4]
Generation 1210: Best = 1.283340e+11  FES = 121100  Diversity = 9.8830e-01  k_t
= [1, 1, 1, 1]
Generation 1220: Best = 1.281888e+11  FES = 122100  Diversity = 9.6881e-01  k_t
= [2, 4, 4, 4]
Generation 1230: Best = 1.281330e+11  FES = 123100  Diversity = 1.0021e+00  k_t
= [1, 1, 1, 1]
Generation 1240: Best = 1.279614e+11  FES = 124100  Diversity = 1.0147e+00  k_t
= [2, 3, 4, 4]
Generation 1250: Best = 1.279099e+11  FES = 125100  Diversity = 1.0038e+00  k_t
= [1, 1, 1, 1]
Generation 1260: Best = 1.277536e+11  FES = 126100  Diversity = 1.0333e+00  k_t
= [4, 3, 4, 2]
Generation 1270: Best = 1.275242e+11  FES = 127100  Diversity = 1.0479e+00  k_t
= [1, 1, 1, 1]
Generation 1280: Best = 1.274620e+11  FES = 128100  Diversity = 1.0571e+00  k_t
= [4, 4, 4, 3]
Generation 1290: Best = 1.274076e+11  FES = 129100  Diversity = 1.0290e+00  k_t
= [1, 1, 1, 1]
Generation 1300: Best = 1.273159e+11  FES = 130100  Diversity = 9.9512e-01  k_t

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= [2, 3, 3, 2]
Generation 1310: Best = 1.272712e+11  FES = 131100  Diversity = 9.4086e-01  k_t
= [1, 1, 1, 1]
Generation 1320: Best = 1.271586e+11  FES = 132100  Diversity = 9.8213e-01  k_t
= [3, 4, 2, 2]
Generation 1330: Best = 1.270400e+11  FES = 133100  Diversity = 1.0399e+00  k_t
= [1, 1, 1, 1]
Generation 1340: Best = 1.270027e+11  FES = 134100  Diversity = 1.0448e+00  k_t
= [2, 3, 3, 3]
Generation 1350: Best = 1.269170e+11  FES = 135100  Diversity = 1.0575e+00  k_t
= [1, 1, 1, 1]
Generation 1360: Best = 1.268092e+11  FES = 136100  Diversity = 1.0105e+00  k_t
= [4, 4, 4, 3]
Generation 1370: Best = 1.266915e+11  FES = 137100  Diversity = 1.0101e+00  k_t
= [1, 1, 1, 1]
Generation 1380: Best = 1.265342e+11  FES = 138100  Diversity = 1.0118e+00  k_t
= [3, 3, 4, 3]
Generation 1390: Best = 1.262877e+11  FES = 139100  Diversity = 9.9311e-01  k_t
= [1, 1, 1, 1]
Generation 1400: Best = 1.259296e+11  FES = 140100  Diversity = 1.0545e+00  k_t
= [2, 2, 4, 4]
Generation 1410: Best = 1.257420e+11  FES = 141100  Diversity = 1.1362e+00  k_t
= [1, 1, 1, 1]
Generation 1420: Best = 1.256517e+11  FES = 142100  Diversity = 1.1399e+00  k_t
= [4, 2, 4, 3]
Generation 1430: Best = 1.254600e+11  FES = 143100  Diversity = 1.1237e+00  k_t
= [1, 1, 1, 1]
Generation 1440: Best = 1.252733e+11  FES = 144100  Diversity = 1.0664e+00  k_t
= [2, 4, 2, 4]
Generation 1450: Best = 1.251388e+11  FES = 145100  Diversity = 1.0753e+00  k_t
= [1, 1, 1, 1]
Generation 1460: Best = 1.250110e+11  FES = 146100  Diversity = 1.0836e+00  k_t
= [4, 2, 2, 3]
Generation 1470: Best = 1.249419e+11  FES = 147100  Diversity = 1.0957e+00  k_t
= [1, 1, 1, 1]
Generation 1480: Best = 1.247431e+11  FES = 148100  Diversity = 1.0798e+00  k_t
= [4, 3, 3, 4]
Generation 1490: Best = 1.246210e+11  FES = 149100  Diversity = 1.1167e+00  k_t
= [1, 1, 1, 1]
Generation 1500: Best = 1.244728e+11  FES = 150100  Diversity = 1.1528e+00  k_t
= [4, 3, 4, 3]
Generation 1510: Best = 1.242950e+11  FES = 151100  Diversity = 1.1456e+00  k_t
= [1, 1, 1, 1]
Generation 1520: Best = 1.241294e+11  FES = 152100  Diversity = 1.1214e+00  k_t
= [4, 4, 4, 3]
Generation 1530: Best = 1.240960e+11  FES = 153100  Diversity = 1.1532e+00  k_t
= [1, 1, 1, 1]
Generation 1540: Best = 1.240767e+11  FES = 154100  Diversity = 1.0857e+00  k_t
= [4, 3, 2, 3]
Generation 1550: Best = 1.240286e+11  FES = 155100  Diversity = 1.0654e+00  k_t
= [1, 1, 1, 1]
Generation 1560: Best = 1.239259e+11  FES = 156100  Diversity = 1.0620e+00  k_t
= [3, 4, 2, 3]
Generation 1570: Best = 1.238350e+11  FES = 157100  Diversity = 1.0748e+00  k_t

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= [1, 1, 1, 1]
Generation 1580: Best = 1.237597e+11 FES = 158100 Diversity = 1.1117e+00 k_t
= [4, 2, 2, 3]
Generation 1590: Best = 1.236683e+11 FES = 159100 Diversity = 1.0570e+00 k_t
= [1, 1, 1, 1]
Generation 1600: Best = 1.236300e+11 FES = 160100 Diversity = 1.0578e+00 k_t
= [4, 3, 2, 4]
Generation 1610: Best = 1.235899e+11 FES = 161100 Diversity = 1.0523e+00 k_t
= [1, 1, 1, 1]
Generation 1620: Best = 1.235497e+11 FES = 162100 Diversity = 1.0538e+00 k_t
= [2, 3, 4, 4]
Generation 1630: Best = 1.235197e+11 FES = 163100 Diversity = 1.0105e+00 k_t
= [1, 1, 1, 1]
Generation 1640: Best = 1.234835e+11 FES = 164100 Diversity = 1.0332e+00 k_t
= [3, 3, 2, 3]
Generation 1650: Best = 1.234670e+11 FES = 165100 Diversity = 1.0510e+00 k_t
= [1, 1, 1, 1]
Generation 1660: Best = 1.234486e+11 FES = 166100 Diversity = 1.0305e+00 k_t
= [2, 3, 3, 2]
Generation 1670: Best = 1.234350e+11 FES = 167100 Diversity = 1.0614e+00 k_t
= [1, 1, 1, 1]
Generation 1680: Best = 1.234248e+11 FES = 168100 Diversity = 1.0313e+00 k_t
= [3, 3, 3, 4]
Generation 1690: Best = 1.234138e+11 FES = 169100 Diversity = 1.0790e+00 k_t
= [1, 1, 1, 1]
Generation 1700: Best = 1.234056e+11 FES = 170100 Diversity = 1.0703e+00 k_t
= [3, 2, 2, 3]
Generation 1710: Best = 1.233924e+11 FES = 171100 Diversity = 1.0286e+00 k_t
= [1, 1, 1, 1]
Generation 1720: Best = 1.233834e+11 FES = 172100 Diversity = 1.0110e+00 k_t
= [3, 4, 4, 2]
Generation 1730: Best = 1.233715e+11 FES = 173100 Diversity = 1.0954e+00 k_t
= [1, 1, 1, 1]
Generation 1740: Best = 1.233614e+11 FES = 174100 Diversity = 1.1446e+00 k_t
= [3, 3, 4, 3]
Generation 1750: Best = 1.233440e+11 FES = 175100 Diversity = 1.1393e+00 k_t
= [1, 1, 1, 1]
Generation 1760: Best = 1.233397e+11 FES = 176100 Diversity = 1.1179e+00 k_t
= [4, 3, 2, 2]
Generation 1770: Best = 1.233345e+11 FES = 177100 Diversity = 1.0879e+00 k_t
= [1, 1, 1, 1]
Generation 1780: Best = 1.233312e+11 FES = 178100 Diversity = 1.0893e+00 k_t
= [2, 2, 3, 3]
Generation 1790: Best = 1.233285e+11 FES = 179100 Diversity = 1.0549e+00 k_t
= [1, 1, 1, 1]
Generation 1800: Best = 1.233260e+11 FES = 180100 Diversity = 1.0151e+00 k_t
= [3, 4, 2, 2]
Generation 1810: Best = 1.233249e+11 FES = 181100 Diversity = 9.4433e-01 k_t
= [1, 1, 1, 1]
Generation 1820: Best = 1.233241e+11 FES = 182100 Diversity = 8.8862e-01 k_t
= [4, 2, 4, 2]
Generation 1830: Best = 1.233233e+11 FES = 183100 Diversity = 7.8651e-01 k_t
= [1, 1, 1, 1]
Generation 1840: Best = 1.233218e+11 FES = 184100 Diversity = 8.3257e-01 k_t

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= [2, 4, 4, 4]
Generation 1850: Best = 1.233212e+11  FES = 185100  Diversity = 8.4437e-01  k_t
= [1, 1, 1, 1]
Generation 1860: Best = 1.233207e+11  FES = 186100  Diversity = 7.9713e-01  k_t
= [3, 4, 4, 4]
Generation 1870: Best = 1.233205e+11  FES = 187100  Diversity = 8.4326e-01  k_t
= [1, 1, 1, 1]
Generation 1880: Best = 1.233201e+11  FES = 188100  Diversity = 7.9402e-01  k_t
= [2, 4, 4, 3]
Generation 1890: Best = 1.233199e+11  FES = 189100  Diversity = 7.5558e-01  k_t
= [1, 1, 1, 1]
Generation 1900: Best = 1.233194e+11  FES = 190100  Diversity = 7.1104e-01  k_t
= [2, 4, 2, 4]
Generation 1910: Best = 1.233189e+11  FES = 191100  Diversity = 7.2412e-01  k_t
= [1, 1, 1, 1]
Generation 1920: Best = 1.233187e+11  FES = 192100  Diversity = 7.4449e-01  k_t
= [4, 3, 3, 4]
Generation 1930: Best = 1.233186e+11  FES = 193100  Diversity = 7.5849e-01  k_t
= [1, 1, 1, 1]
Generation 1940: Best = 1.233184e+11  FES = 194100  Diversity = 7.1200e-01  k_t
= [2, 2, 4, 4]
Generation 1950: Best = 1.233182e+11  FES = 195100  Diversity = 7.0170e-01  k_t
= [1, 1, 1, 1]
Generation 1960: Best = 1.233181e+11  FES = 196100  Diversity = 7.1314e-01  k_t
= [2, 2, 4, 2]
Generation 1970: Best = 1.233178e+11  FES = 197100  Diversity = 6.8541e-01  k_t
= [1, 1, 1, 1]
Generation 1980: Best = 1.233175e+11  FES = 198100  Diversity = 6.9540e-01  k_t
= [2, 2, 2, 4]
Generation 1990: Best = 1.233173e+11  FES = 199100  Diversity = 7.0812e-01  k_t
= [1, 1, 1, 1]
Generation 2000: Best = 1.233172e+11  FES = 200100  Diversity = 6.7189e-01  k_t
= [3, 4, 2, 4]
Generation 2010: Best = 1.233171e+11  FES = 201100  Diversity = 6.9493e-01  k_t
= [1, 1, 1, 1]
Generation 2020: Best = 1.233169e+11  FES = 202100  Diversity = 6.5517e-01  k_t
= [4, 3, 4, 2]
Generation 2030: Best = 1.233168e+11  FES = 203100  Diversity = 6.6062e-01  k_t
= [1, 1, 1, 1]
Generation 2040: Best = 1.233168e+11  FES = 204100  Diversity = 6.5490e-01  k_t
= [2, 3, 4, 2]
Generation 2050: Best = 1.233167e+11  FES = 205100  Diversity = 6.4098e-01  k_t
= [1, 1, 1, 1]
Generation 2060: Best = 1.233167e+11  FES = 206100  Diversity = 5.9505e-01  k_t
= [2, 3, 2, 3]
Generation 2070: Best = 1.233166e+11  FES = 207100  Diversity = 6.6925e-01  k_t
= [1, 1, 1, 1]
Generation 2080: Best = 1.233166e+11  FES = 208100  Diversity = 7.2594e-01  k_t
= [2, 2, 4, 2]
Generation 2090: Best = 1.233165e+11  FES = 209100  Diversity = 7.2703e-01  k_t
= [1, 1, 1, 1]
Generation 2100: Best = 1.233165e+11  FES = 210100  Diversity = 7.1959e-01  k_t
= [2, 3, 4, 3]
Generation 2110: Best = 1.233165e+11  FES = 211100  Diversity = 6.5886e-01  k_t

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= [1, 1, 1, 1]
Generation 2120: Best = 1.233165e+11  FES = 212100  Diversity = 6.2177e-01  k_t
= [4, 3, 2, 2]
Generation 2130: Best = 1.233165e+11  FES = 213100  Diversity = 6.0793e-01  k_t
= [1, 1, 1, 1]
Generation 2140: Best = 1.233164e+11  FES = 214100  Diversity = 5.8755e-01  k_t
= [4, 4, 3, 2]
Generation 2150: Best = 1.233164e+11  FES = 215100  Diversity = 5.6755e-01  k_t
= [1, 1, 1, 1]
Generation 2160: Best = 1.233164e+11  FES = 216100  Diversity = 5.7695e-01  k_t
= [2, 4, 3, 2]
Generation 2170: Best = 1.233164e+11  FES = 217100  Diversity = 6.2531e-01  k_t
= [1, 1, 1, 1]
Generation 2180: Best = 1.233164e+11  FES = 218100  Diversity = 6.6656e-01  k_t
= [2, 2, 3, 4]
Generation 2190: Best = 1.233164e+11  FES = 219100  Diversity = 6.6215e-01  k_t
= [1, 1, 1, 1]
Generation 2200: Best = 1.233163e+11  FES = 220100  Diversity = 6.2880e-01  k_t
= [4, 2, 3, 2]
Generation 2210: Best = 1.233163e+11  FES = 221100  Diversity = 6.2446e-01  k_t
= [1, 1, 1, 1]
Generation 2220: Best = 1.233163e+11  FES = 222100  Diversity = 6.2597e-01  k_t
= [3, 3, 3, 3]
Generation 2230: Best = 1.233163e+11  FES = 223100  Diversity = 6.3746e-01  k_t
= [1, 1, 1, 1]
Generation 2240: Best = 1.233162e+11  FES = 224100  Diversity = 6.3719e-01  k_t
= [4, 2, 2, 4]
Generation 2250: Best = 1.233162e+11  FES = 225100  Diversity = 5.9054e-01  k_t
= [1, 1, 1, 1]
Generation 2260: Best = 1.233162e+11  FES = 226100  Diversity = 5.9804e-01  k_t
= [2, 2, 3, 3]
Generation 2270: Best = 1.233162e+11  FES = 227100  Diversity = 5.4379e-01  k_t
= [1, 1, 1, 1]
Generation 2280: Best = 1.233162e+11  FES = 228100  Diversity = 5.6346e-01  k_t
= [4, 4, 4, 2]
Generation 2290: Best = 1.233162e+11  FES = 229100  Diversity = 5.5657e-01  k_t
= [1, 1, 1, 1]
Generation 2300: Best = 1.233162e+11  FES = 230100  Diversity = 5.4405e-01  k_t
= [2, 3, 3, 4]
Generation 2310: Best = 1.233162e+11  FES = 231100  Diversity = 5.4412e-01  k_t
= [1, 1, 1, 1]
Generation 2320: Best = 1.233162e+11  FES = 232100  Diversity = 5.0610e-01  k_t
= [4, 3, 4, 4]
Generation 2330: Best = 1.233162e+11  FES = 233100  Diversity = 4.8152e-01  k_t
= [1, 1, 1, 1]
Generation 2340: Best = 1.233162e+11  FES = 234100  Diversity = 4.7880e-01  k_t
= [4, 4, 2, 2]
Generation 2350: Best = 1.233162e+11  FES = 235100  Diversity = 4.9648e-01  k_t
= [1, 1, 1, 1]
Generation 2360: Best = 1.233162e+11  FES = 236100  Diversity = 4.9384e-01  k_t
= [3, 3, 2, 3]
Generation 2370: Best = 1.233162e+11  FES = 237100  Diversity = 5.1925e-01  k_t
= [1, 1, 1, 1]
Generation 2380: Best = 1.233162e+11  FES = 238100  Diversity = 5.0621e-01  k_t

```

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= [4, 3, 4, 2]
Generation 2390: Best = 1.233162e+11  FES = 239100  Diversity = 4.9499e-01  k_t
= [1, 1, 1, 1]
Generation 2400: Best = 1.233162e+11  FES = 240100  Diversity = 4.9470e-01  k_t
= [2, 2, 2, 3]
Generation 2410: Best = 1.233162e+11  FES = 241100  Diversity = 4.6278e-01  k_t
= [1, 1, 1, 1]
Generation 2420: Best = 1.233162e+11  FES = 242100  Diversity = 4.9240e-01  k_t
= [3, 4, 2, 2]
Generation 2430: Best = 1.233162e+11  FES = 243100  Diversity = 4.8338e-01  k_t
= [1, 1, 1, 1]
Generation 2440: Best = 1.233162e+11  FES = 244100  Diversity = 4.6104e-01  k_t
= [2, 4, 3, 2]
Generation 2450: Best = 1.233162e+11  FES = 245100  Diversity = 4.6224e-01  k_t
= [1, 1, 1, 1]
Generation 2460: Best = 1.233162e+11  FES = 246100  Diversity = 4.8300e-01  k_t
= [4, 4, 2, 2]
Generation 2470: Best = 1.233162e+11  FES = 247100  Diversity = 4.9195e-01  k_t
= [1, 1, 1, 1]
Generation 2480: Best = 1.233162e+11  FES = 248100  Diversity = 4.8909e-01  k_t
= [2, 3, 2, 4]
Generation 2490: Best = 1.233162e+11  FES = 249100  Diversity = 4.8159e-01  k_t
= [1, 1, 1, 1]
Generation 2500: Best = 1.233162e+11  FES = 250100  Diversity = 4.4683e-01  k_t
= [3, 2, 2, 4]
Generation 2510: Best = 1.233162e+11  FES = 251100  Diversity = 4.7222e-01  k_t
= [1, 1, 1, 1]
Generation 2520: Best = 1.233162e+11  FES = 252100  Diversity = 4.8878e-01  k_t
= [2, 3, 3, 2]
Generation 2530: Best = 1.233162e+11  FES = 253100  Diversity = 4.7186e-01  k_t
= [1, 1, 1, 1]
Generation 2540: Best = 1.233162e+11  FES = 254100  Diversity = 4.7324e-01  k_t
= [3, 4, 3, 2]
Generation 2550: Best = 1.233162e+11  FES = 255100  Diversity = 4.7384e-01  k_t
= [1, 1, 1, 1]
Generation 2560: Best = 1.233162e+11  FES = 256100  Diversity = 4.8819e-01  k_t
= [3, 3, 2, 2]
Generation 2570: Best = 1.233162e+11  FES = 257100  Diversity = 5.0141e-01  k_t
= [1, 1, 1, 1]
Generation 2580: Best = 1.233162e+11  FES = 258100  Diversity = 4.9955e-01  k_t
= [3, 2, 2, 2]
Generation 2590: Best = 1.233162e+11  FES = 259100  Diversity = 5.2319e-01  k_t
= [1, 1, 1, 1]
Generation 2600: Best = 1.233162e+11  FES = 260100  Diversity = 4.8844e-01  k_t
= [4, 4, 4, 4]
Generation 2610: Best = 1.233162e+11  FES = 261100  Diversity = 4.8404e-01  k_t
= [1, 1, 1, 1]
Generation 2620: Best = 1.233162e+11  FES = 262100  Diversity = 4.6331e-01  k_t
= [3, 4, 2, 4]
Generation 2630: Best = 1.233162e+11  FES = 263100  Diversity = 4.4123e-01  k_t
= [1, 1, 1, 1]
Generation 2640: Best = 1.233162e+11  FES = 264100  Diversity = 4.3478e-01  k_t
= [4, 2, 3, 2]
Generation 2650: Best = 1.233162e+11  FES = 265100  Diversity = 4.5178e-01  k_t

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= [1, 1, 1, 1]
Generation 2660: Best = 1.233162e+11  FES = 266100  Diversity = 4.8220e-01  k_t
= [2, 4, 3, 3]
Generation 2670: Best = 1.233162e+11  FES = 267100  Diversity = 4.6715e-01  k_t
= [1, 1, 1, 1]
Generation 2680: Best = 1.233162e+11  FES = 268100  Diversity = 4.4974e-01  k_t
= [4, 4, 3, 2]
Generation 2690: Best = 1.233162e+11  FES = 269100  Diversity = 4.2728e-01  k_t
= [1, 1, 1, 1]
Generation 2700: Best = 1.233162e+11  FES = 270100  Diversity = 4.3920e-01  k_t
= [3, 4, 3, 3]
Generation 2710: Best = 1.233162e+11  FES = 271100  Diversity = 4.9125e-01  k_t
= [1, 1, 1, 1]
Generation 2720: Best = 1.233162e+11  FES = 272100  Diversity = 5.0325e-01  k_t
= [3, 3, 4, 4]
Generation 2730: Best = 1.233162e+11  FES = 273100  Diversity = 4.7840e-01  k_t
= [1, 1, 1, 1]
Generation 2740: Best = 1.233162e+11  FES = 274100  Diversity = 4.7537e-01  k_t
= [2, 2, 2, 4]
Generation 2750: Best = 1.233162e+11  FES = 275100  Diversity = 4.6142e-01  k_t
= [1, 1, 1, 1]
Generation 2760: Best = 1.233162e+11  FES = 276100  Diversity = 4.3059e-01  k_t
= [4, 4, 2, 4]
Generation 2770: Best = 1.233162e+11  FES = 277100  Diversity = 4.0273e-01  k_t
= [1, 1, 1, 1]
Generation 2780: Best = 1.233162e+11  FES = 278100  Diversity = 4.1029e-01  k_t
= [2, 2, 4, 4]
Generation 2790: Best = 1.233162e+11  FES = 279100  Diversity = 4.1374e-01  k_t
= [1, 1, 1, 1]
Generation 2800: Best = 1.233162e+11  FES = 280100  Diversity = 4.1403e-01  k_t
= [2, 4, 2, 4]
Generation 2810: Best = 1.233162e+11  FES = 281100  Diversity = 4.2272e-01  k_t
= [1, 1, 1, 1]
Generation 2820: Best = 1.233162e+11  FES = 282100  Diversity = 4.4320e-01  k_t
= [3, 2, 3, 4]
Generation 2830: Best = 1.233162e+11  FES = 283100  Diversity = 4.1824e-01  k_t
= [1, 1, 1, 1]
Generation 2840: Best = 1.233162e+11  FES = 284100  Diversity = 4.3229e-01  k_t
= [3, 3, 2, 3]
Generation 2850: Best = 1.233162e+11  FES = 285100  Diversity = 4.1491e-01  k_t
= [1, 1, 1, 1]
Generation 2860: Best = 1.233162e+11  FES = 286100  Diversity = 3.9858e-01  k_t
= [3, 2, 4, 4]
Generation 2870: Best = 1.233162e+11  FES = 287100  Diversity = 3.9296e-01  k_t
= [1, 1, 1, 1]
Generation 2880: Best = 1.233162e+11  FES = 288100  Diversity = 3.9723e-01  k_t
= [2, 2, 2, 3]
Generation 2890: Best = 1.233162e+11  FES = 289100  Diversity = 3.8711e-01  k_t
= [1, 1, 1, 1]
Generation 2900: Best = 1.233162e+11  FES = 290100  Diversity = 3.6356e-01  k_t
= [4, 3, 4, 2]
Generation 2910: Best = 1.233162e+11  FES = 291100  Diversity = 3.6243e-01  k_t
= [1, 1, 1, 1]
Generation 2920: Best = 1.233162e+11  FES = 292100  Diversity = 3.4785e-01  k_t

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```

= [4, 2, 4, 4]
Generation 2930: Best = 1.233162e+11  FES = 293100  Diversity = 3.5126e-01  k_t
= [1, 1, 1, 1]
Generation 2940: Best = 1.233162e+11  FES = 294100  Diversity = 3.4278e-01  k_t
= [3, 3, 3, 4]
Generation 2950: Best = 1.233162e+11  FES = 295100  Diversity = 3.4855e-01  k_t
= [1, 1, 1, 1]
Generation 2960: Best = 1.233162e+11  FES = 296100  Diversity = 3.7438e-01  k_t
= [3, 4, 2, 2]
Generation 2970: Best = 1.233162e+11  FES = 297100  Diversity = 3.5899e-01  k_t
= [1, 1, 1, 1]
Generation 2980: Best = 1.233162e+11  FES = 298100  Diversity = 3.5852e-01  k_t
= [3, 4, 4, 3]
Generation 2990: Best = 1.233162e+11  FES = 299100  Diversity = 3.7400e-01  k_t
= [1, 1, 1, 1]

```

Optimization completed!

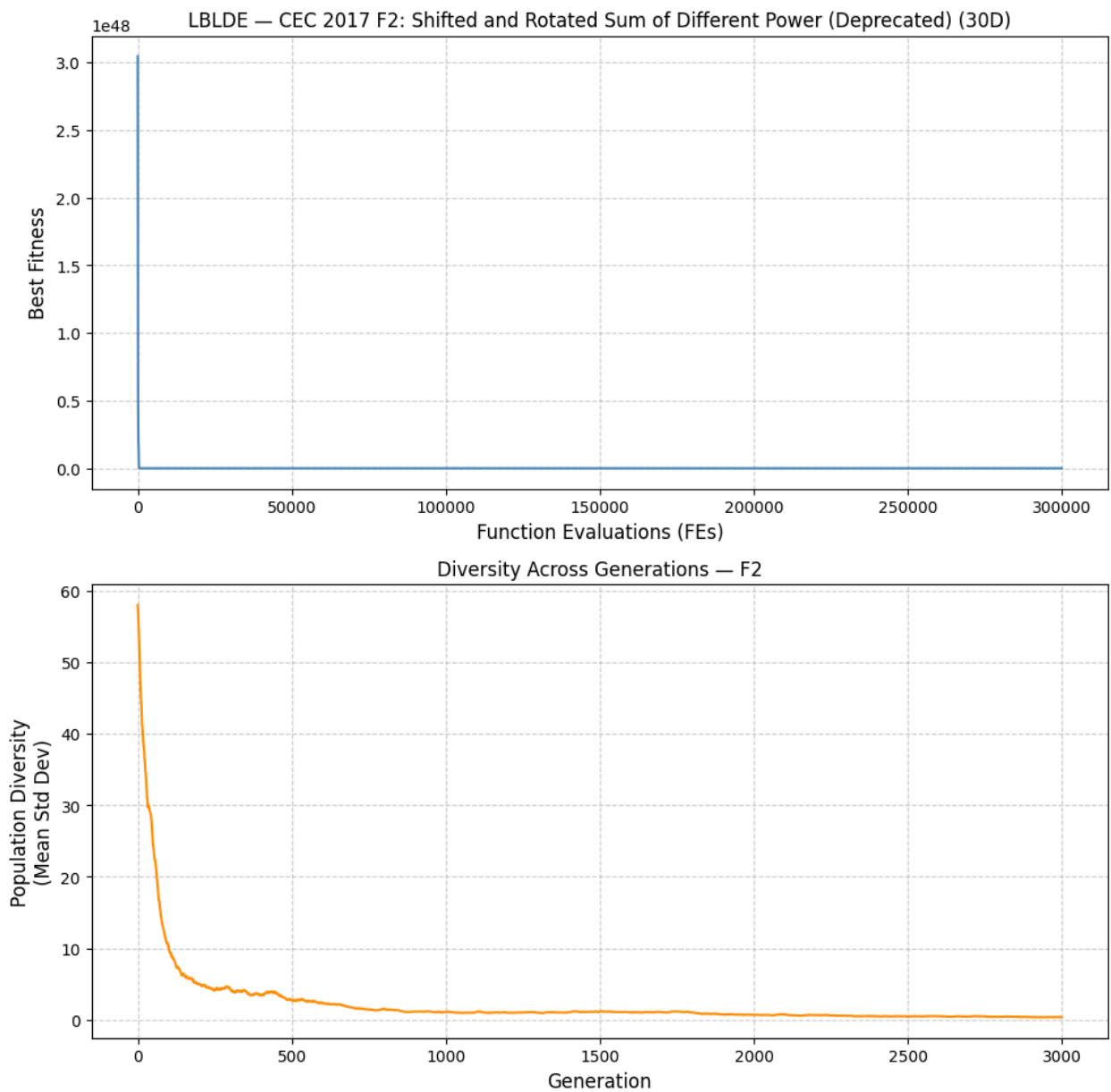
Final Best Fitness = 1.233162e+11

Total Function Evaluations = 300000

```

=====
=
Results:
  Best Fitness          : 1.233162e+11
  Global Optimum        : 2.000000e+02
  Error (|best-optimum|) : 1.233162e+11
  Best Solution (5 dim)  : [-34.46238708 -17.70901662 -75.56532453 -0.55328742
4.31962084]
  D^(0) (init. diversity): 5.7976e+01
=====
=

```



Graph saved as convergence_F2.png

```
In [13]: # =====
# F3 - Shifted and Rotated Zakharov
# =====
run_and_plot('F3', make_wrapper(simple.f3), 300.0, 'Shifted and Rotated Zakharov')
```

```

=====
=
LBLDE_s on CEC 2017 F3 – Single Run
=====
=
Function      : Shifted and Rotated Zakharov (30D)
Bounds       : [-100, 100]
Global optimum : 300.0
Initial D^(0) = 5.7976e+01
Generation 0: Best Fitness = 4.672334e+07 | Diversity = 5.7976e+01
Generation 10: Best = 1.884224e+05 FES = 1100 Diversity = 5.4212e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 1.398868e+05 FES = 2100 Diversity = 5.1475e+01 k_t
= [3, 3, 2, 2]
Generation 30: Best = 1.053721e+05 FES = 3100 Diversity = 4.7492e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 1.053721e+05 FES = 4100 Diversity = 4.2401e+01 k_t
= [3, 4, 3, 4]
Generation 50: Best = 8.979987e+04 FES = 5100 Diversity = 3.4351e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 6.904335e+04 FES = 6100 Diversity = 2.6548e+01 k_t
= [2, 3, 4, 3]
Generation 70: Best = 6.417745e+04 FES = 7100 Diversity = 1.9258e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 5.602134e+04 FES = 8100 Diversity = 1.5080e+01 k_t
= [4, 3, 4, 3]
Generation 90: Best = 5.192698e+04 FES = 9100 Diversity = 1.2137e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 4.895714e+04 FES = 10100 Diversity = 9.5715e+00 k_t
= [2, 4, 4, 3]
Generation 110: Best = 4.763727e+04 FES = 11100 Diversity = 7.5120e+00 k_t
= [1, 1, 1, 1]
Generation 120: Best = 4.249408e+04 FES = 12100 Diversity = 6.1099e+00 k_t
= [4, 2, 4, 3]
Generation 130: Best = 4.249408e+04 FES = 13100 Diversity = 5.8360e+00 k_t
= [1, 1, 1, 1]
Generation 140: Best = 4.180655e+04 FES = 14100 Diversity = 5.2362e+00 k_t
= [4, 2, 2, 3]
Generation 150: Best = 4.135187e+04 FES = 15100 Diversity = 4.9688e+00 k_t
= [1, 1, 1, 1]
Generation 160: Best = 4.035640e+04 FES = 16100 Diversity = 4.8101e+00 k_t
= [3, 2, 4, 2]
Generation 170: Best = 3.875858e+04 FES = 17100 Diversity = 5.1529e+00 k_t
= [1, 1, 1, 1]
Generation 180: Best = 3.655084e+04 FES = 18100 Diversity = 5.1017e+00 k_t
= [3, 2, 4, 3]
Generation 190: Best = 3.506396e+04 FES = 19100 Diversity = 4.9940e+00 k_t
= [1, 1, 1, 1]
Generation 200: Best = 3.458497e+04 FES = 20100 Diversity = 4.3351e+00 k_t
= [4, 2, 3, 4]
Generation 210: Best = 3.329958e+04 FES = 21100 Diversity = 4.5287e+00 k_t
= [1, 1, 1, 1]
Generation 220: Best = 3.166650e+04 FES = 22100 Diversity = 4.6227e+00 k_t
= [2, 2, 4, 2]

```

Generation 230:	Best = 3.070609e+04	FES = 23100	Diversity = 4.5393e+00	k_t
= [1, 1, 1, 1]				
Generation 240:	Best = 2.916927e+04	FES = 24100	Diversity = 4.7196e+00	k_t
= [4, 4, 2, 3]				
Generation 250:	Best = 2.700168e+04	FES = 25100	Diversity = 4.6566e+00	k_t
= [1, 1, 1, 1]				
Generation 260:	Best = 2.575356e+04	FES = 26100	Diversity = 4.4182e+00	k_t
= [2, 2, 3, 2]				
Generation 270:	Best = 2.417647e+04	FES = 27100	Diversity = 4.9720e+00	k_t
= [1, 1, 1, 1]				
Generation 280:	Best = 2.361432e+04	FES = 28100	Diversity = 4.6603e+00	k_t
= [3, 3, 4, 2]				
Generation 290:	Best = 2.298605e+04	FES = 29100	Diversity = 4.3765e+00	k_t
= [1, 1, 1, 1]				
Generation 300:	Best = 2.231000e+04	FES = 30100	Diversity = 4.2437e+00	k_t
= [4, 2, 2, 2]				
Generation 310:	Best = 2.121623e+04	FES = 31100	Diversity = 4.2920e+00	k_t
= [1, 1, 1, 1]				
Generation 320:	Best = 2.071881e+04	FES = 32100	Diversity = 4.1759e+00	k_t
= [3, 3, 2, 4]				
Generation 330:	Best = 1.982879e+04	FES = 33100	Diversity = 4.1997e+00	k_t
= [1, 1, 1, 1]				
Generation 340:	Best = 1.935436e+04	FES = 34100	Diversity = 4.0143e+00	k_t
= [2, 4, 2, 2]				
Generation 350:	Best = 1.802095e+04	FES = 35100	Diversity = 4.3007e+00	k_t
= [1, 1, 1, 1]				
Generation 360:	Best = 1.802095e+04	FES = 36100	Diversity = 4.1948e+00	k_t
= [3, 4, 3, 4]				
Generation 370:	Best = 1.765017e+04	FES = 37100	Diversity = 3.9395e+00	k_t
= [1, 1, 1, 1]				
Generation 380:	Best = 1.665626e+04	FES = 38100	Diversity = 3.8552e+00	k_t
= [4, 3, 3, 2]				
Generation 390:	Best = 1.547405e+04	FES = 39100	Diversity = 4.0013e+00	k_t
= [1, 1, 1, 1]				
Generation 400:	Best = 1.533785e+04	FES = 40100	Diversity = 3.9798e+00	k_t
= [4, 2, 3, 3]				
Generation 410:	Best = 1.455059e+04	FES = 41100	Diversity = 3.6292e+00	k_t
= [1, 1, 1, 1]				
Generation 420:	Best = 1.401571e+04	FES = 42100	Diversity = 3.1662e+00	k_t
= [4, 3, 3, 2]				
Generation 430:	Best = 1.390315e+04	FES = 43100	Diversity = 2.9688e+00	k_t
= [1, 1, 1, 1]				
Generation 440:	Best = 1.275388e+04	FES = 44100	Diversity = 2.7148e+00	k_t
= [4, 4, 4, 2]				
Generation 450:	Best = 1.273692e+04	FES = 45100	Diversity = 2.7711e+00	k_t
= [1, 1, 1, 1]				
Generation 460:	Best = 1.250009e+04	FES = 46100	Diversity = 2.8639e+00	k_t
= [2, 2, 2, 2]				
Generation 470:	Best = 1.188657e+04	FES = 47100	Diversity = 2.8785e+00	k_t
= [1, 1, 1, 1]				
Generation 480:	Best = 1.076418e+04	FES = 48100	Diversity = 2.8327e+00	k_t
= [3, 2, 2, 4]				
Generation 490:	Best = 1.030291e+04	FES = 49100	Diversity = 3.0915e+00	k_t
= [1, 1, 1, 1]				

Generation 500:	Best = 1.014665e+04	FES = 50100	Diversity = 3.4159e+00	k_t
= [3, 4, 3, 3]				
Generation 510:	Best = 9.755361e+03	FES = 51100	Diversity = 3.1300e+00	k_t
= [1, 1, 1, 1]				
Generation 520:	Best = 9.245160e+03	FES = 52100	Diversity = 3.1031e+00	k_t
= [3, 4, 3, 3]				
Generation 530:	Best = 9.056694e+03	FES = 53100	Diversity = 3.1111e+00	k_t
= [1, 1, 1, 1]				
Generation 540:	Best = 8.087275e+03	FES = 54100	Diversity = 3.3402e+00	k_t
= [2, 4, 2, 2]				
Generation 550:	Best = 7.268593e+03	FES = 55100	Diversity = 3.4001e+00	k_t
= [1, 1, 1, 1]				
Generation 560:	Best = 7.268593e+03	FES = 56100	Diversity = 3.3719e+00	k_t
= [3, 3, 2, 3]				
Generation 570:	Best = 6.442323e+03	FES = 57100	Diversity = 3.0688e+00	k_t
= [1, 1, 1, 1]				
Generation 580:	Best = 5.955353e+03	FES = 58100	Diversity = 3.2433e+00	k_t
= [4, 2, 3, 4]				
Generation 590:	Best = 5.663462e+03	FES = 59100	Diversity = 3.0694e+00	k_t
= [1, 1, 1, 1]				
Generation 600:	Best = 5.106069e+03	FES = 60100	Diversity = 3.1751e+00	k_t
= [3, 2, 2, 4]				
Generation 610:	Best = 4.661701e+03	FES = 61100	Diversity = 3.1528e+00	k_t
= [1, 1, 1, 1]				
Generation 620:	Best = 4.173676e+03	FES = 62100	Diversity = 2.9636e+00	k_t
= [3, 2, 3, 2]				
Generation 630:	Best = 4.057506e+03	FES = 63100	Diversity = 2.7086e+00	k_t
= [1, 1, 1, 1]				
Generation 640:	Best = 3.846192e+03	FES = 64100	Diversity = 2.3150e+00	k_t
= [4, 4, 4, 3]				
Generation 650:	Best = 3.587605e+03	FES = 65100	Diversity = 2.1294e+00	k_t
= [1, 1, 1, 1]				
Generation 660:	Best = 3.377739e+03	FES = 66100	Diversity = 2.0087e+00	k_t
= [2, 3, 3, 4]				
Generation 670:	Best = 3.182889e+03	FES = 67100	Diversity = 1.8346e+00	k_t
= [1, 1, 1, 1]				
Generation 680:	Best = 2.958128e+03	FES = 68100	Diversity = 1.9675e+00	k_t
= [4, 2, 2, 4]				
Generation 690:	Best = 2.729688e+03	FES = 69100	Diversity = 1.8357e+00	k_t
= [1, 1, 1, 1]				
Generation 700:	Best = 2.597073e+03	FES = 70100	Diversity = 1.7679e+00	k_t
= [2, 4, 3, 2]				
Generation 710:	Best = 2.507903e+03	FES = 71100	Diversity = 1.6771e+00	k_t
= [1, 1, 1, 1]				
Generation 720:	Best = 2.310757e+03	FES = 72100	Diversity = 1.5982e+00	k_t
= [3, 3, 4, 3]				
Generation 730:	Best = 2.222037e+03	FES = 73100	Diversity = 1.6482e+00	k_t
= [1, 1, 1, 1]				
Generation 740:	Best = 1.994886e+03	FES = 74100	Diversity = 1.7273e+00	k_t
= [4, 2, 2, 2]				
Generation 750:	Best = 1.829372e+03	FES = 75100	Diversity = 1.5986e+00	k_t
= [1, 1, 1, 1]				
Generation 760:	Best = 1.723899e+03	FES = 76100	Diversity = 1.7144e+00	k_t
= [2, 3, 4, 3]				

Generation 770:	Best = 1.719038e+03	FES = 77100	Diversity = 1.6347e+00	k_t = [1, 1, 1, 1]
Generation 780:	Best = 1.609072e+03	FES = 78100	Diversity = 1.5865e+00	k_t = [2, 2, 4, 4]
Generation 790:	Best = 1.451720e+03	FES = 79100	Diversity = 1.4738e+00	k_t = [1, 1, 1, 1]
Generation 800:	Best = 1.451659e+03	FES = 80100	Diversity = 1.4007e+00	k_t = [4, 3, 4, 4]
Generation 810:	Best = 1.381207e+03	FES = 81100	Diversity = 1.3150e+00	k_t = [1, 1, 1, 1]
Generation 820:	Best = 1.327549e+03	FES = 82100	Diversity = 1.2157e+00	k_t = [3, 4, 4, 4]
Generation 830:	Best = 1.279432e+03	FES = 83100	Diversity = 1.1101e+00	k_t = [1, 1, 1, 1]
Generation 840:	Best = 1.242885e+03	FES = 84100	Diversity = 1.0825e+00	k_t = [2, 3, 2, 3]
Generation 850:	Best = 1.208819e+03	FES = 85100	Diversity = 9.8707e-01	k_t = [1, 1, 1, 1]
Generation 860:	Best = 1.179209e+03	FES = 86100	Diversity = 8.3960e-01	k_t = [2, 3, 2, 3]
Generation 870:	Best = 1.137784e+03	FES = 87100	Diversity = 8.5390e-01	k_t = [1, 1, 1, 1]
Generation 880:	Best = 1.076258e+03	FES = 88100	Diversity = 8.1906e-01	k_t = [4, 2, 2, 4]
Generation 890:	Best = 1.026882e+03	FES = 89100	Diversity = 8.6447e-01	k_t = [1, 1, 1, 1]
Generation 900:	Best = 1.003522e+03	FES = 90100	Diversity = 8.7077e-01	k_t = [2, 2, 2, 3]
Generation 910:	Best = 9.577576e+02	FES = 91100	Diversity = 8.4215e-01	k_t = [1, 1, 1, 1]
Generation 920:	Best = 9.307098e+02	FES = 92100	Diversity = 7.3358e-01	k_t = [3, 2, 3, 3]
Generation 930:	Best = 9.016796e+02	FES = 93100	Diversity = 6.8792e-01	k_t = [1, 1, 1, 1]
Generation 940:	Best = 8.489443e+02	FES = 94100	Diversity = 6.9013e-01	k_t = [3, 4, 2, 4]
Generation 950:	Best = 8.196244e+02	FES = 95100	Diversity = 7.6435e-01	k_t = [1, 1, 1, 1]
Generation 960:	Best = 7.881503e+02	FES = 96100	Diversity = 7.7016e-01	k_t = [3, 4, 3, 2]
Generation 970:	Best = 7.642993e+02	FES = 97100	Diversity = 7.1508e-01	k_t = [1, 1, 1, 1]
Generation 980:	Best = 7.529849e+02	FES = 98100	Diversity = 7.2484e-01	k_t = [4, 3, 2, 3]
Generation 990:	Best = 7.448769e+02	FES = 99100	Diversity = 6.3903e-01	k_t = [1, 1, 1, 1]
Generation 1000:	Best = 7.218943e+02	FES = 100100	Diversity = 6.3650e-01	k_t = [4, 3, 3, 4]
Generation 1010:	Best = 6.985813e+02	FES = 101100	Diversity = 6.2715e-01	k_t = [1, 1, 1, 1]
Generation 1020:	Best = 6.683021e+02	FES = 102100	Diversity = 6.4315e-01	k_t = [4, 3, 4, 4]
Generation 1030:	Best = 6.289066e+02	FES = 103100	Diversity = 7.0822e-01	k_t = [1, 1, 1, 1]

Generation 1040: Best = 6.100831e+02	FES = 104100	Diversity = 6.9971e-01	k_t
= [3, 3, 4, 2]			
Generation 1050: Best = 5.828139e+02	FES = 105100	Diversity = 6.9213e-01	k_t
= [1, 1, 1, 1]			
Generation 1060: Best = 5.585599e+02	FES = 106100	Diversity = 7.4375e-01	k_t
= [3, 3, 4, 2]			
Generation 1070: Best = 5.419477e+02	FES = 107100	Diversity = 7.2349e-01	k_t
= [1, 1, 1, 1]			
Generation 1080: Best = 5.259714e+02	FES = 108100	Diversity = 7.7901e-01	k_t
= [3, 3, 2, 3]			
Generation 1090: Best = 4.998748e+02	FES = 109100	Diversity = 6.8535e-01	k_t
= [1, 1, 1, 1]			
Generation 1100: Best = 4.895479e+02	FES = 110100	Diversity = 6.7684e-01	k_t
= [4, 4, 2, 4]			
Generation 1110: Best = 4.714566e+02	FES = 111100	Diversity = 6.2491e-01	k_t
= [1, 1, 1, 1]			
Generation 1120: Best = 4.616348e+02	FES = 112100	Diversity = 5.5128e-01	k_t
= [3, 4, 4, 4]			
Generation 1130: Best = 4.210380e+02	FES = 113100	Diversity = 5.1023e-01	k_t
= [1, 1, 1, 1]			
Generation 1140: Best = 4.210380e+02	FES = 114100	Diversity = 5.4883e-01	k_t
= [2, 2, 3, 4]			
Generation 1150: Best = 4.146232e+02	FES = 115100	Diversity = 5.4614e-01	k_t
= [1, 1, 1, 1]			
Generation 1160: Best = 4.021047e+02	FES = 116100	Diversity = 5.1294e-01	k_t
= [2, 4, 3, 2]			
Generation 1170: Best = 3.876407e+02	FES = 117100	Diversity = 4.8655e-01	k_t
= [1, 1, 1, 1]			
Generation 1180: Best = 3.861758e+02	FES = 118100	Diversity = 4.6966e-01	k_t
= [3, 2, 3, 4]			
Generation 1190: Best = 3.816388e+02	FES = 119100	Diversity = 4.2970e-01	k_t
= [1, 1, 1, 1]			
Generation 1200: Best = 3.718040e+02	FES = 120100	Diversity = 4.0517e-01	k_t
= [2, 3, 3, 2]			
Generation 1210: Best = 3.662689e+02	FES = 121100	Diversity = 3.8247e-01	k_t
= [1, 1, 1, 1]			
Generation 1220: Best = 3.592927e+02	FES = 122100	Diversity = 3.5349e-01	k_t
= [3, 2, 4, 4]			
Generation 1230: Best = 3.539930e+02	FES = 123100	Diversity = 3.3881e-01	k_t
= [1, 1, 1, 1]			
Generation 1240: Best = 3.406801e+02	FES = 124100	Diversity = 3.2249e-01	k_t
= [2, 4, 3, 2]			
Generation 1250: Best = 3.377826e+02	FES = 125100	Diversity = 3.6838e-01	k_t
= [1, 1, 1, 1]			
Generation 1260: Best = 3.310705e+02	FES = 126100	Diversity = 3.6323e-01	k_t
= [4, 4, 3, 3]			
Generation 1270: Best = 3.282493e+02	FES = 127100	Diversity = 3.6270e-01	k_t
= [1, 1, 1, 1]			
Generation 1280: Best = 3.262961e+02	FES = 128100	Diversity = 3.4542e-01	k_t
= [2, 3, 2, 2]			
Generation 1290: Best = 3.225553e+02	FES = 129100	Diversity = 3.0948e-01	k_t
= [1, 1, 1, 1]			
Generation 1300: Best = 3.176408e+02	FES = 130100	Diversity = 2.7984e-01	k_t
= [2, 4, 2, 3]			

Generation 1310: Best = 3.171857e+02	FES = 131100	Diversity = 2.4311e-01	k_t
= [1, 1, 1, 1]			
Generation 1320: Best = 3.163542e+02	FES = 132100	Diversity = 2.1122e-01	k_t
= [4, 3, 2, 4]			
Generation 1330: Best = 3.150777e+02	FES = 133100	Diversity = 1.8804e-01	k_t
= [1, 1, 1, 1]			
Generation 1340: Best = 3.132299e+02	FES = 134100	Diversity = 1.7252e-01	k_t
= [2, 3, 3, 4]			
Generation 1350: Best = 3.119735e+02	FES = 135100	Diversity = 1.7559e-01	k_t
= [1, 1, 1, 1]			
Generation 1360: Best = 3.101465e+02	FES = 136100	Diversity = 1.6670e-01	k_t
= [4, 2, 4, 2]			
Generation 1370: Best = 3.088392e+02	FES = 137100	Diversity = 1.5178e-01	k_t
= [1, 1, 1, 1]			
Generation 1380: Best = 3.088101e+02	FES = 138100	Diversity = 1.3714e-01	k_t
= [3, 2, 4, 4]			
Generation 1390: Best = 3.082441e+02	FES = 139100	Diversity = 1.2891e-01	k_t
= [1, 1, 1, 1]			
Generation 1400: Best = 3.074335e+02	FES = 140100	Diversity = 1.1343e-01	k_t
= [2, 2, 4, 3]			
Generation 1410: Best = 3.071561e+02	FES = 141100	Diversity = 1.0857e-01	k_t
= [1, 1, 1, 1]			
Generation 1420: Best = 3.065626e+02	FES = 142100	Diversity = 9.6347e-02	k_t
= [3, 2, 4, 2]			
Generation 1430: Best = 3.059651e+02	FES = 143100	Diversity = 9.0076e-02	k_t
= [1, 1, 1, 1]			
Generation 1440: Best = 3.054399e+02	FES = 144100	Diversity = 9.4839e-02	k_t
= [2, 4, 2, 4]			
Generation 1450: Best = 3.046296e+02	FES = 145100	Diversity = 9.1416e-02	k_t
= [1, 1, 1, 1]			
Generation 1460: Best = 3.044233e+02	FES = 146100	Diversity = 8.8636e-02	k_t
= [3, 3, 4, 3]			
Generation 1470: Best = 3.042281e+02	FES = 147100	Diversity = 8.6548e-02	k_t
= [1, 1, 1, 1]			
Generation 1480: Best = 3.035493e+02	FES = 148100	Diversity = 8.0721e-02	k_t
= [4, 3, 3, 3]			
Generation 1490: Best = 3.033546e+02	FES = 149100	Diversity = 7.6065e-02	k_t
= [1, 1, 1, 1]			
Generation 1500: Best = 3.030259e+02	FES = 150100	Diversity = 7.5650e-02	k_t
= [3, 2, 2, 2]			
Generation 1510: Best = 3.028258e+02	FES = 151100	Diversity = 7.7530e-02	k_t
= [1, 1, 1, 1]			
Generation 1520: Best = 3.025551e+02	FES = 152100	Diversity = 7.5740e-02	k_t
= [2, 3, 4, 3]			
Generation 1530: Best = 3.020988e+02	FES = 153100	Diversity = 7.6854e-02	k_t
= [1, 1, 1, 1]			
Generation 1540: Best = 3.020177e+02	FES = 154100	Diversity = 7.1734e-02	k_t
= [2, 4, 3, 3]			
Generation 1550: Best = 3.019116e+02	FES = 155100	Diversity = 7.0338e-02	k_t
= [1, 1, 1, 1]			
Generation 1560: Best = 3.017735e+02	FES = 156100	Diversity = 6.4766e-02	k_t
= [2, 3, 4, 4]			
Generation 1570: Best = 3.015911e+02	FES = 157100	Diversity = 6.0053e-02	k_t
= [1, 1, 1, 1]			

Generation 1580: Best = 3.014555e+02	FES = 158100	Diversity = 6.0568e-02	k_t
= [3, 2, 2, 4]			
Generation 1590: Best = 3.013477e+02	FES = 159100	Diversity = 5.6912e-02	k_t
= [1, 1, 1, 1]			
Generation 1600: Best = 3.010710e+02	FES = 160100	Diversity = 6.1500e-02	k_t
= [2, 4, 3, 3]			
Generation 1610: Best = 3.010260e+02	FES = 161100	Diversity = 5.9239e-02	k_t
= [1, 1, 1, 1]			
Generation 1620: Best = 3.009733e+02	FES = 162100	Diversity = 5.3472e-02	k_t
= [3, 3, 4, 3]			
Generation 1630: Best = 3.008475e+02	FES = 163100	Diversity = 4.8861e-02	k_t
= [1, 1, 1, 1]			
Generation 1640: Best = 3.007942e+02	FES = 164100	Diversity = 4.4174e-02	k_t
= [4, 4, 4, 3]			
Generation 1650: Best = 3.007396e+02	FES = 165100	Diversity = 4.3635e-02	k_t
= [1, 1, 1, 1]			
Generation 1660: Best = 3.006916e+02	FES = 166100	Diversity = 3.7869e-02	k_t
= [4, 3, 4, 2]			
Generation 1670: Best = 3.006240e+02	FES = 167100	Diversity = 3.4047e-02	k_t
= [1, 1, 1, 1]			
Generation 1680: Best = 3.005657e+02	FES = 168100	Diversity = 3.1877e-02	k_t
= [3, 4, 2, 2]			
Generation 1690: Best = 3.005131e+02	FES = 169100	Diversity = 3.2323e-02	k_t
= [1, 1, 1, 1]			
Generation 1700: Best = 3.005120e+02	FES = 170100	Diversity = 2.7912e-02	k_t
= [3, 2, 2, 3]			
Generation 1710: Best = 3.004844e+02	FES = 171100	Diversity = 2.6691e-02	k_t
= [1, 1, 1, 1]			
Generation 1720: Best = 3.004380e+02	FES = 172100	Diversity = 2.6150e-02	k_t
= [3, 4, 4, 4]			
Generation 1730: Best = 3.004252e+02	FES = 173100	Diversity = 2.4318e-02	k_t
= [1, 1, 1, 1]			
Generation 1740: Best = 3.004076e+02	FES = 174100	Diversity = 2.0574e-02	k_t
= [4, 2, 2, 2]			
Generation 1750: Best = 3.003902e+02	FES = 175100	Diversity = 1.9539e-02	k_t
= [1, 1, 1, 1]			
Generation 1760: Best = 3.003589e+02	FES = 176100	Diversity = 1.8823e-02	k_t
= [3, 4, 2, 4]			
Generation 1770: Best = 3.003369e+02	FES = 177100	Diversity = 1.9321e-02	k_t
= [1, 1, 1, 1]			
Generation 1780: Best = 3.003023e+02	FES = 178100	Diversity = 1.9105e-02	k_t
= [3, 4, 2, 3]			
Generation 1790: Best = 3.002782e+02	FES = 179100	Diversity = 1.9156e-02	k_t
= [1, 1, 1, 1]			
Generation 1800: Best = 3.002693e+02	FES = 180100	Diversity = 1.9402e-02	k_t
= [4, 3, 3, 3]			
Generation 1810: Best = 3.002415e+02	FES = 181100	Diversity = 1.8556e-02	k_t
= [1, 1, 1, 1]			
Generation 1820: Best = 3.002314e+02	FES = 182100	Diversity = 1.7539e-02	k_t
= [3, 3, 3, 4]			
Generation 1830: Best = 3.002037e+02	FES = 183100	Diversity = 1.8153e-02	k_t
= [1, 1, 1, 1]			
Generation 1840: Best = 3.001876e+02	FES = 184100	Diversity = 1.9073e-02	k_t
= [2, 2, 4, 3]			

Generation 1850: Best = 3.001816e+02 = [1, 1, 1, 1]	FES = 185100	Diversity = 1.8215e-02	k_t
Generation 1860: Best = 3.001649e+02 = [2, 4, 3, 3]	FES = 186100	Diversity = 1.7514e-02	k_t
Generation 1870: Best = 3.001517e+02 = [1, 1, 1, 1]	FES = 187100	Diversity = 1.5805e-02	k_t
Generation 1880: Best = 3.001406e+02 = [3, 2, 3, 2]	FES = 188100	Diversity = 1.5409e-02	k_t
Generation 1890: Best = 3.001359e+02 = [1, 1, 1, 1]	FES = 189100	Diversity = 1.4427e-02	k_t
Generation 1900: Best = 3.001172e+02 = [2, 3, 4, 3]	FES = 190100	Diversity = 1.4141e-02	k_t
Generation 1910: Best = 3.001071e+02 = [1, 1, 1, 1]	FES = 191100	Diversity = 1.4382e-02	k_t
Generation 1920: Best = 3.000934e+02 = [3, 4, 3, 3]	FES = 192100	Diversity = 1.3993e-02	k_t
Generation 1930: Best = 3.000909e+02 = [1, 1, 1, 1]	FES = 193100	Diversity = 1.6453e-02	k_t
Generation 1940: Best = 3.000822e+02 = [4, 4, 3, 3]	FES = 194100	Diversity = 1.4188e-02	k_t
Generation 1950: Best = 3.000787e+02 = [1, 1, 1, 1]	FES = 195100	Diversity = 1.3187e-02	k_t
Generation 1960: Best = 3.000710e+02 = [2, 3, 2, 2]	FES = 196100	Diversity = 1.2319e-02	k_t
Generation 1970: Best = 3.000688e+02 = [1, 1, 1, 1]	FES = 197100	Diversity = 1.1590e-02	k_t
Generation 1980: Best = 3.000630e+02 = [4, 2, 2, 2]	FES = 198100	Diversity = 1.0701e-02	k_t
Generation 1990: Best = 3.000587e+02 = [1, 1, 1, 1]	FES = 199100	Diversity = 1.0091e-02	k_t
Generation 2000: Best = 3.000567e+02 = [3, 4, 3, 4]	FES = 200100	Diversity = 9.4492e-03	k_t
Generation 2010: Best = 3.000531e+02 = [1, 1, 1, 1]	FES = 201100	Diversity = 8.0797e-03	k_t
Generation 2020: Best = 3.000508e+02 = [4, 2, 2, 2]	FES = 202100	Diversity = 7.6029e-03	k_t
Generation 2030: Best = 3.000491e+02 = [1, 1, 1, 1]	FES = 203100	Diversity = 7.2377e-03	k_t
Generation 2040: Best = 3.000484e+02 = [2, 4, 2, 4]	FES = 204100	Diversity = 6.5271e-03	k_t
Generation 2050: Best = 3.000460e+02 = [1, 1, 1, 1]	FES = 205100	Diversity = 5.8643e-03	k_t
Generation 2060: Best = 3.000447e+02 = [4, 4, 3, 2]	FES = 206100	Diversity = 5.4350e-03	k_t
Generation 2070: Best = 3.000424e+02 = [1, 1, 1, 1]	FES = 207100	Diversity = 5.7815e-03	k_t
Generation 2080: Best = 3.000407e+02 = [2, 2, 4, 4]	FES = 208100	Diversity = 5.4526e-03	k_t
Generation 2090: Best = 3.000393e+02 = [1, 1, 1, 1]	FES = 209100	Diversity = 5.6043e-03	k_t
Generation 2100: Best = 3.000375e+02 = [4, 3, 3, 4]	FES = 210100	Diversity = 5.5346e-03	k_t
Generation 2110: Best = 3.000349e+02 = [1, 1, 1, 1]	FES = 211100	Diversity = 5.7102e-03	k_t

Generation 2120: Best = 3.000332e+02 = [2, 2, 3, 3]	FES = 212100	Diversity = 5.7979e-03	k_t
Generation 2130: Best = 3.000322e+02 = [1, 1, 1, 1]	FES = 213100	Diversity = 5.8484e-03	k_t
Generation 2140: Best = 3.000289e+02 = [4, 2, 4, 4]	FES = 214100	Diversity = 5.9984e-03	k_t
Generation 2150: Best = 3.000275e+02 = [1, 1, 1, 1]	FES = 215100	Diversity = 6.1855e-03	k_t
Generation 2160: Best = 3.000263e+02 = [3, 3, 4, 4]	FES = 216100	Diversity = 5.9068e-03	k_t
Generation 2170: Best = 3.000252e+02 = [1, 1, 1, 1]	FES = 217100	Diversity = 5.4419e-03	k_t
Generation 2180: Best = 3.000229e+02 = [4, 3, 2, 3]	FES = 218100	Diversity = 4.9243e-03	k_t
Generation 2190: Best = 3.000224e+02 = [1, 1, 1, 1]	FES = 219100	Diversity = 4.7470e-03	k_t
Generation 2200: Best = 3.000215e+02 = [2, 3, 4, 4]	FES = 220100	Diversity = 4.7179e-03	k_t
Generation 2210: Best = 3.000189e+02 = [1, 1, 1, 1]	FES = 221100	Diversity = 4.7577e-03	k_t
Generation 2220: Best = 3.000185e+02 = [3, 3, 4, 3]	FES = 222100	Diversity = 4.8830e-03	k_t
Generation 2230: Best = 3.000173e+02 = [1, 1, 1, 1]	FES = 223100	Diversity = 4.9305e-03	k_t
Generation 2240: Best = 3.000165e+02 = [2, 2, 4, 4]	FES = 224100	Diversity = 4.9764e-03	k_t
Generation 2250: Best = 3.000152e+02 = [1, 1, 1, 1]	FES = 225100	Diversity = 4.9891e-03	k_t
Generation 2260: Best = 3.000137e+02 = [4, 3, 2, 3]	FES = 226100	Diversity = 4.8981e-03	k_t
Generation 2270: Best = 3.000130e+02 = [1, 1, 1, 1]	FES = 227100	Diversity = 4.9361e-03	k_t
Generation 2280: Best = 3.000116e+02 = [4, 3, 4, 3]	FES = 228100	Diversity = 4.8381e-03	k_t
Generation 2290: Best = 3.000105e+02 = [1, 1, 1, 1]	FES = 229100	Diversity = 4.6932e-03	k_t
Generation 2300: Best = 3.000098e+02 = [3, 2, 4, 2]	FES = 230100	Diversity = 4.3609e-03	k_t
Generation 2310: Best = 3.000094e+02 = [1, 1, 1, 1]	FES = 231100	Diversity = 3.9847e-03	k_t
Generation 2320: Best = 3.000085e+02 = [3, 3, 3, 3]	FES = 232100	Diversity = 4.0721e-03	k_t
Generation 2330: Best = 3.000077e+02 = [1, 1, 1, 1]	FES = 233100	Diversity = 3.9738e-03	k_t
Generation 2340: Best = 3.000069e+02 = [3, 4, 2, 2]	FES = 234100	Diversity = 4.1282e-03	k_t
Generation 2350: Best = 3.000063e+02 = [1, 1, 1, 1]	FES = 235100	Diversity = 4.2814e-03	k_t
Generation 2360: Best = 3.000058e+02 = [3, 4, 3, 3]	FES = 236100	Diversity = 4.1377e-03	k_t
Generation 2370: Best = 3.000054e+02 = [1, 1, 1, 1]	FES = 237100	Diversity = 3.6362e-03	k_t
Generation 2380: Best = 3.000050e+02 = [2, 3, 3, 4]	FES = 238100	Diversity = 3.5143e-03	k_t

Generation 2390: Best = 3.000046e+02 = [1, 1, 1, 1]	FES = 239100	Diversity = 3.2766e-03	k_t
Generation 2400: Best = 3.000041e+02 = [3, 3, 3, 2]	FES = 240100	Diversity = 2.9434e-03	k_t
Generation 2410: Best = 3.000035e+02 = [1, 1, 1, 1]	FES = 241100	Diversity = 2.7070e-03	k_t
Generation 2420: Best = 3.000033e+02 = [4, 4, 2, 2]	FES = 242100	Diversity = 2.5967e-03	k_t
Generation 2430: Best = 3.000031e+02 = [1, 1, 1, 1]	FES = 243100	Diversity = 2.4756e-03	k_t
Generation 2440: Best = 3.000028e+02 = [2, 4, 4, 3]	FES = 244100	Diversity = 2.3107e-03	k_t
Generation 2450: Best = 3.000026e+02 = [1, 1, 1, 1]	FES = 245100	Diversity = 2.0827e-03	k_t
Generation 2460: Best = 3.000020e+02 = [3, 2, 2, 2]	FES = 246100	Diversity = 2.2360e-03	k_t
Generation 2470: Best = 3.000020e+02 = [1, 1, 1, 1]	FES = 247100	Diversity = 2.5407e-03	k_t
Generation 2480: Best = 3.000018e+02 = [2, 3, 2, 3]	FES = 248100	Diversity = 2.3813e-03	k_t
Generation 2490: Best = 3.000015e+02 = [1, 1, 1, 1]	FES = 249100	Diversity = 2.2465e-03	k_t
Generation 2500: Best = 3.000011e+02 = [3, 3, 4, 3]	FES = 250100	Diversity = 2.2969e-03	k_t
Generation 2510: Best = 3.000009e+02 = [1, 1, 1, 1]	FES = 251100	Diversity = 2.2518e-03	k_t
Generation 2520: Best = 3.000009e+02 = [4, 2, 4, 3]	FES = 252100	Diversity = 2.1082e-03	k_t
Generation 2530: Best = 3.000007e+02 = [1, 1, 1, 1]	FES = 253100	Diversity = 1.9669e-03	k_t
Generation 2540: Best = 3.000007e+02 = [4, 3, 3, 2]	FES = 254100	Diversity = 1.6047e-03	k_t
Generation 2550: Best = 3.000006e+02 = [1, 1, 1, 1]	FES = 255100	Diversity = 1.3391e-03	k_t
Generation 2560: Best = 3.000006e+02 = [2, 4, 2, 2]	FES = 256100	Diversity = 1.1615e-03	k_t
Generation 2570: Best = 3.000005e+02 = [1, 1, 1, 1]	FES = 257100	Diversity = 1.0542e-03	k_t
Generation 2580: Best = 3.000005e+02 = [2, 4, 2, 4]	FES = 258100	Diversity = 8.8327e-04	k_t
Generation 2590: Best = 3.000005e+02 = [1, 1, 1, 1]	FES = 259100	Diversity = 7.9441e-04	k_t
Generation 2600: Best = 3.000004e+02 = [3, 2, 2, 4]	FES = 260100	Diversity = 7.4790e-04	k_t
Generation 2610: Best = 3.000004e+02 = [1, 1, 1, 1]	FES = 261100	Diversity = 6.9356e-04	k_t
Generation 2620: Best = 3.000004e+02 = [2, 4, 2, 4]	FES = 262100	Diversity = 6.7138e-04	k_t
Generation 2630: Best = 3.000004e+02 = [1, 1, 1, 1]	FES = 263100	Diversity = 6.6407e-04	k_t
Generation 2640: Best = 3.000003e+02 = [2, 2, 2, 3]	FES = 264100	Diversity = 6.2386e-04	k_t
Generation 2650: Best = 3.000003e+02 = [1, 1, 1, 1]	FES = 265100	Diversity = 6.1433e-04	k_t

Generation 2660: Best = 3.000003e+02 = [2, 4, 2, 4]	FES = 266100	Diversity = 5.6725e-04	k_t
Generation 2670: Best = 3.000003e+02 = [1, 1, 1, 1]	FES = 267100	Diversity = 5.5167e-04	k_t
Generation 2680: Best = 3.000003e+02 = [4, 2, 3, 3]	FES = 268100	Diversity = 5.6279e-04	k_t
Generation 2690: Best = 3.000002e+02 = [1, 1, 1, 1]	FES = 269100	Diversity = 5.9387e-04	k_t
Generation 2700: Best = 3.000002e+02 = [4, 2, 3, 3]	FES = 270100	Diversity = 5.8202e-04	k_t
Generation 2710: Best = 3.000002e+02 = [1, 1, 1, 1]	FES = 271100	Diversity = 5.9722e-04	k_t
Generation 2720: Best = 3.000002e+02 = [4, 4, 2, 2]	FES = 272100	Diversity = 6.3008e-04	k_t
Generation 2730: Best = 3.000002e+02 = [1, 1, 1, 1]	FES = 273100	Diversity = 5.4880e-04	k_t
Generation 2740: Best = 3.000002e+02 = [3, 4, 4, 2]	FES = 274100	Diversity = 5.4333e-04	k_t
Generation 2750: Best = 3.000002e+02 = [1, 1, 1, 1]	FES = 275100	Diversity = 5.1766e-04	k_t
Generation 2760: Best = 3.000002e+02 = [4, 3, 3, 4]	FES = 276100	Diversity = 5.0619e-04	k_t
Generation 2770: Best = 3.000002e+02 = [1, 1, 1, 1]	FES = 277100	Diversity = 4.4432e-04	k_t
Generation 2780: Best = 3.000001e+02 = [3, 4, 2, 3]	FES = 278100	Diversity = 4.0134e-04	k_t
Generation 2790: Best = 3.000001e+02 = [1, 1, 1, 1]	FES = 279100	Diversity = 3.7183e-04	k_t
Generation 2800: Best = 3.000001e+02 = [4, 4, 4, 3]	FES = 280100	Diversity = 3.6187e-04	k_t
Generation 2810: Best = 3.000001e+02 = [1, 1, 1, 1]	FES = 281100	Diversity = 3.4087e-04	k_t
Generation 2820: Best = 3.000001e+02 = [3, 4, 4, 4]	FES = 282100	Diversity = 3.2661e-04	k_t
Generation 2830: Best = 3.000001e+02 = [1, 1, 1, 1]	FES = 283100	Diversity = 3.1714e-04	k_t
Generation 2840: Best = 3.000001e+02 = [3, 3, 2, 3]	FES = 284100	Diversity = 3.2847e-04	k_t
Generation 2850: Best = 3.000001e+02 = [1, 1, 1, 1]	FES = 285100	Diversity = 3.1518e-04	k_t
Generation 2860: Best = 3.000001e+02 = [3, 2, 3, 2]	FES = 286100	Diversity = 3.9162e-04	k_t
Generation 2870: Best = 3.000001e+02 = [1, 1, 1, 1]	FES = 287100	Diversity = 4.3509e-04	k_t
Generation 2880: Best = 3.000001e+02 = [3, 2, 3, 4]	FES = 288100	Diversity = 4.5300e-04	k_t
Generation 2890: Best = 3.000001e+02 = [1, 1, 1, 1]	FES = 289100	Diversity = 4.2080e-04	k_t
Generation 2900: Best = 3.000001e+02 = [4, 3, 3, 4]	FES = 290100	Diversity = 3.7522e-04	k_t
Generation 2910: Best = 3.000000e+02 = [1, 1, 1, 1]	FES = 291100	Diversity = 3.7421e-04	k_t
Generation 2920: Best = 3.000000e+02 = [4, 4, 3, 3]	FES = 292100	Diversity = 3.4212e-04	k_t

Generation 2930: Best = 3.000000e+02 FES = 293100 Diversity = 3.0799e-04 k_t
 = [1, 1, 1, 1]
 Generation 2940: Best = 3.000000e+02 FES = 294100 Diversity = 2.8295e-04 k_t
 = [3, 4, 2, 2]
 Generation 2950: Best = 3.000000e+02 FES = 295100 Diversity = 2.5315e-04 k_t
 = [1, 1, 1, 1]
 Generation 2960: Best = 3.000000e+02 FES = 296100 Diversity = 2.2194e-04 k_t
 = [2, 3, 3, 2]
 Generation 2970: Best = 3.000000e+02 FES = 297100 Diversity = 2.2553e-04 k_t
 = [1, 1, 1, 1]
 Generation 2980: Best = 3.000000e+02 FES = 298100 Diversity = 2.0450e-04 k_t
 = [3, 2, 2, 2]
 Generation 2990: Best = 3.000000e+02 FES = 299100 Diversity = 1.7957e-04 k_t
 = [1, 1, 1, 1]

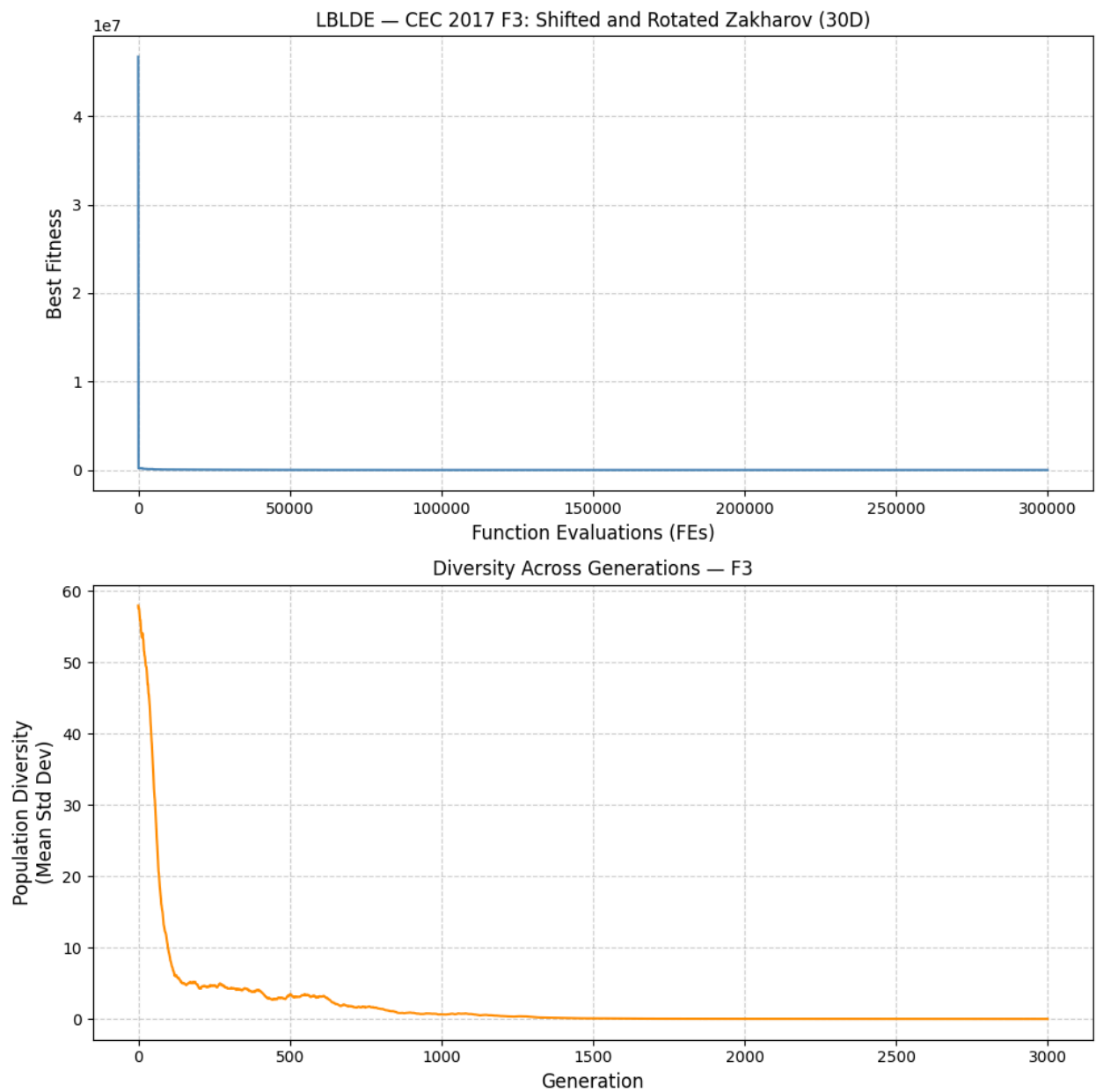
Optimization completed!

Final Best Fitness = 3.000000e+02

Total Function Evaluations = 300000

```

=====
=
Results:
  Best Fitness           : 3.000000e+02
  Global Optimum         : 3.000000e+02
  Error (|best-optimum|) : 2.667123e-05
  Best Solution (5 dim)  : [-55.93790118   4.54359863  35.30141568   8.27998747
-47.43681467]
  D^(0) (init. diversity): 5.7976e+01
=====
=
  
```



Graph saved as convergence_F3.png

6.2 Multimodal Functions (F4-F10)

```
In [14]: # =====
# F4 — Shifted and Rotated Rosenbrock
# =====
run_and_plot('F4', make_wrapper(simple.f4), 400.0, "Shifted and Rotated Rosenb
```



```

=====
=
LBLDE_s on CEC 2017 F4 – Single Run
=====
=
Function      : Shifted and Rotated Rosenbrock's (30D)
Bounds       : [-100, 100]
Global optimum : 400.0
Initial D^(0) = 5.7976e+01
Generation 0: Best Fitness = 3.421845e+04 | Diversity = 5.7976e+01
Generation 10: Best = 4.973895e+03 FES = 1100 Diversity = 3.6319e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 2.023230e+03 FES = 2100 Diversity = 2.2609e+01 k_t
= [3, 3, 2, 2]
Generation 30: Best = 1.159589e+03 FES = 3100 Diversity = 1.6742e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 8.213169e+02 FES = 4100 Diversity = 1.2487e+01 k_t
= [3, 4, 3, 4]
Generation 50: Best = 6.420526e+02 FES = 5100 Diversity = 9.0516e+00 k_t
= [1, 1, 1, 1]
Generation 60: Best = 6.078267e+02 FES = 6100 Diversity = 6.9536e+00 k_t
= [2, 3, 4, 3]
Generation 70: Best = 5.593988e+02 FES = 7100 Diversity = 5.5271e+00 k_t
= [1, 1, 1, 1]
Generation 80: Best = 5.176695e+02 FES = 8100 Diversity = 4.1299e+00 k_t
= [4, 3, 4, 3]
Generation 90: Best = 5.081890e+02 FES = 9100 Diversity = 3.1218e+00 k_t
= [1, 1, 1, 1]
Generation 100: Best = 4.923973e+02 FES = 10100 Diversity = 2.0601e+00 k_t
= [2, 4, 4, 3]
Generation 110: Best = 4.845643e+02 FES = 11100 Diversity = 1.2459e+00 k_t
= [1, 1, 1, 1]
Generation 120: Best = 4.797427e+02 FES = 12100 Diversity = 8.7884e-01 k_t
= [3, 3, 3, 4]
Generation 130: Best = 4.766793e+02 FES = 13100 Diversity = 7.1160e-01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 4.748747e+02 FES = 14100 Diversity = 5.5886e-01 k_t
= [2, 3, 3, 3]
Generation 150: Best = 4.729548e+02 FES = 15100 Diversity = 4.2260e-01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 4.722347e+02 FES = 16100 Diversity = 3.2618e-01 k_t
= [2, 2, 2, 2]
Generation 170: Best = 4.712210e+02 FES = 17100 Diversity = 2.8427e-01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 4.703339e+02 FES = 18100 Diversity = 2.4895e-01 k_t
= [3, 3, 3, 2]
Generation 190: Best = 4.694203e+02 FES = 19100 Diversity = 2.8484e-01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 4.684268e+02 FES = 20100 Diversity = 2.8656e-01 k_t
= [4, 4, 3, 3]
Generation 210: Best = 4.671479e+02 FES = 21100 Diversity = 3.3416e-01 k_t
= [1, 1, 1, 1]
Generation 220: Best = 4.650703e+02 FES = 22100 Diversity = 3.7657e-01 k_t
= [3, 4, 2, 2]

```

Generation 230:	Best = 4.634531e+02	FES = 23100	Diversity = 3.9657e-01	k_t = [1, 1, 1, 1]
Generation 240:	Best = 4.610027e+02	FES = 24100	Diversity = 4.6059e-01	k_t = [2, 3, 4, 4]
Generation 250:	Best = 4.577287e+02	FES = 25100	Diversity = 5.1593e-01	k_t = [1, 1, 1, 1]
Generation 260:	Best = 4.528027e+02	FES = 26100	Diversity = 5.9057e-01	k_t = [2, 4, 2, 3]
Generation 270:	Best = 4.492245e+02	FES = 27100	Diversity = 5.8075e-01	k_t = [1, 1, 1, 1]
Generation 280:	Best = 4.419859e+02	FES = 28100	Diversity = 6.3982e-01	k_t = [2, 4, 2, 3]
Generation 290:	Best = 4.373132e+02	FES = 29100	Diversity = 7.1122e-01	k_t = [1, 1, 1, 1]
Generation 300:	Best = 4.308398e+02	FES = 30100	Diversity = 7.2011e-01	k_t = [4, 3, 4, 4]
Generation 310:	Best = 4.249353e+02	FES = 31100	Diversity = 7.1733e-01	k_t = [1, 1, 1, 1]
Generation 320:	Best = 4.205932e+02	FES = 32100	Diversity = 6.4702e-01	k_t = [3, 4, 4, 3]
Generation 330:	Best = 4.157254e+02	FES = 33100	Diversity = 6.0453e-01	k_t = [1, 1, 1, 1]
Generation 340:	Best = 4.116069e+02	FES = 34100	Diversity = 5.6452e-01	k_t = [3, 3, 2, 4]
Generation 350:	Best = 4.100090e+02	FES = 35100	Diversity = 4.7083e-01	k_t = [1, 1, 1, 1]
Generation 360:	Best = 4.075335e+02	FES = 36100	Diversity = 4.2229e-01	k_t = [3, 3, 2, 3]
Generation 370:	Best = 4.067816e+02	FES = 37100	Diversity = 3.4711e-01	k_t = [1, 1, 1, 1]
Generation 380:	Best = 4.056878e+02	FES = 38100	Diversity = 2.8439e-01	k_t = [2, 2, 2, 4]
Generation 390:	Best = 4.052504e+02	FES = 39100	Diversity = 2.1771e-01	k_t = [1, 1, 1, 1]
Generation 400:	Best = 4.048486e+02	FES = 40100	Diversity = 1.8347e-01	k_t = [3, 4, 4, 3]
Generation 410:	Best = 4.046743e+02	FES = 41100	Diversity = 1.4897e-01	k_t = [1, 1, 1, 1]
Generation 420:	Best = 4.045342e+02	FES = 42100	Diversity = 1.1539e-01	k_t = [2, 3, 3, 3]
Generation 430:	Best = 4.044542e+02	FES = 43100	Diversity = 9.0645e-02	k_t = [1, 1, 1, 1]
Generation 440:	Best = 4.044114e+02	FES = 44100	Diversity = 7.0906e-02	k_t = [2, 4, 2, 4]
Generation 450:	Best = 4.043729e+02	FES = 45100	Diversity = 4.9649e-02	k_t = [1, 1, 1, 1]
Generation 460:	Best = 4.043540e+02	FES = 46100	Diversity = 4.0563e-02	k_t = [2, 3, 2, 3]
Generation 470:	Best = 4.043442e+02	FES = 47100	Diversity = 3.3810e-02	k_t = [1, 1, 1, 1]
Generation 480:	Best = 4.043341e+02	FES = 48100	Diversity = 3.0324e-02	k_t = [4, 2, 3, 4]
Generation 490:	Best = 4.043268e+02	FES = 49100	Diversity = 2.5277e-02	k_t = [1, 1, 1, 1]

Generation 500:	Best = 4.043223e+02	FES = 50100	Diversity = 1.7594e-02	k_t = [4, 3, 2, 4]
Generation 510:	Best = 4.043207e+02	FES = 51100	Diversity = 1.1998e-02	k_t = [1, 1, 1, 1]
Generation 520:	Best = 4.043193e+02	FES = 52100	Diversity = 8.8924e-03	k_t = [3, 4, 3, 2]
Generation 530:	Best = 4.043181e+02	FES = 53100	Diversity = 8.5657e-03	k_t = [1, 1, 1, 1]
Generation 540:	Best = 4.043176e+02	FES = 54100	Diversity = 6.7094e-03	k_t = [2, 2, 2, 2]
Generation 550:	Best = 4.043170e+02	FES = 55100	Diversity = 5.8043e-03	k_t = [1, 1, 1, 1]
Generation 560:	Best = 4.043167e+02	FES = 56100	Diversity = 4.9733e-03	k_t = [2, 3, 2, 4]
Generation 570:	Best = 4.043164e+02	FES = 57100	Diversity = 4.0373e-03	k_t = [1, 1, 1, 1]
Generation 580:	Best = 4.043161e+02	FES = 58100	Diversity = 4.2121e-03	k_t = [4, 2, 3, 2]
Generation 590:	Best = 4.043159e+02	FES = 59100	Diversity = 3.8151e-03	k_t = [1, 1, 1, 1]
Generation 600:	Best = 4.043157e+02	FES = 60100	Diversity = 3.5499e-03	k_t = [2, 2, 3, 3]
Generation 610:	Best = 4.043155e+02	FES = 61100	Diversity = 3.1787e-03	k_t = [1, 1, 1, 1]
Generation 620:	Best = 4.043153e+02	FES = 62100	Diversity = 3.1066e-03	k_t = [2, 2, 3, 4]
Generation 630:	Best = 4.043152e+02	FES = 63100	Diversity = 2.8249e-03	k_t = [1, 1, 1, 1]
Generation 640:	Best = 4.043150e+02	FES = 64100	Diversity = 2.6732e-03	k_t = [2, 3, 4, 4]
Generation 650:	Best = 4.043149e+02	FES = 65100	Diversity = 2.6654e-03	k_t = [1, 1, 1, 1]
Generation 660:	Best = 4.043148e+02	FES = 66100	Diversity = 2.2479e-03	k_t = [4, 2, 4, 3]
Generation 670:	Best = 4.043147e+02	FES = 67100	Diversity = 2.1787e-03	k_t = [1, 1, 1, 1]
Generation 680:	Best = 4.043146e+02	FES = 68100	Diversity = 2.0903e-03	k_t = [4, 3, 3, 3]
Generation 690:	Best = 4.043144e+02	FES = 69100	Diversity = 2.5122e-03	k_t = [1, 1, 1, 1]
Generation 700:	Best = 4.043143e+02	FES = 70100	Diversity = 2.3326e-03	k_t = [4, 3, 4, 3]
Generation 710:	Best = 4.043142e+02	FES = 71100	Diversity = 2.4031e-03	k_t = [1, 1, 1, 1]
Generation 720:	Best = 4.043141e+02	FES = 72100	Diversity = 2.5128e-03	k_t = [3, 2, 2, 2]
Generation 730:	Best = 4.043139e+02	FES = 73100	Diversity = 2.7733e-03	k_t = [1, 1, 1, 1]
Generation 740:	Best = 4.043137e+02	FES = 74100	Diversity = 2.7497e-03	k_t = [4, 4, 3, 2]
Generation 750:	Best = 4.043136e+02	FES = 75100	Diversity = 2.5312e-03	k_t = [1, 1, 1, 1]
Generation 760:	Best = 4.043135e+02	FES = 76100	Diversity = 2.2718e-03	k_t = [3, 4, 3, 2]

Generation 770:	Best = 4.043134e+02	FES = 77100	Diversity = 2.5022e-03	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 4.043133e+02	FES = 78100	Diversity = 2.3747e-03	k_t
= [3, 2, 2, 2]				
Generation 790:	Best = 4.043131e+02	FES = 79100	Diversity = 2.5311e-03	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 4.043130e+02	FES = 80100	Diversity = 2.6236e-03	k_t
= [2, 4, 4, 3]				
Generation 810:	Best = 4.043129e+02	FES = 81100	Diversity = 2.3165e-03	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 4.043128e+02	FES = 82100	Diversity = 2.4772e-03	k_t
= [3, 2, 3, 4]				
Generation 830:	Best = 4.043127e+02	FES = 83100	Diversity = 2.7462e-03	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 4.043126e+02	FES = 84100	Diversity = 2.6689e-03	k_t
= [4, 4, 2, 2]				
Generation 850:	Best = 4.043124e+02	FES = 85100	Diversity = 2.8173e-03	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 4.043123e+02	FES = 86100	Diversity = 2.7854e-03	k_t
= [4, 2, 2, 2]				
Generation 870:	Best = 4.043122e+02	FES = 87100	Diversity = 2.6839e-03	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 4.043120e+02	FES = 88100	Diversity = 2.5925e-03	k_t
= [3, 4, 2, 3]				
Generation 890:	Best = 4.043119e+02	FES = 89100	Diversity = 2.3212e-03	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 4.043118e+02	FES = 90100	Diversity = 2.1696e-03	k_t
= [2, 2, 3, 2]				
Generation 910:	Best = 4.043117e+02	FES = 91100	Diversity = 2.3526e-03	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 4.043116e+02	FES = 92100	Diversity = 2.4817e-03	k_t
= [3, 3, 4, 3]				
Generation 930:	Best = 4.043115e+02	FES = 93100	Diversity = 2.3335e-03	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 4.043114e+02	FES = 94100	Diversity = 2.2717e-03	k_t
= [3, 4, 2, 4]				
Generation 950:	Best = 4.043112e+02	FES = 95100	Diversity = 2.4763e-03	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 4.043111e+02	FES = 96100	Diversity = 2.4136e-03	k_t
= [2, 3, 4, 4]				
Generation 970:	Best = 4.043109e+02	FES = 97100	Diversity = 2.7998e-03	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 4.043108e+02	FES = 98100	Diversity = 2.7207e-03	k_t
= [2, 2, 2, 2]				
Generation 990:	Best = 4.043107e+02	FES = 99100	Diversity = 2.2831e-03	k_t
= [1, 1, 1, 1]				
Generation 1000:	Best = 4.043106e+02	FES = 100100	Diversity = 2.1335e-03	k_t
= [3, 3, 3, 2]				
Generation 1010:	Best = 4.043105e+02	FES = 101100	Diversity = 2.0018e-03	k_t
= [1, 1, 1, 1]				
Generation 1020:	Best = 4.043104e+02	FES = 102100	Diversity = 2.2208e-03	k_t
= [2, 2, 4, 4]				
Generation 1030:	Best = 4.043103e+02	FES = 103100	Diversity = 2.0040e-03	k_t
= [1, 1, 1, 1]				

Generation 1040: Best = 4.043102e+02	FES = 104100	Diversity = 1.9503e-03	k_t
= [3, 3, 3, 4]			
Generation 1050: Best = 4.043101e+02	FES = 105100	Diversity = 1.9728e-03	k_t
= [1, 1, 1, 1]			
Generation 1060: Best = 4.043100e+02	FES = 106100	Diversity = 1.9396e-03	k_t
= [4, 2, 4, 4]			
Generation 1070: Best = 4.043099e+02	FES = 107100	Diversity = 1.9919e-03	k_t
= [1, 1, 1, 1]			
Generation 1080: Best = 4.043098e+02	FES = 108100	Diversity = 2.3371e-03	k_t
= [2, 3, 3, 4]			
Generation 1090: Best = 4.043097e+02	FES = 109100	Diversity = 2.2936e-03	k_t
= [1, 1, 1, 1]			
Generation 1100: Best = 4.043095e+02	FES = 110100	Diversity = 2.5250e-03	k_t
= [3, 2, 2, 2]			
Generation 1110: Best = 4.043094e+02	FES = 111100	Diversity = 2.2751e-03	k_t
= [1, 1, 1, 1]			
Generation 1120: Best = 4.043093e+02	FES = 112100	Diversity = 2.5345e-03	k_t
= [3, 4, 4, 2]			
Generation 1130: Best = 4.043091e+02	FES = 113100	Diversity = 2.5764e-03	k_t
= [1, 1, 1, 1]			
Generation 1140: Best = 4.043090e+02	FES = 114100	Diversity = 2.6767e-03	k_t
= [4, 2, 2, 2]			
Generation 1150: Best = 4.043089e+02	FES = 115100	Diversity = 2.5667e-03	k_t
= [1, 1, 1, 1]			
Generation 1160: Best = 4.043087e+02	FES = 116100	Diversity = 2.7730e-03	k_t
= [3, 3, 4, 2]			
Generation 1170: Best = 4.043085e+02	FES = 117100	Diversity = 3.1901e-03	k_t
= [1, 1, 1, 1]			
Generation 1180: Best = 4.043083e+02	FES = 118100	Diversity = 2.8398e-03	k_t
= [3, 3, 3, 4]			
Generation 1190: Best = 4.043082e+02	FES = 119100	Diversity = 2.6015e-03	k_t
= [1, 1, 1, 1]			
Generation 1200: Best = 4.043081e+02	FES = 120100	Diversity = 2.2678e-03	k_t
= [2, 2, 3, 3]			
Generation 1210: Best = 4.043080e+02	FES = 121100	Diversity = 2.0283e-03	k_t
= [1, 1, 1, 1]			
Generation 1220: Best = 4.043079e+02	FES = 122100	Diversity = 2.2248e-03	k_t
= [2, 3, 4, 4]			
Generation 1230: Best = 4.043078e+02	FES = 123100	Diversity = 2.4953e-03	k_t
= [1, 1, 1, 1]			
Generation 1240: Best = 4.043077e+02	FES = 124100	Diversity = 2.4737e-03	k_t
= [3, 2, 3, 4]			
Generation 1250: Best = 4.043075e+02	FES = 125100	Diversity = 2.8107e-03	k_t
= [1, 1, 1, 1]			
Generation 1260: Best = 4.043074e+02	FES = 126100	Diversity = 2.7600e-03	k_t
= [4, 2, 2, 4]			
Generation 1270: Best = 4.043072e+02	FES = 127100	Diversity = 2.5318e-03	k_t
= [1, 1, 1, 1]			
Generation 1280: Best = 4.043070e+02	FES = 128100	Diversity = 2.6430e-03	k_t
= [4, 4, 4, 3]			
Generation 1290: Best = 4.043069e+02	FES = 129100	Diversity = 2.8986e-03	k_t
= [1, 1, 1, 1]			
Generation 1300: Best = 4.043067e+02	FES = 130100	Diversity = 2.6400e-03	k_t
= [3, 2, 3, 3]			

Generation 1310: Best = 4.043066e+02	FES = 131100	Diversity = 2.2980e-03	k_t
= [1, 1, 1, 1]			
Generation 1320: Best = 4.043065e+02	FES = 132100	Diversity = 2.2173e-03	k_t
= [3, 2, 4, 3]			
Generation 1330: Best = 4.043064e+02	FES = 133100	Diversity = 2.1492e-03	k_t
= [1, 1, 1, 1]			
Generation 1340: Best = 4.043062e+02	FES = 134100	Diversity = 2.2305e-03	k_t
= [3, 2, 4, 2]			
Generation 1350: Best = 4.043061e+02	FES = 135100	Diversity = 2.0767e-03	k_t
= [1, 1, 1, 1]			
Generation 1360: Best = 4.043060e+02	FES = 136100	Diversity = 2.2968e-03	k_t
= [2, 2, 3, 2]			
Generation 1370: Best = 4.043059e+02	FES = 137100	Diversity = 2.3864e-03	k_t
= [1, 1, 1, 1]			
Generation 1380: Best = 4.043058e+02	FES = 138100	Diversity = 2.0976e-03	k_t
= [3, 4, 2, 3]			
Generation 1390: Best = 4.043057e+02	FES = 139100	Diversity = 1.8538e-03	k_t
= [1, 1, 1, 1]			
Generation 1400: Best = 4.043056e+02	FES = 140100	Diversity = 1.9379e-03	k_t
= [4, 4, 3, 2]			
Generation 1410: Best = 4.043054e+02	FES = 141100	Diversity = 2.0044e-03	k_t
= [1, 1, 1, 1]			
Generation 1420: Best = 4.043053e+02	FES = 142100	Diversity = 2.1962e-03	k_t
= [2, 4, 3, 3]			
Generation 1430: Best = 4.043052e+02	FES = 143100	Diversity = 2.4101e-03	k_t
= [1, 1, 1, 1]			
Generation 1440: Best = 4.043050e+02	FES = 144100	Diversity = 2.5643e-03	k_t
= [4, 4, 3, 3]			
Generation 1450: Best = 4.043048e+02	FES = 145100	Diversity = 2.7725e-03	k_t
= [1, 1, 1, 1]			
Generation 1460: Best = 4.043047e+02	FES = 146100	Diversity = 2.5587e-03	k_t
= [3, 4, 3, 2]			
Generation 1470: Best = 4.043046e+02	FES = 147100	Diversity = 2.4786e-03	k_t
= [1, 1, 1, 1]			
Generation 1480: Best = 4.043044e+02	FES = 148100	Diversity = 2.5383e-03	k_t
= [3, 3, 3, 4]			
Generation 1490: Best = 4.043043e+02	FES = 149100	Diversity = 2.4049e-03	k_t
= [1, 1, 1, 1]			
Generation 1500: Best = 4.043042e+02	FES = 150100	Diversity = 2.1710e-03	k_t
= [3, 2, 4, 2]			
Generation 1510: Best = 4.043041e+02	FES = 151100	Diversity = 2.0384e-03	k_t
= [1, 1, 1, 1]			
Generation 1520: Best = 4.043040e+02	FES = 152100	Diversity = 1.7052e-03	k_t
= [3, 4, 4, 4]			
Generation 1530: Best = 4.043039e+02	FES = 153100	Diversity = 1.6060e-03	k_t
= [1, 1, 1, 1]			
Generation 1540: Best = 4.043038e+02	FES = 154100	Diversity = 1.7092e-03	k_t
= [3, 3, 3, 3]			
Generation 1550: Best = 4.043038e+02	FES = 155100	Diversity = 1.8702e-03	k_t
= [1, 1, 1, 1]			
Generation 1560: Best = 4.043037e+02	FES = 156100	Diversity = 1.9827e-03	k_t
= [2, 2, 2, 3]			
Generation 1570: Best = 4.043035e+02	FES = 157100	Diversity = 2.0741e-03	k_t
= [1, 1, 1, 1]			

Generation 1580: Best = 4.043034e+02	FES = 158100	Diversity = 1.8501e-03	k_t
= [3, 3, 4, 2]			
Generation 1590: Best = 4.043034e+02	FES = 159100	Diversity = 1.6272e-03	k_t
= [1, 1, 1, 1]			
Generation 1600: Best = 4.043033e+02	FES = 160100	Diversity = 1.6921e-03	k_t
= [4, 3, 4, 3]			
Generation 1610: Best = 4.043032e+02	FES = 161100	Diversity = 1.9952e-03	k_t
= [1, 1, 1, 1]			
Generation 1620: Best = 4.043030e+02	FES = 162100	Diversity = 1.9604e-03	k_t
= [4, 3, 2, 3]			
Generation 1630: Best = 4.043030e+02	FES = 163100	Diversity = 1.8598e-03	k_t
= [1, 1, 1, 1]			
Generation 1640: Best = 4.043029e+02	FES = 164100	Diversity = 1.7975e-03	k_t
= [4, 2, 3, 3]			
Generation 1650: Best = 4.043028e+02	FES = 165100	Diversity = 1.7470e-03	k_t
= [1, 1, 1, 1]			
Generation 1660: Best = 4.043027e+02	FES = 166100	Diversity = 1.7883e-03	k_t
= [4, 2, 2, 2]			
Generation 1670: Best = 4.043025e+02	FES = 167100	Diversity = 2.1971e-03	k_t
= [1, 1, 1, 1]			
Generation 1680: Best = 4.043023e+02	FES = 168100	Diversity = 2.5157e-03	k_t
= [3, 3, 4, 4]			
Generation 1690: Best = 4.043022e+02	FES = 169100	Diversity = 2.2702e-03	k_t
= [1, 1, 1, 1]			
Generation 1700: Best = 4.043021e+02	FES = 170100	Diversity = 1.8067e-03	k_t
= [4, 3, 4, 2]			
Generation 1710: Best = 4.043020e+02	FES = 171100	Diversity = 1.6141e-03	k_t
= [1, 1, 1, 1]			
Generation 1720: Best = 4.043019e+02	FES = 172100	Diversity = 1.9813e-03	k_t
= [4, 2, 2, 2]			
Generation 1730: Best = 4.043018e+02	FES = 173100	Diversity = 1.8855e-03	k_t
= [1, 1, 1, 1]			
Generation 1740: Best = 4.043017e+02	FES = 174100	Diversity = 1.7275e-03	k_t
= [2, 3, 4, 2]			
Generation 1750: Best = 4.043016e+02	FES = 175100	Diversity = 1.4243e-03	k_t
= [1, 1, 1, 1]			
Generation 1760: Best = 4.043015e+02	FES = 176100	Diversity = 1.4902e-03	k_t
= [3, 3, 3, 4]			
Generation 1770: Best = 4.043014e+02	FES = 177100	Diversity = 1.6982e-03	k_t
= [1, 1, 1, 1]			
Generation 1780: Best = 4.043014e+02	FES = 178100	Diversity = 1.5825e-03	k_t
= [2, 4, 4, 2]			
Generation 1790: Best = 4.043012e+02	FES = 179100	Diversity = 1.8659e-03	k_t
= [1, 1, 1, 1]			
Generation 1800: Best = 4.043012e+02	FES = 180100	Diversity = 1.5933e-03	k_t
= [2, 2, 3, 2]			
Generation 1810: Best = 4.043011e+02	FES = 181100	Diversity = 1.7030e-03	k_t
= [1, 1, 1, 1]			
Generation 1820: Best = 4.043010e+02	FES = 182100	Diversity = 1.6134e-03	k_t
= [4, 2, 2, 2]			
Generation 1830: Best = 4.043009e+02	FES = 183100	Diversity = 1.9196e-03	k_t
= [1, 1, 1, 1]			
Generation 1840: Best = 4.043008e+02	FES = 184100	Diversity = 1.6730e-03	k_t
= [4, 4, 4, 2]			

Generation 1850: Best = 4.043007e+02 = [1, 1, 1, 1]	FES = 185100	Diversity = 2.0086e-03	k_t
Generation 1860: Best = 4.043006e+02 = [2, 3, 2, 4]	FES = 186100	Diversity = 1.7590e-03	k_t
Generation 1870: Best = 4.043005e+02 = [1, 1, 1, 1]	FES = 187100	Diversity = 2.1185e-03	k_t
Generation 1880: Best = 4.043004e+02 = [4, 3, 4, 4]	FES = 188100	Diversity = 1.9369e-03	k_t
Generation 1890: Best = 4.043002e+02 = [1, 1, 1, 1]	FES = 189100	Diversity = 2.1945e-03	k_t
Generation 1900: Best = 4.043000e+02 = [2, 2, 4, 3]	FES = 190100	Diversity = 2.3594e-03	k_t
Generation 1910: Best = 4.042999e+02 = [1, 1, 1, 1]	FES = 191100	Diversity = 2.5251e-03	k_t
Generation 1920: Best = 4.042997e+02 = [4, 2, 4, 4]	FES = 192100	Diversity = 2.5718e-03	k_t
Generation 1930: Best = 4.042995e+02 = [1, 1, 1, 1]	FES = 193100	Diversity = 2.5387e-03	k_t
Generation 1940: Best = 4.042995e+02 = [3, 2, 2, 2]	FES = 194100	Diversity = 1.9795e-03	k_t
Generation 1950: Best = 4.042994e+02 = [1, 1, 1, 1]	FES = 195100	Diversity = 1.7734e-03	k_t
Generation 1960: Best = 4.042993e+02 = [3, 3, 4, 4]	FES = 196100	Diversity = 1.4910e-03	k_t
Generation 1970: Best = 4.042992e+02 = [1, 1, 1, 1]	FES = 197100	Diversity = 1.3080e-03	k_t
Generation 1980: Best = 4.042992e+02 = [3, 3, 4, 2]	FES = 198100	Diversity = 1.5206e-03	k_t
Generation 1990: Best = 4.042991e+02 = [1, 1, 1, 1]	FES = 199100	Diversity = 1.4139e-03	k_t
Generation 2000: Best = 4.042990e+02 = [4, 4, 2, 2]	FES = 200100	Diversity = 1.9187e-03	k_t
Generation 2010: Best = 4.042989e+02 = [1, 1, 1, 1]	FES = 201100	Diversity = 1.9733e-03	k_t
Generation 2020: Best = 4.042988e+02 = [4, 3, 4, 3]	FES = 202100	Diversity = 1.8662e-03	k_t
Generation 2030: Best = 4.042987e+02 = [1, 1, 1, 1]	FES = 203100	Diversity = 1.6366e-03	k_t
Generation 2040: Best = 4.042986e+02 = [2, 3, 2, 3]	FES = 204100	Diversity = 1.3883e-03	k_t
Generation 2050: Best = 4.042986e+02 = [1, 1, 1, 1]	FES = 205100	Diversity = 1.3744e-03	k_t
Generation 2060: Best = 4.042985e+02 = [3, 4, 4, 4]	FES = 206100	Diversity = 1.6411e-03	k_t
Generation 2070: Best = 4.042984e+02 = [1, 1, 1, 1]	FES = 207100	Diversity = 1.4871e-03	k_t
Generation 2080: Best = 4.042984e+02 = [3, 2, 3, 3]	FES = 208100	Diversity = 1.1857e-03	k_t
Generation 2090: Best = 4.042983e+02 = [1, 1, 1, 1]	FES = 209100	Diversity = 1.3619e-03	k_t
Generation 2100: Best = 4.042983e+02 = [3, 2, 3, 4]	FES = 210100	Diversity = 1.4335e-03	k_t
Generation 2110: Best = 4.042982e+02 = [1, 1, 1, 1]	FES = 211100	Diversity = 1.3631e-03	k_t

Generation 2120: Best = 4.042982e+02 = [3, 4, 4, 2]	FES = 212100	Diversity = 1.2047e-03	k_t
Generation 2130: Best = 4.042981e+02 = [1, 1, 1, 1]	FES = 213100	Diversity = 1.3983e-03	k_t
Generation 2140: Best = 4.042981e+02 = [3, 3, 4, 3]	FES = 214100	Diversity = 1.5548e-03	k_t
Generation 2150: Best = 4.042980e+02 = [1, 1, 1, 1]	FES = 215100	Diversity = 1.8736e-03	k_t
Generation 2160: Best = 4.042979e+02 = [3, 2, 3, 3]	FES = 216100	Diversity = 1.7781e-03	k_t
Generation 2170: Best = 4.042977e+02 = [1, 1, 1, 1]	FES = 217100	Diversity = 1.8252e-03	k_t
Generation 2180: Best = 4.042976e+02 = [2, 2, 4, 4]	FES = 218100	Diversity = 2.0340e-03	k_t
Generation 2190: Best = 4.042975e+02 = [1, 1, 1, 1]	FES = 219100	Diversity = 2.0503e-03	k_t
Generation 2200: Best = 4.042975e+02 = [3, 3, 3, 3]	FES = 220100	Diversity = 1.5684e-03	k_t
Generation 2210: Best = 4.042974e+02 = [1, 1, 1, 1]	FES = 221100	Diversity = 1.5442e-03	k_t
Generation 2220: Best = 4.042973e+02 = [3, 3, 4, 3]	FES = 222100	Diversity = 1.6935e-03	k_t
Generation 2230: Best = 4.042972e+02 = [1, 1, 1, 1]	FES = 223100	Diversity = 1.7004e-03	k_t
Generation 2240: Best = 4.042971e+02 = [3, 4, 3, 2]	FES = 224100	Diversity = 1.6411e-03	k_t
Generation 2250: Best = 4.042971e+02 = [1, 1, 1, 1]	FES = 225100	Diversity = 1.4848e-03	k_t
Generation 2260: Best = 4.042970e+02 = [2, 2, 2, 3]	FES = 226100	Diversity = 1.5169e-03	k_t
Generation 2270: Best = 4.042969e+02 = [1, 1, 1, 1]	FES = 227100	Diversity = 1.8598e-03	k_t
Generation 2280: Best = 4.042968e+02 = [4, 2, 3, 3]	FES = 228100	Diversity = 1.8620e-03	k_t
Generation 2290: Best = 4.042967e+02 = [1, 1, 1, 1]	FES = 229100	Diversity = 1.9297e-03	k_t
Generation 2300: Best = 4.042966e+02 = [4, 2, 2, 2]	FES = 230100	Diversity = 1.7966e-03	k_t
Generation 2310: Best = 4.042965e+02 = [1, 1, 1, 1]	FES = 231100	Diversity = 1.5222e-03	k_t
Generation 2320: Best = 4.042965e+02 = [4, 2, 4, 4]	FES = 232100	Diversity = 1.2957e-03	k_t
Generation 2330: Best = 4.042964e+02 = [1, 1, 1, 1]	FES = 233100	Diversity = 1.8461e-03	k_t
Generation 2340: Best = 4.042963e+02 = [2, 2, 4, 3]	FES = 234100	Diversity = 1.9155e-03	k_t
Generation 2350: Best = 4.042961e+02 = [1, 1, 1, 1]	FES = 235100	Diversity = 2.0227e-03	k_t
Generation 2360: Best = 4.042960e+02 = [3, 2, 2, 3]	FES = 236100	Diversity = 1.6897e-03	k_t
Generation 2370: Best = 4.042959e+02 = [1, 1, 1, 1]	FES = 237100	Diversity = 1.8622e-03	k_t
Generation 2380: Best = 4.042958e+02 = [4, 2, 4, 3]	FES = 238100	Diversity = 2.6311e-03	k_t

Generation 2390: Best = 4.042956e+02	FES = 239100	Diversity = 2.5441e-03	k_t
= [1, 1, 1, 1]			
Generation 2400: Best = 4.042955e+02	FES = 240100	Diversity = 2.0908e-03	k_t
= [2, 2, 2, 3]			
Generation 2410: Best = 4.042954e+02	FES = 241100	Diversity = 2.1225e-03	k_t
= [1, 1, 1, 1]			
Generation 2420: Best = 4.042953e+02	FES = 242100	Diversity = 1.8105e-03	k_t
= [2, 3, 3, 2]			
Generation 2430: Best = 4.042951e+02	FES = 243100	Diversity = 1.9407e-03	k_t
= [1, 1, 1, 1]			
Generation 2440: Best = 4.042951e+02	FES = 244100	Diversity = 1.9080e-03	k_t
= [3, 4, 2, 2]			
Generation 2450: Best = 4.042950e+02	FES = 245100	Diversity = 1.8293e-03	k_t
= [1, 1, 1, 1]			
Generation 2460: Best = 4.042948e+02	FES = 246100	Diversity = 1.9468e-03	k_t
= [3, 2, 4, 2]			
Generation 2470: Best = 4.042947e+02	FES = 247100	Diversity = 1.8948e-03	k_t
= [1, 1, 1, 1]			
Generation 2480: Best = 4.042946e+02	FES = 248100	Diversity = 2.2713e-03	k_t
= [3, 3, 2, 3]			
Generation 2490: Best = 4.042945e+02	FES = 249100	Diversity = 1.8129e-03	k_t
= [1, 1, 1, 1]			
Generation 2500: Best = 4.042944e+02	FES = 250100	Diversity = 1.7878e-03	k_t
= [4, 2, 3, 3]			
Generation 2510: Best = 4.042943e+02	FES = 251100	Diversity = 1.6419e-03	k_t
= [1, 1, 1, 1]			
Generation 2520: Best = 4.042942e+02	FES = 252100	Diversity = 1.3854e-03	k_t
= [4, 4, 3, 3]			
Generation 2530: Best = 4.042941e+02	FES = 253100	Diversity = 1.4371e-03	k_t
= [1, 1, 1, 1]			
Generation 2540: Best = 4.042941e+02	FES = 254100	Diversity = 1.2554e-03	k_t
= [4, 3, 2, 4]			
Generation 2550: Best = 4.042940e+02	FES = 255100	Diversity = 1.2403e-03	k_t
= [1, 1, 1, 1]			
Generation 2560: Best = 4.042939e+02	FES = 256100	Diversity = 1.7448e-03	k_t
= [3, 3, 3, 4]			
Generation 2570: Best = 4.042938e+02	FES = 257100	Diversity = 1.9205e-03	k_t
= [1, 1, 1, 1]			
Generation 2580: Best = 4.042937e+02	FES = 258100	Diversity = 1.8853e-03	k_t
= [2, 4, 2, 2]			
Generation 2590: Best = 4.042936e+02	FES = 259100	Diversity = 1.4444e-03	k_t
= [1, 1, 1, 1]			
Generation 2600: Best = 4.042935e+02	FES = 260100	Diversity = 1.5929e-03	k_t
= [2, 4, 4, 4]			
Generation 2610: Best = 4.042935e+02	FES = 261100	Diversity = 1.4588e-03	k_t
= [1, 1, 1, 1]			
Generation 2620: Best = 4.042934e+02	FES = 262100	Diversity = 1.5651e-03	k_t
= [2, 2, 4, 3]			
Generation 2630: Best = 4.042933e+02	FES = 263100	Diversity = 1.6895e-03	k_t
= [1, 1, 1, 1]			
Generation 2640: Best = 4.042932e+02	FES = 264100	Diversity = 1.7785e-03	k_t
= [3, 4, 2, 2]			
Generation 2650: Best = 4.042931e+02	FES = 265100	Diversity = 1.7272e-03	k_t
= [1, 1, 1, 1]			

Generation 2660: Best = 4.042930e+02 = [2, 2, 3, 4]	FES = 266100	Diversity = 2.4558e-03	k_t
Generation 2670: Best = 4.042928e+02 = [1, 1, 1, 1]	FES = 267100	Diversity = 2.5710e-03	k_t
Generation 2680: Best = 4.042927e+02 = [4, 2, 2, 4]	FES = 268100	Diversity = 2.5414e-03	k_t
Generation 2690: Best = 4.042926e+02 = [1, 1, 1, 1]	FES = 269100	Diversity = 2.2538e-03	k_t
Generation 2700: Best = 4.042925e+02 = [4, 3, 3, 4]	FES = 270100	Diversity = 1.6553e-03	k_t
Generation 2710: Best = 4.042924e+02 = [1, 1, 1, 1]	FES = 271100	Diversity = 1.7940e-03	k_t
Generation 2720: Best = 4.042923e+02 = [2, 3, 2, 2]	FES = 272100	Diversity = 1.7431e-03	k_t
Generation 2730: Best = 4.042921e+02 = [1, 1, 1, 1]	FES = 273100	Diversity = 1.9955e-03	k_t
Generation 2740: Best = 4.042920e+02 = [4, 4, 4, 2]	FES = 274100	Diversity = 2.0408e-03	k_t
Generation 2750: Best = 4.042919e+02 = [1, 1, 1, 1]	FES = 275100	Diversity = 1.9320e-03	k_t
Generation 2760: Best = 4.042918e+02 = [2, 2, 4, 4]	FES = 276100	Diversity = 1.7810e-03	k_t
Generation 2770: Best = 4.042917e+02 = [1, 1, 1, 1]	FES = 277100	Diversity = 1.8645e-03	k_t
Generation 2780: Best = 4.042916e+02 = [3, 2, 2, 2]	FES = 278100	Diversity = 1.9412e-03	k_t
Generation 2790: Best = 4.042915e+02 = [1, 1, 1, 1]	FES = 279100	Diversity = 1.5296e-03	k_t
Generation 2800: Best = 4.042914e+02 = [2, 4, 3, 3]	FES = 280100	Diversity = 1.9533e-03	k_t
Generation 2810: Best = 4.042913e+02 = [1, 1, 1, 1]	FES = 281100	Diversity = 2.2450e-03	k_t
Generation 2820: Best = 4.042912e+02 = [3, 4, 3, 3]	FES = 282100	Diversity = 1.8602e-03	k_t
Generation 2830: Best = 4.042912e+02 = [1, 1, 1, 1]	FES = 283100	Diversity = 1.3838e-03	k_t
Generation 2840: Best = 4.042911e+02 = [3, 3, 2, 4]	FES = 284100	Diversity = 1.3553e-03	k_t
Generation 2850: Best = 4.042910e+02 = [1, 1, 1, 1]	FES = 285100	Diversity = 1.6088e-03	k_t
Generation 2860: Best = 4.042909e+02 = [2, 4, 2, 2]	FES = 286100	Diversity = 1.8428e-03	k_t
Generation 2870: Best = 4.042908e+02 = [1, 1, 1, 1]	FES = 287100	Diversity = 1.9852e-03	k_t
Generation 2880: Best = 4.042907e+02 = [3, 4, 3, 4]	FES = 288100	Diversity = 2.1024e-03	k_t
Generation 2890: Best = 4.042905e+02 = [1, 1, 1, 1]	FES = 289100	Diversity = 2.7526e-03	k_t
Generation 2900: Best = 4.042904e+02 = [3, 3, 3, 3]	FES = 290100	Diversity = 2.0136e-03	k_t
Generation 2910: Best = 4.042903e+02 = [1, 1, 1, 1]	FES = 291100	Diversity = 2.1191e-03	k_t
Generation 2920: Best = 4.042901e+02 = [4, 2, 3, 3]	FES = 292100	Diversity = 2.2593e-03	k_t

Generation 2930: Best = 4.042900e+02 FES = 293100 Diversity = 2.0995e-03 k_t
 = [1, 1, 1, 1]
 Generation 2940: Best = 4.042898e+02 FES = 294100 Diversity = 2.5393e-03 k_t
 = [4, 3, 2, 4]
 Generation 2950: Best = 4.042897e+02 FES = 295100 Diversity = 1.9575e-03 k_t
 = [1, 1, 1, 1]
 Generation 2960: Best = 4.042896e+02 FES = 296100 Diversity = 1.8815e-03 k_t
 = [4, 4, 2, 3]
 Generation 2970: Best = 4.042895e+02 FES = 297100 Diversity = 1.7520e-03 k_t
 = [1, 1, 1, 1]
 Generation 2980: Best = 4.042894e+02 FES = 298100 Diversity = 2.2182e-03 k_t
 = [4, 4, 3, 2]
 Generation 2990: Best = 4.042892e+02 FES = 299100 Diversity = 2.2102e-03 k_t
 = [1, 1, 1, 1]

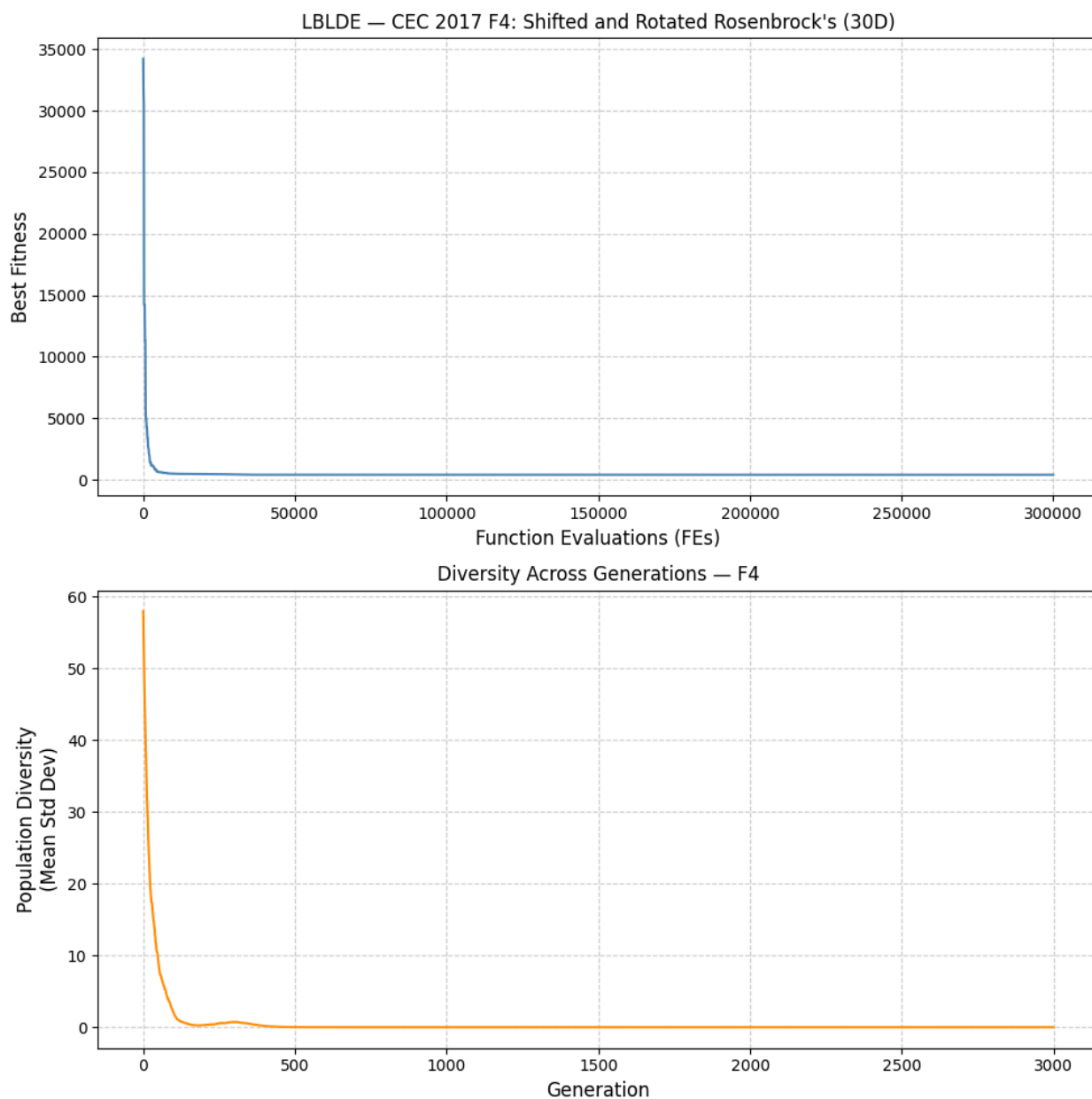
Optimization completed!

Final Best Fitness = 4.042891e+02

Total Function Evaluations = 300000

```

=====
=
Results:
  Best Fitness           : 4.042891e+02
  Global Optimum         : 4.000000e+02
  Error (|best-optimum|) : 4.289085e+00
  Best Solution (5 dim)  : [ 71.71513883   7.32100437 -41.07715158  96.71870618
31.5578265 ]
  D^(0) (init. diversity): 5.7976e+01
=====
=
  
```



Graph saved as convergence_F4.png

```
In [15]: # =====
# F5 – Shifted and Rotated Rastrigin
# =====
run_and_plot('F5', make_wrapper(simple.f5), 500.0, "Shifted and Rotated Rastrigin")
```

```

=====
=
LBLDE_s on CEC 2017 F5 – Single Run
=====
=
Function      : Shifted and Rotated Rastrigin's (30D)
Bounds       : [-100, 100]
Global optimum : 500.0
Initial D^(0) = 5.7976e+01
Generation 0: Best Fitness = 1.177228e+03 | Diversity = 5.7976e+01
Generation 10: Best = 8.736848e+02 FES = 1100 Diversity = 3.6008e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 7.788720e+02 FES = 2100 Diversity = 2.7213e+01 k_t
= [3, 3, 2, 2]
Generation 30: Best = 7.211696e+02 FES = 3100 Diversity = 2.3929e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 7.211696e+02 FES = 4100 Diversity = 2.2494e+01 k_t
= [3, 4, 3, 4]
Generation 50: Best = 7.211696e+02 FES = 5100 Diversity = 2.1702e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 7.094231e+02 FES = 6100 Diversity = 2.1040e+01 k_t
= [3, 2, 3, 3]
Generation 70: Best = 7.094231e+02 FES = 7100 Diversity = 2.0614e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 6.992564e+02 FES = 8100 Diversity = 2.0281e+01 k_t
= [3, 2, 3, 4]
Generation 90: Best = 6.992564e+02 FES = 9100 Diversity = 1.9786e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 6.932204e+02 FES = 10100 Diversity = 1.9757e+01 k_t
= [4, 3, 2, 2]
Generation 110: Best = 6.853405e+02 FES = 11100 Diversity = 1.9641e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 6.853405e+02 FES = 12100 Diversity = 1.9479e+01 k_t
= [2, 3, 3, 2]
Generation 130: Best = 6.853405e+02 FES = 13100 Diversity = 1.9381e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 6.745698e+02 FES = 14100 Diversity = 1.9497e+01 k_t
= [4, 4, 2, 2]
Generation 150: Best = 6.485942e+02 FES = 15100 Diversity = 1.9425e+01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 6.469029e+02 FES = 16100 Diversity = 1.9092e+01 k_t
= [2, 2, 2, 4]
Generation 170: Best = 6.469029e+02 FES = 17100 Diversity = 1.9161e+01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 6.469029e+02 FES = 18100 Diversity = 1.9088e+01 k_t
= [4, 4, 2, 3]
Generation 190: Best = 6.462102e+02 FES = 19100 Diversity = 1.8916e+01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 6.462102e+02 FES = 20100 Diversity = 1.8864e+01 k_t
= [3, 4, 2, 2]
Generation 210: Best = 6.462102e+02 FES = 21100 Diversity = 1.8782e+01 k_t
= [1, 1, 1, 1]
Generation 220: Best = 6.462102e+02 FES = 22100 Diversity = 1.8881e+01 k_t
= [3, 2, 2, 3]

```

Generation 230:	Best = 6.462102e+02	FES = 23100	Diversity = 1.8843e+01	k_t = [1, 1, 1, 1]
Generation 240:	Best = 6.462102e+02	FES = 24100	Diversity = 1.8771e+01	k_t = [3, 2, 3, 4]
Generation 250:	Best = 6.462102e+02	FES = 25100	Diversity = 1.8793e+01	k_t = [1, 1, 1, 1]
Generation 260:	Best = 6.462102e+02	FES = 26100	Diversity = 1.8803e+01	k_t = [2, 2, 3, 2]
Generation 270:	Best = 6.462102e+02	FES = 27100	Diversity = 1.8811e+01	k_t = [1, 1, 1, 1]
Generation 280:	Best = 6.390187e+02	FES = 28100	Diversity = 1.8690e+01	k_t = [4, 4, 3, 3]
Generation 290:	Best = 6.382178e+02	FES = 29100	Diversity = 1.8710e+01	k_t = [1, 1, 1, 1]
Generation 300:	Best = 6.349138e+02	FES = 30100	Diversity = 1.8748e+01	k_t = [4, 4, 2, 2]
Generation 310:	Best = 6.349138e+02	FES = 31100	Diversity = 1.8732e+01	k_t = [1, 1, 1, 1]
Generation 320:	Best = 6.349138e+02	FES = 32100	Diversity = 1.8420e+01	k_t = [3, 2, 4, 2]
Generation 330:	Best = 6.349138e+02	FES = 33100	Diversity = 1.8310e+01	k_t = [1, 1, 1, 1]
Generation 340:	Best = 6.252935e+02	FES = 34100	Diversity = 1.8225e+01	k_t = [2, 2, 4, 2]
Generation 350:	Best = 6.252935e+02	FES = 35100	Diversity = 1.8090e+01	k_t = [1, 1, 1, 1]
Generation 360:	Best = 6.221862e+02	FES = 36100	Diversity = 1.8069e+01	k_t = [2, 4, 2, 2]
Generation 370:	Best = 6.221862e+02	FES = 37100	Diversity = 1.7947e+01	k_t = [1, 1, 1, 1]
Generation 380:	Best = 6.211445e+02	FES = 38100	Diversity = 1.7783e+01	k_t = [2, 4, 4, 3]
Generation 390:	Best = 6.211445e+02	FES = 39100	Diversity = 1.7735e+01	k_t = [1, 1, 1, 1]
Generation 400:	Best = 6.211445e+02	FES = 40100	Diversity = 1.7722e+01	k_t = [3, 4, 2, 4]
Generation 410:	Best = 6.211445e+02	FES = 41100	Diversity = 1.7758e+01	k_t = [1, 1, 1, 1]
Generation 420:	Best = 6.211445e+02	FES = 42100	Diversity = 1.7719e+01	k_t = [3, 2, 2, 3]
Generation 430:	Best = 6.151601e+02	FES = 43100	Diversity = 1.7721e+01	k_t = [1, 1, 1, 1]
Generation 440:	Best = 6.149230e+02	FES = 44100	Diversity = 1.7642e+01	k_t = [3, 3, 2, 2]
Generation 450:	Best = 6.149230e+02	FES = 45100	Diversity = 1.7606e+01	k_t = [1, 1, 1, 1]
Generation 460:	Best = 6.093766e+02	FES = 46100	Diversity = 1.7499e+01	k_t = [4, 4, 2, 4]
Generation 470:	Best = 6.052445e+02	FES = 47100	Diversity = 1.7360e+01	k_t = [1, 1, 1, 1]
Generation 480:	Best = 6.052445e+02	FES = 48100	Diversity = 1.7352e+01	k_t = [2, 3, 2, 4]
Generation 490:	Best = 6.052445e+02	FES = 49100	Diversity = 1.7344e+01	k_t = [1, 1, 1, 1]

Generation 500:	Best = 6.052445e+02	FES = 50100	Diversity = 1.7375e+01	k_t = [2, 4, 2, 4]
Generation 510:	Best = 6.052445e+02	FES = 51100	Diversity = 1.7317e+01	k_t = [1, 1, 1, 1]
Generation 520:	Best = 6.052445e+02	FES = 52100	Diversity = 1.7276e+01	k_t = [4, 3, 3, 3]
Generation 530:	Best = 6.052445e+02	FES = 53100	Diversity = 1.7330e+01	k_t = [1, 1, 1, 1]
Generation 540:	Best = 6.052445e+02	FES = 54100	Diversity = 1.7319e+01	k_t = [4, 2, 3, 4]
Generation 550:	Best = 6.052445e+02	FES = 55100	Diversity = 1.7306e+01	k_t = [1, 1, 1, 1]
Generation 560:	Best = 6.052445e+02	FES = 56100	Diversity = 1.7298e+01	k_t = [2, 2, 3, 3]
Generation 570:	Best = 6.034237e+02	FES = 57100	Diversity = 1.7261e+01	k_t = [1, 1, 1, 1]
Generation 580:	Best = 6.006013e+02	FES = 58100	Diversity = 1.7255e+01	k_t = [2, 4, 4, 3]
Generation 590:	Best = 5.986483e+02	FES = 59100	Diversity = 1.7256e+01	k_t = [1, 1, 1, 1]
Generation 600:	Best = 5.986483e+02	FES = 60100	Diversity = 1.7241e+01	k_t = [4, 4, 4, 3]
Generation 610:	Best = 5.986483e+02	FES = 61100	Diversity = 1.7276e+01	k_t = [1, 1, 1, 1]
Generation 620:	Best = 5.986483e+02	FES = 62100	Diversity = 1.7252e+01	k_t = [3, 4, 4, 3]
Generation 630:	Best = 5.986483e+02	FES = 63100	Diversity = 1.7242e+01	k_t = [1, 1, 1, 1]
Generation 640:	Best = 5.986483e+02	FES = 64100	Diversity = 1.7217e+01	k_t = [3, 2, 2, 4]
Generation 650:	Best = 5.971819e+02	FES = 65100	Diversity = 1.7217e+01	k_t = [1, 1, 1, 1]
Generation 660:	Best = 5.963155e+02	FES = 66100	Diversity = 1.7209e+01	k_t = [3, 2, 2, 3]
Generation 670:	Best = 5.889241e+02	FES = 67100	Diversity = 1.7228e+01	k_t = [1, 1, 1, 1]
Generation 680:	Best = 5.889241e+02	FES = 68100	Diversity = 1.7278e+01	k_t = [3, 4, 3, 3]
Generation 690:	Best = 5.889241e+02	FES = 69100	Diversity = 1.7181e+01	k_t = [1, 1, 1, 1]
Generation 700:	Best = 5.887360e+02	FES = 70100	Diversity = 1.7059e+01	k_t = [4, 2, 2, 3]
Generation 710:	Best = 5.886854e+02	FES = 71100	Diversity = 1.7036e+01	k_t = [1, 1, 1, 1]
Generation 720:	Best = 5.879963e+02	FES = 72100	Diversity = 1.7033e+01	k_t = [2, 4, 3, 4]
Generation 730:	Best = 5.879963e+02	FES = 73100	Diversity = 1.7012e+01	k_t = [1, 1, 1, 1]
Generation 740:	Best = 5.879963e+02	FES = 74100	Diversity = 1.6983e+01	k_t = [2, 2, 3, 4]
Generation 750:	Best = 5.872406e+02	FES = 75100	Diversity = 1.6989e+01	k_t = [1, 1, 1, 1]
Generation 760:	Best = 5.867744e+02	FES = 76100	Diversity = 1.6990e+01	k_t = [4, 3, 2, 4]

Generation 770:	Best = 5.857711e+02	FES = 77100	Diversity = 1.6965e+01	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 5.857711e+02	FES = 78100	Diversity = 1.6962e+01	k_t
= [2, 2, 3, 4]				
Generation 790:	Best = 5.857711e+02	FES = 79100	Diversity = 1.6944e+01	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 5.852695e+02	FES = 80100	Diversity = 1.6887e+01	k_t
= [3, 3, 2, 4]				
Generation 810:	Best = 5.852352e+02	FES = 81100	Diversity = 1.6825e+01	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 5.780874e+02	FES = 82100	Diversity = 1.6798e+01	k_t
= [2, 4, 3, 2]				
Generation 830:	Best = 5.780280e+02	FES = 83100	Diversity = 1.6782e+01	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 5.780280e+02	FES = 84100	Diversity = 1.6738e+01	k_t
= [2, 3, 3, 4]				
Generation 850:	Best = 5.683592e+02	FES = 85100	Diversity = 1.6729e+01	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 5.683592e+02	FES = 86100	Diversity = 1.6724e+01	k_t
= [3, 4, 4, 4]				
Generation 870:	Best = 5.683592e+02	FES = 87100	Diversity = 1.6724e+01	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 5.683592e+02	FES = 88100	Diversity = 1.6739e+01	k_t
= [2, 2, 2, 2]				
Generation 890:	Best = 5.683592e+02	FES = 89100	Diversity = 1.6743e+01	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 5.683592e+02	FES = 90100	Diversity = 1.6692e+01	k_t
= [3, 2, 3, 4]				
Generation 910:	Best = 5.683592e+02	FES = 91100	Diversity = 1.6703e+01	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 5.683592e+02	FES = 92100	Diversity = 1.6701e+01	k_t
= [4, 2, 4, 2]				
Generation 930:	Best = 5.683592e+02	FES = 93100	Diversity = 1.6666e+01	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 5.683592e+02	FES = 94100	Diversity = 1.6685e+01	k_t
= [3, 3, 2, 3]				
Generation 950:	Best = 5.683592e+02	FES = 95100	Diversity = 1.6738e+01	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 5.673454e+02	FES = 96100	Diversity = 1.6778e+01	k_t
= [3, 4, 4, 4]				
Generation 970:	Best = 5.673454e+02	FES = 97100	Diversity = 1.6775e+01	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 5.664701e+02	FES = 98100	Diversity = 1.6767e+01	k_t
= [4, 3, 2, 2]				
Generation 990:	Best = 5.663948e+02	FES = 99100	Diversity = 1.6766e+01	k_t
= [1, 1, 1, 1]				
Generation 1000:	Best = 5.653358e+02	FES = 100100	Diversity = 1.6647e+01	k_t
= [3, 4, 4, 4]				
Generation 1010:	Best = 5.653358e+02	FES = 101100	Diversity = 1.6614e+01	k_t
= [1, 1, 1, 1]				
Generation 1020:	Best = 5.637324e+02	FES = 102100	Diversity = 1.6594e+01	k_t
= [2, 3, 4, 3]				
Generation 1030:	Best = 5.632054e+02	FES = 103100	Diversity = 1.6567e+01	k_t
= [1, 1, 1, 1]				

Generation 1040: Best = 5.628665e+02 = [2, 2, 2, 4]	FES = 104100	Diversity = 1.6563e+01	k_t
Generation 1050: Best = 5.620278e+02 = [1, 1, 1, 1]	FES = 105100	Diversity = 1.6552e+01	k_t
Generation 1060: Best = 5.611628e+02 = [4, 4, 3, 3]	FES = 106100	Diversity = 1.6479e+01	k_t
Generation 1070: Best = 5.596922e+02 = [1, 1, 1, 1]	FES = 107100	Diversity = 1.6493e+01	k_t
Generation 1080: Best = 5.596922e+02 = [2, 3, 3, 4]	FES = 108100	Diversity = 1.6502e+01	k_t
Generation 1090: Best = 5.593264e+02 = [1, 1, 1, 1]	FES = 109100	Diversity = 1.6503e+01	k_t
Generation 1100: Best = 5.566693e+02 = [2, 4, 2, 3]	FES = 110100	Diversity = 1.6492e+01	k_t
Generation 1110: Best = 5.566693e+02 = [1, 1, 1, 1]	FES = 111100	Diversity = 1.6403e+01	k_t
Generation 1120: Best = 5.566188e+02 = [2, 2, 2, 4]	FES = 112100	Diversity = 1.6397e+01	k_t
Generation 1130: Best = 5.559913e+02 = [1, 1, 1, 1]	FES = 113100	Diversity = 1.6411e+01	k_t
Generation 1140: Best = 5.559913e+02 = [2, 4, 2, 3]	FES = 114100	Diversity = 1.6407e+01	k_t
Generation 1150: Best = 5.559913e+02 = [1, 1, 1, 1]	FES = 115100	Diversity = 1.6403e+01	k_t
Generation 1160: Best = 5.553809e+02 = [2, 3, 4, 4]	FES = 116100	Diversity = 1.6338e+01	k_t
Generation 1170: Best = 5.551717e+02 = [1, 1, 1, 1]	FES = 117100	Diversity = 1.6322e+01	k_t
Generation 1180: Best = 5.551717e+02 = [3, 2, 3, 3]	FES = 118100	Diversity = 1.6312e+01	k_t
Generation 1190: Best = 5.551717e+02 = [1, 1, 1, 1]	FES = 119100	Diversity = 1.6305e+01	k_t
Generation 1200: Best = 5.544762e+02 = [3, 4, 3, 2]	FES = 120100	Diversity = 1.6350e+01	k_t
Generation 1210: Best = 5.537609e+02 = [1, 1, 1, 1]	FES = 121100	Diversity = 1.6326e+01	k_t
Generation 1220: Best = 5.537609e+02 = [3, 4, 4, 4]	FES = 122100	Diversity = 1.6299e+01	k_t
Generation 1230: Best = 5.535677e+02 = [1, 1, 1, 1]	FES = 123100	Diversity = 1.6300e+01	k_t
Generation 1240: Best = 5.531490e+02 = [2, 3, 3, 4]	FES = 124100	Diversity = 1.6330e+01	k_t
Generation 1250: Best = 5.501156e+02 = [1, 1, 1, 1]	FES = 125100	Diversity = 1.6163e+01	k_t
Generation 1260: Best = 5.501156e+02 = [2, 2, 4, 2]	FES = 126100	Diversity = 1.6062e+01	k_t
Generation 1270: Best = 5.501156e+02 = [1, 1, 1, 1]	FES = 127100	Diversity = 1.5843e+01	k_t
Generation 1280: Best = 5.477360e+02 = [4, 3, 4, 2]	FES = 128100	Diversity = 1.5775e+01	k_t
Generation 1290: Best = 5.467098e+02 = [1, 1, 1, 1]	FES = 129100	Diversity = 1.5569e+01	k_t
Generation 1300: Best = 5.466775e+02 = [2, 2, 2, 2]	FES = 130100	Diversity = 1.5277e+01	k_t

Generation 1310: Best = 5.466775e+02	FES = 131100	Diversity = 1.5142e+01	k_t
= [1, 1, 1, 1]			
Generation 1320: Best = 5.466275e+02	FES = 132100	Diversity = 1.4841e+01	k_t
= [2, 3, 4, 2]			
Generation 1330: Best = 5.466275e+02	FES = 133100	Diversity = 1.4708e+01	k_t
= [1, 1, 1, 1]			
Generation 1340: Best = 5.466275e+02	FES = 134100	Diversity = 1.4627e+01	k_t
= [4, 2, 4, 4]			
Generation 1350: Best = 5.466275e+02	FES = 135100	Diversity = 1.4250e+01	k_t
= [1, 1, 1, 1]			
Generation 1360: Best = 5.466275e+02	FES = 136100	Diversity = 1.4089e+01	k_t
= [4, 2, 2, 4]			
Generation 1370: Best = 5.466275e+02	FES = 137100	Diversity = 1.3738e+01	k_t
= [1, 1, 1, 1]			
Generation 1380: Best = 5.451882e+02	FES = 138100	Diversity = 1.3338e+01	k_t
= [2, 3, 2, 4]			
Generation 1390: Best = 5.428544e+02	FES = 139100	Diversity = 1.2896e+01	k_t
= [1, 1, 1, 1]			
Generation 1400: Best = 5.428544e+02	FES = 140100	Diversity = 1.2456e+01	k_t
= [2, 3, 3, 2]			
Generation 1410: Best = 5.400495e+02	FES = 141100	Diversity = 1.2297e+01	k_t
= [1, 1, 1, 1]			
Generation 1420: Best = 5.398383e+02	FES = 142100	Diversity = 1.1564e+01	k_t
= [3, 3, 4, 2]			
Generation 1430: Best = 5.397800e+02	FES = 143100	Diversity = 1.0643e+01	k_t
= [1, 1, 1, 1]			
Generation 1440: Best = 5.388314e+02	FES = 144100	Diversity = 9.6986e+00	k_t
= [3, 2, 3, 4]			
Generation 1450: Best = 5.388314e+02	FES = 145100	Diversity = 8.3345e+00	k_t
= [1, 1, 1, 1]			
Generation 1460: Best = 5.377754e+02	FES = 146100	Diversity = 6.5129e+00	k_t
= [2, 3, 2, 2]			
Generation 1470: Best = 5.376827e+02	FES = 147100	Diversity = 4.8463e+00	k_t
= [1, 1, 1, 1]			
Generation 1480: Best = 5.369603e+02	FES = 148100	Diversity = 3.1234e+00	k_t
= [2, 4, 3, 4]			
Generation 1490: Best = 5.363228e+02	FES = 149100	Diversity = 1.6731e+00	k_t
= [1, 1, 1, 1]			
Generation 1500: Best = 5.356304e+02	FES = 150100	Diversity = 9.0867e-01	k_t
= [2, 3, 3, 4]			
Generation 1510: Best = 5.356304e+02	FES = 151100	Diversity = 7.4305e-01	k_t
= [1, 1, 1, 1]			
Generation 1520: Best = 5.353548e+02	FES = 152100	Diversity = 5.8347e-01	k_t
= [3, 2, 3, 4]			
Generation 1530: Best = 5.351702e+02	FES = 153100	Diversity = 3.7718e-01	k_t
= [1, 1, 1, 1]			
Generation 1540: Best = 5.350214e+02	FES = 154100	Diversity = 1.3023e-01	k_t
= [3, 3, 2, 4]			
Generation 1550: Best = 5.349230e+02	FES = 155100	Diversity = 9.0464e-02	k_t
= [1, 1, 1, 1]			
Generation 1560: Best = 5.348728e+02	FES = 156100	Diversity = 6.5354e-02	k_t
= [4, 3, 3, 2]			
Generation 1570: Best = 5.348498e+02	FES = 157100	Diversity = 4.8077e-02	k_t
= [1, 1, 1, 1]			

Generation 1580: Best = 5.348306e+02	FES = 158100	Diversity = 3.1566e-02	k_t
= [2, 3, 3, 4]			
Generation 1590: Best = 5.348291e+02	FES = 159100	Diversity = 2.1710e-02	k_t
= [1, 1, 1, 1]			
Generation 1600: Best = 5.348260e+02	FES = 160100	Diversity = 1.4077e-02	k_t
= [2, 3, 2, 3]			
Generation 1610: Best = 5.348247e+02	FES = 161100	Diversity = 9.3134e-03	k_t
= [1, 1, 1, 1]			
Generation 1620: Best = 5.348241e+02	FES = 162100	Diversity = 6.1475e-03	k_t
= [2, 2, 4, 3]			
Generation 1630: Best = 5.348237e+02	FES = 163100	Diversity = 3.8527e-03	k_t
= [1, 1, 1, 1]			
Generation 1640: Best = 5.348236e+02	FES = 164100	Diversity = 2.5481e-03	k_t
= [3, 4, 4, 2]			
Generation 1650: Best = 5.348236e+02	FES = 165100	Diversity = 1.6103e-03	k_t
= [1, 1, 1, 1]			
Generation 1660: Best = 5.348235e+02	FES = 166100	Diversity = 1.0648e-03	k_t
= [4, 3, 3, 2]			
Generation 1670: Best = 5.348235e+02	FES = 167100	Diversity = 6.9572e-04	k_t
= [1, 1, 1, 1]			
Generation 1680: Best = 5.348235e+02	FES = 168100	Diversity = 4.5231e-04	k_t
= [2, 2, 2, 3]			
Generation 1690: Best = 5.348235e+02	FES = 169100	Diversity = 2.9055e-04	k_t
= [1, 1, 1, 1]			
Generation 1700: Best = 5.348235e+02	FES = 170100	Diversity = 1.8446e-04	k_t
= [4, 2, 3, 4]			
Generation 1710: Best = 5.348235e+02	FES = 171100	Diversity = 1.1774e-04	k_t
= [1, 1, 1, 1]			
Generation 1720: Best = 5.348235e+02	FES = 172100	Diversity = 8.2844e-05	k_t
= [3, 2, 2, 2]			
Generation 1730: Best = 5.348235e+02	FES = 173100	Diversity = 5.2123e-05	k_t
= [1, 1, 1, 1]			
Generation 1740: Best = 5.348235e+02	FES = 174100	Diversity = 3.2784e-05	k_t
= [3, 2, 4, 3]			
Generation 1750: Best = 5.348235e+02	FES = 175100	Diversity = 2.0485e-05	k_t
= [1, 1, 1, 1]			
Generation 1760: Best = 5.348235e+02	FES = 176100	Diversity = 1.2742e-05	k_t
= [4, 3, 2, 2]			
Generation 1770: Best = 5.348235e+02	FES = 177100	Diversity = 8.5995e-06	k_t
= [1, 1, 1, 1]			
Generation 1780: Best = 5.348235e+02	FES = 178100	Diversity = 5.2545e-06	k_t
= [4, 3, 3, 2]			
Generation 1790: Best = 5.348235e+02	FES = 179100	Diversity = 3.2611e-06	k_t
= [1, 1, 1, 1]			
Generation 1800: Best = 5.348235e+02	FES = 180100	Diversity = 2.0343e-06	k_t
= [3, 3, 3, 2]			
Generation 1810: Best = 5.348235e+02	FES = 181100	Diversity = 1.3176e-06	k_t
= [1, 1, 1, 1]			
Generation 1820: Best = 5.348235e+02	FES = 182100	Diversity = 9.0356e-07	k_t
= [4, 3, 3, 2]			
Generation 1830: Best = 5.348235e+02	FES = 183100	Diversity = 6.3380e-07	k_t
= [1, 1, 1, 1]			
Generation 1840: Best = 5.348235e+02	FES = 184100	Diversity = 4.1255e-07	k_t
= [3, 4, 2, 3]			

Generation 1850: Best = 5.348235e+02 = [1, 1, 1, 1]	FES = 185100	Diversity = 2.8478e-07	k_t
Generation 1860: Best = 5.348235e+02 = [3, 4, 2, 4]	FES = 186100	Diversity = 1.9612e-07	k_t
Generation 1870: Best = 5.348235e+02 = [1, 1, 1, 1]	FES = 187100	Diversity = 1.4220e-07	k_t
Generation 1880: Best = 5.348235e+02 = [2, 2, 3, 3]	FES = 188100	Diversity = 1.0365e-07	k_t
Generation 1890: Best = 5.348235e+02 = [1, 1, 1, 1]	FES = 189100	Diversity = 8.4516e-08	k_t
Generation 1900: Best = 5.348235e+02 = [3, 4, 4, 3]	FES = 190100	Diversity = 6.8837e-08	k_t
Generation 1910: Best = 5.348235e+02 = [1, 1, 1, 1]	FES = 191100	Diversity = 6.6427e-08	k_t
Generation 1920: Best = 5.348235e+02 = [4, 3, 2, 2]	FES = 192100	Diversity = 6.6058e-08	k_t
Generation 1930: Best = 5.348235e+02 = [1, 1, 1, 1]	FES = 193100	Diversity = 6.8329e-08	k_t
Generation 1940: Best = 5.348235e+02 = [4, 3, 4, 3]	FES = 194100	Diversity = 6.7517e-08	k_t
Generation 1950: Best = 5.348235e+02 = [1, 1, 1, 1]	FES = 195100	Diversity = 6.7422e-08	k_t
Generation 1960: Best = 5.348235e+02 = [3, 4, 3, 3]	FES = 196100	Diversity = 6.1997e-08	k_t
Generation 1970: Best = 5.348235e+02 = [1, 1, 1, 1]	FES = 197100	Diversity = 5.5599e-08	k_t
Generation 1980: Best = 5.348235e+02 = [3, 2, 4, 3]	FES = 198100	Diversity = 5.0411e-08	k_t
Generation 1990: Best = 5.348235e+02 = [1, 1, 1, 1]	FES = 199100	Diversity = 5.0201e-08	k_t
Generation 2000: Best = 5.348235e+02 = [3, 4, 4, 3]	FES = 200100	Diversity = 4.9460e-08	k_t
Generation 2010: Best = 5.348235e+02 = [1, 1, 1, 1]	FES = 201100	Diversity = 4.7962e-08	k_t
Generation 2020: Best = 5.348235e+02 = [2, 2, 4, 3]	FES = 202100	Diversity = 4.6968e-08	k_t
Generation 2030: Best = 5.348235e+02 = [1, 1, 1, 1]	FES = 203100	Diversity = 4.7251e-08	k_t
Generation 2040: Best = 5.348235e+02 = [4, 3, 3, 4]	FES = 204100	Diversity = 4.7307e-08	k_t
Generation 2050: Best = 5.348235e+02 = [1, 1, 1, 1]	FES = 205100	Diversity = 4.7646e-08	k_t
Generation 2060: Best = 5.348235e+02 = [3, 2, 2, 4]	FES = 206100	Diversity = 4.7569e-08	k_t
Generation 2070: Best = 5.348235e+02 = [1, 1, 1, 1]	FES = 207100	Diversity = 4.8975e-08	k_t
Generation 2080: Best = 5.348235e+02 = [3, 2, 4, 3]	FES = 208100	Diversity = 4.8791e-08	k_t
Generation 2090: Best = 5.348235e+02 = [1, 1, 1, 1]	FES = 209100	Diversity = 4.9129e-08	k_t
Generation 2100: Best = 5.348235e+02 = [4, 2, 3, 4]	FES = 210100	Diversity = 4.9491e-08	k_t
Generation 2110: Best = 5.348235e+02 = [1, 1, 1, 1]	FES = 211100	Diversity = 4.8515e-08	k_t

Generation 2120: Best = 5.348235e+02 = [2, 2, 4, 3]	FES = 212100	Diversity = 4.7489e-08	k_t
Generation 2130: Best = 5.348235e+02 = [1, 1, 1, 1]	FES = 213100	Diversity = 4.7498e-08	k_t
Generation 2140: Best = 5.348235e+02 = [3, 4, 4, 4]	FES = 214100	Diversity = 4.7107e-08	k_t
Generation 2150: Best = 5.348235e+02 = [1, 1, 1, 1]	FES = 215100	Diversity = 4.6666e-08	k_t
Generation 2160: Best = 5.348235e+02 = [3, 4, 3, 2]	FES = 216100	Diversity = 4.7835e-08	k_t
Generation 2170: Best = 5.348235e+02 = [1, 1, 1, 1]	FES = 217100	Diversity = 4.7740e-08	k_t
Generation 2180: Best = 5.348235e+02 = [3, 2, 2, 2]	FES = 218100	Diversity = 4.8219e-08	k_t
Generation 2190: Best = 5.348235e+02 = [1, 1, 1, 1]	FES = 219100	Diversity = 4.9062e-08	k_t
Generation 2200: Best = 5.348235e+02 = [2, 2, 3, 2]	FES = 220100	Diversity = 4.8885e-08	k_t
Generation 2210: Best = 5.348235e+02 = [1, 1, 1, 1]	FES = 221100	Diversity = 4.7482e-08	k_t
Generation 2220: Best = 5.348235e+02 = [2, 2, 2, 2]	FES = 222100	Diversity = 4.7161e-08	k_t
Generation 2230: Best = 5.348235e+02 = [1, 1, 1, 1]	FES = 223100	Diversity = 4.7545e-08	k_t
Generation 2240: Best = 5.348235e+02 = [3, 3, 3, 4]	FES = 224100	Diversity = 4.7938e-08	k_t
Generation 2250: Best = 5.348235e+02 = [1, 1, 1, 1]	FES = 225100	Diversity = 4.7399e-08	k_t
Generation 2260: Best = 5.348235e+02 = [2, 3, 4, 4]	FES = 226100	Diversity = 4.7181e-08	k_t
Generation 2270: Best = 5.348235e+02 = [1, 1, 1, 1]	FES = 227100	Diversity = 4.7619e-08	k_t
Generation 2280: Best = 5.348235e+02 = [4, 3, 3, 2]	FES = 228100	Diversity = 4.8011e-08	k_t
Generation 2290: Best = 5.348235e+02 = [1, 1, 1, 1]	FES = 229100	Diversity = 4.7625e-08	k_t
Generation 2300: Best = 5.348235e+02 = [3, 3, 3, 3]	FES = 230100	Diversity = 4.7230e-08	k_t
Generation 2310: Best = 5.348235e+02 = [1, 1, 1, 1]	FES = 231100	Diversity = 4.6071e-08	k_t
Generation 2320: Best = 5.348235e+02 = [2, 3, 4, 3]	FES = 232100	Diversity = 4.6454e-08	k_t
Generation 2330: Best = 5.348235e+02 = [1, 1, 1, 1]	FES = 233100	Diversity = 4.7001e-08	k_t
Generation 2340: Best = 5.348235e+02 = [3, 4, 2, 2]	FES = 234100	Diversity = 4.6674e-08	k_t
Generation 2350: Best = 5.348235e+02 = [1, 1, 1, 1]	FES = 235100	Diversity = 4.5910e-08	k_t
Generation 2360: Best = 5.348235e+02 = [3, 3, 2, 4]	FES = 236100	Diversity = 4.6670e-08	k_t
Generation 2370: Best = 5.348235e+02 = [1, 1, 1, 1]	FES = 237100	Diversity = 4.7529e-08	k_t
Generation 2380: Best = 5.348235e+02 = [2, 3, 3, 3]	FES = 238100	Diversity = 4.6486e-08	k_t

Generation 2390: Best = 5.348235e+02	FES = 239100	Diversity = 4.6187e-08	k_t
= [1, 1, 1, 1]			
Generation 2400: Best = 5.348235e+02	FES = 240100	Diversity = 4.7636e-08	k_t
= [2, 2, 2, 2]			
Generation 2410: Best = 5.348235e+02	FES = 241100	Diversity = 4.7931e-08	k_t
= [1, 1, 1, 1]			
Generation 2420: Best = 5.348235e+02	FES = 242100	Diversity = 4.8406e-08	k_t
= [2, 4, 2, 3]			
Generation 2430: Best = 5.348235e+02	FES = 243100	Diversity = 4.7755e-08	k_t
= [1, 1, 1, 1]			
Generation 2440: Best = 5.348235e+02	FES = 244100	Diversity = 4.6952e-08	k_t
= [2, 2, 4, 3]			
Generation 2450: Best = 5.348235e+02	FES = 245100	Diversity = 4.6160e-08	k_t
= [1, 1, 1, 1]			
Generation 2460: Best = 5.348235e+02	FES = 246100	Diversity = 4.5570e-08	k_t
= [3, 3, 4, 3]			
Generation 2470: Best = 5.348235e+02	FES = 247100	Diversity = 4.6140e-08	k_t
= [1, 1, 1, 1]			
Generation 2480: Best = 5.348235e+02	FES = 248100	Diversity = 4.7410e-08	k_t
= [3, 2, 2, 3]			
Generation 2490: Best = 5.348235e+02	FES = 249100	Diversity = 4.6146e-08	k_t
= [1, 1, 1, 1]			
Generation 2500: Best = 5.348235e+02	FES = 250100	Diversity = 4.6135e-08	k_t
= [3, 3, 2, 4]			
Generation 2510: Best = 5.348235e+02	FES = 251100	Diversity = 4.7460e-08	k_t
= [1, 1, 1, 1]			
Generation 2520: Best = 5.348235e+02	FES = 252100	Diversity = 4.7044e-08	k_t
= [4, 3, 4, 2]			
Generation 2530: Best = 5.348235e+02	FES = 253100	Diversity = 4.6723e-08	k_t
= [1, 1, 1, 1]			
Generation 2540: Best = 5.348235e+02	FES = 254100	Diversity = 4.5219e-08	k_t
= [2, 4, 2, 3]			
Generation 2550: Best = 5.348235e+02	FES = 255100	Diversity = 4.5666e-08	k_t
= [1, 1, 1, 1]			
Generation 2560: Best = 5.348235e+02	FES = 256100	Diversity = 4.3694e-08	k_t
= [3, 4, 4, 2]			
Generation 2570: Best = 5.348235e+02	FES = 257100	Diversity = 4.3215e-08	k_t
= [1, 1, 1, 1]			
Generation 2580: Best = 5.348235e+02	FES = 258100	Diversity = 4.3559e-08	k_t
= [3, 3, 3, 3]			
Generation 2590: Best = 5.348235e+02	FES = 259100	Diversity = 4.3541e-08	k_t
= [1, 1, 1, 1]			
Generation 2600: Best = 5.348235e+02	FES = 260100	Diversity = 4.4543e-08	k_t
= [3, 2, 3, 4]			
Generation 2610: Best = 5.348235e+02	FES = 261100	Diversity = 4.5068e-08	k_t
= [1, 1, 1, 1]			
Generation 2620: Best = 5.348235e+02	FES = 262100	Diversity = 4.5656e-08	k_t
= [3, 2, 2, 3]			
Generation 2630: Best = 5.348235e+02	FES = 263100	Diversity = 4.7369e-08	k_t
= [1, 1, 1, 1]			
Generation 2640: Best = 5.348235e+02	FES = 264100	Diversity = 4.8341e-08	k_t
= [3, 4, 2, 4]			
Generation 2650: Best = 5.348235e+02	FES = 265100	Diversity = 4.8102e-08	k_t
= [1, 1, 1, 1]			

Generation 2660: Best = 5.348235e+02	FES = 266100	Diversity = 4.8411e-08	k_t
= [4, 4, 2, 4]			
Generation 2670: Best = 5.348235e+02	FES = 267100	Diversity = 4.7172e-08	k_t
= [1, 1, 1, 1]			
Generation 2680: Best = 5.348235e+02	FES = 268100	Diversity = 4.7066e-08	k_t
= [3, 4, 2, 4]			
Generation 2690: Best = 5.348235e+02	FES = 269100	Diversity = 4.6793e-08	k_t
= [1, 1, 1, 1]			
Generation 2700: Best = 5.348235e+02	FES = 270100	Diversity = 4.7681e-08	k_t
= [4, 4, 4, 4]			
Generation 2710: Best = 5.348235e+02	FES = 271100	Diversity = 4.8520e-08	k_t
= [1, 1, 1, 1]			
Generation 2720: Best = 5.348235e+02	FES = 272100	Diversity = 4.7671e-08	k_t
= [3, 2, 4, 2]			
Generation 2730: Best = 5.348235e+02	FES = 273100	Diversity = 4.9224e-08	k_t
= [1, 1, 1, 1]			
Generation 2740: Best = 5.348235e+02	FES = 274100	Diversity = 4.9531e-08	k_t
= [3, 3, 3, 2]			
Generation 2750: Best = 5.348235e+02	FES = 275100	Diversity = 4.8513e-08	k_t
= [1, 1, 1, 1]			
Generation 2760: Best = 5.348235e+02	FES = 276100	Diversity = 4.9730e-08	k_t
= [2, 3, 2, 3]			
Generation 2770: Best = 5.348235e+02	FES = 277100	Diversity = 5.0907e-08	k_t
= [1, 1, 1, 1]			
Generation 2780: Best = 5.348235e+02	FES = 278100	Diversity = 5.1379e-08	k_t
= [4, 2, 2, 4]			
Generation 2790: Best = 5.348235e+02	FES = 279100	Diversity = 5.1081e-08	k_t
= [1, 1, 1, 1]			
Generation 2800: Best = 5.348235e+02	FES = 280100	Diversity = 5.1264e-08	k_t
= [3, 3, 2, 2]			
Generation 2810: Best = 5.348235e+02	FES = 281100	Diversity = 4.9640e-08	k_t
= [1, 1, 1, 1]			
Generation 2820: Best = 5.348235e+02	FES = 282100	Diversity = 4.9658e-08	k_t
= [2, 3, 3, 3]			
Generation 2830: Best = 5.348235e+02	FES = 283100	Diversity = 5.0196e-08	k_t
= [1, 1, 1, 1]			
Generation 2840: Best = 5.348235e+02	FES = 284100	Diversity = 5.0556e-08	k_t
= [4, 4, 2, 4]			
Generation 2850: Best = 5.348235e+02	FES = 285100	Diversity = 5.0371e-08	k_t
= [1, 1, 1, 1]			
Generation 2860: Best = 5.348235e+02	FES = 286100	Diversity = 5.1055e-08	k_t
= [4, 2, 2, 3]			
Generation 2870: Best = 5.348235e+02	FES = 287100	Diversity = 5.0371e-08	k_t
= [1, 1, 1, 1]			
Generation 2880: Best = 5.348235e+02	FES = 288100	Diversity = 4.9578e-08	k_t
= [3, 2, 2, 4]			
Generation 2890: Best = 5.348235e+02	FES = 289100	Diversity = 4.9219e-08	k_t
= [1, 1, 1, 1]			
Generation 2900: Best = 5.348235e+02	FES = 290100	Diversity = 4.8856e-08	k_t
= [2, 4, 4, 2]			
Generation 2910: Best = 5.348235e+02	FES = 291100	Diversity = 4.7509e-08	k_t
= [1, 1, 1, 1]			
Generation 2920: Best = 5.348235e+02	FES = 292100	Diversity = 4.8436e-08	k_t
= [2, 3, 3, 2]			


```

Generation 2930: Best = 5.348235e+02  FES = 293100  Diversity = 4.7964e-08  k_t
= [1, 1, 1, 1]
Generation 2940: Best = 5.348235e+02  FES = 294100  Diversity = 4.7361e-08  k_t
= [3, 3, 3, 3]
Generation 2950: Best = 5.348235e+02  FES = 295100  Diversity = 4.6434e-08  k_t
= [1, 1, 1, 1]
Generation 2960: Best = 5.348235e+02  FES = 296100  Diversity = 4.6277e-08  k_t
= [2, 2, 2, 3]
Generation 2970: Best = 5.348235e+02  FES = 297100  Diversity = 4.5898e-08  k_t
= [1, 1, 1, 1]
Generation 2980: Best = 5.348235e+02  FES = 298100  Diversity = 4.7103e-08  k_t
= [3, 3, 3, 3]
Generation 2990: Best = 5.348235e+02  FES = 299100  Diversity = 4.7652e-08  k_t
= [1, 1, 1, 1]

```

Optimization completed!

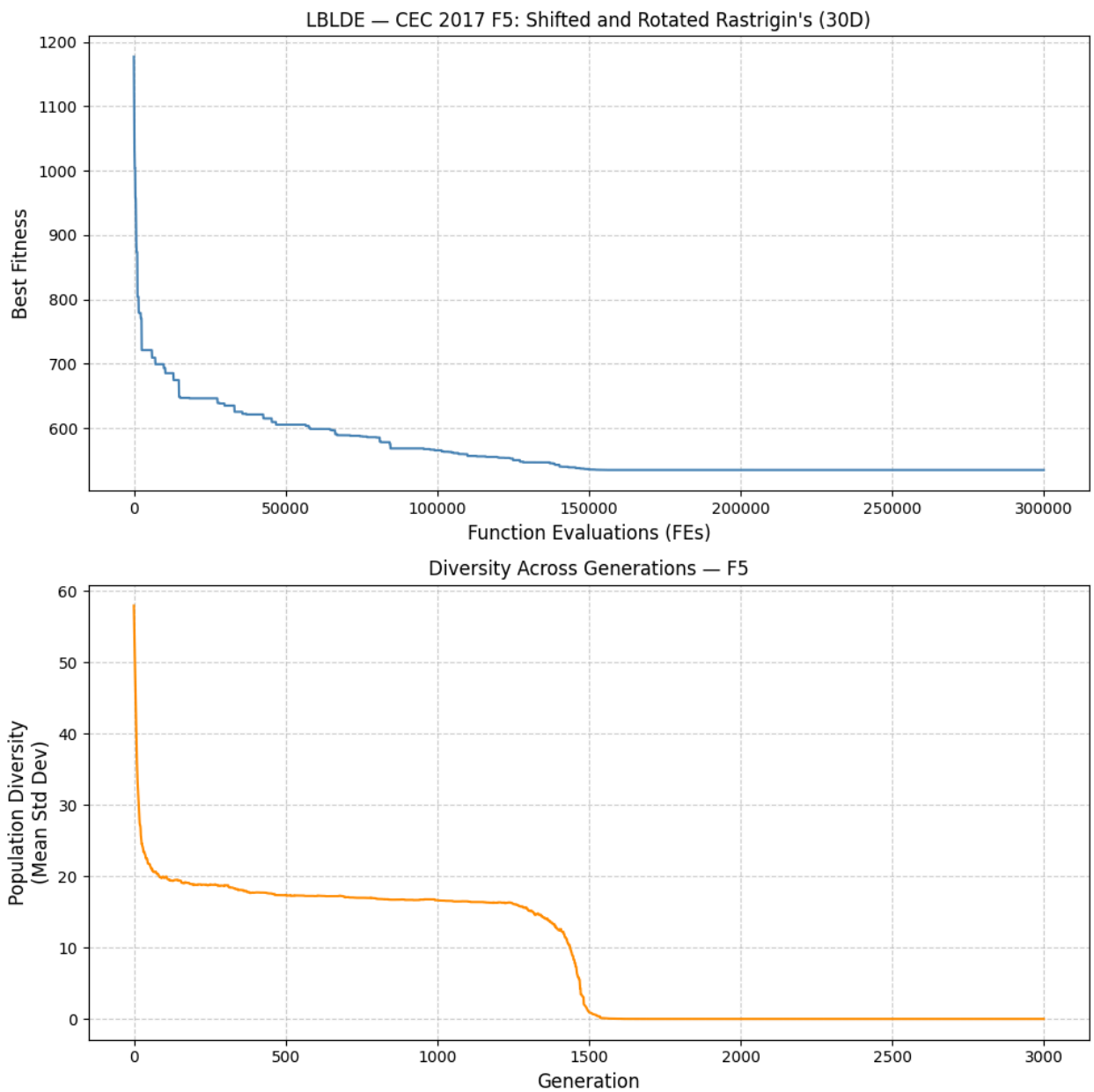
Final Best Fitness = 5.348235e+02

Total Function Evaluations = 300000

```

=====
=
Results:
  Best Fitness           : 5.348235e+02
  Global Optimum         : 5.000000e+02
  Error (|best-optimum|) : 3.482354e+01
  Best Solution (5 dim)  : [ 11.88015543  72.42284987 -33.74837096 -22.13993385
27.97884413]
  D^(0) (init. diversity): 5.7976e+01
=====
=

```



Graph saved as convergence_F5.png

```
In [16]: # =====
# F6 – Shifted and Rotated Schaffer's F7
# =====
run_and_plot('F6', make_wrapper(simple.f6), 600.0, "Shifted and Rotated Schaff
```

```

=====
=
LBLDE_s on CEC 2017 F6 – Single Run
=====
=
Function      : Shifted and Rotated Schaffer's F7 (30D)
Bounds       : [-100, 100]
Global optimum : 600.0
Initial D^(0) = 5.7976e+01
Generation 0: Best Fitness = 7.816536e+02 | Diversity = 5.7976e+01
Generation 10: Best = 6.997247e+02 FES = 1100 Diversity = 4.3209e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 6.629638e+02 FES = 2100 Diversity = 3.1107e+01 k_t
= [3, 3, 2, 4]
Generation 30: Best = 6.495104e+02 FES = 3100 Diversity = 2.2897e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 6.388207e+02 FES = 4100 Diversity = 1.7954e+01 k_t
= [3, 4, 3, 4]
Generation 50: Best = 6.259374e+02 FES = 5100 Diversity = 1.3864e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 6.175343e+02 FES = 6100 Diversity = 1.0884e+01 k_t
= [2, 3, 4, 3]
Generation 70: Best = 6.175343e+02 FES = 7100 Diversity = 8.1923e+00 k_t
= [1, 1, 1, 1]
Generation 80: Best = 6.138062e+02 FES = 8100 Diversity = 6.0717e+00 k_t
= [4, 3, 4, 3]
Generation 90: Best = 6.086658e+02 FES = 9100 Diversity = 4.7772e+00 k_t
= [1, 1, 1, 1]
Generation 100: Best = 6.072356e+02 FES = 10100 Diversity = 3.4107e+00 k_t
= [2, 4, 4, 3]
Generation 110: Best = 6.056218e+02 FES = 11100 Diversity = 2.7449e+00 k_t
= [1, 1, 1, 1]
Generation 120: Best = 6.045882e+02 FES = 12100 Diversity = 2.0587e+00 k_t
= [3, 3, 3, 4]
Generation 130: Best = 6.035068e+02 FES = 13100 Diversity = 1.6790e+00 k_t
= [1, 1, 1, 1]
Generation 140: Best = 6.030108e+02 FES = 14100 Diversity = 1.3843e+00 k_t
= [2, 3, 3, 3]
Generation 150: Best = 6.023857e+02 FES = 15100 Diversity = 1.1865e+00 k_t
= [1, 1, 1, 1]
Generation 160: Best = 6.019762e+02 FES = 16100 Diversity = 9.6352e-01 k_t
= [2, 4, 4, 4]
Generation 170: Best = 6.016281e+02 FES = 17100 Diversity = 7.9359e-01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 6.012995e+02 FES = 18100 Diversity = 6.8578e-01 k_t
= [3, 3, 3, 2]
Generation 190: Best = 6.009178e+02 FES = 19100 Diversity = 5.4270e-01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 6.009178e+02 FES = 20100 Diversity = 4.5839e-01 k_t
= [4, 4, 3, 3]
Generation 210: Best = 6.008188e+02 FES = 21100 Diversity = 4.1224e-01 k_t
= [1, 1, 1, 1]
Generation 220: Best = 6.007062e+02 FES = 22100 Diversity = 3.7531e-01 k_t
= [3, 3, 3, 4]

```

Generation 230:	Best = 6.006407e+02	FES = 23100	Diversity = 3.4339e-01	k_t = [1, 1, 1, 1]
Generation 240:	Best = 6.006204e+02	FES = 24100	Diversity = 3.0624e-01	k_t = [3, 3, 3, 2]
Generation 250:	Best = 6.005473e+02	FES = 25100	Diversity = 2.9160e-01	k_t = [1, 1, 1, 1]
Generation 260:	Best = 6.005124e+02	FES = 26100	Diversity = 2.8319e-01	k_t = [2, 3, 2, 4]
Generation 270:	Best = 6.003963e+02	FES = 27100	Diversity = 2.6957e-01	k_t = [1, 1, 1, 1]
Generation 280:	Best = 6.003963e+02	FES = 28100	Diversity = 2.5040e-01	k_t = [4, 3, 3, 3]
Generation 290:	Best = 6.003219e+02	FES = 29100	Diversity = 2.4002e-01	k_t = [1, 1, 1, 1]
Generation 300:	Best = 6.002884e+02	FES = 30100	Diversity = 2.3183e-01	k_t = [4, 4, 3, 3]
Generation 310:	Best = 6.002884e+02	FES = 31100	Diversity = 2.1677e-01	k_t = [1, 1, 1, 1]
Generation 320:	Best = 6.002884e+02	FES = 32100	Diversity = 2.0302e-01	k_t = [4, 2, 3, 4]
Generation 330:	Best = 6.002182e+02	FES = 33100	Diversity = 1.9893e-01	k_t = [1, 1, 1, 1]
Generation 340:	Best = 6.002154e+02	FES = 34100	Diversity = 2.0183e-01	k_t = [3, 2, 4, 3]
Generation 350:	Best = 6.002053e+02	FES = 35100	Diversity = 1.9687e-01	k_t = [1, 1, 1, 1]
Generation 360:	Best = 6.002046e+02	FES = 36100	Diversity = 1.8608e-01	k_t = [3, 2, 3, 4]
Generation 370:	Best = 6.001797e+02	FES = 37100	Diversity = 1.7829e-01	k_t = [1, 1, 1, 1]
Generation 380:	Best = 6.001684e+02	FES = 38100	Diversity = 1.7107e-01	k_t = [2, 4, 2, 4]
Generation 390:	Best = 6.001562e+02	FES = 39100	Diversity = 1.6669e-01	k_t = [1, 1, 1, 1]
Generation 400:	Best = 6.001547e+02	FES = 40100	Diversity = 1.5486e-01	k_t = [3, 2, 2, 4]
Generation 410:	Best = 6.001496e+02	FES = 41100	Diversity = 1.4543e-01	k_t = [1, 1, 1, 1]
Generation 420:	Best = 6.001261e+02	FES = 42100	Diversity = 1.3112e-01	k_t = [3, 4, 4, 3]
Generation 430:	Best = 6.001199e+02	FES = 43100	Diversity = 1.2037e-01	k_t = [1, 1, 1, 1]
Generation 440:	Best = 6.001172e+02	FES = 44100	Diversity = 1.1679e-01	k_t = [3, 4, 4, 4]
Generation 450:	Best = 6.001160e+02	FES = 45100	Diversity = 1.1256e-01	k_t = [1, 1, 1, 1]
Generation 460:	Best = 6.000993e+02	FES = 46100	Diversity = 1.0416e-01	k_t = [4, 3, 2, 2]
Generation 470:	Best = 6.000980e+02	FES = 47100	Diversity = 9.5260e-02	k_t = [1, 1, 1, 1]
Generation 480:	Best = 6.000980e+02	FES = 48100	Diversity = 9.0433e-02	k_t = [2, 3, 3, 2]
Generation 490:	Best = 6.000961e+02	FES = 49100	Diversity = 8.4962e-02	k_t = [1, 1, 1, 1]

Generation 500:	Best = 6.000929e+02	FES = 50100	Diversity = 7.8619e-02	k_t = [4, 3, 3, 2]
Generation 510:	Best = 6.000820e+02	FES = 51100	Diversity = 7.0835e-02	k_t = [1, 1, 1, 1]
Generation 520:	Best = 6.000769e+02	FES = 52100	Diversity = 6.9017e-02	k_t = [3, 4, 3, 4]
Generation 530:	Best = 6.000689e+02	FES = 53100	Diversity = 6.3585e-02	k_t = [1, 1, 1, 1]
Generation 540:	Best = 6.000634e+02	FES = 54100	Diversity = 6.2595e-02	k_t = [4, 2, 4, 3]
Generation 550:	Best = 6.000590e+02	FES = 55100	Diversity = 5.8828e-02	k_t = [1, 1, 1, 1]
Generation 560:	Best = 6.000590e+02	FES = 56100	Diversity = 5.5169e-02	k_t = [3, 4, 2, 4]
Generation 570:	Best = 6.000466e+02	FES = 57100	Diversity = 4.9495e-02	k_t = [1, 1, 1, 1]
Generation 580:	Best = 6.000466e+02	FES = 58100	Diversity = 4.5252e-02	k_t = [4, 3, 2, 4]
Generation 590:	Best = 6.000432e+02	FES = 59100	Diversity = 4.1859e-02	k_t = [1, 1, 1, 1]
Generation 600:	Best = 6.000432e+02	FES = 60100	Diversity = 3.8759e-02	k_t = [2, 3, 2, 4]
Generation 610:	Best = 6.000405e+02	FES = 61100	Diversity = 3.5916e-02	k_t = [1, 1, 1, 1]
Generation 620:	Best = 6.000359e+02	FES = 62100	Diversity = 3.2684e-02	k_t = [4, 4, 2, 2]
Generation 630:	Best = 6.000354e+02	FES = 63100	Diversity = 2.9988e-02	k_t = [1, 1, 1, 1]
Generation 640:	Best = 6.000298e+02	FES = 64100	Diversity = 2.7520e-02	k_t = [2, 2, 3, 3]
Generation 650:	Best = 6.000285e+02	FES = 65100	Diversity = 2.5904e-02	k_t = [1, 1, 1, 1]
Generation 660:	Best = 6.000269e+02	FES = 66100	Diversity = 2.3878e-02	k_t = [4, 2, 4, 2]
Generation 670:	Best = 6.000259e+02	FES = 67100	Diversity = 2.1622e-02	k_t = [1, 1, 1, 1]
Generation 680:	Best = 6.000191e+02	FES = 68100	Diversity = 1.9384e-02	k_t = [4, 2, 2, 3]
Generation 690:	Best = 6.000186e+02	FES = 69100	Diversity = 1.7561e-02	k_t = [1, 1, 1, 1]
Generation 700:	Best = 6.000186e+02	FES = 70100	Diversity = 1.5352e-02	k_t = [2, 4, 3, 4]
Generation 710:	Best = 6.000186e+02	FES = 71100	Diversity = 1.2558e-02	k_t = [1, 1, 1, 1]
Generation 720:	Best = 6.000173e+02	FES = 72100	Diversity = 1.1514e-02	k_t = [3, 3, 3, 2]
Generation 730:	Best = 6.000136e+02	FES = 73100	Diversity = 1.0450e-02	k_t = [1, 1, 1, 1]
Generation 740:	Best = 6.000127e+02	FES = 74100	Diversity = 9.0792e-03	k_t = [2, 3, 2, 2]
Generation 750:	Best = 6.000108e+02	FES = 75100	Diversity = 8.1000e-03	k_t = [1, 1, 1, 1]
Generation 760:	Best = 6.000080e+02	FES = 76100	Diversity = 7.1367e-03	k_t = [3, 4, 3, 3]

Generation 770:	Best = 6.000080e+02	FES = 77100	Diversity = 6.7754e-03	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 6.000077e+02	FES = 78100	Diversity = 6.5269e-03	k_t
= [4, 2, 3, 4]				
Generation 790:	Best = 6.000077e+02	FES = 79100	Diversity = 5.6737e-03	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 6.000073e+02	FES = 80100	Diversity = 5.0814e-03	k_t
= [3, 4, 2, 4]				
Generation 810:	Best = 6.000068e+02	FES = 81100	Diversity = 4.8279e-03	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 6.000068e+02	FES = 82100	Diversity = 4.5092e-03	k_t
= [4, 3, 4, 2]				
Generation 830:	Best = 6.000067e+02	FES = 83100	Diversity = 4.2176e-03	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 6.000060e+02	FES = 84100	Diversity = 4.0146e-03	k_t
= [2, 4, 4, 4]				
Generation 850:	Best = 6.000060e+02	FES = 85100	Diversity = 3.7679e-03	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 6.000057e+02	FES = 86100	Diversity = 3.6509e-03	k_t
= [4, 4, 2, 2]				
Generation 870:	Best = 6.000056e+02	FES = 87100	Diversity = 3.5420e-03	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 6.000055e+02	FES = 88100	Diversity = 3.4663e-03	k_t
= [4, 3, 4, 2]				
Generation 890:	Best = 6.000053e+02	FES = 89100	Diversity = 3.3057e-03	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 6.000048e+02	FES = 90100	Diversity = 3.2239e-03	k_t
= [3, 2, 4, 3]				
Generation 910:	Best = 6.000047e+02	FES = 91100	Diversity = 3.2474e-03	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 6.000045e+02	FES = 92100	Diversity = 3.1792e-03	k_t
= [4, 2, 2, 2]				
Generation 930:	Best = 6.000045e+02	FES = 93100	Diversity = 3.2537e-03	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 6.000043e+02	FES = 94100	Diversity = 3.1952e-03	k_t
= [2, 4, 2, 4]				
Generation 950:	Best = 6.000043e+02	FES = 95100	Diversity = 3.1144e-03	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 6.000042e+02	FES = 96100	Diversity = 3.0396e-03	k_t
= [4, 2, 3, 2]				
Generation 970:	Best = 6.000040e+02	FES = 97100	Diversity = 2.8605e-03	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 6.000040e+02	FES = 98100	Diversity = 2.8855e-03	k_t
= [2, 3, 4, 3]				
Generation 990:	Best = 6.000035e+02	FES = 99100	Diversity = 2.8740e-03	k_t
= [1, 1, 1, 1]				
Generation 1000:	Best = 6.000035e+02	FES = 100100	Diversity = 2.8579e-03	k_t
= [4, 2, 2, 4]				
Generation 1010:	Best = 6.000035e+02	FES = 101100	Diversity = 2.7859e-03	k_t
= [1, 1, 1, 1]				
Generation 1020:	Best = 6.000034e+02	FES = 102100	Diversity = 2.7024e-03	k_t
= [4, 3, 2, 3]				
Generation 1030:	Best = 6.000034e+02	FES = 103100	Diversity = 2.7206e-03	k_t
= [1, 1, 1, 1]				

Generation 1040: Best = 6.000029e+02 = [4, 3, 2, 4]	FES = 104100	Diversity = 2.6851e-03	k_t
Generation 1050: Best = 6.000029e+02 = [1, 1, 1, 1]	FES = 105100	Diversity = 2.6826e-03	k_t
Generation 1060: Best = 6.000029e+02 = [2, 2, 4, 2]	FES = 106100	Diversity = 2.7060e-03	k_t
Generation 1070: Best = 6.000029e+02 = [1, 1, 1, 1]	FES = 107100	Diversity = 2.6592e-03	k_t
Generation 1080: Best = 6.000029e+02 = [4, 3, 4, 2]	FES = 108100	Diversity = 2.6167e-03	k_t
Generation 1090: Best = 6.000028e+02 = [1, 1, 1, 1]	FES = 109100	Diversity = 2.5661e-03	k_t
Generation 1100: Best = 6.000028e+02 = [3, 4, 3, 3]	FES = 110100	Diversity = 2.4461e-03	k_t
Generation 1110: Best = 6.000025e+02 = [1, 1, 1, 1]	FES = 111100	Diversity = 2.3600e-03	k_t
Generation 1120: Best = 6.000025e+02 = [4, 3, 4, 4]	FES = 112100	Diversity = 2.3777e-03	k_t
Generation 1130: Best = 6.000022e+02 = [1, 1, 1, 1]	FES = 113100	Diversity = 2.2613e-03	k_t
Generation 1140: Best = 6.000019e+02 = [3, 3, 3, 2]	FES = 114100	Diversity = 2.1584e-03	k_t
Generation 1150: Best = 6.000018e+02 = [1, 1, 1, 1]	FES = 115100	Diversity = 2.0643e-03	k_t
Generation 1160: Best = 6.000016e+02 = [4, 2, 3, 2]	FES = 116100	Diversity = 1.9848e-03	k_t
Generation 1170: Best = 6.000016e+02 = [1, 1, 1, 1]	FES = 117100	Diversity = 1.9138e-03	k_t
Generation 1180: Best = 6.000016e+02 = [2, 3, 4, 2]	FES = 118100	Diversity = 1.8223e-03	k_t
Generation 1190: Best = 6.000016e+02 = [1, 1, 1, 1]	FES = 119100	Diversity = 1.7288e-03	k_t
Generation 1200: Best = 6.000015e+02 = [2, 2, 2, 3]	FES = 120100	Diversity = 1.5988e-03	k_t
Generation 1210: Best = 6.000015e+02 = [1, 1, 1, 1]	FES = 121100	Diversity = 1.5868e-03	k_t
Generation 1220: Best = 6.000015e+02 = [2, 3, 2, 4]	FES = 122100	Diversity = 1.5745e-03	k_t
Generation 1230: Best = 6.000014e+02 = [1, 1, 1, 1]	FES = 123100	Diversity = 1.5183e-03	k_t
Generation 1240: Best = 6.000013e+02 = [2, 2, 3, 3]	FES = 124100	Diversity = 1.4485e-03	k_t
Generation 1250: Best = 6.000013e+02 = [1, 1, 1, 1]	FES = 125100	Diversity = 1.3977e-03	k_t
Generation 1260: Best = 6.000013e+02 = [4, 4, 3, 2]	FES = 126100	Diversity = 1.3027e-03	k_t
Generation 1270: Best = 6.000012e+02 = [1, 1, 1, 1]	FES = 127100	Diversity = 1.2900e-03	k_t
Generation 1280: Best = 6.000011e+02 = [2, 2, 2, 2]	FES = 128100	Diversity = 1.1765e-03	k_t
Generation 1290: Best = 6.000008e+02 = [1, 1, 1, 1]	FES = 129100	Diversity = 1.0782e-03	k_t
Generation 1300: Best = 6.000008e+02 = [2, 2, 2, 2]	FES = 130100	Diversity = 9.9200e-04	k_t

Generation 1310: Best = 6.000008e+02 = [1, 1, 1, 1]	FES = 131100	Diversity = 9.3227e-04	k_t
Generation 1320: Best = 6.000007e+02 = [2, 4, 4, 4]	FES = 132100	Diversity = 8.4867e-04	k_t
Generation 1330: Best = 6.000007e+02 = [1, 1, 1, 1]	FES = 133100	Diversity = 7.9551e-04	k_t
Generation 1340: Best = 6.000007e+02 = [2, 4, 2, 2]	FES = 134100	Diversity = 7.0652e-04	k_t
Generation 1350: Best = 6.000007e+02 = [1, 1, 1, 1]	FES = 135100	Diversity = 6.7057e-04	k_t
Generation 1360: Best = 6.000007e+02 = [3, 4, 2, 2]	FES = 136100	Diversity = 5.3656e-04	k_t
Generation 1370: Best = 6.000007e+02 = [1, 1, 1, 1]	FES = 137100	Diversity = 4.1352e-04	k_t
Generation 1380: Best = 6.000006e+02 = [3, 2, 2, 4]	FES = 138100	Diversity = 3.7909e-04	k_t
Generation 1390: Best = 6.000006e+02 = [1, 1, 1, 1]	FES = 139100	Diversity = 3.5038e-04	k_t
Generation 1400: Best = 6.000006e+02 = [3, 2, 2, 3]	FES = 140100	Diversity = 3.3251e-04	k_t
Generation 1410: Best = 6.000006e+02 = [1, 1, 1, 1]	FES = 141100	Diversity = 3.2692e-04	k_t
Generation 1420: Best = 6.000005e+02 = [3, 2, 4, 3]	FES = 142100	Diversity = 3.0147e-04	k_t
Generation 1430: Best = 6.000005e+02 = [1, 1, 1, 1]	FES = 143100	Diversity = 2.8306e-04	k_t
Generation 1440: Best = 6.000005e+02 = [4, 2, 4, 4]	FES = 144100	Diversity = 2.6073e-04	k_t
Generation 1450: Best = 6.000005e+02 = [1, 1, 1, 1]	FES = 145100	Diversity = 2.4582e-04	k_t
Generation 1460: Best = 6.000005e+02 = [4, 3, 2, 3]	FES = 146100	Diversity = 2.2986e-04	k_t
Generation 1470: Best = 6.000004e+02 = [1, 1, 1, 1]	FES = 147100	Diversity = 2.1906e-04	k_t
Generation 1480: Best = 6.000004e+02 = [4, 2, 2, 2]	FES = 148100	Diversity = 2.0505e-04	k_t
Generation 1490: Best = 6.000004e+02 = [1, 1, 1, 1]	FES = 149100	Diversity = 2.0366e-04	k_t
Generation 1500: Best = 6.000004e+02 = [3, 3, 4, 4]	FES = 150100	Diversity = 1.9652e-04	k_t
Generation 1510: Best = 6.000004e+02 = [1, 1, 1, 1]	FES = 151100	Diversity = 1.9296e-04	k_t
Generation 1520: Best = 6.000004e+02 = [4, 2, 2, 4]	FES = 152100	Diversity = 1.8695e-04	k_t
Generation 1530: Best = 6.000003e+02 = [1, 1, 1, 1]	FES = 153100	Diversity = 1.8145e-04	k_t
Generation 1540: Best = 6.000003e+02 = [2, 2, 2, 3]	FES = 154100	Diversity = 1.7780e-04	k_t
Generation 1550: Best = 6.000003e+02 = [1, 1, 1, 1]	FES = 155100	Diversity = 1.6526e-04	k_t
Generation 1560: Best = 6.000003e+02 = [2, 4, 3, 4]	FES = 156100	Diversity = 1.5733e-04	k_t
Generation 1570: Best = 6.000003e+02 = [1, 1, 1, 1]	FES = 157100	Diversity = 1.5418e-04	k_t

Generation 1580: Best = 6.000003e+02 = [4, 3, 3, 3]	FES = 158100	Diversity = 1.4447e-04	k_t
Generation 1590: Best = 6.000003e+02 = [1, 1, 1, 1]	FES = 159100	Diversity = 1.3968e-04	k_t
Generation 1600: Best = 6.000003e+02 = [2, 3, 2, 4]	FES = 160100	Diversity = 1.3684e-04	k_t
Generation 1610: Best = 6.000003e+02 = [1, 1, 1, 1]	FES = 161100	Diversity = 1.3214e-04	k_t
Generation 1620: Best = 6.000003e+02 = [2, 4, 4, 2]	FES = 162100	Diversity = 1.2704e-04	k_t
Generation 1630: Best = 6.000003e+02 = [1, 1, 1, 1]	FES = 163100	Diversity = 1.2323e-04	k_t
Generation 1640: Best = 6.000003e+02 = [2, 2, 3, 3]	FES = 164100	Diversity = 1.1877e-04	k_t
Generation 1650: Best = 6.000003e+02 = [1, 1, 1, 1]	FES = 165100	Diversity = 1.1386e-04	k_t
Generation 1660: Best = 6.000002e+02 = [2, 4, 3, 3]	FES = 166100	Diversity = 1.0837e-04	k_t
Generation 1670: Best = 6.000002e+02 = [1, 1, 1, 1]	FES = 167100	Diversity = 1.0683e-04	k_t
Generation 1680: Best = 6.000002e+02 = [3, 4, 4, 4]	FES = 168100	Diversity = 1.0529e-04	k_t
Generation 1690: Best = 6.000002e+02 = [1, 1, 1, 1]	FES = 169100	Diversity = 1.0254e-04	k_t
Generation 1700: Best = 6.000002e+02 = [2, 4, 2, 2]	FES = 170100	Diversity = 9.8974e-05	k_t
Generation 1710: Best = 6.000002e+02 = [1, 1, 1, 1]	FES = 171100	Diversity = 9.2460e-05	k_t
Generation 1720: Best = 6.000002e+02 = [4, 4, 4, 2]	FES = 172100	Diversity = 8.6874e-05	k_t
Generation 1730: Best = 6.000002e+02 = [1, 1, 1, 1]	FES = 173100	Diversity = 8.2550e-05	k_t
Generation 1740: Best = 6.000002e+02 = [4, 2, 4, 4]	FES = 174100	Diversity = 7.6356e-05	k_t
Generation 1750: Best = 6.000002e+02 = [1, 1, 1, 1]	FES = 175100	Diversity = 6.9591e-05	k_t
Generation 1760: Best = 6.000002e+02 = [2, 2, 2, 3]	FES = 176100	Diversity = 6.4019e-05	k_t
Generation 1770: Best = 6.000002e+02 = [1, 1, 1, 1]	FES = 177100	Diversity = 5.8363e-05	k_t
Generation 1780: Best = 6.000002e+02 = [4, 3, 2, 3]	FES = 178100	Diversity = 5.4842e-05	k_t
Generation 1790: Best = 6.000002e+02 = [1, 1, 1, 1]	FES = 179100	Diversity = 4.7239e-05	k_t
Generation 1800: Best = 6.000001e+02 = [4, 4, 2, 3]	FES = 180100	Diversity = 4.4580e-05	k_t
Generation 1810: Best = 6.000001e+02 = [1, 1, 1, 1]	FES = 181100	Diversity = 3.8284e-05	k_t
Generation 1820: Best = 6.000001e+02 = [4, 3, 2, 3]	FES = 182100	Diversity = 3.5264e-05	k_t
Generation 1830: Best = 6.000001e+02 = [1, 1, 1, 1]	FES = 183100	Diversity = 3.3149e-05	k_t
Generation 1840: Best = 6.000001e+02 = [4, 3, 3, 3]	FES = 184100	Diversity = 3.2301e-05	k_t

Generation 1850: Best = 6.000001e+02 = [1, 1, 1, 1]	FES = 185100	Diversity = 3.0721e-05	k_t
Generation 1860: Best = 6.000001e+02 = [4, 4, 4, 2]	FES = 186100	Diversity = 2.8224e-05	k_t
Generation 1870: Best = 6.000001e+02 = [1, 1, 1, 1]	FES = 187100	Diversity = 2.5709e-05	k_t
Generation 1880: Best = 6.000001e+02 = [3, 4, 4, 3]	FES = 188100	Diversity = 2.4221e-05	k_t
Generation 1890: Best = 6.000001e+02 = [1, 1, 1, 1]	FES = 189100	Diversity = 2.2844e-05	k_t
Generation 1900: Best = 6.000001e+02 = [2, 4, 4, 4]	FES = 190100	Diversity = 2.3053e-05	k_t
Generation 1910: Best = 6.000001e+02 = [1, 1, 1, 1]	FES = 191100	Diversity = 2.2130e-05	k_t
Generation 1920: Best = 6.000001e+02 = [3, 3, 4, 4]	FES = 192100	Diversity = 2.0906e-05	k_t
Generation 1930: Best = 6.000001e+02 = [1, 1, 1, 1]	FES = 193100	Diversity = 1.9736e-05	k_t
Generation 1940: Best = 6.000001e+02 = [4, 2, 3, 3]	FES = 194100	Diversity = 1.9341e-05	k_t
Generation 1950: Best = 6.000001e+02 = [1, 1, 1, 1]	FES = 195100	Diversity = 1.9651e-05	k_t
Generation 1960: Best = 6.000001e+02 = [2, 4, 3, 4]	FES = 196100	Diversity = 1.8731e-05	k_t
Generation 1970: Best = 6.000001e+02 = [1, 1, 1, 1]	FES = 197100	Diversity = 1.8521e-05	k_t
Generation 1980: Best = 6.000001e+02 = [4, 3, 4, 2]	FES = 198100	Diversity = 1.8578e-05	k_t
Generation 1990: Best = 6.000001e+02 = [1, 1, 1, 1]	FES = 199100	Diversity = 1.7773e-05	k_t
Generation 2000: Best = 6.000001e+02 = [2, 3, 4, 2]	FES = 200100	Diversity = 1.7286e-05	k_t
Generation 2010: Best = 6.000001e+02 = [1, 1, 1, 1]	FES = 201100	Diversity = 1.6572e-05	k_t
Generation 2020: Best = 6.000001e+02 = [3, 3, 2, 3]	FES = 202100	Diversity = 1.6581e-05	k_t
Generation 2030: Best = 6.000001e+02 = [1, 1, 1, 1]	FES = 203100	Diversity = 1.6336e-05	k_t
Generation 2040: Best = 6.000001e+02 = [3, 4, 4, 3]	FES = 204100	Diversity = 1.5772e-05	k_t
Generation 2050: Best = 6.000001e+02 = [1, 1, 1, 1]	FES = 205100	Diversity = 1.5522e-05	k_t
Generation 2060: Best = 6.000001e+02 = [2, 4, 3, 3]	FES = 206100	Diversity = 1.5276e-05	k_t
Generation 2070: Best = 6.000001e+02 = [1, 1, 1, 1]	FES = 207100	Diversity = 1.5124e-05	k_t
Generation 2080: Best = 6.000001e+02 = [4, 2, 3, 2]	FES = 208100	Diversity = 1.5081e-05	k_t
Generation 2090: Best = 6.000001e+02 = [1, 1, 1, 1]	FES = 209100	Diversity = 1.5331e-05	k_t
Generation 2100: Best = 6.000001e+02 = [2, 2, 2, 4]	FES = 210100	Diversity = 1.4967e-05	k_t
Generation 2110: Best = 6.000001e+02 = [1, 1, 1, 1]	FES = 211100	Diversity = 1.5051e-05	k_t

Generation 2120: Best = 6.000001e+02 = [3, 2, 4, 4]	FES = 212100	Diversity = 1.4429e-05	k_t
Generation 2130: Best = 6.000001e+02 = [1, 1, 1, 1]	FES = 213100	Diversity = 1.4458e-05	k_t
Generation 2140: Best = 6.000001e+02 = [4, 2, 3, 2]	FES = 214100	Diversity = 1.4043e-05	k_t
Generation 2150: Best = 6.000001e+02 = [1, 1, 1, 1]	FES = 215100	Diversity = 1.3758e-05	k_t
Generation 2160: Best = 6.000001e+02 = [4, 2, 3, 3]	FES = 216100	Diversity = 1.3587e-05	k_t
Generation 2170: Best = 6.000001e+02 = [1, 1, 1, 1]	FES = 217100	Diversity = 1.3339e-05	k_t
Generation 2180: Best = 6.000001e+02 = [2, 2, 3, 3]	FES = 218100	Diversity = 1.3201e-05	k_t
Generation 2190: Best = 6.000001e+02 = [1, 1, 1, 1]	FES = 219100	Diversity = 1.2516e-05	k_t
Generation 2200: Best = 6.000001e+02 = [4, 2, 4, 2]	FES = 220100	Diversity = 1.2577e-05	k_t
Generation 2210: Best = 6.000001e+02 = [1, 1, 1, 1]	FES = 221100	Diversity = 1.2395e-05	k_t
Generation 2220: Best = 6.000001e+02 = [3, 4, 2, 4]	FES = 222100	Diversity = 1.2339e-05	k_t
Generation 2230: Best = 6.000001e+02 = [1, 1, 1, 1]	FES = 223100	Diversity = 1.2140e-05	k_t
Generation 2240: Best = 6.000001e+02 = [4, 2, 3, 2]	FES = 224100	Diversity = 1.1538e-05	k_t
Generation 2250: Best = 6.000001e+02 = [1, 1, 1, 1]	FES = 225100	Diversity = 1.0616e-05	k_t
Generation 2260: Best = 6.000001e+02 = [3, 3, 2, 2]	FES = 226100	Diversity = 9.6324e-06	k_t
Generation 2270: Best = 6.000001e+02 = [1, 1, 1, 1]	FES = 227100	Diversity = 9.0405e-06	k_t
Generation 2280: Best = 6.000001e+02 = [2, 2, 4, 4]	FES = 228100	Diversity = 7.8042e-06	k_t
Generation 2290: Best = 6.000001e+02 = [1, 1, 1, 1]	FES = 229100	Diversity = 7.2609e-06	k_t
Generation 2300: Best = 6.000001e+02 = [3, 3, 3, 3]	FES = 230100	Diversity = 6.3110e-06	k_t
Generation 2310: Best = 6.000001e+02 = [1, 1, 1, 1]	FES = 231100	Diversity = 5.7060e-06	k_t
Generation 2320: Best = 6.000001e+02 = [2, 4, 4, 2]	FES = 232100	Diversity = 4.4104e-06	k_t
Generation 2330: Best = 6.000001e+02 = [1, 1, 1, 1]	FES = 233100	Diversity = 3.9455e-06	k_t
Generation 2340: Best = 6.000001e+02 = [3, 4, 3, 3]	FES = 234100	Diversity = 3.3362e-06	k_t
Generation 2350: Best = 6.000001e+02 = [1, 1, 1, 1]	FES = 235100	Diversity = 2.7942e-06	k_t
Generation 2360: Best = 6.000000e+02 = [4, 2, 3, 4]	FES = 236100	Diversity = 2.2102e-06	k_t
Generation 2370: Best = 6.000000e+02 = [1, 1, 1, 1]	FES = 237100	Diversity = 1.9081e-06	k_t
Generation 2380: Best = 6.000000e+02 = [4, 2, 2, 2]	FES = 238100	Diversity = 1.7315e-06	k_t

Generation 2390: Best = 6.000000e+02 = [1, 1, 1, 1]	FES = 239100	Diversity = 1.6049e-06	k_t
Generation 2400: Best = 6.000000e+02 = [3, 3, 4, 2]	FES = 240100	Diversity = 1.4534e-06	k_t
Generation 2410: Best = 6.000000e+02 = [1, 1, 1, 1]	FES = 241100	Diversity = 1.0505e-06	k_t
Generation 2420: Best = 6.000000e+02 = [2, 4, 2, 3]	FES = 242100	Diversity = 9.3284e-07	k_t
Generation 2430: Best = 6.000000e+02 = [1, 1, 1, 1]	FES = 243100	Diversity = 8.8045e-07	k_t
Generation 2440: Best = 6.000000e+02 = [3, 3, 4, 2]	FES = 244100	Diversity = 7.8072e-07	k_t
Generation 2450: Best = 6.000000e+02 = [1, 1, 1, 1]	FES = 245100	Diversity = 7.2797e-07	k_t
Generation 2460: Best = 6.000000e+02 = [4, 2, 2, 4]	FES = 246100	Diversity = 6.6360e-07	k_t
Generation 2470: Best = 6.000000e+02 = [1, 1, 1, 1]	FES = 247100	Diversity = 6.5645e-07	k_t
Generation 2480: Best = 6.000000e+02 = [3, 2, 3, 3]	FES = 248100	Diversity = 5.5665e-07	k_t
Generation 2490: Best = 6.000000e+02 = [1, 1, 1, 1]	FES = 249100	Diversity = 6.0059e-07	k_t
Generation 2500: Best = 6.000000e+02 = [3, 3, 2, 3]	FES = 250100	Diversity = 5.6818e-07	k_t
Generation 2510: Best = 6.000000e+02 = [1, 1, 1, 1]	FES = 251100	Diversity = 5.1103e-07	k_t
Generation 2520: Best = 6.000000e+02 = [4, 4, 2, 4]	FES = 252100	Diversity = 4.3802e-07	k_t
Generation 2530: Best = 6.000000e+02 = [1, 1, 1, 1]	FES = 253100	Diversity = 4.1758e-07	k_t
Generation 2540: Best = 6.000000e+02 = [3, 4, 2, 2]	FES = 254100	Diversity = 4.1837e-07	k_t
Generation 2550: Best = 6.000000e+02 = [1, 1, 1, 1]	FES = 255100	Diversity = 3.4566e-07	k_t
Generation 2560: Best = 6.000000e+02 = [3, 3, 4, 3]	FES = 256100	Diversity = 2.9897e-07	k_t
Generation 2570: Best = 6.000000e+02 = [1, 1, 1, 1]	FES = 257100	Diversity = 2.7417e-07	k_t
Generation 2580: Best = 6.000000e+02 = [2, 4, 4, 2]	FES = 258100	Diversity = 2.7243e-07	k_t
Generation 2590: Best = 6.000000e+02 = [1, 1, 1, 1]	FES = 259100	Diversity = 2.5163e-07	k_t
Generation 2600: Best = 6.000000e+02 = [2, 4, 4, 2]	FES = 260100	Diversity = 2.2184e-07	k_t
Generation 2610: Best = 6.000000e+02 = [1, 1, 1, 1]	FES = 261100	Diversity = 1.8702e-07	k_t
Generation 2620: Best = 6.000000e+02 = [3, 2, 4, 4]	FES = 262100	Diversity = 1.6156e-07	k_t
Generation 2630: Best = 6.000000e+02 = [1, 1, 1, 1]	FES = 263100	Diversity = 1.4716e-07	k_t
Generation 2640: Best = 6.000000e+02 = [2, 2, 4, 2]	FES = 264100	Diversity = 1.1921e-07	k_t
Generation 2650: Best = 6.000000e+02 = [1, 1, 1, 1]	FES = 265100	Diversity = 9.6357e-08	k_t

Generation 2660: Best = 6.000000e+02 = [3, 4, 3, 4]	FES = 266100	Diversity = 8.7993e-08	k_t
Generation 2670: Best = 6.000000e+02 = [1, 1, 1, 1]	FES = 267100	Diversity = 7.7363e-08	k_t
Generation 2680: Best = 6.000000e+02 = [2, 3, 4, 3]	FES = 268100	Diversity = 6.6024e-08	k_t
Generation 2690: Best = 6.000000e+02 = [1, 1, 1, 1]	FES = 269100	Diversity = 5.3540e-08	k_t
Generation 2700: Best = 6.000000e+02 = [4, 3, 4, 3]	FES = 270100	Diversity = 5.2678e-08	k_t
Generation 2710: Best = 6.000000e+02 = [1, 1, 1, 1]	FES = 271100	Diversity = 4.1494e-08	k_t
Generation 2720: Best = 6.000000e+02 = [3, 2, 4, 4]	FES = 272100	Diversity = 3.5457e-08	k_t
Generation 2730: Best = 6.000000e+02 = [1, 1, 1, 1]	FES = 273100	Diversity = 3.4522e-08	k_t
Generation 2740: Best = 6.000000e+02 = [4, 2, 3, 2]	FES = 274100	Diversity = 3.0108e-08	k_t
Generation 2750: Best = 6.000000e+02 = [1, 1, 1, 1]	FES = 275100	Diversity = 3.2768e-08	k_t
Generation 2760: Best = 6.000000e+02 = [3, 2, 4, 3]	FES = 276100	Diversity = 3.5673e-08	k_t
Generation 2770: Best = 6.000000e+02 = [1, 1, 1, 1]	FES = 277100	Diversity = 3.3393e-08	k_t
Generation 2780: Best = 6.000000e+02 = [3, 3, 4, 2]	FES = 278100	Diversity = 2.6214e-08	k_t
Generation 2790: Best = 6.000000e+02 = [1, 1, 1, 1]	FES = 279100	Diversity = 2.7450e-08	k_t
Generation 2800: Best = 6.000000e+02 = [3, 3, 2, 2]	FES = 280100	Diversity = 2.7432e-08	k_t
Generation 2810: Best = 6.000000e+02 = [1, 1, 1, 1]	FES = 281100	Diversity = 2.5516e-08	k_t
Generation 2820: Best = 6.000000e+02 = [3, 2, 4, 4]	FES = 282100	Diversity = 1.9345e-08	k_t
Generation 2830: Best = 6.000000e+02 = [1, 1, 1, 1]	FES = 283100	Diversity = 1.6731e-08	k_t
Generation 2840: Best = 6.000000e+02 = [2, 2, 4, 3]	FES = 284100	Diversity = 1.3933e-08	k_t
Generation 2850: Best = 6.000000e+02 = [1, 1, 1, 1]	FES = 285100	Diversity = 1.5256e-08	k_t
Generation 2860: Best = 6.000000e+02 = [4, 4, 4, 4]	FES = 286100	Diversity = 1.2497e-08	k_t
Generation 2870: Best = 6.000000e+02 = [1, 1, 1, 1]	FES = 287100	Diversity = 1.1038e-08	k_t
Generation 2880: Best = 6.000000e+02 = [2, 4, 2, 4]	FES = 288100	Diversity = 1.2461e-08	k_t
Generation 2890: Best = 6.000000e+02 = [1, 1, 1, 1]	FES = 289100	Diversity = 1.4018e-08	k_t
Generation 2900: Best = 6.000000e+02 = [2, 4, 2, 2]	FES = 290100	Diversity = 9.8651e-09	k_t
Generation 2910: Best = 6.000000e+02 = [1, 1, 1, 1]	FES = 291100	Diversity = 9.3207e-09	k_t
Generation 2920: Best = 6.000000e+02 = [3, 2, 2, 2]	FES = 292100	Diversity = 7.9216e-09	k_t

Generation 2930: Best = 6.000000e+02 FES = 293100 Diversity = 7.4292e-09 k_t
 = [1, 1, 1, 1]
 Generation 2940: Best = 6.000000e+02 FES = 294100 Diversity = 7.5773e-09 k_t
 = [2, 4, 4, 4]
 Generation 2950: Best = 6.000000e+02 FES = 295100 Diversity = 6.6420e-09 k_t
 = [1, 1, 1, 1]
 Generation 2960: Best = 6.000000e+02 FES = 296100 Diversity = 5.8340e-09 k_t
 = [4, 3, 4, 4]
 Generation 2970: Best = 6.000000e+02 FES = 297100 Diversity = 5.9510e-09 k_t
 = [1, 1, 1, 1]
 Generation 2980: Best = 6.000000e+02 FES = 298100 Diversity = 5.4721e-09 k_t
 = [4, 3, 4, 3]
 Generation 2990: Best = 6.000000e+02 FES = 299100 Diversity = 5.7716e-09 k_t
 = [1, 1, 1, 1]

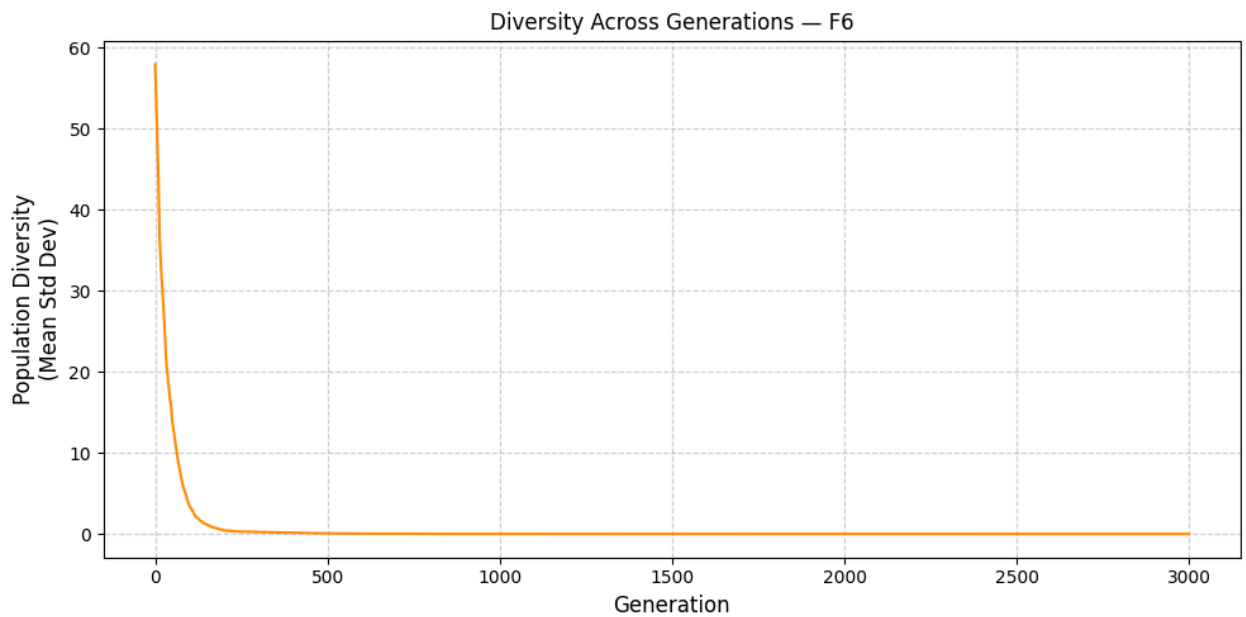
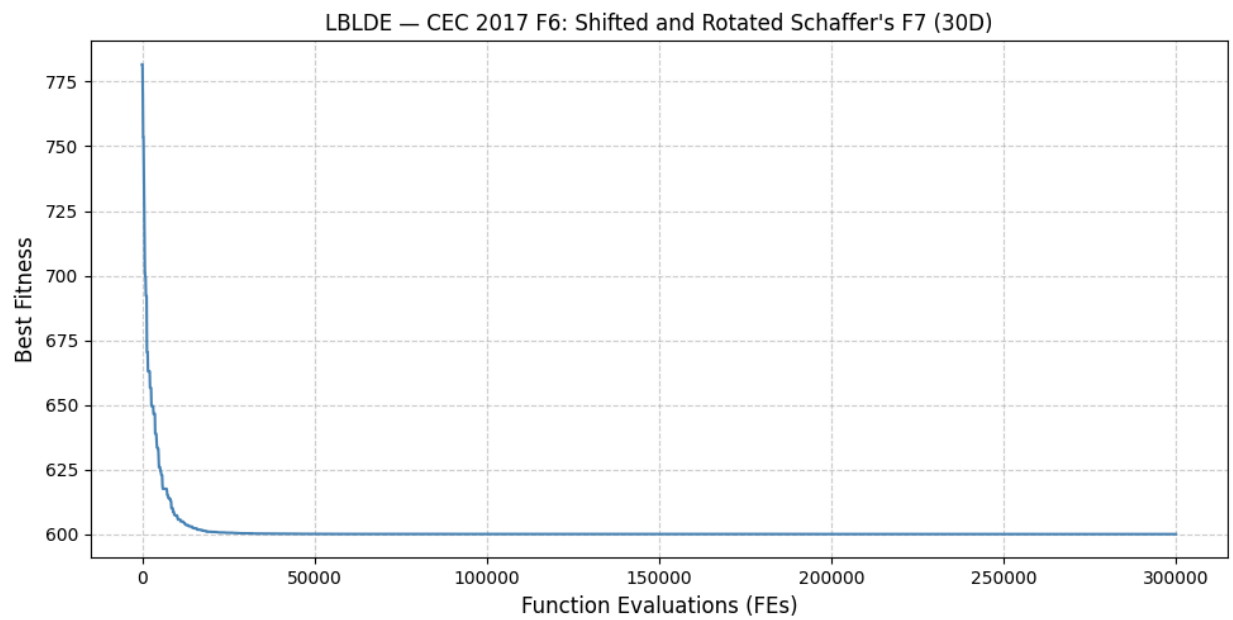
Optimization completed!

Final Best Fitness = 6.000000e+02

Total Function Evaluations = 300000

```

=====
=
Results:
  Best Fitness           : 6.000000e+02
  Global Optimum         : 6.000000e+02
  Error (|best-optimum|) : 3.819354e-05
  Best Solution (5 dim)  : [ 79.08951256 -24.5706694 -34.67392688 -44.68157807
75.99787996]
  D^(0) (init. diversity): 5.7976e+01
=====
=
  
```



Graph saved as convergence_F6.png

```
In [17]: # =====
# F7 - Shifted and Rotated Lunacek Bi-Rastrigin
# =====
run_and_plot('F7', make_wrapper(simple.f7), 700.0, "Shifted and Rotated Lunacek Bi-Rastrigin")
```

```

=====
=
LBLDE_s on CEC 2017 F7 – Single Run
=====
=
Function      : Shifted and Rotated Lunacek Bi-Rastrigin's (30D)
Bounds       : [-100, 100]
Global optimum : 700.0
Initial D^(0) = 5.7976e+01
Generation 0: Best Fitness = 3.268478e+03 | Diversity = 5.7976e+01
Generation 10: Best = 1.660499e+03 FES = 1100 Diversity = 2.9022e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 1.197889e+03 FES = 2100 Diversity = 1.6059e+01 k_t
= [3, 3, 2, 4]
Generation 30: Best = 1.057100e+03 FES = 3100 Diversity = 1.0971e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 9.978248e+02 FES = 4100 Diversity = 8.8189e+00 k_t
= [3, 4, 3, 4]
Generation 50: Best = 9.700483e+02 FES = 5100 Diversity = 8.1418e+00 k_t
= [1, 1, 1, 1]
Generation 60: Best = 9.573145e+02 FES = 6100 Diversity = 7.3869e+00 k_t
= [2, 3, 4, 3]
Generation 70: Best = 9.573145e+02 FES = 7100 Diversity = 7.2833e+00 k_t
= [1, 1, 1, 1]
Generation 80: Best = 9.510012e+02 FES = 8100 Diversity = 7.0864e+00 k_t
= [2, 4, 2, 3]
Generation 90: Best = 9.201723e+02 FES = 9100 Diversity = 6.8231e+00 k_t
= [1, 1, 1, 1]
Generation 100: Best = 9.201723e+02 FES = 10100 Diversity = 6.7368e+00 k_t
= [4, 2, 2, 3]
Generation 110: Best = 9.050011e+02 FES = 11100 Diversity = 6.6933e+00 k_t
= [1, 1, 1, 1]
Generation 120: Best = 9.050011e+02 FES = 12100 Diversity = 6.5934e+00 k_t
= [3, 4, 2, 3]
Generation 130: Best = 9.050011e+02 FES = 13100 Diversity = 6.5305e+00 k_t
= [1, 1, 1, 1]
Generation 140: Best = 8.843826e+02 FES = 14100 Diversity = 6.4693e+00 k_t
= [3, 2, 4, 2]
Generation 150: Best = 8.843826e+02 FES = 15100 Diversity = 6.3274e+00 k_t
= [1, 1, 1, 1]
Generation 160: Best = 8.739359e+02 FES = 16100 Diversity = 6.3385e+00 k_t
= [4, 2, 3, 3]
Generation 170: Best = 8.739359e+02 FES = 17100 Diversity = 6.3134e+00 k_t
= [1, 1, 1, 1]
Generation 180: Best = 8.739359e+02 FES = 18100 Diversity = 6.2151e+00 k_t
= [3, 2, 4, 3]
Generation 190: Best = 8.739359e+02 FES = 19100 Diversity = 6.2172e+00 k_t
= [1, 1, 1, 1]
Generation 200: Best = 8.739359e+02 FES = 20100 Diversity = 6.2269e+00 k_t
= [4, 3, 3, 4]
Generation 210: Best = 8.676724e+02 FES = 21100 Diversity = 6.2569e+00 k_t
= [1, 1, 1, 1]
Generation 220: Best = 8.676724e+02 FES = 22100 Diversity = 6.1912e+00 k_t
= [4, 3, 2, 4]

```


Generation 230:	Best = 8.676724e+02	FES = 23100	Diversity = 6.1741e+00	k_t = [1, 1, 1, 1]
Generation 240:	Best = 8.667633e+02	FES = 24100	Diversity = 6.2306e+00	k_t = [3, 4, 4, 4]
Generation 250:	Best = 8.667633e+02	FES = 25100	Diversity = 6.2195e+00	k_t = [1, 1, 1, 1]
Generation 260:	Best = 8.455739e+02	FES = 26100	Diversity = 6.0882e+00	k_t = [2, 4, 4, 3]
Generation 270:	Best = 8.455739e+02	FES = 27100	Diversity = 6.0101e+00	k_t = [1, 1, 1, 1]
Generation 280:	Best = 8.455739e+02	FES = 28100	Diversity = 5.9993e+00	k_t = [4, 3, 2, 4]
Generation 290:	Best = 8.455739e+02	FES = 29100	Diversity = 5.9655e+00	k_t = [1, 1, 1, 1]
Generation 300:	Best = 8.455739e+02	FES = 30100	Diversity = 5.9471e+00	k_t = [3, 3, 4, 3]
Generation 310:	Best = 8.455739e+02	FES = 31100	Diversity = 5.8675e+00	k_t = [1, 1, 1, 1]
Generation 320:	Best = 8.455739e+02	FES = 32100	Diversity = 5.7976e+00	k_t = [4, 2, 4, 3]
Generation 330:	Best = 8.363345e+02	FES = 33100	Diversity = 5.8033e+00	k_t = [1, 1, 1, 1]
Generation 340:	Best = 8.363345e+02	FES = 34100	Diversity = 5.7146e+00	k_t = [3, 4, 2, 3]
Generation 350:	Best = 8.363345e+02	FES = 35100	Diversity = 5.6621e+00	k_t = [1, 1, 1, 1]
Generation 360:	Best = 8.363345e+02	FES = 36100	Diversity = 5.6468e+00	k_t = [3, 3, 3, 4]
Generation 370:	Best = 8.341394e+02	FES = 37100	Diversity = 5.6451e+00	k_t = [1, 1, 1, 1]
Generation 380:	Best = 8.341394e+02	FES = 38100	Diversity = 5.5419e+00	k_t = [2, 4, 3, 2]
Generation 390:	Best = 8.341394e+02	FES = 39100	Diversity = 5.5775e+00	k_t = [1, 1, 1, 1]
Generation 400:	Best = 8.314161e+02	FES = 40100	Diversity = 5.5518e+00	k_t = [3, 2, 2, 3]
Generation 410:	Best = 8.314161e+02	FES = 41100	Diversity = 5.5747e+00	k_t = [1, 1, 1, 1]
Generation 420:	Best = 8.314161e+02	FES = 42100	Diversity = 5.5733e+00	k_t = [3, 3, 2, 4]
Generation 430:	Best = 8.314161e+02	FES = 43100	Diversity = 5.5165e+00	k_t = [1, 1, 1, 1]
Generation 440:	Best = 8.314161e+02	FES = 44100	Diversity = 5.5246e+00	k_t = [4, 3, 3, 4]
Generation 450:	Best = 8.314161e+02	FES = 45100	Diversity = 5.5102e+00	k_t = [1, 1, 1, 1]
Generation 460:	Best = 8.298889e+02	FES = 46100	Diversity = 5.5061e+00	k_t = [3, 4, 2, 2]
Generation 470:	Best = 8.298889e+02	FES = 47100	Diversity = 5.5251e+00	k_t = [1, 1, 1, 1]
Generation 480:	Best = 8.298889e+02	FES = 48100	Diversity = 5.5035e+00	k_t = [2, 3, 2, 3]
Generation 490:	Best = 8.224260e+02	FES = 49100	Diversity = 5.4996e+00	k_t = [1, 1, 1, 1]

Generation 500:	Best = 8.224260e+02	FES = 50100	Diversity = 5.4973e+00	k_t
= [3, 2, 4, 3]				
Generation 510:	Best = 8.224260e+02	FES = 51100	Diversity = 5.4730e+00	k_t
= [1, 1, 1, 1]				
Generation 520:	Best = 8.224260e+02	FES = 52100	Diversity = 5.4734e+00	k_t
= [4, 4, 2, 2]				
Generation 530:	Best = 8.224260e+02	FES = 53100	Diversity = 5.4450e+00	k_t
= [1, 1, 1, 1]				
Generation 540:	Best = 8.224260e+02	FES = 54100	Diversity = 5.4437e+00	k_t
= [3, 4, 3, 4]				
Generation 550:	Best = 8.224260e+02	FES = 55100	Diversity = 5.4460e+00	k_t
= [1, 1, 1, 1]				
Generation 560:	Best = 8.219977e+02	FES = 56100	Diversity = 5.4479e+00	k_t
= [3, 3, 4, 2]				
Generation 570:	Best = 8.212563e+02	FES = 57100	Diversity = 5.4237e+00	k_t
= [1, 1, 1, 1]				
Generation 580:	Best = 8.191627e+02	FES = 58100	Diversity = 5.4004e+00	k_t
= [4, 3, 3, 2]				
Generation 590:	Best = 8.191627e+02	FES = 59100	Diversity = 5.3290e+00	k_t
= [1, 1, 1, 1]				
Generation 600:	Best = 8.187123e+02	FES = 60100	Diversity = 5.2307e+00	k_t
= [2, 3, 2, 4]				
Generation 610:	Best = 8.187123e+02	FES = 61100	Diversity = 5.2327e+00	k_t
= [1, 1, 1, 1]				
Generation 620:	Best = 8.187123e+02	FES = 62100	Diversity = 5.0836e+00	k_t
= [3, 4, 4, 2]				
Generation 630:	Best = 8.133346e+02	FES = 63100	Diversity = 5.0000e+00	k_t
= [1, 1, 1, 1]				
Generation 640:	Best = 8.133346e+02	FES = 64100	Diversity = 4.9917e+00	k_t
= [2, 2, 4, 3]				
Generation 650:	Best = 8.133346e+02	FES = 65100	Diversity = 4.9568e+00	k_t
= [1, 1, 1, 1]				
Generation 660:	Best = 8.116378e+02	FES = 66100	Diversity = 4.9256e+00	k_t
= [4, 3, 4, 3]				
Generation 670:	Best = 8.116378e+02	FES = 67100	Diversity = 4.8701e+00	k_t
= [1, 1, 1, 1]				
Generation 680:	Best = 8.041585e+02	FES = 68100	Diversity = 4.8294e+00	k_t
= [2, 4, 3, 4]				
Generation 690:	Best = 8.004894e+02	FES = 69100	Diversity = 4.8000e+00	k_t
= [1, 1, 1, 1]				
Generation 700:	Best = 7.865436e+02	FES = 70100	Diversity = 4.8171e+00	k_t
= [4, 4, 4, 3]				
Generation 710:	Best = 7.865436e+02	FES = 71100	Diversity = 4.8045e+00	k_t
= [1, 1, 1, 1]				
Generation 720:	Best = 7.857131e+02	FES = 72100	Diversity = 4.7621e+00	k_t
= [4, 4, 2, 3]				
Generation 730:	Best = 7.857131e+02	FES = 73100	Diversity = 4.7695e+00	k_t
= [1, 1, 1, 1]				
Generation 740:	Best = 7.857131e+02	FES = 74100	Diversity = 4.7480e+00	k_t
= [4, 3, 4, 3]				
Generation 750:	Best = 7.857131e+02	FES = 75100	Diversity = 4.7299e+00	k_t
= [1, 1, 1, 1]				
Generation 760:	Best = 7.824553e+02	FES = 76100	Diversity = 4.7492e+00	k_t
= [2, 4, 2, 3]				

Generation 770:	Best = 7.772508e+02	FES = 77100	Diversity = 4.6833e+00	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 7.758395e+02	FES = 78100	Diversity = 4.6323e+00	k_t
= [3, 2, 2, 4]				
Generation 790:	Best = 7.755253e+02	FES = 79100	Diversity = 4.3576e+00	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 7.751362e+02	FES = 80100	Diversity = 4.2106e+00	k_t
= [3, 4, 4, 4]				
Generation 810:	Best = 7.677310e+02	FES = 81100	Diversity = 3.4519e+00	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 7.662138e+02	FES = 82100	Diversity = 1.4468e+00	k_t
= [4, 2, 3, 4]				
Generation 830:	Best = 7.637377e+02	FES = 83100	Diversity = 4.4421e-01	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 7.619572e+02	FES = 84100	Diversity = 3.7359e-01	k_t
= [3, 3, 4, 2]				
Generation 850:	Best = 7.588803e+02	FES = 85100	Diversity = 3.2081e-01	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 7.581151e+02	FES = 86100	Diversity = 2.5854e-01	k_t
= [3, 2, 3, 3]				
Generation 870:	Best = 7.575524e+02	FES = 87100	Diversity = 2.3225e-01	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 7.574616e+02	FES = 88100	Diversity = 2.0540e-01	k_t
= [4, 3, 2, 4]				
Generation 890:	Best = 7.568902e+02	FES = 89100	Diversity = 1.6998e-01	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 7.566693e+02	FES = 90100	Diversity = 1.1233e-01	k_t
= [4, 3, 4, 4]				
Generation 910:	Best = 7.565598e+02	FES = 91100	Diversity = 7.2837e-02	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 7.564397e+02	FES = 92100	Diversity = 2.8744e-02	k_t
= [4, 3, 2, 2]				
Generation 930:	Best = 7.563801e+02	FES = 93100	Diversity = 2.0599e-02	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 7.563250e+02	FES = 94100	Diversity = 1.4555e-02	k_t
= [3, 3, 2, 2]				
Generation 950:	Best = 7.563122e+02	FES = 95100	Diversity = 1.0100e-02	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 7.563005e+02	FES = 96100	Diversity = 7.1206e-03	k_t
= [4, 2, 4, 2]				
Generation 970:	Best = 7.562947e+02	FES = 97100	Diversity = 4.9219e-03	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 7.562921e+02	FES = 98100	Diversity = 3.2775e-03	k_t
= [2, 4, 2, 3]				
Generation 990:	Best = 7.562914e+02	FES = 99100	Diversity = 2.2172e-03	k_t
= [1, 1, 1, 1]				
Generation 1000:	Best = 7.562909e+02	FES = 100100	Diversity = 1.4648e-03	k_t
= [4, 4, 4, 3]				
Generation 1010:	Best = 7.562907e+02	FES = 101100	Diversity = 9.8813e-04	k_t
= [1, 1, 1, 1]				
Generation 1020:	Best = 7.562905e+02	FES = 102100	Diversity = 6.5947e-04	k_t
= [4, 2, 4, 4]				
Generation 1030:	Best = 7.562905e+02	FES = 103100	Diversity = 4.2746e-04	k_t
= [1, 1, 1, 1]				

Generation 1040: Best = 7.562905e+02 = [3, 3, 2, 4]	FES = 104100	Diversity = 2.8132e-04	k_t
Generation 1050: Best = 7.562905e+02 = [1, 1, 1, 1]	FES = 105100	Diversity = 1.7799e-04	k_t
Generation 1060: Best = 7.562905e+02 = [2, 2, 4, 2]	FES = 106100	Diversity = 1.1250e-04	k_t
Generation 1070: Best = 7.562905e+02 = [1, 1, 1, 1]	FES = 107100	Diversity = 7.3467e-05	k_t
Generation 1080: Best = 7.562905e+02 = [4, 4, 4, 3]	FES = 108100	Diversity = 4.8490e-05	k_t
Generation 1090: Best = 7.562905e+02 = [1, 1, 1, 1]	FES = 109100	Diversity = 3.1893e-05	k_t
Generation 1100: Best = 7.562905e+02 = [4, 3, 4, 3]	FES = 110100	Diversity = 2.1266e-05	k_t
Generation 1110: Best = 7.562905e+02 = [1, 1, 1, 1]	FES = 111100	Diversity = 1.4163e-05	k_t
Generation 1120: Best = 7.562905e+02 = [4, 4, 4, 4]	FES = 112100	Diversity = 9.5985e-06	k_t
Generation 1130: Best = 7.562905e+02 = [1, 1, 1, 1]	FES = 113100	Diversity = 6.5718e-06	k_t
Generation 1140: Best = 7.562905e+02 = [4, 4, 3, 2]	FES = 114100	Diversity = 4.3244e-06	k_t
Generation 1150: Best = 7.562905e+02 = [1, 1, 1, 1]	FES = 115100	Diversity = 2.8273e-06	k_t
Generation 1160: Best = 7.562905e+02 = [3, 3, 3, 4]	FES = 116100	Diversity = 1.7913e-06	k_t
Generation 1170: Best = 7.562905e+02 = [1, 1, 1, 1]	FES = 117100	Diversity = 1.2038e-06	k_t
Generation 1180: Best = 7.562905e+02 = [2, 4, 3, 4]	FES = 118100	Diversity = 8.2993e-07	k_t
Generation 1190: Best = 7.562905e+02 = [1, 1, 1, 1]	FES = 119100	Diversity = 5.3045e-07	k_t
Generation 1200: Best = 7.562905e+02 = [3, 4, 4, 3]	FES = 120100	Diversity = 3.6233e-07	k_t
Generation 1210: Best = 7.562905e+02 = [1, 1, 1, 1]	FES = 121100	Diversity = 2.3355e-07	k_t
Generation 1220: Best = 7.562905e+02 = [4, 4, 2, 2]	FES = 122100	Diversity = 1.5515e-07	k_t
Generation 1230: Best = 7.562905e+02 = [1, 1, 1, 1]	FES = 123100	Diversity = 1.0315e-07	k_t
Generation 1240: Best = 7.562905e+02 = [4, 3, 4, 3]	FES = 124100	Diversity = 6.9974e-08	k_t
Generation 1250: Best = 7.562905e+02 = [1, 1, 1, 1]	FES = 125100	Diversity = 4.9265e-08	k_t
Generation 1260: Best = 7.562905e+02 = [3, 3, 3, 4]	FES = 126100	Diversity = 3.5741e-08	k_t
Generation 1270: Best = 7.562905e+02 = [1, 1, 1, 1]	FES = 127100	Diversity = 2.7436e-08	k_t
Generation 1280: Best = 7.562905e+02 = [3, 3, 3, 4]	FES = 128100	Diversity = 2.3055e-08	k_t
Generation 1290: Best = 7.562905e+02 = [1, 1, 1, 1]	FES = 129100	Diversity = 2.1044e-08	k_t
Generation 1300: Best = 7.562905e+02 = [4, 4, 3, 4]	FES = 130100	Diversity = 1.9054e-08	k_t

Generation 1310: Best = 7.562905e+02 = [1, 1, 1, 1]	FES = 131100	Diversity = 1.8234e-08	k_t
Generation 1320: Best = 7.562905e+02 = [4, 4, 2, 4]	FES = 132100	Diversity = 1.7625e-08	k_t
Generation 1330: Best = 7.562905e+02 = [1, 1, 1, 1]	FES = 133100	Diversity = 1.6430e-08	k_t
Generation 1340: Best = 7.562905e+02 = [4, 3, 3, 2]	FES = 134100	Diversity = 1.6033e-08	k_t
Generation 1350: Best = 7.562905e+02 = [1, 1, 1, 1]	FES = 135100	Diversity = 1.6026e-08	k_t
Generation 1360: Best = 7.562905e+02 = [3, 3, 2, 2]	FES = 136100	Diversity = 1.5916e-08	k_t
Generation 1370: Best = 7.562905e+02 = [1, 1, 1, 1]	FES = 137100	Diversity = 1.6138e-08	k_t
Generation 1380: Best = 7.562905e+02 = [2, 3, 3, 2]	FES = 138100	Diversity = 1.6584e-08	k_t
Generation 1390: Best = 7.562905e+02 = [1, 1, 1, 1]	FES = 139100	Diversity = 1.6483e-08	k_t
Generation 1400: Best = 7.562905e+02 = [4, 3, 3, 2]	FES = 140100	Diversity = 1.6911e-08	k_t
Generation 1410: Best = 7.562905e+02 = [1, 1, 1, 1]	FES = 141100	Diversity = 1.7157e-08	k_t
Generation 1420: Best = 7.562905e+02 = [2, 3, 4, 3]	FES = 142100	Diversity = 1.7025e-08	k_t
Generation 1430: Best = 7.562905e+02 = [1, 1, 1, 1]	FES = 143100	Diversity = 1.7355e-08	k_t
Generation 1440: Best = 7.562905e+02 = [2, 3, 3, 3]	FES = 144100	Diversity = 1.7219e-08	k_t
Generation 1450: Best = 7.562905e+02 = [1, 1, 1, 1]	FES = 145100	Diversity = 1.7008e-08	k_t
Generation 1460: Best = 7.562905e+02 = [2, 3, 4, 4]	FES = 146100	Diversity = 1.7364e-08	k_t
Generation 1470: Best = 7.562905e+02 = [1, 1, 1, 1]	FES = 147100	Diversity = 1.7332e-08	k_t
Generation 1480: Best = 7.562905e+02 = [2, 3, 3, 4]	FES = 148100	Diversity = 1.7130e-08	k_t
Generation 1490: Best = 7.562905e+02 = [1, 1, 1, 1]	FES = 149100	Diversity = 1.7231e-08	k_t
Generation 1500: Best = 7.562905e+02 = [4, 4, 4, 2]	FES = 150100	Diversity = 1.6751e-08	k_t
Generation 1510: Best = 7.562905e+02 = [1, 1, 1, 1]	FES = 151100	Diversity = 1.6433e-08	k_t
Generation 1520: Best = 7.562905e+02 = [4, 4, 3, 4]	FES = 152100	Diversity = 1.6503e-08	k_t
Generation 1530: Best = 7.562905e+02 = [1, 1, 1, 1]	FES = 153100	Diversity = 1.6859e-08	k_t
Generation 1540: Best = 7.562905e+02 = [2, 2, 3, 3]	FES = 154100	Diversity = 1.6703e-08	k_t
Generation 1550: Best = 7.562905e+02 = [1, 1, 1, 1]	FES = 155100	Diversity = 1.6767e-08	k_t
Generation 1560: Best = 7.562905e+02 = [3, 2, 4, 4]	FES = 156100	Diversity = 1.6691e-08	k_t
Generation 1570: Best = 7.562905e+02 = [1, 1, 1, 1]	FES = 157100	Diversity = 1.6665e-08	k_t

Generation 1580: Best = 7.562905e+02	FES = 158100	Diversity = 1.6751e-08	k_t
= [4, 4, 3, 3]			
Generation 1590: Best = 7.562905e+02	FES = 159100	Diversity = 1.6880e-08	k_t
= [1, 1, 1, 1]			
Generation 1600: Best = 7.562905e+02	FES = 160100	Diversity = 1.7020e-08	k_t
= [3, 3, 3, 2]			
Generation 1610: Best = 7.562905e+02	FES = 161100	Diversity = 1.7121e-08	k_t
= [1, 1, 1, 1]			
Generation 1620: Best = 7.562905e+02	FES = 162100	Diversity = 1.6762e-08	k_t
= [2, 3, 4, 4]			
Generation 1630: Best = 7.562905e+02	FES = 163100	Diversity = 1.6808e-08	k_t
= [1, 1, 1, 1]			
Generation 1640: Best = 7.562905e+02	FES = 164100	Diversity = 1.6507e-08	k_t
= [3, 3, 3, 4]			
Generation 1650: Best = 7.562905e+02	FES = 165100	Diversity = 1.6726e-08	k_t
= [1, 1, 1, 1]			
Generation 1660: Best = 7.562905e+02	FES = 166100	Diversity = 1.6621e-08	k_t
= [4, 3, 4, 2]			
Generation 1670: Best = 7.562905e+02	FES = 167100	Diversity = 1.6799e-08	k_t
= [1, 1, 1, 1]			
Generation 1680: Best = 7.562905e+02	FES = 168100	Diversity = 1.6691e-08	k_t
= [4, 3, 3, 2]			
Generation 1690: Best = 7.562905e+02	FES = 169100	Diversity = 1.6360e-08	k_t
= [1, 1, 1, 1]			
Generation 1700: Best = 7.562905e+02	FES = 170100	Diversity = 1.6178e-08	k_t
= [3, 4, 2, 2]			
Generation 1710: Best = 7.562905e+02	FES = 171100	Diversity = 1.5918e-08	k_t
= [1, 1, 1, 1]			
Generation 1720: Best = 7.562905e+02	FES = 172100	Diversity = 1.6383e-08	k_t
= [3, 2, 4, 4]			
Generation 1730: Best = 7.562905e+02	FES = 173100	Diversity = 1.6519e-08	k_t
= [1, 1, 1, 1]			
Generation 1740: Best = 7.562905e+02	FES = 174100	Diversity = 1.6374e-08	k_t
= [2, 3, 3, 3]			
Generation 1750: Best = 7.562905e+02	FES = 175100	Diversity = 1.6300e-08	k_t
= [1, 1, 1, 1]			
Generation 1760: Best = 7.562905e+02	FES = 176100	Diversity = 1.6206e-08	k_t
= [4, 4, 4, 2]			
Generation 1770: Best = 7.562905e+02	FES = 177100	Diversity = 1.6062e-08	k_t
= [1, 1, 1, 1]			
Generation 1780: Best = 7.562905e+02	FES = 178100	Diversity = 1.6156e-08	k_t
= [4, 2, 3, 4]			
Generation 1790: Best = 7.562905e+02	FES = 179100	Diversity = 1.6579e-08	k_t
= [1, 1, 1, 1]			
Generation 1800: Best = 7.562905e+02	FES = 180100	Diversity = 1.6123e-08	k_t
= [4, 3, 2, 3]			
Generation 1810: Best = 7.562905e+02	FES = 181100	Diversity = 1.6055e-08	k_t
= [1, 1, 1, 1]			
Generation 1820: Best = 7.562905e+02	FES = 182100	Diversity = 1.5974e-08	k_t
= [2, 3, 4, 3]			
Generation 1830: Best = 7.562905e+02	FES = 183100	Diversity = 1.5717e-08	k_t
= [1, 1, 1, 1]			
Generation 1840: Best = 7.562905e+02	FES = 184100	Diversity = 1.4900e-08	k_t
= [4, 2, 4, 4]			

Generation 1850: Best = 7.562905e+02 = [1, 1, 1, 1]	FES = 185100	Diversity = 1.4631e-08	k_t
Generation 1860: Best = 7.562905e+02 = [2, 3, 2, 4]	FES = 186100	Diversity = 1.4672e-08	k_t
Generation 1870: Best = 7.562905e+02 = [1, 1, 1, 1]	FES = 187100	Diversity = 1.5372e-08	k_t
Generation 1880: Best = 7.562905e+02 = [3, 4, 3, 2]	FES = 188100	Diversity = 1.5524e-08	k_t
Generation 1890: Best = 7.562905e+02 = [1, 1, 1, 1]	FES = 189100	Diversity = 1.5073e-08	k_t
Generation 1900: Best = 7.562905e+02 = [2, 4, 4, 4]	FES = 190100	Diversity = 1.5152e-08	k_t
Generation 1910: Best = 7.562905e+02 = [1, 1, 1, 1]	FES = 191100	Diversity = 1.5576e-08	k_t
Generation 1920: Best = 7.562905e+02 = [4, 4, 3, 4]	FES = 192100	Diversity = 1.5836e-08	k_t
Generation 1930: Best = 7.562905e+02 = [1, 1, 1, 1]	FES = 193100	Diversity = 1.5789e-08	k_t
Generation 1940: Best = 7.562905e+02 = [4, 4, 2, 3]	FES = 194100	Diversity = 1.5406e-08	k_t
Generation 1950: Best = 7.562905e+02 = [1, 1, 1, 1]	FES = 195100	Diversity = 1.5503e-08	k_t
Generation 1960: Best = 7.562905e+02 = [3, 2, 4, 4]	FES = 196100	Diversity = 1.5367e-08	k_t
Generation 1970: Best = 7.562905e+02 = [1, 1, 1, 1]	FES = 197100	Diversity = 1.5166e-08	k_t
Generation 1980: Best = 7.562905e+02 = [4, 4, 2, 2]	FES = 198100	Diversity = 1.5264e-08	k_t
Generation 1990: Best = 7.562905e+02 = [1, 1, 1, 1]	FES = 199100	Diversity = 1.4879e-08	k_t
Generation 2000: Best = 7.562905e+02 = [4, 2, 2, 2]	FES = 200100	Diversity = 1.4994e-08	k_t
Generation 2010: Best = 7.562905e+02 = [1, 1, 1, 1]	FES = 201100	Diversity = 1.5223e-08	k_t
Generation 2020: Best = 7.562905e+02 = [3, 3, 2, 2]	FES = 202100	Diversity = 1.5243e-08	k_t
Generation 2030: Best = 7.562905e+02 = [1, 1, 1, 1]	FES = 203100	Diversity = 1.5134e-08	k_t
Generation 2040: Best = 7.562905e+02 = [2, 4, 3, 4]	FES = 204100	Diversity = 1.4568e-08	k_t
Generation 2050: Best = 7.562905e+02 = [1, 1, 1, 1]	FES = 205100	Diversity = 1.4223e-08	k_t
Generation 2060: Best = 7.562905e+02 = [2, 4, 2, 3]	FES = 206100	Diversity = 1.4242e-08	k_t
Generation 2070: Best = 7.562905e+02 = [1, 1, 1, 1]	FES = 207100	Diversity = 1.4288e-08	k_t
Generation 2080: Best = 7.562905e+02 = [2, 2, 4, 2]	FES = 208100	Diversity = 1.4467e-08	k_t
Generation 2090: Best = 7.562905e+02 = [1, 1, 1, 1]	FES = 209100	Diversity = 1.4149e-08	k_t
Generation 2100: Best = 7.562905e+02 = [4, 2, 2, 3]	FES = 210100	Diversity = 1.4005e-08	k_t
Generation 2110: Best = 7.562905e+02 = [1, 1, 1, 1]	FES = 211100	Diversity = 1.3814e-08	k_t

Generation 2120: Best = 7.562905e+02 = [3, 4, 3, 3]	FES = 212100	Diversity = 1.4096e-08	k_t
Generation 2130: Best = 7.562905e+02 = [1, 1, 1, 1]	FES = 213100	Diversity = 1.4289e-08	k_t
Generation 2140: Best = 7.562905e+02 = [4, 2, 4, 4]	FES = 214100	Diversity = 1.4116e-08	k_t
Generation 2150: Best = 7.562905e+02 = [1, 1, 1, 1]	FES = 215100	Diversity = 1.4037e-08	k_t
Generation 2160: Best = 7.562905e+02 = [3, 3, 4, 4]	FES = 216100	Diversity = 1.3712e-08	k_t
Generation 2170: Best = 7.562905e+02 = [1, 1, 1, 1]	FES = 217100	Diversity = 1.3792e-08	k_t
Generation 2180: Best = 7.562905e+02 = [3, 3, 2, 2]	FES = 218100	Diversity = 1.3343e-08	k_t
Generation 2190: Best = 7.562905e+02 = [1, 1, 1, 1]	FES = 219100	Diversity = 1.3149e-08	k_t
Generation 2200: Best = 7.562905e+02 = [3, 3, 2, 3]	FES = 220100	Diversity = 1.3165e-08	k_t
Generation 2210: Best = 7.562905e+02 = [1, 1, 1, 1]	FES = 221100	Diversity = 1.3040e-08	k_t
Generation 2220: Best = 7.562905e+02 = [4, 3, 3, 2]	FES = 222100	Diversity = 1.2692e-08	k_t
Generation 2230: Best = 7.562905e+02 = [1, 1, 1, 1]	FES = 223100	Diversity = 1.2834e-08	k_t
Generation 2240: Best = 7.562905e+02 = [4, 3, 4, 2]	FES = 224100	Diversity = 1.2731e-08	k_t
Generation 2250: Best = 7.562905e+02 = [1, 1, 1, 1]	FES = 225100	Diversity = 1.2475e-08	k_t
Generation 2260: Best = 7.562905e+02 = [2, 3, 3, 3]	FES = 226100	Diversity = 1.2048e-08	k_t
Generation 2270: Best = 7.562905e+02 = [1, 1, 1, 1]	FES = 227100	Diversity = 1.1803e-08	k_t
Generation 2280: Best = 7.562905e+02 = [4, 3, 4, 4]	FES = 228100	Diversity = 1.2273e-08	k_t
Generation 2290: Best = 7.562905e+02 = [1, 1, 1, 1]	FES = 229100	Diversity = 1.1886e-08	k_t
Generation 2300: Best = 7.562905e+02 = [2, 3, 4, 2]	FES = 230100	Diversity = 1.2037e-08	k_t
Generation 2310: Best = 7.562905e+02 = [1, 1, 1, 1]	FES = 231100	Diversity = 1.2118e-08	k_t
Generation 2320: Best = 7.562905e+02 = [4, 4, 4, 2]	FES = 232100	Diversity = 1.2337e-08	k_t
Generation 2330: Best = 7.562905e+02 = [1, 1, 1, 1]	FES = 233100	Diversity = 1.2059e-08	k_t
Generation 2340: Best = 7.562905e+02 = [3, 2, 3, 3]	FES = 234100	Diversity = 1.2293e-08	k_t
Generation 2350: Best = 7.562905e+02 = [1, 1, 1, 1]	FES = 235100	Diversity = 1.2060e-08	k_t
Generation 2360: Best = 7.562905e+02 = [2, 4, 2, 4]	FES = 236100	Diversity = 1.2150e-08	k_t
Generation 2370: Best = 7.562905e+02 = [1, 1, 1, 1]	FES = 237100	Diversity = 1.1768e-08	k_t
Generation 2380: Best = 7.562905e+02 = [3, 4, 3, 4]	FES = 238100	Diversity = 1.2424e-08	k_t

Generation 2390: Best = 7.562905e+02	FES = 239100	Diversity = 1.2074e-08	k_t
= [1, 1, 1, 1]			
Generation 2400: Best = 7.562905e+02	FES = 240100	Diversity = 1.1785e-08	k_t
= [4, 4, 2, 2]			
Generation 2410: Best = 7.562905e+02	FES = 241100	Diversity = 1.1785e-08	k_t
= [1, 1, 1, 1]			
Generation 2420: Best = 7.562905e+02	FES = 242100	Diversity = 1.1737e-08	k_t
= [4, 4, 3, 3]			
Generation 2430: Best = 7.562905e+02	FES = 243100	Diversity = 1.1632e-08	k_t
= [1, 1, 1, 1]			
Generation 2440: Best = 7.562905e+02	FES = 244100	Diversity = 1.1688e-08	k_t
= [4, 4, 3, 2]			
Generation 2450: Best = 7.562905e+02	FES = 245100	Diversity = 1.1863e-08	k_t
= [1, 1, 1, 1]			
Generation 2460: Best = 7.562905e+02	FES = 246100	Diversity = 1.1554e-08	k_t
= [4, 4, 3, 4]			
Generation 2470: Best = 7.562905e+02	FES = 247100	Diversity = 1.1998e-08	k_t
= [1, 1, 1, 1]			
Generation 2480: Best = 7.562905e+02	FES = 248100	Diversity = 1.2225e-08	k_t
= [4, 3, 3, 4]			
Generation 2490: Best = 7.562905e+02	FES = 249100	Diversity = 1.1832e-08	k_t
= [1, 1, 1, 1]			
Generation 2500: Best = 7.562905e+02	FES = 250100	Diversity = 1.1822e-08	k_t
= [4, 4, 4, 3]			
Generation 2510: Best = 7.562905e+02	FES = 251100	Diversity = 1.2047e-08	k_t
= [1, 1, 1, 1]			
Generation 2520: Best = 7.562905e+02	FES = 252100	Diversity = 1.1762e-08	k_t
= [2, 4, 3, 4]			
Generation 2530: Best = 7.562905e+02	FES = 253100	Diversity = 1.1902e-08	k_t
= [1, 1, 1, 1]			
Generation 2540: Best = 7.562905e+02	FES = 254100	Diversity = 1.2033e-08	k_t
= [2, 2, 3, 2]			
Generation 2550: Best = 7.562905e+02	FES = 255100	Diversity = 1.1975e-08	k_t
= [1, 1, 1, 1]			
Generation 2560: Best = 7.562905e+02	FES = 256100	Diversity = 1.1900e-08	k_t
= [2, 2, 4, 2]			
Generation 2570: Best = 7.562905e+02	FES = 257100	Diversity = 1.2084e-08	k_t
= [1, 1, 1, 1]			
Generation 2580: Best = 7.562905e+02	FES = 258100	Diversity = 1.1965e-08	k_t
= [4, 4, 3, 3]			
Generation 2590: Best = 7.562905e+02	FES = 259100	Diversity = 1.1978e-08	k_t
= [1, 1, 1, 1]			
Generation 2600: Best = 7.562905e+02	FES = 260100	Diversity = 1.1952e-08	k_t
= [2, 2, 3, 2]			
Generation 2610: Best = 7.562905e+02	FES = 261100	Diversity = 1.1969e-08	k_t
= [1, 1, 1, 1]			
Generation 2620: Best = 7.562905e+02	FES = 262100	Diversity = 1.1985e-08	k_t
= [2, 4, 4, 2]			
Generation 2630: Best = 7.562905e+02	FES = 263100	Diversity = 1.1877e-08	k_t
= [1, 1, 1, 1]			
Generation 2640: Best = 7.562905e+02	FES = 264100	Diversity = 1.1988e-08	k_t
= [2, 2, 2, 2]			
Generation 2650: Best = 7.562905e+02	FES = 265100	Diversity = 1.2018e-08	k_t
= [1, 1, 1, 1]			

Generation 2660: Best = 7.562905e+02	FES = 266100	Diversity = 1.2240e-08	k_t
= [3, 4, 3, 2]			
Generation 2670: Best = 7.562905e+02	FES = 267100	Diversity = 1.2281e-08	k_t
= [1, 1, 1, 1]			
Generation 2680: Best = 7.562905e+02	FES = 268100	Diversity = 1.2233e-08	k_t
= [4, 3, 4, 3]			
Generation 2690: Best = 7.562905e+02	FES = 269100	Diversity = 1.2224e-08	k_t
= [1, 1, 1, 1]			
Generation 2700: Best = 7.562905e+02	FES = 270100	Diversity = 1.2247e-08	k_t
= [4, 4, 2, 3]			
Generation 2710: Best = 7.562905e+02	FES = 271100	Diversity = 1.2185e-08	k_t
= [1, 1, 1, 1]			
Generation 2720: Best = 7.562905e+02	FES = 272100	Diversity = 1.2049e-08	k_t
= [4, 3, 4, 4]			
Generation 2730: Best = 7.562905e+02	FES = 273100	Diversity = 1.2280e-08	k_t
= [1, 1, 1, 1]			
Generation 2740: Best = 7.562905e+02	FES = 274100	Diversity = 1.2343e-08	k_t
= [4, 2, 2, 3]			
Generation 2750: Best = 7.562905e+02	FES = 275100	Diversity = 1.2256e-08	k_t
= [1, 1, 1, 1]			
Generation 2760: Best = 7.562905e+02	FES = 276100	Diversity = 1.2277e-08	k_t
= [3, 4, 3, 4]			
Generation 2770: Best = 7.562905e+02	FES = 277100	Diversity = 1.2359e-08	k_t
= [1, 1, 1, 1]			
Generation 2780: Best = 7.562905e+02	FES = 278100	Diversity = 1.2433e-08	k_t
= [3, 4, 4, 2]			
Generation 2790: Best = 7.562905e+02	FES = 279100	Diversity = 1.2519e-08	k_t
= [1, 1, 1, 1]			
Generation 2800: Best = 7.562905e+02	FES = 280100	Diversity = 1.2547e-08	k_t
= [2, 4, 4, 4]			
Generation 2810: Best = 7.562905e+02	FES = 281100	Diversity = 1.2618e-08	k_t
= [1, 1, 1, 1]			
Generation 2820: Best = 7.562905e+02	FES = 282100	Diversity = 1.2636e-08	k_t
= [2, 4, 2, 3]			
Generation 2830: Best = 7.562905e+02	FES = 283100	Diversity = 1.2498e-08	k_t
= [1, 1, 1, 1]			
Generation 2840: Best = 7.562905e+02	FES = 284100	Diversity = 1.2589e-08	k_t
= [3, 3, 3, 4]			
Generation 2850: Best = 7.562905e+02	FES = 285100	Diversity = 1.2568e-08	k_t
= [1, 1, 1, 1]			
Generation 2860: Best = 7.562905e+02	FES = 286100	Diversity = 1.2513e-08	k_t
= [3, 2, 3, 2]			
Generation 2870: Best = 7.562905e+02	FES = 287100	Diversity = 1.2535e-08	k_t
= [1, 1, 1, 1]			
Generation 2880: Best = 7.562905e+02	FES = 288100	Diversity = 1.2611e-08	k_t
= [2, 4, 3, 3]			
Generation 2890: Best = 7.562905e+02	FES = 289100	Diversity = 1.2577e-08	k_t
= [1, 1, 1, 1]			
Generation 2900: Best = 7.562905e+02	FES = 290100	Diversity = 1.2517e-08	k_t
= [4, 2, 4, 3]			
Generation 2910: Best = 7.562905e+02	FES = 291100	Diversity = 1.2532e-08	k_t
= [1, 1, 1, 1]			
Generation 2920: Best = 7.562905e+02	FES = 292100	Diversity = 1.2507e-08	k_t
= [3, 2, 4, 2]			

Generation 2930: Best = 7.562905e+02 FES = 293100 Diversity = 1.2513e-08 k_t
 = [1, 1, 1, 1]
 Generation 2940: Best = 7.562905e+02 FES = 294100 Diversity = 1.2584e-08 k_t
 = [3, 3, 3, 3]
 Generation 2950: Best = 7.562905e+02 FES = 295100 Diversity = 1.2653e-08 k_t
 = [1, 1, 1, 1]
 Generation 2960: Best = 7.562905e+02 FES = 296100 Diversity = 1.2592e-08 k_t
 = [4, 3, 2, 2]
 Generation 2970: Best = 7.562905e+02 FES = 297100 Diversity = 1.2798e-08 k_t
 = [1, 1, 1, 1]
 Generation 2980: Best = 7.562905e+02 FES = 298100 Diversity = 1.2767e-08 k_t
 = [4, 3, 4, 3]
 Generation 2990: Best = 7.562905e+02 FES = 299100 Diversity = 1.2823e-08 k_t
 = [1, 1, 1, 1]

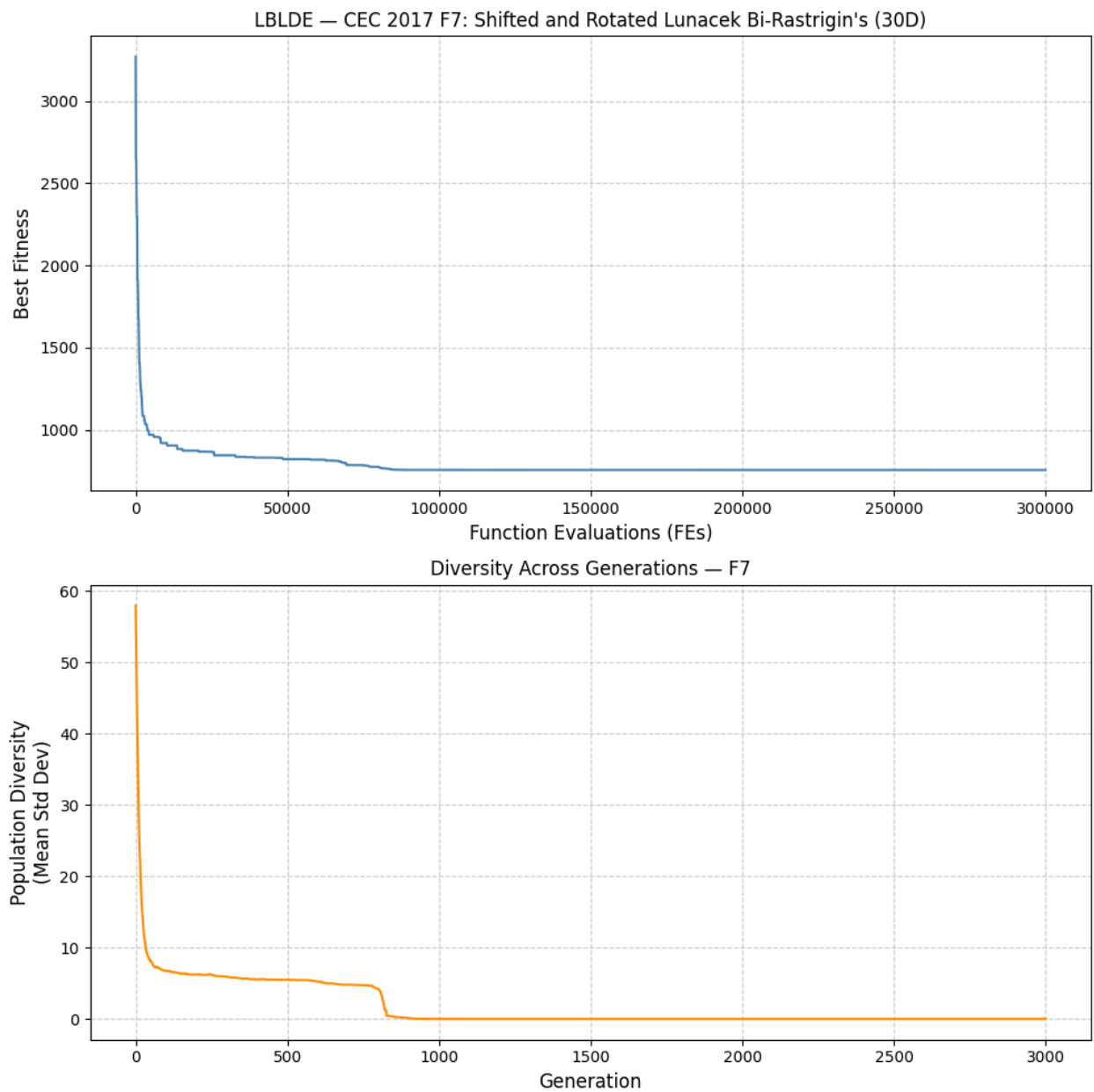
Optimization completed!

Final Best Fitness = 7.562905e+02

Total Function Evaluations = 300000

```

=====
=
Results:
  Best Fitness           : 7.562905e+02
  Global Optimum         : 7.000000e+02
  Error (|best-optimum|) : 5.629047e+01
  Best Solution (5 dim)  : [-15.959025    13.31143041 -30.0533728   -2.15566078
20.09408264]
  D^(0) (init. diversity): 5.7976e+01
=====
=
  
```



Graph saved as convergence_F7.png

```
In [18]: # =====
# F8 — Shifted and Rotated Non-Continuous Rastrigin
# =====
run_and_plot('F8', make_wrapper(simple.f8), 800.0, "Shifted and Rotated Non-Co
```

```

=====
=
LBLDE_s on CEC 2017 F8 – Single Run
=====
=
Function      : Shifted and Rotated Non-Continuous Rastrigin's (30D)
Bounds       : [-100, 100]
Global optimum : 800.0
Initial D^(0) = 5.7976e+01
Generation 0: Best Fitness = 1.442304e+03 | Diversity = 5.7976e+01
Generation 10: Best = 1.157478e+03 FES = 1100 Diversity = 3.6939e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 1.062477e+03 FES = 2100 Diversity = 2.8152e+01 k_t
= [3, 3, 2, 4]
Generation 30: Best = 1.060977e+03 FES = 3100 Diversity = 2.4973e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 1.056226e+03 FES = 4100 Diversity = 2.3176e+01 k_t
= [3, 4, 3, 4]
Generation 50: Best = 1.012444e+03 FES = 5100 Diversity = 2.2711e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 1.011534e+03 FES = 6100 Diversity = 2.2097e+01 k_t
= [2, 3, 4, 3]
Generation 70: Best = 1.000392e+03 FES = 7100 Diversity = 2.1451e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 9.967641e+02 FES = 8100 Diversity = 2.0983e+01 k_t
= [4, 2, 4, 4]
Generation 90: Best = 9.967641e+02 FES = 9100 Diversity = 2.0673e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 9.835677e+02 FES = 10100 Diversity = 2.0571e+01 k_t
= [2, 3, 4, 3]
Generation 110: Best = 9.835677e+02 FES = 11100 Diversity = 2.0534e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 9.820922e+02 FES = 12100 Diversity = 2.0424e+01 k_t
= [2, 3, 3, 2]
Generation 130: Best = 9.772600e+02 FES = 13100 Diversity = 2.0397e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 9.772600e+02 FES = 14100 Diversity = 2.0225e+01 k_t
= [4, 4, 4, 4]
Generation 150: Best = 9.526311e+02 FES = 15100 Diversity = 2.0228e+01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 9.526311e+02 FES = 16100 Diversity = 1.9794e+01 k_t
= [3, 3, 2, 2]
Generation 170: Best = 9.450310e+02 FES = 17100 Diversity = 1.9725e+01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 9.446418e+02 FES = 18100 Diversity = 1.9319e+01 k_t
= [4, 4, 2, 3]
Generation 190: Best = 9.446418e+02 FES = 19100 Diversity = 1.9208e+01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 9.238566e+02 FES = 20100 Diversity = 1.9346e+01 k_t
= [3, 4, 4, 3]
Generation 210: Best = 9.238566e+02 FES = 21100 Diversity = 1.9369e+01 k_t
= [1, 1, 1, 1]
Generation 220: Best = 9.238566e+02 FES = 22100 Diversity = 1.9263e+01 k_t
= [3, 4, 3, 4]

```

Generation 230:	Best = 9.238566e+02	FES = 23100	Diversity = 1.9318e+01	k_t = [1, 1, 1, 1]
Generation 240:	Best = 9.193548e+02	FES = 24100	Diversity = 1.9201e+01	k_t = [4, 3, 2, 3]
Generation 250:	Best = 9.193548e+02	FES = 25100	Diversity = 1.9166e+01	k_t = [1, 1, 1, 1]
Generation 260:	Best = 9.193548e+02	FES = 26100	Diversity = 1.9172e+01	k_t = [3, 3, 3, 3]
Generation 270:	Best = 9.193548e+02	FES = 27100	Diversity = 1.9118e+01	k_t = [1, 1, 1, 1]
Generation 280:	Best = 9.193548e+02	FES = 28100	Diversity = 1.9110e+01	k_t = [4, 3, 4, 2]
Generation 290:	Best = 9.120759e+02	FES = 29100	Diversity = 1.8842e+01	k_t = [1, 1, 1, 1]
Generation 300:	Best = 9.120759e+02	FES = 30100	Diversity = 1.8831e+01	k_t = [3, 2, 2, 2]
Generation 310:	Best = 9.120759e+02	FES = 31100	Diversity = 1.8789e+01	k_t = [1, 1, 1, 1]
Generation 320:	Best = 9.120759e+02	FES = 32100	Diversity = 1.8754e+01	k_t = [3, 3, 4, 2]
Generation 330:	Best = 9.108680e+02	FES = 33100	Diversity = 1.8724e+01	k_t = [1, 1, 1, 1]
Generation 340:	Best = 9.101006e+02	FES = 34100	Diversity = 1.8792e+01	k_t = [3, 4, 2, 2]
Generation 350:	Best = 9.101006e+02	FES = 35100	Diversity = 1.8807e+01	k_t = [1, 1, 1, 1]
Generation 360:	Best = 9.071282e+02	FES = 36100	Diversity = 1.8759e+01	k_t = [3, 2, 4, 2]
Generation 370:	Best = 9.071282e+02	FES = 37100	Diversity = 1.8714e+01	k_t = [1, 1, 1, 1]
Generation 380:	Best = 9.037795e+02	FES = 38100	Diversity = 1.8548e+01	k_t = [3, 4, 2, 4]
Generation 390:	Best = 9.037795e+02	FES = 39100	Diversity = 1.8459e+01	k_t = [1, 1, 1, 1]
Generation 400:	Best = 9.037795e+02	FES = 40100	Diversity = 1.8494e+01	k_t = [3, 4, 2, 4]
Generation 410:	Best = 9.034380e+02	FES = 41100	Diversity = 1.8497e+01	k_t = [1, 1, 1, 1]
Generation 420:	Best = 9.020308e+02	FES = 42100	Diversity = 1.8464e+01	k_t = [4, 2, 2, 3]
Generation 430:	Best = 9.020308e+02	FES = 43100	Diversity = 1.8249e+01	k_t = [1, 1, 1, 1]
Generation 440:	Best = 8.956538e+02	FES = 44100	Diversity = 1.8234e+01	k_t = [3, 4, 3, 4]
Generation 450:	Best = 8.922299e+02	FES = 45100	Diversity = 1.8198e+01	k_t = [1, 1, 1, 1]
Generation 460:	Best = 8.922299e+02	FES = 46100	Diversity = 1.8185e+01	k_t = [4, 3, 4, 3]
Generation 470:	Best = 8.922038e+02	FES = 47100	Diversity = 1.8123e+01	k_t = [1, 1, 1, 1]
Generation 480:	Best = 8.922038e+02	FES = 48100	Diversity = 1.8168e+01	k_t = [3, 4, 2, 4]
Generation 490:	Best = 8.918212e+02	FES = 49100	Diversity = 1.8027e+01	k_t = [1, 1, 1, 1]

Generation 500:	Best = 8.918212e+02	FES = 50100	Diversity = 1.7924e+01	k_t = [2, 3, 4, 2]
Generation 510:	Best = 8.918212e+02	FES = 51100	Diversity = 1.7842e+01	k_t = [1, 1, 1, 1]
Generation 520:	Best = 8.918212e+02	FES = 52100	Diversity = 1.7826e+01	k_t = [3, 2, 4, 3]
Generation 530:	Best = 8.918212e+02	FES = 53100	Diversity = 1.7885e+01	k_t = [1, 1, 1, 1]
Generation 540:	Best = 8.887003e+02	FES = 54100	Diversity = 1.7855e+01	k_t = [2, 3, 3, 3]
Generation 550:	Best = 8.887003e+02	FES = 55100	Diversity = 1.7831e+01	k_t = [1, 1, 1, 1]
Generation 560:	Best = 8.887003e+02	FES = 56100	Diversity = 1.7853e+01	k_t = [4, 4, 4, 2]
Generation 570:	Best = 8.887003e+02	FES = 57100	Diversity = 1.7812e+01	k_t = [1, 1, 1, 1]
Generation 580:	Best = 8.865989e+02	FES = 58100	Diversity = 1.7879e+01	k_t = [2, 4, 2, 4]
Generation 590:	Best = 8.865989e+02	FES = 59100	Diversity = 1.7707e+01	k_t = [1, 1, 1, 1]
Generation 600:	Best = 8.849046e+02	FES = 60100	Diversity = 1.7413e+01	k_t = [3, 3, 4, 4]
Generation 610:	Best = 8.849046e+02	FES = 61100	Diversity = 1.7221e+01	k_t = [1, 1, 1, 1]
Generation 620:	Best = 8.849046e+02	FES = 62100	Diversity = 1.7140e+01	k_t = [2, 3, 4, 4]
Generation 630:	Best = 8.849046e+02	FES = 63100	Diversity = 1.7069e+01	k_t = [1, 1, 1, 1]
Generation 640:	Best = 8.848278e+02	FES = 64100	Diversity = 1.6965e+01	k_t = [3, 2, 2, 2]
Generation 650:	Best = 8.848278e+02	FES = 65100	Diversity = 1.6848e+01	k_t = [1, 1, 1, 1]
Generation 660:	Best = 8.848278e+02	FES = 66100	Diversity = 1.6835e+01	k_t = [2, 4, 2, 3]
Generation 670:	Best = 8.848278e+02	FES = 67100	Diversity = 1.6719e+01	k_t = [1, 1, 1, 1]
Generation 680:	Best = 8.848278e+02	FES = 68100	Diversity = 1.6705e+01	k_t = [4, 3, 4, 2]
Generation 690:	Best = 8.847950e+02	FES = 69100	Diversity = 1.6665e+01	k_t = [1, 1, 1, 1]
Generation 700:	Best = 8.834060e+02	FES = 70100	Diversity = 1.6560e+01	k_t = [4, 3, 3, 4]
Generation 710:	Best = 8.695858e+02	FES = 71100	Diversity = 1.6500e+01	k_t = [1, 1, 1, 1]
Generation 720:	Best = 8.695858e+02	FES = 72100	Diversity = 1.6566e+01	k_t = [2, 3, 2, 3]
Generation 730:	Best = 8.695858e+02	FES = 73100	Diversity = 1.6463e+01	k_t = [1, 1, 1, 1]
Generation 740:	Best = 8.695858e+02	FES = 74100	Diversity = 1.6428e+01	k_t = [3, 2, 4, 2]
Generation 750:	Best = 8.672828e+02	FES = 75100	Diversity = 1.6535e+01	k_t = [1, 1, 1, 1]
Generation 760:	Best = 8.672828e+02	FES = 76100	Diversity = 1.6429e+01	k_t = [3, 3, 3, 4]

Generation 770:	Best = 8.672828e+02	FES = 77100	Diversity = 1.6423e+01	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 8.672828e+02	FES = 78100	Diversity = 1.6394e+01	k_t
= [3, 3, 2, 3]				
Generation 790:	Best = 8.672828e+02	FES = 79100	Diversity = 1.6365e+01	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 8.640350e+02	FES = 80100	Diversity = 1.6365e+01	k_t
= [4, 3, 4, 2]				
Generation 810:	Best = 8.640350e+02	FES = 81100	Diversity = 1.6113e+01	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 8.640350e+02	FES = 82100	Diversity = 1.6104e+01	k_t
= [4, 2, 4, 3]				
Generation 830:	Best = 8.635478e+02	FES = 83100	Diversity = 1.6109e+01	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 8.630308e+02	FES = 84100	Diversity = 1.5930e+01	k_t
= [3, 4, 3, 4]				
Generation 850:	Best = 8.630308e+02	FES = 85100	Diversity = 1.5895e+01	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 8.630308e+02	FES = 86100	Diversity = 1.5788e+01	k_t
= [2, 2, 4, 2]				
Generation 870:	Best = 8.630308e+02	FES = 87100	Diversity = 1.5743e+01	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 8.630308e+02	FES = 88100	Diversity = 1.5741e+01	k_t
= [4, 3, 4, 2]				
Generation 890:	Best = 8.566375e+02	FES = 89100	Diversity = 1.5702e+01	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 8.566375e+02	FES = 90100	Diversity = 1.5663e+01	k_t
= [3, 2, 2, 2]				
Generation 910:	Best = 8.566375e+02	FES = 91100	Diversity = 1.5637e+01	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 8.566375e+02	FES = 92100	Diversity = 1.5607e+01	k_t
= [4, 3, 4, 2]				
Generation 930:	Best = 8.566375e+02	FES = 93100	Diversity = 1.5591e+01	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 8.566375e+02	FES = 94100	Diversity = 1.5591e+01	k_t
= [3, 3, 2, 4]				
Generation 950:	Best = 8.549253e+02	FES = 95100	Diversity = 1.5586e+01	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 8.549253e+02	FES = 96100	Diversity = 1.5590e+01	k_t
= [4, 3, 2, 2]				
Generation 970:	Best = 8.549253e+02	FES = 97100	Diversity = 1.5535e+01	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 8.549253e+02	FES = 98100	Diversity = 1.5474e+01	k_t
= [2, 4, 3, 2]				
Generation 990:	Best = 8.546399e+02	FES = 99100	Diversity = 1.5469e+01	k_t
= [1, 1, 1, 1]				
Generation 1000:	Best = 8.543117e+02	FES = 100100	Diversity = 1.5470e+01	k_t
= [3, 4, 3, 3]				
Generation 1010:	Best = 8.537703e+02	FES = 101100	Diversity = 1.5468e+01	k_t
= [1, 1, 1, 1]				
Generation 1020:	Best = 8.536642e+02	FES = 102100	Diversity = 1.5435e+01	k_t
= [4, 4, 2, 2]				
Generation 1030:	Best = 8.513148e+02	FES = 103100	Diversity = 1.5403e+01	k_t
= [1, 1, 1, 1]				

Generation 1040: Best = 8.513148e+02	FES = 104100	Diversity = 1.5205e+01	k_t
= [2, 2, 3, 4]			
Generation 1050: Best = 8.513148e+02	FES = 105100	Diversity = 1.5160e+01	k_t
= [1, 1, 1, 1]			
Generation 1060: Best = 8.513148e+02	FES = 106100	Diversity = 1.5159e+01	k_t
= [2, 4, 3, 2]			
Generation 1070: Best = 8.492340e+02	FES = 107100	Diversity = 1.5059e+01	k_t
= [1, 1, 1, 1]			
Generation 1080: Best = 8.492340e+02	FES = 108100	Diversity = 1.5066e+01	k_t
= [4, 3, 4, 2]			
Generation 1090: Best = 8.455863e+02	FES = 109100	Diversity = 1.5067e+01	k_t
= [1, 1, 1, 1]			
Generation 1100: Best = 8.455863e+02	FES = 110100	Diversity = 1.4979e+01	k_t
= [3, 4, 4, 4]			
Generation 1110: Best = 8.455863e+02	FES = 111100	Diversity = 1.4813e+01	k_t
= [1, 1, 1, 1]			
Generation 1120: Best = 8.455863e+02	FES = 112100	Diversity = 1.4813e+01	k_t
= [3, 4, 4, 3]			
Generation 1130: Best = 8.455863e+02	FES = 113100	Diversity = 1.4813e+01	k_t
= [1, 1, 1, 1]			
Generation 1140: Best = 8.455863e+02	FES = 114100	Diversity = 1.4658e+01	k_t
= [2, 2, 3, 4]			
Generation 1150: Best = 8.455863e+02	FES = 115100	Diversity = 1.4658e+01	k_t
= [1, 1, 1, 1]			
Generation 1160: Best = 8.448323e+02	FES = 116100	Diversity = 1.4561e+01	k_t
= [4, 3, 3, 3]			
Generation 1170: Best = 8.442424e+02	FES = 117100	Diversity = 1.4578e+01	k_t
= [1, 1, 1, 1]			
Generation 1180: Best = 8.442424e+02	FES = 118100	Diversity = 1.4620e+01	k_t
= [3, 4, 3, 2]			
Generation 1190: Best = 8.442349e+02	FES = 119100	Diversity = 1.4579e+01	k_t
= [1, 1, 1, 1]			
Generation 1200: Best = 8.442349e+02	FES = 120100	Diversity = 1.4396e+01	k_t
= [2, 2, 4, 4]			
Generation 1210: Best = 8.441186e+02	FES = 121100	Diversity = 1.4193e+01	k_t
= [1, 1, 1, 1]			
Generation 1220: Best = 8.372642e+02	FES = 122100	Diversity = 1.3970e+01	k_t
= [4, 4, 4, 2]			
Generation 1230: Best = 8.371778e+02	FES = 123100	Diversity = 1.3795e+01	k_t
= [1, 1, 1, 1]			
Generation 1240: Best = 8.367787e+02	FES = 124100	Diversity = 1.3670e+01	k_t
= [2, 2, 4, 2]			
Generation 1250: Best = 8.357151e+02	FES = 125100	Diversity = 1.3305e+01	k_t
= [1, 1, 1, 1]			
Generation 1260: Best = 8.357151e+02	FES = 126100	Diversity = 1.2779e+01	k_t
= [2, 3, 3, 3]			
Generation 1270: Best = 8.355767e+02	FES = 127100	Diversity = 1.1972e+01	k_t
= [1, 1, 1, 1]			
Generation 1280: Best = 8.352852e+02	FES = 128100	Diversity = 1.1686e+01	k_t
= [3, 2, 4, 3]			
Generation 1290: Best = 8.334816e+02	FES = 129100	Diversity = 1.0325e+01	k_t
= [1, 1, 1, 1]			
Generation 1300: Best = 8.316897e+02	FES = 130100	Diversity = 8.6550e+00	k_t
= [4, 4, 3, 2]			

Generation 1310: Best = 8.303761e+02	FES = 131100	Diversity = 4.7242e+00	k_t
= [1, 1, 1, 1]			
Generation 1320: Best = 8.302694e+02	FES = 132100	Diversity = 2.0036e+00	k_t
= [4, 3, 3, 3]			
Generation 1330: Best = 8.297177e+02	FES = 133100	Diversity = 7.4939e-01	k_t
= [1, 1, 1, 1]			
Generation 1340: Best = 8.288280e+02	FES = 134100	Diversity = 4.7217e-01	k_t
= [2, 3, 4, 4]			
Generation 1350: Best = 8.283921e+02	FES = 135100	Diversity = 3.1635e-01	k_t
= [1, 1, 1, 1]			
Generation 1360: Best = 8.282114e+02	FES = 136100	Diversity = 2.6037e-01	k_t
= [3, 4, 4, 3]			
Generation 1370: Best = 8.279638e+02	FES = 137100	Diversity = 1.2298e-01	k_t
= [1, 1, 1, 1]			
Generation 1380: Best = 8.279345e+02	FES = 138100	Diversity = 8.5493e-02	k_t
= [2, 2, 4, 4]			
Generation 1390: Best = 8.278941e+02	FES = 139100	Diversity = 5.7551e-02	k_t
= [1, 1, 1, 1]			
Generation 1400: Best = 8.278746e+02	FES = 140100	Diversity = 3.9318e-02	k_t
= [4, 4, 3, 3]			
Generation 1410: Best = 8.278667e+02	FES = 141100	Diversity = 2.5083e-02	k_t
= [1, 1, 1, 1]			
Generation 1420: Best = 8.278618e+02	FES = 142100	Diversity = 1.6951e-02	k_t
= [4, 2, 2, 3]			
Generation 1430: Best = 8.278605e+02	FES = 143100	Diversity = 1.1286e-02	k_t
= [1, 1, 1, 1]			
Generation 1440: Best = 8.278594e+02	FES = 144100	Diversity = 7.5865e-03	k_t
= [2, 4, 3, 2]			
Generation 1450: Best = 8.278591e+02	FES = 145100	Diversity = 4.9184e-03	k_t
= [1, 1, 1, 1]			
Generation 1460: Best = 8.278589e+02	FES = 146100	Diversity = 3.2242e-03	k_t
= [3, 4, 3, 4]			
Generation 1470: Best = 8.278589e+02	FES = 147100	Diversity = 2.1011e-03	k_t
= [1, 1, 1, 1]			
Generation 1480: Best = 8.278589e+02	FES = 148100	Diversity = 1.3363e-03	k_t
= [2, 4, 2, 2]			
Generation 1490: Best = 8.278588e+02	FES = 149100	Diversity = 8.0754e-04	k_t
= [1, 1, 1, 1]			
Generation 1500: Best = 8.278588e+02	FES = 150100	Diversity = 5.1117e-04	k_t
= [2, 4, 2, 3]			
Generation 1510: Best = 8.278588e+02	FES = 151100	Diversity = 3.0801e-04	k_t
= [1, 1, 1, 1]			
Generation 1520: Best = 8.278588e+02	FES = 152100	Diversity = 1.9968e-04	k_t
= [3, 2, 3, 3]			
Generation 1530: Best = 8.278588e+02	FES = 153100	Diversity = 1.1704e-04	k_t
= [1, 1, 1, 1]			
Generation 1540: Best = 8.278588e+02	FES = 154100	Diversity = 7.7298e-05	k_t
= [4, 2, 2, 4]			
Generation 1550: Best = 8.278588e+02	FES = 155100	Diversity = 5.1502e-05	k_t
= [1, 1, 1, 1]			
Generation 1560: Best = 8.278588e+02	FES = 156100	Diversity = 3.5667e-05	k_t
= [3, 3, 2, 4]			
Generation 1570: Best = 8.278588e+02	FES = 157100	Diversity = 2.3132e-05	k_t
= [1, 1, 1, 1]			

Generation 1580: Best = 8.278588e+02	FES = 158100	Diversity = 1.4584e-05	k_t
= [4, 4, 2, 3]			
Generation 1590: Best = 8.278588e+02	FES = 159100	Diversity = 9.3989e-06	k_t
= [1, 1, 1, 1]			
Generation 1600: Best = 8.278588e+02	FES = 160100	Diversity = 6.1839e-06	k_t
= [2, 3, 2, 2]			
Generation 1610: Best = 8.278588e+02	FES = 161100	Diversity = 3.7468e-06	k_t
= [1, 1, 1, 1]			
Generation 1620: Best = 8.278588e+02	FES = 162100	Diversity = 2.3863e-06	k_t
= [3, 2, 4, 3]			
Generation 1630: Best = 8.278588e+02	FES = 163100	Diversity = 1.4389e-06	k_t
= [1, 1, 1, 1]			
Generation 1640: Best = 8.278588e+02	FES = 164100	Diversity = 9.0982e-07	k_t
= [4, 3, 4, 3]			
Generation 1650: Best = 8.278588e+02	FES = 165100	Diversity = 5.7672e-07	k_t
= [1, 1, 1, 1]			
Generation 1660: Best = 8.278588e+02	FES = 166100	Diversity = 3.4793e-07	k_t
= [3, 2, 2, 3]			
Generation 1670: Best = 8.278588e+02	FES = 167100	Diversity = 2.4684e-07	k_t
= [1, 1, 1, 1]			
Generation 1680: Best = 8.278588e+02	FES = 168100	Diversity = 1.7143e-07	k_t
= [3, 2, 4, 4]			
Generation 1690: Best = 8.278588e+02	FES = 169100	Diversity = 1.2708e-07	k_t
= [1, 1, 1, 1]			
Generation 1700: Best = 8.278588e+02	FES = 170100	Diversity = 1.0027e-07	k_t
= [4, 3, 4, 4]			
Generation 1710: Best = 8.278588e+02	FES = 171100	Diversity = 9.1518e-08	k_t
= [1, 1, 1, 1]			
Generation 1720: Best = 8.278588e+02	FES = 172100	Diversity = 8.3380e-08	k_t
= [4, 3, 3, 4]			
Generation 1730: Best = 8.278588e+02	FES = 173100	Diversity = 7.3532e-08	k_t
= [1, 1, 1, 1]			
Generation 1740: Best = 8.278588e+02	FES = 174100	Diversity = 7.0674e-08	k_t
= [3, 4, 4, 3]			
Generation 1750: Best = 8.278588e+02	FES = 175100	Diversity = 6.9053e-08	k_t
= [1, 1, 1, 1]			
Generation 1760: Best = 8.278588e+02	FES = 176100	Diversity = 7.0712e-08	k_t
= [2, 2, 2, 4]			
Generation 1770: Best = 8.278588e+02	FES = 177100	Diversity = 7.0227e-08	k_t
= [1, 1, 1, 1]			
Generation 1780: Best = 8.278588e+02	FES = 178100	Diversity = 6.9532e-08	k_t
= [3, 2, 3, 2]			
Generation 1790: Best = 8.278588e+02	FES = 179100	Diversity = 7.0230e-08	k_t
= [1, 1, 1, 1]			
Generation 1800: Best = 8.278588e+02	FES = 180100	Diversity = 7.0306e-08	k_t
= [4, 3, 3, 2]			
Generation 1810: Best = 8.278588e+02	FES = 181100	Diversity = 7.0088e-08	k_t
= [1, 1, 1, 1]			
Generation 1820: Best = 8.278588e+02	FES = 182100	Diversity = 7.2812e-08	k_t
= [3, 3, 2, 4]			
Generation 1830: Best = 8.278588e+02	FES = 183100	Diversity = 7.4209e-08	k_t
= [1, 1, 1, 1]			
Generation 1840: Best = 8.278588e+02	FES = 184100	Diversity = 7.3773e-08	k_t
= [4, 2, 3, 4]			

Generation 1850: Best = 8.278588e+02 = [1, 1, 1, 1]	FES = 185100	Diversity = 7.0696e-08	k_t
Generation 1860: Best = 8.278588e+02 = [3, 3, 2, 4]	FES = 186100	Diversity = 6.7997e-08	k_t
Generation 1870: Best = 8.278588e+02 = [1, 1, 1, 1]	FES = 187100	Diversity = 5.8915e-08	k_t
Generation 1880: Best = 8.278588e+02 = [2, 2, 4, 4]	FES = 188100	Diversity = 5.4067e-08	k_t
Generation 1890: Best = 8.278588e+02 = [1, 1, 1, 1]	FES = 189100	Diversity = 4.7131e-08	k_t
Generation 1900: Best = 8.278588e+02 = [2, 2, 4, 3]	FES = 190100	Diversity = 4.5728e-08	k_t
Generation 1910: Best = 8.278588e+02 = [1, 1, 1, 1]	FES = 191100	Diversity = 4.6401e-08	k_t
Generation 1920: Best = 8.278588e+02 = [4, 3, 2, 2]	FES = 192100	Diversity = 4.6001e-08	k_t
Generation 1930: Best = 8.278588e+02 = [1, 1, 1, 1]	FES = 193100	Diversity = 4.4507e-08	k_t
Generation 1940: Best = 8.278588e+02 = [3, 4, 2, 4]	FES = 194100	Diversity = 4.4879e-08	k_t
Generation 1950: Best = 8.278588e+02 = [1, 1, 1, 1]	FES = 195100	Diversity = 4.4565e-08	k_t
Generation 1960: Best = 8.278588e+02 = [2, 3, 2, 2]	FES = 196100	Diversity = 4.4074e-08	k_t
Generation 1970: Best = 8.278588e+02 = [1, 1, 1, 1]	FES = 197100	Diversity = 4.4738e-08	k_t
Generation 1980: Best = 8.278588e+02 = [4, 2, 4, 2]	FES = 198100	Diversity = 4.5780e-08	k_t
Generation 1990: Best = 8.278588e+02 = [1, 1, 1, 1]	FES = 199100	Diversity = 4.6150e-08	k_t
Generation 2000: Best = 8.278588e+02 = [3, 4, 2, 4]	FES = 200100	Diversity = 4.6229e-08	k_t
Generation 2010: Best = 8.278588e+02 = [1, 1, 1, 1]	FES = 201100	Diversity = 4.7687e-08	k_t
Generation 2020: Best = 8.278588e+02 = [3, 3, 3, 2]	FES = 202100	Diversity = 4.6630e-08	k_t
Generation 2030: Best = 8.278588e+02 = [1, 1, 1, 1]	FES = 203100	Diversity = 4.7454e-08	k_t
Generation 2040: Best = 8.278588e+02 = [2, 4, 4, 4]	FES = 204100	Diversity = 4.5985e-08	k_t
Generation 2050: Best = 8.278588e+02 = [1, 1, 1, 1]	FES = 205100	Diversity = 4.4812e-08	k_t
Generation 2060: Best = 8.278588e+02 = [4, 3, 2, 3]	FES = 206100	Diversity = 4.4801e-08	k_t
Generation 2070: Best = 8.278588e+02 = [1, 1, 1, 1]	FES = 207100	Diversity = 4.4984e-08	k_t
Generation 2080: Best = 8.278588e+02 = [2, 3, 2, 3]	FES = 208100	Diversity = 4.4413e-08	k_t
Generation 2090: Best = 8.278588e+02 = [1, 1, 1, 1]	FES = 209100	Diversity = 4.3090e-08	k_t
Generation 2100: Best = 8.278588e+02 = [3, 2, 2, 4]	FES = 210100	Diversity = 4.1913e-08	k_t
Generation 2110: Best = 8.278588e+02 = [1, 1, 1, 1]	FES = 211100	Diversity = 4.2260e-08	k_t

Generation 2120: Best = 8.278588e+02 = [2, 4, 2, 2]	FES = 212100	Diversity = 4.3194e-08	k_t
Generation 2130: Best = 8.278588e+02 = [1, 1, 1, 1]	FES = 213100	Diversity = 4.3665e-08	k_t
Generation 2140: Best = 8.278588e+02 = [4, 4, 2, 2]	FES = 214100	Diversity = 4.4192e-08	k_t
Generation 2150: Best = 8.278588e+02 = [1, 1, 1, 1]	FES = 215100	Diversity = 4.4180e-08	k_t
Generation 2160: Best = 8.278588e+02 = [2, 2, 2, 2]	FES = 216100	Diversity = 4.5025e-08	k_t
Generation 2170: Best = 8.278588e+02 = [1, 1, 1, 1]	FES = 217100	Diversity = 4.5212e-08	k_t
Generation 2180: Best = 8.278588e+02 = [2, 4, 2, 2]	FES = 218100	Diversity = 4.3956e-08	k_t
Generation 2190: Best = 8.278588e+02 = [1, 1, 1, 1]	FES = 219100	Diversity = 4.3869e-08	k_t
Generation 2200: Best = 8.278588e+02 = [3, 4, 4, 2]	FES = 220100	Diversity = 4.4152e-08	k_t
Generation 2210: Best = 8.278588e+02 = [1, 1, 1, 1]	FES = 221100	Diversity = 4.3458e-08	k_t
Generation 2220: Best = 8.278588e+02 = [2, 4, 4, 4]	FES = 222100	Diversity = 4.2415e-08	k_t
Generation 2230: Best = 8.278588e+02 = [1, 1, 1, 1]	FES = 223100	Diversity = 4.3106e-08	k_t
Generation 2240: Best = 8.278588e+02 = [3, 4, 2, 3]	FES = 224100	Diversity = 4.2086e-08	k_t
Generation 2250: Best = 8.278588e+02 = [1, 1, 1, 1]	FES = 225100	Diversity = 4.3014e-08	k_t
Generation 2260: Best = 8.278588e+02 = [4, 3, 3, 2]	FES = 226100	Diversity = 4.2874e-08	k_t
Generation 2270: Best = 8.278588e+02 = [1, 1, 1, 1]	FES = 227100	Diversity = 4.2275e-08	k_t
Generation 2280: Best = 8.278588e+02 = [3, 2, 2, 3]	FES = 228100	Diversity = 4.3091e-08	k_t
Generation 2290: Best = 8.278588e+02 = [1, 1, 1, 1]	FES = 229100	Diversity = 4.3114e-08	k_t
Generation 2300: Best = 8.278588e+02 = [4, 2, 2, 3]	FES = 230100	Diversity = 4.3891e-08	k_t
Generation 2310: Best = 8.278588e+02 = [1, 1, 1, 1]	FES = 231100	Diversity = 4.3680e-08	k_t
Generation 2320: Best = 8.278588e+02 = [3, 3, 3, 3]	FES = 232100	Diversity = 4.4375e-08	k_t
Generation 2330: Best = 8.278588e+02 = [1, 1, 1, 1]	FES = 233100	Diversity = 4.4541e-08	k_t
Generation 2340: Best = 8.278588e+02 = [2, 2, 4, 2]	FES = 234100	Diversity = 4.5449e-08	k_t
Generation 2350: Best = 8.278588e+02 = [1, 1, 1, 1]	FES = 235100	Diversity = 4.4501e-08	k_t
Generation 2360: Best = 8.278588e+02 = [4, 4, 4, 4]	FES = 236100	Diversity = 4.4885e-08	k_t
Generation 2370: Best = 8.278588e+02 = [1, 1, 1, 1]	FES = 237100	Diversity = 4.3870e-08	k_t
Generation 2380: Best = 8.278588e+02 = [2, 2, 3, 2]	FES = 238100	Diversity = 4.4180e-08	k_t

Generation 2390: Best = 8.278588e+02	FES = 239100	Diversity = 4.3651e-08	k_t
= [1, 1, 1, 1]			
Generation 2400: Best = 8.278588e+02	FES = 240100	Diversity = 4.4716e-08	k_t
= [4, 3, 2, 3]			
Generation 2410: Best = 8.278588e+02	FES = 241100	Diversity = 4.5142e-08	k_t
= [1, 1, 1, 1]			
Generation 2420: Best = 8.278588e+02	FES = 242100	Diversity = 4.6143e-08	k_t
= [4, 2, 3, 2]			
Generation 2430: Best = 8.278588e+02	FES = 243100	Diversity = 4.5576e-08	k_t
= [1, 1, 1, 1]			
Generation 2440: Best = 8.278588e+02	FES = 244100	Diversity = 4.5820e-08	k_t
= [3, 4, 2, 4]			
Generation 2450: Best = 8.278588e+02	FES = 245100	Diversity = 4.4485e-08	k_t
= [1, 1, 1, 1]			
Generation 2460: Best = 8.278588e+02	FES = 246100	Diversity = 4.5915e-08	k_t
= [3, 4, 2, 4]			
Generation 2470: Best = 8.278588e+02	FES = 247100	Diversity = 4.5049e-08	k_t
= [1, 1, 1, 1]			
Generation 2480: Best = 8.278588e+02	FES = 248100	Diversity = 4.4977e-08	k_t
= [4, 2, 4, 3]			
Generation 2490: Best = 8.278588e+02	FES = 249100	Diversity = 4.3279e-08	k_t
= [1, 1, 1, 1]			
Generation 2500: Best = 8.278588e+02	FES = 250100	Diversity = 4.4659e-08	k_t
= [4, 4, 2, 3]			
Generation 2510: Best = 8.278588e+02	FES = 251100	Diversity = 4.4205e-08	k_t
= [1, 1, 1, 1]			
Generation 2520: Best = 8.278588e+02	FES = 252100	Diversity = 4.5033e-08	k_t
= [4, 4, 4, 3]			
Generation 2530: Best = 8.278588e+02	FES = 253100	Diversity = 4.5034e-08	k_t
= [1, 1, 1, 1]			
Generation 2540: Best = 8.278588e+02	FES = 254100	Diversity = 4.5317e-08	k_t
= [2, 2, 2, 4]			
Generation 2550: Best = 8.278588e+02	FES = 255100	Diversity = 4.4833e-08	k_t
= [1, 1, 1, 1]			
Generation 2560: Best = 8.278588e+02	FES = 256100	Diversity = 4.5347e-08	k_t
= [4, 4, 2, 4]			
Generation 2570: Best = 8.278588e+02	FES = 257100	Diversity = 4.5899e-08	k_t
= [1, 1, 1, 1]			
Generation 2580: Best = 8.278588e+02	FES = 258100	Diversity = 4.6434e-08	k_t
= [4, 3, 4, 3]			
Generation 2590: Best = 8.278588e+02	FES = 259100	Diversity = 4.6596e-08	k_t
= [1, 1, 1, 1]			
Generation 2600: Best = 8.278588e+02	FES = 260100	Diversity = 4.6884e-08	k_t
= [3, 4, 2, 3]			
Generation 2610: Best = 8.278588e+02	FES = 261100	Diversity = 4.6845e-08	k_t
= [1, 1, 1, 1]			
Generation 2620: Best = 8.278588e+02	FES = 262100	Diversity = 4.6121e-08	k_t
= [3, 2, 2, 3]			
Generation 2630: Best = 8.278588e+02	FES = 263100	Diversity = 4.5011e-08	k_t
= [1, 1, 1, 1]			
Generation 2640: Best = 8.278588e+02	FES = 264100	Diversity = 4.6502e-08	k_t
= [2, 3, 3, 4]			
Generation 2650: Best = 8.278588e+02	FES = 265100	Diversity = 4.6263e-08	k_t
= [1, 1, 1, 1]			

Generation 2660: Best = 8.278588e+02	FES = 266100	Diversity = 4.7143e-08	k_t
= [4, 4, 2, 3]			
Generation 2670: Best = 8.278588e+02	FES = 267100	Diversity = 4.5284e-08	k_t
= [1, 1, 1, 1]			
Generation 2680: Best = 8.278588e+02	FES = 268100	Diversity = 4.4824e-08	k_t
= [2, 3, 2, 2]			
Generation 2690: Best = 8.278588e+02	FES = 269100	Diversity = 4.5375e-08	k_t
= [1, 1, 1, 1]			
Generation 2700: Best = 8.278588e+02	FES = 270100	Diversity = 4.4935e-08	k_t
= [2, 4, 3, 4]			
Generation 2710: Best = 8.278588e+02	FES = 271100	Diversity = 4.4602e-08	k_t
= [1, 1, 1, 1]			
Generation 2720: Best = 8.278588e+02	FES = 272100	Diversity = 4.3728e-08	k_t
= [4, 2, 4, 2]			
Generation 2730: Best = 8.278588e+02	FES = 273100	Diversity = 4.3945e-08	k_t
= [1, 1, 1, 1]			
Generation 2740: Best = 8.278588e+02	FES = 274100	Diversity = 4.3069e-08	k_t
= [3, 3, 3, 4]			
Generation 2750: Best = 8.278588e+02	FES = 275100	Diversity = 4.3304e-08	k_t
= [1, 1, 1, 1]			
Generation 2760: Best = 8.278588e+02	FES = 276100	Diversity = 4.2793e-08	k_t
= [3, 2, 3, 3]			
Generation 2770: Best = 8.278588e+02	FES = 277100	Diversity = 4.3387e-08	k_t
= [1, 1, 1, 1]			
Generation 2780: Best = 8.278588e+02	FES = 278100	Diversity = 4.3568e-08	k_t
= [2, 2, 2, 3]			
Generation 2790: Best = 8.278588e+02	FES = 279100	Diversity = 4.4439e-08	k_t
= [1, 1, 1, 1]			
Generation 2800: Best = 8.278588e+02	FES = 280100	Diversity = 4.6304e-08	k_t
= [2, 3, 3, 3]			
Generation 2810: Best = 8.278588e+02	FES = 281100	Diversity = 4.6081e-08	k_t
= [1, 1, 1, 1]			
Generation 2820: Best = 8.278588e+02	FES = 282100	Diversity = 4.5940e-08	k_t
= [3, 3, 4, 4]			
Generation 2830: Best = 8.278588e+02	FES = 283100	Diversity = 4.5927e-08	k_t
= [1, 1, 1, 1]			
Generation 2840: Best = 8.278588e+02	FES = 284100	Diversity = 4.5800e-08	k_t
= [3, 4, 2, 3]			
Generation 2850: Best = 8.278588e+02	FES = 285100	Diversity = 4.7083e-08	k_t
= [1, 1, 1, 1]			
Generation 2860: Best = 8.278588e+02	FES = 286100	Diversity = 4.7491e-08	k_t
= [2, 2, 2, 3]			
Generation 2870: Best = 8.278588e+02	FES = 287100	Diversity = 4.6310e-08	k_t
= [1, 1, 1, 1]			
Generation 2880: Best = 8.278588e+02	FES = 288100	Diversity = 4.6455e-08	k_t
= [3, 4, 2, 2]			
Generation 2890: Best = 8.278588e+02	FES = 289100	Diversity = 4.6229e-08	k_t
= [1, 1, 1, 1]			
Generation 2900: Best = 8.278588e+02	FES = 290100	Diversity = 4.5606e-08	k_t
= [2, 4, 2, 4]			
Generation 2910: Best = 8.278588e+02	FES = 291100	Diversity = 4.5206e-08	k_t
= [1, 1, 1, 1]			
Generation 2920: Best = 8.278588e+02	FES = 292100	Diversity = 4.3905e-08	k_t
= [3, 3, 3, 2]			

Generation 2930: Best = 8.278588e+02 FES = 293100 Diversity = 4.5632e-08 k_t
 = [1, 1, 1, 1]
 Generation 2940: Best = 8.278588e+02 FES = 294100 Diversity = 4.5219e-08 k_t
 = [2, 3, 2, 2]
 Generation 2950: Best = 8.278588e+02 FES = 295100 Diversity = 4.6785e-08 k_t
 = [1, 1, 1, 1]
 Generation 2960: Best = 8.278588e+02 FES = 296100 Diversity = 4.6883e-08 k_t
 = [3, 3, 3, 3]
 Generation 2970: Best = 8.278588e+02 FES = 297100 Diversity = 4.7182e-08 k_t
 = [1, 1, 1, 1]
 Generation 2980: Best = 8.278588e+02 FES = 298100 Diversity = 4.6533e-08 k_t
 = [3, 3, 4, 4]
 Generation 2990: Best = 8.278588e+02 FES = 299100 Diversity = 4.6332e-08 k_t
 = [1, 1, 1, 1]

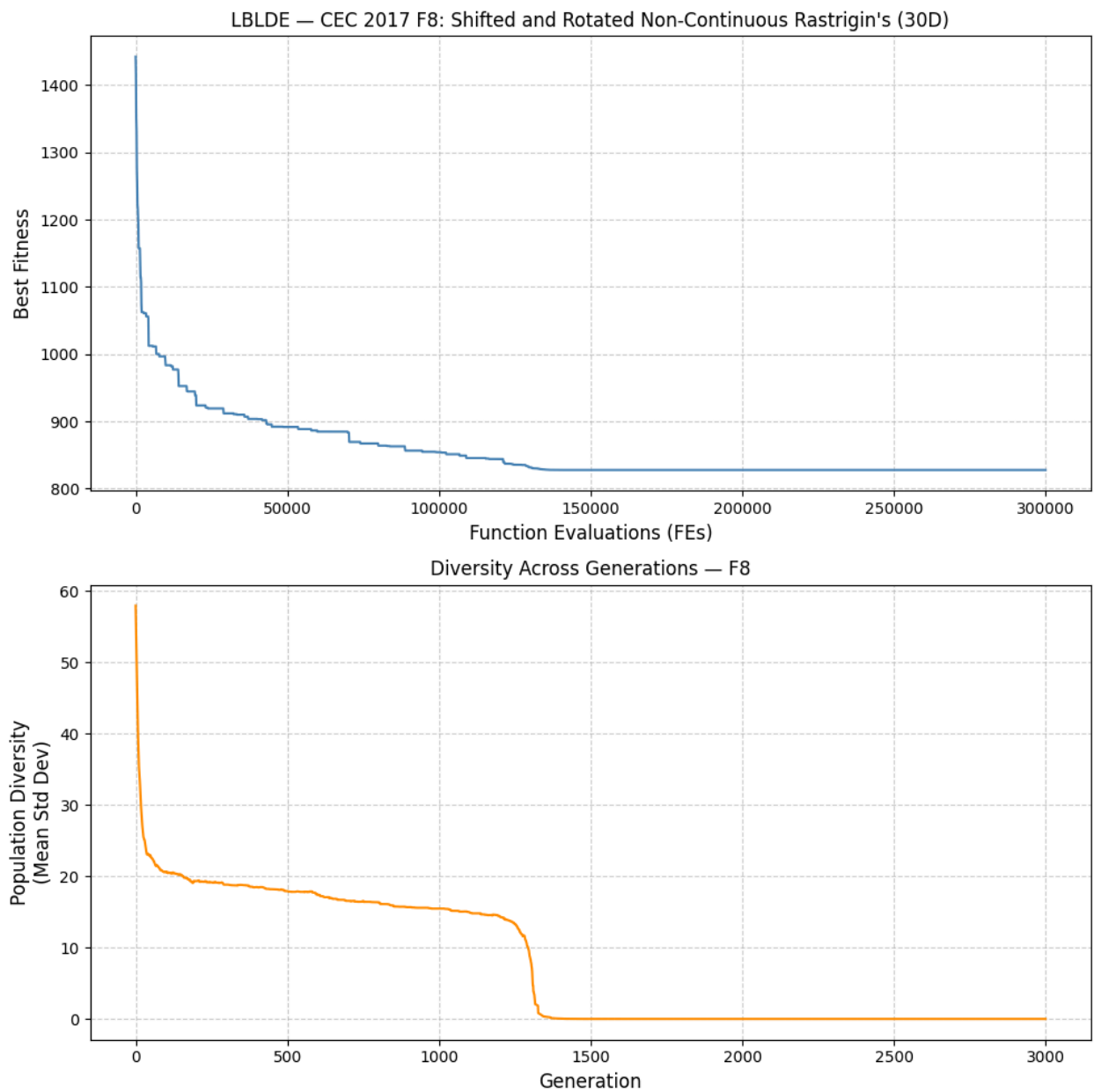
Optimization completed!

Final Best Fitness = 8.278588e+02

Total Function Evaluations = 300000

```

=====
=
Results:
  Best Fitness           : 8.278588e+02
  Global Optimum         : 8.000000e+02
  Error (|best-optimum|) : 2.785884e+01
  Best Solution (5 dim)  : [ 32.1642194  -55.96501395  56.3649851  -26.66857156
-40.16556528]
  D^(0) (init. diversity): 5.7976e+01
=====
=
  
```

Graph saved as convergence_F8.png

```
In [19]: # =====
# F9 - Shifted and Rotated Levy
# =====
run_and_plot('F9', make_wrapper(simple.f9), 900.0, 'Shifted and Rotated Levy')
```

```

=====
=
LBLDE_s on CEC 2017 F9 – Single Run
=====
=
Function      : Shifted and Rotated Levy (30D)
Bounds       : [-100, 100]
Global optimum : 900.0
Initial D^(0) = 5.7976e+01
Generation 0: Best Fitness = 2.774722e+04 | Diversity = 5.7976e+01
Generation 10: Best = 1.238079e+04 FES = 1100 Diversity = 4.0478e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 5.637628e+03 FES = 2100 Diversity = 2.6447e+01 k_t
= [3, 3, 2, 2]
Generation 30: Best = 4.052886e+03 FES = 3100 Diversity = 2.0116e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 2.078136e+03 FES = 4100 Diversity = 1.5237e+01 k_t
= [3, 4, 3, 4]
Generation 50: Best = 2.078136e+03 FES = 5100 Diversity = 1.2103e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 1.453385e+03 FES = 6100 Diversity = 9.4678e+00 k_t
= [2, 3, 4, 3]
Generation 70: Best = 1.208098e+03 FES = 7100 Diversity = 6.4433e+00 k_t
= [1, 1, 1, 1]
Generation 80: Best = 1.011215e+03 FES = 8100 Diversity = 4.4797e+00 k_t
= [4, 3, 4, 3]
Generation 90: Best = 9.691724e+02 FES = 9100 Diversity = 3.1656e+00 k_t
= [1, 1, 1, 1]
Generation 100: Best = 9.457646e+02 FES = 10100 Diversity = 2.2078e+00 k_t
= [2, 4, 4, 3]
Generation 110: Best = 9.231729e+02 FES = 11100 Diversity = 1.7775e+00 k_t
= [1, 1, 1, 1]
Generation 120: Best = 9.160925e+02 FES = 12100 Diversity = 1.4095e+00 k_t
= [3, 3, 3, 4]
Generation 130: Best = 9.073486e+02 FES = 13100 Diversity = 1.0810e+00 k_t
= [1, 1, 1, 1]
Generation 140: Best = 9.039200e+02 FES = 14100 Diversity = 7.7896e-01 k_t
= [2, 3, 3, 3]
Generation 150: Best = 9.022654e+02 FES = 15100 Diversity = 5.0932e-01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 9.014928e+02 FES = 16100 Diversity = 3.4547e-01 k_t
= [2, 4, 4, 4]
Generation 170: Best = 9.012163e+02 FES = 17100 Diversity = 2.4016e-01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 9.010010e+02 FES = 18100 Diversity = 1.7278e-01 k_t
= [2, 4, 2, 3]
Generation 190: Best = 9.006977e+02 FES = 19100 Diversity = 1.2026e-01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 9.005812e+02 FES = 20100 Diversity = 7.3382e-02 k_t
= [2, 2, 4, 4]
Generation 210: Best = 9.005192e+02 FES = 21100 Diversity = 4.5831e-02 k_t
= [1, 1, 1, 1]
Generation 220: Best = 9.004835e+02 FES = 22100 Diversity = 3.2025e-02 k_t
= [4, 3, 2, 3]

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Generation 230:	Best = 9.004708e+02	FES = 23100	Diversity = 2.3046e-02	k_t = [1, 1, 1, 1]
Generation 240:	Best = 9.004629e+02	FES = 24100	Diversity = 1.5802e-02	k_t = [4, 2, 4, 4]
Generation 250:	Best = 9.004579e+02	FES = 25100	Diversity = 1.1444e-02	k_t = [1, 1, 1, 1]
Generation 260:	Best = 9.004558e+02	FES = 26100	Diversity = 8.4228e-03	k_t = [3, 2, 3, 4]
Generation 270:	Best = 9.004548e+02	FES = 27100	Diversity = 5.6606e-03	k_t = [1, 1, 1, 1]
Generation 280:	Best = 9.004546e+02	FES = 28100	Diversity = 3.6377e-03	k_t = [3, 4, 4, 3]
Generation 290:	Best = 9.004544e+02	FES = 29100	Diversity = 2.3429e-03	k_t = [1, 1, 1, 1]
Generation 300:	Best = 9.004544e+02	FES = 30100	Diversity = 1.3727e-03	k_t = [4, 3, 4, 4]
Generation 310:	Best = 9.004543e+02	FES = 31100	Diversity = 8.0160e-04	k_t = [1, 1, 1, 1]
Generation 320:	Best = 9.004543e+02	FES = 32100	Diversity = 4.8045e-04	k_t = [3, 3, 3, 4]
Generation 330:	Best = 9.004543e+02	FES = 33100	Diversity = 2.8644e-04	k_t = [1, 1, 1, 1]
Generation 340:	Best = 9.004543e+02	FES = 34100	Diversity = 1.6732e-04	k_t = [2, 2, 4, 3]
Generation 350:	Best = 9.004543e+02	FES = 35100	Diversity = 1.1484e-04	k_t = [1, 1, 1, 1]
Generation 360:	Best = 9.004543e+02	FES = 36100	Diversity = 8.4338e-05	k_t = [3, 3, 3, 4]
Generation 370:	Best = 9.004543e+02	FES = 37100	Diversity = 5.5981e-05	k_t = [1, 1, 1, 1]
Generation 380:	Best = 9.004543e+02	FES = 38100	Diversity = 4.6284e-05	k_t = [4, 3, 3, 3]
Generation 390:	Best = 9.004543e+02	FES = 39100	Diversity = 3.4845e-05	k_t = [1, 1, 1, 1]
Generation 400:	Best = 9.004543e+02	FES = 40100	Diversity = 2.2812e-05	k_t = [2, 3, 4, 3]
Generation 410:	Best = 9.004543e+02	FES = 41100	Diversity = 1.4562e-05	k_t = [1, 1, 1, 1]
Generation 420:	Best = 9.004543e+02	FES = 42100	Diversity = 9.7526e-06	k_t = [3, 3, 2, 4]
Generation 430:	Best = 9.004543e+02	FES = 43100	Diversity = 7.4047e-06	k_t = [1, 1, 1, 1]
Generation 440:	Best = 9.004543e+02	FES = 44100	Diversity = 4.9645e-06	k_t = [4, 4, 3, 3]
Generation 450:	Best = 9.004543e+02	FES = 45100	Diversity = 3.4644e-06	k_t = [1, 1, 1, 1]
Generation 460:	Best = 9.004543e+02	FES = 46100	Diversity = 2.2682e-06	k_t = [2, 3, 2, 4]
Generation 470:	Best = 9.004543e+02	FES = 47100	Diversity = 1.3952e-06	k_t = [1, 1, 1, 1]
Generation 480:	Best = 9.004543e+02	FES = 48100	Diversity = 8.7778e-07	k_t = [4, 4, 2, 4]
Generation 490:	Best = 9.004543e+02	FES = 49100	Diversity = 4.8631e-07	k_t = [1, 1, 1, 1]

Generation 500:	Best = 9.004543e+02	FES = 50100	Diversity = 2.9939e-07	k_t = [3, 2, 2, 3]
Generation 510:	Best = 9.004543e+02	FES = 51100	Diversity = 1.7669e-07	k_t = [1, 1, 1, 1]
Generation 520:	Best = 9.004543e+02	FES = 52100	Diversity = 1.1428e-07	k_t = [4, 4, 3, 4]
Generation 530:	Best = 9.004543e+02	FES = 53100	Diversity = 8.7582e-08	k_t = [1, 1, 1, 1]
Generation 540:	Best = 9.004543e+02	FES = 54100	Diversity = 7.4882e-08	k_t = [4, 2, 3, 4]
Generation 550:	Best = 9.004543e+02	FES = 55100	Diversity = 7.2148e-08	k_t = [1, 1, 1, 1]
Generation 560:	Best = 9.004543e+02	FES = 56100	Diversity = 6.2435e-08	k_t = [2, 4, 4, 4]
Generation 570:	Best = 9.004543e+02	FES = 57100	Diversity = 5.3528e-08	k_t = [1, 1, 1, 1]
Generation 580:	Best = 9.004543e+02	FES = 58100	Diversity = 5.0292e-08	k_t = [4, 2, 3, 4]
Generation 590:	Best = 9.004543e+02	FES = 59100	Diversity = 5.2447e-08	k_t = [1, 1, 1, 1]
Generation 600:	Best = 9.004543e+02	FES = 60100	Diversity = 5.2051e-08	k_t = [4, 4, 2, 4]
Generation 610:	Best = 9.004543e+02	FES = 61100	Diversity = 4.8996e-08	k_t = [1, 1, 1, 1]
Generation 620:	Best = 9.004543e+02	FES = 62100	Diversity = 4.7015e-08	k_t = [2, 3, 4, 2]
Generation 630:	Best = 9.004543e+02	FES = 63100	Diversity = 4.7124e-08	k_t = [1, 1, 1, 1]
Generation 640:	Best = 9.004543e+02	FES = 64100	Diversity = 4.4845e-08	k_t = [4, 2, 3, 2]
Generation 650:	Best = 9.004543e+02	FES = 65100	Diversity = 4.6797e-08	k_t = [1, 1, 1, 1]
Generation 660:	Best = 9.004543e+02	FES = 66100	Diversity = 4.7819e-08	k_t = [2, 2, 3, 2]
Generation 670:	Best = 9.004543e+02	FES = 67100	Diversity = 4.8374e-08	k_t = [1, 1, 1, 1]
Generation 680:	Best = 9.004543e+02	FES = 68100	Diversity = 4.7911e-08	k_t = [2, 4, 4, 2]
Generation 690:	Best = 9.004543e+02	FES = 69100	Diversity = 4.5720e-08	k_t = [1, 1, 1, 1]
Generation 700:	Best = 9.004543e+02	FES = 70100	Diversity = 4.5996e-08	k_t = [4, 3, 3, 3]
Generation 710:	Best = 9.004543e+02	FES = 71100	Diversity = 4.7525e-08	k_t = [1, 1, 1, 1]
Generation 720:	Best = 9.004543e+02	FES = 72100	Diversity = 4.6698e-08	k_t = [3, 2, 3, 4]
Generation 730:	Best = 9.004543e+02	FES = 73100	Diversity = 4.9234e-08	k_t = [1, 1, 1, 1]
Generation 740:	Best = 9.004543e+02	FES = 74100	Diversity = 4.9522e-08	k_t = [4, 2, 2, 3]
Generation 750:	Best = 9.004543e+02	FES = 75100	Diversity = 4.9392e-08	k_t = [1, 1, 1, 1]
Generation 760:	Best = 9.004543e+02	FES = 76100	Diversity = 4.9050e-08	k_t = [2, 3, 4, 3]

Generation 770:	Best = 9.004543e+02	FES = 77100	Diversity = 4.7845e-08	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 9.004543e+02	FES = 78100	Diversity = 4.5821e-08	k_t
= [3, 4, 4, 4]				
Generation 790:	Best = 9.004543e+02	FES = 79100	Diversity = 4.6145e-08	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 9.004543e+02	FES = 80100	Diversity = 5.0258e-08	k_t
= [3, 3, 4, 2]				
Generation 810:	Best = 9.004543e+02	FES = 81100	Diversity = 5.3190e-08	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 9.004543e+02	FES = 82100	Diversity = 5.3841e-08	k_t
= [3, 3, 4, 3]				
Generation 830:	Best = 9.004543e+02	FES = 83100	Diversity = 5.3912e-08	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 9.004543e+02	FES = 84100	Diversity = 5.3085e-08	k_t
= [3, 3, 3, 4]				
Generation 850:	Best = 9.004543e+02	FES = 85100	Diversity = 5.2655e-08	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 9.004543e+02	FES = 86100	Diversity = 5.0844e-08	k_t
= [2, 4, 3, 3]				
Generation 870:	Best = 9.004543e+02	FES = 87100	Diversity = 5.2150e-08	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 9.004543e+02	FES = 88100	Diversity = 5.0795e-08	k_t
= [2, 2, 3, 3]				
Generation 890:	Best = 9.004543e+02	FES = 89100	Diversity = 5.0878e-08	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 9.004543e+02	FES = 90100	Diversity = 5.0604e-08	k_t
= [2, 4, 3, 2]				
Generation 910:	Best = 9.004543e+02	FES = 91100	Diversity = 5.0410e-08	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 9.004543e+02	FES = 92100	Diversity = 5.0403e-08	k_t
= [4, 2, 2, 4]				
Generation 930:	Best = 9.004543e+02	FES = 93100	Diversity = 4.9824e-08	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 9.004543e+02	FES = 94100	Diversity = 5.1167e-08	k_t
= [2, 3, 4, 4]				
Generation 950:	Best = 9.004543e+02	FES = 95100	Diversity = 5.2185e-08	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 9.004543e+02	FES = 96100	Diversity = 5.0867e-08	k_t
= [2, 2, 2, 4]				
Generation 970:	Best = 9.004543e+02	FES = 97100	Diversity = 5.1628e-08	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 9.004543e+02	FES = 98100	Diversity = 5.1515e-08	k_t
= [4, 2, 3, 3]				
Generation 990:	Best = 9.004543e+02	FES = 99100	Diversity = 4.9094e-08	k_t
= [1, 1, 1, 1]				
Generation 1000:	Best = 9.004543e+02	FES = 100100	Diversity = 5.1735e-08	k_t
= [3, 4, 3, 2]				
Generation 1010:	Best = 9.004543e+02	FES = 101100	Diversity = 5.1169e-08	k_t
= [1, 1, 1, 1]				
Generation 1020:	Best = 9.004543e+02	FES = 102100	Diversity = 5.2005e-08	k_t
= [3, 2, 2, 4]				
Generation 1030:	Best = 9.004543e+02	FES = 103100	Diversity = 5.2475e-08	k_t
= [1, 1, 1, 1]				

Generation 1040: Best = 9.004543e+02 = [2, 3, 4, 2]	FES = 104100	Diversity = 5.1586e-08	k_t
Generation 1050: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 105100	Diversity = 5.0667e-08	k_t
Generation 1060: Best = 9.004543e+02 = [4, 3, 3, 3]	FES = 106100	Diversity = 4.9975e-08	k_t
Generation 1070: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 107100	Diversity = 4.8968e-08	k_t
Generation 1080: Best = 9.004543e+02 = [4, 2, 2, 3]	FES = 108100	Diversity = 4.8435e-08	k_t
Generation 1090: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 109100	Diversity = 4.8400e-08	k_t
Generation 1100: Best = 9.004543e+02 = [4, 3, 2, 4]	FES = 110100	Diversity = 4.9491e-08	k_t
Generation 1110: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 111100	Diversity = 5.0661e-08	k_t
Generation 1120: Best = 9.004543e+02 = [2, 2, 2, 3]	FES = 112100	Diversity = 5.0446e-08	k_t
Generation 1130: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 113100	Diversity = 4.9886e-08	k_t
Generation 1140: Best = 9.004543e+02 = [3, 4, 3, 4]	FES = 114100	Diversity = 4.9735e-08	k_t
Generation 1150: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 115100	Diversity = 4.9784e-08	k_t
Generation 1160: Best = 9.004543e+02 = [2, 3, 4, 2]	FES = 116100	Diversity = 4.9271e-08	k_t
Generation 1170: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 117100	Diversity = 4.9588e-08	k_t
Generation 1180: Best = 9.004543e+02 = [2, 3, 2, 4]	FES = 118100	Diversity = 5.1371e-08	k_t
Generation 1190: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 119100	Diversity = 5.2316e-08	k_t
Generation 1200: Best = 9.004543e+02 = [4, 4, 4, 3]	FES = 120100	Diversity = 5.2122e-08	k_t
Generation 1210: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 121100	Diversity = 5.3109e-08	k_t
Generation 1220: Best = 9.004543e+02 = [3, 4, 3, 3]	FES = 122100	Diversity = 5.4814e-08	k_t
Generation 1230: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 123100	Diversity = 5.3515e-08	k_t
Generation 1240: Best = 9.004543e+02 = [4, 2, 4, 2]	FES = 124100	Diversity = 5.3300e-08	k_t
Generation 1250: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 125100	Diversity = 5.1719e-08	k_t
Generation 1260: Best = 9.004543e+02 = [4, 3, 2, 4]	FES = 126100	Diversity = 5.0535e-08	k_t
Generation 1270: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 127100	Diversity = 5.1138e-08	k_t
Generation 1280: Best = 9.004543e+02 = [3, 4, 2, 3]	FES = 128100	Diversity = 5.0856e-08	k_t
Generation 1290: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 129100	Diversity = 5.2656e-08	k_t
Generation 1300: Best = 9.004543e+02 = [4, 2, 4, 2]	FES = 130100	Diversity = 5.3819e-08	k_t

Generation 1310: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 131100	Diversity = 5.2142e-08	k_t
Generation 1320: Best = 9.004543e+02 = [3, 2, 3, 3]	FES = 132100	Diversity = 5.1907e-08	k_t
Generation 1330: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 133100	Diversity = 5.2793e-08	k_t
Generation 1340: Best = 9.004543e+02 = [2, 3, 2, 4]	FES = 134100	Diversity = 5.2062e-08	k_t
Generation 1350: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 135100	Diversity = 5.2508e-08	k_t
Generation 1360: Best = 9.004543e+02 = [2, 4, 2, 3]	FES = 136100	Diversity = 5.2245e-08	k_t
Generation 1370: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 137100	Diversity = 5.0619e-08	k_t
Generation 1380: Best = 9.004543e+02 = [4, 2, 2, 3]	FES = 138100	Diversity = 5.1151e-08	k_t
Generation 1390: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 139100	Diversity = 5.2051e-08	k_t
Generation 1400: Best = 9.004543e+02 = [4, 2, 3, 4]	FES = 140100	Diversity = 5.2221e-08	k_t
Generation 1410: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 141100	Diversity = 5.2140e-08	k_t
Generation 1420: Best = 9.004543e+02 = [4, 4, 2, 2]	FES = 142100	Diversity = 5.4513e-08	k_t
Generation 1430: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 143100	Diversity = 5.2871e-08	k_t
Generation 1440: Best = 9.004543e+02 = [2, 4, 2, 4]	FES = 144100	Diversity = 5.2108e-08	k_t
Generation 1450: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 145100	Diversity = 5.3529e-08	k_t
Generation 1460: Best = 9.004543e+02 = [2, 4, 2, 2]	FES = 146100	Diversity = 5.2553e-08	k_t
Generation 1470: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 147100	Diversity = 5.3015e-08	k_t
Generation 1480: Best = 9.004543e+02 = [4, 2, 3, 4]	FES = 148100	Diversity = 5.3731e-08	k_t
Generation 1490: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 149100	Diversity = 5.3670e-08	k_t
Generation 1500: Best = 9.004543e+02 = [4, 2, 2, 2]	FES = 150100	Diversity = 5.4596e-08	k_t
Generation 1510: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 151100	Diversity = 5.4595e-08	k_t
Generation 1520: Best = 9.004543e+02 = [4, 3, 2, 2]	FES = 152100	Diversity = 5.3462e-08	k_t
Generation 1530: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 153100	Diversity = 5.1361e-08	k_t
Generation 1540: Best = 9.004543e+02 = [3, 4, 2, 2]	FES = 154100	Diversity = 5.2752e-08	k_t
Generation 1550: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 155100	Diversity = 5.2575e-08	k_t
Generation 1560: Best = 9.004543e+02 = [2, 3, 3, 4]	FES = 156100	Diversity = 5.2446e-08	k_t
Generation 1570: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 157100	Diversity = 5.3159e-08	k_t

Generation 1580: Best = 9.004543e+02	FES = 158100	Diversity = 5.2152e-08	k_t
= [3, 2, 3, 4]			
Generation 1590: Best = 9.004543e+02	FES = 159100	Diversity = 5.2469e-08	k_t
= [1, 1, 1, 1]			
Generation 1600: Best = 9.004543e+02	FES = 160100	Diversity = 5.3819e-08	k_t
= [3, 2, 4, 3]			
Generation 1610: Best = 9.004543e+02	FES = 161100	Diversity = 5.1753e-08	k_t
= [1, 1, 1, 1]			
Generation 1620: Best = 9.004543e+02	FES = 162100	Diversity = 5.0316e-08	k_t
= [4, 2, 2, 2]			
Generation 1630: Best = 9.004543e+02	FES = 163100	Diversity = 5.1359e-08	k_t
= [1, 1, 1, 1]			
Generation 1640: Best = 9.004543e+02	FES = 164100	Diversity = 5.1565e-08	k_t
= [4, 4, 4, 2]			
Generation 1650: Best = 9.004543e+02	FES = 165100	Diversity = 5.3851e-08	k_t
= [1, 1, 1, 1]			
Generation 1660: Best = 9.004543e+02	FES = 166100	Diversity = 5.3311e-08	k_t
= [4, 2, 2, 2]			
Generation 1670: Best = 9.004543e+02	FES = 167100	Diversity = 5.1827e-08	k_t
= [1, 1, 1, 1]			
Generation 1680: Best = 9.004543e+02	FES = 168100	Diversity = 5.2093e-08	k_t
= [2, 2, 4, 3]			
Generation 1690: Best = 9.004543e+02	FES = 169100	Diversity = 5.2174e-08	k_t
= [1, 1, 1, 1]			
Generation 1700: Best = 9.004543e+02	FES = 170100	Diversity = 5.1552e-08	k_t
= [2, 2, 3, 3]			
Generation 1710: Best = 9.004543e+02	FES = 171100	Diversity = 5.2563e-08	k_t
= [1, 1, 1, 1]			
Generation 1720: Best = 9.004543e+02	FES = 172100	Diversity = 5.1504e-08	k_t
= [2, 2, 2, 4]			
Generation 1730: Best = 9.004543e+02	FES = 173100	Diversity = 5.1927e-08	k_t
= [1, 1, 1, 1]			
Generation 1740: Best = 9.004543e+02	FES = 174100	Diversity = 5.2348e-08	k_t
= [3, 2, 3, 2]			
Generation 1750: Best = 9.004543e+02	FES = 175100	Diversity = 5.2096e-08	k_t
= [1, 1, 1, 1]			
Generation 1760: Best = 9.004543e+02	FES = 176100	Diversity = 5.2200e-08	k_t
= [4, 3, 4, 2]			
Generation 1770: Best = 9.004543e+02	FES = 177100	Diversity = 5.1112e-08	k_t
= [1, 1, 1, 1]			
Generation 1780: Best = 9.004543e+02	FES = 178100	Diversity = 5.2017e-08	k_t
= [4, 3, 3, 3]			
Generation 1790: Best = 9.004543e+02	FES = 179100	Diversity = 5.2308e-08	k_t
= [1, 1, 1, 1]			
Generation 1800: Best = 9.004543e+02	FES = 180100	Diversity = 5.1725e-08	k_t
= [3, 2, 2, 4]			
Generation 1810: Best = 9.004543e+02	FES = 181100	Diversity = 5.2059e-08	k_t
= [1, 1, 1, 1]			
Generation 1820: Best = 9.004543e+02	FES = 182100	Diversity = 5.1476e-08	k_t
= [3, 4, 4, 4]			
Generation 1830: Best = 9.004543e+02	FES = 183100	Diversity = 5.0760e-08	k_t
= [1, 1, 1, 1]			
Generation 1840: Best = 9.004543e+02	FES = 184100	Diversity = 4.9991e-08	k_t
= [4, 3, 3, 4]			

Generation 1850: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 185100	Diversity = 5.0477e-08	k_t
Generation 1860: Best = 9.004543e+02 = [3, 3, 3, 2]	FES = 186100	Diversity = 5.1134e-08	k_t
Generation 1870: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 187100	Diversity = 5.0793e-08	k_t
Generation 1880: Best = 9.004543e+02 = [2, 4, 3, 3]	FES = 188100	Diversity = 5.2019e-08	k_t
Generation 1890: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 189100	Diversity = 5.2041e-08	k_t
Generation 1900: Best = 9.004543e+02 = [3, 3, 3, 3]	FES = 190100	Diversity = 5.2026e-08	k_t
Generation 1910: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 191100	Diversity = 5.1843e-08	k_t
Generation 1920: Best = 9.004543e+02 = [2, 3, 4, 3]	FES = 192100	Diversity = 5.0855e-08	k_t
Generation 1930: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 193100	Diversity = 5.1847e-08	k_t
Generation 1940: Best = 9.004543e+02 = [3, 3, 4, 4]	FES = 194100	Diversity = 5.2973e-08	k_t
Generation 1950: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 195100	Diversity = 5.1579e-08	k_t
Generation 1960: Best = 9.004543e+02 = [4, 4, 4, 3]	FES = 196100	Diversity = 5.1809e-08	k_t
Generation 1970: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 197100	Diversity = 5.0744e-08	k_t
Generation 1980: Best = 9.004543e+02 = [3, 4, 2, 3]	FES = 198100	Diversity = 5.0834e-08	k_t
Generation 1990: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 199100	Diversity = 4.9770e-08	k_t
Generation 2000: Best = 9.004543e+02 = [4, 4, 3, 3]	FES = 200100	Diversity = 5.0118e-08	k_t
Generation 2010: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 201100	Diversity = 5.0129e-08	k_t
Generation 2020: Best = 9.004543e+02 = [3, 2, 2, 3]	FES = 202100	Diversity = 5.0909e-08	k_t
Generation 2030: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 203100	Diversity = 5.0919e-08	k_t
Generation 2040: Best = 9.004543e+02 = [4, 2, 3, 3]	FES = 204100	Diversity = 4.8975e-08	k_t
Generation 2050: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 205100	Diversity = 4.8014e-08	k_t
Generation 2060: Best = 9.004543e+02 = [3, 4, 2, 4]	FES = 206100	Diversity = 4.9034e-08	k_t
Generation 2070: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 207100	Diversity = 4.8753e-08	k_t
Generation 2080: Best = 9.004543e+02 = [4, 2, 3, 3]	FES = 208100	Diversity = 4.9457e-08	k_t
Generation 2090: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 209100	Diversity = 5.0693e-08	k_t
Generation 2100: Best = 9.004543e+02 = [4, 3, 2, 4]	FES = 210100	Diversity = 5.0745e-08	k_t
Generation 2110: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 211100	Diversity = 5.0254e-08	k_t

Generation 2120: Best = 9.004543e+02	FES = 212100	Diversity = 4.8368e-08	k_t
= [2, 4, 3, 2]			
Generation 2130: Best = 9.004543e+02	FES = 213100	Diversity = 4.9348e-08	k_t
= [1, 1, 1, 1]			
Generation 2140: Best = 9.004543e+02	FES = 214100	Diversity = 4.8322e-08	k_t
= [4, 4, 4, 3]			
Generation 2150: Best = 9.004543e+02	FES = 215100	Diversity = 4.8090e-08	k_t
= [1, 1, 1, 1]			
Generation 2160: Best = 9.004543e+02	FES = 216100	Diversity = 5.0056e-08	k_t
= [3, 3, 2, 3]			
Generation 2170: Best = 9.004543e+02	FES = 217100	Diversity = 4.8638e-08	k_t
= [1, 1, 1, 1]			
Generation 2180: Best = 9.004543e+02	FES = 218100	Diversity = 5.0141e-08	k_t
= [3, 2, 2, 3]			
Generation 2190: Best = 9.004543e+02	FES = 219100	Diversity = 5.0250e-08	k_t
= [1, 1, 1, 1]			
Generation 2200: Best = 9.004543e+02	FES = 220100	Diversity = 5.0387e-08	k_t
= [3, 4, 2, 4]			
Generation 2210: Best = 9.004543e+02	FES = 221100	Diversity = 5.0180e-08	k_t
= [1, 1, 1, 1]			
Generation 2220: Best = 9.004543e+02	FES = 222100	Diversity = 5.0137e-08	k_t
= [2, 4, 3, 3]			
Generation 2230: Best = 9.004543e+02	FES = 223100	Diversity = 5.0205e-08	k_t
= [1, 1, 1, 1]			
Generation 2240: Best = 9.004543e+02	FES = 224100	Diversity = 5.1771e-08	k_t
= [3, 3, 2, 3]			
Generation 2250: Best = 9.004543e+02	FES = 225100	Diversity = 5.1673e-08	k_t
= [1, 1, 1, 1]			
Generation 2260: Best = 9.004543e+02	FES = 226100	Diversity = 5.2447e-08	k_t
= [3, 4, 3, 3]			
Generation 2270: Best = 9.004543e+02	FES = 227100	Diversity = 5.2586e-08	k_t
= [1, 1, 1, 1]			
Generation 2280: Best = 9.004543e+02	FES = 228100	Diversity = 5.5083e-08	k_t
= [3, 4, 3, 4]			
Generation 2290: Best = 9.004543e+02	FES = 229100	Diversity = 5.3217e-08	k_t
= [1, 1, 1, 1]			
Generation 2300: Best = 9.004543e+02	FES = 230100	Diversity = 5.2839e-08	k_t
= [4, 4, 2, 4]			
Generation 2310: Best = 9.004543e+02	FES = 231100	Diversity = 5.1811e-08	k_t
= [1, 1, 1, 1]			
Generation 2320: Best = 9.004543e+02	FES = 232100	Diversity = 4.9940e-08	k_t
= [4, 2, 4, 2]			
Generation 2330: Best = 9.004543e+02	FES = 233100	Diversity = 4.9565e-08	k_t
= [1, 1, 1, 1]			
Generation 2340: Best = 9.004543e+02	FES = 234100	Diversity = 4.9310e-08	k_t
= [3, 4, 3, 2]			
Generation 2350: Best = 9.004543e+02	FES = 235100	Diversity = 5.1210e-08	k_t
= [1, 1, 1, 1]			
Generation 2360: Best = 9.004543e+02	FES = 236100	Diversity = 5.2047e-08	k_t
= [2, 3, 3, 3]			
Generation 2370: Best = 9.004543e+02	FES = 237100	Diversity = 5.3645e-08	k_t
= [1, 1, 1, 1]			
Generation 2380: Best = 9.004543e+02	FES = 238100	Diversity = 5.2453e-08	k_t
= [2, 2, 2, 4]			

Generation 2390: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 239100	Diversity = 5.1786e-08	k_t
Generation 2400: Best = 9.004543e+02 = [4, 3, 4, 3]	FES = 240100	Diversity = 5.2615e-08	k_t
Generation 2410: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 241100	Diversity = 5.4063e-08	k_t
Generation 2420: Best = 9.004543e+02 = [3, 2, 2, 3]	FES = 242100	Diversity = 5.2386e-08	k_t
Generation 2430: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 243100	Diversity = 5.2071e-08	k_t
Generation 2440: Best = 9.004543e+02 = [2, 2, 3, 4]	FES = 244100	Diversity = 5.0417e-08	k_t
Generation 2450: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 245100	Diversity = 4.9017e-08	k_t
Generation 2460: Best = 9.004543e+02 = [4, 3, 2, 4]	FES = 246100	Diversity = 4.9745e-08	k_t
Generation 2470: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 247100	Diversity = 5.0864e-08	k_t
Generation 2480: Best = 9.004543e+02 = [4, 4, 2, 4]	FES = 248100	Diversity = 5.3580e-08	k_t
Generation 2490: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 249100	Diversity = 5.3166e-08	k_t
Generation 2500: Best = 9.004543e+02 = [2, 2, 2, 3]	FES = 250100	Diversity = 5.2394e-08	k_t
Generation 2510: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 251100	Diversity = 5.4081e-08	k_t
Generation 2520: Best = 9.004543e+02 = [2, 2, 2, 4]	FES = 252100	Diversity = 5.4053e-08	k_t
Generation 2530: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 253100	Diversity = 5.3409e-08	k_t
Generation 2540: Best = 9.004543e+02 = [3, 2, 2, 3]	FES = 254100	Diversity = 5.0827e-08	k_t
Generation 2550: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 255100	Diversity = 4.9834e-08	k_t
Generation 2560: Best = 9.004543e+02 = [2, 3, 2, 2]	FES = 256100	Diversity = 4.8618e-08	k_t
Generation 2570: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 257100	Diversity = 4.8557e-08	k_t
Generation 2580: Best = 9.004543e+02 = [3, 3, 4, 4]	FES = 258100	Diversity = 4.9177e-08	k_t
Generation 2590: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 259100	Diversity = 4.9223e-08	k_t
Generation 2600: Best = 9.004543e+02 = [2, 3, 4, 3]	FES = 260100	Diversity = 4.9753e-08	k_t
Generation 2610: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 261100	Diversity = 5.0086e-08	k_t
Generation 2620: Best = 9.004543e+02 = [2, 3, 3, 4]	FES = 262100	Diversity = 4.9253e-08	k_t
Generation 2630: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 263100	Diversity = 5.0791e-08	k_t
Generation 2640: Best = 9.004543e+02 = [4, 4, 3, 3]	FES = 264100	Diversity = 5.0485e-08	k_t
Generation 2650: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 265100	Diversity = 5.0782e-08	k_t

Generation 2660: Best = 9.004543e+02 = [3, 2, 3, 4]	FES = 266100	Diversity = 5.1053e-08	k_t
Generation 2670: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 267100	Diversity = 5.0196e-08	k_t
Generation 2680: Best = 9.004543e+02 = [3, 2, 2, 2]	FES = 268100	Diversity = 4.9539e-08	k_t
Generation 2690: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 269100	Diversity = 4.8648e-08	k_t
Generation 2700: Best = 9.004543e+02 = [4, 3, 4, 2]	FES = 270100	Diversity = 4.8132e-08	k_t
Generation 2710: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 271100	Diversity = 4.9264e-08	k_t
Generation 2720: Best = 9.004543e+02 = [4, 4, 4, 3]	FES = 272100	Diversity = 4.9317e-08	k_t
Generation 2730: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 273100	Diversity = 4.8344e-08	k_t
Generation 2740: Best = 9.004543e+02 = [2, 2, 4, 2]	FES = 274100	Diversity = 4.9110e-08	k_t
Generation 2750: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 275100	Diversity = 4.9247e-08	k_t
Generation 2760: Best = 9.004543e+02 = [2, 4, 3, 2]	FES = 276100	Diversity = 5.0967e-08	k_t
Generation 2770: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 277100	Diversity = 4.9988e-08	k_t
Generation 2780: Best = 9.004543e+02 = [2, 3, 3, 2]	FES = 278100	Diversity = 4.7806e-08	k_t
Generation 2790: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 279100	Diversity = 4.9230e-08	k_t
Generation 2800: Best = 9.004543e+02 = [2, 4, 3, 4]	FES = 280100	Diversity = 5.0227e-08	k_t
Generation 2810: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 281100	Diversity = 5.0964e-08	k_t
Generation 2820: Best = 9.004543e+02 = [3, 3, 4, 3]	FES = 282100	Diversity = 5.1121e-08	k_t
Generation 2830: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 283100	Diversity = 4.9764e-08	k_t
Generation 2840: Best = 9.004543e+02 = [2, 4, 3, 4]	FES = 284100	Diversity = 4.8781e-08	k_t
Generation 2850: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 285100	Diversity = 4.8201e-08	k_t
Generation 2860: Best = 9.004543e+02 = [2, 2, 4, 4]	FES = 286100	Diversity = 4.9212e-08	k_t
Generation 2870: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 287100	Diversity = 4.9784e-08	k_t
Generation 2880: Best = 9.004543e+02 = [4, 2, 4, 3]	FES = 288100	Diversity = 4.8709e-08	k_t
Generation 2890: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 289100	Diversity = 4.9635e-08	k_t
Generation 2900: Best = 9.004543e+02 = [4, 2, 4, 2]	FES = 290100	Diversity = 5.0175e-08	k_t
Generation 2910: Best = 9.004543e+02 = [1, 1, 1, 1]	FES = 291100	Diversity = 5.1365e-08	k_t
Generation 2920: Best = 9.004543e+02 = [2, 4, 3, 4]	FES = 292100	Diversity = 5.1267e-08	k_t

Generation 2930: Best = 9.004543e+02 FES = 293100 Diversity = 5.0647e-08 k_t
 = [1, 1, 1, 1]
 Generation 2940: Best = 9.004543e+02 FES = 294100 Diversity = 5.1598e-08 k_t
 = [3, 4, 3, 3]
 Generation 2950: Best = 9.004543e+02 FES = 295100 Diversity = 5.1254e-08 k_t
 = [1, 1, 1, 1]
 Generation 2960: Best = 9.004543e+02 FES = 296100 Diversity = 5.2068e-08 k_t
 = [2, 4, 3, 2]
 Generation 2970: Best = 9.004543e+02 FES = 297100 Diversity = 5.2023e-08 k_t
 = [1, 1, 1, 1]
 Generation 2980: Best = 9.004543e+02 FES = 298100 Diversity = 5.3396e-08 k_t
 = [3, 3, 3, 2]
 Generation 2990: Best = 9.004543e+02 FES = 299100 Diversity = 5.2693e-08 k_t
 = [1, 1, 1, 1]

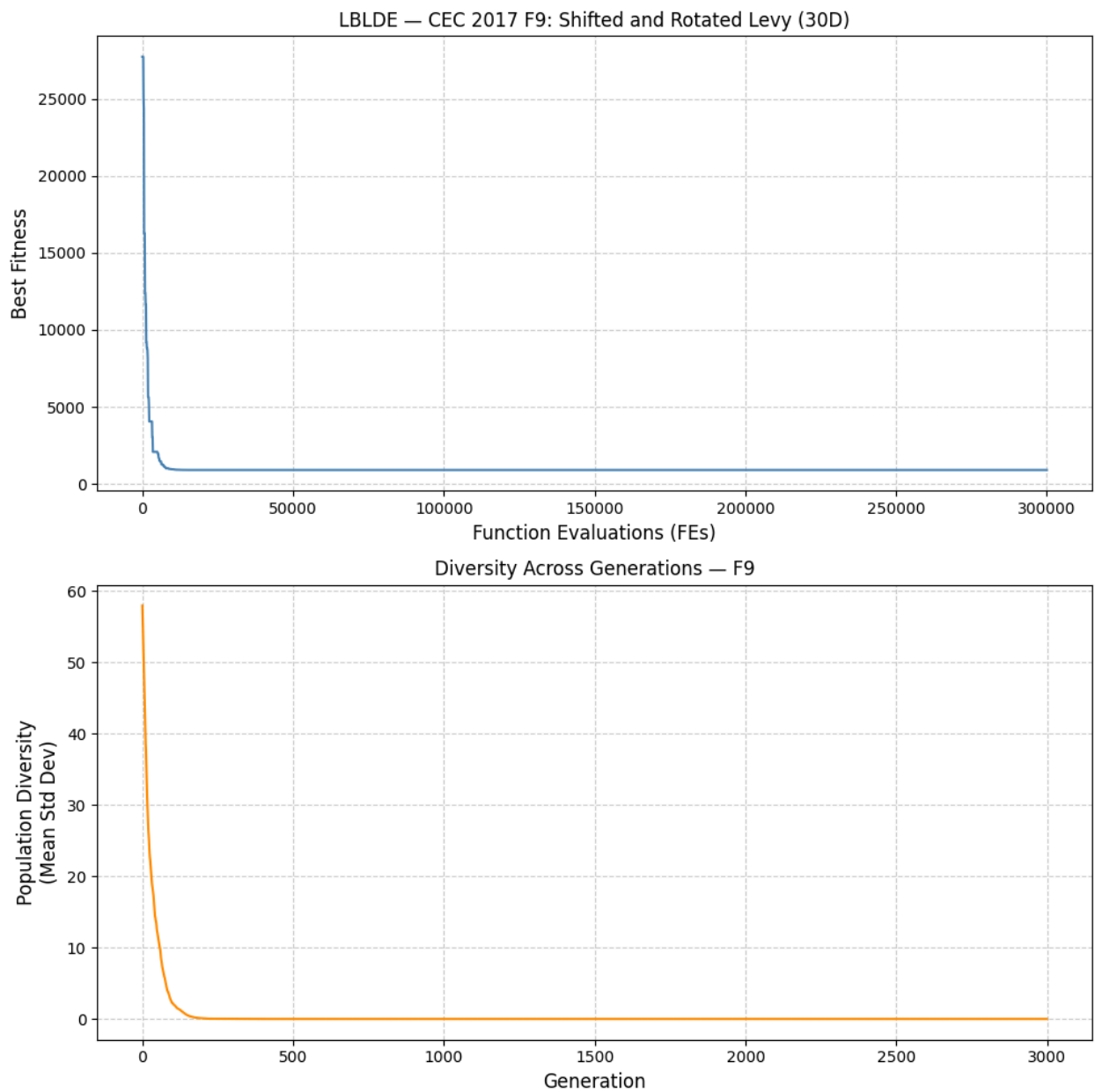
Optimization completed!

Final Best Fitness = 9.004543e+02

Total Function Evaluations = 300000

```

=====
=
Results:
  Best Fitness           : 9.004543e+02
  Global Optimum         : 9.000000e+02
  Error (|best-optimum|) : 4.543240e-01
  Best Solution (5 dim)  : [-22.77149429   1.61188536   21.68061479   51.93725496
-16.88374926]
  D^(0) (init. diversity): 5.7976e+01
=====
=
  
```



Graph saved as convergence_F9.png

```
In [20]: # =====
# F10 - Shifted and Rotated Schwefel
# =====
run_and_plot('F10', make_wrapper(simple.f10), 1000.0, "Shifted and Rotated Sch
```

```

=====
=
LBLDE_s on CEC 2017 F10 – Single Run
=====
=
Function      : Shifted and Rotated Schwefel's (30D)
Bounds       : [-100, 100]
Global optimum : 1000.0
Initial D^(0) = 5.7976e+01
Generation 0: Best Fitness = 9.881110e+03 | Diversity = 5.7976e+01
Generation 10: Best = 9.681592e+03 FES = 1100 Diversity = 5.6253e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 9.108367e+03 FES = 2100 Diversity = 5.7450e+01 k_t
= [3, 3, 2, 4]
Generation 30: Best = 8.800221e+03 FES = 3100 Diversity = 5.8104e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 8.800221e+03 FES = 4100 Diversity = 5.8349e+01 k_t
= [2, 2, 4, 3]
Generation 50: Best = 8.800221e+03 FES = 5100 Diversity = 5.8483e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 8.440353e+03 FES = 6100 Diversity = 5.9013e+01 k_t
= [4, 4, 3, 2]
Generation 70: Best = 8.030715e+03 FES = 7100 Diversity = 5.9526e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 8.030715e+03 FES = 8100 Diversity = 5.9453e+01 k_t
= [3, 2, 2, 3]
Generation 90: Best = 7.943852e+03 FES = 9100 Diversity = 6.0280e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 7.460203e+03 FES = 10100 Diversity = 6.0705e+01 k_t
= [4, 2, 2, 2]
Generation 110: Best = 7.460203e+03 FES = 11100 Diversity = 6.0995e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 7.460203e+03 FES = 12100 Diversity = 6.0950e+01 k_t
= [2, 2, 3, 2]
Generation 130: Best = 7.425649e+03 FES = 13100 Diversity = 6.1021e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 7.103629e+03 FES = 14100 Diversity = 6.1025e+01 k_t
= [2, 3, 3, 2]
Generation 150: Best = 7.103629e+03 FES = 15100 Diversity = 6.0857e+01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 7.103629e+03 FES = 16100 Diversity = 6.0400e+01 k_t
= [2, 2, 2, 2]
Generation 170: Best = 7.103629e+03 FES = 17100 Diversity = 5.9979e+01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 7.103629e+03 FES = 18100 Diversity = 5.9569e+01 k_t
= [3, 2, 4, 4]
Generation 190: Best = 6.966574e+03 FES = 19100 Diversity = 5.9176e+01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 5.963914e+03 FES = 20100 Diversity = 5.9466e+01 k_t
= [2, 3, 4, 2]
Generation 210: Best = 5.963914e+03 FES = 21100 Diversity = 5.9507e+01 k_t
= [1, 1, 1, 1]
Generation 220: Best = 5.963914e+03 FES = 22100 Diversity = 5.9030e+01 k_t
= [4, 2, 3, 2]

```

Generation 230:	Best = 5.963914e+03	FES = 23100	Diversity = 5.8879e+01	k_t
= [1, 1, 1, 1]				
Generation 240:	Best = 5.963914e+03	FES = 24100	Diversity = 5.9012e+01	k_t
= [2, 3, 4, 2]				
Generation 250:	Best = 5.963914e+03	FES = 25100	Diversity = 5.9038e+01	k_t
= [1, 1, 1, 1]				
Generation 260:	Best = 5.963914e+03	FES = 26100	Diversity = 5.8830e+01	k_t
= [3, 4, 4, 4]				
Generation 270:	Best = 5.963914e+03	FES = 27100	Diversity = 5.8418e+01	k_t
= [1, 1, 1, 1]				
Generation 280:	Best = 5.963914e+03	FES = 28100	Diversity = 5.8508e+01	k_t
= [3, 4, 4, 3]				
Generation 290:	Best = 5.963914e+03	FES = 29100	Diversity = 5.8366e+01	k_t
= [1, 1, 1, 1]				
Generation 300:	Best = 5.753626e+03	FES = 30100	Diversity = 5.8359e+01	k_t
= [2, 4, 3, 3]				
Generation 310:	Best = 5.753626e+03	FES = 31100	Diversity = 5.8542e+01	k_t
= [1, 1, 1, 1]				
Generation 320:	Best = 5.753626e+03	FES = 32100	Diversity = 5.8570e+01	k_t
= [3, 4, 3, 2]				
Generation 330:	Best = 5.753626e+03	FES = 33100	Diversity = 5.8666e+01	k_t
= [1, 1, 1, 1]				
Generation 340:	Best = 5.753626e+03	FES = 34100	Diversity = 5.8484e+01	k_t
= [4, 3, 3, 2]				
Generation 350:	Best = 5.620173e+03	FES = 35100	Diversity = 5.8686e+01	k_t
= [1, 1, 1, 1]				
Generation 360:	Best = 5.620173e+03	FES = 36100	Diversity = 5.8862e+01	k_t
= [2, 2, 3, 4]				
Generation 370:	Best = 5.296432e+03	FES = 37100	Diversity = 5.8930e+01	k_t
= [1, 1, 1, 1]				
Generation 380:	Best = 5.296432e+03	FES = 38100	Diversity = 5.8795e+01	k_t
= [3, 2, 3, 2]				
Generation 390:	Best = 5.296432e+03	FES = 39100	Diversity = 5.8780e+01	k_t
= [1, 1, 1, 1]				
Generation 400:	Best = 5.296432e+03	FES = 40100	Diversity = 5.8486e+01	k_t
= [4, 3, 3, 4]				
Generation 410:	Best = 5.296432e+03	FES = 41100	Diversity = 5.8605e+01	k_t
= [1, 1, 1, 1]				
Generation 420:	Best = 5.231516e+03	FES = 42100	Diversity = 5.8469e+01	k_t
= [3, 2, 3, 3]				
Generation 430:	Best = 5.231516e+03	FES = 43100	Diversity = 5.8311e+01	k_t
= [1, 1, 1, 1]				
Generation 440:	Best = 5.231516e+03	FES = 44100	Diversity = 5.7540e+01	k_t
= [4, 4, 4, 2]				
Generation 450:	Best = 5.183691e+03	FES = 45100	Diversity = 5.7520e+01	k_t
= [1, 1, 1, 1]				
Generation 460:	Best = 5.183691e+03	FES = 46100	Diversity = 5.7115e+01	k_t
= [4, 2, 4, 2]				
Generation 470:	Best = 5.115101e+03	FES = 47100	Diversity = 5.5367e+01	k_t
= [1, 1, 1, 1]				
Generation 480:	Best = 4.893586e+03	FES = 48100	Diversity = 5.4545e+01	k_t
= [4, 2, 2, 2]				
Generation 490:	Best = 4.743658e+03	FES = 49100	Diversity = 5.3813e+01	k_t
= [1, 1, 1, 1]				

Generation 500:	Best = 4.552943e+03	FES = 50100	Diversity = 5.1787e+01	k_t = [2, 2, 3, 4]
Generation 510:	Best = 4.522140e+03	FES = 51100	Diversity = 5.1187e+01	k_t = [1, 1, 1, 1]
Generation 520:	Best = 4.508043e+03	FES = 52100	Diversity = 4.2671e+01	k_t = [2, 4, 4, 4]
Generation 530:	Best = 4.350905e+03	FES = 53100	Diversity = 3.7964e+01	k_t = [1, 1, 1, 1]
Generation 540:	Best = 4.230336e+03	FES = 54100	Diversity = 3.3868e+01	k_t = [4, 2, 2, 2]
Generation 550:	Best = 4.230336e+03	FES = 55100	Diversity = 3.0264e+01	k_t = [1, 1, 1, 1]
Generation 560:	Best = 4.069096e+03	FES = 56100	Diversity = 1.9694e+01	k_t = [2, 2, 3, 3]
Generation 570:	Best = 4.046232e+03	FES = 57100	Diversity = 1.6525e+01	k_t = [1, 1, 1, 1]
Generation 580:	Best = 4.015557e+03	FES = 58100	Diversity = 1.3023e+01	k_t = [4, 4, 3, 4]
Generation 590:	Best = 4.015557e+03	FES = 59100	Diversity = 1.2525e+01	k_t = [1, 1, 1, 1]
Generation 600:	Best = 3.874208e+03	FES = 60100	Diversity = 1.2385e+01	k_t = [2, 3, 2, 2]
Generation 610:	Best = 3.857859e+03	FES = 61100	Diversity = 1.2296e+01	k_t = [1, 1, 1, 1]
Generation 620:	Best = 3.843612e+03	FES = 62100	Diversity = 1.2149e+01	k_t = [2, 4, 3, 2]
Generation 630:	Best = 3.796087e+03	FES = 63100	Diversity = 1.1792e+01	k_t = [1, 1, 1, 1]
Generation 640:	Best = 3.625301e+03	FES = 64100	Diversity = 1.1738e+01	k_t = [2, 2, 2, 3]
Generation 650:	Best = 3.513098e+03	FES = 65100	Diversity = 1.1668e+01	k_t = [1, 1, 1, 1]
Generation 660:	Best = 3.461241e+03	FES = 66100	Diversity = 1.1691e+01	k_t = [3, 3, 4, 4]
Generation 670:	Best = 3.376761e+03	FES = 67100	Diversity = 1.1867e+01	k_t = [1, 1, 1, 1]
Generation 680:	Best = 3.188119e+03	FES = 68100	Diversity = 1.1859e+01	k_t = [4, 3, 2, 4]
Generation 690:	Best = 3.141149e+03	FES = 69100	Diversity = 1.2034e+01	k_t = [1, 1, 1, 1]
Generation 700:	Best = 3.075650e+03	FES = 70100	Diversity = 1.2140e+01	k_t = [2, 2, 4, 2]
Generation 710:	Best = 3.075650e+03	FES = 71100	Diversity = 1.2497e+01	k_t = [1, 1, 1, 1]
Generation 720:	Best = 3.014464e+03	FES = 72100	Diversity = 1.2840e+01	k_t = [2, 3, 3, 2]
Generation 730:	Best = 2.933379e+03	FES = 73100	Diversity = 1.2957e+01	k_t = [1, 1, 1, 1]
Generation 740:	Best = 2.899166e+03	FES = 74100	Diversity = 1.2777e+01	k_t = [2, 3, 3, 4]
Generation 750:	Best = 2.835619e+03	FES = 75100	Diversity = 1.2235e+01	k_t = [1, 1, 1, 1]
Generation 760:	Best = 2.797518e+03	FES = 76100	Diversity = 8.6855e+00	k_t = [3, 2, 3, 4]

Generation 770:	Best = 2.762052e+03	FES = 77100	Diversity = 9.0426e-01	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 2.698086e+03	FES = 78100	Diversity = 6.0257e-01	k_t
= [2, 3, 2, 4]				
Generation 790:	Best = 2.684017e+03	FES = 79100	Diversity = 3.6578e-01	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 2.666470e+03	FES = 80100	Diversity = 2.7637e-01	k_t
= [3, 4, 4, 4]				
Generation 810:	Best = 2.651211e+03	FES = 81100	Diversity = 1.9560e-01	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 2.643978e+03	FES = 82100	Diversity = 1.4011e-01	k_t
= [3, 3, 2, 3]				
Generation 830:	Best = 2.640499e+03	FES = 83100	Diversity = 9.8735e-02	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 2.638909e+03	FES = 84100	Diversity = 6.4992e-02	k_t
= [2, 4, 2, 3]				
Generation 850:	Best = 2.637588e+03	FES = 85100	Diversity = 4.5768e-02	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 2.637046e+03	FES = 86100	Diversity = 3.3546e-02	k_t
= [3, 3, 3, 4]				
Generation 870:	Best = 2.636676e+03	FES = 87100	Diversity = 2.4870e-02	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 2.636459e+03	FES = 88100	Diversity = 1.7139e-02	k_t
= [3, 4, 3, 3]				
Generation 890:	Best = 2.636373e+03	FES = 89100	Diversity = 1.1857e-02	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 2.636328e+03	FES = 90100	Diversity = 8.3236e-03	k_t
= [2, 2, 4, 3]				
Generation 910:	Best = 2.636307e+03	FES = 91100	Diversity = 6.1898e-03	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 2.636288e+03	FES = 92100	Diversity = 4.3322e-03	k_t
= [2, 2, 3, 2]				
Generation 930:	Best = 2.636281e+03	FES = 93100	Diversity = 3.2204e-03	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 2.636278e+03	FES = 94100	Diversity = 2.3768e-03	k_t
= [4, 3, 4, 4]				
Generation 950:	Best = 2.636276e+03	FES = 95100	Diversity = 1.7235e-03	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 2.636275e+03	FES = 96100	Diversity = 1.2746e-03	k_t
= [3, 2, 2, 2]				
Generation 970:	Best = 2.636274e+03	FES = 97100	Diversity = 9.2681e-04	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 2.636274e+03	FES = 98100	Diversity = 6.2542e-04	k_t
= [2, 2, 2, 2]				
Generation 990:	Best = 2.636273e+03	FES = 99100	Diversity = 4.4846e-04	k_t
= [1, 1, 1, 1]				
Generation 1000:	Best = 2.636273e+03	FES = 100100	Diversity = 3.1555e-04	k_t
= [2, 3, 2, 4]				
Generation 1010:	Best = 2.636273e+03	FES = 101100	Diversity = 2.3375e-04	k_t
= [1, 1, 1, 1]				
Generation 1020:	Best = 2.636273e+03	FES = 102100	Diversity = 1.5722e-04	k_t
= [2, 3, 4, 3]				
Generation 1030:	Best = 2.636273e+03	FES = 103100	Diversity = 1.1615e-04	k_t
= [1, 1, 1, 1]				

Generation 1040: Best = 2.636273e+03 = [3, 2, 3, 4]	FES = 104100	Diversity = 8.2161e-05	k_t
Generation 1050: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 105100	Diversity = 5.7984e-05	k_t
Generation 1060: Best = 2.636273e+03 = [3, 2, 3, 4]	FES = 106100	Diversity = 4.0748e-05	k_t
Generation 1070: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 107100	Diversity = 2.8217e-05	k_t
Generation 1080: Best = 2.636273e+03 = [4, 2, 4, 3]	FES = 108100	Diversity = 2.0671e-05	k_t
Generation 1090: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 109100	Diversity = 1.5198e-05	k_t
Generation 1100: Best = 2.636273e+03 = [2, 4, 4, 3]	FES = 110100	Diversity = 1.0436e-05	k_t
Generation 1110: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 111100	Diversity = 6.9824e-06	k_t
Generation 1120: Best = 2.636273e+03 = [2, 2, 4, 2]	FES = 112100	Diversity = 4.9314e-06	k_t
Generation 1130: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 113100	Diversity = 3.6861e-06	k_t
Generation 1140: Best = 2.636273e+03 = [3, 3, 2, 2]	FES = 114100	Diversity = 2.7304e-06	k_t
Generation 1150: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 115100	Diversity = 1.9039e-06	k_t
Generation 1160: Best = 2.636273e+03 = [3, 4, 2, 3]	FES = 116100	Diversity = 1.3752e-06	k_t
Generation 1170: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 117100	Diversity = 1.0134e-06	k_t
Generation 1180: Best = 2.636273e+03 = [2, 2, 2, 4]	FES = 118100	Diversity = 7.2798e-07	k_t
Generation 1190: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 119100	Diversity = 5.5172e-07	k_t
Generation 1200: Best = 2.636273e+03 = [3, 4, 4, 3]	FES = 120100	Diversity = 3.9249e-07	k_t
Generation 1210: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 121100	Diversity = 2.8032e-07	k_t
Generation 1220: Best = 2.636273e+03 = [4, 2, 2, 2]	FES = 122100	Diversity = 1.8431e-07	k_t
Generation 1230: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 123100	Diversity = 1.3316e-07	k_t
Generation 1240: Best = 2.636273e+03 = [2, 3, 2, 3]	FES = 124100	Diversity = 1.0133e-07	k_t
Generation 1250: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 125100	Diversity = 8.7123e-08	k_t
Generation 1260: Best = 2.636273e+03 = [3, 2, 2, 2]	FES = 126100	Diversity = 7.6813e-08	k_t
Generation 1270: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 127100	Diversity = 7.2014e-08	k_t
Generation 1280: Best = 2.636273e+03 = [3, 2, 2, 3]	FES = 128100	Diversity = 6.5587e-08	k_t
Generation 1290: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 129100	Diversity = 6.4779e-08	k_t
Generation 1300: Best = 2.636273e+03 = [3, 3, 2, 3]	FES = 130100	Diversity = 6.2182e-08	k_t

Generation 1310: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 131100	Diversity = 6.0746e-08	k_t
Generation 1320: Best = 2.636273e+03 = [2, 4, 3, 3]	FES = 132100	Diversity = 5.9559e-08	k_t
Generation 1330: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 133100	Diversity = 5.8578e-08	k_t
Generation 1340: Best = 2.636273e+03 = [3, 4, 2, 4]	FES = 134100	Diversity = 5.7759e-08	k_t
Generation 1350: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 135100	Diversity = 5.7140e-08	k_t
Generation 1360: Best = 2.636273e+03 = [4, 2, 2, 3]	FES = 136100	Diversity = 5.7582e-08	k_t
Generation 1370: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 137100	Diversity = 5.6954e-08	k_t
Generation 1380: Best = 2.636273e+03 = [4, 2, 3, 3]	FES = 138100	Diversity = 5.6469e-08	k_t
Generation 1390: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 139100	Diversity = 5.6117e-08	k_t
Generation 1400: Best = 2.636273e+03 = [3, 2, 4, 4]	FES = 140100	Diversity = 5.5475e-08	k_t
Generation 1410: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 141100	Diversity = 5.3722e-08	k_t
Generation 1420: Best = 2.636273e+03 = [4, 2, 4, 2]	FES = 142100	Diversity = 5.3994e-08	k_t
Generation 1430: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 143100	Diversity = 5.4326e-08	k_t
Generation 1440: Best = 2.636273e+03 = [3, 4, 4, 3]	FES = 144100	Diversity = 5.4510e-08	k_t
Generation 1450: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 145100	Diversity = 5.4296e-08	k_t
Generation 1460: Best = 2.636273e+03 = [3, 3, 3, 4]	FES = 146100	Diversity = 5.4986e-08	k_t
Generation 1470: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 147100	Diversity = 5.6303e-08	k_t
Generation 1480: Best = 2.636273e+03 = [4, 4, 2, 2]	FES = 148100	Diversity = 5.5950e-08	k_t
Generation 1490: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 149100	Diversity = 5.5381e-08	k_t
Generation 1500: Best = 2.636273e+03 = [2, 3, 2, 2]	FES = 150100	Diversity = 5.4784e-08	k_t
Generation 1510: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 151100	Diversity = 5.4834e-08	k_t
Generation 1520: Best = 2.636273e+03 = [4, 2, 4, 2]	FES = 152100	Diversity = 5.5108e-08	k_t
Generation 1530: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 153100	Diversity = 5.4407e-08	k_t
Generation 1540: Best = 2.636273e+03 = [2, 2, 3, 4]	FES = 154100	Diversity = 5.4730e-08	k_t
Generation 1550: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 155100	Diversity = 5.4614e-08	k_t
Generation 1560: Best = 2.636273e+03 = [2, 4, 2, 4]	FES = 156100	Diversity = 5.4338e-08	k_t
Generation 1570: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 157100	Diversity = 5.4374e-08	k_t

Generation 1580: Best = 2.636273e+03 = [3, 4, 3, 4]	FES = 158100	Diversity = 5.4592e-08	k_t
Generation 1590: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 159100	Diversity = 5.4674e-08	k_t
Generation 1600: Best = 2.636273e+03 = [4, 4, 3, 4]	FES = 160100	Diversity = 5.5364e-08	k_t
Generation 1610: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 161100	Diversity = 5.5849e-08	k_t
Generation 1620: Best = 2.636273e+03 = [4, 2, 2, 4]	FES = 162100	Diversity = 5.5951e-08	k_t
Generation 1630: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 163100	Diversity = 5.5227e-08	k_t
Generation 1640: Best = 2.636273e+03 = [4, 4, 4, 2]	FES = 164100	Diversity = 5.5215e-08	k_t
Generation 1650: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 165100	Diversity = 5.5978e-08	k_t
Generation 1660: Best = 2.636273e+03 = [4, 4, 3, 2]	FES = 166100	Diversity = 5.5406e-08	k_t
Generation 1670: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 167100	Diversity = 5.5635e-08	k_t
Generation 1680: Best = 2.636273e+03 = [3, 3, 4, 4]	FES = 168100	Diversity = 5.5236e-08	k_t
Generation 1690: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 169100	Diversity = 5.4849e-08	k_t
Generation 1700: Best = 2.636273e+03 = [3, 4, 3, 2]	FES = 170100	Diversity = 5.5208e-08	k_t
Generation 1710: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 171100	Diversity = 5.5268e-08	k_t
Generation 1720: Best = 2.636273e+03 = [4, 2, 4, 2]	FES = 172100	Diversity = 5.4424e-08	k_t
Generation 1730: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 173100	Diversity = 5.4386e-08	k_t
Generation 1740: Best = 2.636273e+03 = [4, 2, 4, 3]	FES = 174100	Diversity = 5.4279e-08	k_t
Generation 1750: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 175100	Diversity = 5.3959e-08	k_t
Generation 1760: Best = 2.636273e+03 = [2, 4, 3, 4]	FES = 176100	Diversity = 5.3362e-08	k_t
Generation 1770: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 177100	Diversity = 5.3121e-08	k_t
Generation 1780: Best = 2.636273e+03 = [4, 4, 4, 3]	FES = 178100	Diversity = 5.2245e-08	k_t
Generation 1790: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 179100	Diversity = 5.1967e-08	k_t
Generation 1800: Best = 2.636273e+03 = [4, 3, 4, 4]	FES = 180100	Diversity = 5.1898e-08	k_t
Generation 1810: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 181100	Diversity = 5.1185e-08	k_t
Generation 1820: Best = 2.636273e+03 = [2, 4, 2, 3]	FES = 182100	Diversity = 5.1742e-08	k_t
Generation 1830: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 183100	Diversity = 5.1337e-08	k_t
Generation 1840: Best = 2.636273e+03 = [2, 2, 2, 2]	FES = 184100	Diversity = 5.1252e-08	k_t

Generation 1850: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 185100	Diversity = 5.1000e-08	k_t
Generation 1860: Best = 2.636273e+03 = [3, 4, 3, 2]	FES = 186100	Diversity = 5.0719e-08	k_t
Generation 1870: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 187100	Diversity = 5.0758e-08	k_t
Generation 1880: Best = 2.636273e+03 = [3, 2, 2, 3]	FES = 188100	Diversity = 5.0825e-08	k_t
Generation 1890: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 189100	Diversity = 5.0649e-08	k_t
Generation 1900: Best = 2.636273e+03 = [2, 3, 3, 3]	FES = 190100	Diversity = 5.0238e-08	k_t
Generation 1910: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 191100	Diversity = 5.0363e-08	k_t
Generation 1920: Best = 2.636273e+03 = [4, 2, 3, 3]	FES = 192100	Diversity = 5.0316e-08	k_t
Generation 1930: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 193100	Diversity = 5.0453e-08	k_t
Generation 1940: Best = 2.636273e+03 = [4, 4, 3, 4]	FES = 194100	Diversity = 4.9989e-08	k_t
Generation 1950: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 195100	Diversity = 4.9917e-08	k_t
Generation 1960: Best = 2.636273e+03 = [4, 4, 2, 3]	FES = 196100	Diversity = 4.9809e-08	k_t
Generation 1970: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 197100	Diversity = 4.9164e-08	k_t
Generation 1980: Best = 2.636273e+03 = [2, 3, 2, 4]	FES = 198100	Diversity = 4.8860e-08	k_t
Generation 1990: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 199100	Diversity = 4.8827e-08	k_t
Generation 2000: Best = 2.636273e+03 = [2, 3, 3, 2]	FES = 200100	Diversity = 4.8667e-08	k_t
Generation 2010: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 201100	Diversity = 4.8193e-08	k_t
Generation 2020: Best = 2.636273e+03 = [3, 2, 4, 2]	FES = 202100	Diversity = 4.8042e-08	k_t
Generation 2030: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 203100	Diversity = 4.8082e-08	k_t
Generation 2040: Best = 2.636273e+03 = [3, 3, 3, 3]	FES = 204100	Diversity = 4.8013e-08	k_t
Generation 2050: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 205100	Diversity = 4.8001e-08	k_t
Generation 2060: Best = 2.636273e+03 = [4, 3, 3, 3]	FES = 206100	Diversity = 4.7413e-08	k_t
Generation 2070: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 207100	Diversity = 4.7472e-08	k_t
Generation 2080: Best = 2.636273e+03 = [4, 4, 4, 3]	FES = 208100	Diversity = 4.7696e-08	k_t
Generation 2090: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 209100	Diversity = 4.7655e-08	k_t
Generation 2100: Best = 2.636273e+03 = [2, 3, 2, 4]	FES = 210100	Diversity = 4.7688e-08	k_t
Generation 2110: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 211100	Diversity = 4.7282e-08	k_t

Generation 2120: Best = 2.636273e+03 = [2, 4, 2, 4]	FES = 212100	Diversity = 4.7318e-08	k_t
Generation 2130: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 213100	Diversity = 4.7232e-08	k_t
Generation 2140: Best = 2.636273e+03 = [2, 2, 2, 3]	FES = 214100	Diversity = 4.7377e-08	k_t
Generation 2150: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 215100	Diversity = 4.7094e-08	k_t
Generation 2160: Best = 2.636273e+03 = [2, 2, 4, 2]	FES = 216100	Diversity = 4.7258e-08	k_t
Generation 2170: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 217100	Diversity = 4.7083e-08	k_t
Generation 2180: Best = 2.636273e+03 = [3, 2, 2, 3]	FES = 218100	Diversity = 4.7216e-08	k_t
Generation 2190: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 219100	Diversity = 4.7408e-08	k_t
Generation 2200: Best = 2.636273e+03 = [2, 2, 4, 3]	FES = 220100	Diversity = 4.7317e-08	k_t
Generation 2210: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 221100	Diversity = 4.7375e-08	k_t
Generation 2220: Best = 2.636273e+03 = [3, 3, 2, 2]	FES = 222100	Diversity = 4.7427e-08	k_t
Generation 2230: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 223100	Diversity = 4.6983e-08	k_t
Generation 2240: Best = 2.636273e+03 = [4, 4, 3, 4]	FES = 224100	Diversity = 4.7380e-08	k_t
Generation 2250: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 225100	Diversity = 4.7167e-08	k_t
Generation 2260: Best = 2.636273e+03 = [2, 2, 3, 4]	FES = 226100	Diversity = 4.6934e-08	k_t
Generation 2270: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 227100	Diversity = 4.7209e-08	k_t
Generation 2280: Best = 2.636273e+03 = [3, 3, 4, 2]	FES = 228100	Diversity = 4.6913e-08	k_t
Generation 2290: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 229100	Diversity = 4.6919e-08	k_t
Generation 2300: Best = 2.636273e+03 = [2, 3, 2, 4]	FES = 230100	Diversity = 4.6985e-08	k_t
Generation 2310: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 231100	Diversity = 4.6746e-08	k_t
Generation 2320: Best = 2.636273e+03 = [3, 4, 2, 3]	FES = 232100	Diversity = 4.6765e-08	k_t
Generation 2330: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 233100	Diversity = 4.6561e-08	k_t
Generation 2340: Best = 2.636273e+03 = [2, 4, 4, 2]	FES = 234100	Diversity = 4.6141e-08	k_t
Generation 2350: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 235100	Diversity = 4.6649e-08	k_t
Generation 2360: Best = 2.636273e+03 = [2, 4, 3, 3]	FES = 236100	Diversity = 4.6395e-08	k_t
Generation 2370: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 237100	Diversity = 4.6376e-08	k_t
Generation 2380: Best = 2.636273e+03 = [3, 3, 4, 3]	FES = 238100	Diversity = 4.5649e-08	k_t

Generation 2390: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 239100	Diversity = 4.5563e-08	k_t
Generation 2400: Best = 2.636273e+03 = [2, 3, 3, 2]	FES = 240100	Diversity = 4.5202e-08	k_t
Generation 2410: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 241100	Diversity = 4.5197e-08	k_t
Generation 2420: Best = 2.636273e+03 = [4, 4, 4, 2]	FES = 242100	Diversity = 4.5416e-08	k_t
Generation 2430: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 243100	Diversity = 4.5168e-08	k_t
Generation 2440: Best = 2.636273e+03 = [4, 2, 4, 4]	FES = 244100	Diversity = 4.4753e-08	k_t
Generation 2450: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 245100	Diversity = 4.4908e-08	k_t
Generation 2460: Best = 2.636273e+03 = [3, 4, 4, 4]	FES = 246100	Diversity = 4.4507e-08	k_t
Generation 2470: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 247100	Diversity = 4.4908e-08	k_t
Generation 2480: Best = 2.636273e+03 = [3, 2, 3, 3]	FES = 248100	Diversity = 4.4997e-08	k_t
Generation 2490: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 249100	Diversity = 4.5349e-08	k_t
Generation 2500: Best = 2.636273e+03 = [2, 4, 4, 3]	FES = 250100	Diversity = 4.5070e-08	k_t
Generation 2510: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 251100	Diversity = 4.5270e-08	k_t
Generation 2520: Best = 2.636273e+03 = [2, 4, 4, 4]	FES = 252100	Diversity = 4.5434e-08	k_t
Generation 2530: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 253100	Diversity = 4.5368e-08	k_t
Generation 2540: Best = 2.636273e+03 = [2, 3, 2, 2]	FES = 254100	Diversity = 4.4774e-08	k_t
Generation 2550: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 255100	Diversity = 4.4719e-08	k_t
Generation 2560: Best = 2.636273e+03 = [2, 2, 4, 4]	FES = 256100	Diversity = 4.4658e-08	k_t
Generation 2570: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 257100	Diversity = 4.4710e-08	k_t
Generation 2580: Best = 2.636273e+03 = [4, 2, 4, 3]	FES = 258100	Diversity = 4.4425e-08	k_t
Generation 2590: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 259100	Diversity = 4.4205e-08	k_t
Generation 2600: Best = 2.636273e+03 = [2, 4, 2, 2]	FES = 260100	Diversity = 4.3909e-08	k_t
Generation 2610: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 261100	Diversity = 4.3418e-08	k_t
Generation 2620: Best = 2.636273e+03 = [4, 4, 3, 4]	FES = 262100	Diversity = 4.3288e-08	k_t
Generation 2630: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 263100	Diversity = 4.3208e-08	k_t
Generation 2640: Best = 2.636273e+03 = [2, 2, 4, 3]	FES = 264100	Diversity = 4.2658e-08	k_t
Generation 2650: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 265100	Diversity = 4.2781e-08	k_t

Generation 2660: Best = 2.636273e+03 = [3, 3, 4, 2]	FES = 266100	Diversity = 4.2686e-08	k_t
Generation 2670: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 267100	Diversity = 4.2242e-08	k_t
Generation 2680: Best = 2.636273e+03 = [2, 2, 3, 3]	FES = 268100	Diversity = 4.2205e-08	k_t
Generation 2690: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 269100	Diversity = 4.2736e-08	k_t
Generation 2700: Best = 2.636273e+03 = [4, 2, 2, 4]	FES = 270100	Diversity = 4.2828e-08	k_t
Generation 2710: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 271100	Diversity = 4.3189e-08	k_t
Generation 2720: Best = 2.636273e+03 = [2, 2, 2, 2]	FES = 272100	Diversity = 4.3143e-08	k_t
Generation 2730: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 273100	Diversity = 4.2770e-08	k_t
Generation 2740: Best = 2.636273e+03 = [4, 4, 3, 2]	FES = 274100	Diversity = 4.2980e-08	k_t
Generation 2750: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 275100	Diversity = 4.2484e-08	k_t
Generation 2760: Best = 2.636273e+03 = [2, 3, 2, 4]	FES = 276100	Diversity = 4.2264e-08	k_t
Generation 2770: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 277100	Diversity = 4.2063e-08	k_t
Generation 2780: Best = 2.636273e+03 = [4, 3, 3, 3]	FES = 278100	Diversity = 4.2393e-08	k_t
Generation 2790: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 279100	Diversity = 4.2629e-08	k_t
Generation 2800: Best = 2.636273e+03 = [4, 3, 3, 3]	FES = 280100	Diversity = 4.2773e-08	k_t
Generation 2810: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 281100	Diversity = 4.2909e-08	k_t
Generation 2820: Best = 2.636273e+03 = [4, 2, 2, 3]	FES = 282100	Diversity = 4.3081e-08	k_t
Generation 2830: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 283100	Diversity = 4.3084e-08	k_t
Generation 2840: Best = 2.636273e+03 = [3, 2, 4, 2]	FES = 284100	Diversity = 4.3246e-08	k_t
Generation 2850: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 285100	Diversity = 4.3021e-08	k_t
Generation 2860: Best = 2.636273e+03 = [4, 2, 2, 4]	FES = 286100	Diversity = 4.3436e-08	k_t
Generation 2870: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 287100	Diversity = 4.3050e-08	k_t
Generation 2880: Best = 2.636273e+03 = [2, 3, 4, 4]	FES = 288100	Diversity = 4.2705e-08	k_t
Generation 2890: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 289100	Diversity = 4.2816e-08	k_t
Generation 2900: Best = 2.636273e+03 = [2, 4, 2, 2]	FES = 290100	Diversity = 4.2578e-08	k_t
Generation 2910: Best = 2.636273e+03 = [1, 1, 1, 1]	FES = 291100	Diversity = 4.2416e-08	k_t
Generation 2920: Best = 2.636273e+03 = [2, 2, 4, 2]	FES = 292100	Diversity = 4.2254e-08	k_t

Generation 2930: Best = 2.636273e+03 FES = 293100 Diversity = 4.2167e-08 k_t
 = [1, 1, 1, 1]
 Generation 2940: Best = 2.636273e+03 FES = 294100 Diversity = 4.2327e-08 k_t
 = [3, 2, 2, 3]
 Generation 2950: Best = 2.636273e+03 FES = 295100 Diversity = 4.2185e-08 k_t
 = [1, 1, 1, 1]
 Generation 2960: Best = 2.636273e+03 FES = 296100 Diversity = 4.2167e-08 k_t
 = [3, 4, 3, 4]
 Generation 2970: Best = 2.636273e+03 FES = 297100 Diversity = 4.2510e-08 k_t
 = [1, 1, 1, 1]
 Generation 2980: Best = 2.636273e+03 FES = 298100 Diversity = 4.2331e-08 k_t
 = [3, 4, 4, 2]
 Generation 2990: Best = 2.636273e+03 FES = 299100 Diversity = 4.1878e-08 k_t
 = [1, 1, 1, 1]

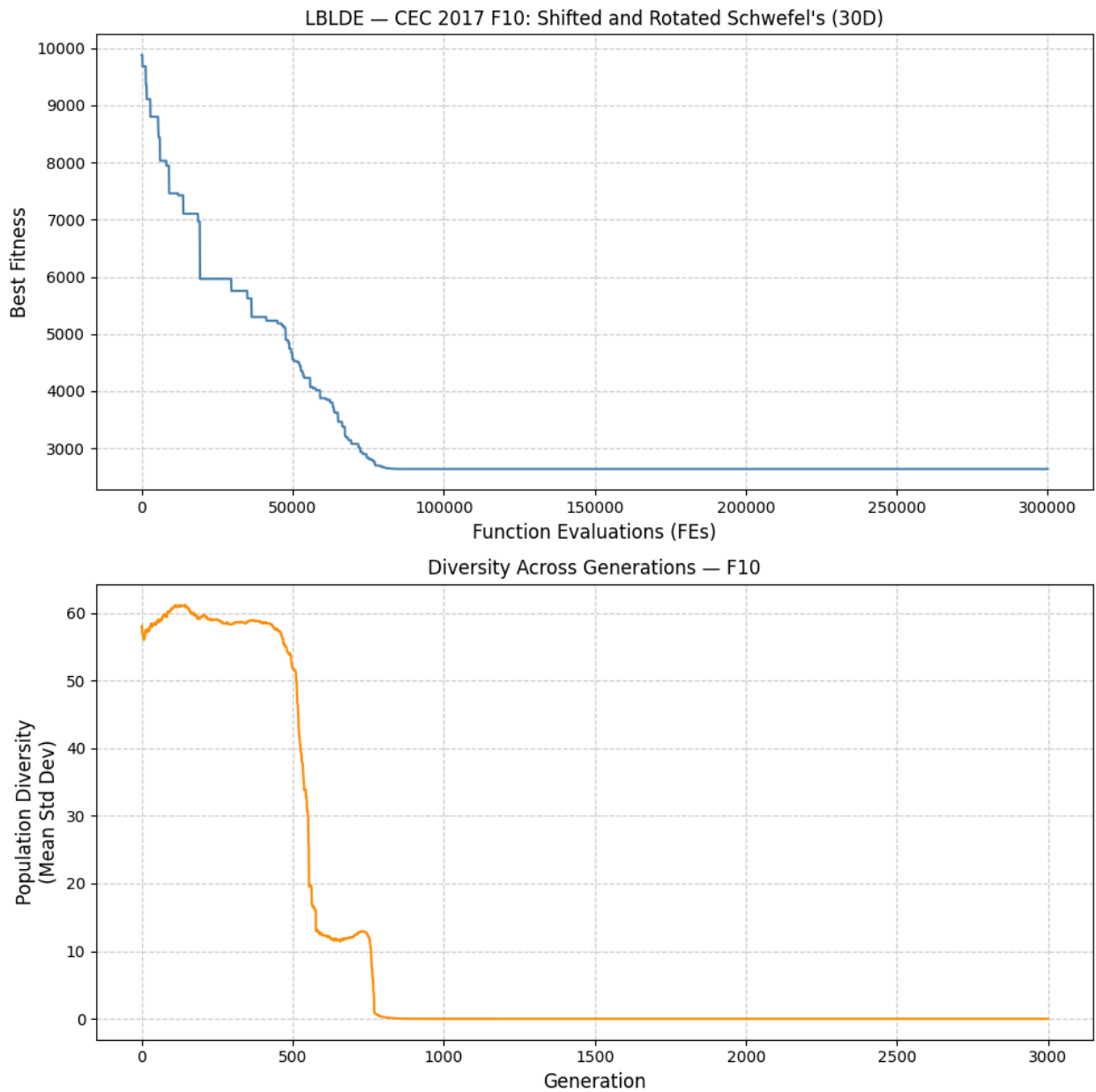
Optimization completed!

Final Best Fitness = 2.636273e+03

Total Function Evaluations = 300000

```

=====
=
Results:
  Best Fitness           : 2.636273e+03
  Global Optimum         : 1.000000e+03
  Error (|best-optimum|) : 1.636273e+03
  Best Solution (5 dim)  : [ 7.24734722  41.27534398  76.63929686 100.
-36.72957929]
  D^(0) (init. diversity): 5.7976e+01
=====
=
  
```



Graph saved as convergence_F10.png

6.3 Hybrid Functions (F11-F20)

```
In [21]: # =====
# F11 – Hybrid Function 1 (Zakharov; Rosenbrock; Rastrigin)
# =====
run_and_plot('F11', make_wrapper(hybrid.f11), 1100.0, 'Hybrid: Zakharov; Roser
```

```

=====
=
LBLDE_s on CEC 2017 F11 – Single Run
=====
=
Function      : Hybrid: Zakharov; Rosenbrock; Rastrigin (30D)
Bounds       : [-100, 100]
Global optimum : 1100.0
Initial D^(0) = 5.7976e+01
Generation 0: Best Fitness = 1.525880e+04 | Diversity = 5.7976e+01
Generation 10: Best = 9.433525e+03 FES = 1100 Diversity = 5.2752e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 5.883661e+03 FES = 2100 Diversity = 4.1358e+01 k_t
= [3, 3, 2, 2]
Generation 30: Best = 3.513822e+03 FES = 3100 Diversity = 3.5102e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 2.016492e+03 FES = 4100 Diversity = 3.0246e+01 k_t
= [4, 2, 4, 4]
Generation 50: Best = 1.552210e+03 FES = 5100 Diversity = 2.5972e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 1.491826e+03 FES = 6100 Diversity = 2.2305e+01 k_t
= [3, 3, 2, 3]
Generation 70: Best = 1.417214e+03 FES = 7100 Diversity = 1.8456e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 1.326290e+03 FES = 8100 Diversity = 1.5166e+01 k_t
= [2, 3, 4, 2]
Generation 90: Best = 1.315170e+03 FES = 9100 Diversity = 1.2930e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 1.281060e+03 FES = 10100 Diversity = 1.1691e+01 k_t
= [2, 4, 4, 4]
Generation 110: Best = 1.261144e+03 FES = 11100 Diversity = 1.1069e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 1.255745e+03 FES = 12100 Diversity = 1.0790e+01 k_t
= [4, 2, 4, 2]
Generation 130: Best = 1.247999e+03 FES = 13100 Diversity = 1.0034e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 1.247999e+03 FES = 14100 Diversity = 9.6385e+00 k_t
= [3, 2, 4, 3]
Generation 150: Best = 1.244308e+03 FES = 15100 Diversity = 9.4130e+00 k_t
= [1, 1, 1, 1]
Generation 160: Best = 1.233710e+03 FES = 16100 Diversity = 9.3640e+00 k_t
= [4, 2, 2, 4]
Generation 170: Best = 1.233710e+03 FES = 17100 Diversity = 9.2116e+00 k_t
= [1, 1, 1, 1]
Generation 180: Best = 1.232667e+03 FES = 18100 Diversity = 9.2310e+00 k_t
= [3, 3, 2, 3]
Generation 190: Best = 1.228622e+03 FES = 19100 Diversity = 9.1608e+00 k_t
= [1, 1, 1, 1]
Generation 200: Best = 1.228622e+03 FES = 20100 Diversity = 9.1514e+00 k_t
= [3, 4, 3, 2]
Generation 210: Best = 1.227011e+03 FES = 21100 Diversity = 9.2931e+00 k_t
= [1, 1, 1, 1]
Generation 220: Best = 1.227011e+03 FES = 22100 Diversity = 9.2571e+00 k_t
= [2, 4, 4, 4]

```

Generation 230:	Best = 1.227011e+03	FES = 23100	Diversity = 9.3676e+00	k_t = [1, 1, 1, 1]
Generation 240:	Best = 1.223081e+03	FES = 24100	Diversity = 9.2525e+00	k_t = [3, 3, 4, 4]
Generation 250:	Best = 1.221660e+03	FES = 25100	Diversity = 9.2575e+00	k_t = [1, 1, 1, 1]
Generation 260:	Best = 1.220161e+03	FES = 26100	Diversity = 9.1324e+00	k_t = [4, 4, 3, 3]
Generation 270:	Best = 1.220161e+03	FES = 27100	Diversity = 9.1042e+00	k_t = [1, 1, 1, 1]
Generation 280:	Best = 1.220154e+03	FES = 28100	Diversity = 9.0279e+00	k_t = [3, 2, 2, 4]
Generation 290:	Best = 1.220024e+03	FES = 29100	Diversity = 9.2160e+00	k_t = [1, 1, 1, 1]
Generation 300:	Best = 1.216181e+03	FES = 30100	Diversity = 9.1218e+00	k_t = [2, 4, 4, 4]
Generation 310:	Best = 1.216181e+03	FES = 31100	Diversity = 9.0315e+00	k_t = [1, 1, 1, 1]
Generation 320:	Best = 1.216181e+03	FES = 32100	Diversity = 8.9191e+00	k_t = [2, 3, 2, 3]
Generation 330:	Best = 1.210351e+03	FES = 33100	Diversity = 8.9304e+00	k_t = [1, 1, 1, 1]
Generation 340:	Best = 1.210351e+03	FES = 34100	Diversity = 8.8845e+00	k_t = [4, 4, 3, 3]
Generation 350:	Best = 1.208295e+03	FES = 35100	Diversity = 8.8430e+00	k_t = [1, 1, 1, 1]
Generation 360:	Best = 1.207906e+03	FES = 36100	Diversity = 8.8062e+00	k_t = [4, 3, 3, 3]
Generation 370:	Best = 1.207906e+03	FES = 37100	Diversity = 8.7781e+00	k_t = [1, 1, 1, 1]
Generation 380:	Best = 1.203871e+03	FES = 38100	Diversity = 8.6492e+00	k_t = [4, 4, 4, 3]
Generation 390:	Best = 1.203871e+03	FES = 39100	Diversity = 8.7548e+00	k_t = [1, 1, 1, 1]
Generation 400:	Best = 1.203168e+03	FES = 40100	Diversity = 8.8239e+00	k_t = [3, 3, 3, 4]
Generation 410:	Best = 1.203168e+03	FES = 41100	Diversity = 8.7429e+00	k_t = [1, 1, 1, 1]
Generation 420:	Best = 1.203168e+03	FES = 42100	Diversity = 8.7355e+00	k_t = [2, 3, 2, 4]
Generation 430:	Best = 1.197954e+03	FES = 43100	Diversity = 8.6363e+00	k_t = [1, 1, 1, 1]
Generation 440:	Best = 1.189955e+03	FES = 44100	Diversity = 8.6165e+00	k_t = [2, 2, 4, 3]
Generation 450:	Best = 1.189955e+03	FES = 45100	Diversity = 8.5570e+00	k_t = [1, 1, 1, 1]
Generation 460:	Best = 1.189112e+03	FES = 46100	Diversity = 8.6976e+00	k_t = [3, 3, 3, 2]
Generation 470:	Best = 1.186662e+03	FES = 47100	Diversity = 8.7332e+00	k_t = [1, 1, 1, 1]
Generation 480:	Best = 1.182409e+03	FES = 48100	Diversity = 8.5635e+00	k_t = [2, 4, 3, 2]
Generation 490:	Best = 1.175529e+03	FES = 49100	Diversity = 8.6692e+00	k_t = [1, 1, 1, 1]

Generation 500:	Best = 1.175529e+03	FES = 50100	Diversity = 8.6507e+00	k_t = [2, 4, 3, 2]
Generation 510:	Best = 1.174404e+03	FES = 51100	Diversity = 8.7287e+00	k_t = [1, 1, 1, 1]
Generation 520:	Best = 1.171298e+03	FES = 52100	Diversity = 8.4500e+00	k_t = [3, 4, 3, 4]
Generation 530:	Best = 1.168326e+03	FES = 53100	Diversity = 8.3350e+00	k_t = [1, 1, 1, 1]
Generation 540:	Best = 1.164577e+03	FES = 54100	Diversity = 7.9821e+00	k_t = [3, 4, 3, 4]
Generation 550:	Best = 1.162586e+03	FES = 55100	Diversity = 6.4874e+00	k_t = [1, 1, 1, 1]
Generation 560:	Best = 1.153719e+03	FES = 56100	Diversity = 3.8866e+00	k_t = [4, 2, 3, 4]
Generation 570:	Best = 1.142537e+03	FES = 57100	Diversity = 1.8508e+00	k_t = [1, 1, 1, 1]
Generation 580:	Best = 1.137019e+03	FES = 58100	Diversity = 1.8166e+00	k_t = [4, 3, 2, 3]
Generation 590:	Best = 1.129617e+03	FES = 59100	Diversity = 1.5752e+00	k_t = [1, 1, 1, 1]
Generation 600:	Best = 1.123516e+03	FES = 60100	Diversity = 1.3071e+00	k_t = [2, 3, 4, 4]
Generation 610:	Best = 1.120393e+03	FES = 61100	Diversity = 1.1474e+00	k_t = [1, 1, 1, 1]
Generation 620:	Best = 1.118875e+03	FES = 62100	Diversity = 8.9133e-01	k_t = [4, 3, 2, 2]
Generation 630:	Best = 1.118398e+03	FES = 63100	Diversity = 7.1547e-01	k_t = [1, 1, 1, 1]
Generation 640:	Best = 1.117415e+03	FES = 64100	Diversity = 5.7078e-01	k_t = [3, 3, 2, 2]
Generation 650:	Best = 1.116948e+03	FES = 65100	Diversity = 4.3510e-01	k_t = [1, 1, 1, 1]
Generation 660:	Best = 1.116684e+03	FES = 66100	Diversity = 3.2136e-01	k_t = [4, 4, 4, 3]
Generation 670:	Best = 1.116586e+03	FES = 67100	Diversity = 2.4240e-01	k_t = [1, 1, 1, 1]
Generation 680:	Best = 1.116464e+03	FES = 68100	Diversity = 1.8926e-01	k_t = [2, 4, 4, 2]
Generation 690:	Best = 1.116384e+03	FES = 69100	Diversity = 1.5642e-01	k_t = [1, 1, 1, 1]
Generation 700:	Best = 1.116332e+03	FES = 70100	Diversity = 1.2337e-01	k_t = [4, 3, 4, 3]
Generation 710:	Best = 1.116287e+03	FES = 71100	Diversity = 9.8001e-02	k_t = [1, 1, 1, 1]
Generation 720:	Best = 1.116245e+03	FES = 72100	Diversity = 8.9724e-02	k_t = [2, 3, 4, 2]
Generation 730:	Best = 1.116224e+03	FES = 73100	Diversity = 8.2851e-02	k_t = [1, 1, 1, 1]
Generation 740:	Best = 1.116195e+03	FES = 74100	Diversity = 7.1421e-02	k_t = [4, 4, 4, 4]
Generation 750:	Best = 1.116179e+03	FES = 75100	Diversity = 5.8142e-02	k_t = [1, 1, 1, 1]
Generation 760:	Best = 1.116160e+03	FES = 76100	Diversity = 5.0723e-02	k_t = [4, 3, 2, 4]

Generation 770:	Best = 1.116140e+03	FES = 77100	Diversity = 4.6659e-02	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 1.116122e+03	FES = 78100	Diversity = 4.2654e-02	k_t
= [2, 4, 3, 2]				
Generation 790:	Best = 1.116111e+03	FES = 79100	Diversity = 3.8307e-02	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 1.116101e+03	FES = 80100	Diversity = 3.4658e-02	k_t
= [3, 3, 2, 4]				
Generation 810:	Best = 1.116094e+03	FES = 81100	Diversity = 2.9630e-02	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 1.116086e+03	FES = 82100	Diversity = 2.4636e-02	k_t
= [2, 3, 4, 4]				
Generation 830:	Best = 1.116081e+03	FES = 83100	Diversity = 2.1034e-02	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 1.116076e+03	FES = 84100	Diversity = 2.1805e-02	k_t
= [2, 3, 4, 4]				
Generation 850:	Best = 1.116068e+03	FES = 85100	Diversity = 2.0950e-02	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 1.116063e+03	FES = 86100	Diversity = 2.2034e-02	k_t
= [3, 3, 2, 2]				
Generation 870:	Best = 1.116055e+03	FES = 87100	Diversity = 2.1997e-02	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 1.116051e+03	FES = 88100	Diversity = 2.3562e-02	k_t
= [2, 3, 4, 2]				
Generation 890:	Best = 1.116044e+03	FES = 89100	Diversity = 2.4066e-02	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 1.116036e+03	FES = 90100	Diversity = 2.3467e-02	k_t
= [2, 2, 4, 3]				
Generation 910:	Best = 1.116030e+03	FES = 91100	Diversity = 2.5655e-02	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 1.116021e+03	FES = 92100	Diversity = 2.8187e-02	k_t
= [4, 4, 2, 4]				
Generation 930:	Best = 1.116015e+03	FES = 93100	Diversity = 2.7454e-02	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 1.116007e+03	FES = 94100	Diversity = 2.6966e-02	k_t
= [3, 3, 3, 2]				
Generation 950:	Best = 1.115995e+03	FES = 95100	Diversity = 2.9581e-02	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 1.115988e+03	FES = 96100	Diversity = 3.2012e-02	k_t
= [3, 2, 2, 2]				
Generation 970:	Best = 1.115979e+03	FES = 97100	Diversity = 2.9664e-02	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 1.115967e+03	FES = 98100	Diversity = 3.1305e-02	k_t
= [2, 4, 4, 3]				
Generation 990:	Best = 1.115961e+03	FES = 99100	Diversity = 3.0480e-02	k_t
= [1, 1, 1, 1]				
Generation 1000:	Best = 1.115951e+03	FES = 100100	Diversity = 3.4353e-02	k_t
= [2, 4, 3, 4]				
Generation 1010:	Best = 1.115936e+03	FES = 101100	Diversity = 3.2640e-02	k_t
= [1, 1, 1, 1]				
Generation 1020:	Best = 1.115930e+03	FES = 102100	Diversity = 3.5241e-02	k_t
= [4, 3, 2, 2]				
Generation 1030:	Best = 1.115916e+03	FES = 103100	Diversity = 3.4606e-02	k_t
= [1, 1, 1, 1]				

Generation 1040: Best = 1.115907e+03 = [3, 2, 2, 4]	FES = 104100	Diversity = 3.5584e-02	k_t
Generation 1050: Best = 1.115902e+03 = [1, 1, 1, 1]	FES = 105100	Diversity = 3.3235e-02	k_t
Generation 1060: Best = 1.115891e+03 = [4, 2, 4, 4]	FES = 106100	Diversity = 3.2457e-02	k_t
Generation 1070: Best = 1.115883e+03 = [1, 1, 1, 1]	FES = 107100	Diversity = 3.1488e-02	k_t
Generation 1080: Best = 1.115873e+03 = [2, 2, 3, 2]	FES = 108100	Diversity = 2.9438e-02	k_t
Generation 1090: Best = 1.115869e+03 = [1, 1, 1, 1]	FES = 109100	Diversity = 2.6126e-02	k_t
Generation 1100: Best = 1.115862e+03 = [3, 3, 2, 2]	FES = 110100	Diversity = 2.4633e-02	k_t
Generation 1110: Best = 1.115855e+03 = [1, 1, 1, 1]	FES = 111100	Diversity = 2.4485e-02	k_t
Generation 1120: Best = 1.115848e+03 = [4, 2, 4, 4]	FES = 112100	Diversity = 2.4680e-02	k_t
Generation 1130: Best = 1.115844e+03 = [1, 1, 1, 1]	FES = 113100	Diversity = 2.3433e-02	k_t
Generation 1140: Best = 1.115838e+03 = [4, 4, 4, 3]	FES = 114100	Diversity = 2.5132e-02	k_t
Generation 1150: Best = 1.115831e+03 = [1, 1, 1, 1]	FES = 115100	Diversity = 2.3504e-02	k_t
Generation 1160: Best = 1.115824e+03 = [2, 2, 3, 2]	FES = 116100	Diversity = 2.5201e-02	k_t
Generation 1170: Best = 1.115815e+03 = [1, 1, 1, 1]	FES = 117100	Diversity = 2.7880e-02	k_t
Generation 1180: Best = 1.115806e+03 = [2, 2, 3, 2]	FES = 118100	Diversity = 2.9629e-02	k_t
Generation 1190: Best = 1.115797e+03 = [1, 1, 1, 1]	FES = 119100	Diversity = 2.9951e-02	k_t
Generation 1200: Best = 1.115788e+03 = [2, 2, 4, 3]	FES = 120100	Diversity = 2.9689e-02	k_t
Generation 1210: Best = 1.115777e+03 = [1, 1, 1, 1]	FES = 121100	Diversity = 3.0503e-02	k_t
Generation 1220: Best = 1.115770e+03 = [4, 4, 4, 3]	FES = 122100	Diversity = 3.2070e-02	k_t
Generation 1230: Best = 1.115756e+03 = [1, 1, 1, 1]	FES = 123100	Diversity = 3.1879e-02	k_t
Generation 1240: Best = 1.115750e+03 = [2, 2, 4, 2]	FES = 124100	Diversity = 3.3358e-02	k_t
Generation 1250: Best = 1.115740e+03 = [1, 1, 1, 1]	FES = 125100	Diversity = 3.1556e-02	k_t
Generation 1260: Best = 1.115727e+03 = [2, 4, 2, 4]	FES = 126100	Diversity = 3.1088e-02	k_t
Generation 1270: Best = 1.115715e+03 = [1, 1, 1, 1]	FES = 127100	Diversity = 3.1202e-02	k_t
Generation 1280: Best = 1.115710e+03 = [3, 2, 2, 4]	FES = 128100	Diversity = 3.3076e-02	k_t
Generation 1290: Best = 1.115700e+03 = [1, 1, 1, 1]	FES = 129100	Diversity = 2.8549e-02	k_t
Generation 1300: Best = 1.115691e+03 = [4, 2, 4, 3]	FES = 130100	Diversity = 2.5731e-02	k_t

Generation 1310: Best = 1.115686e+03 = [1, 1, 1, 1]	FES = 131100	Diversity = 2.4850e-02	k_t
Generation 1320: Best = 1.115677e+03 = [2, 2, 2, 2]	FES = 132100	Diversity = 2.5315e-02	k_t
Generation 1330: Best = 1.115670e+03 = [1, 1, 1, 1]	FES = 133100	Diversity = 2.4929e-02	k_t
Generation 1340: Best = 1.115661e+03 = [2, 3, 3, 4]	FES = 134100	Diversity = 2.4337e-02	k_t
Generation 1350: Best = 1.115654e+03 = [1, 1, 1, 1]	FES = 135100	Diversity = 2.2793e-02	k_t
Generation 1360: Best = 1.115649e+03 = [2, 4, 2, 4]	FES = 136100	Diversity = 2.1768e-02	k_t
Generation 1370: Best = 1.115638e+03 = [1, 1, 1, 1]	FES = 137100	Diversity = 2.6192e-02	k_t
Generation 1380: Best = 1.115633e+03 = [2, 3, 2, 4]	FES = 138100	Diversity = 2.5806e-02	k_t
Generation 1390: Best = 1.115626e+03 = [1, 1, 1, 1]	FES = 139100	Diversity = 2.4196e-02	k_t
Generation 1400: Best = 1.115617e+03 = [2, 4, 2, 2]	FES = 140100	Diversity = 2.7710e-02	k_t
Generation 1410: Best = 1.115611e+03 = [1, 1, 1, 1]	FES = 141100	Diversity = 2.7619e-02	k_t
Generation 1420: Best = 1.115601e+03 = [3, 4, 2, 4]	FES = 142100	Diversity = 2.4944e-02	k_t
Generation 1430: Best = 1.115595e+03 = [1, 1, 1, 1]	FES = 143100	Diversity = 2.4192e-02	k_t
Generation 1440: Best = 1.115589e+03 = [4, 4, 3, 2]	FES = 144100	Diversity = 2.2758e-02	k_t
Generation 1450: Best = 1.115582e+03 = [1, 1, 1, 1]	FES = 145100	Diversity = 2.3705e-02	k_t
Generation 1460: Best = 1.115575e+03 = [2, 3, 2, 3]	FES = 146100	Diversity = 2.2702e-02	k_t
Generation 1470: Best = 1.115568e+03 = [1, 1, 1, 1]	FES = 147100	Diversity = 2.2760e-02	k_t
Generation 1480: Best = 1.115562e+03 = [3, 3, 3, 2]	FES = 148100	Diversity = 2.6706e-02	k_t
Generation 1490: Best = 1.115554e+03 = [1, 1, 1, 1]	FES = 149100	Diversity = 2.7794e-02	k_t
Generation 1500: Best = 1.115545e+03 = [4, 2, 2, 4]	FES = 150100	Diversity = 2.6838e-02	k_t
Generation 1510: Best = 1.115540e+03 = [1, 1, 1, 1]	FES = 151100	Diversity = 2.7187e-02	k_t
Generation 1520: Best = 1.115526e+03 = [4, 4, 3, 3]	FES = 152100	Diversity = 2.7811e-02	k_t
Generation 1530: Best = 1.115522e+03 = [1, 1, 1, 1]	FES = 153100	Diversity = 2.8135e-02	k_t
Generation 1540: Best = 1.115515e+03 = [4, 3, 3, 3]	FES = 154100	Diversity = 2.6909e-02	k_t
Generation 1550: Best = 1.115509e+03 = [1, 1, 1, 1]	FES = 155100	Diversity = 2.5134e-02	k_t
Generation 1560: Best = 1.115501e+03 = [3, 3, 2, 2]	FES = 156100	Diversity = 2.4615e-02	k_t
Generation 1570: Best = 1.115492e+03 = [1, 1, 1, 1]	FES = 157100	Diversity = 2.8741e-02	k_t

Generation 1580: Best = 1.115486e+03	FES = 158100	Diversity = 2.7012e-02	k_t
= [3, 3, 4, 3]			
Generation 1590: Best = 1.115478e+03	FES = 159100	Diversity = 2.5635e-02	k_t
= [1, 1, 1, 1]			
Generation 1600: Best = 1.115471e+03	FES = 160100	Diversity = 2.4544e-02	k_t
= [4, 2, 4, 3]			
Generation 1610: Best = 1.115465e+03	FES = 161100	Diversity = 2.1828e-02	k_t
= [1, 1, 1, 1]			
Generation 1620: Best = 1.115459e+03	FES = 162100	Diversity = 1.9481e-02	k_t
= [4, 4, 3, 4]			
Generation 1630: Best = 1.115454e+03	FES = 163100	Diversity = 1.9800e-02	k_t
= [1, 1, 1, 1]			
Generation 1640: Best = 1.115448e+03	FES = 164100	Diversity = 1.7872e-02	k_t
= [2, 3, 2, 2]			
Generation 1650: Best = 1.115444e+03	FES = 165100	Diversity = 1.8348e-02	k_t
= [1, 1, 1, 1]			
Generation 1660: Best = 1.115435e+03	FES = 166100	Diversity = 2.1969e-02	k_t
= [4, 4, 4, 3]			
Generation 1670: Best = 1.115426e+03	FES = 167100	Diversity = 2.3023e-02	k_t
= [1, 1, 1, 1]			
Generation 1680: Best = 1.115421e+03	FES = 168100	Diversity = 2.2921e-02	k_t
= [2, 2, 2, 4]			
Generation 1690: Best = 1.115415e+03	FES = 169100	Diversity = 2.1810e-02	k_t
= [1, 1, 1, 1]			
Generation 1700: Best = 1.115410e+03	FES = 170100	Diversity = 2.2075e-02	k_t
= [2, 2, 3, 3]			
Generation 1710: Best = 1.115403e+03	FES = 171100	Diversity = 2.0660e-02	k_t
= [1, 1, 1, 1]			
Generation 1720: Best = 1.115397e+03	FES = 172100	Diversity = 2.1647e-02	k_t
= [2, 2, 2, 2]			
Generation 1730: Best = 1.115391e+03	FES = 173100	Diversity = 2.3546e-02	k_t
= [1, 1, 1, 1]			
Generation 1740: Best = 1.115380e+03	FES = 174100	Diversity = 2.5280e-02	k_t
= [4, 3, 4, 4]			
Generation 1750: Best = 1.115372e+03	FES = 175100	Diversity = 2.4524e-02	k_t
= [1, 1, 1, 1]			
Generation 1760: Best = 1.115366e+03	FES = 176100	Diversity = 2.3694e-02	k_t
= [3, 2, 4, 2]			
Generation 1770: Best = 1.115359e+03	FES = 177100	Diversity = 2.6259e-02	k_t
= [1, 1, 1, 1]			
Generation 1780: Best = 1.115351e+03	FES = 178100	Diversity = 2.5902e-02	k_t
= [3, 2, 4, 3]			
Generation 1790: Best = 1.115343e+03	FES = 179100	Diversity = 2.4005e-02	k_t
= [1, 1, 1, 1]			
Generation 1800: Best = 1.115338e+03	FES = 180100	Diversity = 2.4099e-02	k_t
= [3, 4, 4, 2]			
Generation 1810: Best = 1.115331e+03	FES = 181100	Diversity = 2.2804e-02	k_t
= [1, 1, 1, 1]			
Generation 1820: Best = 1.115324e+03	FES = 182100	Diversity = 2.2649e-02	k_t
= [2, 3, 4, 3]			
Generation 1830: Best = 1.115319e+03	FES = 183100	Diversity = 2.2405e-02	k_t
= [1, 1, 1, 1]			
Generation 1840: Best = 1.115312e+03	FES = 184100	Diversity = 2.1728e-02	k_t
= [3, 4, 2, 3]			

Generation 1850: Best = 1.115308e+03 = [1, 1, 1, 1]	FES = 185100	Diversity = 2.0630e-02	k_t
Generation 1860: Best = 1.115300e+03 = [2, 3, 4, 4]	FES = 186100	Diversity = 2.3793e-02	k_t
Generation 1870: Best = 1.115295e+03 = [1, 1, 1, 1]	FES = 187100	Diversity = 2.5358e-02	k_t
Generation 1880: Best = 1.115290e+03 = [2, 2, 2, 3]	FES = 188100	Diversity = 2.3074e-02	k_t
Generation 1890: Best = 1.115284e+03 = [1, 1, 1, 1]	FES = 189100	Diversity = 2.2693e-02	k_t
Generation 1900: Best = 1.115277e+03 = [2, 3, 2, 2]	FES = 190100	Diversity = 2.4246e-02	k_t
Generation 1910: Best = 1.115269e+03 = [1, 1, 1, 1]	FES = 191100	Diversity = 2.4644e-02	k_t
Generation 1920: Best = 1.115263e+03 = [4, 4, 2, 3]	FES = 192100	Diversity = 2.6240e-02	k_t
Generation 1930: Best = 1.115254e+03 = [1, 1, 1, 1]	FES = 193100	Diversity = 2.5114e-02	k_t
Generation 1940: Best = 1.115247e+03 = [2, 4, 2, 2]	FES = 194100	Diversity = 2.3213e-02	k_t
Generation 1950: Best = 1.115241e+03 = [1, 1, 1, 1]	FES = 195100	Diversity = 2.2053e-02	k_t
Generation 1960: Best = 1.115234e+03 = [4, 3, 2, 2]	FES = 196100	Diversity = 2.2010e-02	k_t
Generation 1970: Best = 1.115228e+03 = [1, 1, 1, 1]	FES = 197100	Diversity = 1.9495e-02	k_t
Generation 1980: Best = 1.115224e+03 = [3, 4, 3, 2]	FES = 198100	Diversity = 1.9630e-02	k_t
Generation 1990: Best = 1.115219e+03 = [1, 1, 1, 1]	FES = 199100	Diversity = 1.9187e-02	k_t
Generation 2000: Best = 1.115213e+03 = [4, 4, 4, 4]	FES = 200100	Diversity = 1.9796e-02	k_t
Generation 2010: Best = 1.115209e+03 = [1, 1, 1, 1]	FES = 201100	Diversity = 2.0741e-02	k_t
Generation 2020: Best = 1.115201e+03 = [2, 3, 4, 3]	FES = 202100	Diversity = 2.2411e-02	k_t
Generation 2030: Best = 1.115198e+03 = [1, 1, 1, 1]	FES = 203100	Diversity = 2.0089e-02	k_t
Generation 2040: Best = 1.115190e+03 = [2, 2, 4, 3]	FES = 204100	Diversity = 1.9639e-02	k_t
Generation 2050: Best = 1.115182e+03 = [1, 1, 1, 1]	FES = 205100	Diversity = 2.3269e-02	k_t
Generation 2060: Best = 1.115177e+03 = [2, 4, 4, 4]	FES = 206100	Diversity = 2.3246e-02	k_t
Generation 2070: Best = 1.115170e+03 = [1, 1, 1, 1]	FES = 207100	Diversity = 2.2364e-02	k_t
Generation 2080: Best = 1.115163e+03 = [3, 2, 3, 2]	FES = 208100	Diversity = 2.3257e-02	k_t
Generation 2090: Best = 1.115157e+03 = [1, 1, 1, 1]	FES = 209100	Diversity = 2.2893e-02	k_t
Generation 2100: Best = 1.115148e+03 = [4, 3, 2, 2]	FES = 210100	Diversity = 2.2362e-02	k_t
Generation 2110: Best = 1.115143e+03 = [1, 1, 1, 1]	FES = 211100	Diversity = 2.2926e-02	k_t

Generation 2120: Best = 1.115138e+03 = [2, 3, 3, 3]	FES = 212100	Diversity = 2.1168e-02	k_t
Generation 2130: Best = 1.115133e+03 = [1, 1, 1, 1]	FES = 213100	Diversity = 2.0426e-02	k_t
Generation 2140: Best = 1.115127e+03 = [3, 2, 2, 2]	FES = 214100	Diversity = 1.9662e-02	k_t
Generation 2150: Best = 1.115122e+03 = [1, 1, 1, 1]	FES = 215100	Diversity = 1.8453e-02	k_t
Generation 2160: Best = 1.115117e+03 = [2, 2, 4, 2]	FES = 216100	Diversity = 1.8188e-02	k_t
Generation 2170: Best = 1.115113e+03 = [1, 1, 1, 1]	FES = 217100	Diversity = 1.9124e-02	k_t
Generation 2180: Best = 1.115106e+03 = [3, 4, 3, 3]	FES = 218100	Diversity = 1.8862e-02	k_t
Generation 2190: Best = 1.115101e+03 = [1, 1, 1, 1]	FES = 219100	Diversity = 2.1231e-02	k_t
Generation 2200: Best = 1.115096e+03 = [4, 4, 4, 3]	FES = 220100	Diversity = 2.0194e-02	k_t
Generation 2210: Best = 1.115090e+03 = [1, 1, 1, 1]	FES = 221100	Diversity = 2.0995e-02	k_t
Generation 2220: Best = 1.115082e+03 = [4, 3, 2, 3]	FES = 222100	Diversity = 2.1424e-02	k_t
Generation 2230: Best = 1.115076e+03 = [1, 1, 1, 1]	FES = 223100	Diversity = 2.3848e-02	k_t
Generation 2240: Best = 1.115069e+03 = [3, 3, 3, 4]	FES = 224100	Diversity = 2.2606e-02	k_t
Generation 2250: Best = 1.115064e+03 = [1, 1, 1, 1]	FES = 225100	Diversity = 2.0289e-02	k_t
Generation 2260: Best = 1.115061e+03 = [2, 2, 2, 3]	FES = 226100	Diversity = 1.7416e-02	k_t
Generation 2270: Best = 1.115056e+03 = [1, 1, 1, 1]	FES = 227100	Diversity = 1.7795e-02	k_t
Generation 2280: Best = 1.115051e+03 = [4, 2, 2, 4]	FES = 228100	Diversity = 1.6871e-02	k_t
Generation 2290: Best = 1.115046e+03 = [1, 1, 1, 1]	FES = 229100	Diversity = 1.8643e-02	k_t
Generation 2300: Best = 1.115042e+03 = [2, 4, 3, 3]	FES = 230100	Diversity = 1.8168e-02	k_t
Generation 2310: Best = 1.115038e+03 = [1, 1, 1, 1]	FES = 231100	Diversity = 1.8623e-02	k_t
Generation 2320: Best = 1.115028e+03 = [4, 2, 2, 3]	FES = 232100	Diversity = 1.9805e-02	k_t
Generation 2330: Best = 1.115023e+03 = [1, 1, 1, 1]	FES = 233100	Diversity = 2.2298e-02	k_t
Generation 2340: Best = 1.115016e+03 = [4, 3, 4, 2]	FES = 234100	Diversity = 2.1506e-02	k_t
Generation 2350: Best = 1.115010e+03 = [1, 1, 1, 1]	FES = 235100	Diversity = 2.0963e-02	k_t
Generation 2360: Best = 1.115005e+03 = [4, 4, 3, 4]	FES = 236100	Diversity = 1.9874e-02	k_t
Generation 2370: Best = 1.114999e+03 = [1, 1, 1, 1]	FES = 237100	Diversity = 2.2133e-02	k_t
Generation 2380: Best = 1.114993e+03 = [3, 4, 3, 4]	FES = 238100	Diversity = 2.0956e-02	k_t

Generation 2390: Best = 1.114990e+03 = [1, 1, 1, 1]	FES = 239100	Diversity = 1.9919e-02	k_t
Generation 2400: Best = 1.114983e+03 = [3, 2, 2, 2]	FES = 240100	Diversity = 2.0492e-02	k_t
Generation 2410: Best = 1.114977e+03 = [1, 1, 1, 1]	FES = 241100	Diversity = 2.0375e-02	k_t
Generation 2420: Best = 1.114972e+03 = [3, 3, 4, 2]	FES = 242100	Diversity = 2.0201e-02	k_t
Generation 2430: Best = 1.114966e+03 = [1, 1, 1, 1]	FES = 243100	Diversity = 1.9812e-02	k_t
Generation 2440: Best = 1.114960e+03 = [3, 2, 2, 2]	FES = 244100	Diversity = 2.0105e-02	k_t
Generation 2450: Best = 1.114955e+03 = [1, 1, 1, 1]	FES = 245100	Diversity = 1.7701e-02	k_t
Generation 2460: Best = 1.114952e+03 = [3, 4, 4, 3]	FES = 246100	Diversity = 1.7400e-02	k_t
Generation 2470: Best = 1.114948e+03 = [1, 1, 1, 1]	FES = 247100	Diversity = 1.4832e-02	k_t
Generation 2480: Best = 1.114945e+03 = [4, 2, 2, 2]	FES = 248100	Diversity = 1.4675e-02	k_t
Generation 2490: Best = 1.114942e+03 = [1, 1, 1, 1]	FES = 249100	Diversity = 1.3489e-02	k_t
Generation 2500: Best = 1.114938e+03 = [3, 2, 4, 2]	FES = 250100	Diversity = 1.4891e-02	k_t
Generation 2510: Best = 1.114933e+03 = [1, 1, 1, 1]	FES = 251100	Diversity = 1.5622e-02	k_t
Generation 2520: Best = 1.114929e+03 = [2, 4, 3, 2]	FES = 252100	Diversity = 1.5506e-02	k_t
Generation 2530: Best = 1.114926e+03 = [1, 1, 1, 1]	FES = 253100	Diversity = 1.7774e-02	k_t
Generation 2540: Best = 1.114921e+03 = [4, 2, 3, 2]	FES = 254100	Diversity = 1.7097e-02	k_t
Generation 2550: Best = 1.114916e+03 = [1, 1, 1, 1]	FES = 255100	Diversity = 1.5524e-02	k_t
Generation 2560: Best = 1.114912e+03 = [3, 2, 4, 2]	FES = 256100	Diversity = 1.4532e-02	k_t
Generation 2570: Best = 1.114908e+03 = [1, 1, 1, 1]	FES = 257100	Diversity = 1.6058e-02	k_t
Generation 2580: Best = 1.114904e+03 = [3, 4, 3, 2]	FES = 258100	Diversity = 1.5397e-02	k_t
Generation 2590: Best = 1.114900e+03 = [1, 1, 1, 1]	FES = 259100	Diversity = 1.6567e-02	k_t
Generation 2600: Best = 1.114895e+03 = [3, 3, 3, 3]	FES = 260100	Diversity = 1.4652e-02	k_t
Generation 2610: Best = 1.114892e+03 = [1, 1, 1, 1]	FES = 261100	Diversity = 1.4013e-02	k_t
Generation 2620: Best = 1.114888e+03 = [3, 2, 2, 2]	FES = 262100	Diversity = 1.7612e-02	k_t
Generation 2630: Best = 1.114882e+03 = [1, 1, 1, 1]	FES = 263100	Diversity = 1.9189e-02	k_t
Generation 2640: Best = 1.114876e+03 = [4, 2, 4, 2]	FES = 264100	Diversity = 2.2698e-02	k_t
Generation 2650: Best = 1.114867e+03 = [1, 1, 1, 1]	FES = 265100	Diversity = 2.3519e-02	k_t

Generation 2660: Best = 1.114863e+03	FES = 266100	Diversity = 2.1861e-02	k_t
= [2, 2, 2, 3]			
Generation 2670: Best = 1.114856e+03	FES = 267100	Diversity = 1.9841e-02	k_t
= [1, 1, 1, 1]			
Generation 2680: Best = 1.114852e+03	FES = 268100	Diversity = 1.9563e-02	k_t
= [3, 4, 3, 2]			
Generation 2690: Best = 1.114846e+03	FES = 269100	Diversity = 1.7361e-02	k_t
= [1, 1, 1, 1]			
Generation 2700: Best = 1.114841e+03	FES = 270100	Diversity = 1.7410e-02	k_t
= [2, 2, 3, 3]			
Generation 2710: Best = 1.114838e+03	FES = 271100	Diversity = 1.5056e-02	k_t
= [1, 1, 1, 1]			
Generation 2720: Best = 1.114835e+03	FES = 272100	Diversity = 1.3984e-02	k_t
= [3, 4, 2, 3]			
Generation 2730: Best = 1.114829e+03	FES = 273100	Diversity = 1.4991e-02	k_t
= [1, 1, 1, 1]			
Generation 2740: Best = 1.114824e+03	FES = 274100	Diversity = 1.4672e-02	k_t
= [4, 2, 2, 4]			
Generation 2750: Best = 1.114822e+03	FES = 275100	Diversity = 1.6112e-02	k_t
= [1, 1, 1, 1]			
Generation 2760: Best = 1.114819e+03	FES = 276100	Diversity = 1.3896e-02	k_t
= [4, 3, 4, 2]			
Generation 2770: Best = 1.114816e+03	FES = 277100	Diversity = 1.3574e-02	k_t
= [1, 1, 1, 1]			
Generation 2780: Best = 1.114812e+03	FES = 278100	Diversity = 1.3333e-02	k_t
= [3, 4, 4, 4]			
Generation 2790: Best = 1.114809e+03	FES = 279100	Diversity = 1.3630e-02	k_t
= [1, 1, 1, 1]			
Generation 2800: Best = 1.114805e+03	FES = 280100	Diversity = 1.3962e-02	k_t
= [4, 3, 4, 4]			
Generation 2810: Best = 1.114802e+03	FES = 281100	Diversity = 1.3265e-02	k_t
= [1, 1, 1, 1]			
Generation 2820: Best = 1.114800e+03	FES = 282100	Diversity = 1.2320e-02	k_t
= [4, 4, 3, 2]			
Generation 2830: Best = 1.114797e+03	FES = 283100	Diversity = 1.3371e-02	k_t
= [1, 1, 1, 1]			
Generation 2840: Best = 1.114793e+03	FES = 284100	Diversity = 1.4484e-02	k_t
= [3, 2, 3, 3]			
Generation 2850: Best = 1.114790e+03	FES = 285100	Diversity = 1.5288e-02	k_t
= [1, 1, 1, 1]			
Generation 2860: Best = 1.114785e+03	FES = 286100	Diversity = 1.5929e-02	k_t
= [3, 2, 4, 4]			
Generation 2870: Best = 1.114781e+03	FES = 287100	Diversity = 1.7044e-02	k_t
= [1, 1, 1, 1]			
Generation 2880: Best = 1.114777e+03	FES = 288100	Diversity = 1.5827e-02	k_t
= [4, 3, 4, 3]			
Generation 2890: Best = 1.114773e+03	FES = 289100	Diversity = 1.3226e-02	k_t
= [1, 1, 1, 1]			
Generation 2900: Best = 1.114770e+03	FES = 290100	Diversity = 1.3299e-02	k_t
= [4, 4, 3, 3]			
Generation 2910: Best = 1.114765e+03	FES = 291100	Diversity = 1.5392e-02	k_t
= [1, 1, 1, 1]			
Generation 2920: Best = 1.114759e+03	FES = 292100	Diversity = 1.8725e-02	k_t
= [4, 2, 2, 2]			

Generation 2930: Best = 1.114754e+03 FES = 293100 Diversity = 1.7049e-02 k_t
 = [1, 1, 1, 1]
 Generation 2940: Best = 1.114750e+03 FES = 294100 Diversity = 1.8071e-02 k_t
 = [2, 4, 2, 2]
 Generation 2950: Best = 1.114743e+03 FES = 295100 Diversity = 2.1002e-02 k_t
 = [1, 1, 1, 1]
 Generation 2960: Best = 1.114740e+03 FES = 296100 Diversity = 1.9181e-02 k_t
 = [4, 4, 3, 3]
 Generation 2970: Best = 1.114736e+03 FES = 297100 Diversity = 1.6628e-02 k_t
 = [1, 1, 1, 1]
 Generation 2980: Best = 1.114732e+03 FES = 298100 Diversity = 1.4621e-02 k_t
 = [2, 3, 4, 4]
 Generation 2990: Best = 1.114729e+03 FES = 299100 Diversity = 1.3718e-02 k_t
 = [1, 1, 1, 1]

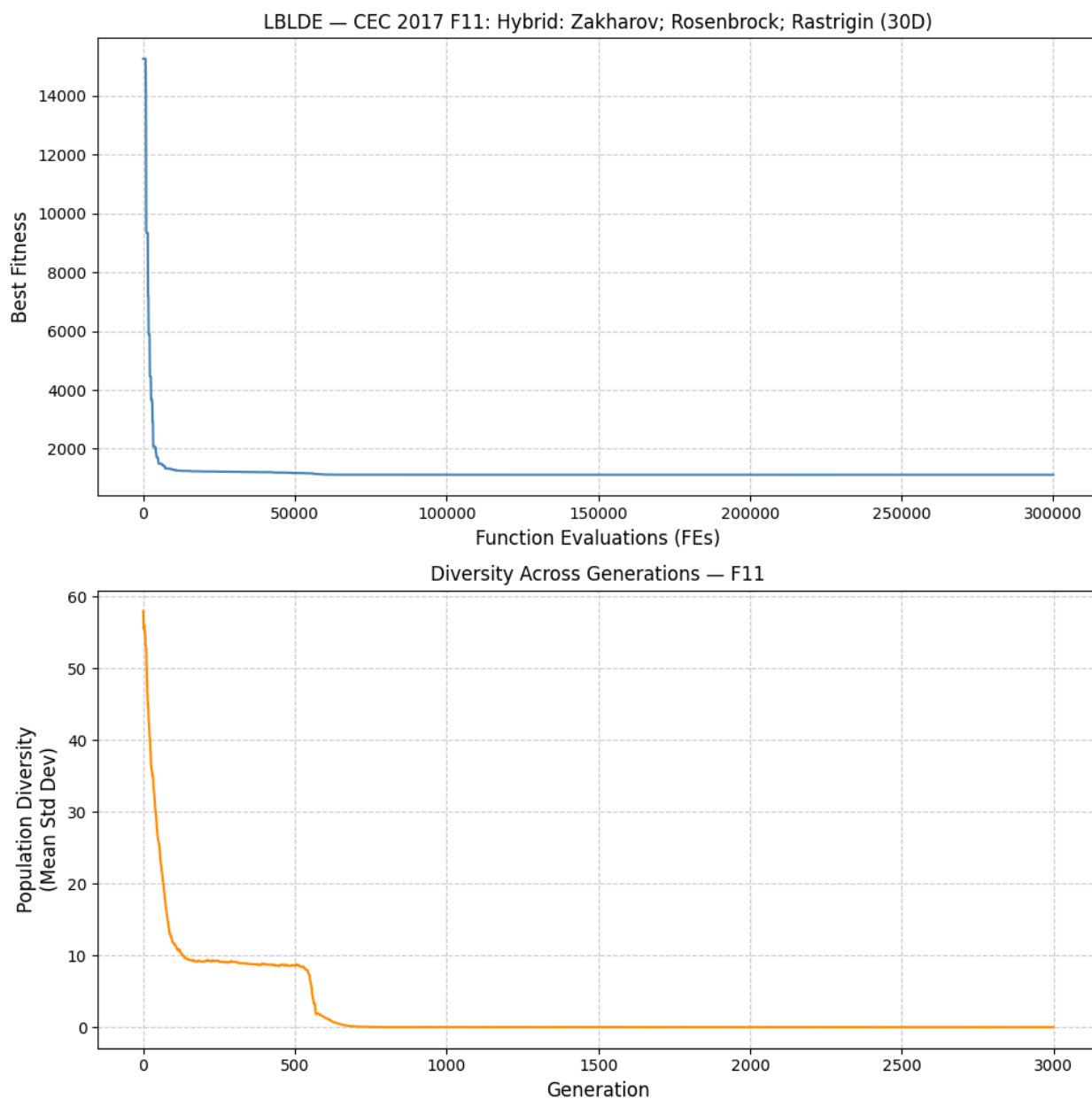
Optimization completed!

Final Best Fitness = 1.114726e+03

Total Function Evaluations = 300000

```

=====
=
Results:
  Best Fitness           : 1.114726e+03
  Global Optimum         : 1.100000e+03
  Error (|best-optimum|) : 1.472576e+01
  Best Solution (5 dim)  : [-26.03831399  -7.72829451 -45.13095334  45.9210751
-34.08024445]
  D^(0) (init. diversity): 5.7976e+01
=====
=
  
```



Graph saved as convergence_F11.png

```
In [22]: # =====
# F12 - Hybrid Function 2 (Elliptic; Schwefel; Bent Cigar)
# =====
run_and_plot('F12', make_wrapper(hybrid.f12), 1200.0, 'Hybrid: Elliptic; Schwe
```



```

=====
=
LBLDE_s on CEC 2017 F12 – Single Run
=====
=
Function      : Hybrid: Elliptic; Schwefel; Bent Cigar (30D)
Bounds       : [-100, 100]
Global optimum : 1200.0
Initial D^(0) = 5.7976e+01
Generation 0: Best Fitness = 1.806776e+11 | Diversity = 5.7976e+01
Generation 10: Best = 2.532706e+10 FES = 1100 Diversity = 4.0022e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 5.168994e+09 FES = 2100 Diversity = 3.1180e+01 k_t
= [3, 3, 2, 2]
Generation 30: Best = 1.579755e+09 FES = 3100 Diversity = 2.7128e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 6.598350e+08 FES = 4100 Diversity = 2.3813e+01 k_t
= [2, 3, 2, 3]
Generation 50: Best = 1.906642e+08 FES = 5100 Diversity = 2.1265e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 1.231671e+08 FES = 6100 Diversity = 1.8700e+01 k_t
= [2, 2, 3, 4]
Generation 70: Best = 5.590993e+07 FES = 7100 Diversity = 1.5783e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 2.943845e+07 FES = 8100 Diversity = 1.3652e+01 k_t
= [4, 3, 4, 3]
Generation 90: Best = 1.958592e+07 FES = 9100 Diversity = 1.1579e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 1.214753e+07 FES = 10100 Diversity = 9.7115e+00 k_t
= [2, 4, 4, 3]
Generation 110: Best = 6.895274e+06 FES = 11100 Diversity = 9.1345e+00 k_t
= [1, 1, 1, 1]
Generation 120: Best = 4.616272e+06 FES = 12100 Diversity = 7.6339e+00 k_t
= [3, 3, 3, 4]
Generation 130: Best = 3.668107e+06 FES = 13100 Diversity = 6.5266e+00 k_t
= [1, 1, 1, 1]
Generation 140: Best = 2.392806e+06 FES = 14100 Diversity = 5.8573e+00 k_t
= [3, 2, 3, 2]
Generation 150: Best = 1.759405e+06 FES = 15100 Diversity = 5.1989e+00 k_t
= [1, 1, 1, 1]
Generation 160: Best = 1.420384e+06 FES = 16100 Diversity = 4.6731e+00 k_t
= [2, 2, 3, 2]
Generation 170: Best = 1.023444e+06 FES = 17100 Diversity = 3.7932e+00 k_t
= [1, 1, 1, 1]
Generation 180: Best = 9.604183e+05 FES = 18100 Diversity = 3.5069e+00 k_t
= [2, 3, 2, 2]
Generation 190: Best = 8.584110e+05 FES = 19100 Diversity = 3.4474e+00 k_t
= [1, 1, 1, 1]
Generation 200: Best = 8.253690e+05 FES = 20100 Diversity = 3.0218e+00 k_t
= [4, 2, 3, 2]
Generation 210: Best = 7.010683e+05 FES = 21100 Diversity = 3.3638e+00 k_t
= [1, 1, 1, 1]
Generation 220: Best = 6.696486e+05 FES = 22100 Diversity = 3.8743e+00 k_t
= [2, 3, 4, 2]

```

Generation 230:	Best = 5.623695e+05	FES = 23100	Diversity = 4.6647e+00	k_t = [1, 1, 1, 1]
Generation 240:	Best = 5.407066e+05	FES = 24100	Diversity = 4.9742e+00	k_t = [2, 2, 3, 4]
Generation 250:	Best = 5.050674e+05	FES = 25100	Diversity = 4.7816e+00	k_t = [1, 1, 1, 1]
Generation 260:	Best = 4.953425e+05	FES = 26100	Diversity = 4.5943e+00	k_t = [4, 3, 4, 3]
Generation 270:	Best = 4.415697e+05	FES = 27100	Diversity = 4.5901e+00	k_t = [1, 1, 1, 1]
Generation 280:	Best = 4.214045e+05	FES = 28100	Diversity = 5.0013e+00	k_t = [3, 3, 2, 3]
Generation 290:	Best = 3.985805e+05	FES = 29100	Diversity = 5.5589e+00	k_t = [1, 1, 1, 1]
Generation 300:	Best = 3.881046e+05	FES = 30100	Diversity = 5.3824e+00	k_t = [3, 4, 3, 2]
Generation 310:	Best = 3.658533e+05	FES = 31100	Diversity = 5.0982e+00	k_t = [1, 1, 1, 1]
Generation 320:	Best = 3.611596e+05	FES = 32100	Diversity = 4.6437e+00	k_t = [2, 2, 3, 4]
Generation 330:	Best = 3.482301e+05	FES = 33100	Diversity = 4.8458e+00	k_t = [1, 1, 1, 1]
Generation 340:	Best = 3.359833e+05	FES = 34100	Diversity = 4.4830e+00	k_t = [2, 4, 3, 4]
Generation 350:	Best = 3.229765e+05	FES = 35100	Diversity = 5.4578e+00	k_t = [1, 1, 1, 1]
Generation 360:	Best = 3.072330e+05	FES = 36100	Diversity = 6.3843e+00	k_t = [4, 2, 4, 2]
Generation 370:	Best = 2.870467e+05	FES = 37100	Diversity = 7.2815e+00	k_t = [1, 1, 1, 1]
Generation 380:	Best = 2.779454e+05	FES = 38100	Diversity = 6.5394e+00	k_t = [3, 3, 2, 4]
Generation 390:	Best = 2.694015e+05	FES = 39100	Diversity = 6.6047e+00	k_t = [1, 1, 1, 1]
Generation 400:	Best = 2.595148e+05	FES = 40100	Diversity = 5.9017e+00	k_t = [3, 4, 2, 4]
Generation 410:	Best = 2.456860e+05	FES = 41100	Diversity = 6.4343e+00	k_t = [1, 1, 1, 1]
Generation 420:	Best = 2.383034e+05	FES = 42100	Diversity = 6.7029e+00	k_t = [3, 3, 2, 4]
Generation 430:	Best = 2.210689e+05	FES = 43100	Diversity = 6.4024e+00	k_t = [1, 1, 1, 1]
Generation 440:	Best = 2.155356e+05	FES = 44100	Diversity = 6.0445e+00	k_t = [3, 3, 4, 4]
Generation 450:	Best = 2.078641e+05	FES = 45100	Diversity = 6.2538e+00	k_t = [1, 1, 1, 1]
Generation 460:	Best = 1.962073e+05	FES = 46100	Diversity = 6.5510e+00	k_t = [4, 4, 2, 3]
Generation 470:	Best = 1.869138e+05	FES = 47100	Diversity = 6.9666e+00	k_t = [1, 1, 1, 1]
Generation 480:	Best = 1.734562e+05	FES = 48100	Diversity = 8.3573e+00	k_t = [4, 2, 4, 3]
Generation 490:	Best = 1.734562e+05	FES = 49100	Diversity = 7.7819e+00	k_t = [1, 1, 1, 1]

Generation 500:	Best = 1.659226e+05	FES = 50100	Diversity = 7.7656e+00	k_t = [2, 4, 3, 2]
Generation 510:	Best = 1.597664e+05	FES = 51100	Diversity = 7.4986e+00	k_t = [1, 1, 1, 1]
Generation 520:	Best = 1.563283e+05	FES = 52100	Diversity = 7.5518e+00	k_t = [3, 3, 4, 3]
Generation 530:	Best = 1.544058e+05	FES = 53100	Diversity = 7.1424e+00	k_t = [1, 1, 1, 1]
Generation 540:	Best = 1.485686e+05	FES = 54100	Diversity = 7.3021e+00	k_t = [3, 3, 4, 2]
Generation 550:	Best = 1.449777e+05	FES = 55100	Diversity = 7.0823e+00	k_t = [1, 1, 1, 1]
Generation 560:	Best = 1.420503e+05	FES = 56100	Diversity = 7.1605e+00	k_t = [3, 3, 3, 4]
Generation 570:	Best = 1.395925e+05	FES = 57100	Diversity = 7.2534e+00	k_t = [1, 1, 1, 1]
Generation 580:	Best = 1.361385e+05	FES = 58100	Diversity = 8.0651e+00	k_t = [2, 3, 3, 2]
Generation 590:	Best = 1.332805e+05	FES = 59100	Diversity = 8.0405e+00	k_t = [1, 1, 1, 1]
Generation 600:	Best = 1.277603e+05	FES = 60100	Diversity = 8.8433e+00	k_t = [4, 4, 2, 4]
Generation 610:	Best = 1.257405e+05	FES = 61100	Diversity = 8.8380e+00	k_t = [1, 1, 1, 1]
Generation 620:	Best = 1.249659e+05	FES = 62100	Diversity = 8.6219e+00	k_t = [3, 4, 4, 2]
Generation 630:	Best = 1.228005e+05	FES = 63100	Diversity = 8.7748e+00	k_t = [1, 1, 1, 1]
Generation 640:	Best = 1.166519e+05	FES = 64100	Diversity = 9.3239e+00	k_t = [4, 4, 2, 4]
Generation 650:	Best = 1.152975e+05	FES = 65100	Diversity = 9.7493e+00	k_t = [1, 1, 1, 1]
Generation 660:	Best = 1.125943e+05	FES = 66100	Diversity = 1.0642e+01	k_t = [4, 2, 3, 4]
Generation 670:	Best = 1.100745e+05	FES = 67100	Diversity = 1.0025e+01	k_t = [1, 1, 1, 1]
Generation 680:	Best = 1.079869e+05	FES = 68100	Diversity = 9.8458e+00	k_t = [2, 4, 3, 3]
Generation 690:	Best = 1.069683e+05	FES = 69100	Diversity = 9.6612e+00	k_t = [1, 1, 1, 1]
Generation 700:	Best = 1.058509e+05	FES = 70100	Diversity = 1.0189e+01	k_t = [3, 3, 4, 3]
Generation 710:	Best = 1.014074e+05	FES = 71100	Diversity = 1.0054e+01	k_t = [1, 1, 1, 1]
Generation 720:	Best = 1.009493e+05	FES = 72100	Diversity = 1.0572e+01	k_t = [2, 2, 2, 2]
Generation 730:	Best = 9.891007e+04	FES = 73100	Diversity = 1.0264e+01	k_t = [1, 1, 1, 1]
Generation 740:	Best = 9.752202e+04	FES = 74100	Diversity = 9.4163e+00	k_t = [4, 2, 4, 2]
Generation 750:	Best = 9.463180e+04	FES = 75100	Diversity = 1.0116e+01	k_t = [1, 1, 1, 1]
Generation 760:	Best = 9.198663e+04	FES = 76100	Diversity = 1.0042e+01	k_t = [2, 3, 4, 2]

Generation 770:	Best = 9.002271e+04	FES = 77100	Diversity = 1.0838e+01	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 8.256186e+04	FES = 78100	Diversity = 1.0887e+01	k_t
= [3, 4, 3, 4]				
Generation 790:	Best = 7.961604e+04	FES = 79100	Diversity = 1.1669e+01	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 7.628382e+04	FES = 80100	Diversity = 1.1938e+01	k_t
= [3, 4, 3, 2]				
Generation 810:	Best = 7.419049e+04	FES = 81100	Diversity = 1.1601e+01	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 7.190954e+04	FES = 82100	Diversity = 1.0994e+01	k_t
= [2, 3, 2, 4]				
Generation 830:	Best = 6.986135e+04	FES = 83100	Diversity = 1.1072e+01	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 6.752422e+04	FES = 84100	Diversity = 1.0678e+01	k_t
= [4, 2, 2, 4]				
Generation 850:	Best = 6.469264e+04	FES = 85100	Diversity = 1.0939e+01	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 6.451282e+04	FES = 86100	Diversity = 1.1296e+01	k_t
= [2, 3, 3, 3]				
Generation 870:	Best = 6.064757e+04	FES = 87100	Diversity = 1.2010e+01	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 5.965120e+04	FES = 88100	Diversity = 1.2562e+01	k_t
= [2, 4, 2, 2]				
Generation 890:	Best = 5.761715e+04	FES = 89100	Diversity = 1.1852e+01	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 5.664350e+04	FES = 90100	Diversity = 1.1631e+01	k_t
= [4, 3, 2, 3]				
Generation 910:	Best = 5.487959e+04	FES = 91100	Diversity = 1.2118e+01	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 5.352726e+04	FES = 92100	Diversity = 1.2028e+01	k_t
= [3, 3, 4, 4]				
Generation 930:	Best = 5.352726e+04	FES = 93100	Diversity = 1.1562e+01	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 5.160340e+04	FES = 94100	Diversity = 1.1749e+01	k_t
= [4, 2, 3, 2]				
Generation 950:	Best = 5.147859e+04	FES = 95100	Diversity = 1.2923e+01	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 4.881287e+04	FES = 96100	Diversity = 1.3842e+01	k_t
= [3, 2, 3, 2]				
Generation 970:	Best = 4.790192e+04	FES = 97100	Diversity = 1.3943e+01	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 4.764672e+04	FES = 98100	Diversity = 1.2878e+01	k_t
= [4, 2, 4, 2]				
Generation 990:	Best = 4.680009e+04	FES = 99100	Diversity = 1.2172e+01	k_t
= [1, 1, 1, 1]				
Generation 1000:	Best = 4.630343e+04	FES = 100100	Diversity = 1.2102e+01	k_t
= [4, 4, 4, 3]				
Generation 1010:	Best = 4.439428e+04	FES = 101100	Diversity = 1.1534e+01	k_t
= [1, 1, 1, 1]				
Generation 1020:	Best = 4.398682e+04	FES = 102100	Diversity = 1.0723e+01	k_t
= [4, 4, 2, 4]				
Generation 1030:	Best = 4.319505e+04	FES = 103100	Diversity = 1.0404e+01	k_t
= [1, 1, 1, 1]				

Generation 1040: Best = 4.261305e+04 = [4, 4, 2, 3]	FES = 104100	Diversity = 1.0668e+01	k_t
Generation 1050: Best = 4.137271e+04 = [1, 1, 1, 1]	FES = 105100	Diversity = 1.0463e+01	k_t
Generation 1060: Best = 4.104409e+04 = [2, 3, 3, 3]	FES = 106100	Diversity = 9.8127e+00	k_t
Generation 1070: Best = 3.991985e+04 = [1, 1, 1, 1]	FES = 107100	Diversity = 9.6469e+00	k_t
Generation 1080: Best = 3.934640e+04 = [2, 2, 3, 4]	FES = 108100	Diversity = 1.0208e+01	k_t
Generation 1090: Best = 3.920911e+04 = [1, 1, 1, 1]	FES = 109100	Diversity = 1.1041e+01	k_t
Generation 1100: Best = 3.812517e+04 = [4, 4, 3, 4]	FES = 110100	Diversity = 1.3287e+01	k_t
Generation 1110: Best = 3.798983e+04 = [1, 1, 1, 1]	FES = 111100	Diversity = 1.3116e+01	k_t
Generation 1120: Best = 3.694100e+04 = [2, 4, 4, 4]	FES = 112100	Diversity = 1.3233e+01	k_t
Generation 1130: Best = 3.654989e+04 = [1, 1, 1, 1]	FES = 113100	Diversity = 1.3723e+01	k_t
Generation 1140: Best = 3.498536e+04 = [2, 3, 3, 4]	FES = 114100	Diversity = 1.4091e+01	k_t
Generation 1150: Best = 3.410811e+04 = [1, 1, 1, 1]	FES = 115100	Diversity = 1.3782e+01	k_t
Generation 1160: Best = 3.267953e+04 = [3, 3, 4, 2]	FES = 116100	Diversity = 1.3989e+01	k_t
Generation 1170: Best = 3.156449e+04 = [1, 1, 1, 1]	FES = 117100	Diversity = 1.4184e+01	k_t
Generation 1180: Best = 3.119875e+04 = [2, 2, 4, 3]	FES = 118100	Diversity = 1.4273e+01	k_t
Generation 1190: Best = 3.095712e+04 = [1, 1, 1, 1]	FES = 119100	Diversity = 1.3170e+01	k_t
Generation 1200: Best = 3.045254e+04 = [3, 3, 2, 3]	FES = 120100	Diversity = 1.3099e+01	k_t
Generation 1210: Best = 2.984983e+04 = [1, 1, 1, 1]	FES = 121100	Diversity = 1.2283e+01	k_t
Generation 1220: Best = 2.984983e+04 = [3, 3, 2, 4]	FES = 122100	Diversity = 1.2018e+01	k_t
Generation 1230: Best = 2.964716e+04 = [1, 1, 1, 1]	FES = 123100	Diversity = 1.1833e+01	k_t
Generation 1240: Best = 2.900391e+04 = [3, 2, 4, 3]	FES = 124100	Diversity = 1.1456e+01	k_t
Generation 1250: Best = 2.826599e+04 = [1, 1, 1, 1]	FES = 125100	Diversity = 1.2851e+01	k_t
Generation 1260: Best = 2.579252e+04 = [4, 2, 2, 3]	FES = 126100	Diversity = 1.4916e+01	k_t
Generation 1270: Best = 2.488911e+04 = [1, 1, 1, 1]	FES = 127100	Diversity = 1.5308e+01	k_t
Generation 1280: Best = 2.321336e+04 = [2, 2, 3, 4]	FES = 128100	Diversity = 1.4808e+01	k_t
Generation 1290: Best = 2.281831e+04 = [1, 1, 1, 1]	FES = 129100	Diversity = 1.4583e+01	k_t
Generation 1300: Best = 2.246144e+04 = [3, 2, 2, 4]	FES = 130100	Diversity = 1.3779e+01	k_t

Generation 1310: Best = 2.089894e+04 = [1, 1, 1, 1]	FES = 131100	Diversity = 1.2402e+01	k_t
Generation 1320: Best = 2.016552e+04 = [3, 3, 3, 2]	FES = 132100	Diversity = 1.2349e+01	k_t
Generation 1330: Best = 1.959724e+04 = [1, 1, 1, 1]	FES = 133100	Diversity = 1.2412e+01	k_t
Generation 1340: Best = 1.861811e+04 = [3, 3, 4, 3]	FES = 134100	Diversity = 1.2920e+01	k_t
Generation 1350: Best = 1.861811e+04 = [1, 1, 1, 1]	FES = 135100	Diversity = 1.2562e+01	k_t
Generation 1360: Best = 1.812700e+04 = [3, 3, 2, 3]	FES = 136100	Diversity = 1.2938e+01	k_t
Generation 1370: Best = 1.812700e+04 = [1, 1, 1, 1]	FES = 137100	Diversity = 1.2863e+01	k_t
Generation 1380: Best = 1.784904e+04 = [3, 4, 2, 2]	FES = 138100	Diversity = 1.2165e+01	k_t
Generation 1390: Best = 1.699742e+04 = [1, 1, 1, 1]	FES = 139100	Diversity = 1.1748e+01	k_t
Generation 1400: Best = 1.699742e+04 = [3, 3, 3, 3]	FES = 140100	Diversity = 1.1614e+01	k_t
Generation 1410: Best = 1.657785e+04 = [1, 1, 1, 1]	FES = 141100	Diversity = 1.2610e+01	k_t
Generation 1420: Best = 1.657785e+04 = [2, 4, 4, 4]	FES = 142100	Diversity = 1.2535e+01	k_t
Generation 1430: Best = 1.577665e+04 = [1, 1, 1, 1]	FES = 143100	Diversity = 1.2291e+01	k_t
Generation 1440: Best = 1.451590e+04 = [4, 3, 4, 4]	FES = 144100	Diversity = 1.2788e+01	k_t
Generation 1450: Best = 1.450802e+04 = [1, 1, 1, 1]	FES = 145100	Diversity = 1.3156e+01	k_t
Generation 1460: Best = 1.315735e+04 = [3, 2, 4, 3]	FES = 146100	Diversity = 1.2967e+01	k_t
Generation 1470: Best = 1.237533e+04 = [1, 1, 1, 1]	FES = 147100	Diversity = 1.3020e+01	k_t
Generation 1480: Best = 1.090212e+04 = [2, 2, 3, 3]	FES = 148100	Diversity = 1.3172e+01	k_t
Generation 1490: Best = 1.084757e+04 = [1, 1, 1, 1]	FES = 149100	Diversity = 1.3331e+01	k_t
Generation 1500: Best = 1.077057e+04 = [4, 3, 4, 4]	FES = 150100	Diversity = 1.2602e+01	k_t
Generation 1510: Best = 1.057090e+04 = [1, 1, 1, 1]	FES = 151100	Diversity = 1.2306e+01	k_t
Generation 1520: Best = 1.036630e+04 = [3, 2, 2, 3]	FES = 152100	Diversity = 1.1809e+01	k_t
Generation 1530: Best = 9.843415e+03 = [1, 1, 1, 1]	FES = 153100	Diversity = 1.1177e+01	k_t
Generation 1540: Best = 8.831941e+03 = [4, 4, 4, 3]	FES = 154100	Diversity = 1.0883e+01	k_t
Generation 1550: Best = 8.266282e+03 = [1, 1, 1, 1]	FES = 155100	Diversity = 1.1587e+01	k_t
Generation 1560: Best = 8.023621e+03 = [3, 2, 2, 2]	FES = 156100	Diversity = 1.1779e+01	k_t
Generation 1570: Best = 7.888712e+03 = [1, 1, 1, 1]	FES = 157100	Diversity = 1.2090e+01	k_t

Generation 1580: Best = 7.888712e+03 = [4, 2, 4, 3]	FES = 158100	Diversity = 1.2067e+01	k_t
Generation 1590: Best = 7.451301e+03 = [1, 1, 1, 1]	FES = 159100	Diversity = 1.2051e+01	k_t
Generation 1600: Best = 7.451301e+03 = [3, 3, 3, 4]	FES = 160100	Diversity = 1.1036e+01	k_t
Generation 1610: Best = 7.249794e+03 = [1, 1, 1, 1]	FES = 161100	Diversity = 1.1044e+01	k_t
Generation 1620: Best = 6.882741e+03 = [4, 4, 3, 4]	FES = 162100	Diversity = 1.0852e+01	k_t
Generation 1630: Best = 6.882741e+03 = [1, 1, 1, 1]	FES = 163100	Diversity = 1.0464e+01	k_t
Generation 1640: Best = 6.882741e+03 = [4, 3, 3, 4]	FES = 164100	Diversity = 1.0363e+01	k_t
Generation 1650: Best = 6.882741e+03 = [1, 1, 1, 1]	FES = 165100	Diversity = 1.0153e+01	k_t
Generation 1660: Best = 6.358180e+03 = [4, 2, 3, 3]	FES = 166100	Diversity = 1.0221e+01	k_t
Generation 1670: Best = 6.358180e+03 = [1, 1, 1, 1]	FES = 167100	Diversity = 1.0329e+01	k_t
Generation 1680: Best = 6.358180e+03 = [3, 3, 2, 3]	FES = 168100	Diversity = 1.0376e+01	k_t
Generation 1690: Best = 6.358180e+03 = [1, 1, 1, 1]	FES = 169100	Diversity = 1.0848e+01	k_t
Generation 1700: Best = 6.299403e+03 = [3, 2, 3, 4]	FES = 170100	Diversity = 1.0479e+01	k_t
Generation 1710: Best = 6.263752e+03 = [1, 1, 1, 1]	FES = 171100	Diversity = 1.0807e+01	k_t
Generation 1720: Best = 6.263752e+03 = [2, 2, 2, 3]	FES = 172100	Diversity = 1.1045e+01	k_t
Generation 1730: Best = 6.263752e+03 = [1, 1, 1, 1]	FES = 173100	Diversity = 1.1311e+01	k_t
Generation 1740: Best = 5.592755e+03 = [3, 2, 2, 2]	FES = 174100	Diversity = 1.1225e+01	k_t
Generation 1750: Best = 5.592755e+03 = [1, 1, 1, 1]	FES = 175100	Diversity = 1.1137e+01	k_t
Generation 1760: Best = 5.592755e+03 = [3, 2, 4, 3]	FES = 176100	Diversity = 1.1054e+01	k_t
Generation 1770: Best = 5.350501e+03 = [1, 1, 1, 1]	FES = 177100	Diversity = 1.0956e+01	k_t
Generation 1780: Best = 5.350501e+03 = [3, 4, 3, 2]	FES = 178100	Diversity = 1.0896e+01	k_t
Generation 1790: Best = 5.252578e+03 = [1, 1, 1, 1]	FES = 179100	Diversity = 1.0815e+01	k_t
Generation 1800: Best = 5.252578e+03 = [2, 2, 4, 2]	FES = 180100	Diversity = 1.0885e+01	k_t
Generation 1810: Best = 5.252578e+03 = [1, 1, 1, 1]	FES = 181100	Diversity = 1.0832e+01	k_t
Generation 1820: Best = 5.252578e+03 = [3, 4, 4, 4]	FES = 182100	Diversity = 1.0738e+01	k_t
Generation 1830: Best = 5.252578e+03 = [1, 1, 1, 1]	FES = 183100	Diversity = 1.0781e+01	k_t
Generation 1840: Best = 5.252578e+03 = [2, 2, 3, 3]	FES = 184100	Diversity = 1.0838e+01	k_t

Generation 1850: Best = 5.252578e+03 = [1, 1, 1, 1]	FES = 185100	Diversity = 1.0668e+01	k_t
Generation 1860: Best = 5.235374e+03 = [4, 4, 3, 2]	FES = 186100	Diversity = 1.0373e+01	k_t
Generation 1870: Best = 5.079010e+03 = [1, 1, 1, 1]	FES = 187100	Diversity = 1.0342e+01	k_t
Generation 1880: Best = 4.887573e+03 = [4, 4, 3, 2]	FES = 188100	Diversity = 9.5378e+00	k_t
Generation 1890: Best = 4.887573e+03 = [1, 1, 1, 1]	FES = 189100	Diversity = 9.4619e+00	k_t
Generation 1900: Best = 4.887573e+03 = [4, 3, 4, 2]	FES = 190100	Diversity = 9.3916e+00	k_t
Generation 1910: Best = 4.887573e+03 = [1, 1, 1, 1]	FES = 191100	Diversity = 9.2001e+00	k_t
Generation 1920: Best = 4.779632e+03 = [2, 2, 4, 3]	FES = 192100	Diversity = 9.2261e+00	k_t
Generation 1930: Best = 4.779632e+03 = [1, 1, 1, 1]	FES = 193100	Diversity = 9.2087e+00	k_t
Generation 1940: Best = 4.779632e+03 = [3, 3, 2, 3]	FES = 194100	Diversity = 9.2043e+00	k_t
Generation 1950: Best = 4.632164e+03 = [1, 1, 1, 1]	FES = 195100	Diversity = 9.1902e+00	k_t
Generation 1960: Best = 4.632164e+03 = [2, 4, 3, 3]	FES = 196100	Diversity = 9.1013e+00	k_t
Generation 1970: Best = 4.632164e+03 = [1, 1, 1, 1]	FES = 197100	Diversity = 9.0267e+00	k_t
Generation 1980: Best = 4.451335e+03 = [4, 4, 4, 2]	FES = 198100	Diversity = 9.0254e+00	k_t
Generation 1990: Best = 4.451335e+03 = [1, 1, 1, 1]	FES = 199100	Diversity = 8.8981e+00	k_t
Generation 2000: Best = 4.451335e+03 = [2, 2, 4, 4]	FES = 200100	Diversity = 8.8167e+00	k_t
Generation 2010: Best = 4.451335e+03 = [1, 1, 1, 1]	FES = 201100	Diversity = 8.7969e+00	k_t
Generation 2020: Best = 4.451335e+03 = [2, 2, 3, 3]	FES = 202100	Diversity = 8.6496e+00	k_t
Generation 2030: Best = 4.451335e+03 = [1, 1, 1, 1]	FES = 203100	Diversity = 8.7368e+00	k_t
Generation 2040: Best = 4.451335e+03 = [3, 3, 4, 4]	FES = 204100	Diversity = 8.6336e+00	k_t
Generation 2050: Best = 4.451335e+03 = [1, 1, 1, 1]	FES = 205100	Diversity = 8.5658e+00	k_t
Generation 2060: Best = 4.451335e+03 = [4, 4, 2, 2]	FES = 206100	Diversity = 8.3704e+00	k_t
Generation 2070: Best = 4.451335e+03 = [1, 1, 1, 1]	FES = 207100	Diversity = 8.2196e+00	k_t
Generation 2080: Best = 4.451335e+03 = [2, 3, 3, 3]	FES = 208100	Diversity = 8.0344e+00	k_t
Generation 2090: Best = 4.451335e+03 = [1, 1, 1, 1]	FES = 209100	Diversity = 7.9179e+00	k_t
Generation 2100: Best = 4.451335e+03 = [3, 2, 2, 4]	FES = 210100	Diversity = 7.8634e+00	k_t
Generation 2110: Best = 4.451335e+03 = [1, 1, 1, 1]	FES = 211100	Diversity = 7.6808e+00	k_t

Generation 2120: Best = 4.451335e+03 = [3, 2, 4, 3]	FES = 212100	Diversity = 7.6523e+00	k_t
Generation 2130: Best = 4.395456e+03 = [1, 1, 1, 1]	FES = 213100	Diversity = 7.7101e+00	k_t
Generation 2140: Best = 4.372472e+03 = [2, 4, 2, 2]	FES = 214100	Diversity = 7.6854e+00	k_t
Generation 2150: Best = 4.372472e+03 = [1, 1, 1, 1]	FES = 215100	Diversity = 7.6523e+00	k_t
Generation 2160: Best = 4.372472e+03 = [4, 2, 3, 2]	FES = 216100	Diversity = 7.4951e+00	k_t
Generation 2170: Best = 4.372472e+03 = [1, 1, 1, 1]	FES = 217100	Diversity = 7.5365e+00	k_t
Generation 2180: Best = 4.372472e+03 = [2, 4, 2, 4]	FES = 218100	Diversity = 7.5039e+00	k_t
Generation 2190: Best = 4.170881e+03 = [1, 1, 1, 1]	FES = 219100	Diversity = 7.3306e+00	k_t
Generation 2200: Best = 4.170881e+03 = [2, 2, 3, 4]	FES = 220100	Diversity = 7.3072e+00	k_t
Generation 2210: Best = 4.170881e+03 = [1, 1, 1, 1]	FES = 221100	Diversity = 7.1710e+00	k_t
Generation 2220: Best = 4.170881e+03 = [3, 2, 4, 4]	FES = 222100	Diversity = 7.1483e+00	k_t
Generation 2230: Best = 4.169556e+03 = [1, 1, 1, 1]	FES = 223100	Diversity = 7.0648e+00	k_t
Generation 2240: Best = 4.148847e+03 = [4, 3, 4, 4]	FES = 224100	Diversity = 7.1000e+00	k_t
Generation 2250: Best = 4.148847e+03 = [1, 1, 1, 1]	FES = 225100	Diversity = 7.0132e+00	k_t
Generation 2260: Best = 4.148847e+03 = [4, 2, 2, 4]	FES = 226100	Diversity = 6.9697e+00	k_t
Generation 2270: Best = 4.129360e+03 = [1, 1, 1, 1]	FES = 227100	Diversity = 6.9097e+00	k_t
Generation 2280: Best = 4.129360e+03 = [3, 2, 4, 2]	FES = 228100	Diversity = 6.8715e+00	k_t
Generation 2290: Best = 4.129360e+03 = [1, 1, 1, 1]	FES = 229100	Diversity = 6.8365e+00	k_t
Generation 2300: Best = 4.129360e+03 = [3, 4, 3, 3]	FES = 230100	Diversity = 6.8160e+00	k_t
Generation 2310: Best = 4.104473e+03 = [1, 1, 1, 1]	FES = 231100	Diversity = 6.8198e+00	k_t
Generation 2320: Best = 3.893294e+03 = [2, 3, 2, 2]	FES = 232100	Diversity = 6.7333e+00	k_t
Generation 2330: Best = 3.893294e+03 = [1, 1, 1, 1]	FES = 233100	Diversity = 6.6611e+00	k_t
Generation 2340: Best = 3.893294e+03 = [2, 2, 2, 2]	FES = 234100	Diversity = 6.6047e+00	k_t
Generation 2350: Best = 3.893294e+03 = [1, 1, 1, 1]	FES = 235100	Diversity = 6.5848e+00	k_t
Generation 2360: Best = 3.893294e+03 = [2, 4, 4, 3]	FES = 236100	Diversity = 6.5787e+00	k_t
Generation 2370: Best = 3.882614e+03 = [1, 1, 1, 1]	FES = 237100	Diversity = 6.4992e+00	k_t
Generation 2380: Best = 3.877828e+03 = [4, 4, 4, 2]	FES = 238100	Diversity = 6.4014e+00	k_t

Generation 2390: Best = 3.877828e+03 = [1, 1, 1, 1]	FES = 239100	Diversity = 6.4103e+00	k_t
Generation 2400: Best = 3.873315e+03 = [3, 3, 3, 4]	FES = 240100	Diversity = 6.2743e+00	k_t
Generation 2410: Best = 3.873315e+03 = [1, 1, 1, 1]	FES = 241100	Diversity = 6.2463e+00	k_t
Generation 2420: Best = 3.873315e+03 = [2, 2, 3, 3]	FES = 242100	Diversity = 6.2266e+00	k_t
Generation 2430: Best = 3.873315e+03 = [1, 1, 1, 1]	FES = 243100	Diversity = 6.2318e+00	k_t
Generation 2440: Best = 3.734782e+03 = [2, 4, 4, 2]	FES = 244100	Diversity = 6.2177e+00	k_t
Generation 2450: Best = 3.734782e+03 = [1, 1, 1, 1]	FES = 245100	Diversity = 6.1762e+00	k_t
Generation 2460: Best = 3.732625e+03 = [3, 3, 3, 3]	FES = 246100	Diversity = 6.2495e+00	k_t
Generation 2470: Best = 3.732625e+03 = [1, 1, 1, 1]	FES = 247100	Diversity = 6.2163e+00	k_t
Generation 2480: Best = 3.708716e+03 = [4, 2, 2, 4]	FES = 248100	Diversity = 6.2245e+00	k_t
Generation 2490: Best = 3.708716e+03 = [1, 1, 1, 1]	FES = 249100	Diversity = 6.1447e+00	k_t
Generation 2500: Best = 3.708716e+03 = [4, 4, 3, 3]	FES = 250100	Diversity = 5.9294e+00	k_t
Generation 2510: Best = 3.708716e+03 = [1, 1, 1, 1]	FES = 251100	Diversity = 5.8110e+00	k_t
Generation 2520: Best = 3.682445e+03 = [3, 3, 3, 3]	FES = 252100	Diversity = 5.5428e+00	k_t
Generation 2530: Best = 3.652183e+03 = [1, 1, 1, 1]	FES = 253100	Diversity = 5.4658e+00	k_t
Generation 2540: Best = 3.652183e+03 = [4, 3, 3, 4]	FES = 254100	Diversity = 5.4592e+00	k_t
Generation 2550: Best = 3.652183e+03 = [1, 1, 1, 1]	FES = 255100	Diversity = 5.5219e+00	k_t
Generation 2560: Best = 3.640154e+03 = [2, 3, 4, 2]	FES = 256100	Diversity = 5.4931e+00	k_t
Generation 2570: Best = 3.635380e+03 = [1, 1, 1, 1]	FES = 257100	Diversity = 5.4735e+00	k_t
Generation 2580: Best = 3.635380e+03 = [4, 2, 2, 2]	FES = 258100	Diversity = 5.4872e+00	k_t
Generation 2590: Best = 3.634517e+03 = [1, 1, 1, 1]	FES = 259100	Diversity = 5.4194e+00	k_t
Generation 2600: Best = 3.632553e+03 = [2, 2, 2, 4]	FES = 260100	Diversity = 5.3935e+00	k_t
Generation 2610: Best = 3.632553e+03 = [1, 1, 1, 1]	FES = 261100	Diversity = 5.3251e+00	k_t
Generation 2620: Best = 3.632224e+03 = [4, 3, 2, 3]	FES = 262100	Diversity = 5.2761e+00	k_t
Generation 2630: Best = 3.632224e+03 = [1, 1, 1, 1]	FES = 263100	Diversity = 5.2305e+00	k_t
Generation 2640: Best = 3.632224e+03 = [3, 4, 3, 2]	FES = 264100	Diversity = 5.2290e+00	k_t
Generation 2650: Best = 3.632224e+03 = [1, 1, 1, 1]	FES = 265100	Diversity = 5.1359e+00	k_t

Generation 2660: Best = 3.614769e+03 = [3, 3, 4, 3]	FES = 266100	Diversity = 5.1353e+00	k_t
Generation 2670: Best = 3.613544e+03 = [1, 1, 1, 1]	FES = 267100	Diversity = 5.1279e+00	k_t
Generation 2680: Best = 3.613544e+03 = [4, 2, 2, 4]	FES = 268100	Diversity = 5.0053e+00	k_t
Generation 2690: Best = 3.556702e+03 = [1, 1, 1, 1]	FES = 269100	Diversity = 4.9841e+00	k_t
Generation 2700: Best = 3.453174e+03 = [4, 4, 4, 2]	FES = 270100	Diversity = 4.9860e+00	k_t
Generation 2710: Best = 3.433683e+03 = [1, 1, 1, 1]	FES = 271100	Diversity = 4.8236e+00	k_t
Generation 2720: Best = 3.430998e+03 = [4, 3, 3, 4]	FES = 272100	Diversity = 4.8262e+00	k_t
Generation 2730: Best = 3.416469e+03 = [1, 1, 1, 1]	FES = 273100	Diversity = 4.7794e+00	k_t
Generation 2740: Best = 3.411215e+03 = [3, 3, 2, 2]	FES = 274100	Diversity = 4.7818e+00	k_t
Generation 2750: Best = 3.411215e+03 = [1, 1, 1, 1]	FES = 275100	Diversity = 4.5748e+00	k_t
Generation 2760: Best = 3.411215e+03 = [4, 2, 2, 2]	FES = 276100	Diversity = 4.4325e+00	k_t
Generation 2770: Best = 3.411215e+03 = [1, 1, 1, 1]	FES = 277100	Diversity = 4.4128e+00	k_t
Generation 2780: Best = 3.409502e+03 = [4, 4, 3, 4]	FES = 278100	Diversity = 4.4124e+00	k_t
Generation 2790: Best = 3.409502e+03 = [1, 1, 1, 1]	FES = 279100	Diversity = 4.2200e+00	k_t
Generation 2800: Best = 3.286938e+03 = [4, 4, 4, 4]	FES = 280100	Diversity = 4.1678e+00	k_t
Generation 2810: Best = 3.286938e+03 = [1, 1, 1, 1]	FES = 281100	Diversity = 4.1708e+00	k_t
Generation 2820: Best = 3.286938e+03 = [2, 2, 4, 2]	FES = 282100	Diversity = 3.9432e+00	k_t
Generation 2830: Best = 3.286938e+03 = [1, 1, 1, 1]	FES = 283100	Diversity = 3.8402e+00	k_t
Generation 2840: Best = 3.286938e+03 = [3, 2, 3, 3]	FES = 284100	Diversity = 3.6423e+00	k_t
Generation 2850: Best = 3.286938e+03 = [1, 1, 1, 1]	FES = 285100	Diversity = 3.4902e+00	k_t
Generation 2860: Best = 3.286938e+03 = [3, 4, 2, 3]	FES = 286100	Diversity = 3.3779e+00	k_t
Generation 2870: Best = 3.286938e+03 = [1, 1, 1, 1]	FES = 287100	Diversity = 3.0883e+00	k_t
Generation 2880: Best = 3.286938e+03 = [4, 3, 4, 4]	FES = 288100	Diversity = 3.0125e+00	k_t
Generation 2890: Best = 3.286938e+03 = [1, 1, 1, 1]	FES = 289100	Diversity = 2.9869e+00	k_t
Generation 2900: Best = 3.286938e+03 = [2, 4, 3, 2]	FES = 290100	Diversity = 2.7589e+00	k_t
Generation 2910: Best = 3.286938e+03 = [1, 1, 1, 1]	FES = 291100	Diversity = 2.6720e+00	k_t
Generation 2920: Best = 3.285837e+03 = [3, 2, 2, 4]	FES = 292100	Diversity = 2.3030e+00	k_t

Generation 2930: Best = 3.285837e+03 FES = 293100 Diversity = 2.1414e+00 k_t
 = [1, 1, 1, 1]
 Generation 2940: Best = 3.285837e+03 FES = 294100 Diversity = 1.9732e+00 k_t
 = [3, 4, 2, 3]
 Generation 2950: Best = 3.285837e+03 FES = 295100 Diversity = 1.8066e+00 k_t
 = [1, 1, 1, 1]
 Generation 2960: Best = 3.285837e+03 FES = 296100 Diversity = 1.2874e+00 k_t
 = [2, 4, 4, 4]
 Generation 2970: Best = 3.261800e+03 FES = 297100 Diversity = 9.2730e-01 k_t
 = [1, 1, 1, 1]
 Generation 2980: Best = 3.199298e+03 FES = 298100 Diversity = 4.2767e-01 k_t
 = [4, 4, 2, 4]
 Generation 2990: Best = 3.184543e+03 FES = 299100 Diversity = 3.9103e-01 k_t
 = [1, 1, 1, 1]

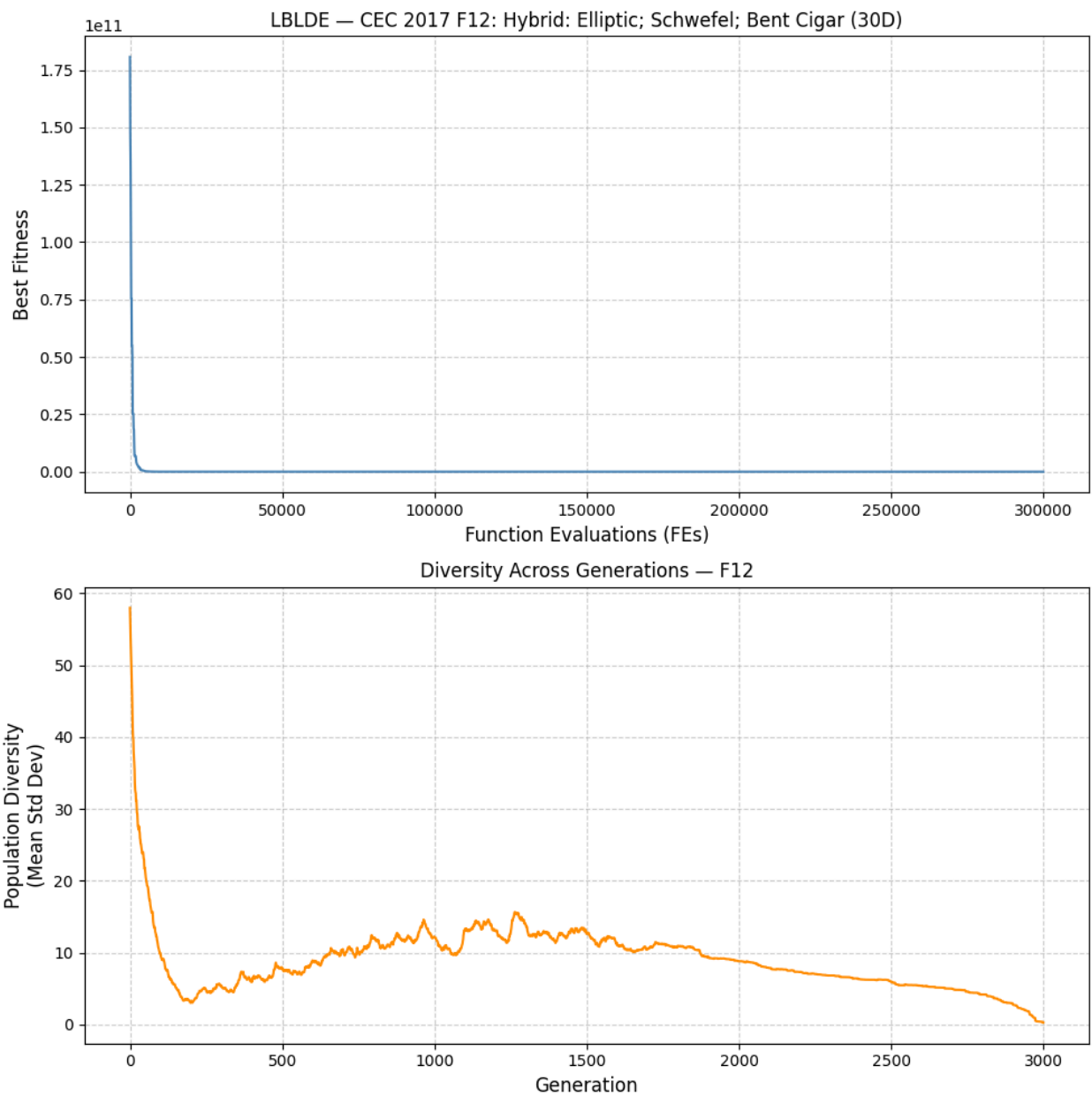
Optimization completed!

Final Best Fitness = 3.164549e+03

Total Function Evaluations = 300000

```

=====
=
Results:
  Best Fitness           : 3.164549e+03
  Global Optimum         : 1.200000e+03
  Error (|best-optimum|) : 1.964549e+03
  Best Solution (5 dim)  : [ 58.12612048 -55.08976004 -77.09186949  24.77467444
-36.83052111]
  D^(0) (init. diversity): 5.7976e+01
=====
=
  
```



Graph saved as convergence_F12.png

```
In [23]: # =====
# F13 – Hybrid Function 3 (Bent Cigar; Rosenbrock; Bi-Rastrigin)
# =====
run_and_plot('F13', make_wrapper(hybrid.f13), 1300.0, 'Hybrid: Bent Cigar; Ros
```

```

=====
=
LBLDE_s on CEC 2017 F13 – Single Run
=====
=
Function      : Hybrid: Bent Cigar; Rosenbrock; Bi-Rastrigin (30D)
Bounds       : [-100, 100]
Global optimum : 1300.0
Initial D^(0) = 5.7976e+01
Generation 0: Best Fitness = 1.700763e+11 | Diversity = 5.7976e+01
Generation 10: Best = 1.846562e+10 FES = 1100 Diversity = 4.3124e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 3.801518e+09 FES = 2100 Diversity = 3.8723e+01 k_t
= [3, 3, 2, 2]
Generation 30: Best = 5.658831e+08 FES = 3100 Diversity = 3.5355e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 2.802517e+08 FES = 4100 Diversity = 3.3367e+01 k_t
= [2, 3, 2, 3]
Generation 50: Best = 6.065754e+07 FES = 5100 Diversity = 3.2077e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 4.270846e+07 FES = 6100 Diversity = 3.0313e+01 k_t
= [4, 2, 3, 2]
Generation 70: Best = 9.406373e+06 FES = 7100 Diversity = 2.9075e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 9.389100e+06 FES = 8100 Diversity = 3.0662e+01 k_t
= [2, 3, 2, 2]
Generation 90: Best = 1.452260e+06 FES = 9100 Diversity = 3.0128e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 1.452260e+06 FES = 10100 Diversity = 3.0583e+01 k_t
= [2, 2, 3, 2]
Generation 110: Best = 5.572136e+05 FES = 11100 Diversity = 3.2047e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 4.863760e+05 FES = 12100 Diversity = 3.0645e+01 k_t
= [2, 3, 3, 2]
Generation 130: Best = 2.138919e+05 FES = 13100 Diversity = 3.0144e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 1.365395e+05 FES = 14100 Diversity = 3.1111e+01 k_t
= [2, 4, 3, 2]
Generation 150: Best = 6.333009e+04 FES = 15100 Diversity = 3.2728e+01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 6.027268e+04 FES = 16100 Diversity = 3.2624e+01 k_t
= [4, 2, 3, 4]
Generation 170: Best = 3.256537e+04 FES = 17100 Diversity = 3.1483e+01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 2.306140e+04 FES = 18100 Diversity = 3.0765e+01 k_t
= [3, 3, 3, 2]
Generation 190: Best = 1.514498e+04 FES = 19100 Diversity = 2.8606e+01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 1.081059e+04 FES = 20100 Diversity = 2.6416e+01 k_t
= [2, 2, 3, 2]
Generation 210: Best = 7.071690e+03 FES = 21100 Diversity = 2.3897e+01 k_t
= [1, 1, 1, 1]
Generation 220: Best = 5.298290e+03 FES = 22100 Diversity = 2.1435e+01 k_t
= [4, 3, 2, 2]

```

Generation 230:	Best = 3.205249e+03	FES = 23100	Diversity = 1.9471e+01	k_t = [1, 1, 1, 1]
Generation 240:	Best = 3.155012e+03	FES = 24100	Diversity = 1.7560e+01	k_t = [2, 2, 3, 4]
Generation 250:	Best = 2.781723e+03	FES = 25100	Diversity = 1.4960e+01	k_t = [1, 1, 1, 1]
Generation 260:	Best = 2.339561e+03	FES = 26100	Diversity = 1.3248e+01	k_t = [2, 4, 4, 3]
Generation 270:	Best = 2.051908e+03	FES = 27100	Diversity = 1.1495e+01	k_t = [1, 1, 1, 1]
Generation 280:	Best = 1.967470e+03	FES = 28100	Diversity = 9.7445e+00	k_t = [4, 2, 4, 3]
Generation 290:	Best = 1.840883e+03	FES = 29100	Diversity = 8.3182e+00	k_t = [1, 1, 1, 1]
Generation 300:	Best = 1.671699e+03	FES = 30100	Diversity = 7.2775e+00	k_t = [2, 4, 2, 4]
Generation 310:	Best = 1.622904e+03	FES = 31100	Diversity = 5.6125e+00	k_t = [1, 1, 1, 1]
Generation 320:	Best = 1.547825e+03	FES = 32100	Diversity = 4.9212e+00	k_t = [2, 2, 2, 4]
Generation 330:	Best = 1.547825e+03	FES = 33100	Diversity = 4.2851e+00	k_t = [1, 1, 1, 1]
Generation 340:	Best = 1.523086e+03	FES = 34100	Diversity = 3.8417e+00	k_t = [2, 3, 2, 3]
Generation 350:	Best = 1.491385e+03	FES = 35100	Diversity = 3.4084e+00	k_t = [1, 1, 1, 1]
Generation 360:	Best = 1.488731e+03	FES = 36100	Diversity = 2.9966e+00	k_t = [3, 2, 4, 2]
Generation 370:	Best = 1.478932e+03	FES = 37100	Diversity = 3.0293e+00	k_t = [1, 1, 1, 1]
Generation 380:	Best = 1.478932e+03	FES = 38100	Diversity = 2.8838e+00	k_t = [3, 4, 4, 2]
Generation 390:	Best = 1.463500e+03	FES = 39100	Diversity = 2.8418e+00	k_t = [1, 1, 1, 1]
Generation 400:	Best = 1.447133e+03	FES = 40100	Diversity = 2.7266e+00	k_t = [3, 2, 2, 3]
Generation 410:	Best = 1.443117e+03	FES = 41100	Diversity = 2.6954e+00	k_t = [1, 1, 1, 1]
Generation 420:	Best = 1.443117e+03	FES = 42100	Diversity = 2.6377e+00	k_t = [2, 2, 4, 4]
Generation 430:	Best = 1.443117e+03	FES = 43100	Diversity = 2.5382e+00	k_t = [1, 1, 1, 1]
Generation 440:	Best = 1.424450e+03	FES = 44100	Diversity = 2.5024e+00	k_t = [3, 4, 2, 3]
Generation 450:	Best = 1.424415e+03	FES = 45100	Diversity = 2.4863e+00	k_t = [1, 1, 1, 1]
Generation 460:	Best = 1.424024e+03	FES = 46100	Diversity = 2.4721e+00	k_t = [3, 4, 2, 2]
Generation 470:	Best = 1.422138e+03	FES = 47100	Diversity = 2.5141e+00	k_t = [1, 1, 1, 1]
Generation 480:	Best = 1.422138e+03	FES = 48100	Diversity = 2.5356e+00	k_t = [4, 2, 2, 4]
Generation 490:	Best = 1.413774e+03	FES = 49100	Diversity = 2.5618e+00	k_t = [1, 1, 1, 1]

Generation 500:	Best = 1.413774e+03	FES = 50100	Diversity = 2.5440e+00	k_t
= [4, 4, 3, 2]				
Generation 510:	Best = 1.403722e+03	FES = 51100	Diversity = 2.4517e+00	k_t
= [1, 1, 1, 1]				
Generation 520:	Best = 1.403722e+03	FES = 52100	Diversity = 2.4639e+00	k_t
= [4, 2, 2, 2]				
Generation 530:	Best = 1.392946e+03	FES = 53100	Diversity = 2.4763e+00	k_t
= [1, 1, 1, 1]				
Generation 540:	Best = 1.392946e+03	FES = 54100	Diversity = 2.4699e+00	k_t
= [3, 3, 3, 2]				
Generation 550:	Best = 1.392946e+03	FES = 55100	Diversity = 2.4802e+00	k_t
= [1, 1, 1, 1]				
Generation 560:	Best = 1.392946e+03	FES = 56100	Diversity = 2.5085e+00	k_t
= [2, 2, 3, 4]				
Generation 570:	Best = 1.391812e+03	FES = 57100	Diversity = 2.5071e+00	k_t
= [1, 1, 1, 1]				
Generation 580:	Best = 1.391812e+03	FES = 58100	Diversity = 2.4764e+00	k_t
= [3, 3, 4, 2]				
Generation 590:	Best = 1.390741e+03	FES = 59100	Diversity = 2.5200e+00	k_t
= [1, 1, 1, 1]				
Generation 600:	Best = 1.387639e+03	FES = 60100	Diversity = 2.5187e+00	k_t
= [2, 4, 4, 3]				
Generation 610:	Best = 1.383932e+03	FES = 61100	Diversity = 2.5202e+00	k_t
= [1, 1, 1, 1]				
Generation 620:	Best = 1.383932e+03	FES = 62100	Diversity = 2.5583e+00	k_t
= [2, 2, 3, 2]				
Generation 630:	Best = 1.377260e+03	FES = 63100	Diversity = 2.5376e+00	k_t
= [1, 1, 1, 1]				
Generation 640:	Best = 1.377260e+03	FES = 64100	Diversity = 2.4954e+00	k_t
= [4, 4, 4, 2]				
Generation 650:	Best = 1.377260e+03	FES = 65100	Diversity = 2.5415e+00	k_t
= [1, 1, 1, 1]				
Generation 660:	Best = 1.377260e+03	FES = 66100	Diversity = 2.5836e+00	k_t
= [3, 3, 2, 3]				
Generation 670:	Best = 1.377260e+03	FES = 67100	Diversity = 2.5599e+00	k_t
= [1, 1, 1, 1]				
Generation 680:	Best = 1.377260e+03	FES = 68100	Diversity = 2.5410e+00	k_t
= [3, 4, 3, 3]				
Generation 690:	Best = 1.374973e+03	FES = 69100	Diversity = 2.5505e+00	k_t
= [1, 1, 1, 1]				
Generation 700:	Best = 1.374973e+03	FES = 70100	Diversity = 2.4749e+00	k_t
= [2, 4, 2, 2]				
Generation 710:	Best = 1.374973e+03	FES = 71100	Diversity = 2.4047e+00	k_t
= [1, 1, 1, 1]				
Generation 720:	Best = 1.374973e+03	FES = 72100	Diversity = 2.3988e+00	k_t
= [2, 4, 4, 2]				
Generation 730:	Best = 1.374074e+03	FES = 73100	Diversity = 2.3822e+00	k_t
= [1, 1, 1, 1]				
Generation 740:	Best = 1.374074e+03	FES = 74100	Diversity = 2.3786e+00	k_t
= [2, 2, 4, 2]				
Generation 750:	Best = 1.373483e+03	FES = 75100	Diversity = 2.3840e+00	k_t
= [1, 1, 1, 1]				
Generation 760:	Best = 1.373483e+03	FES = 76100	Diversity = 2.4295e+00	k_t
= [2, 3, 4, 2]				

Generation 770:	Best = 1.371499e+03	FES = 77100	Diversity = 2.3893e+00	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 1.362259e+03	FES = 78100	Diversity = 2.3922e+00	k_t
= [2, 4, 2, 3]				
Generation 790:	Best = 1.362259e+03	FES = 79100	Diversity = 2.3090e+00	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 1.362259e+03	FES = 80100	Diversity = 2.3205e+00	k_t
= [3, 4, 4, 4]				
Generation 810:	Best = 1.361920e+03	FES = 81100	Diversity = 2.3115e+00	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 1.361920e+03	FES = 82100	Diversity = 2.2298e+00	k_t
= [3, 4, 4, 4]				
Generation 830:	Best = 1.361920e+03	FES = 83100	Diversity = 2.2038e+00	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 1.359279e+03	FES = 84100	Diversity = 2.1643e+00	k_t
= [4, 2, 4, 3]				
Generation 850:	Best = 1.359279e+03	FES = 85100	Diversity = 2.1034e+00	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 1.358054e+03	FES = 86100	Diversity = 2.1064e+00	k_t
= [4, 3, 3, 3]				
Generation 870:	Best = 1.358054e+03	FES = 87100	Diversity = 2.0878e+00	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 1.358054e+03	FES = 88100	Diversity = 2.0432e+00	k_t
= [3, 4, 2, 3]				
Generation 890:	Best = 1.355913e+03	FES = 89100	Diversity = 2.0071e+00	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 1.355913e+03	FES = 90100	Diversity = 2.0173e+00	k_t
= [3, 2, 3, 3]				
Generation 910:	Best = 1.355370e+03	FES = 91100	Diversity = 1.9836e+00	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 1.355370e+03	FES = 92100	Diversity = 1.9902e+00	k_t
= [4, 2, 2, 3]				
Generation 930:	Best = 1.354859e+03	FES = 93100	Diversity = 1.9582e+00	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 1.351829e+03	FES = 94100	Diversity = 1.9401e+00	k_t
= [2, 2, 4, 4]				
Generation 950:	Best = 1.347693e+03	FES = 95100	Diversity = 1.9067e+00	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 1.346618e+03	FES = 96100	Diversity = 1.8751e+00	k_t
= [2, 3, 2, 2]				
Generation 970:	Best = 1.346003e+03	FES = 97100	Diversity = 1.8649e+00	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 1.344810e+03	FES = 98100	Diversity = 1.8423e+00	k_t
= [2, 4, 3, 2]				
Generation 990:	Best = 1.344810e+03	FES = 99100	Diversity = 1.7817e+00	k_t
= [1, 1, 1, 1]				
Generation 1000:	Best = 1.338864e+03	FES = 100100	Diversity = 1.7240e+00	k_t
= [4, 2, 4, 4]				
Generation 1010:	Best = 1.338864e+03	FES = 101100	Diversity = 1.6864e+00	k_t
= [1, 1, 1, 1]				
Generation 1020:	Best = 1.335202e+03	FES = 102100	Diversity = 1.5737e+00	k_t
= [3, 3, 4, 2]				
Generation 1030:	Best = 1.335080e+03	FES = 103100	Diversity = 1.5057e+00	k_t
= [1, 1, 1, 1]				

Generation 1040: Best = 1.334203e+03 = [2, 2, 3, 4]	FES = 104100	Diversity = 1.4501e+00	k_t
Generation 1050: Best = 1.332383e+03 = [1, 1, 1, 1]	FES = 105100	Diversity = 1.3410e+00	k_t
Generation 1060: Best = 1.331155e+03 = [3, 4, 3, 3]	FES = 106100	Diversity = 1.3041e+00	k_t
Generation 1070: Best = 1.330342e+03 = [1, 1, 1, 1]	FES = 107100	Diversity = 1.2070e+00	k_t
Generation 1080: Best = 1.329182e+03 = [4, 4, 3, 4]	FES = 108100	Diversity = 1.0226e+00	k_t
Generation 1090: Best = 1.325889e+03 = [1, 1, 1, 1]	FES = 109100	Diversity = 8.0019e-01	k_t
Generation 1100: Best = 1.325370e+03 = [2, 3, 2, 4]	FES = 110100	Diversity = 5.4125e-01	k_t
Generation 1110: Best = 1.322273e+03 = [1, 1, 1, 1]	FES = 111100	Diversity = 4.5453e-01	k_t
Generation 1120: Best = 1.320898e+03 = [2, 2, 3, 3]	FES = 112100	Diversity = 4.2583e-01	k_t
Generation 1130: Best = 1.319832e+03 = [1, 1, 1, 1]	FES = 113100	Diversity = 3.7137e-01	k_t
Generation 1140: Best = 1.318676e+03 = [4, 3, 2, 4]	FES = 114100	Diversity = 3.2098e-01	k_t
Generation 1150: Best = 1.317727e+03 = [1, 1, 1, 1]	FES = 115100	Diversity = 2.8141e-01	k_t
Generation 1160: Best = 1.317370e+03 = [3, 4, 4, 2]	FES = 116100	Diversity = 2.5011e-01	k_t
Generation 1170: Best = 1.317049e+03 = [1, 1, 1, 1]	FES = 117100	Diversity = 2.1531e-01	k_t
Generation 1180: Best = 1.316658e+03 = [4, 4, 4, 4]	FES = 118100	Diversity = 1.9729e-01	k_t
Generation 1190: Best = 1.316198e+03 = [1, 1, 1, 1]	FES = 119100	Diversity = 1.6583e-01	k_t
Generation 1200: Best = 1.315520e+03 = [2, 3, 3, 2]	FES = 120100	Diversity = 1.4885e-01	k_t
Generation 1210: Best = 1.315520e+03 = [1, 1, 1, 1]	FES = 121100	Diversity = 1.3349e-01	k_t
Generation 1220: Best = 1.315188e+03 = [3, 2, 4, 4]	FES = 122100	Diversity = 1.2009e-01	k_t
Generation 1230: Best = 1.314867e+03 = [1, 1, 1, 1]	FES = 123100	Diversity = 1.1089e-01	k_t
Generation 1240: Best = 1.314613e+03 = [4, 3, 2, 4]	FES = 124100	Diversity = 9.8674e-02	k_t
Generation 1250: Best = 1.314473e+03 = [1, 1, 1, 1]	FES = 125100	Diversity = 9.0225e-02	k_t
Generation 1260: Best = 1.314414e+03 = [2, 2, 2, 2]	FES = 126100	Diversity = 8.8201e-02	k_t
Generation 1270: Best = 1.314300e+03 = [1, 1, 1, 1]	FES = 127100	Diversity = 8.0406e-02	k_t
Generation 1280: Best = 1.314190e+03 = [3, 2, 3, 2]	FES = 128100	Diversity = 7.5759e-02	k_t
Generation 1290: Best = 1.314021e+03 = [1, 1, 1, 1]	FES = 129100	Diversity = 7.0884e-02	k_t
Generation 1300: Best = 1.313858e+03 = [4, 4, 4, 4]	FES = 130100	Diversity = 7.5346e-02	k_t

Generation 1310: Best = 1.313847e+03 = [1, 1, 1, 1]	FES = 131100	Diversity = 6.9114e-02	k_t
Generation 1320: Best = 1.313771e+03 = [4, 4, 4, 4]	FES = 132100	Diversity = 7.0775e-02	k_t
Generation 1330: Best = 1.313715e+03 = [1, 1, 1, 1]	FES = 133100	Diversity = 6.3693e-02	k_t
Generation 1340: Best = 1.313630e+03 = [2, 4, 4, 3]	FES = 134100	Diversity = 6.3952e-02	k_t
Generation 1350: Best = 1.313551e+03 = [1, 1, 1, 1]	FES = 135100	Diversity = 5.9548e-02	k_t
Generation 1360: Best = 1.313518e+03 = [2, 4, 3, 3]	FES = 136100	Diversity = 5.8257e-02	k_t
Generation 1370: Best = 1.313440e+03 = [1, 1, 1, 1]	FES = 137100	Diversity = 5.5309e-02	k_t
Generation 1380: Best = 1.313381e+03 = [4, 3, 2, 2]	FES = 138100	Diversity = 5.5196e-02	k_t
Generation 1390: Best = 1.313358e+03 = [1, 1, 1, 1]	FES = 139100	Diversity = 5.6944e-02	k_t
Generation 1400: Best = 1.313310e+03 = [4, 4, 2, 3]	FES = 140100	Diversity = 5.3847e-02	k_t
Generation 1410: Best = 1.313284e+03 = [1, 1, 1, 1]	FES = 141100	Diversity = 4.8636e-02	k_t
Generation 1420: Best = 1.313262e+03 = [2, 3, 2, 3]	FES = 142100	Diversity = 4.5727e-02	k_t
Generation 1430: Best = 1.313245e+03 = [1, 1, 1, 1]	FES = 143100	Diversity = 4.6912e-02	k_t
Generation 1440: Best = 1.313226e+03 = [4, 2, 2, 4]	FES = 144100	Diversity = 4.0508e-02	k_t
Generation 1450: Best = 1.313219e+03 = [1, 1, 1, 1]	FES = 145100	Diversity = 3.4580e-02	k_t
Generation 1460: Best = 1.313203e+03 = [2, 4, 2, 2]	FES = 146100	Diversity = 3.1693e-02	k_t
Generation 1470: Best = 1.313180e+03 = [1, 1, 1, 1]	FES = 147100	Diversity = 3.1174e-02	k_t
Generation 1480: Best = 1.313176e+03 = [4, 3, 4, 3]	FES = 148100	Diversity = 3.0973e-02	k_t
Generation 1490: Best = 1.313140e+03 = [1, 1, 1, 1]	FES = 149100	Diversity = 3.3301e-02	k_t
Generation 1500: Best = 1.313125e+03 = [2, 4, 3, 3]	FES = 150100	Diversity = 3.6843e-02	k_t
Generation 1510: Best = 1.313110e+03 = [1, 1, 1, 1]	FES = 151100	Diversity = 3.5912e-02	k_t
Generation 1520: Best = 1.313101e+03 = [3, 4, 3, 3]	FES = 152100	Diversity = 3.4535e-02	k_t
Generation 1530: Best = 1.313089e+03 = [1, 1, 1, 1]	FES = 153100	Diversity = 3.1274e-02	k_t
Generation 1540: Best = 1.313071e+03 = [4, 4, 2, 4]	FES = 154100	Diversity = 2.8953e-02	k_t
Generation 1550: Best = 1.313067e+03 = [1, 1, 1, 1]	FES = 155100	Diversity = 2.7258e-02	k_t
Generation 1560: Best = 1.313053e+03 = [2, 4, 4, 3]	FES = 156100	Diversity = 2.4784e-02	k_t
Generation 1570: Best = 1.313046e+03 = [1, 1, 1, 1]	FES = 157100	Diversity = 2.4211e-02	k_t

Generation 1580: Best = 1.313038e+03 = [4, 3, 3, 4]	FES = 158100	Diversity = 2.1538e-02	k_t
Generation 1590: Best = 1.313034e+03 = [1, 1, 1, 1]	FES = 159100	Diversity = 1.9088e-02	k_t
Generation 1600: Best = 1.313026e+03 = [4, 2, 3, 3]	FES = 160100	Diversity = 1.7445e-02	k_t
Generation 1610: Best = 1.313020e+03 = [1, 1, 1, 1]	FES = 161100	Diversity = 1.6719e-02	k_t
Generation 1620: Best = 1.313011e+03 = [2, 4, 4, 2]	FES = 162100	Diversity = 1.6951e-02	k_t
Generation 1630: Best = 1.312998e+03 = [1, 1, 1, 1]	FES = 163100	Diversity = 2.2095e-02	k_t
Generation 1640: Best = 1.312983e+03 = [4, 3, 3, 4]	FES = 164100	Diversity = 2.1120e-02	k_t
Generation 1650: Best = 1.312983e+03 = [1, 1, 1, 1]	FES = 165100	Diversity = 1.9172e-02	k_t
Generation 1660: Best = 1.312977e+03 = [3, 2, 4, 4]	FES = 166100	Diversity = 1.7877e-02	k_t
Generation 1670: Best = 1.312969e+03 = [1, 1, 1, 1]	FES = 167100	Diversity = 2.1757e-02	k_t
Generation 1680: Best = 1.312958e+03 = [3, 4, 3, 4]	FES = 168100	Diversity = 2.2102e-02	k_t
Generation 1690: Best = 1.312952e+03 = [1, 1, 1, 1]	FES = 169100	Diversity = 2.1177e-02	k_t
Generation 1700: Best = 1.312947e+03 = [3, 3, 2, 3]	FES = 170100	Diversity = 2.0308e-02	k_t
Generation 1710: Best = 1.312940e+03 = [1, 1, 1, 1]	FES = 171100	Diversity = 1.9402e-02	k_t
Generation 1720: Best = 1.312933e+03 = [2, 2, 2, 2]	FES = 172100	Diversity = 1.9504e-02	k_t
Generation 1730: Best = 1.312925e+03 = [1, 1, 1, 1]	FES = 173100	Diversity = 1.8300e-02	k_t
Generation 1740: Best = 1.312915e+03 = [3, 4, 2, 3]	FES = 174100	Diversity = 1.9370e-02	k_t
Generation 1750: Best = 1.312908e+03 = [1, 1, 1, 1]	FES = 175100	Diversity = 2.0919e-02	k_t
Generation 1760: Best = 1.312896e+03 = [3, 2, 2, 2]	FES = 176100	Diversity = 2.1233e-02	k_t
Generation 1770: Best = 1.312893e+03 = [1, 1, 1, 1]	FES = 177100	Diversity = 2.0994e-02	k_t
Generation 1780: Best = 1.312889e+03 = [3, 4, 4, 3]	FES = 178100	Diversity = 1.9311e-02	k_t
Generation 1790: Best = 1.312881e+03 = [1, 1, 1, 1]	FES = 179100	Diversity = 1.6518e-02	k_t
Generation 1800: Best = 1.312874e+03 = [4, 3, 4, 2]	FES = 180100	Diversity = 1.5459e-02	k_t
Generation 1810: Best = 1.312869e+03 = [1, 1, 1, 1]	FES = 181100	Diversity = 1.6641e-02	k_t
Generation 1820: Best = 1.312864e+03 = [3, 2, 2, 4]	FES = 182100	Diversity = 1.5971e-02	k_t
Generation 1830: Best = 1.312858e+03 = [1, 1, 1, 1]	FES = 183100	Diversity = 1.6729e-02	k_t
Generation 1840: Best = 1.312851e+03 = [3, 3, 4, 4]	FES = 184100	Diversity = 1.6388e-02	k_t

Generation 1850: Best = 1.312849e+03 = [1, 1, 1, 1]	FES = 185100	Diversity = 1.7153e-02	k_t
Generation 1860: Best = 1.312838e+03 = [4, 2, 3, 2]	FES = 186100	Diversity = 1.7077e-02	k_t
Generation 1870: Best = 1.312833e+03 = [1, 1, 1, 1]	FES = 187100	Diversity = 1.9792e-02	k_t
Generation 1880: Best = 1.312821e+03 = [4, 3, 2, 2]	FES = 188100	Diversity = 2.1662e-02	k_t
Generation 1890: Best = 1.312816e+03 = [1, 1, 1, 1]	FES = 189100	Diversity = 2.2666e-02	k_t
Generation 1900: Best = 1.312814e+03 = [3, 2, 3, 3]	FES = 190100	Diversity = 2.0006e-02	k_t
Generation 1910: Best = 1.312801e+03 = [1, 1, 1, 1]	FES = 191100	Diversity = 1.9921e-02	k_t
Generation 1920: Best = 1.312790e+03 = [3, 3, 3, 3]	FES = 192100	Diversity = 2.0404e-02	k_t
Generation 1930: Best = 1.312780e+03 = [1, 1, 1, 1]	FES = 193100	Diversity = 2.1554e-02	k_t
Generation 1940: Best = 1.312765e+03 = [4, 4, 2, 4]	FES = 194100	Diversity = 2.4772e-02	k_t
Generation 1950: Best = 1.312755e+03 = [1, 1, 1, 1]	FES = 195100	Diversity = 2.6021e-02	k_t
Generation 1960: Best = 1.312746e+03 = [4, 4, 2, 3]	FES = 196100	Diversity = 2.7802e-02	k_t
Generation 1970: Best = 1.312731e+03 = [1, 1, 1, 1]	FES = 197100	Diversity = 2.5968e-02	k_t
Generation 1980: Best = 1.312724e+03 = [4, 4, 4, 2]	FES = 198100	Diversity = 2.4587e-02	k_t
Generation 1990: Best = 1.312716e+03 = [1, 1, 1, 1]	FES = 199100	Diversity = 2.2047e-02	k_t
Generation 2000: Best = 1.312705e+03 = [2, 3, 4, 3]	FES = 200100	Diversity = 2.2692e-02	k_t
Generation 2010: Best = 1.312699e+03 = [1, 1, 1, 1]	FES = 201100	Diversity = 2.1973e-02	k_t
Generation 2020: Best = 1.312689e+03 = [2, 2, 4, 4]	FES = 202100	Diversity = 2.1274e-02	k_t
Generation 2030: Best = 1.312674e+03 = [1, 1, 1, 1]	FES = 203100	Diversity = 2.3577e-02	k_t
Generation 2040: Best = 1.312663e+03 = [2, 4, 3, 2]	FES = 204100	Diversity = 2.5578e-02	k_t
Generation 2050: Best = 1.312657e+03 = [1, 1, 1, 1]	FES = 205100	Diversity = 2.5054e-02	k_t
Generation 2060: Best = 1.312653e+03 = [2, 3, 2, 3]	FES = 206100	Diversity = 2.2041e-02	k_t
Generation 2070: Best = 1.312643e+03 = [1, 1, 1, 1]	FES = 207100	Diversity = 2.0458e-02	k_t
Generation 2080: Best = 1.312635e+03 = [3, 3, 2, 4]	FES = 208100	Diversity = 2.0260e-02	k_t
Generation 2090: Best = 1.312633e+03 = [1, 1, 1, 1]	FES = 209100	Diversity = 2.2431e-02	k_t
Generation 2100: Best = 1.312617e+03 = [4, 3, 2, 2]	FES = 210100	Diversity = 2.3951e-02	k_t
Generation 2110: Best = 1.312610e+03 = [1, 1, 1, 1]	FES = 211100	Diversity = 2.4832e-02	k_t

Generation 2120: Best = 1.312601e+03 = [3, 2, 3, 4]	FES = 212100	Diversity = 2.5507e-02	k_t
Generation 2130: Best = 1.312581e+03 = [1, 1, 1, 1]	FES = 213100	Diversity = 2.3494e-02	k_t
Generation 2140: Best = 1.312574e+03 = [4, 2, 4, 2]	FES = 214100	Diversity = 2.3508e-02	k_t
Generation 2150: Best = 1.312567e+03 = [1, 1, 1, 1]	FES = 215100	Diversity = 2.4718e-02	k_t
Generation 2160: Best = 1.312560e+03 = [4, 2, 4, 2]	FES = 216100	Diversity = 2.0820e-02	k_t
Generation 2170: Best = 1.312556e+03 = [1, 1, 1, 1]	FES = 217100	Diversity = 1.9836e-02	k_t
Generation 2180: Best = 1.312548e+03 = [3, 2, 3, 4]	FES = 218100	Diversity = 1.7929e-02	k_t
Generation 2190: Best = 1.312542e+03 = [1, 1, 1, 1]	FES = 219100	Diversity = 1.6397e-02	k_t
Generation 2200: Best = 1.312536e+03 = [2, 2, 2, 3]	FES = 220100	Diversity = 1.5992e-02	k_t
Generation 2210: Best = 1.312530e+03 = [1, 1, 1, 1]	FES = 221100	Diversity = 1.6603e-02	k_t
Generation 2220: Best = 1.312524e+03 = [3, 2, 4, 3]	FES = 222100	Diversity = 1.7335e-02	k_t
Generation 2230: Best = 1.312516e+03 = [1, 1, 1, 1]	FES = 223100	Diversity = 1.8033e-02	k_t
Generation 2240: Best = 1.312513e+03 = [2, 2, 3, 4]	FES = 224100	Diversity = 1.6086e-02	k_t
Generation 2250: Best = 1.312507e+03 = [1, 1, 1, 1]	FES = 225100	Diversity = 1.6397e-02	k_t
Generation 2260: Best = 1.312500e+03 = [2, 2, 2, 4]	FES = 226100	Diversity = 1.5761e-02	k_t
Generation 2270: Best = 1.312496e+03 = [1, 1, 1, 1]	FES = 227100	Diversity = 1.4840e-02	k_t
Generation 2280: Best = 1.312490e+03 = [2, 4, 2, 2]	FES = 228100	Diversity = 1.5568e-02	k_t
Generation 2290: Best = 1.312480e+03 = [1, 1, 1, 1]	FES = 229100	Diversity = 1.6919e-02	k_t
Generation 2300: Best = 1.312477e+03 = [3, 3, 2, 2]	FES = 230100	Diversity = 1.7152e-02	k_t
Generation 2310: Best = 1.312465e+03 = [1, 1, 1, 1]	FES = 231100	Diversity = 1.9733e-02	k_t
Generation 2320: Best = 1.312453e+03 = [4, 2, 4, 4]	FES = 232100	Diversity = 2.4800e-02	k_t
Generation 2330: Best = 1.312435e+03 = [1, 1, 1, 1]	FES = 233100	Diversity = 2.9174e-02	k_t
Generation 2340: Best = 1.312414e+03 = [2, 3, 2, 2]	FES = 234100	Diversity = 3.1448e-02	k_t
Generation 2350: Best = 1.312410e+03 = [1, 1, 1, 1]	FES = 235100	Diversity = 3.0849e-02	k_t
Generation 2360: Best = 1.312392e+03 = [3, 4, 2, 2]	FES = 236100	Diversity = 2.8503e-02	k_t
Generation 2370: Best = 1.312378e+03 = [1, 1, 1, 1]	FES = 237100	Diversity = 2.8481e-02	k_t
Generation 2380: Best = 1.312374e+03 = [4, 2, 2, 2]	FES = 238100	Diversity = 2.5209e-02	k_t

Generation 2390: Best = 1.312362e+03 = [1, 1, 1, 1]	FES = 239100	Diversity = 2.0949e-02	k_t
Generation 2400: Best = 1.312353e+03 = [3, 2, 3, 3]	FES = 240100	Diversity = 2.1393e-02	k_t
Generation 2410: Best = 1.312342e+03 = [1, 1, 1, 1]	FES = 241100	Diversity = 1.9986e-02	k_t
Generation 2420: Best = 1.312325e+03 = [4, 3, 3, 3]	FES = 242100	Diversity = 2.1567e-02	k_t
Generation 2430: Best = 1.312322e+03 = [1, 1, 1, 1]	FES = 243100	Diversity = 2.2351e-02	k_t
Generation 2440: Best = 1.312314e+03 = [4, 2, 3, 3]	FES = 244100	Diversity = 2.1633e-02	k_t
Generation 2450: Best = 1.312309e+03 = [1, 1, 1, 1]	FES = 245100	Diversity = 2.1044e-02	k_t
Generation 2460: Best = 1.312302e+03 = [3, 2, 2, 2]	FES = 246100	Diversity = 1.9096e-02	k_t
Generation 2470: Best = 1.312298e+03 = [1, 1, 1, 1]	FES = 247100	Diversity = 1.6484e-02	k_t
Generation 2480: Best = 1.312292e+03 = [2, 2, 2, 3]	FES = 248100	Diversity = 1.5228e-02	k_t
Generation 2490: Best = 1.312284e+03 = [1, 1, 1, 1]	FES = 249100	Diversity = 1.4079e-02	k_t
Generation 2500: Best = 1.312278e+03 = [3, 4, 2, 4]	FES = 250100	Diversity = 1.5681e-02	k_t
Generation 2510: Best = 1.312271e+03 = [1, 1, 1, 1]	FES = 251100	Diversity = 1.5123e-02	k_t
Generation 2520: Best = 1.312261e+03 = [3, 4, 4, 2]	FES = 252100	Diversity = 1.6867e-02	k_t
Generation 2530: Best = 1.312259e+03 = [1, 1, 1, 1]	FES = 253100	Diversity = 1.7454e-02	k_t
Generation 2540: Best = 1.312255e+03 = [4, 3, 2, 3]	FES = 254100	Diversity = 1.6192e-02	k_t
Generation 2550: Best = 1.312248e+03 = [1, 1, 1, 1]	FES = 255100	Diversity = 1.6558e-02	k_t
Generation 2560: Best = 1.312245e+03 = [4, 4, 2, 2]	FES = 256100	Diversity = 1.6808e-02	k_t
Generation 2570: Best = 1.312241e+03 = [1, 1, 1, 1]	FES = 257100	Diversity = 1.6926e-02	k_t
Generation 2580: Best = 1.312234e+03 = [3, 3, 4, 2]	FES = 258100	Diversity = 1.7535e-02	k_t
Generation 2590: Best = 1.312224e+03 = [1, 1, 1, 1]	FES = 259100	Diversity = 1.6054e-02	k_t
Generation 2600: Best = 1.312221e+03 = [2, 3, 2, 4]	FES = 260100	Diversity = 1.6232e-02	k_t
Generation 2610: Best = 1.312215e+03 = [1, 1, 1, 1]	FES = 261100	Diversity = 1.6541e-02	k_t
Generation 2620: Best = 1.312211e+03 = [3, 3, 3, 2]	FES = 262100	Diversity = 1.7377e-02	k_t
Generation 2630: Best = 1.312205e+03 = [1, 1, 1, 1]	FES = 263100	Diversity = 1.7867e-02	k_t
Generation 2640: Best = 1.312201e+03 = [4, 4, 2, 4]	FES = 264100	Diversity = 1.5516e-02	k_t
Generation 2650: Best = 1.312194e+03 = [1, 1, 1, 1]	FES = 265100	Diversity = 1.6273e-02	k_t

Generation 2660: Best = 1.312189e+03 = [3, 4, 3, 2]	FES = 266100	Diversity = 1.6871e-02	k_t
Generation 2670: Best = 1.312180e+03 = [1, 1, 1, 1]	FES = 267100	Diversity = 1.5847e-02	k_t
Generation 2680: Best = 1.312172e+03 = [2, 2, 3, 4]	FES = 268100	Diversity = 1.6845e-02	k_t
Generation 2690: Best = 1.312169e+03 = [1, 1, 1, 1]	FES = 269100	Diversity = 1.6692e-02	k_t
Generation 2700: Best = 1.312162e+03 = [3, 4, 2, 3]	FES = 270100	Diversity = 1.5845e-02	k_t
Generation 2710: Best = 1.312156e+03 = [1, 1, 1, 1]	FES = 271100	Diversity = 1.6486e-02	k_t
Generation 2720: Best = 1.312148e+03 = [3, 4, 4, 2]	FES = 272100	Diversity = 1.7719e-02	k_t
Generation 2730: Best = 1.312135e+03 = [1, 1, 1, 1]	FES = 273100	Diversity = 1.8698e-02	k_t
Generation 2740: Best = 1.312131e+03 = [4, 3, 2, 4]	FES = 274100	Diversity = 1.7660e-02	k_t
Generation 2750: Best = 1.312125e+03 = [1, 1, 1, 1]	FES = 275100	Diversity = 1.6550e-02	k_t
Generation 2760: Best = 1.312117e+03 = [3, 2, 4, 4]	FES = 276100	Diversity = 1.5422e-02	k_t
Generation 2770: Best = 1.312107e+03 = [1, 1, 1, 1]	FES = 277100	Diversity = 1.4337e-02	k_t
Generation 2780: Best = 1.312102e+03 = [2, 3, 3, 3]	FES = 278100	Diversity = 1.3476e-02	k_t
Generation 2790: Best = 1.312097e+03 = [1, 1, 1, 1]	FES = 279100	Diversity = 1.3005e-02	k_t
Generation 2800: Best = 1.312088e+03 = [3, 4, 4, 3]	FES = 280100	Diversity = 1.5253e-02	k_t
Generation 2810: Best = 1.312086e+03 = [1, 1, 1, 1]	FES = 281100	Diversity = 1.5068e-02	k_t
Generation 2820: Best = 1.312080e+03 = [2, 4, 4, 2]	FES = 282100	Diversity = 1.5563e-02	k_t
Generation 2830: Best = 1.312078e+03 = [1, 1, 1, 1]	FES = 283100	Diversity = 1.5154e-02	k_t
Generation 2840: Best = 1.312072e+03 = [4, 4, 2, 3]	FES = 284100	Diversity = 1.4697e-02	k_t
Generation 2850: Best = 1.312066e+03 = [1, 1, 1, 1]	FES = 285100	Diversity = 1.3413e-02	k_t
Generation 2860: Best = 1.312062e+03 = [2, 4, 3, 2]	FES = 286100	Diversity = 1.2644e-02	k_t
Generation 2870: Best = 1.312058e+03 = [1, 1, 1, 1]	FES = 287100	Diversity = 1.3090e-02	k_t
Generation 2880: Best = 1.312052e+03 = [3, 4, 2, 4]	FES = 288100	Diversity = 1.2363e-02	k_t
Generation 2890: Best = 1.312051e+03 = [1, 1, 1, 1]	FES = 289100	Diversity = 1.1182e-02	k_t
Generation 2900: Best = 1.312046e+03 = [2, 3, 2, 4]	FES = 290100	Diversity = 1.0998e-02	k_t
Generation 2910: Best = 1.312043e+03 = [1, 1, 1, 1]	FES = 291100	Diversity = 1.1610e-02	k_t
Generation 2920: Best = 1.312036e+03 = [3, 3, 2, 4]	FES = 292100	Diversity = 1.2957e-02	k_t

Generation 2930: Best = 1.312035e+03 FES = 293100 Diversity = 1.2458e-02 k_t
 = [1, 1, 1, 1]
 Generation 2940: Best = 1.312028e+03 FES = 294100 Diversity = 1.2436e-02 k_t
 = [4, 2, 2, 3]
 Generation 2950: Best = 1.312025e+03 FES = 295100 Diversity = 1.2271e-02 k_t
 = [1, 1, 1, 1]
 Generation 2960: Best = 1.312018e+03 FES = 296100 Diversity = 1.1831e-02 k_t
 = [3, 4, 2, 4]
 Generation 2970: Best = 1.312013e+03 FES = 297100 Diversity = 1.1573e-02 k_t
 = [1, 1, 1, 1]
 Generation 2980: Best = 1.312011e+03 FES = 298100 Diversity = 1.0958e-02 k_t
 = [4, 4, 2, 2]
 Generation 2990: Best = 1.312008e+03 FES = 299100 Diversity = 1.0957e-02 k_t
 = [1, 1, 1, 1]

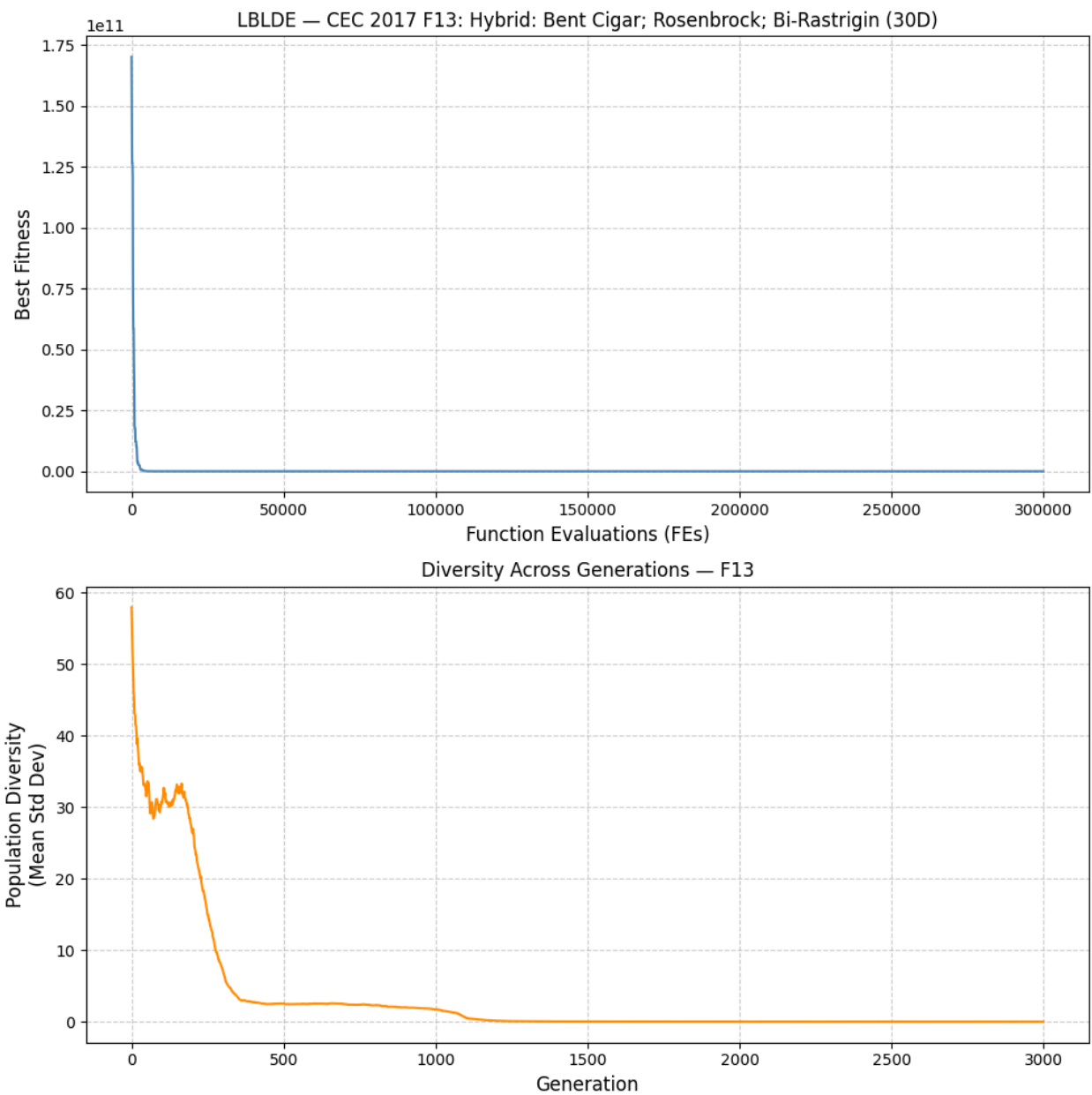
Optimization completed!

Final Best Fitness = 1.312004e+03

Total Function Evaluations = 300000

```

=====
=
Results:
  Best Fitness           : 1.312004e+03
  Global Optimum         : 1.300000e+03
  Error (|best-optimum|) : 1.200392e+01
  Best Solution (5 dim)  : [ 4.58164077 -83.69530904  27.8331362  -16.83661644
56.35155655]
  D^(0) (init. diversity): 5.7976e+01
=====
=
  
```



Graph saved as convergence_F13.png

```
In [24]: # =====
# F14 – Hybrid Function 4 (Elliptic; Ackley; Schaffer; Rastrigin)
# =====
run_and_plot('F14', make_wrapper(hybrid.f14), 1400.0, 'Hybrid: Elliptic; Ackle
```

```

=====
=
LBLDE_s on CEC 2017 F14 – Single Run
=====
=
Function      : Hybrid: Elliptic; Ackley; Schaffer; Rastrigin (30D)
Bounds       : [-100, 100]
Global optimum : 1400.0
Initial D^(0) = 5.7976e+01
Generation 0: Best Fitness = 3.706702e+07 | Diversity = 5.7976e+01
Generation 10: Best = 1.840470e+06 FES = 1100 Diversity = 5.2579e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 3.249273e+05 FES = 2100 Diversity = 4.8654e+01 k_t
= [3, 3, 2, 2]
Generation 30: Best = 1.543161e+05 FES = 3100 Diversity = 4.7630e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 2.542973e+04 FES = 4100 Diversity = 4.8424e+01 k_t
= [4, 2, 4, 3]
Generation 50: Best = 1.013074e+04 FES = 5100 Diversity = 5.2198e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 5.260752e+03 FES = 6100 Diversity = 4.9154e+01 k_t
= [2, 2, 2, 2]
Generation 70: Best = 5.052405e+03 FES = 7100 Diversity = 5.1679e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 2.561405e+03 FES = 8100 Diversity = 4.9279e+01 k_t
= [3, 4, 3, 2]
Generation 90: Best = 2.217910e+03 FES = 9100 Diversity = 4.6951e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 1.945160e+03 FES = 10100 Diversity = 4.4548e+01 k_t
= [4, 2, 2, 2]
Generation 110: Best = 1.899908e+03 FES = 11100 Diversity = 4.3736e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 1.809777e+03 FES = 12100 Diversity = 4.1754e+01 k_t
= [4, 2, 3, 3]
Generation 130: Best = 1.787992e+03 FES = 13100 Diversity = 3.8942e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 1.692591e+03 FES = 14100 Diversity = 3.7231e+01 k_t
= [4, 4, 3, 3]
Generation 150: Best = 1.641219e+03 FES = 15100 Diversity = 3.6006e+01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 1.619594e+03 FES = 16100 Diversity = 3.3970e+01 k_t
= [4, 3, 2, 2]
Generation 170: Best = 1.580243e+03 FES = 17100 Diversity = 3.0638e+01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 1.578246e+03 FES = 18100 Diversity = 2.7982e+01 k_t
= [3, 4, 3, 4]
Generation 190: Best = 1.565251e+03 FES = 19100 Diversity = 2.7425e+01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 1.539756e+03 FES = 20100 Diversity = 2.5881e+01 k_t
= [2, 2, 3, 2]
Generation 210: Best = 1.539756e+03 FES = 21100 Diversity = 2.3781e+01 k_t
= [1, 1, 1, 1]
Generation 220: Best = 1.528043e+03 FES = 22100 Diversity = 2.2426e+01 k_t
= [4, 3, 2, 2]

```

Generation 230:	Best = 1.528043e+03	FES = 23100	Diversity = 2.1184e+01	k_t = [1, 1, 1, 1]
Generation 240:	Best = 1.518956e+03	FES = 24100	Diversity = 2.0530e+01	k_t = [2, 2, 2, 3]
Generation 250:	Best = 1.507319e+03	FES = 25100	Diversity = 2.0163e+01	k_t = [1, 1, 1, 1]
Generation 260:	Best = 1.507319e+03	FES = 26100	Diversity = 1.9885e+01	k_t = [2, 4, 4, 3]
Generation 270:	Best = 1.507319e+03	FES = 27100	Diversity = 1.9672e+01	k_t = [1, 1, 1, 1]
Generation 280:	Best = 1.504465e+03	FES = 28100	Diversity = 1.9200e+01	k_t = [2, 2, 3, 2]
Generation 290:	Best = 1.493536e+03	FES = 29100	Diversity = 1.9414e+01	k_t = [1, 1, 1, 1]
Generation 300:	Best = 1.493536e+03	FES = 30100	Diversity = 1.9199e+01	k_t = [3, 3, 4, 3]
Generation 310:	Best = 1.493536e+03	FES = 31100	Diversity = 1.8949e+01	k_t = [1, 1, 1, 1]
Generation 320:	Best = 1.491180e+03	FES = 32100	Diversity = 1.9005e+01	k_t = [4, 3, 4, 2]
Generation 330:	Best = 1.482547e+03	FES = 33100	Diversity = 1.9130e+01	k_t = [1, 1, 1, 1]
Generation 340:	Best = 1.482547e+03	FES = 34100	Diversity = 1.8764e+01	k_t = [3, 4, 4, 3]
Generation 350:	Best = 1.482547e+03	FES = 35100	Diversity = 1.8702e+01	k_t = [1, 1, 1, 1]
Generation 360:	Best = 1.481586e+03	FES = 36100	Diversity = 1.8581e+01	k_t = [3, 3, 2, 4]
Generation 370:	Best = 1.480028e+03	FES = 37100	Diversity = 1.8450e+01	k_t = [1, 1, 1, 1]
Generation 380:	Best = 1.480028e+03	FES = 38100	Diversity = 1.8363e+01	k_t = [3, 2, 4, 2]
Generation 390:	Best = 1.480028e+03	FES = 39100	Diversity = 1.8511e+01	k_t = [1, 1, 1, 1]
Generation 400:	Best = 1.478576e+03	FES = 40100	Diversity = 1.8229e+01	k_t = [4, 2, 2, 3]
Generation 410:	Best = 1.478576e+03	FES = 41100	Diversity = 1.8296e+01	k_t = [1, 1, 1, 1]
Generation 420:	Best = 1.478576e+03	FES = 42100	Diversity = 1.8058e+01	k_t = [2, 4, 3, 2]
Generation 430:	Best = 1.477414e+03	FES = 43100	Diversity = 1.8094e+01	k_t = [1, 1, 1, 1]
Generation 440:	Best = 1.477414e+03	FES = 44100	Diversity = 1.8144e+01	k_t = [3, 2, 4, 3]
Generation 450:	Best = 1.477414e+03	FES = 45100	Diversity = 1.8080e+01	k_t = [1, 1, 1, 1]
Generation 460:	Best = 1.476862e+03	FES = 46100	Diversity = 1.7926e+01	k_t = [4, 2, 3, 2]
Generation 470:	Best = 1.475984e+03	FES = 47100	Diversity = 1.7850e+01	k_t = [1, 1, 1, 1]
Generation 480:	Best = 1.475326e+03	FES = 48100	Diversity = 1.7680e+01	k_t = [3, 2, 4, 2]
Generation 490:	Best = 1.474995e+03	FES = 49100	Diversity = 1.7624e+01	k_t = [1, 1, 1, 1]

Generation 500:	Best = 1.474822e+03	FES = 50100	Diversity = 1.7659e+01	k_t = [4, 4, 4, 3]
Generation 510:	Best = 1.474822e+03	FES = 51100	Diversity = 1.7528e+01	k_t = [1, 1, 1, 1]
Generation 520:	Best = 1.474423e+03	FES = 52100	Diversity = 1.7491e+01	k_t = [3, 2, 2, 4]
Generation 530:	Best = 1.464063e+03	FES = 53100	Diversity = 1.7507e+01	k_t = [1, 1, 1, 1]
Generation 540:	Best = 1.456431e+03	FES = 54100	Diversity = 1.7565e+01	k_t = [4, 4, 4, 2]
Generation 550:	Best = 1.455792e+03	FES = 55100	Diversity = 1.7690e+01	k_t = [1, 1, 1, 1]
Generation 560:	Best = 1.455553e+03	FES = 56100	Diversity = 1.7494e+01	k_t = [2, 2, 4, 4]
Generation 570:	Best = 1.451926e+03	FES = 57100	Diversity = 1.7463e+01	k_t = [1, 1, 1, 1]
Generation 580:	Best = 1.449328e+03	FES = 58100	Diversity = 1.7371e+01	k_t = [2, 2, 3, 3]
Generation 590:	Best = 1.446993e+03	FES = 59100	Diversity = 1.6715e+01	k_t = [1, 1, 1, 1]
Generation 600:	Best = 1.446198e+03	FES = 60100	Diversity = 1.6433e+01	k_t = [2, 2, 2, 4]
Generation 610:	Best = 1.443344e+03	FES = 61100	Diversity = 1.5881e+01	k_t = [1, 1, 1, 1]
Generation 620:	Best = 1.443344e+03	FES = 62100	Diversity = 1.5263e+01	k_t = [4, 4, 4, 3]
Generation 630:	Best = 1.443344e+03	FES = 63100	Diversity = 1.3636e+01	k_t = [1, 1, 1, 1]
Generation 640:	Best = 1.443344e+03	FES = 64100	Diversity = 1.1528e+01	k_t = [4, 3, 4, 2]
Generation 650:	Best = 1.440528e+03	FES = 65100	Diversity = 1.0032e+01	k_t = [1, 1, 1, 1]
Generation 660:	Best = 1.438814e+03	FES = 66100	Diversity = 8.7370e+00	k_t = [4, 4, 2, 3]
Generation 670:	Best = 1.437732e+03	FES = 67100	Diversity = 8.1228e+00	k_t = [1, 1, 1, 1]
Generation 680:	Best = 1.435808e+03	FES = 68100	Diversity = 7.9018e+00	k_t = [3, 4, 2, 4]
Generation 690:	Best = 1.434450e+03	FES = 69100	Diversity = 7.8331e+00	k_t = [1, 1, 1, 1]
Generation 700:	Best = 1.433952e+03	FES = 70100	Diversity = 7.6035e+00	k_t = [4, 2, 4, 2]
Generation 710:	Best = 1.433726e+03	FES = 71100	Diversity = 7.6118e+00	k_t = [1, 1, 1, 1]
Generation 720:	Best = 1.433663e+03	FES = 72100	Diversity = 7.4795e+00	k_t = [3, 4, 3, 2]
Generation 730:	Best = 1.433486e+03	FES = 73100	Diversity = 6.8890e+00	k_t = [1, 1, 1, 1]
Generation 740:	Best = 1.432033e+03	FES = 74100	Diversity = 7.0053e+00	k_t = [4, 2, 4, 3]
Generation 750:	Best = 1.432033e+03	FES = 75100	Diversity = 6.7629e+00	k_t = [1, 1, 1, 1]
Generation 760:	Best = 1.432033e+03	FES = 76100	Diversity = 6.7762e+00	k_t = [2, 2, 2, 2]

Generation 770:	Best = 1.431999e+03	FES = 77100	Diversity = 6.3744e+00	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 1.431999e+03	FES = 78100	Diversity = 6.2414e+00	k_t
= [3, 4, 4, 4]				
Generation 790:	Best = 1.431956e+03	FES = 79100	Diversity = 6.3908e+00	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 1.431897e+03	FES = 80100	Diversity = 6.2889e+00	k_t
= [3, 3, 4, 4]				
Generation 810:	Best = 1.431763e+03	FES = 81100	Diversity = 6.3148e+00	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 1.431642e+03	FES = 82100	Diversity = 6.2605e+00	k_t
= [4, 4, 3, 2]				
Generation 830:	Best = 1.431430e+03	FES = 83100	Diversity = 6.4116e+00	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 1.431325e+03	FES = 84100	Diversity = 6.4428e+00	k_t
= [4, 4, 4, 4]				
Generation 850:	Best = 1.431325e+03	FES = 85100	Diversity = 6.3395e+00	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 1.430945e+03	FES = 86100	Diversity = 6.4687e+00	k_t
= [4, 4, 3, 4]				
Generation 870:	Best = 1.430945e+03	FES = 87100	Diversity = 6.5878e+00	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 1.430729e+03	FES = 88100	Diversity = 6.7067e+00	k_t
= [3, 2, 3, 2]				
Generation 890:	Best = 1.430720e+03	FES = 89100	Diversity = 6.7217e+00	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 1.430720e+03	FES = 90100	Diversity = 6.7656e+00	k_t
= [4, 4, 4, 4]				
Generation 910:	Best = 1.430720e+03	FES = 91100	Diversity = 6.7033e+00	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 1.430720e+03	FES = 92100	Diversity = 6.6794e+00	k_t
= [2, 3, 2, 3]				
Generation 930:	Best = 1.430621e+03	FES = 93100	Diversity = 6.8461e+00	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 1.430621e+03	FES = 94100	Diversity = 6.8985e+00	k_t
= [3, 2, 2, 2]				
Generation 950:	Best = 1.430618e+03	FES = 95100	Diversity = 7.1217e+00	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 1.430618e+03	FES = 96100	Diversity = 7.2010e+00	k_t
= [2, 4, 2, 4]				
Generation 970:	Best = 1.430618e+03	FES = 97100	Diversity = 7.1974e+00	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 1.430552e+03	FES = 98100	Diversity = 7.1697e+00	k_t
= [3, 4, 2, 3]				
Generation 990:	Best = 1.430548e+03	FES = 99100	Diversity = 7.3441e+00	k_t
= [1, 1, 1, 1]				
Generation 1000:	Best = 1.430548e+03	FES = 100100	Diversity = 7.3051e+00	k_t
= [2, 3, 3, 2]				
Generation 1010:	Best = 1.430548e+03	FES = 101100	Diversity = 7.2721e+00	k_t
= [1, 1, 1, 1]				
Generation 1020:	Best = 1.430548e+03	FES = 102100	Diversity = 7.3778e+00	k_t
= [3, 2, 3, 2]				
Generation 1030:	Best = 1.430548e+03	FES = 103100	Diversity = 7.4905e+00	k_t
= [1, 1, 1, 1]				

Generation 1040: Best = 1.430533e+03 = [2, 2, 4, 4]	FES = 104100	Diversity = 7.5134e+00	k_t
Generation 1050: Best = 1.430492e+03 = [1, 1, 1, 1]	FES = 105100	Diversity = 7.6269e+00	k_t
Generation 1060: Best = 1.430492e+03 = [2, 4, 4, 3]	FES = 106100	Diversity = 7.6004e+00	k_t
Generation 1070: Best = 1.430459e+03 = [1, 1, 1, 1]	FES = 107100	Diversity = 7.6531e+00	k_t
Generation 1080: Best = 1.430399e+03 = [4, 4, 3, 4]	FES = 108100	Diversity = 7.7240e+00	k_t
Generation 1090: Best = 1.430365e+03 = [1, 1, 1, 1]	FES = 109100	Diversity = 7.4825e+00	k_t
Generation 1100: Best = 1.430361e+03 = [4, 2, 4, 3]	FES = 110100	Diversity = 7.6981e+00	k_t
Generation 1110: Best = 1.430361e+03 = [1, 1, 1, 1]	FES = 111100	Diversity = 7.6272e+00	k_t
Generation 1120: Best = 1.430361e+03 = [3, 4, 3, 3]	FES = 112100	Diversity = 7.7337e+00	k_t
Generation 1130: Best = 1.430361e+03 = [1, 1, 1, 1]	FES = 113100	Diversity = 7.8489e+00	k_t
Generation 1140: Best = 1.430268e+03 = [3, 4, 3, 2]	FES = 114100	Diversity = 7.8734e+00	k_t
Generation 1150: Best = 1.430073e+03 = [1, 1, 1, 1]	FES = 115100	Diversity = 7.8754e+00	k_t
Generation 1160: Best = 1.430068e+03 = [2, 4, 4, 4]	FES = 116100	Diversity = 8.0087e+00	k_t
Generation 1170: Best = 1.430066e+03 = [1, 1, 1, 1]	FES = 117100	Diversity = 7.9701e+00	k_t
Generation 1180: Best = 1.430066e+03 = [3, 4, 3, 2]	FES = 118100	Diversity = 7.9948e+00	k_t
Generation 1190: Best = 1.430066e+03 = [1, 1, 1, 1]	FES = 119100	Diversity = 7.8880e+00	k_t
Generation 1200: Best = 1.430066e+03 = [2, 4, 3, 4]	FES = 120100	Diversity = 7.8367e+00	k_t
Generation 1210: Best = 1.430066e+03 = [1, 1, 1, 1]	FES = 121100	Diversity = 8.1342e+00	k_t
Generation 1220: Best = 1.430066e+03 = [3, 2, 3, 2]	FES = 122100	Diversity = 8.3657e+00	k_t
Generation 1230: Best = 1.429957e+03 = [1, 1, 1, 1]	FES = 123100	Diversity = 8.2665e+00	k_t
Generation 1240: Best = 1.429923e+03 = [2, 4, 2, 4]	FES = 124100	Diversity = 8.3707e+00	k_t
Generation 1250: Best = 1.429918e+03 = [1, 1, 1, 1]	FES = 125100	Diversity = 8.4294e+00	k_t
Generation 1260: Best = 1.429910e+03 = [2, 2, 4, 3]	FES = 126100	Diversity = 8.7210e+00	k_t
Generation 1270: Best = 1.429907e+03 = [1, 1, 1, 1]	FES = 127100	Diversity = 8.7138e+00	k_t
Generation 1280: Best = 1.429878e+03 = [4, 3, 3, 4]	FES = 128100	Diversity = 8.7796e+00	k_t
Generation 1290: Best = 1.429862e+03 = [1, 1, 1, 1]	FES = 129100	Diversity = 8.6592e+00	k_t
Generation 1300: Best = 1.429833e+03 = [4, 4, 2, 3]	FES = 130100	Diversity = 8.6715e+00	k_t

Generation 1310: Best = 1.429747e+03 = [1, 1, 1, 1]	FES = 131100	Diversity = 8.6625e+00	k_t
Generation 1320: Best = 1.429747e+03 = [2, 2, 4, 3]	FES = 132100	Diversity = 8.6961e+00	k_t
Generation 1330: Best = 1.429711e+03 = [1, 1, 1, 1]	FES = 133100	Diversity = 8.6536e+00	k_t
Generation 1340: Best = 1.429626e+03 = [3, 3, 4, 2]	FES = 134100	Diversity = 8.6899e+00	k_t
Generation 1350: Best = 1.429592e+03 = [1, 1, 1, 1]	FES = 135100	Diversity = 8.7973e+00	k_t
Generation 1360: Best = 1.429592e+03 = [2, 4, 2, 4]	FES = 136100	Diversity = 8.8925e+00	k_t
Generation 1370: Best = 1.429591e+03 = [1, 1, 1, 1]	FES = 137100	Diversity = 9.2071e+00	k_t
Generation 1380: Best = 1.429591e+03 = [3, 2, 2, 2]	FES = 138100	Diversity = 9.0684e+00	k_t
Generation 1390: Best = 1.429543e+03 = [1, 1, 1, 1]	FES = 139100	Diversity = 9.1640e+00	k_t
Generation 1400: Best = 1.429538e+03 = [2, 3, 4, 2]	FES = 140100	Diversity = 9.1249e+00	k_t
Generation 1410: Best = 1.429538e+03 = [1, 1, 1, 1]	FES = 141100	Diversity = 9.1659e+00	k_t
Generation 1420: Best = 1.429514e+03 = [2, 2, 4, 3]	FES = 142100	Diversity = 9.2252e+00	k_t
Generation 1430: Best = 1.429506e+03 = [1, 1, 1, 1]	FES = 143100	Diversity = 9.1824e+00	k_t
Generation 1440: Best = 1.429506e+03 = [3, 4, 4, 2]	FES = 144100	Diversity = 9.3234e+00	k_t
Generation 1450: Best = 1.429506e+03 = [1, 1, 1, 1]	FES = 145100	Diversity = 9.2662e+00	k_t
Generation 1460: Best = 1.429506e+03 = [2, 3, 4, 3]	FES = 146100	Diversity = 9.3693e+00	k_t
Generation 1470: Best = 1.429498e+03 = [1, 1, 1, 1]	FES = 147100	Diversity = 9.3766e+00	k_t
Generation 1480: Best = 1.429498e+03 = [4, 3, 3, 2]	FES = 148100	Diversity = 9.4443e+00	k_t
Generation 1490: Best = 1.429498e+03 = [1, 1, 1, 1]	FES = 149100	Diversity = 9.3342e+00	k_t
Generation 1500: Best = 1.429498e+03 = [3, 3, 4, 2]	FES = 150100	Diversity = 9.3256e+00	k_t
Generation 1510: Best = 1.429498e+03 = [1, 1, 1, 1]	FES = 151100	Diversity = 9.3350e+00	k_t
Generation 1520: Best = 1.429496e+03 = [3, 4, 4, 4]	FES = 152100	Diversity = 9.2183e+00	k_t
Generation 1530: Best = 1.429496e+03 = [1, 1, 1, 1]	FES = 153100	Diversity = 9.2011e+00	k_t
Generation 1540: Best = 1.429473e+03 = [3, 3, 4, 2]	FES = 154100	Diversity = 9.1406e+00	k_t
Generation 1550: Best = 1.429473e+03 = [1, 1, 1, 1]	FES = 155100	Diversity = 9.1118e+00	k_t
Generation 1560: Best = 1.429472e+03 = [2, 3, 2, 2]	FES = 156100	Diversity = 9.1411e+00	k_t
Generation 1570: Best = 1.429472e+03 = [1, 1, 1, 1]	FES = 157100	Diversity = 9.1633e+00	k_t

Generation 1580: Best = 1.429472e+03 = [4, 3, 2, 3]	FES = 158100	Diversity = 9.2677e+00	k_t
Generation 1590: Best = 1.429391e+03 = [1, 1, 1, 1]	FES = 159100	Diversity = 9.5589e+00	k_t
Generation 1600: Best = 1.429386e+03 = [4, 4, 3, 4]	FES = 160100	Diversity = 9.4966e+00	k_t
Generation 1610: Best = 1.429386e+03 = [1, 1, 1, 1]	FES = 161100	Diversity = 9.4760e+00	k_t
Generation 1620: Best = 1.429386e+03 = [2, 2, 3, 4]	FES = 162100	Diversity = 9.5038e+00	k_t
Generation 1630: Best = 1.429386e+03 = [1, 1, 1, 1]	FES = 163100	Diversity = 9.4381e+00	k_t
Generation 1640: Best = 1.429386e+03 = [4, 3, 3, 2]	FES = 164100	Diversity = 9.4380e+00	k_t
Generation 1650: Best = 1.429386e+03 = [1, 1, 1, 1]	FES = 165100	Diversity = 9.4946e+00	k_t
Generation 1660: Best = 1.429386e+03 = [4, 3, 4, 2]	FES = 166100	Diversity = 9.6005e+00	k_t
Generation 1670: Best = 1.429386e+03 = [1, 1, 1, 1]	FES = 167100	Diversity = 9.4728e+00	k_t
Generation 1680: Best = 1.429371e+03 = [3, 2, 2, 3]	FES = 168100	Diversity = 9.5538e+00	k_t
Generation 1690: Best = 1.429369e+03 = [1, 1, 1, 1]	FES = 169100	Diversity = 9.4650e+00	k_t
Generation 1700: Best = 1.429355e+03 = [3, 4, 2, 4]	FES = 170100	Diversity = 9.5113e+00	k_t
Generation 1710: Best = 1.429352e+03 = [1, 1, 1, 1]	FES = 171100	Diversity = 9.5159e+00	k_t
Generation 1720: Best = 1.429352e+03 = [4, 4, 4, 2]	FES = 172100	Diversity = 9.4681e+00	k_t
Generation 1730: Best = 1.429313e+03 = [1, 1, 1, 1]	FES = 173100	Diversity = 9.4487e+00	k_t
Generation 1740: Best = 1.429313e+03 = [3, 4, 4, 2]	FES = 174100	Diversity = 9.3491e+00	k_t
Generation 1750: Best = 1.429288e+03 = [1, 1, 1, 1]	FES = 175100	Diversity = 9.3424e+00	k_t
Generation 1760: Best = 1.429288e+03 = [2, 2, 3, 2]	FES = 176100	Diversity = 9.3672e+00	k_t
Generation 1770: Best = 1.429288e+03 = [1, 1, 1, 1]	FES = 177100	Diversity = 9.3762e+00	k_t
Generation 1780: Best = 1.429288e+03 = [3, 2, 3, 3]	FES = 178100	Diversity = 9.2394e+00	k_t
Generation 1790: Best = 1.429287e+03 = [1, 1, 1, 1]	FES = 179100	Diversity = 9.1402e+00	k_t
Generation 1800: Best = 1.429263e+03 = [4, 2, 2, 2]	FES = 180100	Diversity = 9.0956e+00	k_t
Generation 1810: Best = 1.429262e+03 = [1, 1, 1, 1]	FES = 181100	Diversity = 9.2436e+00	k_t
Generation 1820: Best = 1.429262e+03 = [4, 4, 3, 2]	FES = 182100	Diversity = 8.9119e+00	k_t
Generation 1830: Best = 1.429259e+03 = [1, 1, 1, 1]	FES = 183100	Diversity = 8.9402e+00	k_t
Generation 1840: Best = 1.429259e+03 = [4, 2, 4, 2]	FES = 184100	Diversity = 8.8198e+00	k_t

Generation 1850: Best = 1.429253e+03 = [1, 1, 1, 1]	FES = 185100	Diversity = 8.4588e+00	k_t
Generation 1860: Best = 1.429230e+03 = [2, 2, 4, 3]	FES = 186100	Diversity = 8.0830e+00	k_t
Generation 1870: Best = 1.429230e+03 = [1, 1, 1, 1]	FES = 187100	Diversity = 8.0020e+00	k_t
Generation 1880: Best = 1.429226e+03 = [4, 3, 4, 2]	FES = 188100	Diversity = 7.7980e+00	k_t
Generation 1890: Best = 1.429225e+03 = [1, 1, 1, 1]	FES = 189100	Diversity = 7.2894e+00	k_t
Generation 1900: Best = 1.429225e+03 = [3, 2, 3, 4]	FES = 190100	Diversity = 7.0937e+00	k_t
Generation 1910: Best = 1.429201e+03 = [1, 1, 1, 1]	FES = 191100	Diversity = 6.9346e+00	k_t
Generation 1920: Best = 1.429200e+03 = [2, 3, 4, 4]	FES = 192100	Diversity = 6.2417e+00	k_t
Generation 1930: Best = 1.429198e+03 = [1, 1, 1, 1]	FES = 193100	Diversity = 5.7107e+00	k_t
Generation 1940: Best = 1.429195e+03 = [3, 3, 4, 3]	FES = 194100	Diversity = 5.1641e+00	k_t
Generation 1950: Best = 1.429192e+03 = [1, 1, 1, 1]	FES = 195100	Diversity = 4.4265e+00	k_t
Generation 1960: Best = 1.429151e+03 = [4, 4, 2, 3]	FES = 196100	Diversity = 3.2484e+00	k_t
Generation 1970: Best = 1.429118e+03 = [1, 1, 1, 1]	FES = 197100	Diversity = 2.8518e+00	k_t
Generation 1980: Best = 1.429105e+03 = [4, 3, 3, 2]	FES = 198100	Diversity = 1.5659e+00	k_t
Generation 1990: Best = 1.429089e+03 = [1, 1, 1, 1]	FES = 199100	Diversity = 8.8653e-01	k_t
Generation 2000: Best = 1.429086e+03 = [3, 3, 2, 2]	FES = 200100	Diversity = 8.8432e-01	k_t
Generation 2010: Best = 1.429070e+03 = [1, 1, 1, 1]	FES = 201100	Diversity = 2.5238e-02	k_t
Generation 2020: Best = 1.429060e+03 = [4, 2, 3, 3]	FES = 202100	Diversity = 2.1975e-02	k_t
Generation 2030: Best = 1.429058e+03 = [1, 1, 1, 1]	FES = 203100	Diversity = 1.9037e-02	k_t
Generation 2040: Best = 1.429048e+03 = [3, 3, 4, 4]	FES = 204100	Diversity = 1.5523e-02	k_t
Generation 2050: Best = 1.429044e+03 = [1, 1, 1, 1]	FES = 205100	Diversity = 1.3334e-02	k_t
Generation 2060: Best = 1.429035e+03 = [3, 2, 4, 4]	FES = 206100	Diversity = 1.1837e-02	k_t
Generation 2070: Best = 1.429030e+03 = [1, 1, 1, 1]	FES = 207100	Diversity = 1.1262e-02	k_t
Generation 2080: Best = 1.429016e+03 = [4, 4, 2, 3]	FES = 208100	Diversity = 1.1299e-02	k_t
Generation 2090: Best = 1.428999e+03 = [1, 1, 1, 1]	FES = 209100	Diversity = 1.2459e-02	k_t
Generation 2100: Best = 1.428996e+03 = [3, 4, 4, 2]	FES = 210100	Diversity = 1.1676e-02	k_t
Generation 2110: Best = 1.428992e+03 = [1, 1, 1, 1]	FES = 211100	Diversity = 1.0090e-02	k_t

Generation 2120: Best = 1.428991e+03 = [2, 3, 4, 4]	FES = 212100	Diversity = 9.1917e-03	k_t
Generation 2130: Best = 1.428985e+03 = [1, 1, 1, 1]	FES = 213100	Diversity = 7.9445e-03	k_t
Generation 2140: Best = 1.428984e+03 = [3, 4, 3, 4]	FES = 214100	Diversity = 7.2147e-03	k_t
Generation 2150: Best = 1.428981e+03 = [1, 1, 1, 1]	FES = 215100	Diversity = 6.9772e-03	k_t
Generation 2160: Best = 1.428976e+03 = [2, 4, 4, 3]	FES = 216100	Diversity = 6.9109e-03	k_t
Generation 2170: Best = 1.428976e+03 = [1, 1, 1, 1]	FES = 217100	Diversity = 6.8560e-03	k_t
Generation 2180: Best = 1.428974e+03 = [2, 3, 3, 4]	FES = 218100	Diversity = 6.3974e-03	k_t
Generation 2190: Best = 1.428972e+03 = [1, 1, 1, 1]	FES = 219100	Diversity = 5.8116e-03	k_t
Generation 2200: Best = 1.428970e+03 = [3, 4, 4, 2]	FES = 220100	Diversity = 5.3455e-03	k_t
Generation 2210: Best = 1.428968e+03 = [1, 1, 1, 1]	FES = 221100	Diversity = 5.1097e-03	k_t
Generation 2220: Best = 1.428966e+03 = [4, 4, 2, 4]	FES = 222100	Diversity = 4.8649e-03	k_t
Generation 2230: Best = 1.428966e+03 = [1, 1, 1, 1]	FES = 223100	Diversity = 4.6918e-03	k_t
Generation 2240: Best = 1.428963e+03 = [4, 3, 2, 2]	FES = 224100	Diversity = 4.3300e-03	k_t
Generation 2250: Best = 1.428962e+03 = [1, 1, 1, 1]	FES = 225100	Diversity = 4.3381e-03	k_t
Generation 2260: Best = 1.428959e+03 = [3, 4, 2, 3]	FES = 226100	Diversity = 4.4485e-03	k_t
Generation 2270: Best = 1.428959e+03 = [1, 1, 1, 1]	FES = 227100	Diversity = 4.1660e-03	k_t
Generation 2280: Best = 1.428957e+03 = [4, 4, 2, 4]	FES = 228100	Diversity = 4.4397e-03	k_t
Generation 2290: Best = 1.428953e+03 = [1, 1, 1, 1]	FES = 229100	Diversity = 5.2784e-03	k_t
Generation 2300: Best = 1.428951e+03 = [3, 3, 3, 2]	FES = 230100	Diversity = 5.6103e-03	k_t
Generation 2310: Best = 1.428949e+03 = [1, 1, 1, 1]	FES = 231100	Diversity = 5.6728e-03	k_t
Generation 2320: Best = 1.428945e+03 = [2, 2, 4, 4]	FES = 232100	Diversity = 5.8220e-03	k_t
Generation 2330: Best = 1.428944e+03 = [1, 1, 1, 1]	FES = 233100	Diversity = 5.8165e-03	k_t
Generation 2340: Best = 1.428942e+03 = [3, 4, 2, 3]	FES = 234100	Diversity = 5.6383e-03	k_t
Generation 2350: Best = 1.428940e+03 = [1, 1, 1, 1]	FES = 235100	Diversity = 5.6127e-03	k_t
Generation 2360: Best = 1.428938e+03 = [3, 3, 4, 4]	FES = 236100	Diversity = 5.5199e-03	k_t
Generation 2370: Best = 1.428937e+03 = [1, 1, 1, 1]	FES = 237100	Diversity = 5.7436e-03	k_t
Generation 2380: Best = 1.428935e+03 = [4, 3, 3, 4]	FES = 238100	Diversity = 5.3910e-03	k_t

Generation 2390: Best = 1.428931e+03 = [1, 1, 1, 1]	FES = 239100	Diversity = 5.3761e-03	k_t
Generation 2400: Best = 1.428929e+03 = [3, 3, 2, 3]	FES = 240100	Diversity = 5.9448e-03	k_t
Generation 2410: Best = 1.428925e+03 = [1, 1, 1, 1]	FES = 241100	Diversity = 6.9487e-03	k_t
Generation 2420: Best = 1.428916e+03 = [4, 3, 3, 2]	FES = 242100	Diversity = 8.2666e-03	k_t
Generation 2430: Best = 1.428915e+03 = [1, 1, 1, 1]	FES = 243100	Diversity = 7.6732e-03	k_t
Generation 2440: Best = 1.428907e+03 = [4, 2, 4, 3]	FES = 244100	Diversity = 7.9877e-03	k_t
Generation 2450: Best = 1.428898e+03 = [1, 1, 1, 1]	FES = 245100	Diversity = 8.8685e-03	k_t
Generation 2460: Best = 1.428889e+03 = [4, 4, 3, 3]	FES = 246100	Diversity = 8.8395e-03	k_t
Generation 2470: Best = 1.428886e+03 = [1, 1, 1, 1]	FES = 247100	Diversity = 8.9382e-03	k_t
Generation 2480: Best = 1.428882e+03 = [4, 3, 2, 4]	FES = 248100	Diversity = 9.1379e-03	k_t
Generation 2490: Best = 1.428877e+03 = [1, 1, 1, 1]	FES = 249100	Diversity = 8.5880e-03	k_t
Generation 2500: Best = 1.428873e+03 = [4, 4, 4, 3]	FES = 250100	Diversity = 8.0683e-03	k_t
Generation 2510: Best = 1.428869e+03 = [1, 1, 1, 1]	FES = 251100	Diversity = 7.8700e-03	k_t
Generation 2520: Best = 1.428864e+03 = [2, 2, 3, 4]	FES = 252100	Diversity = 7.8970e-03	k_t
Generation 2530: Best = 1.428861e+03 = [1, 1, 1, 1]	FES = 253100	Diversity = 8.5270e-03	k_t
Generation 2540: Best = 1.428856e+03 = [4, 3, 3, 4]	FES = 254100	Diversity = 9.0659e-03	k_t
Generation 2550: Best = 1.428835e+03 = [1, 1, 1, 1]	FES = 255100	Diversity = 9.4323e-03	k_t
Generation 2560: Best = 1.428831e+03 = [3, 2, 2, 3]	FES = 256100	Diversity = 1.0428e-02	k_t
Generation 2570: Best = 1.428822e+03 = [1, 1, 1, 1]	FES = 257100	Diversity = 1.0594e-02	k_t
Generation 2580: Best = 1.428812e+03 = [3, 3, 3, 2]	FES = 258100	Diversity = 1.1091e-02	k_t
Generation 2590: Best = 1.428801e+03 = [1, 1, 1, 1]	FES = 259100	Diversity = 1.2008e-02	k_t
Generation 2600: Best = 1.428791e+03 = [4, 3, 4, 2]	FES = 260100	Diversity = 1.2449e-02	k_t
Generation 2610: Best = 1.428785e+03 = [1, 1, 1, 1]	FES = 261100	Diversity = 1.3286e-02	k_t
Generation 2620: Best = 1.428771e+03 = [3, 3, 3, 3]	FES = 262100	Diversity = 1.3484e-02	k_t
Generation 2630: Best = 1.428768e+03 = [1, 1, 1, 1]	FES = 263100	Diversity = 1.3168e-02	k_t
Generation 2640: Best = 1.428759e+03 = [2, 3, 3, 2]	FES = 264100	Diversity = 1.2982e-02	k_t
Generation 2650: Best = 1.428749e+03 = [1, 1, 1, 1]	FES = 265100	Diversity = 1.1451e-02	k_t

Generation 2660: Best = 1.428739e+03 = [2, 2, 3, 2]	FES = 266100	Diversity = 1.1882e-02	k_t
Generation 2670: Best = 1.428735e+03 = [1, 1, 1, 1]	FES = 267100	Diversity = 1.1793e-02	k_t
Generation 2680: Best = 1.428730e+03 = [4, 2, 4, 2]	FES = 268100	Diversity = 1.1589e-02	k_t
Generation 2690: Best = 1.428722e+03 = [1, 1, 1, 1]	FES = 269100	Diversity = 1.0667e-02	k_t
Generation 2700: Best = 1.428718e+03 = [4, 3, 3, 2]	FES = 270100	Diversity = 1.0356e-02	k_t
Generation 2710: Best = 1.428707e+03 = [1, 1, 1, 1]	FES = 271100	Diversity = 1.0592e-02	k_t
Generation 2720: Best = 1.428705e+03 = [4, 4, 3, 3]	FES = 272100	Diversity = 1.0752e-02	k_t
Generation 2730: Best = 1.428698e+03 = [1, 1, 1, 1]	FES = 273100	Diversity = 1.0686e-02	k_t
Generation 2740: Best = 1.428685e+03 = [4, 3, 2, 3]	FES = 274100	Diversity = 1.0784e-02	k_t
Generation 2750: Best = 1.428684e+03 = [1, 1, 1, 1]	FES = 275100	Diversity = 1.1869e-02	k_t
Generation 2760: Best = 1.428676e+03 = [3, 4, 3, 4]	FES = 276100	Diversity = 1.1519e-02	k_t
Generation 2770: Best = 1.428673e+03 = [1, 1, 1, 1]	FES = 277100	Diversity = 1.1350e-02	k_t
Generation 2780: Best = 1.428670e+03 = [4, 4, 3, 2]	FES = 278100	Diversity = 1.0559e-02	k_t
Generation 2790: Best = 1.428662e+03 = [1, 1, 1, 1]	FES = 279100	Diversity = 1.0472e-02	k_t
Generation 2800: Best = 1.428658e+03 = [3, 2, 2, 2]	FES = 280100	Diversity = 9.8078e-03	k_t
Generation 2810: Best = 1.428652e+03 = [1, 1, 1, 1]	FES = 281100	Diversity = 9.8776e-03	k_t
Generation 2820: Best = 1.428648e+03 = [4, 2, 4, 2]	FES = 282100	Diversity = 1.0406e-02	k_t
Generation 2830: Best = 1.428638e+03 = [1, 1, 1, 1]	FES = 283100	Diversity = 1.1158e-02	k_t
Generation 2840: Best = 1.428636e+03 = [2, 4, 2, 2]	FES = 284100	Diversity = 1.1058e-02	k_t
Generation 2850: Best = 1.428630e+03 = [1, 1, 1, 1]	FES = 285100	Diversity = 1.1674e-02	k_t
Generation 2860: Best = 1.428623e+03 = [2, 3, 3, 2]	FES = 286100	Diversity = 1.1026e-02	k_t
Generation 2870: Best = 1.428618e+03 = [1, 1, 1, 1]	FES = 287100	Diversity = 1.0500e-02	k_t
Generation 2880: Best = 1.428613e+03 = [3, 4, 2, 4]	FES = 288100	Diversity = 1.0505e-02	k_t
Generation 2890: Best = 1.428605e+03 = [1, 1, 1, 1]	FES = 289100	Diversity = 1.1285e-02	k_t
Generation 2900: Best = 1.428591e+03 = [4, 4, 4, 2]	FES = 290100	Diversity = 1.2363e-02	k_t
Generation 2910: Best = 1.428579e+03 = [1, 1, 1, 1]	FES = 291100	Diversity = 1.2152e-02	k_t
Generation 2920: Best = 1.428574e+03 = [2, 4, 3, 4]	FES = 292100	Diversity = 1.3704e-02	k_t

Generation 2930: Best = 1.428559e+03 FES = 293100 Diversity = 1.3824e-02 k_t
 = [1, 1, 1, 1]
 Generation 2940: Best = 1.428555e+03 FES = 294100 Diversity = 1.5221e-02 k_t
 = [2, 4, 3, 3]
 Generation 2950: Best = 1.428544e+03 FES = 295100 Diversity = 1.5913e-02 k_t
 = [1, 1, 1, 1]
 Generation 2960: Best = 1.428529e+03 FES = 296100 Diversity = 1.7124e-02 k_t
 = [2, 2, 3, 3]
 Generation 2970: Best = 1.428513e+03 FES = 297100 Diversity = 1.7301e-02 k_t
 = [1, 1, 1, 1]
 Generation 2980: Best = 1.428513e+03 FES = 298100 Diversity = 1.7034e-02 k_t
 = [2, 2, 3, 3]
 Generation 2990: Best = 1.428485e+03 FES = 299100 Diversity = 1.7130e-02 k_t
 = [1, 1, 1, 1]

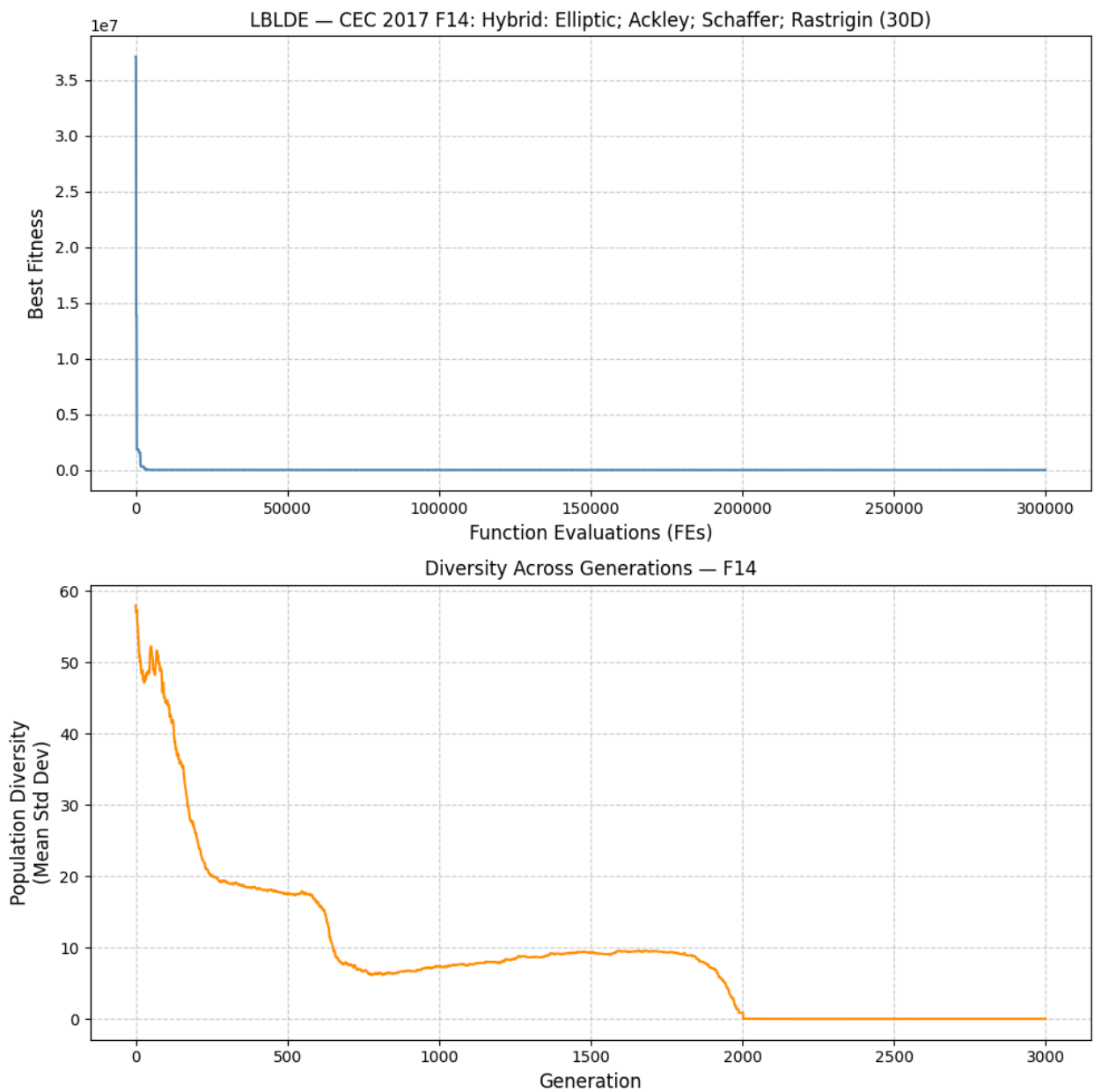
Optimization completed!

Final Best Fitness = 1.428482e+03

Total Function Evaluations = 300000

```

=====
=
Results:
  Best Fitness           : 1.428482e+03
  Global Optimum         : 1.400000e+03
  Error (|best-optimum|) : 2.848173e+01
  Best Solution (5 dim)  : [-36.63695428 -63.11632288  1.72087806  90.270835
61.7128652 ]
  D^(0) (init. diversity): 5.7976e+01
=====
=
  
```



Graph saved as convergence_F14.png

```
In [25]: # =====
# F15 – Hybrid Function 5 (Bent Cigar; HGBat; Rastrigin; Rosenbrock)
# =====
run_and_plot('F15', make_wrapper(hybrid.f15), 1500.0, 'Hybrid: Bent Cigar; HGBat; Rastrigin; Rosenbrock')
```

```

=====
=
LBLDE_s on CEC 2017 F15 – Single Run
=====
=
Function      : Hybrid: Bent Cigar; HGBat; Rastrigin; Rosenbrock (30D)
Bounds       : [-100, 100]
Global optimum : 1500.0
Initial D^(0) = 5.7976e+01
Generation 0: Best Fitness = 4.037230e+10 | Diversity = 5.7976e+01
Generation 10: Best = 2.543494e+09 FES = 1100 Diversity = 4.6628e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 2.521528e+08 FES = 2100 Diversity = 4.3237e+01 k_t
= [3, 3, 2, 2]
Generation 30: Best = 4.332155e+07 FES = 3100 Diversity = 4.3955e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 1.375983e+07 FES = 4100 Diversity = 4.3999e+01 k_t
= [2, 3, 2, 3]
Generation 50: Best = 2.942009e+06 FES = 5100 Diversity = 4.4684e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 1.044930e+06 FES = 6100 Diversity = 4.4929e+01 k_t
= [2, 2, 2, 3]
Generation 70: Best = 7.166060e+05 FES = 7100 Diversity = 4.5029e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 2.022829e+05 FES = 8100 Diversity = 4.8984e+01 k_t
= [2, 3, 4, 2]
Generation 90: Best = 1.289363e+05 FES = 9100 Diversity = 4.9976e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 6.432268e+04 FES = 10100 Diversity = 4.9516e+01 k_t
= [4, 2, 2, 3]
Generation 110: Best = 2.265001e+04 FES = 11100 Diversity = 4.7397e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 1.249320e+04 FES = 12100 Diversity = 4.5029e+01 k_t
= [2, 4, 4, 2]
Generation 130: Best = 3.923712e+03 FES = 13100 Diversity = 4.0617e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 3.923712e+03 FES = 14100 Diversity = 3.5809e+01 k_t
= [3, 2, 2, 2]
Generation 150: Best = 3.923712e+03 FES = 15100 Diversity = 3.3336e+01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 2.803382e+03 FES = 16100 Diversity = 3.1342e+01 k_t
= [3, 2, 3, 4]
Generation 170: Best = 2.405765e+03 FES = 17100 Diversity = 2.8790e+01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 2.107353e+03 FES = 18100 Diversity = 2.8522e+01 k_t
= [2, 2, 2, 4]
Generation 190: Best = 1.974358e+03 FES = 19100 Diversity = 2.4830e+01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 1.831015e+03 FES = 20100 Diversity = 2.1960e+01 k_t
= [3, 3, 2, 4]
Generation 210: Best = 1.791831e+03 FES = 21100 Diversity = 1.9409e+01 k_t
= [1, 1, 1, 1]
Generation 220: Best = 1.732562e+03 FES = 22100 Diversity = 1.7608e+01 k_t
= [4, 3, 4, 3]

```


Generation 230:	Best = 1.692685e+03	FES = 23100	Diversity = 1.5533e+01	k_t = [1, 1, 1, 1]
Generation 240:	Best = 1.642235e+03	FES = 24100	Diversity = 1.3448e+01	k_t = [4, 4, 2, 4]
Generation 250:	Best = 1.642235e+03	FES = 25100	Diversity = 1.2470e+01	k_t = [1, 1, 1, 1]
Generation 260:	Best = 1.623832e+03	FES = 26100	Diversity = 1.1543e+01	k_t = [3, 2, 3, 2]
Generation 270:	Best = 1.613008e+03	FES = 27100	Diversity = 1.0531e+01	k_t = [1, 1, 1, 1]
Generation 280:	Best = 1.601928e+03	FES = 28100	Diversity = 9.8167e+00	k_t = [3, 4, 3, 4]
Generation 290:	Best = 1.596293e+03	FES = 29100	Diversity = 9.0905e+00	k_t = [1, 1, 1, 1]
Generation 300:	Best = 1.588230e+03	FES = 30100	Diversity = 8.7059e+00	k_t = [3, 3, 4, 2]
Generation 310:	Best = 1.575904e+03	FES = 31100	Diversity = 8.1220e+00	k_t = [1, 1, 1, 1]
Generation 320:	Best = 1.575812e+03	FES = 32100	Diversity = 7.8730e+00	k_t = [4, 2, 4, 3]
Generation 330:	Best = 1.575812e+03	FES = 33100	Diversity = 7.7571e+00	k_t = [1, 1, 1, 1]
Generation 340:	Best = 1.565760e+03	FES = 34100	Diversity = 7.7090e+00	k_t = [3, 4, 3, 2]
Generation 350:	Best = 1.565760e+03	FES = 35100	Diversity = 7.6293e+00	k_t = [1, 1, 1, 1]
Generation 360:	Best = 1.565760e+03	FES = 36100	Diversity = 7.5698e+00	k_t = [4, 4, 2, 2]
Generation 370:	Best = 1.559906e+03	FES = 37100	Diversity = 7.4700e+00	k_t = [1, 1, 1, 1]
Generation 380:	Best = 1.559504e+03	FES = 38100	Diversity = 7.4551e+00	k_t = [4, 4, 2, 3]
Generation 390:	Best = 1.554906e+03	FES = 39100	Diversity = 7.3908e+00	k_t = [1, 1, 1, 1]
Generation 400:	Best = 1.554906e+03	FES = 40100	Diversity = 7.3113e+00	k_t = [4, 3, 2, 3]
Generation 410:	Best = 1.554906e+03	FES = 41100	Diversity = 7.3002e+00	k_t = [1, 1, 1, 1]
Generation 420:	Best = 1.546624e+03	FES = 42100	Diversity = 7.1297e+00	k_t = [2, 4, 2, 3]
Generation 430:	Best = 1.545938e+03	FES = 43100	Diversity = 6.9194e+00	k_t = [1, 1, 1, 1]
Generation 440:	Best = 1.545938e+03	FES = 44100	Diversity = 6.8143e+00	k_t = [4, 3, 2, 2]
Generation 450:	Best = 1.545938e+03	FES = 45100	Diversity = 6.7037e+00	k_t = [1, 1, 1, 1]
Generation 460:	Best = 1.545938e+03	FES = 46100	Diversity = 6.7015e+00	k_t = [3, 4, 2, 3]
Generation 470:	Best = 1.545558e+03	FES = 47100	Diversity = 6.5614e+00	k_t = [1, 1, 1, 1]
Generation 480:	Best = 1.537705e+03	FES = 48100	Diversity = 6.5057e+00	k_t = [2, 3, 2, 2]
Generation 490:	Best = 1.537705e+03	FES = 49100	Diversity = 6.4810e+00	k_t = [1, 1, 1, 1]

Generation 500:	Best = 1.537705e+03	FES = 50100	Diversity = 6.3868e+00	k_t
= [3, 2, 2, 2]				
Generation 510:	Best = 1.537414e+03	FES = 51100	Diversity = 6.3173e+00	k_t
= [1, 1, 1, 1]				
Generation 520:	Best = 1.536464e+03	FES = 52100	Diversity = 6.2874e+00	k_t
= [2, 2, 3, 4]				
Generation 530:	Best = 1.536464e+03	FES = 53100	Diversity = 6.3192e+00	k_t
= [1, 1, 1, 1]				
Generation 540:	Best = 1.535567e+03	FES = 54100	Diversity = 6.2393e+00	k_t
= [4, 3, 3, 4]				
Generation 550:	Best = 1.535567e+03	FES = 55100	Diversity = 6.0901e+00	k_t
= [1, 1, 1, 1]				
Generation 560:	Best = 1.535567e+03	FES = 56100	Diversity = 5.9797e+00	k_t
= [4, 4, 2, 4]				
Generation 570:	Best = 1.535260e+03	FES = 57100	Diversity = 5.9169e+00	k_t
= [1, 1, 1, 1]				
Generation 580:	Best = 1.535260e+03	FES = 58100	Diversity = 5.7091e+00	k_t
= [2, 4, 3, 3]				
Generation 590:	Best = 1.531932e+03	FES = 59100	Diversity = 5.7130e+00	k_t
= [1, 1, 1, 1]				
Generation 600:	Best = 1.531932e+03	FES = 60100	Diversity = 5.7115e+00	k_t
= [4, 4, 2, 2]				
Generation 610:	Best = 1.531032e+03	FES = 61100	Diversity = 5.7645e+00	k_t
= [1, 1, 1, 1]				
Generation 620:	Best = 1.531032e+03	FES = 62100	Diversity = 5.7263e+00	k_t
= [3, 4, 4, 4]				
Generation 630:	Best = 1.531032e+03	FES = 63100	Diversity = 5.6203e+00	k_t
= [1, 1, 1, 1]				
Generation 640:	Best = 1.531032e+03	FES = 64100	Diversity = 5.3639e+00	k_t
= [2, 4, 2, 2]				
Generation 650:	Best = 1.526040e+03	FES = 65100	Diversity = 5.2325e+00	k_t
= [1, 1, 1, 1]				
Generation 660:	Best = 1.526040e+03	FES = 66100	Diversity = 5.0929e+00	k_t
= [2, 3, 4, 4]				
Generation 670:	Best = 1.526040e+03	FES = 67100	Diversity = 4.8941e+00	k_t
= [1, 1, 1, 1]				
Generation 680:	Best = 1.525667e+03	FES = 68100	Diversity = 4.9141e+00	k_t
= [3, 2, 2, 2]				
Generation 690:	Best = 1.525667e+03	FES = 69100	Diversity = 4.9091e+00	k_t
= [1, 1, 1, 1]				
Generation 700:	Best = 1.525667e+03	FES = 70100	Diversity = 4.9927e+00	k_t
= [4, 3, 2, 4]				
Generation 710:	Best = 1.525667e+03	FES = 71100	Diversity = 4.9963e+00	k_t
= [1, 1, 1, 1]				
Generation 720:	Best = 1.525388e+03	FES = 72100	Diversity = 5.0056e+00	k_t
= [3, 3, 4, 4]				
Generation 730:	Best = 1.525388e+03	FES = 73100	Diversity = 4.9723e+00	k_t
= [1, 1, 1, 1]				
Generation 740:	Best = 1.525388e+03	FES = 74100	Diversity = 4.9230e+00	k_t
= [2, 4, 4, 2]				
Generation 750:	Best = 1.525388e+03	FES = 75100	Diversity = 4.8634e+00	k_t
= [1, 1, 1, 1]				
Generation 760:	Best = 1.525388e+03	FES = 76100	Diversity = 4.8511e+00	k_t
= [2, 4, 3, 4]				

Generation 770:	Best = 1.525388e+03	FES = 77100	Diversity = 4.8561e+00	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 1.525388e+03	FES = 78100	Diversity = 4.8709e+00	k_t
= [4, 4, 2, 2]				
Generation 790:	Best = 1.525388e+03	FES = 79100	Diversity = 4.8170e+00	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 1.524787e+03	FES = 80100	Diversity = 4.7063e+00	k_t
= [3, 3, 2, 2]				
Generation 810:	Best = 1.524787e+03	FES = 81100	Diversity = 4.6609e+00	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 1.524558e+03	FES = 82100	Diversity = 4.6679e+00	k_t
= [3, 2, 3, 4]				
Generation 830:	Best = 1.524558e+03	FES = 83100	Diversity = 4.7272e+00	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 1.523102e+03	FES = 84100	Diversity = 4.7084e+00	k_t
= [3, 2, 3, 2]				
Generation 850:	Best = 1.522874e+03	FES = 85100	Diversity = 4.6913e+00	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 1.522874e+03	FES = 86100	Diversity = 4.6458e+00	k_t
= [4, 4, 3, 3]				
Generation 870:	Best = 1.522718e+03	FES = 87100	Diversity = 4.6364e+00	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 1.522657e+03	FES = 88100	Diversity = 4.5565e+00	k_t
= [2, 2, 2, 2]				
Generation 890:	Best = 1.522657e+03	FES = 89100	Diversity = 4.5240e+00	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 1.521686e+03	FES = 90100	Diversity = 4.4811e+00	k_t
= [2, 3, 2, 4]				
Generation 910:	Best = 1.520192e+03	FES = 91100	Diversity = 4.4489e+00	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 1.520192e+03	FES = 92100	Diversity = 4.4540e+00	k_t
= [4, 3, 3, 3]				
Generation 930:	Best = 1.520192e+03	FES = 93100	Diversity = 4.4164e+00	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 1.520192e+03	FES = 94100	Diversity = 4.4177e+00	k_t
= [4, 2, 3, 2]				
Generation 950:	Best = 1.520192e+03	FES = 95100	Diversity = 4.3992e+00	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 1.519413e+03	FES = 96100	Diversity = 4.3971e+00	k_t
= [2, 2, 3, 3]				
Generation 970:	Best = 1.518748e+03	FES = 97100	Diversity = 4.3460e+00	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 1.517342e+03	FES = 98100	Diversity = 4.2559e+00	k_t
= [3, 2, 4, 4]				
Generation 990:	Best = 1.517342e+03	FES = 99100	Diversity = 4.2325e+00	k_t
= [1, 1, 1, 1]				
Generation 1000:	Best = 1.517342e+03	FES = 100100	Diversity = 4.1951e+00	k_t
= [3, 2, 4, 2]				
Generation 1010:	Best = 1.516880e+03	FES = 101100	Diversity = 4.0413e+00	k_t
= [1, 1, 1, 1]				
Generation 1020:	Best = 1.516880e+03	FES = 102100	Diversity = 3.9386e+00	k_t
= [4, 4, 2, 4]				
Generation 1030:	Best = 1.516706e+03	FES = 103100	Diversity = 3.8204e+00	k_t
= [1, 1, 1, 1]				

Generation 1040: Best = 1.516706e+03 = [4, 2, 2, 4]	FES = 104100	Diversity = 3.6526e+00	k_t
Generation 1050: Best = 1.516695e+03 = [1, 1, 1, 1]	FES = 105100	Diversity = 3.4370e+00	k_t
Generation 1060: Best = 1.514740e+03 = [4, 2, 2, 4]	FES = 106100	Diversity = 3.2960e+00	k_t
Generation 1070: Best = 1.514730e+03 = [1, 1, 1, 1]	FES = 107100	Diversity = 3.0834e+00	k_t
Generation 1080: Best = 1.514660e+03 = [4, 2, 4, 3]	FES = 108100	Diversity = 2.9186e+00	k_t
Generation 1090: Best = 1.514534e+03 = [1, 1, 1, 1]	FES = 109100	Diversity = 2.7086e+00	k_t
Generation 1100: Best = 1.513480e+03 = [4, 4, 2, 3]	FES = 110100	Diversity = 2.5016e+00	k_t
Generation 1110: Best = 1.513042e+03 = [1, 1, 1, 1]	FES = 111100	Diversity = 2.3109e+00	k_t
Generation 1120: Best = 1.513042e+03 = [2, 2, 4, 4]	FES = 112100	Diversity = 2.1258e+00	k_t
Generation 1130: Best = 1.512805e+03 = [1, 1, 1, 1]	FES = 113100	Diversity = 1.6558e+00	k_t
Generation 1140: Best = 1.512475e+03 = [2, 3, 2, 2]	FES = 114100	Diversity = 1.5791e+00	k_t
Generation 1150: Best = 1.512283e+03 = [1, 1, 1, 1]	FES = 115100	Diversity = 1.5269e+00	k_t
Generation 1160: Best = 1.511735e+03 = [4, 3, 4, 2]	FES = 116100	Diversity = 1.3758e+00	k_t
Generation 1170: Best = 1.511328e+03 = [1, 1, 1, 1]	FES = 117100	Diversity = 1.3868e+00	k_t
Generation 1180: Best = 1.511269e+03 = [2, 4, 3, 2]	FES = 118100	Diversity = 1.3753e+00	k_t
Generation 1190: Best = 1.511263e+03 = [1, 1, 1, 1]	FES = 119100	Diversity = 1.3540e+00	k_t
Generation 1200: Best = 1.511208e+03 = [2, 2, 3, 2]	FES = 120100	Diversity = 1.3198e+00	k_t
Generation 1210: Best = 1.511093e+03 = [1, 1, 1, 1]	FES = 121100	Diversity = 1.2483e+00	k_t
Generation 1220: Best = 1.510926e+03 = [3, 2, 3, 4]	FES = 122100	Diversity = 1.1794e+00	k_t
Generation 1230: Best = 1.510920e+03 = [1, 1, 1, 1]	FES = 123100	Diversity = 1.1235e+00	k_t
Generation 1240: Best = 1.510741e+03 = [2, 4, 4, 4]	FES = 124100	Diversity = 1.0539e+00	k_t
Generation 1250: Best = 1.510468e+03 = [1, 1, 1, 1]	FES = 125100	Diversity = 9.9371e-01	k_t
Generation 1260: Best = 1.510391e+03 = [3, 2, 3, 4]	FES = 126100	Diversity = 8.9895e-01	k_t
Generation 1270: Best = 1.510351e+03 = [1, 1, 1, 1]	FES = 127100	Diversity = 8.2683e-01	k_t
Generation 1280: Best = 1.510187e+03 = [4, 2, 3, 2]	FES = 128100	Diversity = 7.7726e-01	k_t
Generation 1290: Best = 1.510045e+03 = [1, 1, 1, 1]	FES = 129100	Diversity = 6.9358e-01	k_t
Generation 1300: Best = 1.510016e+03 = [2, 4, 3, 3]	FES = 130100	Diversity = 6.3521e-01	k_t

Generation 1310: Best = 1.509872e+03 = [1, 1, 1, 1]	FES = 131100	Diversity = 5.8551e-01	k_t
Generation 1320: Best = 1.509856e+03 = [4, 2, 4, 2]	FES = 132100	Diversity = 5.5384e-01	k_t
Generation 1330: Best = 1.509765e+03 = [1, 1, 1, 1]	FES = 133100	Diversity = 5.0061e-01	k_t
Generation 1340: Best = 1.509693e+03 = [2, 2, 3, 3]	FES = 134100	Diversity = 4.4885e-01	k_t
Generation 1350: Best = 1.509664e+03 = [1, 1, 1, 1]	FES = 135100	Diversity = 4.0444e-01	k_t
Generation 1360: Best = 1.509619e+03 = [4, 2, 2, 3]	FES = 136100	Diversity = 4.0195e-01	k_t
Generation 1370: Best = 1.509578e+03 = [1, 1, 1, 1]	FES = 137100	Diversity = 3.9278e-01	k_t
Generation 1380: Best = 1.509547e+03 = [4, 3, 3, 2]	FES = 138100	Diversity = 3.5627e-01	k_t
Generation 1390: Best = 1.509522e+03 = [1, 1, 1, 1]	FES = 139100	Diversity = 3.4536e-01	k_t
Generation 1400: Best = 1.509485e+03 = [2, 4, 4, 4]	FES = 140100	Diversity = 3.2734e-01	k_t
Generation 1410: Best = 1.509425e+03 = [1, 1, 1, 1]	FES = 141100	Diversity = 2.9579e-01	k_t
Generation 1420: Best = 1.509404e+03 = [4, 2, 3, 2]	FES = 142100	Diversity = 2.8178e-01	k_t
Generation 1430: Best = 1.509385e+03 = [1, 1, 1, 1]	FES = 143100	Diversity = 2.7424e-01	k_t
Generation 1440: Best = 1.509363e+03 = [3, 2, 2, 4]	FES = 144100	Diversity = 2.6146e-01	k_t
Generation 1450: Best = 1.509320e+03 = [1, 1, 1, 1]	FES = 145100	Diversity = 2.5272e-01	k_t
Generation 1460: Best = 1.509268e+03 = [3, 3, 2, 4]	FES = 146100	Diversity = 2.1377e-01	k_t
Generation 1470: Best = 1.509235e+03 = [1, 1, 1, 1]	FES = 147100	Diversity = 2.0704e-01	k_t
Generation 1480: Best = 1.509219e+03 = [4, 3, 2, 2]	FES = 148100	Diversity = 1.9470e-01	k_t
Generation 1490: Best = 1.509207e+03 = [1, 1, 1, 1]	FES = 149100	Diversity = 1.8941e-01	k_t
Generation 1500: Best = 1.509163e+03 = [3, 2, 4, 4]	FES = 150100	Diversity = 1.8684e-01	k_t
Generation 1510: Best = 1.509120e+03 = [1, 1, 1, 1]	FES = 151100	Diversity = 1.7958e-01	k_t
Generation 1520: Best = 1.509096e+03 = [4, 3, 4, 2]	FES = 152100	Diversity = 1.6678e-01	k_t
Generation 1530: Best = 1.509074e+03 = [1, 1, 1, 1]	FES = 153100	Diversity = 1.6684e-01	k_t
Generation 1540: Best = 1.509052e+03 = [2, 2, 2, 4]	FES = 154100	Diversity = 1.4644e-01	k_t
Generation 1550: Best = 1.509036e+03 = [1, 1, 1, 1]	FES = 155100	Diversity = 1.4249e-01	k_t
Generation 1560: Best = 1.508997e+03 = [4, 3, 4, 4]	FES = 156100	Diversity = 1.3690e-01	k_t
Generation 1570: Best = 1.508961e+03 = [1, 1, 1, 1]	FES = 157100	Diversity = 1.5979e-01	k_t

Generation 1580: Best = 1.508942e+03 = [3, 2, 3, 4]	FES = 158100	Diversity = 1.5080e-01	k_t
Generation 1590: Best = 1.508931e+03 = [1, 1, 1, 1]	FES = 159100	Diversity = 1.3792e-01	k_t
Generation 1600: Best = 1.508908e+03 = [2, 2, 3, 4]	FES = 160100	Diversity = 1.4259e-01	k_t
Generation 1610: Best = 1.508888e+03 = [1, 1, 1, 1]	FES = 161100	Diversity = 1.4692e-01	k_t
Generation 1620: Best = 1.508872e+03 = [4, 2, 3, 2]	FES = 162100	Diversity = 1.3255e-01	k_t
Generation 1630: Best = 1.508855e+03 = [1, 1, 1, 1]	FES = 163100	Diversity = 1.3497e-01	k_t
Generation 1640: Best = 1.508843e+03 = [3, 2, 4, 2]	FES = 164100	Diversity = 1.4507e-01	k_t
Generation 1650: Best = 1.508827e+03 = [1, 1, 1, 1]	FES = 165100	Diversity = 1.4466e-01	k_t
Generation 1660: Best = 1.508809e+03 = [3, 2, 2, 3]	FES = 166100	Diversity = 1.3839e-01	k_t
Generation 1670: Best = 1.508788e+03 = [1, 1, 1, 1]	FES = 167100	Diversity = 1.3274e-01	k_t
Generation 1680: Best = 1.508779e+03 = [3, 2, 4, 2]	FES = 168100	Diversity = 1.4174e-01	k_t
Generation 1690: Best = 1.508760e+03 = [1, 1, 1, 1]	FES = 169100	Diversity = 1.3572e-01	k_t
Generation 1700: Best = 1.508747e+03 = [2, 4, 4, 3]	FES = 170100	Diversity = 1.3656e-01	k_t
Generation 1710: Best = 1.508727e+03 = [1, 1, 1, 1]	FES = 171100	Diversity = 1.3853e-01	k_t
Generation 1720: Best = 1.508705e+03 = [2, 2, 4, 2]	FES = 172100	Diversity = 1.2823e-01	k_t
Generation 1730: Best = 1.508704e+03 = [1, 1, 1, 1]	FES = 173100	Diversity = 1.3201e-01	k_t
Generation 1740: Best = 1.508696e+03 = [2, 2, 2, 4]	FES = 174100	Diversity = 1.3149e-01	k_t
Generation 1750: Best = 1.508672e+03 = [1, 1, 1, 1]	FES = 175100	Diversity = 1.2768e-01	k_t
Generation 1760: Best = 1.508659e+03 = [4, 4, 3, 4]	FES = 176100	Diversity = 1.2500e-01	k_t
Generation 1770: Best = 1.508639e+03 = [1, 1, 1, 1]	FES = 177100	Diversity = 1.2289e-01	k_t
Generation 1780: Best = 1.508628e+03 = [3, 4, 4, 4]	FES = 178100	Diversity = 1.3911e-01	k_t
Generation 1790: Best = 1.508616e+03 = [1, 1, 1, 1]	FES = 179100	Diversity = 1.4470e-01	k_t
Generation 1800: Best = 1.508604e+03 = [2, 4, 3, 2]	FES = 180100	Diversity = 1.3930e-01	k_t
Generation 1810: Best = 1.508579e+03 = [1, 1, 1, 1]	FES = 181100	Diversity = 1.3669e-01	k_t
Generation 1820: Best = 1.508569e+03 = [4, 4, 2, 3]	FES = 182100	Diversity = 1.3786e-01	k_t
Generation 1830: Best = 1.508565e+03 = [1, 1, 1, 1]	FES = 183100	Diversity = 1.2853e-01	k_t
Generation 1840: Best = 1.508551e+03 = [4, 4, 2, 2]	FES = 184100	Diversity = 1.2378e-01	k_t

Generation 1850: Best = 1.508548e+03 = [1, 1, 1, 1]	FES = 185100	Diversity = 1.1361e-01	k_t
Generation 1860: Best = 1.508536e+03 = [2, 2, 4, 4]	FES = 186100	Diversity = 1.0796e-01	k_t
Generation 1870: Best = 1.508532e+03 = [1, 1, 1, 1]	FES = 187100	Diversity = 1.0023e-01	k_t
Generation 1880: Best = 1.508526e+03 = [4, 4, 4, 4]	FES = 188100	Diversity = 9.0034e-02	k_t
Generation 1890: Best = 1.508517e+03 = [1, 1, 1, 1]	FES = 189100	Diversity = 7.6795e-02	k_t
Generation 1900: Best = 1.508511e+03 = [3, 2, 3, 4]	FES = 190100	Diversity = 7.5220e-02	k_t
Generation 1910: Best = 1.508497e+03 = [1, 1, 1, 1]	FES = 191100	Diversity = 8.6404e-02	k_t
Generation 1920: Best = 1.508485e+03 = [4, 2, 4, 4]	FES = 192100	Diversity = 8.4250e-02	k_t
Generation 1930: Best = 1.508483e+03 = [1, 1, 1, 1]	FES = 193100	Diversity = 8.4387e-02	k_t
Generation 1940: Best = 1.508471e+03 = [3, 2, 3, 4]	FES = 194100	Diversity = 7.9036e-02	k_t
Generation 1950: Best = 1.508462e+03 = [1, 1, 1, 1]	FES = 195100	Diversity = 8.2311e-02	k_t
Generation 1960: Best = 1.508453e+03 = [2, 3, 4, 4]	FES = 196100	Diversity = 8.7258e-02	k_t
Generation 1970: Best = 1.508443e+03 = [1, 1, 1, 1]	FES = 197100	Diversity = 1.0067e-01	k_t
Generation 1980: Best = 1.508431e+03 = [3, 4, 2, 2]	FES = 198100	Diversity = 1.0747e-01	k_t
Generation 1990: Best = 1.508409e+03 = [1, 1, 1, 1]	FES = 199100	Diversity = 1.0716e-01	k_t
Generation 2000: Best = 1.508407e+03 = [2, 2, 3, 3]	FES = 200100	Diversity = 1.0474e-01	k_t
Generation 2010: Best = 1.508395e+03 = [1, 1, 1, 1]	FES = 201100	Diversity = 9.9408e-02	k_t
Generation 2020: Best = 1.508391e+03 = [2, 3, 4, 4]	FES = 202100	Diversity = 9.7591e-02	k_t
Generation 2030: Best = 1.508379e+03 = [1, 1, 1, 1]	FES = 203100	Diversity = 9.9074e-02	k_t
Generation 2040: Best = 1.508362e+03 = [4, 3, 2, 4]	FES = 204100	Diversity = 1.0159e-01	k_t
Generation 2050: Best = 1.508355e+03 = [1, 1, 1, 1]	FES = 205100	Diversity = 1.0548e-01	k_t
Generation 2060: Best = 1.508346e+03 = [4, 3, 3, 3]	FES = 206100	Diversity = 9.0545e-02	k_t
Generation 2070: Best = 1.508337e+03 = [1, 1, 1, 1]	FES = 207100	Diversity = 8.6068e-02	k_t
Generation 2080: Best = 1.508334e+03 = [4, 2, 2, 4]	FES = 208100	Diversity = 8.3090e-02	k_t
Generation 2090: Best = 1.508329e+03 = [1, 1, 1, 1]	FES = 209100	Diversity = 8.6653e-02	k_t
Generation 2100: Best = 1.508319e+03 = [2, 2, 4, 4]	FES = 210100	Diversity = 8.9783e-02	k_t
Generation 2110: Best = 1.508315e+03 = [1, 1, 1, 1]	FES = 211100	Diversity = 9.3553e-02	k_t

Generation 2120: Best = 1.508303e+03 = [3, 2, 2, 2]	FES = 212100	Diversity = 8.8106e-02	k_t
Generation 2130: Best = 1.508296e+03 = [1, 1, 1, 1]	FES = 213100	Diversity = 9.3707e-02	k_t
Generation 2140: Best = 1.508290e+03 = [2, 4, 2, 2]	FES = 214100	Diversity = 8.8336e-02	k_t
Generation 2150: Best = 1.508282e+03 = [1, 1, 1, 1]	FES = 215100	Diversity = 9.7565e-02	k_t
Generation 2160: Best = 1.508268e+03 = [3, 3, 4, 3]	FES = 216100	Diversity = 9.6038e-02	k_t
Generation 2170: Best = 1.508258e+03 = [1, 1, 1, 1]	FES = 217100	Diversity = 9.1927e-02	k_t
Generation 2180: Best = 1.508247e+03 = [2, 3, 4, 2]	FES = 218100	Diversity = 8.9781e-02	k_t
Generation 2190: Best = 1.508238e+03 = [1, 1, 1, 1]	FES = 219100	Diversity = 8.7782e-02	k_t
Generation 2200: Best = 1.508221e+03 = [3, 4, 4, 3]	FES = 220100	Diversity = 1.0022e-01	k_t
Generation 2210: Best = 1.508203e+03 = [1, 1, 1, 1]	FES = 221100	Diversity = 1.0877e-01	k_t
Generation 2220: Best = 1.508200e+03 = [4, 4, 3, 4]	FES = 222100	Diversity = 9.7794e-02	k_t
Generation 2230: Best = 1.508187e+03 = [1, 1, 1, 1]	FES = 223100	Diversity = 9.3584e-02	k_t
Generation 2240: Best = 1.508162e+03 = [2, 2, 2, 2]	FES = 224100	Diversity = 1.0267e-01	k_t
Generation 2250: Best = 1.508152e+03 = [1, 1, 1, 1]	FES = 225100	Diversity = 1.0256e-01	k_t
Generation 2260: Best = 1.508140e+03 = [3, 4, 3, 4]	FES = 226100	Diversity = 1.1016e-01	k_t
Generation 2270: Best = 1.508124e+03 = [1, 1, 1, 1]	FES = 227100	Diversity = 1.0949e-01	k_t
Generation 2280: Best = 1.508106e+03 = [2, 4, 4, 2]	FES = 228100	Diversity = 1.1308e-01	k_t
Generation 2290: Best = 1.508078e+03 = [1, 1, 1, 1]	FES = 229100	Diversity = 1.2469e-01	k_t
Generation 2300: Best = 1.508072e+03 = [2, 2, 4, 4]	FES = 230100	Diversity = 1.3424e-01	k_t
Generation 2310: Best = 1.508039e+03 = [1, 1, 1, 1]	FES = 231100	Diversity = 1.3667e-01	k_t
Generation 2320: Best = 1.508037e+03 = [2, 3, 4, 2]	FES = 232100	Diversity = 1.1491e-01	k_t
Generation 2330: Best = 1.507995e+03 = [1, 1, 1, 1]	FES = 233100	Diversity = 1.1565e-01	k_t
Generation 2340: Best = 1.507977e+03 = [2, 4, 3, 4]	FES = 234100	Diversity = 1.2961e-01	k_t
Generation 2350: Best = 1.507955e+03 = [1, 1, 1, 1]	FES = 235100	Diversity = 1.3885e-01	k_t
Generation 2360: Best = 1.507913e+03 = [3, 3, 3, 4]	FES = 236100	Diversity = 1.3959e-01	k_t
Generation 2370: Best = 1.507885e+03 = [1, 1, 1, 1]	FES = 237100	Diversity = 1.3899e-01	k_t
Generation 2380: Best = 1.507885e+03 = [4, 4, 3, 4]	FES = 238100	Diversity = 1.3160e-01	k_t

Generation 2390: Best = 1.507863e+03 = [1, 1, 1, 1]	FES = 239100	Diversity = 1.2612e-01	k_t
Generation 2400: Best = 1.507837e+03 = [3, 4, 2, 2]	FES = 240100	Diversity = 1.2458e-01	k_t
Generation 2410: Best = 1.507826e+03 = [1, 1, 1, 1]	FES = 241100	Diversity = 1.2269e-01	k_t
Generation 2420: Best = 1.507811e+03 = [2, 2, 2, 4]	FES = 242100	Diversity = 1.2375e-01	k_t
Generation 2430: Best = 1.507788e+03 = [1, 1, 1, 1]	FES = 243100	Diversity = 1.1610e-01	k_t
Generation 2440: Best = 1.507776e+03 = [3, 3, 2, 4]	FES = 244100	Diversity = 1.1294e-01	k_t
Generation 2450: Best = 1.507769e+03 = [1, 1, 1, 1]	FES = 245100	Diversity = 1.1548e-01	k_t
Generation 2460: Best = 1.507749e+03 = [2, 2, 4, 3]	FES = 246100	Diversity = 1.1830e-01	k_t
Generation 2470: Best = 1.507736e+03 = [1, 1, 1, 1]	FES = 247100	Diversity = 1.2222e-01	k_t
Generation 2480: Best = 1.507705e+03 = [3, 4, 2, 2]	FES = 248100	Diversity = 1.2754e-01	k_t
Generation 2490: Best = 1.507691e+03 = [1, 1, 1, 1]	FES = 249100	Diversity = 1.2270e-01	k_t
Generation 2500: Best = 1.507663e+03 = [4, 2, 3, 3]	FES = 250100	Diversity = 1.1332e-01	k_t
Generation 2510: Best = 1.507659e+03 = [1, 1, 1, 1]	FES = 251100	Diversity = 1.1606e-01	k_t
Generation 2520: Best = 1.507630e+03 = [2, 2, 4, 2]	FES = 252100	Diversity = 1.1553e-01	k_t
Generation 2530: Best = 1.507591e+03 = [1, 1, 1, 1]	FES = 253100	Diversity = 1.3606e-01	k_t
Generation 2540: Best = 1.507570e+03 = [4, 3, 2, 3]	FES = 254100	Diversity = 1.5331e-01	k_t
Generation 2550: Best = 1.507558e+03 = [1, 1, 1, 1]	FES = 255100	Diversity = 1.5583e-01	k_t
Generation 2560: Best = 1.507524e+03 = [2, 4, 2, 2]	FES = 256100	Diversity = 1.4343e-01	k_t
Generation 2570: Best = 1.507509e+03 = [1, 1, 1, 1]	FES = 257100	Diversity = 1.3315e-01	k_t
Generation 2580: Best = 1.507487e+03 = [4, 2, 2, 3]	FES = 258100	Diversity = 1.2969e-01	k_t
Generation 2590: Best = 1.507460e+03 = [1, 1, 1, 1]	FES = 259100	Diversity = 1.2877e-01	k_t
Generation 2600: Best = 1.507426e+03 = [4, 3, 2, 2]	FES = 260100	Diversity = 1.2324e-01	k_t
Generation 2610: Best = 1.507426e+03 = [1, 1, 1, 1]	FES = 261100	Diversity = 1.2216e-01	k_t
Generation 2620: Best = 1.507414e+03 = [4, 2, 4, 4]	FES = 262100	Diversity = 1.1842e-01	k_t
Generation 2630: Best = 1.507395e+03 = [1, 1, 1, 1]	FES = 263100	Diversity = 1.1799e-01	k_t
Generation 2640: Best = 1.507383e+03 = [3, 2, 2, 4]	FES = 264100	Diversity = 1.0540e-01	k_t
Generation 2650: Best = 1.507378e+03 = [1, 1, 1, 1]	FES = 265100	Diversity = 1.0185e-01	k_t

Generation 2660: Best = 1.507363e+03 = [2, 3, 4, 3]	FES = 266100	Diversity = 1.0311e-01	k_t
Generation 2670: Best = 1.507353e+03 = [1, 1, 1, 1]	FES = 267100	Diversity = 1.0408e-01	k_t
Generation 2680: Best = 1.507328e+03 = [3, 2, 4, 4]	FES = 268100	Diversity = 1.0807e-01	k_t
Generation 2690: Best = 1.507317e+03 = [1, 1, 1, 1]	FES = 269100	Diversity = 1.1664e-01	k_t
Generation 2700: Best = 1.507301e+03 = [3, 2, 4, 2]	FES = 270100	Diversity = 1.3029e-01	k_t
Generation 2710: Best = 1.507287e+03 = [1, 1, 1, 1]	FES = 271100	Diversity = 1.3433e-01	k_t
Generation 2720: Best = 1.507277e+03 = [2, 4, 3, 4]	FES = 272100	Diversity = 1.1925e-01	k_t
Generation 2730: Best = 1.507254e+03 = [1, 1, 1, 1]	FES = 273100	Diversity = 1.2715e-01	k_t
Generation 2740: Best = 1.507250e+03 = [4, 2, 4, 2]	FES = 274100	Diversity = 1.3412e-01	k_t
Generation 2750: Best = 1.507237e+03 = [1, 1, 1, 1]	FES = 275100	Diversity = 1.4445e-01	k_t
Generation 2760: Best = 1.507221e+03 = [3, 4, 2, 3]	FES = 276100	Diversity = 1.4697e-01	k_t
Generation 2770: Best = 1.507209e+03 = [1, 1, 1, 1]	FES = 277100	Diversity = 1.4850e-01	k_t
Generation 2780: Best = 1.507186e+03 = [3, 2, 4, 3]	FES = 278100	Diversity = 1.3499e-01	k_t
Generation 2790: Best = 1.507180e+03 = [1, 1, 1, 1]	FES = 279100	Diversity = 1.2638e-01	k_t
Generation 2800: Best = 1.507163e+03 = [2, 3, 3, 3]	FES = 280100	Diversity = 1.1459e-01	k_t
Generation 2810: Best = 1.507162e+03 = [1, 1, 1, 1]	FES = 281100	Diversity = 1.0640e-01	k_t
Generation 2820: Best = 1.507155e+03 = [4, 3, 2, 2]	FES = 282100	Diversity = 9.7648e-02	k_t
Generation 2830: Best = 1.507147e+03 = [1, 1, 1, 1]	FES = 283100	Diversity = 8.9154e-02	k_t
Generation 2840: Best = 1.507135e+03 = [2, 3, 2, 3]	FES = 284100	Diversity = 8.9698e-02	k_t
Generation 2850: Best = 1.507134e+03 = [1, 1, 1, 1]	FES = 285100	Diversity = 8.3738e-02	k_t
Generation 2860: Best = 1.507125e+03 = [2, 2, 4, 4]	FES = 286100	Diversity = 8.0210e-02	k_t
Generation 2870: Best = 1.507118e+03 = [1, 1, 1, 1]	FES = 287100	Diversity = 8.0058e-02	k_t
Generation 2880: Best = 1.507114e+03 = [2, 3, 4, 2]	FES = 288100	Diversity = 8.1366e-02	k_t
Generation 2890: Best = 1.507105e+03 = [1, 1, 1, 1]	FES = 289100	Diversity = 7.9179e-02	k_t
Generation 2900: Best = 1.507094e+03 = [3, 2, 2, 3]	FES = 290100	Diversity = 7.9799e-02	k_t
Generation 2910: Best = 1.507087e+03 = [1, 1, 1, 1]	FES = 291100	Diversity = 7.3199e-02	k_t
Generation 2920: Best = 1.507078e+03 = [3, 4, 4, 3]	FES = 292100	Diversity = 7.8335e-02	k_t

Generation 2930: Best = 1.507070e+03 FES = 293100 Diversity = 7.2419e-02 k_t
 = [1, 1, 1, 1]
 Generation 2940: Best = 1.507067e+03 FES = 294100 Diversity = 7.3180e-02 k_t
 = [2, 3, 2, 3]
 Generation 2950: Best = 1.507057e+03 FES = 295100 Diversity = 7.6464e-02 k_t
 = [1, 1, 1, 1]
 Generation 2960: Best = 1.507046e+03 FES = 296100 Diversity = 7.0056e-02 k_t
 = [3, 4, 2, 3]
 Generation 2970: Best = 1.507043e+03 FES = 297100 Diversity = 7.2574e-02 k_t
 = [1, 1, 1, 1]
 Generation 2980: Best = 1.507034e+03 FES = 298100 Diversity = 8.7535e-02 k_t
 = [2, 2, 3, 4]
 Generation 2990: Best = 1.507018e+03 FES = 299100 Diversity = 9.1075e-02 k_t
 = [1, 1, 1, 1]

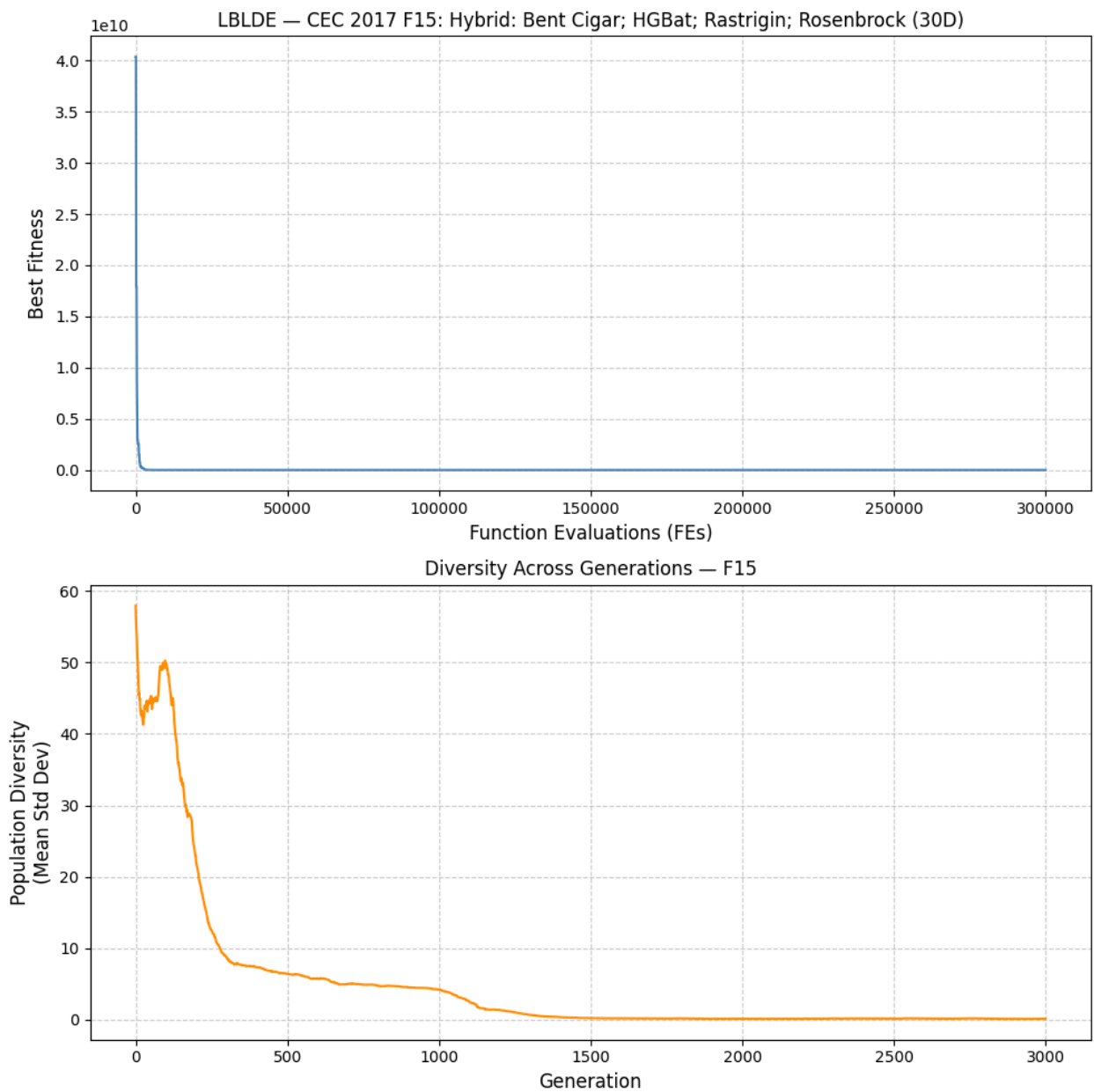
Optimization completed!

Final Best Fitness = 1.507010e+03

Total Function Evaluations = 300000

```

=====
=
Results:
  Best Fitness           : 1.507010e+03
  Global Optimum         : 1.500000e+03
  Error (|best-optimum|) : 7.009543e+00
  Best Solution (5 dim)  : [ 4.98936447   8.36157497  45.72626118 -73.00869858
66.38313155]
  D^(0) (init. diversity): 5.7976e+01
=====
=
  
```



Graph saved as convergence_F15.png

```
In [26]: # =====
# F16 – Hybrid Function 6 (Schaffer; HGBat; Rosenbrock; Schwefel)
# =====
run_and_plot('F16', make_wrapper(hybrid.f16), 1600.0, 'Hybrid: Schaffer; HGBat
```

```

=====
=
LBLDE_s on CEC 2017 F16 – Single Run
=====
=
Function      : Hybrid: Schaffer; HGBat; Rosenbrock; Schwefel (30D)
Bounds       : [-100, 100]
Global optimum : 1600.0
Initial D^(0) = 5.7976e+01
Generation 0: Best Fitness = 8.664946e+03 | Diversity = 5.7976e+01
Generation 10: Best = 4.301402e+03 FES = 1100 Diversity = 4.3390e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 4.083827e+03 FES = 2100 Diversity = 4.0664e+01 k_t
= [3, 3, 2, 2]
Generation 30: Best = 3.979614e+03 FES = 3100 Diversity = 4.0101e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 3.975398e+03 FES = 4100 Diversity = 3.9632e+01 k_t
= [4, 2, 4, 3]
Generation 50: Best = 3.900769e+03 FES = 5100 Diversity = 3.8671e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 3.249189e+03 FES = 6100 Diversity = 3.8520e+01 k_t
= [3, 2, 3, 3]
Generation 70: Best = 3.249189e+03 FES = 7100 Diversity = 3.8035e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 3.232355e+03 FES = 8100 Diversity = 3.7880e+01 k_t
= [2, 3, 2, 4]
Generation 90: Best = 3.232309e+03 FES = 9100 Diversity = 3.7586e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 3.232309e+03 FES = 10100 Diversity = 3.7370e+01 k_t
= [2, 2, 3, 4]
Generation 110: Best = 3.226802e+03 FES = 11100 Diversity = 3.6988e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 3.226802e+03 FES = 12100 Diversity = 3.6765e+01 k_t
= [2, 4, 2, 2]
Generation 130: Best = 3.189862e+03 FES = 13100 Diversity = 3.6641e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 3.122666e+03 FES = 14100 Diversity = 3.6168e+01 k_t
= [4, 3, 2, 4]
Generation 150: Best = 3.122666e+03 FES = 15100 Diversity = 3.6077e+01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 2.991014e+03 FES = 16100 Diversity = 3.5577e+01 k_t
= [3, 3, 2, 4]
Generation 170: Best = 2.977293e+03 FES = 17100 Diversity = 3.5849e+01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 2.970294e+03 FES = 18100 Diversity = 3.5713e+01 k_t
= [4, 4, 2, 4]
Generation 190: Best = 2.970294e+03 FES = 19100 Diversity = 3.5360e+01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 2.886587e+03 FES = 20100 Diversity = 3.5023e+01 k_t
= [2, 4, 4, 3]
Generation 210: Best = 2.886424e+03 FES = 21100 Diversity = 3.4952e+01 k_t
= [1, 1, 1, 1]
Generation 220: Best = 2.883601e+03 FES = 22100 Diversity = 3.4879e+01 k_t
= [3, 4, 2, 2]

```

Generation 230:	Best = 2.883601e+03	FES = 23100	Diversity = 3.4372e+01	k_t
= [1, 1, 1, 1]				
Generation 240:	Best = 2.883601e+03	FES = 24100	Diversity = 3.4265e+01	k_t
= [2, 3, 2, 2]				
Generation 250:	Best = 2.864532e+03	FES = 25100	Diversity = 3.4424e+01	k_t
= [1, 1, 1, 1]				
Generation 260:	Best = 2.853255e+03	FES = 26100	Diversity = 3.4446e+01	k_t
= [2, 2, 3, 2]				
Generation 270:	Best = 2.853177e+03	FES = 27100	Diversity = 3.4584e+01	k_t
= [1, 1, 1, 1]				
Generation 280:	Best = 2.851733e+03	FES = 28100	Diversity = 3.4354e+01	k_t
= [3, 3, 4, 4]				
Generation 290:	Best = 2.840179e+03	FES = 29100	Diversity = 3.4201e+01	k_t
= [1, 1, 1, 1]				
Generation 300:	Best = 2.840179e+03	FES = 30100	Diversity = 3.3841e+01	k_t
= [4, 4, 4, 2]				
Generation 310:	Best = 2.840179e+03	FES = 31100	Diversity = 3.3495e+01	k_t
= [1, 1, 1, 1]				
Generation 320:	Best = 2.835841e+03	FES = 32100	Diversity = 3.3419e+01	k_t
= [3, 2, 4, 4]				
Generation 330:	Best = 2.835453e+03	FES = 33100	Diversity = 3.3404e+01	k_t
= [1, 1, 1, 1]				
Generation 340:	Best = 2.761109e+03	FES = 34100	Diversity = 3.3096e+01	k_t
= [2, 2, 4, 2]				
Generation 350:	Best = 2.761109e+03	FES = 35100	Diversity = 3.3056e+01	k_t
= [1, 1, 1, 1]				
Generation 360:	Best = 2.699688e+03	FES = 36100	Diversity = 3.2887e+01	k_t
= [4, 3, 3, 3]				
Generation 370:	Best = 2.699688e+03	FES = 37100	Diversity = 3.2624e+01	k_t
= [1, 1, 1, 1]				
Generation 380:	Best = 2.696412e+03	FES = 38100	Diversity = 3.2567e+01	k_t
= [4, 4, 2, 2]				
Generation 390:	Best = 2.693834e+03	FES = 39100	Diversity = 3.2548e+01	k_t
= [1, 1, 1, 1]				
Generation 400:	Best = 2.693834e+03	FES = 40100	Diversity = 3.2675e+01	k_t
= [4, 2, 2, 2]				
Generation 410:	Best = 2.693834e+03	FES = 41100	Diversity = 3.2521e+01	k_t
= [1, 1, 1, 1]				
Generation 420:	Best = 2.645319e+03	FES = 42100	Diversity = 3.2484e+01	k_t
= [3, 4, 3, 4]				
Generation 430:	Best = 2.640626e+03	FES = 43100	Diversity = 3.2222e+01	k_t
= [1, 1, 1, 1]				
Generation 440:	Best = 2.640626e+03	FES = 44100	Diversity = 3.2334e+01	k_t
= [3, 4, 2, 2]				
Generation 450:	Best = 2.640513e+03	FES = 45100	Diversity = 3.2135e+01	k_t
= [1, 1, 1, 1]				
Generation 460:	Best = 2.630140e+03	FES = 46100	Diversity = 3.1734e+01	k_t
= [2, 4, 3, 2]				
Generation 470:	Best = 2.630140e+03	FES = 47100	Diversity = 3.2183e+01	k_t
= [1, 1, 1, 1]				
Generation 480:	Best = 2.629893e+03	FES = 48100	Diversity = 3.1989e+01	k_t
= [4, 4, 2, 4]				
Generation 490:	Best = 2.626134e+03	FES = 49100	Diversity = 3.2233e+01	k_t
= [1, 1, 1, 1]				

Generation 500:	Best = 2.626102e+03	FES = 50100	Diversity = 3.2109e+01	k_t
= [3, 2, 2, 2]				
Generation 510:	Best = 2.552021e+03	FES = 51100	Diversity = 3.1776e+01	k_t
= [1, 1, 1, 1]				
Generation 520:	Best = 2.552021e+03	FES = 52100	Diversity = 3.1940e+01	k_t
= [4, 2, 3, 2]				
Generation 530:	Best = 2.550609e+03	FES = 53100	Diversity = 3.1610e+01	k_t
= [1, 1, 1, 1]				
Generation 540:	Best = 2.550609e+03	FES = 54100	Diversity = 3.1452e+01	k_t
= [4, 4, 3, 4]				
Generation 550:	Best = 2.509087e+03	FES = 55100	Diversity = 3.0892e+01	k_t
= [1, 1, 1, 1]				
Generation 560:	Best = 2.509087e+03	FES = 56100	Diversity = 3.0803e+01	k_t
= [2, 4, 3, 2]				
Generation 570:	Best = 2.509087e+03	FES = 57100	Diversity = 3.0873e+01	k_t
= [1, 1, 1, 1]				
Generation 580:	Best = 2.507288e+03	FES = 58100	Diversity = 3.0792e+01	k_t
= [3, 4, 3, 3]				
Generation 590:	Best = 2.505033e+03	FES = 59100	Diversity = 3.0459e+01	k_t
= [1, 1, 1, 1]				
Generation 600:	Best = 2.409934e+03	FES = 60100	Diversity = 3.0130e+01	k_t
= [4, 4, 2, 2]				
Generation 610:	Best = 2.409934e+03	FES = 61100	Diversity = 2.9926e+01	k_t
= [1, 1, 1, 1]				
Generation 620:	Best = 2.409934e+03	FES = 62100	Diversity = 3.0120e+01	k_t
= [4, 2, 3, 4]				
Generation 630:	Best = 2.409920e+03	FES = 63100	Diversity = 3.0447e+01	k_t
= [1, 1, 1, 1]				
Generation 640:	Best = 2.383742e+03	FES = 64100	Diversity = 3.0411e+01	k_t
= [2, 4, 2, 3]				
Generation 650:	Best = 2.331487e+03	FES = 65100	Diversity = 3.0020e+01	k_t
= [1, 1, 1, 1]				
Generation 660:	Best = 2.331487e+03	FES = 66100	Diversity = 3.0420e+01	k_t
= [4, 3, 4, 4]				
Generation 670:	Best = 2.331109e+03	FES = 67100	Diversity = 3.0552e+01	k_t
= [1, 1, 1, 1]				
Generation 680:	Best = 2.331109e+03	FES = 68100	Diversity = 3.0736e+01	k_t
= [2, 4, 4, 2]				
Generation 690:	Best = 2.184341e+03	FES = 69100	Diversity = 3.0264e+01	k_t
= [1, 1, 1, 1]				
Generation 700:	Best = 2.172336e+03	FES = 70100	Diversity = 2.9476e+01	k_t
= [4, 4, 3, 4]				
Generation 710:	Best = 2.136825e+03	FES = 71100	Diversity = 2.7664e+01	k_t
= [1, 1, 1, 1]				
Generation 720:	Best = 2.136825e+03	FES = 72100	Diversity = 2.5559e+01	k_t
= [2, 4, 3, 3]				
Generation 730:	Best = 2.135852e+03	FES = 73100	Diversity = 2.4127e+01	k_t
= [1, 1, 1, 1]				
Generation 740:	Best = 2.125989e+03	FES = 74100	Diversity = 2.2272e+01	k_t
= [2, 3, 4, 4]				
Generation 750:	Best = 2.050429e+03	FES = 75100	Diversity = 2.1022e+01	k_t
= [1, 1, 1, 1]				
Generation 760:	Best = 2.045119e+03	FES = 76100	Diversity = 2.0728e+01	k_t
= [2, 4, 2, 3]				

Generation 770:	Best = 1.980544e+03	FES = 77100	Diversity = 2.0531e+01	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 1.957666e+03	FES = 78100	Diversity = 2.0282e+01	k_t
= [4, 4, 4, 3]				
Generation 790:	Best = 1.946109e+03	FES = 79100	Diversity = 2.0003e+01	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 1.932857e+03	FES = 80100	Diversity = 1.7460e+01	k_t
= [4, 3, 4, 4]				
Generation 810:	Best = 1.923261e+03	FES = 81100	Diversity = 1.5101e+01	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 1.917750e+03	FES = 82100	Diversity = 1.2652e+01	k_t
= [2, 4, 2, 2]				
Generation 830:	Best = 1.915067e+03	FES = 83100	Diversity = 1.2135e+01	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 1.913533e+03	FES = 84100	Diversity = 1.1420e+01	k_t
= [3, 2, 2, 2]				
Generation 850:	Best = 1.913089e+03	FES = 85100	Diversity = 1.1316e+01	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 1.912444e+03	FES = 86100	Diversity = 1.0393e+01	k_t
= [3, 2, 4, 3]				
Generation 870:	Best = 1.911915e+03	FES = 87100	Diversity = 1.0017e+01	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 1.911780e+03	FES = 88100	Diversity = 9.4723e+00	k_t
= [2, 2, 4, 2]				
Generation 890:	Best = 1.911456e+03	FES = 89100	Diversity = 8.6294e+00	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 1.911452e+03	FES = 90100	Diversity = 8.2794e+00	k_t
= [4, 2, 3, 2]				
Generation 910:	Best = 1.911330e+03	FES = 91100	Diversity = 7.8345e+00	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 1.911330e+03	FES = 92100	Diversity = 7.7122e+00	k_t
= [3, 4, 4, 3]				
Generation 930:	Best = 1.911310e+03	FES = 93100	Diversity = 7.6452e+00	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 1.911245e+03	FES = 94100	Diversity = 7.5670e+00	k_t
= [4, 2, 4, 4]				
Generation 950:	Best = 1.911227e+03	FES = 95100	Diversity = 7.4681e+00	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 1.911140e+03	FES = 96100	Diversity = 7.5733e+00	k_t
= [2, 2, 2, 4]				
Generation 970:	Best = 1.911140e+03	FES = 97100	Diversity = 7.5095e+00	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 1.911047e+03	FES = 98100	Diversity = 7.3645e+00	k_t
= [2, 2, 3, 2]				
Generation 990:	Best = 1.911047e+03	FES = 99100	Diversity = 7.2846e+00	k_t
= [1, 1, 1, 1]				
Generation 1000:	Best = 1.911047e+03	FES = 100100	Diversity = 7.2454e+00	k_t
= [4, 2, 4, 2]				
Generation 1010:	Best = 1.911047e+03	FES = 101100	Diversity = 7.3565e+00	k_t
= [1, 1, 1, 1]				
Generation 1020:	Best = 1.911047e+03	FES = 102100	Diversity = 7.2014e+00	k_t
= [2, 2, 4, 3]				
Generation 1030:	Best = 1.911019e+03	FES = 103100	Diversity = 7.1603e+00	k_t
= [1, 1, 1, 1]				

Generation 1040: Best = 1.911018e+03 = [3, 4, 4, 4]	FES = 104100	Diversity = 7.1285e+00	k_t
Generation 1050: Best = 1.911008e+03 = [1, 1, 1, 1]	FES = 105100	Diversity = 7.0898e+00	k_t
Generation 1060: Best = 1.910969e+03 = [2, 4, 3, 4]	FES = 106100	Diversity = 7.1927e+00	k_t
Generation 1070: Best = 1.910943e+03 = [1, 1, 1, 1]	FES = 107100	Diversity = 7.3231e+00	k_t
Generation 1080: Best = 1.910939e+03 = [3, 3, 3, 3]	FES = 108100	Diversity = 7.1480e+00	k_t
Generation 1090: Best = 1.910917e+03 = [1, 1, 1, 1]	FES = 109100	Diversity = 7.0178e+00	k_t
Generation 1100: Best = 1.910738e+03 = [4, 3, 3, 2]	FES = 110100	Diversity = 6.8784e+00	k_t
Generation 1110: Best = 1.910738e+03 = [1, 1, 1, 1]	FES = 111100	Diversity = 6.9333e+00	k_t
Generation 1120: Best = 1.910723e+03 = [2, 4, 3, 4]	FES = 112100	Diversity = 6.7508e+00	k_t
Generation 1130: Best = 1.910713e+03 = [1, 1, 1, 1]	FES = 113100	Diversity = 6.7449e+00	k_t
Generation 1140: Best = 1.910708e+03 = [4, 3, 2, 3]	FES = 114100	Diversity = 6.8023e+00	k_t
Generation 1150: Best = 1.910706e+03 = [1, 1, 1, 1]	FES = 115100	Diversity = 6.8255e+00	k_t
Generation 1160: Best = 1.910706e+03 = [2, 2, 4, 3]	FES = 116100	Diversity = 6.8483e+00	k_t
Generation 1170: Best = 1.910706e+03 = [1, 1, 1, 1]	FES = 117100	Diversity = 6.8682e+00	k_t
Generation 1180: Best = 1.910703e+03 = [4, 2, 4, 3]	FES = 118100	Diversity = 6.6661e+00	k_t
Generation 1190: Best = 1.910702e+03 = [1, 1, 1, 1]	FES = 119100	Diversity = 6.6614e+00	k_t
Generation 1200: Best = 1.910687e+03 = [3, 2, 3, 2]	FES = 120100	Diversity = 6.7457e+00	k_t
Generation 1210: Best = 1.910687e+03 = [1, 1, 1, 1]	FES = 121100	Diversity = 6.6147e+00	k_t
Generation 1220: Best = 1.910687e+03 = [2, 4, 4, 3]	FES = 122100	Diversity = 6.6895e+00	k_t
Generation 1230: Best = 1.910687e+03 = [1, 1, 1, 1]	FES = 123100	Diversity = 6.6244e+00	k_t
Generation 1240: Best = 1.910682e+03 = [3, 3, 4, 2]	FES = 124100	Diversity = 6.6678e+00	k_t
Generation 1250: Best = 1.910678e+03 = [1, 1, 1, 1]	FES = 125100	Diversity = 6.6917e+00	k_t
Generation 1260: Best = 1.910678e+03 = [2, 4, 3, 3]	FES = 126100	Diversity = 6.5543e+00	k_t
Generation 1270: Best = 1.910675e+03 = [1, 1, 1, 1]	FES = 127100	Diversity = 6.4696e+00	k_t
Generation 1280: Best = 1.910675e+03 = [3, 2, 3, 4]	FES = 128100	Diversity = 6.4439e+00	k_t
Generation 1290: Best = 1.910675e+03 = [1, 1, 1, 1]	FES = 129100	Diversity = 6.4951e+00	k_t
Generation 1300: Best = 1.910675e+03 = [3, 4, 3, 2]	FES = 130100	Diversity = 6.6282e+00	k_t

Generation 1310: Best = 1.910675e+03 = [1, 1, 1, 1]	FES = 131100	Diversity = 6.6217e+00	k_t
Generation 1320: Best = 1.910670e+03 = [2, 2, 4, 4]	FES = 132100	Diversity = 6.6647e+00	k_t
Generation 1330: Best = 1.910670e+03 = [1, 1, 1, 1]	FES = 133100	Diversity = 6.6821e+00	k_t
Generation 1340: Best = 1.910670e+03 = [4, 2, 2, 4]	FES = 134100	Diversity = 6.7358e+00	k_t
Generation 1350: Best = 1.910670e+03 = [1, 1, 1, 1]	FES = 135100	Diversity = 6.6038e+00	k_t
Generation 1360: Best = 1.910668e+03 = [4, 2, 4, 2]	FES = 136100	Diversity = 6.6193e+00	k_t
Generation 1370: Best = 1.910640e+03 = [1, 1, 1, 1]	FES = 137100	Diversity = 6.5628e+00	k_t
Generation 1380: Best = 1.910638e+03 = [3, 4, 2, 2]	FES = 138100	Diversity = 6.5770e+00	k_t
Generation 1390: Best = 1.910602e+03 = [1, 1, 1, 1]	FES = 139100	Diversity = 6.3646e+00	k_t
Generation 1400: Best = 1.910602e+03 = [4, 4, 2, 2]	FES = 140100	Diversity = 6.3502e+00	k_t
Generation 1410: Best = 1.910602e+03 = [1, 1, 1, 1]	FES = 141100	Diversity = 6.3436e+00	k_t
Generation 1420: Best = 1.910597e+03 = [4, 3, 3, 3]	FES = 142100	Diversity = 6.3378e+00	k_t
Generation 1430: Best = 1.910595e+03 = [1, 1, 1, 1]	FES = 143100	Diversity = 6.1960e+00	k_t
Generation 1440: Best = 1.910595e+03 = [4, 3, 2, 2]	FES = 144100	Diversity = 6.0961e+00	k_t
Generation 1450: Best = 1.910595e+03 = [1, 1, 1, 1]	FES = 145100	Diversity = 6.0596e+00	k_t
Generation 1460: Best = 1.910593e+03 = [4, 2, 4, 2]	FES = 146100	Diversity = 6.0257e+00	k_t
Generation 1470: Best = 1.910561e+03 = [1, 1, 1, 1]	FES = 147100	Diversity = 5.8730e+00	k_t
Generation 1480: Best = 1.910560e+03 = [2, 2, 2, 4]	FES = 148100	Diversity = 5.7099e+00	k_t
Generation 1490: Best = 1.910560e+03 = [1, 1, 1, 1]	FES = 149100	Diversity = 5.6727e+00	k_t
Generation 1500: Best = 1.910560e+03 = [4, 2, 3, 3]	FES = 150100	Diversity = 5.6120e+00	k_t
Generation 1510: Best = 1.910560e+03 = [1, 1, 1, 1]	FES = 151100	Diversity = 5.5298e+00	k_t
Generation 1520: Best = 1.910558e+03 = [3, 4, 2, 2]	FES = 152100	Diversity = 5.4620e+00	k_t
Generation 1530: Best = 1.910558e+03 = [1, 1, 1, 1]	FES = 153100	Diversity = 5.2898e+00	k_t
Generation 1540: Best = 1.910501e+03 = [4, 4, 2, 2]	FES = 154100	Diversity = 5.2933e+00	k_t
Generation 1550: Best = 1.910501e+03 = [1, 1, 1, 1]	FES = 155100	Diversity = 5.1792e+00	k_t
Generation 1560: Best = 1.910501e+03 = [2, 3, 3, 2]	FES = 156100	Diversity = 5.1623e+00	k_t
Generation 1570: Best = 1.910499e+03 = [1, 1, 1, 1]	FES = 157100	Diversity = 5.1616e+00	k_t

Generation 1580: Best = 1.910443e+03 = [3, 3, 4, 4]	FES = 158100	Diversity = 4.9529e+00	k_t
Generation 1590: Best = 1.910383e+03 = [1, 1, 1, 1]	FES = 159100	Diversity = 4.8173e+00	k_t
Generation 1600: Best = 1.910383e+03 = [3, 2, 3, 3]	FES = 160100	Diversity = 4.7061e+00	k_t
Generation 1610: Best = 1.910383e+03 = [1, 1, 1, 1]	FES = 161100	Diversity = 4.4224e+00	k_t
Generation 1620: Best = 1.910351e+03 = [2, 2, 3, 4]	FES = 162100	Diversity = 4.3281e+00	k_t
Generation 1630: Best = 1.910351e+03 = [1, 1, 1, 1]	FES = 163100	Diversity = 4.2980e+00	k_t
Generation 1640: Best = 1.910338e+03 = [3, 4, 3, 4]	FES = 164100	Diversity = 4.0222e+00	k_t
Generation 1650: Best = 1.910338e+03 = [1, 1, 1, 1]	FES = 165100	Diversity = 3.7722e+00	k_t
Generation 1660: Best = 1.910338e+03 = [4, 2, 2, 3]	FES = 166100	Diversity = 3.5380e+00	k_t
Generation 1670: Best = 1.910333e+03 = [1, 1, 1, 1]	FES = 167100	Diversity = 3.3342e+00	k_t
Generation 1680: Best = 1.910333e+03 = [3, 4, 3, 2]	FES = 168100	Diversity = 3.2209e+00	k_t
Generation 1690: Best = 1.910333e+03 = [1, 1, 1, 1]	FES = 169100	Diversity = 3.1901e+00	k_t
Generation 1700: Best = 1.910330e+03 = [3, 2, 3, 2]	FES = 170100	Diversity = 2.9619e+00	k_t
Generation 1710: Best = 1.910330e+03 = [1, 1, 1, 1]	FES = 171100	Diversity = 2.9234e+00	k_t
Generation 1720: Best = 1.910327e+03 = [4, 3, 2, 4]	FES = 172100	Diversity = 2.5807e+00	k_t
Generation 1730: Best = 1.910327e+03 = [1, 1, 1, 1]	FES = 173100	Diversity = 2.5320e+00	k_t
Generation 1740: Best = 1.910327e+03 = [3, 3, 4, 3]	FES = 174100	Diversity = 2.4629e+00	k_t
Generation 1750: Best = 1.910324e+03 = [1, 1, 1, 1]	FES = 175100	Diversity = 2.4296e+00	k_t
Generation 1760: Best = 1.910316e+03 = [2, 4, 3, 3]	FES = 176100	Diversity = 2.3857e+00	k_t
Generation 1770: Best = 1.910316e+03 = [1, 1, 1, 1]	FES = 177100	Diversity = 2.0315e+00	k_t
Generation 1780: Best = 1.910315e+03 = [4, 2, 2, 3]	FES = 178100	Diversity = 2.0130e+00	k_t
Generation 1790: Best = 1.910314e+03 = [1, 1, 1, 1]	FES = 179100	Diversity = 1.9765e+00	k_t
Generation 1800: Best = 1.910313e+03 = [4, 4, 3, 3]	FES = 180100	Diversity = 1.9528e+00	k_t
Generation 1810: Best = 1.910313e+03 = [1, 1, 1, 1]	FES = 181100	Diversity = 1.9081e+00	k_t
Generation 1820: Best = 1.910313e+03 = [3, 3, 2, 4]	FES = 182100	Diversity = 1.3349e+00	k_t
Generation 1830: Best = 1.910313e+03 = [1, 1, 1, 1]	FES = 183100	Diversity = 1.2749e+00	k_t
Generation 1840: Best = 1.910310e+03 = [2, 3, 3, 3]	FES = 184100	Diversity = 1.2288e+00	k_t

Generation 1850: Best = 1.910308e+03 = [1, 1, 1, 1]	FES = 185100	Diversity = 1.1858e+00	k_t
Generation 1860: Best = 1.910290e+03 = [4, 2, 3, 4]	FES = 186100	Diversity = 1.1371e+00	k_t
Generation 1870: Best = 1.910290e+03 = [1, 1, 1, 1]	FES = 187100	Diversity = 1.0441e+00	k_t
Generation 1880: Best = 1.910288e+03 = [2, 2, 2, 3]	FES = 188100	Diversity = 1.0216e+00	k_t
Generation 1890: Best = 1.910284e+03 = [1, 1, 1, 1]	FES = 189100	Diversity = 9.7220e-01	k_t
Generation 1900: Best = 1.910284e+03 = [4, 2, 4, 4]	FES = 190100	Diversity = 8.6065e-01	k_t
Generation 1910: Best = 1.910284e+03 = [1, 1, 1, 1]	FES = 191100	Diversity = 6.9996e-01	k_t
Generation 1920: Best = 1.910284e+03 = [3, 2, 3, 4]	FES = 192100	Diversity = 5.1563e-01	k_t
Generation 1930: Best = 1.910284e+03 = [1, 1, 1, 1]	FES = 193100	Diversity = 3.8636e-01	k_t
Generation 1940: Best = 1.910284e+03 = [3, 2, 3, 3]	FES = 194100	Diversity = 3.6340e-01	k_t
Generation 1950: Best = 1.910284e+03 = [1, 1, 1, 1]	FES = 195100	Diversity = 3.6546e-01	k_t
Generation 1960: Best = 1.910280e+03 = [4, 3, 3, 2]	FES = 196100	Diversity = 3.0437e-01	k_t
Generation 1970: Best = 1.910280e+03 = [1, 1, 1, 1]	FES = 197100	Diversity = 2.7259e-01	k_t
Generation 1980: Best = 1.910280e+03 = [3, 3, 2, 4]	FES = 198100	Diversity = 2.3913e-01	k_t
Generation 1990: Best = 1.910274e+03 = [1, 1, 1, 1]	FES = 199100	Diversity = 2.2786e-01	k_t
Generation 2000: Best = 1.910274e+03 = [2, 3, 4, 4]	FES = 200100	Diversity = 2.2426e-01	k_t
Generation 2010: Best = 1.910274e+03 = [1, 1, 1, 1]	FES = 201100	Diversity = 2.0799e-01	k_t
Generation 2020: Best = 1.910273e+03 = [4, 4, 3, 3]	FES = 202100	Diversity = 2.1043e-01	k_t
Generation 2030: Best = 1.910273e+03 = [1, 1, 1, 1]	FES = 203100	Diversity = 2.0838e-01	k_t
Generation 2040: Best = 1.910273e+03 = [3, 4, 3, 3]	FES = 204100	Diversity = 2.0726e-01	k_t
Generation 2050: Best = 1.910273e+03 = [1, 1, 1, 1]	FES = 205100	Diversity = 2.0723e-01	k_t
Generation 2060: Best = 1.910272e+03 = [4, 2, 3, 2]	FES = 206100	Diversity = 2.0929e-01	k_t
Generation 2070: Best = 1.910272e+03 = [1, 1, 1, 1]	FES = 207100	Diversity = 2.0735e-01	k_t
Generation 2080: Best = 1.910271e+03 = [4, 4, 4, 3]	FES = 208100	Diversity = 2.0921e-01	k_t
Generation 2090: Best = 1.910271e+03 = [1, 1, 1, 1]	FES = 209100	Diversity = 2.1013e-01	k_t
Generation 2100: Best = 1.910271e+03 = [3, 4, 3, 2]	FES = 210100	Diversity = 2.1060e-01	k_t
Generation 2110: Best = 1.910271e+03 = [1, 1, 1, 1]	FES = 211100	Diversity = 2.1307e-01	k_t

Generation 2120: Best = 1.910271e+03 = [4, 2, 4, 2]	FES = 212100	Diversity = 2.1568e-01	k_t
Generation 2130: Best = 1.910271e+03 = [1, 1, 1, 1]	FES = 213100	Diversity = 2.1495e-01	k_t
Generation 2140: Best = 1.910271e+03 = [4, 2, 2, 3]	FES = 214100	Diversity = 2.1198e-01	k_t
Generation 2150: Best = 1.910271e+03 = [1, 1, 1, 1]	FES = 215100	Diversity = 2.0989e-01	k_t
Generation 2160: Best = 1.910271e+03 = [3, 2, 2, 3]	FES = 216100	Diversity = 2.0741e-01	k_t
Generation 2170: Best = 1.910271e+03 = [1, 1, 1, 1]	FES = 217100	Diversity = 2.0386e-01	k_t
Generation 2180: Best = 1.910271e+03 = [4, 2, 3, 4]	FES = 218100	Diversity = 2.0292e-01	k_t
Generation 2190: Best = 1.910271e+03 = [1, 1, 1, 1]	FES = 219100	Diversity = 2.0203e-01	k_t
Generation 2200: Best = 1.910271e+03 = [3, 2, 4, 2]	FES = 220100	Diversity = 1.9629e-01	k_t
Generation 2210: Best = 1.910265e+03 = [1, 1, 1, 1]	FES = 221100	Diversity = 1.8781e-01	k_t
Generation 2220: Best = 1.910265e+03 = [4, 3, 2, 2]	FES = 222100	Diversity = 1.8439e-01	k_t
Generation 2230: Best = 1.910265e+03 = [1, 1, 1, 1]	FES = 223100	Diversity = 1.8085e-01	k_t
Generation 2240: Best = 1.910265e+03 = [4, 4, 2, 2]	FES = 224100	Diversity = 1.7522e-01	k_t
Generation 2250: Best = 1.910265e+03 = [1, 1, 1, 1]	FES = 225100	Diversity = 1.7268e-01	k_t
Generation 2260: Best = 1.910265e+03 = [4, 3, 3, 2]	FES = 226100	Diversity = 1.7133e-01	k_t
Generation 2270: Best = 1.910265e+03 = [1, 1, 1, 1]	FES = 227100	Diversity = 1.6515e-01	k_t
Generation 2280: Best = 1.910265e+03 = [2, 4, 3, 2]	FES = 228100	Diversity = 1.6158e-01	k_t
Generation 2290: Best = 1.910264e+03 = [1, 1, 1, 1]	FES = 229100	Diversity = 1.5299e-01	k_t
Generation 2300: Best = 1.910264e+03 = [3, 4, 3, 4]	FES = 230100	Diversity = 1.5228e-01	k_t
Generation 2310: Best = 1.910264e+03 = [1, 1, 1, 1]	FES = 231100	Diversity = 1.4977e-01	k_t
Generation 2320: Best = 1.910264e+03 = [2, 3, 4, 3]	FES = 232100	Diversity = 1.4626e-01	k_t
Generation 2330: Best = 1.910263e+03 = [1, 1, 1, 1]	FES = 233100	Diversity = 1.5039e-01	k_t
Generation 2340: Best = 1.910263e+03 = [2, 2, 4, 3]	FES = 234100	Diversity = 1.4934e-01	k_t
Generation 2350: Best = 1.910263e+03 = [1, 1, 1, 1]	FES = 235100	Diversity = 1.4582e-01	k_t
Generation 2360: Best = 1.910263e+03 = [3, 3, 3, 2]	FES = 236100	Diversity = 1.4902e-01	k_t
Generation 2370: Best = 1.910263e+03 = [1, 1, 1, 1]	FES = 237100	Diversity = 1.4781e-01	k_t
Generation 2380: Best = 1.910263e+03 = [4, 3, 2, 2]	FES = 238100	Diversity = 1.4991e-01	k_t

Generation 2390: Best = 1.910262e+03 = [1, 1, 1, 1]	FES = 239100	Diversity = 1.4544e-01	k_t
Generation 2400: Best = 1.910260e+03 = [4, 2, 2, 2]	FES = 240100	Diversity = 1.4598e-01	k_t
Generation 2410: Best = 1.910260e+03 = [1, 1, 1, 1]	FES = 241100	Diversity = 1.4516e-01	k_t
Generation 2420: Best = 1.910260e+03 = [2, 4, 4, 2]	FES = 242100	Diversity = 1.4534e-01	k_t
Generation 2430: Best = 1.910260e+03 = [1, 1, 1, 1]	FES = 243100	Diversity = 1.4371e-01	k_t
Generation 2440: Best = 1.910260e+03 = [4, 2, 4, 3]	FES = 244100	Diversity = 1.4478e-01	k_t
Generation 2450: Best = 1.910260e+03 = [1, 1, 1, 1]	FES = 245100	Diversity = 1.4422e-01	k_t
Generation 2460: Best = 1.910260e+03 = [4, 2, 2, 4]	FES = 246100	Diversity = 1.4285e-01	k_t
Generation 2470: Best = 1.910260e+03 = [1, 1, 1, 1]	FES = 247100	Diversity = 1.3929e-01	k_t
Generation 2480: Best = 1.910260e+03 = [3, 3, 3, 2]	FES = 248100	Diversity = 1.3861e-01	k_t
Generation 2490: Best = 1.910259e+03 = [1, 1, 1, 1]	FES = 249100	Diversity = 1.3960e-01	k_t
Generation 2500: Best = 1.910258e+03 = [2, 2, 3, 2]	FES = 250100	Diversity = 1.3892e-01	k_t
Generation 2510: Best = 1.910258e+03 = [1, 1, 1, 1]	FES = 251100	Diversity = 1.3838e-01	k_t
Generation 2520: Best = 1.910258e+03 = [3, 4, 2, 4]	FES = 252100	Diversity = 1.3786e-01	k_t
Generation 2530: Best = 1.910257e+03 = [1, 1, 1, 1]	FES = 253100	Diversity = 1.3773e-01	k_t
Generation 2540: Best = 1.910257e+03 = [3, 4, 3, 4]	FES = 254100	Diversity = 1.3814e-01	k_t
Generation 2550: Best = 1.910256e+03 = [1, 1, 1, 1]	FES = 255100	Diversity = 1.3710e-01	k_t
Generation 2560: Best = 1.910256e+03 = [4, 4, 3, 4]	FES = 256100	Diversity = 1.3665e-01	k_t
Generation 2570: Best = 1.910256e+03 = [1, 1, 1, 1]	FES = 257100	Diversity = 1.3668e-01	k_t
Generation 2580: Best = 1.910256e+03 = [4, 4, 4, 3]	FES = 258100	Diversity = 1.3411e-01	k_t
Generation 2590: Best = 1.910256e+03 = [1, 1, 1, 1]	FES = 259100	Diversity = 1.3432e-01	k_t
Generation 2600: Best = 1.910256e+03 = [2, 3, 4, 3]	FES = 260100	Diversity = 1.3003e-01	k_t
Generation 2610: Best = 1.910256e+03 = [1, 1, 1, 1]	FES = 261100	Diversity = 1.2882e-01	k_t
Generation 2620: Best = 1.910256e+03 = [2, 4, 4, 3]	FES = 262100	Diversity = 1.2059e-01	k_t
Generation 2630: Best = 1.910256e+03 = [1, 1, 1, 1]	FES = 263100	Diversity = 1.0803e-01	k_t
Generation 2640: Best = 1.910256e+03 = [3, 4, 3, 2]	FES = 264100	Diversity = 9.5334e-02	k_t
Generation 2650: Best = 1.910256e+03 = [1, 1, 1, 1]	FES = 265100	Diversity = 8.1524e-02	k_t

Generation 2660: Best = 1.910255e+03 = [4, 2, 3, 4]	FES = 266100	Diversity = 4.4173e-02	k_t
Generation 2670: Best = 1.910255e+03 = [1, 1, 1, 1]	FES = 267100	Diversity = 4.0225e-02	k_t
Generation 2680: Best = 1.910255e+03 = [3, 2, 4, 2]	FES = 268100	Diversity = 2.4111e-02	k_t
Generation 2690: Best = 1.910255e+03 = [1, 1, 1, 1]	FES = 269100	Diversity = 2.3939e-02	k_t
Generation 2700: Best = 1.910255e+03 = [4, 2, 4, 3]	FES = 270100	Diversity = 2.2374e-02	k_t
Generation 2710: Best = 1.910255e+03 = [1, 1, 1, 1]	FES = 271100	Diversity = 2.1673e-02	k_t
Generation 2720: Best = 1.910255e+03 = [4, 3, 3, 4]	FES = 272100	Diversity = 1.9602e-02	k_t
Generation 2730: Best = 1.910255e+03 = [1, 1, 1, 1]	FES = 273100	Diversity = 1.8413e-02	k_t
Generation 2740: Best = 1.910255e+03 = [3, 2, 4, 4]	FES = 274100	Diversity = 1.6740e-02	k_t
Generation 2750: Best = 1.910255e+03 = [1, 1, 1, 1]	FES = 275100	Diversity = 1.6061e-02	k_t
Generation 2760: Best = 1.910255e+03 = [2, 3, 4, 4]	FES = 276100	Diversity = 1.5836e-02	k_t
Generation 2770: Best = 1.910254e+03 = [1, 1, 1, 1]	FES = 277100	Diversity = 1.5219e-02	k_t
Generation 2780: Best = 1.910254e+03 = [3, 4, 3, 3]	FES = 278100	Diversity = 1.5208e-02	k_t
Generation 2790: Best = 1.910254e+03 = [1, 1, 1, 1]	FES = 279100	Diversity = 1.5089e-02	k_t
Generation 2800: Best = 1.910254e+03 = [3, 3, 3, 3]	FES = 280100	Diversity = 1.4623e-02	k_t
Generation 2810: Best = 1.910254e+03 = [1, 1, 1, 1]	FES = 281100	Diversity = 1.4258e-02	k_t
Generation 2820: Best = 1.910254e+03 = [3, 3, 4, 2]	FES = 282100	Diversity = 1.3889e-02	k_t
Generation 2830: Best = 1.910254e+03 = [1, 1, 1, 1]	FES = 283100	Diversity = 1.3452e-02	k_t
Generation 2840: Best = 1.910254e+03 = [2, 3, 3, 2]	FES = 284100	Diversity = 1.2849e-02	k_t
Generation 2850: Best = 1.910254e+03 = [1, 1, 1, 1]	FES = 285100	Diversity = 1.2775e-02	k_t
Generation 2860: Best = 1.910254e+03 = [2, 2, 2, 4]	FES = 286100	Diversity = 1.2546e-02	k_t
Generation 2870: Best = 1.910254e+03 = [1, 1, 1, 1]	FES = 287100	Diversity = 1.2461e-02	k_t
Generation 2880: Best = 1.910254e+03 = [4, 4, 3, 4]	FES = 288100	Diversity = 1.2678e-02	k_t
Generation 2890: Best = 1.910254e+03 = [1, 1, 1, 1]	FES = 289100	Diversity = 1.2213e-02	k_t
Generation 2900: Best = 1.910254e+03 = [3, 4, 3, 4]	FES = 290100	Diversity = 1.2171e-02	k_t
Generation 2910: Best = 1.910254e+03 = [1, 1, 1, 1]	FES = 291100	Diversity = 1.1716e-02	k_t
Generation 2920: Best = 1.910254e+03 = [3, 4, 2, 2]	FES = 292100	Diversity = 1.1442e-02	k_t

Generation 2930: Best = 1.910254e+03 FES = 293100 Diversity = 1.1302e-02 k_t
 = [1, 1, 1, 1]
 Generation 2940: Best = 1.910254e+03 FES = 294100 Diversity = 1.1381e-02 k_t
 = [2, 2, 4, 4]
 Generation 2950: Best = 1.910254e+03 FES = 295100 Diversity = 1.1477e-02 k_t
 = [1, 1, 1, 1]
 Generation 2960: Best = 1.910254e+03 FES = 296100 Diversity = 1.1252e-02 k_t
 = [3, 2, 3, 4]
 Generation 2970: Best = 1.910254e+03 FES = 297100 Diversity = 1.1035e-02 k_t
 = [1, 1, 1, 1]
 Generation 2980: Best = 1.910254e+03 FES = 298100 Diversity = 1.0999e-02 k_t
 = [2, 3, 2, 2]
 Generation 2990: Best = 1.910254e+03 FES = 299100 Diversity = 1.0688e-02 k_t
 = [1, 1, 1, 1]

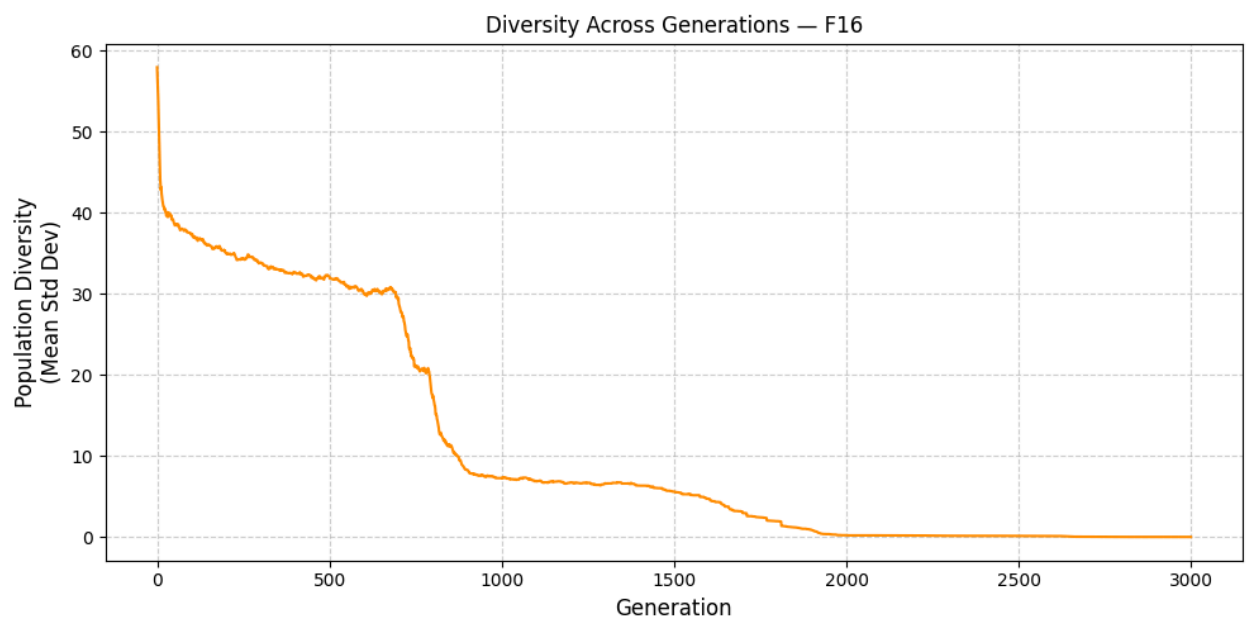
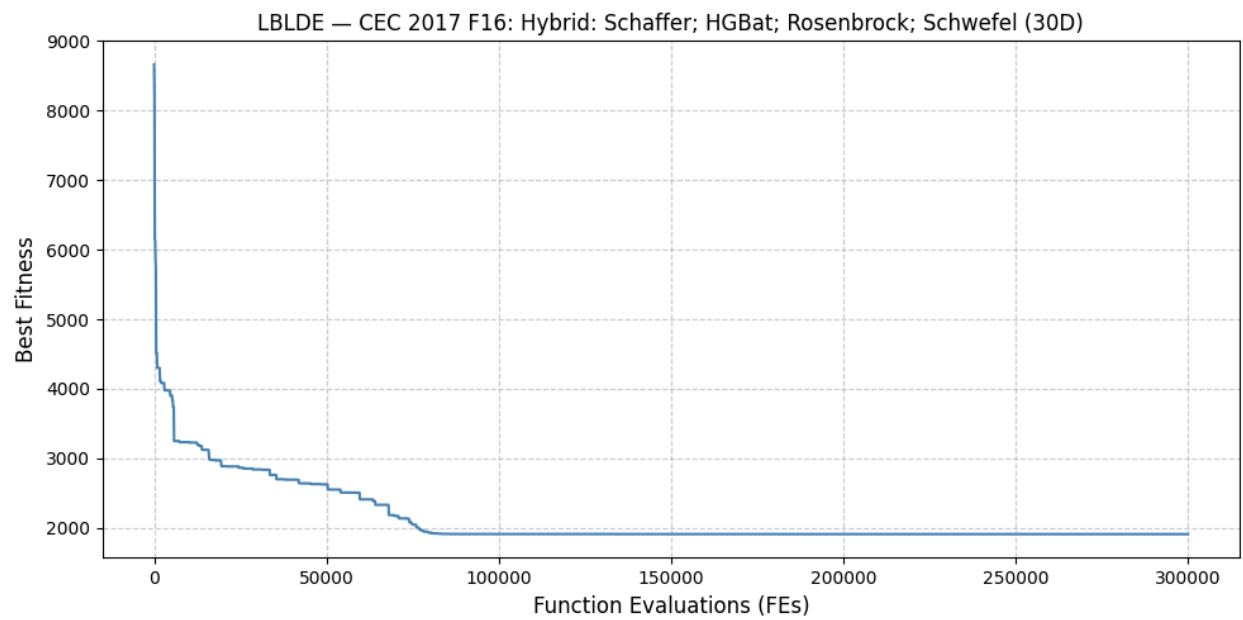
Optimization completed!

Final Best Fitness = 1.910254e+03

Total Function Evaluations = 300000

```

=====
=
Results:
  Best Fitness           : 1.910254e+03
  Global Optimum         : 1.600000e+03
  Error (|best-optimum|) : 3.102544e+02
  Best Solution (5 dim)  : [34.56605389 83.93051401 79.06878202 21.31483954 2
2.16495742]
  D^(0) (init. diversity): 5.7976e+01
=====
=
  
```

Graph saved as convergence_F16.png

```
In [27]: # =====
# F17 – Hybrid Function 7 (Katsuura; Ackley; Griewank+Rosenbrock; Schwefel; Ra
# =====
run_and_plot('F17', make_wrapper(hybrid.f17), 1700.0, 'Hybrid: Katsuura; Ackle
```

```

=====
=
LBLDE_s on CEC 2017 F17 – Single Run
=====
=
Function      : Hybrid: Katsuura; Ackley; Griewank+Rosenbrock; Schwefel; Rastr
igin (30D)
Bounds       : [-100, 100]
Global optimum : 1700.0
Initial D^(0) = 5.7976e+01
Generation 0: Best Fitness = 4.634765e+03 | Diversity = 5.7976e+01
Generation 10: Best = 2.942299e+03 FES = 1100 Diversity = 4.3959e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 2.886648e+03 FES = 2100 Diversity = 4.3028e+01 k_t
= [3, 3, 2, 2]
Generation 30: Best = 2.779265e+03 FES = 3100 Diversity = 4.2059e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 2.565386e+03 FES = 4100 Diversity = 4.2784e+01 k_t
= [2, 2, 4, 3]
Generation 50: Best = 2.565386e+03 FES = 5100 Diversity = 4.2058e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 2.343722e+03 FES = 6100 Diversity = 4.2162e+01 k_t
= [3, 3, 3, 2]
Generation 70: Best = 2.287430e+03 FES = 7100 Diversity = 4.1974e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 2.284552e+03 FES = 8100 Diversity = 4.1598e+01 k_t
= [4, 3, 4, 3]
Generation 90: Best = 2.284307e+03 FES = 9100 Diversity = 4.1497e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 2.242778e+03 FES = 10100 Diversity = 4.1882e+01 k_t
= [2, 2, 3, 3]
Generation 110: Best = 2.236123e+03 FES = 11100 Diversity = 4.2039e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 2.234645e+03 FES = 12100 Diversity = 4.2597e+01 k_t
= [4, 4, 3, 2]
Generation 130: Best = 2.231646e+03 FES = 13100 Diversity = 4.2160e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 2.218725e+03 FES = 14100 Diversity = 4.1889e+01 k_t
= [4, 4, 3, 3]
Generation 150: Best = 2.187034e+03 FES = 15100 Diversity = 4.1538e+01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 2.168626e+03 FES = 16100 Diversity = 4.1679e+01 k_t
= [4, 4, 4, 2]
Generation 170: Best = 2.142482e+03 FES = 17100 Diversity = 4.1271e+01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 2.142482e+03 FES = 18100 Diversity = 4.1439e+01 k_t
= [2, 3, 3, 2]
Generation 190: Best = 2.138525e+03 FES = 19100 Diversity = 4.1421e+01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 2.138525e+03 FES = 20100 Diversity = 4.1304e+01 k_t
= [4, 4, 2, 2]
Generation 210: Best = 2.138525e+03 FES = 21100 Diversity = 4.0721e+01 k_t
= [1, 1, 1, 1]
Generation 220: Best = 2.138102e+03 FES = 22100 Diversity = 4.0624e+01 k_t

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= [3, 4, 3, 4]
Generation 230: Best = 2.115176e+03  FES = 23100  Diversity = 4.0732e+01  k_t
= [1, 1, 1, 1]
Generation 240: Best = 2.115176e+03  FES = 24100  Diversity = 4.0603e+01  k_t
= [4, 2, 2, 4]
Generation 250: Best = 2.078221e+03  FES = 25100  Diversity = 4.0503e+01  k_t
= [1, 1, 1, 1]
Generation 260: Best = 2.075711e+03  FES = 26100  Diversity = 4.0641e+01  k_t
= [4, 3, 4, 4]
Generation 270: Best = 2.075318e+03  FES = 27100  Diversity = 4.0549e+01  k_t
= [1, 1, 1, 1]
Generation 280: Best = 2.075318e+03  FES = 28100  Diversity = 4.0464e+01  k_t
= [4, 4, 3, 3]
Generation 290: Best = 2.067712e+03  FES = 29100  Diversity = 4.0503e+01  k_t
= [1, 1, 1, 1]
Generation 300: Best = 2.064188e+03  FES = 30100  Diversity = 4.0762e+01  k_t
= [2, 3, 3, 4]
Generation 310: Best = 2.063044e+03  FES = 31100  Diversity = 4.0719e+01  k_t
= [1, 1, 1, 1]
Generation 320: Best = 2.063044e+03  FES = 32100  Diversity = 4.0732e+01  k_t
= [2, 3, 4, 2]
Generation 330: Best = 2.045325e+03  FES = 33100  Diversity = 4.0780e+01  k_t
= [1, 1, 1, 1]
Generation 340: Best = 2.040887e+03  FES = 34100  Diversity = 4.0512e+01  k_t
= [3, 4, 4, 3]
Generation 350: Best = 2.040887e+03  FES = 35100  Diversity = 4.0430e+01  k_t
= [1, 1, 1, 1]
Generation 360: Best = 2.040887e+03  FES = 36100  Diversity = 4.0170e+01  k_t
= [3, 2, 2, 2]
Generation 370: Best = 2.040887e+03  FES = 37100  Diversity = 4.0164e+01  k_t
= [1, 1, 1, 1]
Generation 380: Best = 2.034028e+03  FES = 38100  Diversity = 4.0484e+01  k_t
= [4, 3, 4, 3]
Generation 390: Best = 2.032499e+03  FES = 39100  Diversity = 4.0549e+01  k_t
= [1, 1, 1, 1]
Generation 400: Best = 2.029956e+03  FES = 40100  Diversity = 4.1010e+01  k_t
= [4, 3, 3, 4]
Generation 410: Best = 2.004972e+03  FES = 41100  Diversity = 4.0989e+01  k_t
= [1, 1, 1, 1]
Generation 420: Best = 2.004955e+03  FES = 42100  Diversity = 4.0213e+01  k_t
= [3, 2, 3, 4]
Generation 430: Best = 2.004955e+03  FES = 43100  Diversity = 4.0168e+01  k_t
= [1, 1, 1, 1]
Generation 440: Best = 1.998554e+03  FES = 44100  Diversity = 3.9853e+01  k_t
= [3, 2, 2, 4]
Generation 450: Best = 1.995305e+03  FES = 45100  Diversity = 3.9629e+01  k_t
= [1, 1, 1, 1]
Generation 460: Best = 1.995305e+03  FES = 46100  Diversity = 3.8683e+01  k_t
= [3, 2, 3, 4]
Generation 470: Best = 1.989167e+03  FES = 47100  Diversity = 3.8829e+01  k_t
= [1, 1, 1, 1]
Generation 480: Best = 1.985568e+03  FES = 48100  Diversity = 3.8743e+01  k_t
= [2, 3, 2, 2]
Generation 490: Best = 1.975935e+03  FES = 49100  Diversity = 3.8510e+01  k_t

```

```

= [1, 1, 1, 1]
Generation 500: Best = 1.931358e+03 FES = 50100 Diversity = 3.8180e+01 k_t
= [2, 2, 3, 4]
Generation 510: Best = 1.885308e+03 FES = 51100 Diversity = 3.8131e+01 k_t
= [1, 1, 1, 1]
Generation 520: Best = 1.879384e+03 FES = 52100 Diversity = 3.8278e+01 k_t
= [2, 4, 3, 3]
Generation 530: Best = 1.862286e+03 FES = 53100 Diversity = 3.7935e+01 k_t
= [1, 1, 1, 1]
Generation 540: Best = 1.854192e+03 FES = 54100 Diversity = 3.7669e+01 k_t
= [2, 4, 4, 3]
Generation 550: Best = 1.848883e+03 FES = 55100 Diversity = 3.7162e+01 k_t
= [1, 1, 1, 1]
Generation 560: Best = 1.834390e+03 FES = 56100 Diversity = 3.6618e+01 k_t
= [2, 3, 3, 3]
Generation 570: Best = 1.824870e+03 FES = 57100 Diversity = 3.6115e+01 k_t
= [1, 1, 1, 1]
Generation 580: Best = 1.820618e+03 FES = 58100 Diversity = 3.6114e+01 k_t
= [4, 2, 3, 2]
Generation 590: Best = 1.809109e+03 FES = 59100 Diversity = 3.5497e+01 k_t
= [1, 1, 1, 1]
Generation 600: Best = 1.807697e+03 FES = 60100 Diversity = 3.4082e+01 k_t
= [3, 4, 4, 2]
Generation 610: Best = 1.793197e+03 FES = 61100 Diversity = 3.2761e+01 k_t
= [1, 1, 1, 1]
Generation 620: Best = 1.793197e+03 FES = 62100 Diversity = 3.1563e+01 k_t
= [3, 3, 2, 4]
Generation 630: Best = 1.793197e+03 FES = 63100 Diversity = 2.9537e+01 k_t
= [1, 1, 1, 1]
Generation 640: Best = 1.793197e+03 FES = 64100 Diversity = 2.9882e+01 k_t
= [4, 3, 4, 3]
Generation 650: Best = 1.791850e+03 FES = 65100 Diversity = 2.8717e+01 k_t
= [1, 1, 1, 1]
Generation 660: Best = 1.791850e+03 FES = 66100 Diversity = 2.8187e+01 k_t
= [3, 2, 4, 2]
Generation 670: Best = 1.790983e+03 FES = 67100 Diversity = 2.8012e+01 k_t
= [1, 1, 1, 1]
Generation 680: Best = 1.784301e+03 FES = 68100 Diversity = 2.7435e+01 k_t
= [3, 4, 2, 3]
Generation 690: Best = 1.775248e+03 FES = 69100 Diversity = 2.7241e+01 k_t
= [1, 1, 1, 1]
Generation 700: Best = 1.774548e+03 FES = 70100 Diversity = 2.7143e+01 k_t
= [4, 2, 3, 2]
Generation 710: Best = 1.774515e+03 FES = 71100 Diversity = 2.6772e+01 k_t
= [1, 1, 1, 1]
Generation 720: Best = 1.773452e+03 FES = 72100 Diversity = 2.6642e+01 k_t
= [2, 3, 4, 4]
Generation 730: Best = 1.773452e+03 FES = 73100 Diversity = 2.6136e+01 k_t
= [1, 1, 1, 1]
Generation 740: Best = 1.773452e+03 FES = 74100 Diversity = 2.5947e+01 k_t
= [3, 2, 3, 3]
Generation 750: Best = 1.773452e+03 FES = 75100 Diversity = 2.6021e+01 k_t
= [1, 1, 1, 1]
Generation 760: Best = 1.773452e+03 FES = 76100 Diversity = 2.6095e+01 k_t

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= [3, 3, 3, 3]
Generation 770: Best = 1.769015e+03 FES = 77100 Diversity = 2.5979e+01 k_t
= [1, 1, 1, 1]
Generation 780: Best = 1.768066e+03 FES = 78100 Diversity = 2.5983e+01 k_t
= [4, 2, 2, 4]
Generation 790: Best = 1.768066e+03 FES = 79100 Diversity = 2.5987e+01 k_t
= [1, 1, 1, 1]
Generation 800: Best = 1.768066e+03 FES = 80100 Diversity = 2.5962e+01 k_t
= [3, 3, 2, 3]
Generation 810: Best = 1.768066e+03 FES = 81100 Diversity = 2.5955e+01 k_t
= [1, 1, 1, 1]
Generation 820: Best = 1.761986e+03 FES = 82100 Diversity = 2.5759e+01 k_t
= [3, 3, 2, 4]
Generation 830: Best = 1.761986e+03 FES = 83100 Diversity = 2.5896e+01 k_t
= [1, 1, 1, 1]
Generation 840: Best = 1.761971e+03 FES = 84100 Diversity = 2.6264e+01 k_t
= [3, 4, 4, 3]
Generation 850: Best = 1.761519e+03 FES = 85100 Diversity = 2.6346e+01 k_t
= [1, 1, 1, 1]
Generation 860: Best = 1.761287e+03 FES = 86100 Diversity = 2.6296e+01 k_t
= [4, 2, 2, 2]
Generation 870: Best = 1.755858e+03 FES = 87100 Diversity = 2.6101e+01 k_t
= [1, 1, 1, 1]
Generation 880: Best = 1.755851e+03 FES = 88100 Diversity = 2.6057e+01 k_t
= [2, 4, 2, 3]
Generation 890: Best = 1.755851e+03 FES = 89100 Diversity = 2.6218e+01 k_t
= [1, 1, 1, 1]
Generation 900: Best = 1.755662e+03 FES = 90100 Diversity = 2.6344e+01 k_t
= [4, 3, 4, 4]
Generation 910: Best = 1.753271e+03 FES = 91100 Diversity = 2.6284e+01 k_t
= [1, 1, 1, 1]
Generation 920: Best = 1.752979e+03 FES = 92100 Diversity = 2.6079e+01 k_t
= [4, 2, 2, 4]
Generation 930: Best = 1.752805e+03 FES = 93100 Diversity = 2.6151e+01 k_t
= [1, 1, 1, 1]
Generation 940: Best = 1.752105e+03 FES = 94100 Diversity = 2.6214e+01 k_t
= [2, 2, 2, 2]
Generation 950: Best = 1.749885e+03 FES = 95100 Diversity = 2.6068e+01 k_t
= [1, 1, 1, 1]
Generation 960: Best = 1.749463e+03 FES = 96100 Diversity = 2.5701e+01 k_t
= [4, 2, 2, 2]
Generation 970: Best = 1.749348e+03 FES = 97100 Diversity = 2.5330e+01 k_t
= [1, 1, 1, 1]
Generation 980: Best = 1.748184e+03 FES = 98100 Diversity = 2.5339e+01 k_t
= [2, 4, 3, 3]
Generation 990: Best = 1.748184e+03 FES = 99100 Diversity = 2.5253e+01 k_t
= [1, 1, 1, 1]
Generation 1000: Best = 1.746819e+03 FES = 100100 Diversity = 2.5182e+01 k_t
= [4, 2, 3, 4]
Generation 1010: Best = 1.743800e+03 FES = 101100 Diversity = 2.5470e+01 k_t
= [1, 1, 1, 1]
Generation 1020: Best = 1.743780e+03 FES = 102100 Diversity = 2.5227e+01 k_t
= [3, 3, 4, 3]
Generation 1030: Best = 1.743605e+03 FES = 103100 Diversity = 2.4821e+01 k_t

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```

= [1, 1, 1, 1]
Generation 1040: Best = 1.743605e+03 FES = 104100 Diversity = 2.4555e+01 k_t
= [2, 3, 2, 2]
Generation 1050: Best = 1.743573e+03 FES = 105100 Diversity = 2.4324e+01 k_t
= [1, 1, 1, 1]
Generation 1060: Best = 1.743434e+03 FES = 106100 Diversity = 2.4268e+01 k_t
= [4, 4, 4, 2]
Generation 1070: Best = 1.743434e+03 FES = 107100 Diversity = 2.4152e+01 k_t
= [1, 1, 1, 1]
Generation 1080: Best = 1.743270e+03 FES = 108100 Diversity = 2.4203e+01 k_t
= [4, 3, 3, 4]
Generation 1090: Best = 1.742851e+03 FES = 109100 Diversity = 2.4317e+01 k_t
= [1, 1, 1, 1]
Generation 1100: Best = 1.742786e+03 FES = 110100 Diversity = 2.3626e+01 k_t
= [4, 2, 3, 3]
Generation 1110: Best = 1.742407e+03 FES = 111100 Diversity = 2.3345e+01 k_t
= [1, 1, 1, 1]
Generation 1120: Best = 1.742407e+03 FES = 112100 Diversity = 2.2820e+01 k_t
= [3, 2, 4, 3]
Generation 1130: Best = 1.741596e+03 FES = 113100 Diversity = 2.2577e+01 k_t
= [1, 1, 1, 1]
Generation 1140: Best = 1.741192e+03 FES = 114100 Diversity = 2.2340e+01 k_t
= [3, 3, 2, 3]
Generation 1150: Best = 1.741192e+03 FES = 115100 Diversity = 2.1660e+01 k_t
= [1, 1, 1, 1]
Generation 1160: Best = 1.740980e+03 FES = 116100 Diversity = 2.1355e+01 k_t
= [4, 4, 2, 2]
Generation 1170: Best = 1.740875e+03 FES = 117100 Diversity = 2.1176e+01 k_t
= [1, 1, 1, 1]
Generation 1180: Best = 1.740875e+03 FES = 118100 Diversity = 2.0513e+01 k_t
= [4, 4, 4, 3]
Generation 1190: Best = 1.740870e+03 FES = 119100 Diversity = 1.9649e+01 k_t
= [1, 1, 1, 1]
Generation 1200: Best = 1.738695e+03 FES = 120100 Diversity = 1.9434e+01 k_t
= [3, 2, 2, 3]
Generation 1210: Best = 1.738639e+03 FES = 121100 Diversity = 1.9381e+01 k_t
= [1, 1, 1, 1]
Generation 1220: Best = 1.738600e+03 FES = 122100 Diversity = 1.9360e+01 k_t
= [2, 3, 3, 3]
Generation 1230: Best = 1.738556e+03 FES = 123100 Diversity = 1.9483e+01 k_t
= [1, 1, 1, 1]
Generation 1240: Best = 1.738556e+03 FES = 124100 Diversity = 1.9326e+01 k_t
= [2, 3, 3, 2]
Generation 1250: Best = 1.738556e+03 FES = 125100 Diversity = 1.9406e+01 k_t
= [1, 1, 1, 1]
Generation 1260: Best = 1.738554e+03 FES = 126100 Diversity = 1.9495e+01 k_t
= [4, 3, 3, 3]
Generation 1270: Best = 1.738548e+03 FES = 127100 Diversity = 1.9158e+01 k_t
= [1, 1, 1, 1]
Generation 1280: Best = 1.738533e+03 FES = 128100 Diversity = 1.8920e+01 k_t
= [3, 4, 3, 2]
Generation 1290: Best = 1.738533e+03 FES = 129100 Diversity = 1.8479e+01 k_t
= [1, 1, 1, 1]
Generation 1300: Best = 1.738201e+03 FES = 130100 Diversity = 1.8336e+01 k_t

```

```

= [2, 2, 3, 3]
Generation 1310: Best = 1.738035e+03 FES = 131100 Diversity = 1.8331e+01 k_t
= [1, 1, 1, 1]
Generation 1320: Best = 1.738035e+03 FES = 132100 Diversity = 1.8440e+01 k_t
= [4, 3, 4, 2]
Generation 1330: Best = 1.738035e+03 FES = 133100 Diversity = 1.7861e+01 k_t
= [1, 1, 1, 1]
Generation 1340: Best = 1.738033e+03 FES = 134100 Diversity = 1.8006e+01 k_t
= [4, 2, 4, 2]
Generation 1350: Best = 1.737863e+03 FES = 135100 Diversity = 1.8087e+01 k_t
= [1, 1, 1, 1]
Generation 1360: Best = 1.737755e+03 FES = 136100 Diversity = 1.7879e+01 k_t
= [2, 4, 3, 3]
Generation 1370: Best = 1.737655e+03 FES = 137100 Diversity = 1.7896e+01 k_t
= [1, 1, 1, 1]
Generation 1380: Best = 1.737655e+03 FES = 138100 Diversity = 1.7917e+01 k_t
= [4, 4, 3, 2]
Generation 1390: Best = 1.737655e+03 FES = 139100 Diversity = 1.7595e+01 k_t
= [1, 1, 1, 1]
Generation 1400: Best = 1.737655e+03 FES = 140100 Diversity = 1.6845e+01 k_t
= [4, 3, 2, 2]
Generation 1410: Best = 1.737655e+03 FES = 141100 Diversity = 1.6938e+01 k_t
= [1, 1, 1, 1]
Generation 1420: Best = 1.737176e+03 FES = 142100 Diversity = 1.6818e+01 k_t
= [4, 4, 3, 3]
Generation 1430: Best = 1.737176e+03 FES = 143100 Diversity = 1.6771e+01 k_t
= [1, 1, 1, 1]
Generation 1440: Best = 1.737175e+03 FES = 144100 Diversity = 1.6585e+01 k_t
= [4, 3, 4, 4]
Generation 1450: Best = 1.737175e+03 FES = 145100 Diversity = 1.6698e+01 k_t
= [1, 1, 1, 1]
Generation 1460: Best = 1.737175e+03 FES = 146100 Diversity = 1.6671e+01 k_t
= [3, 2, 2, 4]
Generation 1470: Best = 1.737175e+03 FES = 147100 Diversity = 1.6136e+01 k_t
= [1, 1, 1, 1]
Generation 1480: Best = 1.737175e+03 FES = 148100 Diversity = 1.6084e+01 k_t
= [2, 3, 2, 2]
Generation 1490: Best = 1.737175e+03 FES = 149100 Diversity = 1.5851e+01 k_t
= [1, 1, 1, 1]
Generation 1500: Best = 1.737175e+03 FES = 150100 Diversity = 1.5697e+01 k_t
= [2, 3, 3, 3]
Generation 1510: Best = 1.737175e+03 FES = 151100 Diversity = 1.5382e+01 k_t
= [1, 1, 1, 1]
Generation 1520: Best = 1.737175e+03 FES = 152100 Diversity = 1.5111e+01 k_t
= [2, 2, 4, 2]
Generation 1530: Best = 1.737124e+03 FES = 153100 Diversity = 1.4951e+01 k_t
= [1, 1, 1, 1]
Generation 1540: Best = 1.737124e+03 FES = 154100 Diversity = 1.4912e+01 k_t
= [4, 3, 2, 2]
Generation 1550: Best = 1.737012e+03 FES = 155100 Diversity = 1.4790e+01 k_t
= [1, 1, 1, 1]
Generation 1560: Best = 1.736988e+03 FES = 156100 Diversity = 1.4764e+01 k_t
= [2, 2, 2, 4]
Generation 1570: Best = 1.736976e+03 FES = 157100 Diversity = 1.4624e+01 k_t

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= [1, 1, 1, 1]
Generation 1580: Best = 1.736957e+03 FES = 158100 Diversity = 1.4330e+01 k_t
= [3, 2, 3, 3]
Generation 1590: Best = 1.736957e+03 FES = 159100 Diversity = 1.4201e+01 k_t
= [1, 1, 1, 1]
Generation 1600: Best = 1.736957e+03 FES = 160100 Diversity = 1.4141e+01 k_t
= [4, 4, 3, 3]
Generation 1610: Best = 1.736957e+03 FES = 161100 Diversity = 1.4028e+01 k_t
= [1, 1, 1, 1]
Generation 1620: Best = 1.736957e+03 FES = 162100 Diversity = 1.3603e+01 k_t
= [2, 3, 3, 3]
Generation 1630: Best = 1.736901e+03 FES = 163100 Diversity = 1.3626e+01 k_t
= [1, 1, 1, 1]
Generation 1640: Best = 1.736901e+03 FES = 164100 Diversity = 1.3523e+01 k_t
= [3, 2, 2, 4]
Generation 1650: Best = 1.736632e+03 FES = 165100 Diversity = 1.3954e+01 k_t
= [1, 1, 1, 1]
Generation 1660: Best = 1.736625e+03 FES = 166100 Diversity = 1.4040e+01 k_t
= [4, 4, 2, 4]
Generation 1670: Best = 1.736613e+03 FES = 167100 Diversity = 1.3633e+01 k_t
= [1, 1, 1, 1]
Generation 1680: Best = 1.736613e+03 FES = 168100 Diversity = 1.3792e+01 k_t
= [3, 3, 3, 2]
Generation 1690: Best = 1.736613e+03 FES = 169100 Diversity = 1.3893e+01 k_t
= [1, 1, 1, 1]
Generation 1700: Best = 1.736451e+03 FES = 170100 Diversity = 1.4193e+01 k_t
= [3, 3, 3, 3]
Generation 1710: Best = 1.736450e+03 FES = 171100 Diversity = 1.4147e+01 k_t
= [1, 1, 1, 1]
Generation 1720: Best = 1.736110e+03 FES = 172100 Diversity = 1.4528e+01 k_t
= [2, 2, 3, 3]
Generation 1730: Best = 1.736106e+03 FES = 173100 Diversity = 1.4589e+01 k_t
= [1, 1, 1, 1]
Generation 1740: Best = 1.736085e+03 FES = 174100 Diversity = 1.4829e+01 k_t
= [3, 2, 3, 3]
Generation 1750: Best = 1.736085e+03 FES = 175100 Diversity = 1.5058e+01 k_t
= [1, 1, 1, 1]
Generation 1760: Best = 1.736068e+03 FES = 176100 Diversity = 1.5115e+01 k_t
= [3, 4, 3, 4]
Generation 1770: Best = 1.736065e+03 FES = 177100 Diversity = 1.5142e+01 k_t
= [1, 1, 1, 1]
Generation 1780: Best = 1.736032e+03 FES = 178100 Diversity = 1.5381e+01 k_t
= [3, 4, 4, 3]
Generation 1790: Best = 1.735877e+03 FES = 179100 Diversity = 1.5275e+01 k_t
= [1, 1, 1, 1]
Generation 1800: Best = 1.735859e+03 FES = 180100 Diversity = 1.4806e+01 k_t
= [3, 3, 2, 4]
Generation 1810: Best = 1.735823e+03 FES = 181100 Diversity = 1.4857e+01 k_t
= [1, 1, 1, 1]
Generation 1820: Best = 1.735810e+03 FES = 182100 Diversity = 1.4871e+01 k_t
= [2, 2, 2, 3]
Generation 1830: Best = 1.735794e+03 FES = 183100 Diversity = 1.4465e+01 k_t
= [1, 1, 1, 1]
Generation 1840: Best = 1.735789e+03 FES = 184100 Diversity = 1.2358e+01 k_t

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= [3, 2, 4, 3]
Generation 1850: Best = 1.735788e+03 FES = 185100 Diversity = 1.1671e+01 k_t
= [1, 1, 1, 1]
Generation 1860: Best = 1.735708e+03 FES = 186100 Diversity = 1.0228e+01 k_t
= [2, 2, 4, 4]
Generation 1870: Best = 1.735704e+03 FES = 187100 Diversity = 8.5194e+00 k_t
= [1, 1, 1, 1]
Generation 1880: Best = 1.735664e+03 FES = 188100 Diversity = 7.1666e+00 k_t
= [2, 3, 3, 4]
Generation 1890: Best = 1.735664e+03 FES = 189100 Diversity = 5.8057e+00 k_t
= [1, 1, 1, 1]
Generation 1900: Best = 1.735600e+03 FES = 190100 Diversity = 5.3437e+00 k_t
= [3, 3, 2, 3]
Generation 1910: Best = 1.735590e+03 FES = 191100 Diversity = 5.2586e+00 k_t
= [1, 1, 1, 1]
Generation 1920: Best = 1.735521e+03 FES = 192100 Diversity = 4.9669e+00 k_t
= [3, 4, 2, 4]
Generation 1930: Best = 1.735515e+03 FES = 193100 Diversity = 4.6552e+00 k_t
= [1, 1, 1, 1]
Generation 1940: Best = 1.735490e+03 FES = 194100 Diversity = 3.7411e+00 k_t
= [3, 3, 4, 2]
Generation 1950: Best = 1.735445e+03 FES = 195100 Diversity = 3.6685e+00 k_t
= [1, 1, 1, 1]
Generation 1960: Best = 1.735437e+03 FES = 196100 Diversity = 3.4560e+00 k_t
= [4, 3, 2, 2]
Generation 1970: Best = 1.735421e+03 FES = 197100 Diversity = 3.4875e+00 k_t
= [1, 1, 1, 1]
Generation 1980: Best = 1.735387e+03 FES = 198100 Diversity = 3.4634e+00 k_t
= [4, 2, 4, 2]
Generation 1990: Best = 1.735374e+03 FES = 199100 Diversity = 3.4699e+00 k_t
= [1, 1, 1, 1]
Generation 2000: Best = 1.735353e+03 FES = 200100 Diversity = 3.2728e+00 k_t
= [3, 2, 4, 3]
Generation 2010: Best = 1.735333e+03 FES = 201100 Diversity = 3.0894e+00 k_t
= [1, 1, 1, 1]
Generation 2020: Best = 1.735276e+03 FES = 202100 Diversity = 3.0034e+00 k_t
= [3, 2, 3, 3]
Generation 2030: Best = 1.735203e+03 FES = 203100 Diversity = 2.9293e+00 k_t
= [1, 1, 1, 1]
Generation 2040: Best = 1.735146e+03 FES = 204100 Diversity = 3.0984e+00 k_t
= [2, 3, 2, 3]
Generation 2050: Best = 1.735128e+03 FES = 205100 Diversity = 3.0694e+00 k_t
= [1, 1, 1, 1]
Generation 2060: Best = 1.735097e+03 FES = 206100 Diversity = 3.4532e+00 k_t
= [2, 3, 4, 4]
Generation 2070: Best = 1.735070e+03 FES = 207100 Diversity = 3.6671e+00 k_t
= [1, 1, 1, 1]
Generation 2080: Best = 1.735064e+03 FES = 208100 Diversity = 3.5313e+00 k_t
= [3, 2, 2, 2]
Generation 2090: Best = 1.735060e+03 FES = 209100 Diversity = 1.5282e+00 k_t
= [1, 1, 1, 1]
Generation 2100: Best = 1.735052e+03 FES = 210100 Diversity = 9.4863e-01 k_t
= [4, 3, 2, 4]
Generation 2110: Best = 1.735041e+03 FES = 211100 Diversity = 1.6591e-02 k_t

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= [1, 1, 1, 1]
Generation 2120: Best = 1.735030e+03 FES = 212100 Diversity = 1.5576e-02 k_t
= [4, 3, 2, 3]
Generation 2130: Best = 1.735025e+03 FES = 213100 Diversity = 1.2320e-02 k_t
= [1, 1, 1, 1]
Generation 2140: Best = 1.735021e+03 FES = 214100 Diversity = 9.7564e-03 k_t
= [3, 4, 2, 4]
Generation 2150: Best = 1.735019e+03 FES = 215100 Diversity = 7.3335e-03 k_t
= [1, 1, 1, 1]
Generation 2160: Best = 1.735018e+03 FES = 216100 Diversity = 5.2521e-03 k_t
= [3, 3, 3, 2]
Generation 2170: Best = 1.735018e+03 FES = 217100 Diversity = 3.8819e-03 k_t
= [1, 1, 1, 1]
Generation 2180: Best = 1.735017e+03 FES = 218100 Diversity = 2.9166e-03 k_t
= [2, 4, 2, 3]
Generation 2190: Best = 1.735017e+03 FES = 219100 Diversity = 2.2005e-03 k_t
= [1, 1, 1, 1]
Generation 2200: Best = 1.735017e+03 FES = 220100 Diversity = 1.6900e-03 k_t
= [2, 3, 3, 2]
Generation 2210: Best = 1.735017e+03 FES = 221100 Diversity = 1.2596e-03 k_t
= [1, 1, 1, 1]
Generation 2220: Best = 1.735017e+03 FES = 222100 Diversity = 9.1820e-04 k_t
= [2, 2, 4, 4]
Generation 2230: Best = 1.735017e+03 FES = 223100 Diversity = 6.7022e-04 k_t
= [1, 1, 1, 1]
Generation 2240: Best = 1.735017e+03 FES = 224100 Diversity = 4.8632e-04 k_t
= [3, 3, 4, 4]
Generation 2250: Best = 1.735017e+03 FES = 225100 Diversity = 3.4954e-04 k_t
= [1, 1, 1, 1]
Generation 2260: Best = 1.735017e+03 FES = 226100 Diversity = 2.6053e-04 k_t
= [4, 2, 4, 2]
Generation 2270: Best = 1.735017e+03 FES = 227100 Diversity = 1.8767e-04 k_t
= [1, 1, 1, 1]
Generation 2280: Best = 1.735017e+03 FES = 228100 Diversity = 1.2728e-04 k_t
= [3, 3, 4, 2]
Generation 2290: Best = 1.735017e+03 FES = 229100 Diversity = 1.0134e-04 k_t
= [1, 1, 1, 1]
Generation 2300: Best = 1.735017e+03 FES = 230100 Diversity = 7.6784e-05 k_t
= [4, 4, 2, 4]
Generation 2310: Best = 1.735017e+03 FES = 231100 Diversity = 5.6499e-05 k_t
= [1, 1, 1, 1]
Generation 2320: Best = 1.735017e+03 FES = 232100 Diversity = 4.2080e-05 k_t
= [2, 4, 3, 3]
Generation 2330: Best = 1.735017e+03 FES = 233100 Diversity = 2.9837e-05 k_t
= [1, 1, 1, 1]
Generation 2340: Best = 1.735017e+03 FES = 234100 Diversity = 2.3345e-05 k_t
= [2, 3, 4, 3]
Generation 2350: Best = 1.735017e+03 FES = 235100 Diversity = 1.9359e-05 k_t
= [1, 1, 1, 1]
Generation 2360: Best = 1.735017e+03 FES = 236100 Diversity = 1.4944e-05 k_t
= [2, 4, 2, 3]
Generation 2370: Best = 1.735017e+03 FES = 237100 Diversity = 1.1673e-05 k_t
= [1, 1, 1, 1]
Generation 2380: Best = 1.735017e+03 FES = 238100 Diversity = 9.4951e-06 k_t

```

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= [4, 4, 2, 3]
Generation 2390: Best = 1.735017e+03 FES = 239100 Diversity = 7.8663e-06 k_t
= [1, 1, 1, 1]
Generation 2400: Best = 1.735017e+03 FES = 240100 Diversity = 6.5561e-06 k_t
= [2, 2, 3, 3]
Generation 2410: Best = 1.735017e+03 FES = 241100 Diversity = 5.5191e-06 k_t
= [1, 1, 1, 1]
Generation 2420: Best = 1.735017e+03 FES = 242100 Diversity = 4.3551e-06 k_t
= [2, 3, 4, 2]
Generation 2430: Best = 1.735017e+03 FES = 243100 Diversity = 4.1615e-06 k_t
= [1, 1, 1, 1]
Generation 2440: Best = 1.735017e+03 FES = 244100 Diversity = 3.6605e-06 k_t
= [2, 3, 4, 3]
Generation 2450: Best = 1.735017e+03 FES = 245100 Diversity = 3.1381e-06 k_t
= [1, 1, 1, 1]
Generation 2460: Best = 1.735017e+03 FES = 246100 Diversity = 2.9368e-06 k_t
= [2, 4, 4, 3]
Generation 2470: Best = 1.735017e+03 FES = 247100 Diversity = 2.5562e-06 k_t
= [1, 1, 1, 1]
Generation 2480: Best = 1.735017e+03 FES = 248100 Diversity = 2.5229e-06 k_t
= [3, 2, 4, 3]
Generation 2490: Best = 1.735017e+03 FES = 249100 Diversity = 2.5570e-06 k_t
= [1, 1, 1, 1]
Generation 2500: Best = 1.735017e+03 FES = 250100 Diversity = 2.4678e-06 k_t
= [4, 3, 4, 3]
Generation 2510: Best = 1.735017e+03 FES = 251100 Diversity = 2.6201e-06 k_t
= [1, 1, 1, 1]
Generation 2520: Best = 1.735017e+03 FES = 252100 Diversity = 2.5319e-06 k_t
= [2, 4, 4, 3]
Generation 2530: Best = 1.735017e+03 FES = 253100 Diversity = 3.9294e-06 k_t
= [1, 1, 1, 1]
Generation 2540: Best = 1.735017e+03 FES = 254100 Diversity = 5.3825e-06 k_t
= [2, 3, 4, 3]
Generation 2550: Best = 1.735017e+03 FES = 255100 Diversity = 9.9082e-06 k_t
= [1, 1, 1, 1]
Generation 2560: Best = 1.735017e+03 FES = 256100 Diversity = 2.2282e-05 k_t
= [4, 3, 2, 4]
Generation 2570: Best = 1.735017e+03 FES = 257100 Diversity = 4.5474e-05 k_t
= [1, 1, 1, 1]
Generation 2580: Best = 1.735017e+03 FES = 258100 Diversity = 8.0092e-05 k_t
= [4, 4, 4, 3]
Generation 2590: Best = 1.735017e+03 FES = 259100 Diversity = 9.8008e-05 k_t
= [1, 1, 1, 1]
Generation 2600: Best = 1.735017e+03 FES = 260100 Diversity = 1.1469e-04 k_t
= [4, 2, 2, 2]
Generation 2610: Best = 1.735017e+03 FES = 261100 Diversity = 1.7867e-04 k_t
= [1, 1, 1, 1]
Generation 2620: Best = 1.735017e+03 FES = 262100 Diversity = 2.1548e-04 k_t
= [2, 3, 3, 4]
Generation 2630: Best = 1.735017e+03 FES = 263100 Diversity = 3.2244e-04 k_t
= [1, 1, 1, 1]
Generation 2640: Best = 1.735017e+03 FES = 264100 Diversity = 5.6058e-04 k_t
= [2, 2, 3, 3]
Generation 2650: Best = 1.735016e+03 FES = 265100 Diversity = 7.7686e-04 k_t

```

```

= [1, 1, 1, 1]
Generation 2660: Best = 1.735016e+03 FES = 266100 Diversity = 1.3785e-03 k_t
= [4, 2, 3, 3]
Generation 2670: Best = 1.735014e+03 FES = 267100 Diversity = 2.2868e-03 k_t
= [1, 1, 1, 1]
Generation 2680: Best = 1.735005e+03 FES = 268100 Diversity = 4.4191e-03 k_t
= [4, 2, 3, 3]
Generation 2690: Best = 1.734978e+03 FES = 269100 Diversity = 9.5173e-03 k_t
= [1, 1, 1, 1]
Generation 2700: Best = 1.734969e+03 FES = 270100 Diversity = 1.0328e-02 k_t
= [4, 3, 2, 4]
Generation 2710: Best = 1.734967e+03 FES = 271100 Diversity = 6.3341e-03 k_t
= [1, 1, 1, 1]
Generation 2720: Best = 1.734966e+03 FES = 272100 Diversity = 3.7218e-03 k_t
= [4, 3, 3, 2]
Generation 2730: Best = 1.734966e+03 FES = 273100 Diversity = 2.3185e-03 k_t
= [1, 1, 1, 1]
Generation 2740: Best = 1.734966e+03 FES = 274100 Diversity = 1.4742e-03 k_t
= [2, 4, 2, 4]
Generation 2750: Best = 1.734966e+03 FES = 275100 Diversity = 8.8494e-04 k_t
= [1, 1, 1, 1]
Generation 2760: Best = 1.734966e+03 FES = 276100 Diversity = 5.4895e-04 k_t
= [4, 2, 4, 4]
Generation 2770: Best = 1.734966e+03 FES = 277100 Diversity = 3.4192e-04 k_t
= [1, 1, 1, 1]
Generation 2780: Best = 1.734966e+03 FES = 278100 Diversity = 2.2245e-04 k_t
= [3, 3, 3, 2]
Generation 2790: Best = 1.734966e+03 FES = 279100 Diversity = 1.5148e-04 k_t
= [1, 1, 1, 1]
Generation 2800: Best = 1.734966e+03 FES = 280100 Diversity = 1.0018e-04 k_t
= [3, 3, 4, 3]
Generation 2810: Best = 1.734966e+03 FES = 281100 Diversity = 7.2549e-05 k_t
= [1, 1, 1, 1]
Generation 2820: Best = 1.734966e+03 FES = 282100 Diversity = 5.2094e-05 k_t
= [4, 4, 2, 2]
Generation 2830: Best = 1.734966e+03 FES = 283100 Diversity = 3.6307e-05 k_t
= [1, 1, 1, 1]
Generation 2840: Best = 1.734966e+03 FES = 284100 Diversity = 2.8336e-05 k_t
= [4, 3, 3, 3]
Generation 2850: Best = 1.734966e+03 FES = 285100 Diversity = 2.2603e-05 k_t
= [1, 1, 1, 1]
Generation 2860: Best = 1.734966e+03 FES = 286100 Diversity = 1.9232e-05 k_t
= [4, 3, 2, 4]
Generation 2870: Best = 1.734966e+03 FES = 287100 Diversity = 1.5173e-05 k_t
= [1, 1, 1, 1]
Generation 2880: Best = 1.734966e+03 FES = 288100 Diversity = 1.3134e-05 k_t
= [3, 4, 3, 4]
Generation 2890: Best = 1.734966e+03 FES = 289100 Diversity = 1.0822e-05 k_t
= [1, 1, 1, 1]
Generation 2900: Best = 1.734966e+03 FES = 290100 Diversity = 8.4061e-06 k_t
= [4, 4, 2, 2]
Generation 2910: Best = 1.734966e+03 FES = 291100 Diversity = 7.2489e-06 k_t
= [1, 1, 1, 1]
Generation 2920: Best = 1.734966e+03 FES = 292100 Diversity = 6.3549e-06 k_t

```

```

= [3, 3, 4, 2]
Generation 2930: Best = 1.734966e+03  FES = 293100  Diversity = 5.8048e-06  k_t
= [1, 1, 1, 1]
Generation 2940: Best = 1.734966e+03  FES = 294100  Diversity = 5.2175e-06  k_t
= [3, 4, 3, 3]
Generation 2950: Best = 1.734966e+03  FES = 295100  Diversity = 4.4372e-06  k_t
= [1, 1, 1, 1]
Generation 2960: Best = 1.734966e+03  FES = 296100  Diversity = 3.6451e-06  k_t
= [2, 3, 3, 2]
Generation 2970: Best = 1.734966e+03  FES = 297100  Diversity = 3.0326e-06  k_t
= [1, 1, 1, 1]
Generation 2980: Best = 1.734966e+03  FES = 298100  Diversity = 2.7067e-06  k_t
= [2, 3, 4, 4]
Generation 2990: Best = 1.734966e+03  FES = 299100  Diversity = 2.2307e-06  k_t
= [1, 1, 1, 1]

```

Optimization completed!

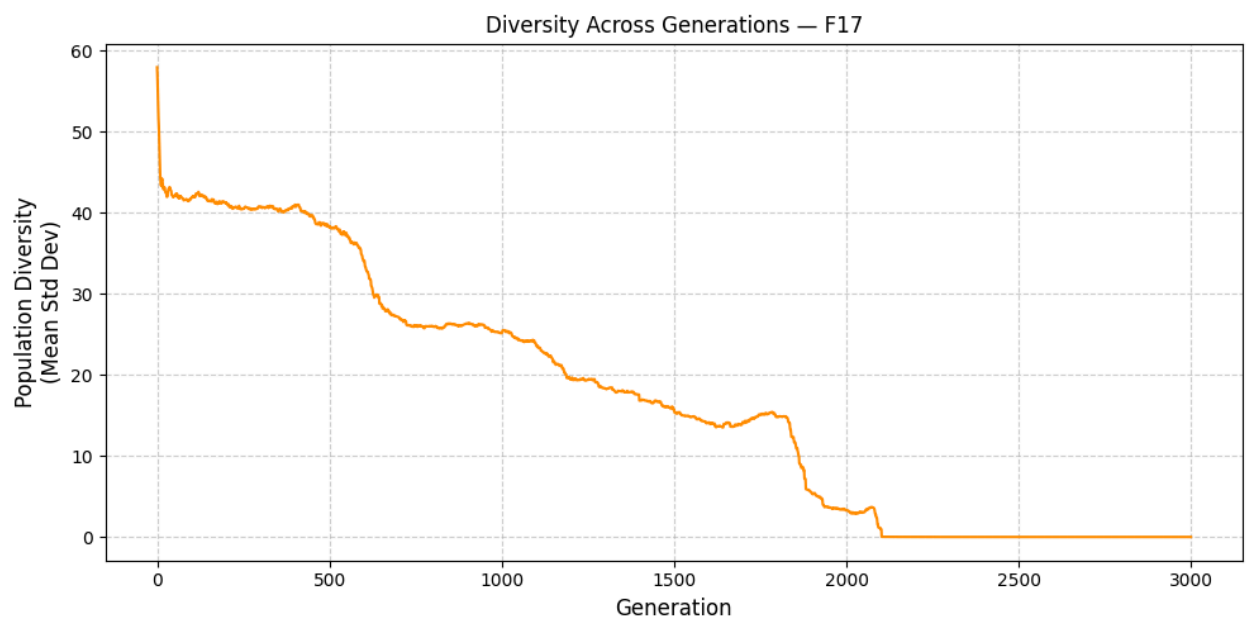
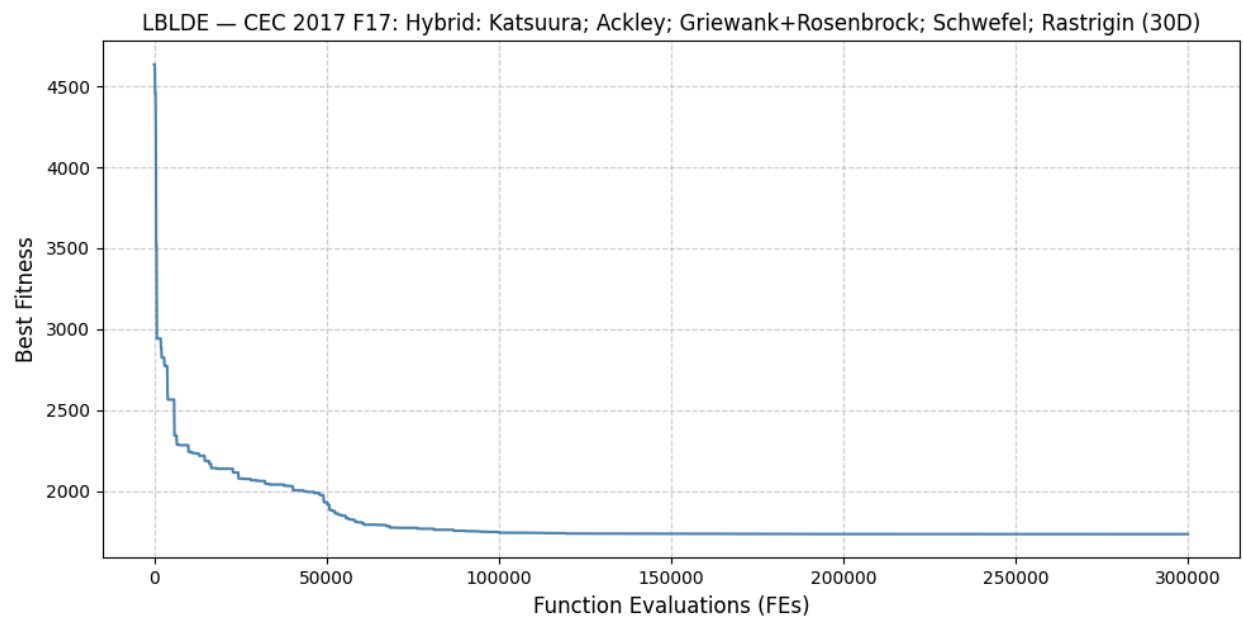
Final Best Fitness = 1.734966e+03

Total Function Evaluations = 300000

```

=====
=
Results:
  Best Fitness          : 1.734966e+03
  Global Optimum        : 1.700000e+03
  Error (|best-optimum|) : 3.496593e+01
  Best Solution (5 dim)  : [-98.87106605   9.24000945 -11.60906749  48.01603901
-45.6588402 ]
  D^(0) (init. diversity): 5.7976e+01
=====
=

```



Graph saved as convergence_F17.png

```
In [28]: # =====
# F18 – Hybrid Function 8 (Elliptic; Ackley; Rastrigin; HGBat; Discus)
# =====
run_and_plot('F18', make_wrapper(hybrid.f18), 1800.0, 'Hybrid: Elliptic; Ackle
```

```

=====
=
LBLDE_s on CEC 2017 F18 – Single Run
=====
=
Function      : Hybrid: Elliptic; Ackley; Rastrigin; HGBat; Discus (30D)
Bounds       : [-100, 100]
Global optimum : 1800.0
Initial D^(0) = 5.7976e+01
Generation 0: Best Fitness = 3.522341e+08 | Diversity = 5.7976e+01
Generation 10: Best = 2.708344e+07 FES = 1100 Diversity = 4.7693e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 1.009996e+07 FES = 2100 Diversity = 4.1803e+01 k_t
= [3, 3, 2, 2]
Generation 30: Best = 9.280974e+05 FES = 3100 Diversity = 4.1936e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 9.280974e+05 FES = 4100 Diversity = 4.0983e+01 k_t
= [2, 2, 4, 2]
Generation 50: Best = 2.106494e+05 FES = 5100 Diversity = 4.0720e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 2.106316e+05 FES = 6100 Diversity = 3.7679e+01 k_t
= [4, 2, 3, 4]
Generation 70: Best = 1.418487e+05 FES = 7100 Diversity = 3.7538e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 3.856181e+04 FES = 8100 Diversity = 3.8687e+01 k_t
= [3, 4, 3, 2]
Generation 90: Best = 3.856181e+04 FES = 9100 Diversity = 3.7212e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 2.349311e+04 FES = 10100 Diversity = 3.5624e+01 k_t
= [4, 4, 2, 4]
Generation 110: Best = 1.686900e+04 FES = 11100 Diversity = 3.0829e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 1.568478e+04 FES = 12100 Diversity = 3.0000e+01 k_t
= [4, 4, 4, 4]
Generation 130: Best = 9.632452e+03 FES = 13100 Diversity = 2.8186e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 7.657575e+03 FES = 14100 Diversity = 2.6701e+01 k_t
= [3, 3, 2, 2]
Generation 150: Best = 4.505177e+03 FES = 15100 Diversity = 2.3571e+01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 4.006116e+03 FES = 16100 Diversity = 2.3938e+01 k_t
= [2, 4, 3, 4]
Generation 170: Best = 3.506948e+03 FES = 17100 Diversity = 2.1586e+01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 2.845425e+03 FES = 18100 Diversity = 1.9596e+01 k_t
= [3, 3, 2, 2]
Generation 190: Best = 2.585946e+03 FES = 19100 Diversity = 1.8197e+01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 2.306354e+03 FES = 20100 Diversity = 1.7807e+01 k_t
= [4, 2, 2, 3]
Generation 210: Best = 2.183731e+03 FES = 21100 Diversity = 1.7093e+01 k_t
= [1, 1, 1, 1]
Generation 220: Best = 2.129854e+03 FES = 22100 Diversity = 1.6536e+01 k_t
= [4, 2, 3, 2]

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Generation 230:	Best = 2.061028e+03	FES = 23100	Diversity = 1.5252e+01	k_t = [1, 1, 1, 1]
Generation 240:	Best = 1.994555e+03	FES = 24100	Diversity = 1.4037e+01	k_t = [2, 2, 4, 2]
Generation 250:	Best = 1.981667e+03	FES = 25100	Diversity = 1.3082e+01	k_t = [1, 1, 1, 1]
Generation 260:	Best = 1.941637e+03	FES = 26100	Diversity = 1.2308e+01	k_t = [2, 3, 3, 4]
Generation 270:	Best = 1.927338e+03	FES = 27100	Diversity = 1.2147e+01	k_t = [1, 1, 1, 1]
Generation 280:	Best = 1.915313e+03	FES = 28100	Diversity = 1.1772e+01	k_t = [2, 4, 4, 2]
Generation 290:	Best = 1.910907e+03	FES = 29100	Diversity = 1.1826e+01	k_t = [1, 1, 1, 1]
Generation 300:	Best = 1.891619e+03	FES = 30100	Diversity = 1.1551e+01	k_t = [2, 3, 3, 3]
Generation 310:	Best = 1.886628e+03	FES = 31100	Diversity = 1.1812e+01	k_t = [1, 1, 1, 1]
Generation 320:	Best = 1.878715e+03	FES = 32100	Diversity = 1.1273e+01	k_t = [4, 2, 2, 2]
Generation 330:	Best = 1.878715e+03	FES = 33100	Diversity = 1.1033e+01	k_t = [1, 1, 1, 1]
Generation 340:	Best = 1.878637e+03	FES = 34100	Diversity = 1.1116e+01	k_t = [4, 3, 2, 4]
Generation 350:	Best = 1.873334e+03	FES = 35100	Diversity = 1.0637e+01	k_t = [1, 1, 1, 1]
Generation 360:	Best = 1.868804e+03	FES = 36100	Diversity = 1.0607e+01	k_t = [4, 4, 4, 3]
Generation 370:	Best = 1.867124e+03	FES = 37100	Diversity = 1.0795e+01	k_t = [1, 1, 1, 1]
Generation 380:	Best = 1.864051e+03	FES = 38100	Diversity = 1.0532e+01	k_t = [3, 4, 2, 4]
Generation 390:	Best = 1.861541e+03	FES = 39100	Diversity = 1.0640e+01	k_t = [1, 1, 1, 1]
Generation 400:	Best = 1.861541e+03	FES = 40100	Diversity = 1.1141e+01	k_t = [3, 2, 2, 4]
Generation 410:	Best = 1.859516e+03	FES = 41100	Diversity = 1.1337e+01	k_t = [1, 1, 1, 1]
Generation 420:	Best = 1.859516e+03	FES = 42100	Diversity = 1.1524e+01	k_t = [4, 3, 4, 4]
Generation 430:	Best = 1.859509e+03	FES = 43100	Diversity = 1.1522e+01	k_t = [1, 1, 1, 1]
Generation 440:	Best = 1.859509e+03	FES = 44100	Diversity = 1.1752e+01	k_t = [4, 4, 3, 3]
Generation 450:	Best = 1.851864e+03	FES = 45100	Diversity = 1.1674e+01	k_t = [1, 1, 1, 1]
Generation 460:	Best = 1.851864e+03	FES = 46100	Diversity = 1.1591e+01	k_t = [3, 3, 3, 3]
Generation 470:	Best = 1.847477e+03	FES = 47100	Diversity = 1.1742e+01	k_t = [1, 1, 1, 1]
Generation 480:	Best = 1.847291e+03	FES = 48100	Diversity = 1.1841e+01	k_t = [3, 3, 4, 4]
Generation 490:	Best = 1.845457e+03	FES = 49100	Diversity = 1.1655e+01	k_t = [1, 1, 1, 1]

Generation 500:	Best = 1.845457e+03	FES = 50100	Diversity = 1.1841e+01	k_t
= [2, 3, 2, 2]				
Generation 510:	Best = 1.844677e+03	FES = 51100	Diversity = 1.1779e+01	k_t
= [1, 1, 1, 1]				
Generation 520:	Best = 1.844497e+03	FES = 52100	Diversity = 1.1678e+01	k_t
= [2, 3, 2, 3]				
Generation 530:	Best = 1.844497e+03	FES = 53100	Diversity = 1.2073e+01	k_t
= [1, 1, 1, 1]				
Generation 540:	Best = 1.844497e+03	FES = 54100	Diversity = 1.1759e+01	k_t
= [2, 2, 3, 2]				
Generation 550:	Best = 1.844497e+03	FES = 55100	Diversity = 1.1467e+01	k_t
= [1, 1, 1, 1]				
Generation 560:	Best = 1.842898e+03	FES = 56100	Diversity = 1.1434e+01	k_t
= [3, 3, 2, 3]				
Generation 570:	Best = 1.842337e+03	FES = 57100	Diversity = 1.1646e+01	k_t
= [1, 1, 1, 1]				
Generation 580:	Best = 1.842236e+03	FES = 58100	Diversity = 1.1382e+01	k_t
= [3, 2, 3, 2]				
Generation 590:	Best = 1.842236e+03	FES = 59100	Diversity = 1.1574e+01	k_t
= [1, 1, 1, 1]				
Generation 600:	Best = 1.842236e+03	FES = 60100	Diversity = 1.1684e+01	k_t
= [3, 4, 2, 2]				
Generation 610:	Best = 1.840398e+03	FES = 61100	Diversity = 1.1868e+01	k_t
= [1, 1, 1, 1]				
Generation 620:	Best = 1.839529e+03	FES = 62100	Diversity = 1.1690e+01	k_t
= [4, 4, 4, 2]				
Generation 630:	Best = 1.839205e+03	FES = 63100	Diversity = 1.2099e+01	k_t
= [1, 1, 1, 1]				
Generation 640:	Best = 1.837906e+03	FES = 64100	Diversity = 1.1752e+01	k_t
= [2, 4, 2, 4]				
Generation 650:	Best = 1.837906e+03	FES = 65100	Diversity = 1.1570e+01	k_t
= [1, 1, 1, 1]				
Generation 660:	Best = 1.837833e+03	FES = 66100	Diversity = 1.1504e+01	k_t
= [2, 3, 4, 2]				
Generation 670:	Best = 1.837400e+03	FES = 67100	Diversity = 1.1579e+01	k_t
= [1, 1, 1, 1]				
Generation 680:	Best = 1.834955e+03	FES = 68100	Diversity = 1.1374e+01	k_t
= [3, 2, 2, 4]				
Generation 690:	Best = 1.834620e+03	FES = 69100	Diversity = 1.1290e+01	k_t
= [1, 1, 1, 1]				
Generation 700:	Best = 1.834620e+03	FES = 70100	Diversity = 1.1386e+01	k_t
= [2, 2, 3, 2]				
Generation 710:	Best = 1.834374e+03	FES = 71100	Diversity = 1.1313e+01	k_t
= [1, 1, 1, 1]				
Generation 720:	Best = 1.834374e+03	FES = 72100	Diversity = 1.1374e+01	k_t
= [2, 3, 2, 3]				
Generation 730:	Best = 1.833575e+03	FES = 73100	Diversity = 1.1670e+01	k_t
= [1, 1, 1, 1]				
Generation 740:	Best = 1.833575e+03	FES = 74100	Diversity = 1.1746e+01	k_t
= [3, 2, 3, 2]				
Generation 750:	Best = 1.832180e+03	FES = 75100	Diversity = 1.1840e+01	k_t
= [1, 1, 1, 1]				
Generation 760:	Best = 1.831771e+03	FES = 76100	Diversity = 1.1745e+01	k_t
= [2, 3, 2, 2]				

Generation 770:	Best = 1.830654e+03	FES = 77100	Diversity = 1.1463e+01	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 1.830654e+03	FES = 78100	Diversity = 1.1392e+01	k_t
= [2, 3, 3, 4]				
Generation 790:	Best = 1.830654e+03	FES = 79100	Diversity = 1.1459e+01	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 1.828907e+03	FES = 80100	Diversity = 1.0948e+01	k_t
= [4, 3, 2, 4]				
Generation 810:	Best = 1.828907e+03	FES = 81100	Diversity = 1.0850e+01	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 1.827601e+03	FES = 82100	Diversity = 1.0383e+01	k_t
= [4, 4, 3, 2]				
Generation 830:	Best = 1.827249e+03	FES = 83100	Diversity = 1.0892e+01	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 1.827125e+03	FES = 84100	Diversity = 1.0129e+01	k_t
= [2, 3, 4, 4]				
Generation 850:	Best = 1.826448e+03	FES = 85100	Diversity = 9.8439e+00	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 1.826037e+03	FES = 86100	Diversity = 9.4408e+00	k_t
= [2, 4, 2, 2]				
Generation 870:	Best = 1.825157e+03	FES = 87100	Diversity = 8.9725e+00	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 1.825080e+03	FES = 88100	Diversity = 8.3905e+00	k_t
= [3, 2, 4, 2]				
Generation 890:	Best = 1.824757e+03	FES = 89100	Diversity = 8.3397e+00	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 1.824665e+03	FES = 90100	Diversity = 7.9505e+00	k_t
= [2, 4, 4, 3]				
Generation 910:	Best = 1.824506e+03	FES = 91100	Diversity = 7.7592e+00	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 1.824327e+03	FES = 92100	Diversity = 7.5941e+00	k_t
= [4, 3, 2, 3]				
Generation 930:	Best = 1.823879e+03	FES = 93100	Diversity = 7.4963e+00	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 1.823879e+03	FES = 94100	Diversity = 7.7219e+00	k_t
= [4, 3, 3, 3]				
Generation 950:	Best = 1.823879e+03	FES = 95100	Diversity = 7.7059e+00	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 1.823879e+03	FES = 96100	Diversity = 7.5946e+00	k_t
= [2, 4, 2, 2]				
Generation 970:	Best = 1.823879e+03	FES = 97100	Diversity = 7.3463e+00	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 1.823710e+03	FES = 98100	Diversity = 7.2540e+00	k_t
= [2, 4, 4, 4]				
Generation 990:	Best = 1.823710e+03	FES = 99100	Diversity = 7.0423e+00	k_t
= [1, 1, 1, 1]				
Generation 1000:	Best = 1.823710e+03	FES = 100100	Diversity = 7.0086e+00	k_t
= [2, 4, 2, 3]				
Generation 1010:	Best = 1.823678e+03	FES = 101100	Diversity = 6.9363e+00	k_t
= [1, 1, 1, 1]				
Generation 1020:	Best = 1.823637e+03	FES = 102100	Diversity = 6.8569e+00	k_t
= [4, 3, 3, 2]				
Generation 1030:	Best = 1.823519e+03	FES = 103100	Diversity = 6.9345e+00	k_t
= [1, 1, 1, 1]				

Generation 1040: Best = 1.823519e+03 = [4, 2, 4, 4]	FES = 104100	Diversity = 6.9679e+00	k_t
Generation 1050: Best = 1.823434e+03 = [1, 1, 1, 1]	FES = 105100	Diversity = 6.9614e+00	k_t
Generation 1060: Best = 1.823434e+03 = [3, 3, 2, 3]	FES = 106100	Diversity = 6.9056e+00	k_t
Generation 1070: Best = 1.823434e+03 = [1, 1, 1, 1]	FES = 107100	Diversity = 6.9803e+00	k_t
Generation 1080: Best = 1.823434e+03 = [2, 2, 4, 4]	FES = 108100	Diversity = 7.0228e+00	k_t
Generation 1090: Best = 1.823263e+03 = [1, 1, 1, 1]	FES = 109100	Diversity = 7.1439e+00	k_t
Generation 1100: Best = 1.823263e+03 = [3, 2, 3, 3]	FES = 110100	Diversity = 6.9950e+00	k_t
Generation 1110: Best = 1.823263e+03 = [1, 1, 1, 1]	FES = 111100	Diversity = 7.0375e+00	k_t
Generation 1120: Best = 1.823263e+03 = [4, 3, 3, 3]	FES = 112100	Diversity = 7.0816e+00	k_t
Generation 1130: Best = 1.823263e+03 = [1, 1, 1, 1]	FES = 113100	Diversity = 7.1512e+00	k_t
Generation 1140: Best = 1.823174e+03 = [4, 3, 3, 2]	FES = 114100	Diversity = 7.2259e+00	k_t
Generation 1150: Best = 1.823174e+03 = [1, 1, 1, 1]	FES = 115100	Diversity = 7.1922e+00	k_t
Generation 1160: Best = 1.823174e+03 = [3, 2, 2, 3]	FES = 116100	Diversity = 7.2212e+00	k_t
Generation 1170: Best = 1.823141e+03 = [1, 1, 1, 1]	FES = 117100	Diversity = 7.2448e+00	k_t
Generation 1180: Best = 1.823133e+03 = [4, 4, 3, 2]	FES = 118100	Diversity = 7.2788e+00	k_t
Generation 1190: Best = 1.823133e+03 = [1, 1, 1, 1]	FES = 119100	Diversity = 7.2608e+00	k_t
Generation 1200: Best = 1.823133e+03 = [4, 4, 3, 4]	FES = 120100	Diversity = 7.2552e+00	k_t
Generation 1210: Best = 1.823133e+03 = [1, 1, 1, 1]	FES = 121100	Diversity = 7.2286e+00	k_t
Generation 1220: Best = 1.823133e+03 = [4, 4, 4, 3]	FES = 122100	Diversity = 7.1229e+00	k_t
Generation 1230: Best = 1.823133e+03 = [1, 1, 1, 1]	FES = 123100	Diversity = 7.1186e+00	k_t
Generation 1240: Best = 1.823133e+03 = [2, 2, 4, 2]	FES = 124100	Diversity = 7.1348e+00	k_t
Generation 1250: Best = 1.823132e+03 = [1, 1, 1, 1]	FES = 125100	Diversity = 7.1843e+00	k_t
Generation 1260: Best = 1.823132e+03 = [4, 2, 2, 2]	FES = 126100	Diversity = 7.0606e+00	k_t
Generation 1270: Best = 1.823132e+03 = [1, 1, 1, 1]	FES = 127100	Diversity = 6.8830e+00	k_t
Generation 1280: Best = 1.823132e+03 = [2, 2, 2, 4]	FES = 128100	Diversity = 6.9686e+00	k_t
Generation 1290: Best = 1.823132e+03 = [1, 1, 1, 1]	FES = 129100	Diversity = 6.9439e+00	k_t
Generation 1300: Best = 1.823132e+03 = [3, 2, 4, 3]	FES = 130100	Diversity = 7.0437e+00	k_t

Generation 1310: Best = 1.823132e+03 = [1, 1, 1, 1]	FES = 131100	Diversity = 6.9319e+00	k_t
Generation 1320: Best = 1.823132e+03 = [4, 4, 4, 4]	FES = 132100	Diversity = 7.0055e+00	k_t
Generation 1330: Best = 1.823132e+03 = [1, 1, 1, 1]	FES = 133100	Diversity = 7.0140e+00	k_t
Generation 1340: Best = 1.823132e+03 = [4, 2, 3, 2]	FES = 134100	Diversity = 7.0145e+00	k_t
Generation 1350: Best = 1.823132e+03 = [1, 1, 1, 1]	FES = 135100	Diversity = 7.0809e+00	k_t
Generation 1360: Best = 1.823131e+03 = [2, 4, 4, 2]	FES = 136100	Diversity = 7.0082e+00	k_t
Generation 1370: Best = 1.823131e+03 = [1, 1, 1, 1]	FES = 137100	Diversity = 7.0166e+00	k_t
Generation 1380: Best = 1.823130e+03 = [4, 4, 2, 4]	FES = 138100	Diversity = 6.9674e+00	k_t
Generation 1390: Best = 1.823117e+03 = [1, 1, 1, 1]	FES = 139100	Diversity = 6.9681e+00	k_t
Generation 1400: Best = 1.823117e+03 = [2, 3, 3, 3]	FES = 140100	Diversity = 6.9709e+00	k_t
Generation 1410: Best = 1.823117e+03 = [1, 1, 1, 1]	FES = 141100	Diversity = 6.8562e+00	k_t
Generation 1420: Best = 1.823117e+03 = [4, 2, 2, 2]	FES = 142100	Diversity = 6.8321e+00	k_t
Generation 1430: Best = 1.823117e+03 = [1, 1, 1, 1]	FES = 143100	Diversity = 6.7622e+00	k_t
Generation 1440: Best = 1.823117e+03 = [2, 4, 3, 4]	FES = 144100	Diversity = 6.7758e+00	k_t
Generation 1450: Best = 1.823117e+03 = [1, 1, 1, 1]	FES = 145100	Diversity = 6.7639e+00	k_t
Generation 1460: Best = 1.823117e+03 = [4, 4, 3, 4]	FES = 146100	Diversity = 6.8043e+00	k_t
Generation 1470: Best = 1.823117e+03 = [1, 1, 1, 1]	FES = 147100	Diversity = 6.8518e+00	k_t
Generation 1480: Best = 1.823117e+03 = [4, 2, 2, 4]	FES = 148100	Diversity = 6.8049e+00	k_t
Generation 1490: Best = 1.823113e+03 = [1, 1, 1, 1]	FES = 149100	Diversity = 6.8228e+00	k_t
Generation 1500: Best = 1.823113e+03 = [3, 2, 4, 4]	FES = 150100	Diversity = 6.8144e+00	k_t
Generation 1510: Best = 1.823113e+03 = [1, 1, 1, 1]	FES = 151100	Diversity = 6.8953e+00	k_t
Generation 1520: Best = 1.823113e+03 = [4, 4, 2, 2]	FES = 152100	Diversity = 6.8679e+00	k_t
Generation 1530: Best = 1.823113e+03 = [1, 1, 1, 1]	FES = 153100	Diversity = 6.8285e+00	k_t
Generation 1540: Best = 1.823113e+03 = [2, 2, 3, 3]	FES = 154100	Diversity = 6.8778e+00	k_t
Generation 1550: Best = 1.823108e+03 = [1, 1, 1, 1]	FES = 155100	Diversity = 6.7757e+00	k_t
Generation 1560: Best = 1.823108e+03 = [3, 4, 4, 4]	FES = 156100	Diversity = 6.8134e+00	k_t
Generation 1570: Best = 1.823108e+03 = [1, 1, 1, 1]	FES = 157100	Diversity = 6.7656e+00	k_t

Generation 1580: Best = 1.823108e+03 = [2, 3, 3, 2]	FES = 158100	Diversity = 6.8042e+00	k_t
Generation 1590: Best = 1.823108e+03 = [1, 1, 1, 1]	FES = 159100	Diversity = 6.8225e+00	k_t
Generation 1600: Best = 1.823108e+03 = [3, 4, 4, 3]	FES = 160100	Diversity = 6.8586e+00	k_t
Generation 1610: Best = 1.823107e+03 = [1, 1, 1, 1]	FES = 161100	Diversity = 6.8834e+00	k_t
Generation 1620: Best = 1.823107e+03 = [4, 4, 4, 3]	FES = 162100	Diversity = 6.6835e+00	k_t
Generation 1630: Best = 1.823069e+03 = [1, 1, 1, 1]	FES = 163100	Diversity = 6.6452e+00	k_t
Generation 1640: Best = 1.823029e+03 = [3, 4, 2, 3]	FES = 164100	Diversity = 6.6359e+00	k_t
Generation 1650: Best = 1.823029e+03 = [1, 1, 1, 1]	FES = 165100	Diversity = 6.5182e+00	k_t
Generation 1660: Best = 1.823029e+03 = [3, 4, 2, 2]	FES = 166100	Diversity = 6.4771e+00	k_t
Generation 1670: Best = 1.823029e+03 = [1, 1, 1, 1]	FES = 167100	Diversity = 6.4650e+00	k_t
Generation 1680: Best = 1.823029e+03 = [4, 4, 3, 2]	FES = 168100	Diversity = 6.4159e+00	k_t
Generation 1690: Best = 1.823029e+03 = [1, 1, 1, 1]	FES = 169100	Diversity = 6.2827e+00	k_t
Generation 1700: Best = 1.823029e+03 = [2, 3, 3, 3]	FES = 170100	Diversity = 6.2154e+00	k_t
Generation 1710: Best = 1.822978e+03 = [1, 1, 1, 1]	FES = 171100	Diversity = 5.9508e+00	k_t
Generation 1720: Best = 1.822978e+03 = [4, 2, 3, 2]	FES = 172100	Diversity = 5.7897e+00	k_t
Generation 1730: Best = 1.822878e+03 = [1, 1, 1, 1]	FES = 173100	Diversity = 5.6083e+00	k_t
Generation 1740: Best = 1.822878e+03 = [3, 3, 3, 3]	FES = 174100	Diversity = 5.5585e+00	k_t
Generation 1750: Best = 1.822847e+03 = [1, 1, 1, 1]	FES = 175100	Diversity = 5.1202e+00	k_t
Generation 1760: Best = 1.822842e+03 = [2, 2, 4, 2]	FES = 176100	Diversity = 4.3774e+00	k_t
Generation 1770: Best = 1.822832e+03 = [1, 1, 1, 1]	FES = 177100	Diversity = 3.5927e+00	k_t
Generation 1780: Best = 1.822822e+03 = [2, 2, 3, 4]	FES = 178100	Diversity = 2.5817e+00	k_t
Generation 1790: Best = 1.822787e+03 = [1, 1, 1, 1]	FES = 179100	Diversity = 2.3375e+00	k_t
Generation 1800: Best = 1.822777e+03 = [4, 3, 2, 2]	FES = 180100	Diversity = 1.8701e+00	k_t
Generation 1810: Best = 1.822741e+03 = [1, 1, 1, 1]	FES = 181100	Diversity = 8.1033e-01	k_t
Generation 1820: Best = 1.822737e+03 = [2, 3, 3, 3]	FES = 182100	Diversity = 3.4741e-01	k_t
Generation 1830: Best = 1.822735e+03 = [1, 1, 1, 1]	FES = 183100	Diversity = 3.6117e-01	k_t
Generation 1840: Best = 1.822735e+03 = [3, 3, 2, 3]	FES = 184100	Diversity = 3.6691e-01	k_t

Generation 1850: Best = 1.822734e+03 = [1, 1, 1, 1]	FES = 185100	Diversity = 3.7082e-01	k_t
Generation 1860: Best = 1.822734e+03 = [3, 3, 2, 3]	FES = 186100	Diversity = 3.7365e-01	k_t
Generation 1870: Best = 1.822732e+03 = [1, 1, 1, 1]	FES = 187100	Diversity = 3.7660e-01	k_t
Generation 1880: Best = 1.822732e+03 = [3, 4, 4, 2]	FES = 188100	Diversity = 3.8262e-01	k_t
Generation 1890: Best = 1.822730e+03 = [1, 1, 1, 1]	FES = 189100	Diversity = 3.8692e-01	k_t
Generation 1900: Best = 1.822729e+03 = [2, 2, 3, 3]	FES = 190100	Diversity = 3.8532e-01	k_t
Generation 1910: Best = 1.822729e+03 = [1, 1, 1, 1]	FES = 191100	Diversity = 3.9131e-01	k_t
Generation 1920: Best = 1.822729e+03 = [3, 2, 4, 2]	FES = 192100	Diversity = 3.8453e-01	k_t
Generation 1930: Best = 1.822729e+03 = [1, 1, 1, 1]	FES = 193100	Diversity = 3.8290e-01	k_t
Generation 1940: Best = 1.822728e+03 = [2, 2, 4, 3]	FES = 194100	Diversity = 3.8323e-01	k_t
Generation 1950: Best = 1.822727e+03 = [1, 1, 1, 1]	FES = 195100	Diversity = 3.8434e-01	k_t
Generation 1960: Best = 1.822724e+03 = [4, 2, 2, 2]	FES = 196100	Diversity = 3.8178e-01	k_t
Generation 1970: Best = 1.822723e+03 = [1, 1, 1, 1]	FES = 197100	Diversity = 3.8689e-01	k_t
Generation 1980: Best = 1.822722e+03 = [3, 3, 4, 3]	FES = 198100	Diversity = 3.8563e-01	k_t
Generation 1990: Best = 1.822722e+03 = [1, 1, 1, 1]	FES = 199100	Diversity = 3.8106e-01	k_t
Generation 2000: Best = 1.822722e+03 = [2, 2, 2, 2]	FES = 200100	Diversity = 3.7314e-01	k_t
Generation 2010: Best = 1.822721e+03 = [1, 1, 1, 1]	FES = 201100	Diversity = 3.7272e-01	k_t
Generation 2020: Best = 1.822721e+03 = [4, 2, 3, 2]	FES = 202100	Diversity = 3.7280e-01	k_t
Generation 2030: Best = 1.822721e+03 = [1, 1, 1, 1]	FES = 203100	Diversity = 3.7322e-01	k_t
Generation 2040: Best = 1.822721e+03 = [3, 4, 2, 2]	FES = 204100	Diversity = 3.5169e-01	k_t
Generation 2050: Best = 1.822721e+03 = [1, 1, 1, 1]	FES = 205100	Diversity = 3.4714e-01	k_t
Generation 2060: Best = 1.822720e+03 = [2, 4, 2, 2]	FES = 206100	Diversity = 3.3317e-01	k_t
Generation 2070: Best = 1.822720e+03 = [1, 1, 1, 1]	FES = 207100	Diversity = 3.2624e-01	k_t
Generation 2080: Best = 1.822720e+03 = [4, 3, 4, 2]	FES = 208100	Diversity = 3.0880e-01	k_t
Generation 2090: Best = 1.822719e+03 = [1, 1, 1, 1]	FES = 209100	Diversity = 3.0626e-01	k_t
Generation 2100: Best = 1.822719e+03 = [3, 4, 2, 3]	FES = 210100	Diversity = 3.0233e-01	k_t
Generation 2110: Best = 1.822719e+03 = [1, 1, 1, 1]	FES = 211100	Diversity = 3.0492e-01	k_t

Generation 2120: Best = 1.822719e+03 = [2, 4, 4, 2]	FES = 212100	Diversity = 2.9301e-01	k_t
Generation 2130: Best = 1.822719e+03 = [1, 1, 1, 1]	FES = 213100	Diversity = 2.8421e-01	k_t
Generation 2140: Best = 1.822714e+03 = [3, 3, 2, 3]	FES = 214100	Diversity = 2.7251e-01	k_t
Generation 2150: Best = 1.822714e+03 = [1, 1, 1, 1]	FES = 215100	Diversity = 2.4712e-01	k_t
Generation 2160: Best = 1.822714e+03 = [3, 2, 4, 4]	FES = 216100	Diversity = 2.2129e-01	k_t
Generation 2170: Best = 1.822714e+03 = [1, 1, 1, 1]	FES = 217100	Diversity = 2.1019e-01	k_t
Generation 2180: Best = 1.822713e+03 = [4, 4, 2, 3]	FES = 218100	Diversity = 2.0392e-01	k_t
Generation 2190: Best = 1.822712e+03 = [1, 1, 1, 1]	FES = 219100	Diversity = 1.8998e-01	k_t
Generation 2200: Best = 1.822710e+03 = [4, 2, 3, 3]	FES = 220100	Diversity = 1.9095e-01	k_t
Generation 2210: Best = 1.822710e+03 = [1, 1, 1, 1]	FES = 221100	Diversity = 1.8288e-01	k_t
Generation 2220: Best = 1.822691e+03 = [2, 4, 2, 3]	FES = 222100	Diversity = 1.7750e-01	k_t
Generation 2230: Best = 1.822691e+03 = [1, 1, 1, 1]	FES = 223100	Diversity = 1.7290e-01	k_t
Generation 2240: Best = 1.822691e+03 = [2, 3, 4, 2]	FES = 224100	Diversity = 1.6948e-01	k_t
Generation 2250: Best = 1.822691e+03 = [1, 1, 1, 1]	FES = 225100	Diversity = 1.6241e-01	k_t
Generation 2260: Best = 1.822691e+03 = [2, 4, 2, 2]	FES = 226100	Diversity = 1.5921e-01	k_t
Generation 2270: Best = 1.822690e+03 = [1, 1, 1, 1]	FES = 227100	Diversity = 1.5506e-01	k_t
Generation 2280: Best = 1.822689e+03 = [2, 3, 3, 2]	FES = 228100	Diversity = 1.3811e-01	k_t
Generation 2290: Best = 1.822687e+03 = [1, 1, 1, 1]	FES = 229100	Diversity = 1.2536e-01	k_t
Generation 2300: Best = 1.822681e+03 = [4, 2, 4, 4]	FES = 230100	Diversity = 1.2504e-01	k_t
Generation 2310: Best = 1.822676e+03 = [1, 1, 1, 1]	FES = 231100	Diversity = 1.1874e-01	k_t
Generation 2320: Best = 1.822655e+03 = [4, 3, 4, 3]	FES = 232100	Diversity = 8.6985e-02	k_t
Generation 2330: Best = 1.822645e+03 = [1, 1, 1, 1]	FES = 233100	Diversity = 6.5979e-02	k_t
Generation 2340: Best = 1.822633e+03 = [2, 3, 2, 2]	FES = 234100	Diversity = 6.7013e-02	k_t
Generation 2350: Best = 1.822615e+03 = [1, 1, 1, 1]	FES = 235100	Diversity = 1.1380e-02	k_t
Generation 2360: Best = 1.822590e+03 = [2, 4, 3, 2]	FES = 236100	Diversity = 1.2002e-02	k_t
Generation 2370: Best = 1.822579e+03 = [1, 1, 1, 1]	FES = 237100	Diversity = 1.2693e-02	k_t
Generation 2380: Best = 1.822562e+03 = [2, 4, 3, 4]	FES = 238100	Diversity = 1.1840e-02	k_t

Generation 2390: Best = 1.822526e+03 = [1, 1, 1, 1]	FES = 239100	Diversity = 1.1784e-02	k_t
Generation 2400: Best = 1.822514e+03 = [2, 4, 3, 2]	FES = 240100	Diversity = 1.1972e-02	k_t
Generation 2410: Best = 1.822500e+03 = [1, 1, 1, 1]	FES = 241100	Diversity = 1.1062e-02	k_t
Generation 2420: Best = 1.822481e+03 = [4, 4, 3, 3]	FES = 242100	Diversity = 1.0739e-02	k_t
Generation 2430: Best = 1.822468e+03 = [1, 1, 1, 1]	FES = 243100	Diversity = 1.0096e-02	k_t
Generation 2440: Best = 1.822438e+03 = [4, 4, 4, 4]	FES = 244100	Diversity = 1.0594e-02	k_t
Generation 2450: Best = 1.822425e+03 = [1, 1, 1, 1]	FES = 245100	Diversity = 1.2924e-02	k_t
Generation 2460: Best = 1.822418e+03 = [4, 3, 3, 4]	FES = 246100	Diversity = 1.2780e-02	k_t
Generation 2470: Best = 1.822396e+03 = [1, 1, 1, 1]	FES = 247100	Diversity = 1.1800e-02	k_t
Generation 2480: Best = 1.822389e+03 = [3, 4, 2, 3]	FES = 248100	Diversity = 1.1818e-02	k_t
Generation 2490: Best = 1.822374e+03 = [1, 1, 1, 1]	FES = 249100	Diversity = 1.1074e-02	k_t
Generation 2500: Best = 1.822370e+03 = [3, 4, 2, 3]	FES = 250100	Diversity = 1.0201e-02	k_t
Generation 2510: Best = 1.822366e+03 = [1, 1, 1, 1]	FES = 251100	Diversity = 7.6411e-03	k_t
Generation 2520: Best = 1.822360e+03 = [4, 2, 2, 2]	FES = 252100	Diversity = 7.5274e-03	k_t
Generation 2530: Best = 1.822353e+03 = [1, 1, 1, 1]	FES = 253100	Diversity = 7.1048e-03	k_t
Generation 2540: Best = 1.822351e+03 = [3, 2, 4, 4]	FES = 254100	Diversity = 6.5168e-03	k_t
Generation 2550: Best = 1.822347e+03 = [1, 1, 1, 1]	FES = 255100	Diversity = 6.0274e-03	k_t
Generation 2560: Best = 1.822343e+03 = [4, 4, 4, 4]	FES = 256100	Diversity = 5.4711e-03	k_t
Generation 2570: Best = 1.822339e+03 = [1, 1, 1, 1]	FES = 257100	Diversity = 4.5248e-03	k_t
Generation 2580: Best = 1.822339e+03 = [3, 3, 4, 3]	FES = 258100	Diversity = 3.8873e-03	k_t
Generation 2590: Best = 1.822335e+03 = [1, 1, 1, 1]	FES = 259100	Diversity = 3.8370e-03	k_t
Generation 2600: Best = 1.822335e+03 = [4, 2, 3, 2]	FES = 260100	Diversity = 3.6094e-03	k_t
Generation 2610: Best = 1.822334e+03 = [1, 1, 1, 1]	FES = 261100	Diversity = 3.5945e-03	k_t
Generation 2620: Best = 1.822333e+03 = [2, 2, 3, 3]	FES = 262100	Diversity = 3.2868e-03	k_t
Generation 2630: Best = 1.822331e+03 = [1, 1, 1, 1]	FES = 263100	Diversity = 3.2082e-03	k_t
Generation 2640: Best = 1.822328e+03 = [2, 3, 3, 4]	FES = 264100	Diversity = 3.0121e-03	k_t
Generation 2650: Best = 1.822327e+03 = [1, 1, 1, 1]	FES = 265100	Diversity = 3.4692e-03	k_t

Generation 2660: Best = 1.822327e+03 = [3, 2, 3, 3]	FES = 266100	Diversity = 3.0175e-03	k_t
Generation 2670: Best = 1.822327e+03 = [1, 1, 1, 1]	FES = 267100	Diversity = 2.7181e-03	k_t
Generation 2680: Best = 1.822326e+03 = [3, 4, 4, 3]	FES = 268100	Diversity = 2.4651e-03	k_t
Generation 2690: Best = 1.822326e+03 = [1, 1, 1, 1]	FES = 269100	Diversity = 2.1286e-03	k_t
Generation 2700: Best = 1.822325e+03 = [4, 2, 3, 3]	FES = 270100	Diversity = 1.8863e-03	k_t
Generation 2710: Best = 1.822325e+03 = [1, 1, 1, 1]	FES = 271100	Diversity = 1.7534e-03	k_t
Generation 2720: Best = 1.822325e+03 = [2, 3, 2, 2]	FES = 272100	Diversity = 1.5267e-03	k_t
Generation 2730: Best = 1.822325e+03 = [1, 1, 1, 1]	FES = 273100	Diversity = 1.3912e-03	k_t
Generation 2740: Best = 1.822324e+03 = [4, 2, 4, 4]	FES = 274100	Diversity = 1.3346e-03	k_t
Generation 2750: Best = 1.822324e+03 = [1, 1, 1, 1]	FES = 275100	Diversity = 1.2324e-03	k_t
Generation 2760: Best = 1.822324e+03 = [3, 3, 3, 3]	FES = 276100	Diversity = 1.1481e-03	k_t
Generation 2770: Best = 1.822324e+03 = [1, 1, 1, 1]	FES = 277100	Diversity = 1.0925e-03	k_t
Generation 2780: Best = 1.822324e+03 = [4, 3, 2, 4]	FES = 278100	Diversity = 1.0546e-03	k_t
Generation 2790: Best = 1.822324e+03 = [1, 1, 1, 1]	FES = 279100	Diversity = 9.8573e-04	k_t
Generation 2800: Best = 1.822324e+03 = [4, 2, 3, 3]	FES = 280100	Diversity = 9.6816e-04	k_t
Generation 2810: Best = 1.822324e+03 = [1, 1, 1, 1]	FES = 281100	Diversity = 9.3883e-04	k_t
Generation 2820: Best = 1.822324e+03 = [4, 2, 3, 3]	FES = 282100	Diversity = 9.1040e-04	k_t
Generation 2830: Best = 1.822324e+03 = [1, 1, 1, 1]	FES = 283100	Diversity = 8.7227e-04	k_t
Generation 2840: Best = 1.822324e+03 = [4, 3, 2, 3]	FES = 284100	Diversity = 8.1832e-04	k_t
Generation 2850: Best = 1.822324e+03 = [1, 1, 1, 1]	FES = 285100	Diversity = 7.8288e-04	k_t
Generation 2860: Best = 1.822324e+03 = [3, 4, 2, 3]	FES = 286100	Diversity = 7.4828e-04	k_t
Generation 2870: Best = 1.822324e+03 = [1, 1, 1, 1]	FES = 287100	Diversity = 7.3963e-04	k_t
Generation 2880: Best = 1.822324e+03 = [3, 4, 4, 2]	FES = 288100	Diversity = 7.2723e-04	k_t
Generation 2890: Best = 1.822324e+03 = [1, 1, 1, 1]	FES = 289100	Diversity = 7.2059e-04	k_t
Generation 2900: Best = 1.822324e+03 = [2, 2, 4, 2]	FES = 290100	Diversity = 7.3149e-04	k_t
Generation 2910: Best = 1.822324e+03 = [1, 1, 1, 1]	FES = 291100	Diversity = 7.4734e-04	k_t
Generation 2920: Best = 1.822324e+03 = [3, 4, 4, 4]	FES = 292100	Diversity = 7.5428e-04	k_t

Generation 2930: Best = 1.822324e+03 FES = 293100 Diversity = 7.8515e-04 k_t
 = [1, 1, 1, 1]
 Generation 2940: Best = 1.822324e+03 FES = 294100 Diversity = 7.5978e-04 k_t
 = [4, 4, 2, 3]
 Generation 2950: Best = 1.822324e+03 FES = 295100 Diversity = 7.6791e-04 k_t
 = [1, 1, 1, 1]
 Generation 2960: Best = 1.822324e+03 FES = 296100 Diversity = 7.6329e-04 k_t
 = [2, 2, 4, 4]
 Generation 2970: Best = 1.822324e+03 FES = 297100 Diversity = 7.9841e-04 k_t
 = [1, 1, 1, 1]
 Generation 2980: Best = 1.822324e+03 FES = 298100 Diversity = 7.9193e-04 k_t
 = [3, 4, 2, 3]
 Generation 2990: Best = 1.822324e+03 FES = 299100 Diversity = 7.7707e-04 k_t
 = [1, 1, 1, 1]

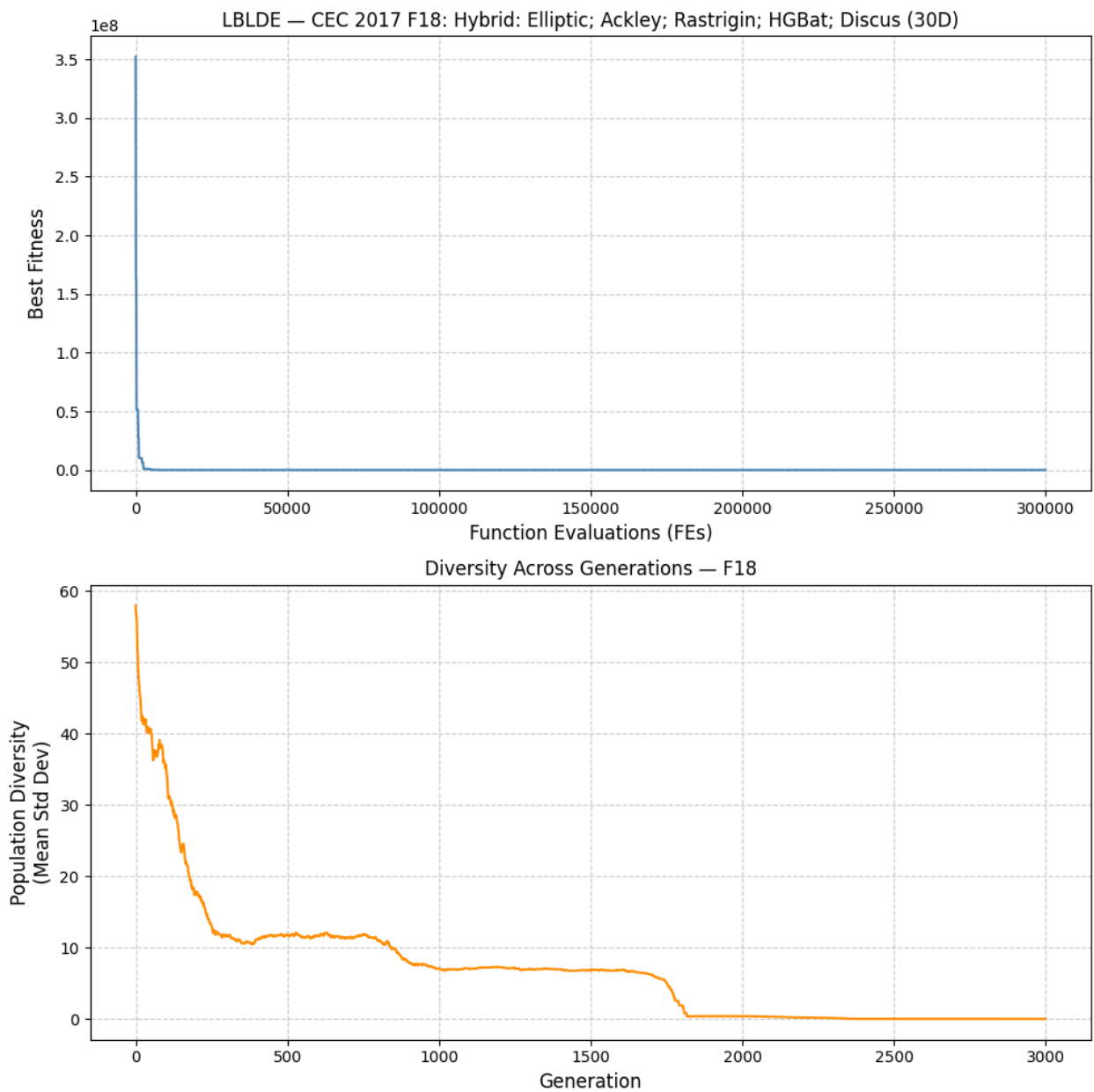
Optimization completed!

Final Best Fitness = 1.822323e+03

Total Function Evaluations = 300000

```

=====
=
Results:
  Best Fitness           : 1.822323e+03
  Global Optimum         : 1.800000e+03
  Error (|best-optimum|) : 2.232350e+01
  Best Solution (5 dim)  : [ 24.06443472  21.93098334  18.08591441 -47.8368534
7.96582126]
  D^(0) (init. diversity): 5.7976e+01
=====
=
  
```



Graph saved as convergence_F18.png

```
In [29]: # =====
# F19 – Hybrid Function 9 (Bent Cigar; Rastrigin; Griewank+Rosenbrock; Weierst
# =====
run_and_plot('F19', make_wrapper(hybrid.f19), 1900.0, 'Hybrid: Bent Cigar; Ras
```

```

=====
=
LBLDE_s on CEC 2017 F19 – Single Run
=====
=
Function      : Hybrid: Bent Cigar; Rastrigin; Griewank+Rosenbrock; Weierstras
s; Schaffer (30D)
Bounds       : [-100, 100]
Global optimum : 1900.0
Initial D^(0) = 5.7976e+01
Generation 0: Best Fitness = 7.567444e+10 | Diversity = 5.7976e+01
Generation 10: Best = 3.479935e+09 FES = 1100 Diversity = 4.4085e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 2.811660e+08 FES = 2100 Diversity = 4.0229e+01 k_t
= [3, 3, 2, 2]
Generation 30: Best = 3.013278e+07 FES = 3100 Diversity = 3.9336e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 5.984378e+06 FES = 4100 Diversity = 3.8939e+01 k_t
= [2, 3, 2, 3]
Generation 50: Best = 3.301444e+06 FES = 5100 Diversity = 3.8376e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 1.142358e+06 FES = 6100 Diversity = 3.5248e+01 k_t
= [2, 2, 2, 3]
Generation 70: Best = 4.162695e+05 FES = 7100 Diversity = 3.6207e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 2.954830e+05 FES = 8100 Diversity = 3.7640e+01 k_t
= [2, 4, 3, 2]
Generation 90: Best = 1.389420e+05 FES = 9100 Diversity = 3.6115e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 5.891997e+04 FES = 10100 Diversity = 3.5760e+01 k_t
= [2, 4, 2, 4]
Generation 110: Best = 4.309837e+04 FES = 11100 Diversity = 3.5444e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 1.325020e+04 FES = 12100 Diversity = 3.6462e+01 k_t
= [2, 4, 4, 2]
Generation 130: Best = 1.199001e+04 FES = 13100 Diversity = 3.5466e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 7.322042e+03 FES = 14100 Diversity = 3.7099e+01 k_t
= [2, 4, 2, 3]
Generation 150: Best = 3.210967e+03 FES = 15100 Diversity = 3.4260e+01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 3.210967e+03 FES = 16100 Diversity = 3.2614e+01 k_t
= [4, 4, 4, 2]
Generation 170: Best = 2.555973e+03 FES = 17100 Diversity = 3.2185e+01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 2.251638e+03 FES = 18100 Diversity = 3.0805e+01 k_t
= [4, 2, 3, 4]
Generation 190: Best = 2.152459e+03 FES = 19100 Diversity = 3.0127e+01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 2.094178e+03 FES = 20100 Diversity = 2.8902e+01 k_t
= [3, 3, 3, 3]
Generation 210: Best = 2.064349e+03 FES = 21100 Diversity = 2.8517e+01 k_t
= [1, 1, 1, 1]
Generation 220: Best = 2.029361e+03 FES = 22100 Diversity = 2.7574e+01 k_t

```

```

= [4, 3, 3, 3]
Generation 230: Best = 2.002752e+03 FES = 23100 Diversity = 2.7648e+01 k_t
= [1, 1, 1, 1]
Generation 240: Best = 1.997210e+03 FES = 24100 Diversity = 2.6955e+01 k_t
= [4, 2, 4, 2]
Generation 250: Best = 1.968151e+03 FES = 25100 Diversity = 2.6531e+01 k_t
= [1, 1, 1, 1]
Generation 260: Best = 1.968151e+03 FES = 26100 Diversity = 2.7154e+01 k_t
= [4, 3, 2, 4]
Generation 270: Best = 1.964198e+03 FES = 27100 Diversity = 2.6856e+01 k_t
= [1, 1, 1, 1]
Generation 280: Best = 1.961489e+03 FES = 28100 Diversity = 2.6726e+01 k_t
= [2, 3, 3, 4]
Generation 290: Best = 1.957324e+03 FES = 29100 Diversity = 2.6542e+01 k_t
= [1, 1, 1, 1]
Generation 300: Best = 1.948206e+03 FES = 30100 Diversity = 2.6283e+01 k_t
= [4, 2, 4, 4]
Generation 310: Best = 1.948012e+03 FES = 31100 Diversity = 2.6067e+01 k_t
= [1, 1, 1, 1]
Generation 320: Best = 1.946609e+03 FES = 32100 Diversity = 2.6261e+01 k_t
= [3, 4, 2, 2]
Generation 330: Best = 1.945761e+03 FES = 33100 Diversity = 2.6135e+01 k_t
= [1, 1, 1, 1]
Generation 340: Best = 1.945761e+03 FES = 34100 Diversity = 2.6151e+01 k_t
= [2, 3, 2, 3]
Generation 350: Best = 1.945761e+03 FES = 35100 Diversity = 2.5940e+01 k_t
= [1, 1, 1, 1]
Generation 360: Best = 1.944224e+03 FES = 36100 Diversity = 2.5999e+01 k_t
= [3, 2, 3, 2]
Generation 370: Best = 1.944224e+03 FES = 37100 Diversity = 2.5683e+01 k_t
= [1, 1, 1, 1]
Generation 380: Best = 1.944219e+03 FES = 38100 Diversity = 2.5795e+01 k_t
= [3, 2, 3, 4]
Generation 390: Best = 1.944219e+03 FES = 39100 Diversity = 2.6382e+01 k_t
= [1, 1, 1, 1]
Generation 400: Best = 1.944219e+03 FES = 40100 Diversity = 2.6695e+01 k_t
= [3, 4, 2, 2]
Generation 410: Best = 1.940703e+03 FES = 41100 Diversity = 2.6708e+01 k_t
= [1, 1, 1, 1]
Generation 420: Best = 1.940601e+03 FES = 42100 Diversity = 2.7109e+01 k_t
= [3, 2, 3, 4]
Generation 430: Best = 1.936268e+03 FES = 43100 Diversity = 2.6965e+01 k_t
= [1, 1, 1, 1]
Generation 440: Best = 1.936268e+03 FES = 44100 Diversity = 2.6892e+01 k_t
= [4, 4, 3, 2]
Generation 450: Best = 1.936268e+03 FES = 45100 Diversity = 2.6725e+01 k_t
= [1, 1, 1, 1]
Generation 460: Best = 1.936268e+03 FES = 46100 Diversity = 2.6585e+01 k_t
= [2, 4, 3, 4]
Generation 470: Best = 1.933524e+03 FES = 47100 Diversity = 2.6930e+01 k_t
= [1, 1, 1, 1]
Generation 480: Best = 1.933524e+03 FES = 48100 Diversity = 2.7116e+01 k_t
= [3, 4, 4, 3]
Generation 490: Best = 1.933524e+03 FES = 49100 Diversity = 2.7121e+01 k_t

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= [1, 1, 1, 1]
Generation 500: Best = 1.932552e+03 FES = 50100 Diversity = 2.6914e+01 k_t
= [2, 3, 4, 2]
Generation 510: Best = 1.932552e+03 FES = 51100 Diversity = 2.7140e+01 k_t
= [1, 1, 1, 1]
Generation 520: Best = 1.931806e+03 FES = 52100 Diversity = 2.7359e+01 k_t
= [4, 3, 3, 2]
Generation 530: Best = 1.931806e+03 FES = 53100 Diversity = 2.7362e+01 k_t
= [1, 1, 1, 1]
Generation 540: Best = 1.931806e+03 FES = 54100 Diversity = 2.7208e+01 k_t
= [4, 2, 4, 3]
Generation 550: Best = 1.931806e+03 FES = 55100 Diversity = 2.7118e+01 k_t
= [1, 1, 1, 1]
Generation 560: Best = 1.931806e+03 FES = 56100 Diversity = 2.6742e+01 k_t
= [4, 3, 2, 4]
Generation 570: Best = 1.931697e+03 FES = 57100 Diversity = 2.6746e+01 k_t
= [1, 1, 1, 1]
Generation 580: Best = 1.931191e+03 FES = 58100 Diversity = 2.6645e+01 k_t
= [4, 4, 3, 3]
Generation 590: Best = 1.931038e+03 FES = 59100 Diversity = 2.6839e+01 k_t
= [1, 1, 1, 1]
Generation 600: Best = 1.929813e+03 FES = 60100 Diversity = 2.6461e+01 k_t
= [4, 2, 3, 2]
Generation 610: Best = 1.929813e+03 FES = 61100 Diversity = 2.6536e+01 k_t
= [1, 1, 1, 1]
Generation 620: Best = 1.929813e+03 FES = 62100 Diversity = 2.6560e+01 k_t
= [2, 3, 2, 2]
Generation 630: Best = 1.929813e+03 FES = 63100 Diversity = 2.6695e+01 k_t
= [1, 1, 1, 1]
Generation 640: Best = 1.928061e+03 FES = 64100 Diversity = 2.6338e+01 k_t
= [3, 4, 2, 2]
Generation 650: Best = 1.927672e+03 FES = 65100 Diversity = 2.6317e+01 k_t
= [1, 1, 1, 1]
Generation 660: Best = 1.927657e+03 FES = 66100 Diversity = 2.6048e+01 k_t
= [2, 4, 3, 2]
Generation 670: Best = 1.927654e+03 FES = 67100 Diversity = 2.5979e+01 k_t
= [1, 1, 1, 1]
Generation 680: Best = 1.925037e+03 FES = 68100 Diversity = 2.5753e+01 k_t
= [2, 3, 4, 4]
Generation 690: Best = 1.924949e+03 FES = 69100 Diversity = 2.5433e+01 k_t
= [1, 1, 1, 1]
Generation 700: Best = 1.924949e+03 FES = 70100 Diversity = 2.5396e+01 k_t
= [2, 2, 2, 3]
Generation 710: Best = 1.924695e+03 FES = 71100 Diversity = 2.5436e+01 k_t
= [1, 1, 1, 1]
Generation 720: Best = 1.924695e+03 FES = 72100 Diversity = 2.5268e+01 k_t
= [2, 4, 4, 3]
Generation 730: Best = 1.924558e+03 FES = 73100 Diversity = 2.5236e+01 k_t
= [1, 1, 1, 1]
Generation 740: Best = 1.921754e+03 FES = 74100 Diversity = 2.5070e+01 k_t
= [2, 3, 2, 4]
Generation 750: Best = 1.921049e+03 FES = 75100 Diversity = 2.5284e+01 k_t
= [1, 1, 1, 1]
Generation 760: Best = 1.921049e+03 FES = 76100 Diversity = 2.5348e+01 k_t

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= [3, 2, 3, 3]
Generation 770: Best = 1.921049e+03 FES = 77100 Diversity = 2.5474e+01 k_t
= [1, 1, 1, 1]
Generation 780: Best = 1.921049e+03 FES = 78100 Diversity = 2.5407e+01 k_t
= [4, 4, 4, 3]
Generation 790: Best = 1.918173e+03 FES = 79100 Diversity = 2.5488e+01 k_t
= [1, 1, 1, 1]
Generation 800: Best = 1.918173e+03 FES = 80100 Diversity = 2.5388e+01 k_t
= [2, 4, 4, 2]
Generation 810: Best = 1.918173e+03 FES = 81100 Diversity = 2.5387e+01 k_t
= [1, 1, 1, 1]
Generation 820: Best = 1.918173e+03 FES = 82100 Diversity = 2.5487e+01 k_t
= [2, 3, 4, 3]
Generation 830: Best = 1.918007e+03 FES = 83100 Diversity = 2.5518e+01 k_t
= [1, 1, 1, 1]
Generation 840: Best = 1.917511e+03 FES = 84100 Diversity = 2.5367e+01 k_t
= [4, 2, 3, 3]
Generation 850: Best = 1.917511e+03 FES = 85100 Diversity = 2.5269e+01 k_t
= [1, 1, 1, 1]
Generation 860: Best = 1.917150e+03 FES = 86100 Diversity = 2.5085e+01 k_t
= [2, 2, 2, 4]
Generation 870: Best = 1.917150e+03 FES = 87100 Diversity = 2.4951e+01 k_t
= [1, 1, 1, 1]
Generation 880: Best = 1.917150e+03 FES = 88100 Diversity = 2.4930e+01 k_t
= [4, 3, 3, 2]
Generation 890: Best = 1.917150e+03 FES = 89100 Diversity = 2.4956e+01 k_t
= [1, 1, 1, 1]
Generation 900: Best = 1.917150e+03 FES = 90100 Diversity = 2.4991e+01 k_t
= [2, 4, 4, 3]
Generation 910: Best = 1.916894e+03 FES = 91100 Diversity = 2.4776e+01 k_t
= [1, 1, 1, 1]
Generation 920: Best = 1.916894e+03 FES = 92100 Diversity = 2.4748e+01 k_t
= [4, 2, 4, 3]
Generation 930: Best = 1.916639e+03 FES = 93100 Diversity = 2.4456e+01 k_t
= [1, 1, 1, 1]
Generation 940: Best = 1.916639e+03 FES = 94100 Diversity = 2.4567e+01 k_t
= [2, 4, 4, 3]
Generation 950: Best = 1.915447e+03 FES = 95100 Diversity = 2.4600e+01 k_t
= [1, 1, 1, 1]
Generation 960: Best = 1.915447e+03 FES = 96100 Diversity = 2.4393e+01 k_t
= [4, 4, 4, 2]
Generation 970: Best = 1.915447e+03 FES = 97100 Diversity = 2.4349e+01 k_t
= [1, 1, 1, 1]
Generation 980: Best = 1.915447e+03 FES = 98100 Diversity = 2.4229e+01 k_t
= [4, 2, 2, 3]
Generation 990: Best = 1.915368e+03 FES = 99100 Diversity = 2.4035e+01 k_t
= [1, 1, 1, 1]
Generation 1000: Best = 1.915234e+03 FES = 100100 Diversity = 2.4197e+01 k_t
= [3, 2, 2, 3]
Generation 1010: Best = 1.915234e+03 FES = 101100 Diversity = 2.3971e+01 k_t
= [1, 1, 1, 1]
Generation 1020: Best = 1.915122e+03 FES = 102100 Diversity = 2.3891e+01 k_t
= [3, 4, 3, 2]
Generation 1030: Best = 1.915122e+03 FES = 103100 Diversity = 2.3658e+01 k_t

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= [1, 1, 1, 1]
Generation 1040: Best = 1.915033e+03 FES = 104100 Diversity = 2.3454e+01 k_t
= [2, 2, 2, 4]
Generation 1050: Best = 1.914573e+03 FES = 105100 Diversity = 2.2999e+01 k_t
= [1, 1, 1, 1]
Generation 1060: Best = 1.914573e+03 FES = 106100 Diversity = 2.2889e+01 k_t
= [4, 4, 2, 3]
Generation 1070: Best = 1.914573e+03 FES = 107100 Diversity = 2.2615e+01 k_t
= [1, 1, 1, 1]
Generation 1080: Best = 1.914573e+03 FES = 108100 Diversity = 2.2423e+01 k_t
= [3, 3, 4, 3]
Generation 1090: Best = 1.914550e+03 FES = 109100 Diversity = 2.1466e+01 k_t
= [1, 1, 1, 1]
Generation 1100: Best = 1.914350e+03 FES = 110100 Diversity = 2.1167e+01 k_t
= [2, 3, 4, 4]
Generation 1110: Best = 1.913878e+03 FES = 111100 Diversity = 2.0964e+01 k_t
= [1, 1, 1, 1]
Generation 1120: Best = 1.913878e+03 FES = 112100 Diversity = 2.0650e+01 k_t
= [2, 2, 4, 2]
Generation 1130: Best = 1.913878e+03 FES = 113100 Diversity = 2.0363e+01 k_t
= [1, 1, 1, 1]
Generation 1140: Best = 1.912737e+03 FES = 114100 Diversity = 1.9512e+01 k_t
= [4, 2, 2, 4]
Generation 1150: Best = 1.912737e+03 FES = 115100 Diversity = 1.9365e+01 k_t
= [1, 1, 1, 1]
Generation 1160: Best = 1.912737e+03 FES = 116100 Diversity = 1.8838e+01 k_t
= [4, 4, 2, 3]
Generation 1170: Best = 1.912675e+03 FES = 117100 Diversity = 1.8381e+01 k_t
= [1, 1, 1, 1]
Generation 1180: Best = 1.912675e+03 FES = 118100 Diversity = 1.8208e+01 k_t
= [4, 3, 3, 3]
Generation 1190: Best = 1.912675e+03 FES = 119100 Diversity = 1.8176e+01 k_t
= [1, 1, 1, 1]
Generation 1200: Best = 1.912582e+03 FES = 120100 Diversity = 1.7428e+01 k_t
= [2, 2, 4, 2]
Generation 1210: Best = 1.912582e+03 FES = 121100 Diversity = 1.7335e+01 k_t
= [1, 1, 1, 1]
Generation 1220: Best = 1.912575e+03 FES = 122100 Diversity = 1.7364e+01 k_t
= [2, 2, 4, 4]
Generation 1230: Best = 1.911986e+03 FES = 123100 Diversity = 1.7091e+01 k_t
= [1, 1, 1, 1]
Generation 1240: Best = 1.911966e+03 FES = 124100 Diversity = 1.7016e+01 k_t
= [2, 3, 4, 3]
Generation 1250: Best = 1.911962e+03 FES = 125100 Diversity = 1.6453e+01 k_t
= [1, 1, 1, 1]
Generation 1260: Best = 1.911471e+03 FES = 126100 Diversity = 1.6474e+01 k_t
= [3, 3, 2, 4]
Generation 1270: Best = 1.911210e+03 FES = 127100 Diversity = 1.5728e+01 k_t
= [1, 1, 1, 1]
Generation 1280: Best = 1.911145e+03 FES = 128100 Diversity = 1.5739e+01 k_t
= [2, 3, 4, 2]
Generation 1290: Best = 1.911145e+03 FES = 129100 Diversity = 1.5386e+01 k_t
= [1, 1, 1, 1]
Generation 1300: Best = 1.911145e+03 FES = 130100 Diversity = 1.5450e+01 k_t

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= [4, 3, 4, 2]
Generation 1310: Best = 1.911145e+03  FES = 131100  Diversity = 1.5119e+01  k_t
= [1, 1, 1, 1]
Generation 1320: Best = 1.911145e+03  FES = 132100  Diversity = 1.5264e+01  k_t
= [4, 3, 3, 2]
Generation 1330: Best = 1.911145e+03  FES = 133100  Diversity = 1.4827e+01  k_t
= [1, 1, 1, 1]
Generation 1340: Best = 1.911145e+03  FES = 134100  Diversity = 1.4733e+01  k_t
= [3, 4, 2, 2]
Generation 1350: Best = 1.910763e+03  FES = 135100  Diversity = 1.4476e+01  k_t
= [1, 1, 1, 1]
Generation 1360: Best = 1.910442e+03  FES = 136100  Diversity = 1.4008e+01  k_t
= [3, 3, 3, 2]
Generation 1370: Best = 1.910442e+03  FES = 137100  Diversity = 1.4081e+01  k_t
= [1, 1, 1, 1]
Generation 1380: Best = 1.910336e+03  FES = 138100  Diversity = 1.4099e+01  k_t
= [3, 2, 4, 2]
Generation 1390: Best = 1.910336e+03  FES = 139100  Diversity = 1.4230e+01  k_t
= [1, 1, 1, 1]
Generation 1400: Best = 1.910231e+03  FES = 140100  Diversity = 1.4225e+01  k_t
= [3, 3, 2, 4]
Generation 1410: Best = 1.910212e+03  FES = 141100  Diversity = 1.3933e+01  k_t
= [1, 1, 1, 1]
Generation 1420: Best = 1.910093e+03  FES = 142100  Diversity = 1.3966e+01  k_t
= [2, 4, 4, 4]
Generation 1430: Best = 1.910093e+03  FES = 143100  Diversity = 1.3776e+01  k_t
= [1, 1, 1, 1]
Generation 1440: Best = 1.909792e+03  FES = 144100  Diversity = 1.3504e+01  k_t
= [4, 2, 3, 4]
Generation 1450: Best = 1.909792e+03  FES = 145100  Diversity = 1.3594e+01  k_t
= [1, 1, 1, 1]
Generation 1460: Best = 1.909792e+03  FES = 146100  Diversity = 1.3527e+01  k_t
= [2, 4, 4, 2]
Generation 1470: Best = 1.909783e+03  FES = 147100  Diversity = 1.3626e+01  k_t
= [1, 1, 1, 1]
Generation 1480: Best = 1.909740e+03  FES = 148100  Diversity = 1.3395e+01  k_t
= [3, 2, 2, 4]
Generation 1490: Best = 1.909740e+03  FES = 149100  Diversity = 1.3146e+01  k_t
= [1, 1, 1, 1]
Generation 1500: Best = 1.909157e+03  FES = 150100  Diversity = 1.3021e+01  k_t
= [2, 3, 2, 3]
Generation 1510: Best = 1.908989e+03  FES = 151100  Diversity = 1.2887e+01  k_t
= [1, 1, 1, 1]
Generation 1520: Best = 1.908699e+03  FES = 152100  Diversity = 1.2908e+01  k_t
= [2, 2, 3, 2]
Generation 1530: Best = 1.908699e+03  FES = 153100  Diversity = 1.2883e+01  k_t
= [1, 1, 1, 1]
Generation 1540: Best = 1.908699e+03  FES = 154100  Diversity = 1.2404e+01  k_t
= [2, 4, 4, 4]
Generation 1550: Best = 1.908699e+03  FES = 155100  Diversity = 1.2138e+01  k_t
= [1, 1, 1, 1]
Generation 1560: Best = 1.908699e+03  FES = 156100  Diversity = 1.1216e+01  k_t
= [3, 4, 3, 2]
Generation 1570: Best = 1.908699e+03  FES = 157100  Diversity = 1.0713e+01  k_t

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= [1, 1, 1, 1]
Generation 1580: Best = 1.908699e+03 FES = 158100 Diversity = 1.0697e+01 k_t
= [2, 2, 2, 3]
Generation 1590: Best = 1.908699e+03 FES = 159100 Diversity = 1.0614e+01 k_t
= [1, 1, 1, 1]
Generation 1600: Best = 1.908598e+03 FES = 160100 Diversity = 1.0182e+01 k_t
= [2, 2, 3, 4]
Generation 1610: Best = 1.908584e+03 FES = 161100 Diversity = 9.8955e+00 k_t
= [1, 1, 1, 1]
Generation 1620: Best = 1.908584e+03 FES = 162100 Diversity = 9.9602e+00 k_t
= [3, 4, 4, 4]
Generation 1630: Best = 1.908584e+03 FES = 163100 Diversity = 9.8752e+00 k_t
= [1, 1, 1, 1]
Generation 1640: Best = 1.908430e+03 FES = 164100 Diversity = 9.4759e+00 k_t
= [3, 4, 3, 3]
Generation 1650: Best = 1.908428e+03 FES = 165100 Diversity = 9.3853e+00 k_t
= [1, 1, 1, 1]
Generation 1660: Best = 1.908428e+03 FES = 166100 Diversity = 9.2744e+00 k_t
= [3, 2, 4, 2]
Generation 1670: Best = 1.908428e+03 FES = 167100 Diversity = 9.2901e+00 k_t
= [1, 1, 1, 1]
Generation 1680: Best = 1.908399e+03 FES = 168100 Diversity = 8.6658e+00 k_t
= [3, 4, 3, 4]
Generation 1690: Best = 1.908399e+03 FES = 169100 Diversity = 8.5380e+00 k_t
= [1, 1, 1, 1]
Generation 1700: Best = 1.908399e+03 FES = 170100 Diversity = 8.4045e+00 k_t
= [3, 3, 4, 4]
Generation 1710: Best = 1.908399e+03 FES = 171100 Diversity = 8.5523e+00 k_t
= [1, 1, 1, 1]
Generation 1720: Best = 1.908399e+03 FES = 172100 Diversity = 8.5283e+00 k_t
= [2, 4, 4, 3]
Generation 1730: Best = 1.908399e+03 FES = 173100 Diversity = 8.1494e+00 k_t
= [1, 1, 1, 1]
Generation 1740: Best = 1.908399e+03 FES = 174100 Diversity = 8.0939e+00 k_t
= [4, 4, 2, 3]
Generation 1750: Best = 1.908399e+03 FES = 175100 Diversity = 8.1264e+00 k_t
= [1, 1, 1, 1]
Generation 1760: Best = 1.908325e+03 FES = 176100 Diversity = 8.0292e+00 k_t
= [4, 4, 3, 3]
Generation 1770: Best = 1.908325e+03 FES = 177100 Diversity = 7.9571e+00 k_t
= [1, 1, 1, 1]
Generation 1780: Best = 1.908325e+03 FES = 178100 Diversity = 7.7620e+00 k_t
= [4, 2, 2, 3]
Generation 1790: Best = 1.908289e+03 FES = 179100 Diversity = 7.6954e+00 k_t
= [1, 1, 1, 1]
Generation 1800: Best = 1.908079e+03 FES = 180100 Diversity = 7.6204e+00 k_t
= [2, 3, 4, 2]
Generation 1810: Best = 1.908079e+03 FES = 181100 Diversity = 7.6292e+00 k_t
= [1, 1, 1, 1]
Generation 1820: Best = 1.908079e+03 FES = 182100 Diversity = 7.4858e+00 k_t
= [2, 3, 4, 4]
Generation 1830: Best = 1.908079e+03 FES = 183100 Diversity = 7.4100e+00 k_t
= [1, 1, 1, 1]
Generation 1840: Best = 1.908079e+03 FES = 184100 Diversity = 7.3760e+00 k_t

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= [2, 4, 3, 2]
Generation 1850: Best = 1.908079e+03 FES = 185100 Diversity = 7.3133e+00 k_t
= [1, 1, 1, 1]
Generation 1860: Best = 1.908079e+03 FES = 186100 Diversity = 7.1702e+00 k_t
= [3, 3, 2, 3]
Generation 1870: Best = 1.908079e+03 FES = 187100 Diversity = 6.9960e+00 k_t
= [1, 1, 1, 1]
Generation 1880: Best = 1.908003e+03 FES = 188100 Diversity = 7.0022e+00 k_t
= [2, 3, 4, 4]
Generation 1890: Best = 1.907979e+03 FES = 189100 Diversity = 6.8747e+00 k_t
= [1, 1, 1, 1]
Generation 1900: Best = 1.907947e+03 FES = 190100 Diversity = 6.8172e+00 k_t
= [3, 3, 4, 2]
Generation 1910: Best = 1.907947e+03 FES = 191100 Diversity = 6.6569e+00 k_t
= [1, 1, 1, 1]
Generation 1920: Best = 1.907947e+03 FES = 192100 Diversity = 6.6680e+00 k_t
= [2, 3, 4, 2]
Generation 1930: Best = 1.907947e+03 FES = 193100 Diversity = 6.5038e+00 k_t
= [1, 1, 1, 1]
Generation 1940: Best = 1.907846e+03 FES = 194100 Diversity = 6.4482e+00 k_t
= [2, 2, 4, 3]
Generation 1950: Best = 1.907841e+03 FES = 195100 Diversity = 6.3366e+00 k_t
= [1, 1, 1, 1]
Generation 1960: Best = 1.907841e+03 FES = 196100 Diversity = 6.2589e+00 k_t
= [2, 3, 2, 3]
Generation 1970: Best = 1.907841e+03 FES = 197100 Diversity = 6.2302e+00 k_t
= [1, 1, 1, 1]
Generation 1980: Best = 1.907828e+03 FES = 198100 Diversity = 6.2337e+00 k_t
= [3, 4, 2, 3]
Generation 1990: Best = 1.907599e+03 FES = 199100 Diversity = 6.1156e+00 k_t
= [1, 1, 1, 1]
Generation 2000: Best = 1.907599e+03 FES = 200100 Diversity = 5.8711e+00 k_t
= [2, 3, 4, 2]
Generation 2010: Best = 1.907599e+03 FES = 201100 Diversity = 5.8677e+00 k_t
= [1, 1, 1, 1]
Generation 2020: Best = 1.907576e+03 FES = 202100 Diversity = 5.9583e+00 k_t
= [2, 3, 4, 3]
Generation 2030: Best = 1.907574e+03 FES = 203100 Diversity = 5.9657e+00 k_t
= [1, 1, 1, 1]
Generation 2040: Best = 1.907567e+03 FES = 204100 Diversity = 5.7401e+00 k_t
= [4, 2, 3, 2]
Generation 2050: Best = 1.907567e+03 FES = 205100 Diversity = 5.7872e+00 k_t
= [1, 1, 1, 1]
Generation 2060: Best = 1.907552e+03 FES = 206100 Diversity = 5.6964e+00 k_t
= [3, 3, 2, 3]
Generation 2070: Best = 1.907378e+03 FES = 207100 Diversity = 5.6511e+00 k_t
= [1, 1, 1, 1]
Generation 2080: Best = 1.907378e+03 FES = 208100 Diversity = 5.6000e+00 k_t
= [2, 2, 2, 2]
Generation 2090: Best = 1.907378e+03 FES = 209100 Diversity = 5.5559e+00 k_t
= [1, 1, 1, 1]
Generation 2100: Best = 1.907377e+03 FES = 210100 Diversity = 5.4839e+00 k_t
= [3, 4, 4, 2]
Generation 2110: Best = 1.907377e+03 FES = 211100 Diversity = 5.4963e+00 k_t

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= [1, 1, 1, 1]
Generation 2120: Best = 1.907377e+03 FES = 212100 Diversity = 5.4462e+00 k_t
= [2, 2, 3, 3]
Generation 2130: Best = 1.907377e+03 FES = 213100 Diversity = 5.2915e+00 k_t
= [1, 1, 1, 1]
Generation 2140: Best = 1.907377e+03 FES = 214100 Diversity = 5.2777e+00 k_t
= [4, 2, 2, 2]
Generation 2150: Best = 1.907339e+03 FES = 215100 Diversity = 5.2439e+00 k_t
= [1, 1, 1, 1]
Generation 2160: Best = 1.907339e+03 FES = 216100 Diversity = 5.0356e+00 k_t
= [2, 2, 4, 4]
Generation 2170: Best = 1.907339e+03 FES = 217100 Diversity = 5.0728e+00 k_t
= [1, 1, 1, 1]
Generation 2180: Best = 1.907336e+03 FES = 218100 Diversity = 4.9686e+00 k_t
= [2, 4, 3, 4]
Generation 2190: Best = 1.907336e+03 FES = 219100 Diversity = 5.0383e+00 k_t
= [1, 1, 1, 1]
Generation 2200: Best = 1.907336e+03 FES = 220100 Diversity = 5.0384e+00 k_t
= [4, 2, 4, 4]
Generation 2210: Best = 1.907310e+03 FES = 221100 Diversity = 5.0166e+00 k_t
= [1, 1, 1, 1]
Generation 2220: Best = 1.907310e+03 FES = 222100 Diversity = 5.0017e+00 k_t
= [3, 3, 4, 4]
Generation 2230: Best = 1.907310e+03 FES = 223100 Diversity = 4.9551e+00 k_t
= [1, 1, 1, 1]
Generation 2240: Best = 1.907308e+03 FES = 224100 Diversity = 4.9762e+00 k_t
= [4, 2, 3, 2]
Generation 2250: Best = 1.907307e+03 FES = 225100 Diversity = 4.8718e+00 k_t
= [1, 1, 1, 1]
Generation 2260: Best = 1.907307e+03 FES = 226100 Diversity = 4.8592e+00 k_t
= [4, 4, 4, 3]
Generation 2270: Best = 1.907307e+03 FES = 227100 Diversity = 4.8581e+00 k_t
= [1, 1, 1, 1]
Generation 2280: Best = 1.907187e+03 FES = 228100 Diversity = 4.6924e+00 k_t
= [3, 2, 2, 3]
Generation 2290: Best = 1.907187e+03 FES = 229100 Diversity = 4.5103e+00 k_t
= [1, 1, 1, 1]
Generation 2300: Best = 1.907187e+03 FES = 230100 Diversity = 4.4554e+00 k_t
= [2, 4, 2, 4]
Generation 2310: Best = 1.907114e+03 FES = 231100 Diversity = 4.4893e+00 k_t
= [1, 1, 1, 1]
Generation 2320: Best = 1.907072e+03 FES = 232100 Diversity = 4.4946e+00 k_t
= [3, 3, 3, 4]
Generation 2330: Best = 1.907046e+03 FES = 233100 Diversity = 4.4855e+00 k_t
= [1, 1, 1, 1]
Generation 2340: Best = 1.907016e+03 FES = 234100 Diversity = 4.3606e+00 k_t
= [2, 4, 4, 2]
Generation 2350: Best = 1.907009e+03 FES = 235100 Diversity = 4.4190e+00 k_t
= [1, 1, 1, 1]
Generation 2360: Best = 1.907009e+03 FES = 236100 Diversity = 4.3353e+00 k_t
= [4, 3, 3, 2]
Generation 2370: Best = 1.906990e+03 FES = 237100 Diversity = 4.2054e+00 k_t
= [1, 1, 1, 1]
Generation 2380: Best = 1.906893e+03 FES = 238100 Diversity = 4.0781e+00 k_t

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= [2, 2, 3, 4]
Generation 2390: Best = 1.906893e+03 FES = 239100 Diversity = 3.8677e+00 k_t
= [1, 1, 1, 1]
Generation 2400: Best = 1.906893e+03 FES = 240100 Diversity = 3.8024e+00 k_t
= [3, 4, 2, 4]
Generation 2410: Best = 1.906862e+03 FES = 241100 Diversity = 3.6685e+00 k_t
= [1, 1, 1, 1]
Generation 2420: Best = 1.906862e+03 FES = 242100 Diversity = 3.4872e+00 k_t
= [2, 2, 3, 4]
Generation 2430: Best = 1.906833e+03 FES = 243100 Diversity = 3.3568e+00 k_t
= [1, 1, 1, 1]
Generation 2440: Best = 1.906806e+03 FES = 244100 Diversity = 3.2088e+00 k_t
= [3, 3, 3, 4]
Generation 2450: Best = 1.906806e+03 FES = 245100 Diversity = 3.1153e+00 k_t
= [1, 1, 1, 1]
Generation 2460: Best = 1.906806e+03 FES = 246100 Diversity = 3.0257e+00 k_t
= [3, 4, 3, 4]
Generation 2470: Best = 1.906806e+03 FES = 247100 Diversity = 3.0189e+00 k_t
= [1, 1, 1, 1]
Generation 2480: Best = 1.906792e+03 FES = 248100 Diversity = 2.8123e+00 k_t
= [2, 3, 4, 4]
Generation 2490: Best = 1.906712e+03 FES = 249100 Diversity = 2.8103e+00 k_t
= [1, 1, 1, 1]
Generation 2500: Best = 1.906699e+03 FES = 250100 Diversity = 2.7877e+00 k_t
= [3, 3, 4, 4]
Generation 2510: Best = 1.906699e+03 FES = 251100 Diversity = 2.7514e+00 k_t
= [1, 1, 1, 1]
Generation 2520: Best = 1.906699e+03 FES = 252100 Diversity = 2.6625e+00 k_t
= [2, 4, 3, 4]
Generation 2530: Best = 1.906676e+03 FES = 253100 Diversity = 2.4109e+00 k_t
= [1, 1, 1, 1]
Generation 2540: Best = 1.906636e+03 FES = 254100 Diversity = 2.3341e+00 k_t
= [4, 3, 3, 4]
Generation 2550: Best = 1.906609e+03 FES = 255100 Diversity = 2.3382e+00 k_t
= [1, 1, 1, 1]
Generation 2560: Best = 1.906609e+03 FES = 256100 Diversity = 2.3246e+00 k_t
= [2, 2, 2, 2]
Generation 2570: Best = 1.906570e+03 FES = 257100 Diversity = 2.1082e+00 k_t
= [1, 1, 1, 1]
Generation 2580: Best = 1.906551e+03 FES = 258100 Diversity = 2.0893e+00 k_t
= [4, 3, 3, 3]
Generation 2590: Best = 1.906551e+03 FES = 259100 Diversity = 2.0924e+00 k_t
= [1, 1, 1, 1]
Generation 2600: Best = 1.906496e+03 FES = 260100 Diversity = 2.0776e+00 k_t
= [3, 4, 3, 4]
Generation 2610: Best = 1.906487e+03 FES = 261100 Diversity = 2.0477e+00 k_t
= [1, 1, 1, 1]
Generation 2620: Best = 1.906487e+03 FES = 262100 Diversity = 2.0292e+00 k_t
= [3, 3, 4, 3]
Generation 2630: Best = 1.906464e+03 FES = 263100 Diversity = 1.9796e+00 k_t
= [1, 1, 1, 1]
Generation 2640: Best = 1.906464e+03 FES = 264100 Diversity = 1.9374e+00 k_t
= [3, 4, 3, 2]
Generation 2650: Best = 1.906464e+03 FES = 265100 Diversity = 1.9185e+00 k_t

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= [1, 1, 1, 1]
Generation 2660: Best = 1.906381e+03 FES = 266100 Diversity = 1.8732e+00 k_t
= [4, 2, 2, 4]
Generation 2670: Best = 1.906381e+03 FES = 267100 Diversity = 1.8669e+00 k_t
= [1, 1, 1, 1]
Generation 2680: Best = 1.906381e+03 FES = 268100 Diversity = 1.8436e+00 k_t
= [2, 2, 3, 3]
Generation 2690: Best = 1.906381e+03 FES = 269100 Diversity = 1.6261e+00 k_t
= [1, 1, 1, 1]
Generation 2700: Best = 1.906374e+03 FES = 270100 Diversity = 1.6176e+00 k_t
= [4, 3, 2, 4]
Generation 2710: Best = 1.906374e+03 FES = 271100 Diversity = 1.6207e+00 k_t
= [1, 1, 1, 1]
Generation 2720: Best = 1.906374e+03 FES = 272100 Diversity = 1.6518e+00 k_t
= [3, 2, 4, 3]
Generation 2730: Best = 1.906374e+03 FES = 273100 Diversity = 1.6344e+00 k_t
= [1, 1, 1, 1]
Generation 2740: Best = 1.906354e+03 FES = 274100 Diversity = 1.6091e+00 k_t
= [4, 3, 4, 4]
Generation 2750: Best = 1.906279e+03 FES = 275100 Diversity = 1.5726e+00 k_t
= [1, 1, 1, 1]
Generation 2760: Best = 1.906279e+03 FES = 276100 Diversity = 1.5521e+00 k_t
= [3, 3, 3, 3]
Generation 2770: Best = 1.906279e+03 FES = 277100 Diversity = 1.5323e+00 k_t
= [1, 1, 1, 1]
Generation 2780: Best = 1.906279e+03 FES = 278100 Diversity = 1.5245e+00 k_t
= [3, 3, 4, 2]
Generation 2790: Best = 1.906279e+03 FES = 279100 Diversity = 1.4973e+00 k_t
= [1, 1, 1, 1]
Generation 2800: Best = 1.906279e+03 FES = 280100 Diversity = 1.4768e+00 k_t
= [2, 4, 3, 3]
Generation 2810: Best = 1.906279e+03 FES = 281100 Diversity = 1.4819e+00 k_t
= [1, 1, 1, 1]
Generation 2820: Best = 1.906262e+03 FES = 282100 Diversity = 1.4709e+00 k_t
= [4, 3, 2, 2]
Generation 2830: Best = 1.906260e+03 FES = 283100 Diversity = 1.4546e+00 k_t
= [1, 1, 1, 1]
Generation 2840: Best = 1.906256e+03 FES = 284100 Diversity = 1.3582e+00 k_t
= [2, 4, 4, 4]
Generation 2850: Best = 1.906252e+03 FES = 285100 Diversity = 1.3157e+00 k_t
= [1, 1, 1, 1]
Generation 2860: Best = 1.906252e+03 FES = 286100 Diversity = 1.2967e+00 k_t
= [2, 3, 3, 4]
Generation 2870: Best = 1.906250e+03 FES = 287100 Diversity = 1.3303e+00 k_t
= [1, 1, 1, 1]
Generation 2880: Best = 1.906237e+03 FES = 288100 Diversity = 1.3183e+00 k_t
= [2, 3, 2, 2]
Generation 2890: Best = 1.906237e+03 FES = 289100 Diversity = 1.3124e+00 k_t
= [1, 1, 1, 1]
Generation 2900: Best = 1.906228e+03 FES = 290100 Diversity = 1.3039e+00 k_t
= [4, 2, 4, 4]
Generation 2910: Best = 1.906223e+03 FES = 291100 Diversity = 1.3052e+00 k_t
= [1, 1, 1, 1]
Generation 2920: Best = 1.906218e+03 FES = 292100 Diversity = 1.2726e+00 k_t

```

```

= [4, 2, 4, 3]
Generation 2930: Best = 1.906216e+03  FES = 293100  Diversity = 1.2667e+00  k_t
= [1, 1, 1, 1]
Generation 2940: Best = 1.906216e+03  FES = 294100  Diversity = 1.2663e+00  k_t
= [3, 2, 2, 3]
Generation 2950: Best = 1.906216e+03  FES = 295100  Diversity = 1.2933e+00  k_t
= [1, 1, 1, 1]
Generation 2960: Best = 1.906192e+03  FES = 296100  Diversity = 1.2810e+00  k_t
= [2, 2, 3, 3]
Generation 2970: Best = 1.906192e+03  FES = 297100  Diversity = 1.2630e+00  k_t
= [1, 1, 1, 1]
Generation 2980: Best = 1.906189e+03  FES = 298100  Diversity = 1.2278e+00  k_t
= [2, 2, 3, 3]
Generation 2990: Best = 1.906185e+03  FES = 299100  Diversity = 1.1966e+00  k_t
= [1, 1, 1, 1]

```

Optimization completed!

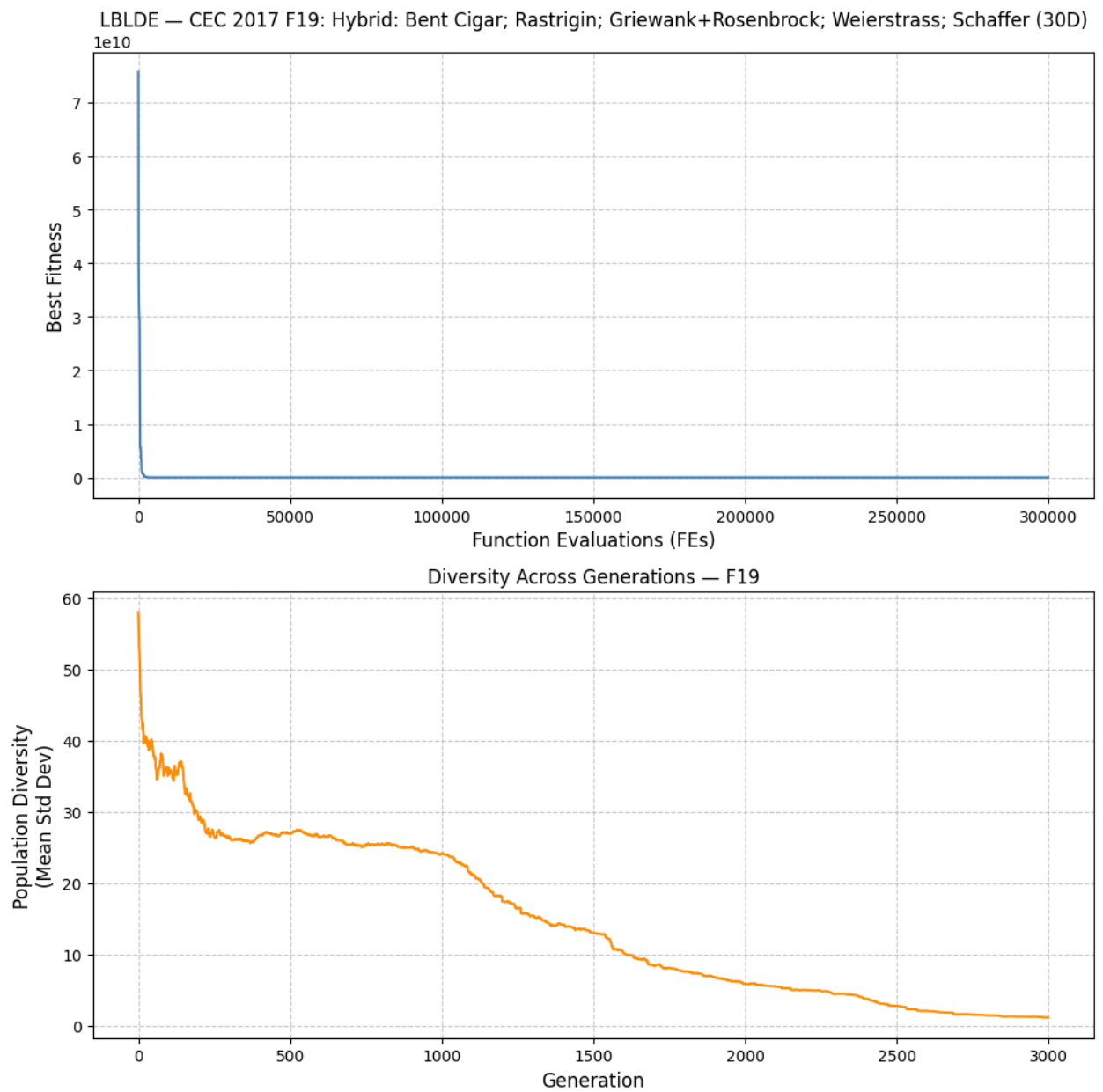
Final Best Fitness = 1.906146e+03

Total Function Evaluations = 300000

```

=====
=
Results:
  Best Fitness           : 1.906146e+03
  Global Optimum         : 1.900000e+03
  Error (|best-optimum|) : 6.146145e+00
  Best Solution (5 dim)  : [-99.90729933  13.43867334 -23.11313744  48.21705708
-46.60706434]
  D^(0) (init. diversity): 5.7976e+01
=====
=

```



Graph saved as convergence_F19.png

```
In [30]: # =====
# F20 — Hybrid Function 10 (HappyCat; Katsuura; Ackley; Rastrigin; Schwefel; S
# =====
run_and_plot('F20', make_wrapper(hybrid.f20), 2000.0, 'Hybrid: HappyCat; Katsu
```



```

=====
=
LBLDE_s on CEC 2017 F20 – Single Run
=====
=
Function      : Hybrid: HappyCat; Katsuura; Ackley; Rastrigin; Schwefel; Schaf
fer (30D)
Bounds       : [-100, 100]
Global optimum : 2000.0
Initial D^(0) = 5.7976e+01
Generation 0: Best Fitness = 3.694893e+03 | Diversity = 5.7976e+01
Generation 10: Best = 3.248700e+03 FES = 1100 Diversity = 5.5080e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 3.201627e+03 FES = 2100 Diversity = 5.4790e+01 k_t
= [3, 3, 2, 4]
Generation 30: Best = 3.080704e+03 FES = 3100 Diversity = 5.4213e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 3.013304e+03 FES = 4100 Diversity = 5.3220e+01 k_t
= [2, 2, 4, 2]
Generation 50: Best = 2.929972e+03 FES = 5100 Diversity = 5.4574e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 2.929972e+03 FES = 6100 Diversity = 5.4502e+01 k_t
= [3, 3, 4, 4]
Generation 70: Best = 2.896990e+03 FES = 7100 Diversity = 5.3906e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 2.846357e+03 FES = 8100 Diversity = 5.3920e+01 k_t
= [3, 4, 4, 3]
Generation 90: Best = 2.803119e+03 FES = 9100 Diversity = 5.4297e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 2.803119e+03 FES = 10100 Diversity = 5.4048e+01 k_t
= [4, 2, 4, 4]
Generation 110: Best = 2.467536e+03 FES = 11100 Diversity = 5.3424e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 2.463691e+03 FES = 12100 Diversity = 5.2318e+01 k_t
= [2, 2, 3, 2]
Generation 130: Best = 2.463691e+03 FES = 13100 Diversity = 5.2267e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 2.457324e+03 FES = 14100 Diversity = 5.1525e+01 k_t
= [3, 4, 4, 4]
Generation 150: Best = 2.432152e+03 FES = 15100 Diversity = 5.0895e+01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 2.432152e+03 FES = 16100 Diversity = 5.0358e+01 k_t
= [4, 2, 4, 2]
Generation 170: Best = 2.425167e+03 FES = 17100 Diversity = 4.9488e+01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 2.409341e+03 FES = 18100 Diversity = 4.8811e+01 k_t
= [3, 3, 4, 2]
Generation 190: Best = 2.409341e+03 FES = 19100 Diversity = 4.8241e+01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 2.409341e+03 FES = 20100 Diversity = 4.8068e+01 k_t
= [2, 3, 4, 4]
Generation 210: Best = 2.409341e+03 FES = 21100 Diversity = 4.7637e+01 k_t
= [1, 1, 1, 1]
Generation 220: Best = 2.409341e+03 FES = 22100 Diversity = 4.7776e+01 k_t

```

```

= [2, 4, 3, 4]
Generation 230: Best = 2.409258e+03 FES = 23100 Diversity = 4.7220e+01 k_t
= [1, 1, 1, 1]
Generation 240: Best = 2.375057e+03 FES = 24100 Diversity = 4.6431e+01 k_t
= [4, 2, 4, 4]
Generation 250: Best = 2.374889e+03 FES = 25100 Diversity = 4.5967e+01 k_t
= [1, 1, 1, 1]
Generation 260: Best = 2.370228e+03 FES = 26100 Diversity = 4.5472e+01 k_t
= [2, 2, 3, 2]
Generation 270: Best = 2.360187e+03 FES = 27100 Diversity = 4.5639e+01 k_t
= [1, 1, 1, 1]
Generation 280: Best = 2.359515e+03 FES = 28100 Diversity = 4.5278e+01 k_t
= [3, 3, 4, 4]
Generation 290: Best = 2.359515e+03 FES = 29100 Diversity = 4.5064e+01 k_t
= [1, 1, 1, 1]
Generation 300: Best = 2.335650e+03 FES = 30100 Diversity = 4.5241e+01 k_t
= [4, 4, 4, 2]
Generation 310: Best = 2.335650e+03 FES = 31100 Diversity = 4.4589e+01 k_t
= [1, 1, 1, 1]
Generation 320: Best = 2.335616e+03 FES = 32100 Diversity = 4.4166e+01 k_t
= [2, 3, 3, 2]
Generation 330: Best = 2.306121e+03 FES = 33100 Diversity = 4.3637e+01 k_t
= [1, 1, 1, 1]
Generation 340: Best = 2.305901e+03 FES = 34100 Diversity = 4.2628e+01 k_t
= [3, 2, 2, 4]
Generation 350: Best = 2.305901e+03 FES = 35100 Diversity = 4.2393e+01 k_t
= [1, 1, 1, 1]
Generation 360: Best = 2.276629e+03 FES = 36100 Diversity = 4.1888e+01 k_t
= [2, 4, 4, 2]
Generation 370: Best = 2.276629e+03 FES = 37100 Diversity = 4.1905e+01 k_t
= [1, 1, 1, 1]
Generation 380: Best = 2.264268e+03 FES = 38100 Diversity = 4.1584e+01 k_t
= [4, 2, 3, 3]
Generation 390: Best = 2.264268e+03 FES = 39100 Diversity = 4.1633e+01 k_t
= [1, 1, 1, 1]
Generation 400: Best = 2.258672e+03 FES = 40100 Diversity = 4.1435e+01 k_t
= [4, 2, 2, 2]
Generation 410: Best = 2.258672e+03 FES = 41100 Diversity = 4.1209e+01 k_t
= [1, 1, 1, 1]
Generation 420: Best = 2.258672e+03 FES = 42100 Diversity = 4.1553e+01 k_t
= [3, 2, 4, 2]
Generation 430: Best = 2.182402e+03 FES = 43100 Diversity = 4.2423e+01 k_t
= [1, 1, 1, 1]
Generation 440: Best = 2.182402e+03 FES = 44100 Diversity = 4.2341e+01 k_t
= [4, 3, 3, 3]
Generation 450: Best = 2.182402e+03 FES = 45100 Diversity = 4.2115e+01 k_t
= [1, 1, 1, 1]
Generation 460: Best = 2.182007e+03 FES = 46100 Diversity = 4.2452e+01 k_t
= [2, 4, 3, 2]
Generation 470: Best = 2.174371e+03 FES = 47100 Diversity = 4.2255e+01 k_t
= [1, 1, 1, 1]
Generation 480: Best = 2.174371e+03 FES = 48100 Diversity = 4.1667e+01 k_t
= [4, 2, 3, 3]
Generation 490: Best = 2.162655e+03 FES = 49100 Diversity = 4.0712e+01 k_t

```

```

= [1, 1, 1, 1]
Generation 500: Best = 2.162655e+03 FES = 50100 Diversity = 3.9962e+01 k_t
= [4, 4, 3, 4]
Generation 510: Best = 2.147413e+03 FES = 51100 Diversity = 3.9222e+01 k_t
= [1, 1, 1, 1]
Generation 520: Best = 2.147413e+03 FES = 52100 Diversity = 3.7527e+01 k_t
= [2, 3, 3, 3]
Generation 530: Best = 2.147413e+03 FES = 53100 Diversity = 3.7063e+01 k_t
= [1, 1, 1, 1]
Generation 540: Best = 2.141560e+03 FES = 54100 Diversity = 3.5988e+01 k_t
= [2, 4, 2, 4]
Generation 550: Best = 2.141560e+03 FES = 55100 Diversity = 3.5216e+01 k_t
= [1, 1, 1, 1]
Generation 560: Best = 2.125611e+03 FES = 56100 Diversity = 3.4132e+01 k_t
= [3, 4, 3, 2]
Generation 570: Best = 2.125004e+03 FES = 57100 Diversity = 3.3893e+01 k_t
= [1, 1, 1, 1]
Generation 580: Best = 2.120847e+03 FES = 58100 Diversity = 3.4401e+01 k_t
= [2, 3, 2, 3]
Generation 590: Best = 2.107000e+03 FES = 59100 Diversity = 3.4209e+01 k_t
= [1, 1, 1, 1]
Generation 600: Best = 2.107000e+03 FES = 60100 Diversity = 3.3327e+01 k_t
= [4, 4, 4, 3]
Generation 610: Best = 2.107000e+03 FES = 61100 Diversity = 3.3213e+01 k_t
= [1, 1, 1, 1]
Generation 620: Best = 2.096244e+03 FES = 62100 Diversity = 3.2667e+01 k_t
= [3, 4, 4, 4]
Generation 630: Best = 2.094119e+03 FES = 63100 Diversity = 3.2562e+01 k_t
= [1, 1, 1, 1]
Generation 640: Best = 2.088972e+03 FES = 64100 Diversity = 3.1966e+01 k_t
= [2, 2, 2, 3]
Generation 650: Best = 2.083876e+03 FES = 65100 Diversity = 3.1389e+01 k_t
= [1, 1, 1, 1]
Generation 660: Best = 2.075422e+03 FES = 66100 Diversity = 3.1210e+01 k_t
= [2, 3, 4, 4]
Generation 670: Best = 2.072585e+03 FES = 67100 Diversity = 3.0563e+01 k_t
= [1, 1, 1, 1]
Generation 680: Best = 2.072585e+03 FES = 68100 Diversity = 3.0161e+01 k_t
= [2, 2, 3, 4]
Generation 690: Best = 2.072585e+03 FES = 69100 Diversity = 3.0084e+01 k_t
= [1, 1, 1, 1]
Generation 700: Best = 2.072585e+03 FES = 70100 Diversity = 2.9911e+01 k_t
= [3, 3, 4, 4]
Generation 710: Best = 2.072585e+03 FES = 71100 Diversity = 2.9279e+01 k_t
= [1, 1, 1, 1]
Generation 720: Best = 2.072585e+03 FES = 72100 Diversity = 2.9049e+01 k_t
= [3, 2, 3, 3]
Generation 730: Best = 2.067635e+03 FES = 73100 Diversity = 2.8409e+01 k_t
= [1, 1, 1, 1]
Generation 740: Best = 2.067635e+03 FES = 74100 Diversity = 2.8300e+01 k_t
= [4, 2, 3, 2]
Generation 750: Best = 2.067635e+03 FES = 75100 Diversity = 2.8221e+01 k_t
= [1, 1, 1, 1]
Generation 760: Best = 2.064906e+03 FES = 76100 Diversity = 2.7905e+01 k_t

```

```

= [4, 4, 3, 3]
Generation 770: Best = 2.061185e+03 FES = 77100 Diversity = 2.7473e+01 k_t
= [1, 1, 1, 1]
Generation 780: Best = 2.059859e+03 FES = 78100 Diversity = 2.7041e+01 k_t
= [4, 2, 3, 4]
Generation 790: Best = 2.059316e+03 FES = 79100 Diversity = 2.6774e+01 k_t
= [1, 1, 1, 1]
Generation 800: Best = 2.057686e+03 FES = 80100 Diversity = 2.6489e+01 k_t
= [3, 2, 3, 2]
Generation 810: Best = 2.057564e+03 FES = 81100 Diversity = 2.6753e+01 k_t
= [1, 1, 1, 1]
Generation 820: Best = 2.056691e+03 FES = 82100 Diversity = 2.6720e+01 k_t
= [4, 2, 4, 2]
Generation 830: Best = 2.053144e+03 FES = 83100 Diversity = 2.6550e+01 k_t
= [1, 1, 1, 1]
Generation 840: Best = 2.051500e+03 FES = 84100 Diversity = 2.6399e+01 k_t
= [2, 3, 4, 2]
Generation 850: Best = 2.051240e+03 FES = 85100 Diversity = 2.5937e+01 k_t
= [1, 1, 1, 1]
Generation 860: Best = 2.051090e+03 FES = 86100 Diversity = 2.5801e+01 k_t
= [3, 3, 3, 4]
Generation 870: Best = 2.051090e+03 FES = 87100 Diversity = 2.5858e+01 k_t
= [1, 1, 1, 1]
Generation 880: Best = 2.051090e+03 FES = 88100 Diversity = 2.6046e+01 k_t
= [2, 4, 3, 3]
Generation 890: Best = 2.049150e+03 FES = 89100 Diversity = 2.5496e+01 k_t
= [1, 1, 1, 1]
Generation 900: Best = 2.039198e+03 FES = 90100 Diversity = 2.5512e+01 k_t
= [4, 3, 2, 2]
Generation 910: Best = 2.039198e+03 FES = 91100 Diversity = 2.5294e+01 k_t
= [1, 1, 1, 1]
Generation 920: Best = 2.039197e+03 FES = 92100 Diversity = 2.5153e+01 k_t
= [4, 3, 3, 2]
Generation 930: Best = 2.039197e+03 FES = 93100 Diversity = 2.4795e+01 k_t
= [1, 1, 1, 1]
Generation 940: Best = 2.039197e+03 FES = 94100 Diversity = 2.4829e+01 k_t
= [4, 3, 4, 3]
Generation 950: Best = 2.038995e+03 FES = 95100 Diversity = 2.4618e+01 k_t
= [1, 1, 1, 1]
Generation 960: Best = 2.038995e+03 FES = 96100 Diversity = 2.4652e+01 k_t
= [3, 4, 2, 2]
Generation 970: Best = 2.038995e+03 FES = 97100 Diversity = 2.4430e+01 k_t
= [1, 1, 1, 1]
Generation 980: Best = 2.038852e+03 FES = 98100 Diversity = 2.4100e+01 k_t
= [3, 4, 2, 4]
Generation 990: Best = 2.038852e+03 FES = 99100 Diversity = 2.3950e+01 k_t
= [1, 1, 1, 1]
Generation 1000: Best = 2.038852e+03 FES = 100100 Diversity = 2.3816e+01 k_t
= [3, 3, 4, 3]
Generation 1010: Best = 2.036476e+03 FES = 101100 Diversity = 2.3374e+01 k_t
= [1, 1, 1, 1]
Generation 1020: Best = 2.036476e+03 FES = 102100 Diversity = 2.3432e+01 k_t
= [3, 4, 3, 2]
Generation 1030: Best = 2.035810e+03 FES = 103100 Diversity = 2.2896e+01 k_t

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= [1, 1, 1, 1]
Generation 1040: Best = 2.035810e+03 FES = 104100 Diversity = 2.2857e+01 k_t
= [4, 3, 4, 2]
Generation 1050: Best = 2.035802e+03 FES = 105100 Diversity = 2.2684e+01 k_t
= [1, 1, 1, 1]
Generation 1060: Best = 2.035752e+03 FES = 106100 Diversity = 2.2813e+01 k_t
= [4, 3, 4, 3]
Generation 1070: Best = 2.035562e+03 FES = 107100 Diversity = 2.2744e+01 k_t
= [1, 1, 1, 1]
Generation 1080: Best = 2.033021e+03 FES = 108100 Diversity = 2.2934e+01 k_t
= [4, 4, 4, 2]
Generation 1090: Best = 2.032887e+03 FES = 109100 Diversity = 2.2815e+01 k_t
= [1, 1, 1, 1]
Generation 1100: Best = 2.032598e+03 FES = 110100 Diversity = 2.2733e+01 k_t
= [2, 4, 3, 4]
Generation 1110: Best = 2.032598e+03 FES = 111100 Diversity = 2.2711e+01 k_t
= [1, 1, 1, 1]
Generation 1120: Best = 2.031873e+03 FES = 112100 Diversity = 2.2720e+01 k_t
= [4, 2, 3, 3]
Generation 1130: Best = 2.031811e+03 FES = 113100 Diversity = 2.2158e+01 k_t
= [1, 1, 1, 1]
Generation 1140: Best = 2.031215e+03 FES = 114100 Diversity = 2.1729e+01 k_t
= [2, 3, 3, 4]
Generation 1150: Best = 2.030660e+03 FES = 115100 Diversity = 2.1441e+01 k_t
= [1, 1, 1, 1]
Generation 1160: Best = 2.030660e+03 FES = 116100 Diversity = 2.1749e+01 k_t
= [2, 2, 4, 3]
Generation 1170: Best = 2.030660e+03 FES = 117100 Diversity = 2.1559e+01 k_t
= [1, 1, 1, 1]
Generation 1180: Best = 2.030660e+03 FES = 118100 Diversity = 2.1158e+01 k_t
= [3, 4, 3, 2]
Generation 1190: Best = 2.030635e+03 FES = 119100 Diversity = 2.1364e+01 k_t
= [1, 1, 1, 1]
Generation 1200: Best = 2.029096e+03 FES = 120100 Diversity = 2.1184e+01 k_t
= [2, 2, 3, 2]
Generation 1210: Best = 2.029096e+03 FES = 121100 Diversity = 2.0992e+01 k_t
= [1, 1, 1, 1]
Generation 1220: Best = 2.029096e+03 FES = 122100 Diversity = 2.0981e+01 k_t
= [4, 3, 3, 2]
Generation 1230: Best = 2.029064e+03 FES = 123100 Diversity = 2.0008e+01 k_t
= [1, 1, 1, 1]
Generation 1240: Best = 2.028527e+03 FES = 124100 Diversity = 1.9220e+01 k_t
= [2, 4, 2, 2]
Generation 1250: Best = 2.028270e+03 FES = 125100 Diversity = 1.8787e+01 k_t
= [1, 1, 1, 1]
Generation 1260: Best = 2.028270e+03 FES = 126100 Diversity = 1.8258e+01 k_t
= [2, 4, 2, 3]
Generation 1270: Best = 2.028246e+03 FES = 127100 Diversity = 1.7582e+01 k_t
= [1, 1, 1, 1]
Generation 1280: Best = 2.028101e+03 FES = 128100 Diversity = 1.7176e+01 k_t
= [3, 2, 2, 4]
Generation 1290: Best = 2.027995e+03 FES = 129100 Diversity = 1.7013e+01 k_t
= [1, 1, 1, 1]
Generation 1300: Best = 2.027884e+03 FES = 130100 Diversity = 1.6661e+01 k_t

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= [4, 4, 4, 3]
Generation 1310: Best = 2.027658e+03  FES = 131100  Diversity = 1.6448e+01  k_t
= [1, 1, 1, 1]
Generation 1320: Best = 2.027658e+03  FES = 132100  Diversity = 1.6227e+01  k_t
= [4, 2, 4, 2]
Generation 1330: Best = 2.026963e+03  FES = 133100  Diversity = 1.5993e+01  k_t
= [1, 1, 1, 1]
Generation 1340: Best = 2.026773e+03  FES = 134100  Diversity = 1.5758e+01  k_t
= [2, 4, 3, 4]
Generation 1350: Best = 2.026675e+03  FES = 135100  Diversity = 1.5751e+01  k_t
= [1, 1, 1, 1]
Generation 1360: Best = 2.026549e+03  FES = 136100  Diversity = 1.5742e+01  k_t
= [2, 4, 2, 3]
Generation 1370: Best = 2.026549e+03  FES = 137100  Diversity = 1.5571e+01  k_t
= [1, 1, 1, 1]
Generation 1380: Best = 2.026540e+03  FES = 138100  Diversity = 1.5586e+01  k_t
= [4, 3, 2, 3]
Generation 1390: Best = 2.026076e+03  FES = 139100  Diversity = 1.5412e+01  k_t
= [1, 1, 1, 1]
Generation 1400: Best = 2.025780e+03  FES = 140100  Diversity = 1.5292e+01  k_t
= [4, 3, 2, 3]
Generation 1410: Best = 2.025772e+03  FES = 141100  Diversity = 1.5508e+01  k_t
= [1, 1, 1, 1]
Generation 1420: Best = 2.025159e+03  FES = 142100  Diversity = 1.5480e+01  k_t
= [4, 2, 2, 3]
Generation 1430: Best = 2.025159e+03  FES = 143100  Diversity = 1.5526e+01  k_t
= [1, 1, 1, 1]
Generation 1440: Best = 2.025155e+03  FES = 144100  Diversity = 1.5111e+01  k_t
= [4, 3, 2, 2]
Generation 1450: Best = 2.025107e+03  FES = 145100  Diversity = 1.4977e+01  k_t
= [1, 1, 1, 1]
Generation 1460: Best = 2.025107e+03  FES = 146100  Diversity = 1.5046e+01  k_t
= [3, 4, 4, 2]
Generation 1470: Best = 2.025107e+03  FES = 147100  Diversity = 1.4959e+01  k_t
= [1, 1, 1, 1]
Generation 1480: Best = 2.025003e+03  FES = 148100  Diversity = 1.4750e+01  k_t
= [2, 4, 3, 4]
Generation 1490: Best = 2.024924e+03  FES = 149100  Diversity = 1.4394e+01  k_t
= [1, 1, 1, 1]
Generation 1500: Best = 2.024924e+03  FES = 150100  Diversity = 1.3969e+01  k_t
= [2, 3, 3, 2]
Generation 1510: Best = 2.024898e+03  FES = 151100  Diversity = 1.3719e+01  k_t
= [1, 1, 1, 1]
Generation 1520: Best = 2.024898e+03  FES = 152100  Diversity = 1.3495e+01  k_t
= [2, 4, 2, 3]
Generation 1530: Best = 2.024898e+03  FES = 153100  Diversity = 1.3154e+01  k_t
= [1, 1, 1, 1]
Generation 1540: Best = 2.024642e+03  FES = 154100  Diversity = 1.3366e+01  k_t
= [2, 3, 3, 4]
Generation 1550: Best = 2.024632e+03  FES = 155100  Diversity = 1.3295e+01  k_t
= [1, 1, 1, 1]
Generation 1560: Best = 2.024632e+03  FES = 156100  Diversity = 1.3018e+01  k_t
= [4, 4, 2, 2]
Generation 1570: Best = 2.024630e+03  FES = 157100  Diversity = 1.2885e+01  k_t

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= [1, 1, 1, 1]
Generation 1580: Best = 2.024630e+03 FES = 158100 Diversity = 1.2803e+01 k_t
= [2, 2, 4, 4]
Generation 1590: Best = 2.024630e+03 FES = 159100 Diversity = 1.2664e+01 k_t
= [1, 1, 1, 1]
Generation 1600: Best = 2.024567e+03 FES = 160100 Diversity = 1.2901e+01 k_t
= [3, 4, 3, 3]
Generation 1610: Best = 2.024557e+03 FES = 161100 Diversity = 1.2278e+01 k_t
= [1, 1, 1, 1]
Generation 1620: Best = 2.024480e+03 FES = 162100 Diversity = 1.2257e+01 k_t
= [4, 4, 2, 4]
Generation 1630: Best = 2.024480e+03 FES = 163100 Diversity = 1.2289e+01 k_t
= [1, 1, 1, 1]
Generation 1640: Best = 2.024468e+03 FES = 164100 Diversity = 1.2285e+01 k_t
= [3, 4, 3, 3]
Generation 1650: Best = 2.024468e+03 FES = 165100 Diversity = 1.2113e+01 k_t
= [1, 1, 1, 1]
Generation 1660: Best = 2.024418e+03 FES = 166100 Diversity = 1.2199e+01 k_t
= [2, 4, 3, 2]
Generation 1670: Best = 2.024406e+03 FES = 167100 Diversity = 1.2072e+01 k_t
= [1, 1, 1, 1]
Generation 1680: Best = 2.024406e+03 FES = 168100 Diversity = 1.2032e+01 k_t
= [4, 2, 2, 4]
Generation 1690: Best = 2.024348e+03 FES = 169100 Diversity = 1.2132e+01 k_t
= [1, 1, 1, 1]
Generation 1700: Best = 2.024111e+03 FES = 170100 Diversity = 1.2075e+01 k_t
= [4, 3, 4, 3]
Generation 1710: Best = 2.024111e+03 FES = 171100 Diversity = 1.2124e+01 k_t
= [1, 1, 1, 1]
Generation 1720: Best = 2.024111e+03 FES = 172100 Diversity = 1.2116e+01 k_t
= [2, 4, 2, 2]
Generation 1730: Best = 2.024111e+03 FES = 173100 Diversity = 1.2010e+01 k_t
= [1, 1, 1, 1]
Generation 1740: Best = 2.024111e+03 FES = 174100 Diversity = 1.1985e+01 k_t
= [4, 3, 2, 2]
Generation 1750: Best = 2.024111e+03 FES = 175100 Diversity = 1.2000e+01 k_t
= [1, 1, 1, 1]
Generation 1760: Best = 2.024111e+03 FES = 176100 Diversity = 1.1997e+01 k_t
= [4, 2, 2, 3]
Generation 1770: Best = 2.024093e+03 FES = 177100 Diversity = 1.2001e+01 k_t
= [1, 1, 1, 1]
Generation 1780: Best = 2.024012e+03 FES = 178100 Diversity = 1.2028e+01 k_t
= [3, 4, 2, 4]
Generation 1790: Best = 2.024012e+03 FES = 179100 Diversity = 1.2053e+01 k_t
= [1, 1, 1, 1]
Generation 1800: Best = 2.024012e+03 FES = 180100 Diversity = 1.1969e+01 k_t
= [4, 4, 4, 3]
Generation 1810: Best = 2.023986e+03 FES = 181100 Diversity = 1.2003e+01 k_t
= [1, 1, 1, 1]
Generation 1820: Best = 2.023986e+03 FES = 182100 Diversity = 1.1883e+01 k_t
= [3, 4, 2, 3]
Generation 1830: Best = 2.023986e+03 FES = 183100 Diversity = 1.1876e+01 k_t
= [1, 1, 1, 1]
Generation 1840: Best = 2.023984e+03 FES = 184100 Diversity = 1.1914e+01 k_t

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= [2, 3, 4, 3]
Generation 1850: Best = 2.023872e+03 FES = 185100 Diversity = 1.1907e+01 k_t
= [1, 1, 1, 1]
Generation 1860: Best = 2.023872e+03 FES = 186100 Diversity = 1.1896e+01 k_t
= [4, 2, 3, 3]
Generation 1870: Best = 2.023872e+03 FES = 187100 Diversity = 1.1837e+01 k_t
= [1, 1, 1, 1]
Generation 1880: Best = 2.023869e+03 FES = 188100 Diversity = 1.1832e+01 k_t
= [2, 2, 4, 2]
Generation 1890: Best = 2.023847e+03 FES = 189100 Diversity = 1.1868e+01 k_t
= [1, 1, 1, 1]
Generation 1900: Best = 2.023847e+03 FES = 190100 Diversity = 1.1666e+01 k_t
= [4, 3, 2, 3]
Generation 1910: Best = 2.023846e+03 FES = 191100 Diversity = 1.1407e+01 k_t
= [1, 1, 1, 1]
Generation 1920: Best = 2.023846e+03 FES = 192100 Diversity = 1.1410e+01 k_t
= [3, 4, 3, 3]
Generation 1930: Best = 2.023846e+03 FES = 193100 Diversity = 1.1381e+01 k_t
= [1, 1, 1, 1]
Generation 1940: Best = 2.023845e+03 FES = 194100 Diversity = 1.1332e+01 k_t
= [3, 4, 2, 2]
Generation 1950: Best = 2.023835e+03 FES = 195100 Diversity = 1.1330e+01 k_t
= [1, 1, 1, 1]
Generation 1960: Best = 2.023835e+03 FES = 196100 Diversity = 1.1158e+01 k_t
= [4, 4, 4, 2]
Generation 1970: Best = 2.023834e+03 FES = 197100 Diversity = 1.1038e+01 k_t
= [1, 1, 1, 1]
Generation 1980: Best = 2.023832e+03 FES = 198100 Diversity = 1.1075e+01 k_t
= [2, 2, 3, 2]
Generation 1990: Best = 2.023829e+03 FES = 199100 Diversity = 1.1053e+01 k_t
= [1, 1, 1, 1]
Generation 2000: Best = 2.023773e+03 FES = 200100 Diversity = 1.1023e+01 k_t
= [2, 4, 3, 2]
Generation 2010: Best = 2.023769e+03 FES = 201100 Diversity = 1.0714e+01 k_t
= [1, 1, 1, 1]
Generation 2020: Best = 2.023760e+03 FES = 202100 Diversity = 1.0474e+01 k_t
= [3, 3, 3, 4]
Generation 2030: Best = 2.023652e+03 FES = 203100 Diversity = 1.0440e+01 k_t
= [1, 1, 1, 1]
Generation 2040: Best = 2.023651e+03 FES = 204100 Diversity = 1.0424e+01 k_t
= [4, 4, 3, 2]
Generation 2050: Best = 2.023651e+03 FES = 205100 Diversity = 1.0446e+01 k_t
= [1, 1, 1, 1]
Generation 2060: Best = 2.023642e+03 FES = 206100 Diversity = 1.0325e+01 k_t
= [4, 3, 4, 3]
Generation 2070: Best = 2.023618e+03 FES = 207100 Diversity = 1.0112e+01 k_t
= [1, 1, 1, 1]
Generation 2080: Best = 2.023613e+03 FES = 208100 Diversity = 9.9024e+00 k_t
= [2, 2, 3, 2]
Generation 2090: Best = 2.023613e+03 FES = 209100 Diversity = 9.7418e+00 k_t
= [1, 1, 1, 1]
Generation 2100: Best = 2.023606e+03 FES = 210100 Diversity = 9.0154e+00 k_t
= [3, 4, 3, 3]
Generation 2110: Best = 2.023587e+03 FES = 211100 Diversity = 8.0296e+00 k_t

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= [1, 1, 1, 1]
Generation 2120: Best = 2.023577e+03 FES = 212100 Diversity = 6.7875e+00 k_t
= [2, 3, 2, 4]
Generation 2130: Best = 2.023577e+03 FES = 213100 Diversity = 5.4053e+00 k_t
= [1, 1, 1, 1]
Generation 2140: Best = 2.023512e+03 FES = 214100 Diversity = 4.5502e+00 k_t
= [4, 2, 4, 4]
Generation 2150: Best = 2.023504e+03 FES = 215100 Diversity = 3.8385e+00 k_t
= [1, 1, 1, 1]
Generation 2160: Best = 2.023490e+03 FES = 216100 Diversity = 2.7443e+00 k_t
= [3, 2, 4, 4]
Generation 2170: Best = 2.023489e+03 FES = 217100 Diversity = 2.3573e+00 k_t
= [1, 1, 1, 1]
Generation 2180: Best = 2.023485e+03 FES = 218100 Diversity = 2.3594e+00 k_t
= [2, 2, 2, 4]
Generation 2190: Best = 2.023484e+03 FES = 219100 Diversity = 2.0663e+00 k_t
= [1, 1, 1, 1]
Generation 2200: Best = 2.023472e+03 FES = 220100 Diversity = 1.9578e+00 k_t
= [4, 3, 3, 2]
Generation 2210: Best = 2.023457e+03 FES = 221100 Diversity = 1.8307e+00 k_t
= [1, 1, 1, 1]
Generation 2220: Best = 2.023427e+03 FES = 222100 Diversity = 1.8342e+00 k_t
= [4, 4, 3, 3]
Generation 2230: Best = 2.023397e+03 FES = 223100 Diversity = 1.8319e+00 k_t
= [1, 1, 1, 1]
Generation 2240: Best = 2.023397e+03 FES = 224100 Diversity = 1.8282e+00 k_t
= [4, 4, 3, 4]
Generation 2250: Best = 2.023379e+03 FES = 225100 Diversity = 1.8227e+00 k_t
= [1, 1, 1, 1]
Generation 2260: Best = 2.023362e+03 FES = 226100 Diversity = 1.0295e+00 k_t
= [3, 4, 3, 3]
Generation 2270: Best = 2.023344e+03 FES = 227100 Diversity = 4.4543e-01 k_t
= [1, 1, 1, 1]
Generation 2280: Best = 2.023342e+03 FES = 228100 Diversity = 4.1589e-01 k_t
= [4, 4, 3, 2]
Generation 2290: Best = 2.023342e+03 FES = 229100 Diversity = 3.9353e-01 k_t
= [1, 1, 1, 1]
Generation 2300: Best = 2.023309e+03 FES = 230100 Diversity = 3.5011e-01 k_t
= [2, 4, 3, 3]
Generation 2310: Best = 2.023302e+03 FES = 231100 Diversity = 2.9928e-01 k_t
= [1, 1, 1, 1]
Generation 2320: Best = 2.023302e+03 FES = 232100 Diversity = 2.4384e-01 k_t
= [2, 4, 2, 4]
Generation 2330: Best = 2.023301e+03 FES = 233100 Diversity = 2.0460e-01 k_t
= [1, 1, 1, 1]
Generation 2340: Best = 2.023295e+03 FES = 234100 Diversity = 1.8132e-01 k_t
= [3, 3, 3, 2]
Generation 2350: Best = 2.023293e+03 FES = 235100 Diversity = 1.5079e-01 k_t
= [1, 1, 1, 1]
Generation 2360: Best = 2.023284e+03 FES = 236100 Diversity = 1.1433e-01 k_t
= [2, 3, 2, 2]
Generation 2370: Best = 2.023282e+03 FES = 237100 Diversity = 1.1359e-01 k_t
= [1, 1, 1, 1]
Generation 2380: Best = 2.023276e+03 FES = 238100 Diversity = 4.8206e-02 k_t

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= [4, 4, 4, 2]
Generation 2390: Best = 2.023272e+03 FES = 239100 Diversity = 4.6973e-02 k_t
= [1, 1, 1, 1]
Generation 2400: Best = 2.023272e+03 FES = 240100 Diversity = 4.5463e-02 k_t
= [3, 3, 3, 3]
Generation 2410: Best = 2.023272e+03 FES = 241100 Diversity = 4.3982e-02 k_t
= [1, 1, 1, 1]
Generation 2420: Best = 2.023272e+03 FES = 242100 Diversity = 4.3266e-02 k_t
= [2, 3, 2, 4]
Generation 2430: Best = 2.023270e+03 FES = 243100 Diversity = 4.1642e-02 k_t
= [1, 1, 1, 1]
Generation 2440: Best = 2.023269e+03 FES = 244100 Diversity = 4.1099e-02 k_t
= [2, 2, 2, 2]
Generation 2450: Best = 2.023265e+03 FES = 245100 Diversity = 3.7408e-02 k_t
= [1, 1, 1, 1]
Generation 2460: Best = 2.023264e+03 FES = 246100 Diversity = 3.5603e-02 k_t
= [3, 3, 2, 4]
Generation 2470: Best = 2.023264e+03 FES = 247100 Diversity = 3.1774e-02 k_t
= [1, 1, 1, 1]
Generation 2480: Best = 2.023262e+03 FES = 248100 Diversity = 3.1454e-02 k_t
= [2, 3, 4, 4]
Generation 2490: Best = 2.023262e+03 FES = 249100 Diversity = 3.0976e-02 k_t
= [1, 1, 1, 1]
Generation 2500: Best = 2.023262e+03 FES = 250100 Diversity = 3.0449e-02 k_t
= [2, 3, 4, 3]
Generation 2510: Best = 2.023262e+03 FES = 251100 Diversity = 3.0125e-02 k_t
= [1, 1, 1, 1]
Generation 2520: Best = 2.023258e+03 FES = 252100 Diversity = 2.9829e-02 k_t
= [2, 4, 3, 4]
Generation 2530: Best = 2.023258e+03 FES = 253100 Diversity = 2.7200e-02 k_t
= [1, 1, 1, 1]
Generation 2540: Best = 2.023258e+03 FES = 254100 Diversity = 2.6808e-02 k_t
= [4, 3, 3, 4]
Generation 2550: Best = 2.023256e+03 FES = 255100 Diversity = 2.6554e-02 k_t
= [1, 1, 1, 1]
Generation 2560: Best = 2.023255e+03 FES = 256100 Diversity = 2.6446e-02 k_t
= [4, 4, 3, 2]
Generation 2570: Best = 2.023254e+03 FES = 257100 Diversity = 2.5868e-02 k_t
= [1, 1, 1, 1]
Generation 2580: Best = 2.023253e+03 FES = 258100 Diversity = 2.5442e-02 k_t
= [3, 4, 2, 2]
Generation 2590: Best = 2.023253e+03 FES = 259100 Diversity = 2.5259e-02 k_t
= [1, 1, 1, 1]
Generation 2600: Best = 2.023253e+03 FES = 260100 Diversity = 2.4998e-02 k_t
= [3, 2, 3, 3]
Generation 2610: Best = 2.023252e+03 FES = 261100 Diversity = 2.4708e-02 k_t
= [1, 1, 1, 1]
Generation 2620: Best = 2.023252e+03 FES = 262100 Diversity = 2.4454e-02 k_t
= [3, 3, 4, 2]
Generation 2630: Best = 2.023252e+03 FES = 263100 Diversity = 2.4358e-02 k_t
= [1, 1, 1, 1]
Generation 2640: Best = 2.023252e+03 FES = 264100 Diversity = 2.4688e-02 k_t
= [3, 4, 3, 3]
Generation 2650: Best = 2.023252e+03 FES = 265100 Diversity = 2.4453e-02 k_t

```

```

= [1, 1, 1, 1]
Generation 2660: Best = 2.023252e+03 FES = 266100 Diversity = 2.4560e-02 k_t
= [3, 4, 4, 4]
Generation 2670: Best = 2.023252e+03 FES = 267100 Diversity = 2.4177e-02 k_t
= [1, 1, 1, 1]
Generation 2680: Best = 2.023252e+03 FES = 268100 Diversity = 2.4327e-02 k_t
= [3, 3, 3, 2]
Generation 2690: Best = 2.023252e+03 FES = 269100 Diversity = 2.4358e-02 k_t
= [1, 1, 1, 1]
Generation 2700: Best = 2.023251e+03 FES = 270100 Diversity = 2.3827e-02 k_t
= [3, 3, 2, 2]
Generation 2710: Best = 2.023251e+03 FES = 271100 Diversity = 2.3238e-02 k_t
= [1, 1, 1, 1]
Generation 2720: Best = 2.023251e+03 FES = 272100 Diversity = 2.2874e-02 k_t
= [4, 2, 2, 4]
Generation 2730: Best = 2.023251e+03 FES = 273100 Diversity = 2.2747e-02 k_t
= [1, 1, 1, 1]
Generation 2740: Best = 2.023251e+03 FES = 274100 Diversity = 2.2780e-02 k_t
= [2, 4, 2, 2]
Generation 2750: Best = 2.023251e+03 FES = 275100 Diversity = 2.2676e-02 k_t
= [1, 1, 1, 1]
Generation 2760: Best = 2.023251e+03 FES = 276100 Diversity = 2.2606e-02 k_t
= [4, 3, 4, 2]
Generation 2770: Best = 2.023250e+03 FES = 277100 Diversity = 2.2265e-02 k_t
= [1, 1, 1, 1]
Generation 2780: Best = 2.023249e+03 FES = 278100 Diversity = 2.2341e-02 k_t
= [4, 4, 4, 2]
Generation 2790: Best = 2.023249e+03 FES = 279100 Diversity = 2.1897e-02 k_t
= [1, 1, 1, 1]
Generation 2800: Best = 2.023249e+03 FES = 280100 Diversity = 2.2095e-02 k_t
= [4, 4, 3, 4]
Generation 2810: Best = 2.023248e+03 FES = 281100 Diversity = 2.1816e-02 k_t
= [1, 1, 1, 1]
Generation 2820: Best = 2.023248e+03 FES = 282100 Diversity = 2.1618e-02 k_t
= [4, 3, 4, 4]
Generation 2830: Best = 2.023248e+03 FES = 283100 Diversity = 2.1280e-02 k_t
= [1, 1, 1, 1]
Generation 2840: Best = 2.023248e+03 FES = 284100 Diversity = 2.1189e-02 k_t
= [4, 3, 2, 3]
Generation 2850: Best = 2.023248e+03 FES = 285100 Diversity = 2.1374e-02 k_t
= [1, 1, 1, 1]
Generation 2860: Best = 2.023247e+03 FES = 286100 Diversity = 2.0959e-02 k_t
= [2, 3, 2, 4]
Generation 2870: Best = 2.023247e+03 FES = 287100 Diversity = 2.1177e-02 k_t
= [1, 1, 1, 1]
Generation 2880: Best = 2.023247e+03 FES = 288100 Diversity = 2.0724e-02 k_t
= [4, 2, 3, 3]
Generation 2890: Best = 2.023246e+03 FES = 289100 Diversity = 2.0608e-02 k_t
= [1, 1, 1, 1]
Generation 2900: Best = 2.023246e+03 FES = 290100 Diversity = 2.0056e-02 k_t
= [4, 3, 2, 4]
Generation 2910: Best = 2.023246e+03 FES = 291100 Diversity = 1.9846e-02 k_t
= [1, 1, 1, 1]
Generation 2920: Best = 2.023245e+03 FES = 292100 Diversity = 1.9474e-02 k_t

```

```

= [3, 4, 4, 4]
Generation 2930: Best = 2.023245e+03  FES = 293100  Diversity = 1.9112e-02  k_t
= [1, 1, 1, 1]
Generation 2940: Best = 2.023245e+03  FES = 294100  Diversity = 1.8499e-02  k_t
= [2, 4, 2, 4]
Generation 2950: Best = 2.023245e+03  FES = 295100  Diversity = 1.8356e-02  k_t
= [1, 1, 1, 1]
Generation 2960: Best = 2.023243e+03  FES = 296100  Diversity = 1.7560e-02  k_t
= [3, 2, 4, 2]
Generation 2970: Best = 2.023241e+03  FES = 297100  Diversity = 1.7331e-02  k_t
= [1, 1, 1, 1]
Generation 2980: Best = 2.023241e+03  FES = 298100  Diversity = 1.7039e-02  k_t
= [3, 3, 3, 4]
Generation 2990: Best = 2.023239e+03  FES = 299100  Diversity = 1.7043e-02  k_t
= [1, 1, 1, 1]

```

Optimization completed!

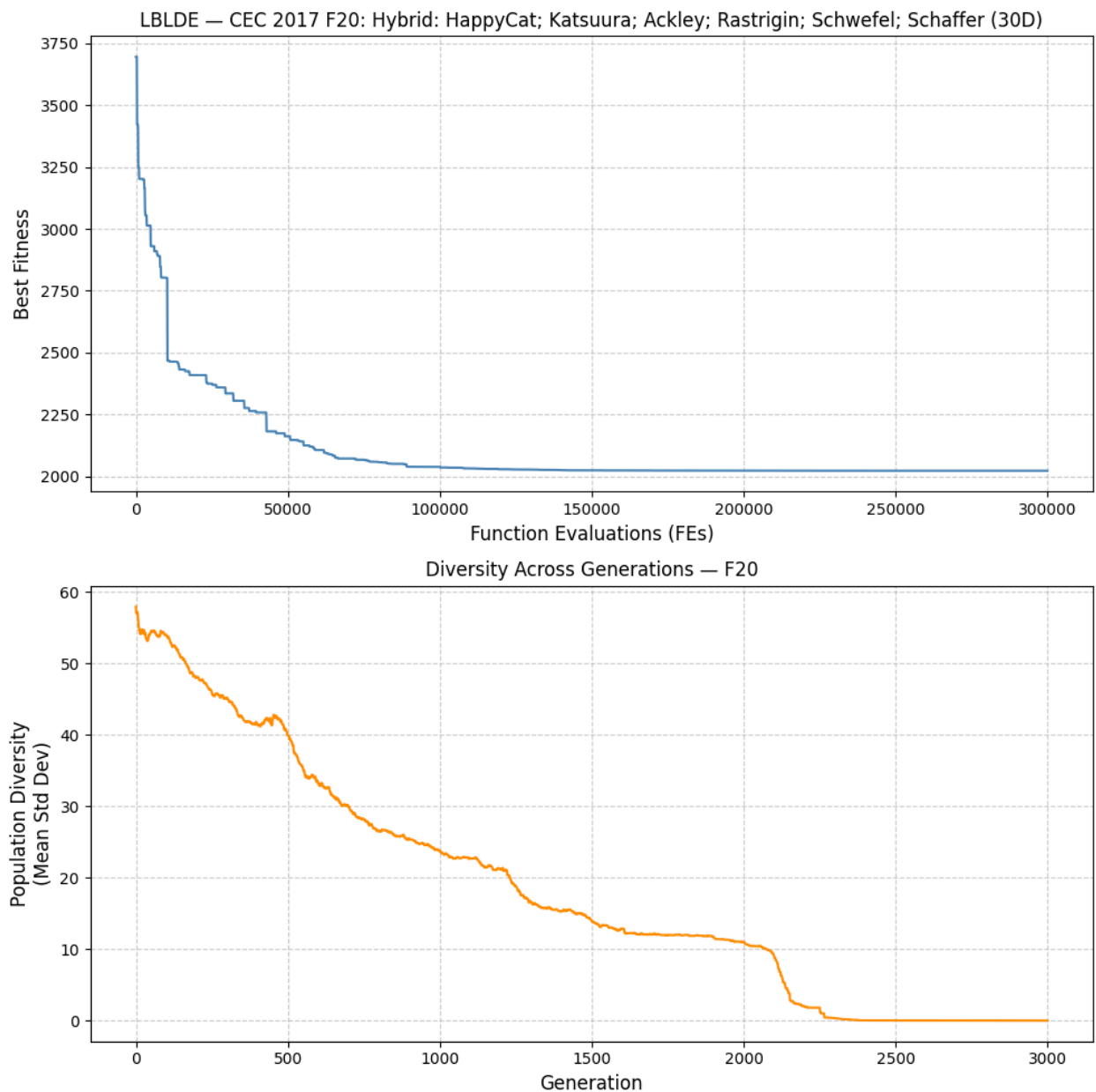
Final Best Fitness = 2.023239e+03

Total Function Evaluations = 300000

```

=====
=
Results:
  Best Fitness      : 2.023239e+03
  Global Optimum    : 2.000000e+03
  Error (|best-optimum|) : 2.323937e+01
  Best Solution (5 dim) : [ 96.77088119  -7.12564129 -29.57300981  52.09585563
62.06182939]
  D^(0) (init. diversity): 5.7976e+01
=====
=

```



Graph saved as convergence_F20.png

6.4 Composite Functions (F21-F30)

```
In [31]: # =====
# F21 — Composition Function 1 (Rosenbrock; Elliptic; Rastrigin)
# =====
run_and_plot('F21', make_wrapper(composition.f21), 2100.0, 'Composition: Roser
```

```

=====
=
LBLDE_s on CEC 2017 F21 – Single Run
=====
=
Function      : Composition: Rosenbrock; Elliptic; Rastrigin (30D)
Bounds       : [-100, 100]
Global optimum : 2100.0
Initial D^(0) = 5.7976e+01
Generation 0: Best Fitness = 2.946639e+03 | Diversity = 5.7976e+01
Generation 10: Best = 2.622762e+03 FES = 1100 Diversity = 3.9296e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 2.540892e+03 FES = 2100 Diversity = 2.9627e+01 k_t
= [3, 3, 2, 2]
Generation 30: Best = 2.540892e+03 FES = 3100 Diversity = 2.7364e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 2.540892e+03 FES = 4100 Diversity = 2.5718e+01 k_t
= [3, 4, 3, 4]
Generation 50: Best = 2.535530e+03 FES = 5100 Diversity = 2.4392e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 2.516653e+03 FES = 6100 Diversity = 2.4059e+01 k_t
= [3, 4, 4, 3]
Generation 70: Best = 2.493673e+03 FES = 7100 Diversity = 2.3656e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 2.492766e+03 FES = 8100 Diversity = 2.3690e+01 k_t
= [2, 3, 3, 2]
Generation 90: Best = 2.492766e+03 FES = 9100 Diversity = 2.3323e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 2.480823e+03 FES = 10100 Diversity = 2.2645e+01 k_t
= [4, 2, 3, 3]
Generation 110: Best = 2.480823e+03 FES = 11100 Diversity = 2.2213e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 2.478180e+03 FES = 12100 Diversity = 2.2090e+01 k_t
= [4, 4, 3, 4]
Generation 130: Best = 2.469155e+03 FES = 13100 Diversity = 2.1541e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 2.469155e+03 FES = 14100 Diversity = 2.1305e+01 k_t
= [3, 2, 3, 3]
Generation 150: Best = 2.469155e+03 FES = 15100 Diversity = 2.1300e+01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 2.465408e+03 FES = 16100 Diversity = 2.1223e+01 k_t
= [2, 4, 2, 4]
Generation 170: Best = 2.465408e+03 FES = 17100 Diversity = 2.1143e+01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 2.465408e+03 FES = 18100 Diversity = 2.1054e+01 k_t
= [3, 3, 4, 2]
Generation 190: Best = 2.443694e+03 FES = 19100 Diversity = 2.1037e+01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 2.443694e+03 FES = 20100 Diversity = 2.1014e+01 k_t
= [4, 3, 4, 2]
Generation 210: Best = 2.443694e+03 FES = 21100 Diversity = 2.1054e+01 k_t
= [1, 1, 1, 1]
Generation 220: Best = 2.443694e+03 FES = 22100 Diversity = 2.0990e+01 k_t
= [4, 3, 3, 4]

```

Generation 230:	Best = 2.442167e+03	FES = 23100	Diversity = 2.0831e+01	k_t = [1, 1, 1, 1]
Generation 240:	Best = 2.441952e+03	FES = 24100	Diversity = 2.0755e+01	k_t = [2, 2, 3, 2]
Generation 250:	Best = 2.441952e+03	FES = 25100	Diversity = 2.0738e+01	k_t = [1, 1, 1, 1]
Generation 260:	Best = 2.441952e+03	FES = 26100	Diversity = 2.0498e+01	k_t = [4, 2, 4, 2]
Generation 270:	Best = 2.435666e+03	FES = 27100	Diversity = 2.0350e+01	k_t = [1, 1, 1, 1]
Generation 280:	Best = 2.435666e+03	FES = 28100	Diversity = 2.0315e+01	k_t = [3, 4, 2, 3]
Generation 290:	Best = 2.427447e+03	FES = 29100	Diversity = 2.0297e+01	k_t = [1, 1, 1, 1]
Generation 300:	Best = 2.427447e+03	FES = 30100	Diversity = 2.0236e+01	k_t = [2, 2, 4, 3]
Generation 310:	Best = 2.427447e+03	FES = 31100	Diversity = 2.0180e+01	k_t = [1, 1, 1, 1]
Generation 320:	Best = 2.427032e+03	FES = 32100	Diversity = 2.0220e+01	k_t = [3, 3, 4, 3]
Generation 330:	Best = 2.417174e+03	FES = 33100	Diversity = 2.0256e+01	k_t = [1, 1, 1, 1]
Generation 340:	Best = 2.417174e+03	FES = 34100	Diversity = 2.0387e+01	k_t = [4, 2, 3, 4]
Generation 350:	Best = 2.417174e+03	FES = 35100	Diversity = 2.0390e+01	k_t = [1, 1, 1, 1]
Generation 360:	Best = 2.417174e+03	FES = 36100	Diversity = 2.0213e+01	k_t = [3, 3, 2, 4]
Generation 370:	Best = 2.415017e+03	FES = 37100	Diversity = 2.0146e+01	k_t = [1, 1, 1, 1]
Generation 380:	Best = 2.411640e+03	FES = 38100	Diversity = 2.0239e+01	k_t = [3, 3, 4, 2]
Generation 390:	Best = 2.405609e+03	FES = 39100	Diversity = 2.0213e+01	k_t = [1, 1, 1, 1]
Generation 400:	Best = 2.402821e+03	FES = 40100	Diversity = 2.0194e+01	k_t = [4, 3, 3, 4]
Generation 410:	Best = 2.402821e+03	FES = 41100	Diversity = 2.0151e+01	k_t = [1, 1, 1, 1]
Generation 420:	Best = 2.402045e+03	FES = 42100	Diversity = 2.0076e+01	k_t = [4, 2, 4, 2]
Generation 430:	Best = 2.402045e+03	FES = 43100	Diversity = 2.0041e+01	k_t = [1, 1, 1, 1]
Generation 440:	Best = 2.398577e+03	FES = 44100	Diversity = 2.0063e+01	k_t = [3, 4, 4, 2]
Generation 450:	Best = 2.398577e+03	FES = 45100	Diversity = 2.0019e+01	k_t = [1, 1, 1, 1]
Generation 460:	Best = 2.398577e+03	FES = 46100	Diversity = 1.9957e+01	k_t = [3, 3, 3, 3]
Generation 470:	Best = 2.397036e+03	FES = 47100	Diversity = 1.9928e+01	k_t = [1, 1, 1, 1]
Generation 480:	Best = 2.395128e+03	FES = 48100	Diversity = 1.9811e+01	k_t = [3, 4, 2, 2]
Generation 490:	Best = 2.395128e+03	FES = 49100	Diversity = 1.9655e+01	k_t = [1, 1, 1, 1]

Generation 500:	Best = 2.395128e+03	FES = 50100	Diversity = 1.9617e+01	k_t = [3, 2, 2, 3]
Generation 510:	Best = 2.395128e+03	FES = 51100	Diversity = 1.9638e+01	k_t = [1, 1, 1, 1]
Generation 520:	Best = 2.391119e+03	FES = 52100	Diversity = 1.9647e+01	k_t = [2, 3, 4, 3]
Generation 530:	Best = 2.391119e+03	FES = 53100	Diversity = 1.9725e+01	k_t = [1, 1, 1, 1]
Generation 540:	Best = 2.391119e+03	FES = 54100	Diversity = 1.9830e+01	k_t = [4, 4, 3, 4]
Generation 550:	Best = 2.391119e+03	FES = 55100	Diversity = 1.9494e+01	k_t = [1, 1, 1, 1]
Generation 560:	Best = 2.390032e+03	FES = 56100	Diversity = 1.9386e+01	k_t = [3, 4, 2, 2]
Generation 570:	Best = 2.390032e+03	FES = 57100	Diversity = 1.9368e+01	k_t = [1, 1, 1, 1]
Generation 580:	Best = 2.390032e+03	FES = 58100	Diversity = 1.9356e+01	k_t = [2, 2, 4, 3]
Generation 590:	Best = 2.388827e+03	FES = 59100	Diversity = 1.9354e+01	k_t = [1, 1, 1, 1]
Generation 600:	Best = 2.388822e+03	FES = 60100	Diversity = 1.9401e+01	k_t = [4, 3, 2, 2]
Generation 610:	Best = 2.387480e+03	FES = 61100	Diversity = 1.9381e+01	k_t = [1, 1, 1, 1]
Generation 620:	Best = 2.386264e+03	FES = 62100	Diversity = 1.9425e+01	k_t = [2, 2, 3, 4]
Generation 630:	Best = 2.386264e+03	FES = 63100	Diversity = 1.9291e+01	k_t = [1, 1, 1, 1]
Generation 640:	Best = 2.373330e+03	FES = 64100	Diversity = 1.9267e+01	k_t = [3, 2, 2, 3]
Generation 650:	Best = 2.373330e+03	FES = 65100	Diversity = 1.9177e+01	k_t = [1, 1, 1, 1]
Generation 660:	Best = 2.372987e+03	FES = 66100	Diversity = 1.9214e+01	k_t = [4, 2, 3, 3]
Generation 670:	Best = 2.372738e+03	FES = 67100	Diversity = 1.9174e+01	k_t = [1, 1, 1, 1]
Generation 680:	Best = 2.372738e+03	FES = 68100	Diversity = 1.9136e+01	k_t = [3, 2, 2, 4]
Generation 690:	Best = 2.372738e+03	FES = 69100	Diversity = 1.9128e+01	k_t = [1, 1, 1, 1]
Generation 700:	Best = 2.371787e+03	FES = 70100	Diversity = 1.9096e+01	k_t = [4, 4, 2, 3]
Generation 710:	Best = 2.369951e+03	FES = 71100	Diversity = 1.9042e+01	k_t = [1, 1, 1, 1]
Generation 720:	Best = 2.369951e+03	FES = 72100	Diversity = 1.8994e+01	k_t = [2, 3, 4, 4]
Generation 730:	Best = 2.369951e+03	FES = 73100	Diversity = 1.8981e+01	k_t = [1, 1, 1, 1]
Generation 740:	Best = 2.369951e+03	FES = 74100	Diversity = 1.8947e+01	k_t = [3, 2, 2, 4]
Generation 750:	Best = 2.366097e+03	FES = 75100	Diversity = 1.8952e+01	k_t = [1, 1, 1, 1]
Generation 760:	Best = 2.366021e+03	FES = 76100	Diversity = 1.8973e+01	k_t = [3, 3, 2, 2]

Generation 770:	Best = 2.366021e+03	FES = 77100	Diversity = 1.8950e+01	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 2.362145e+03	FES = 78100	Diversity = 1.8942e+01	k_t
= [4, 4, 4, 2]				
Generation 790:	Best = 2.362145e+03	FES = 79100	Diversity = 1.8920e+01	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 2.362145e+03	FES = 80100	Diversity = 1.8908e+01	k_t
= [4, 3, 3, 3]				
Generation 810:	Best = 2.362069e+03	FES = 81100	Diversity = 1.8863e+01	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 2.362069e+03	FES = 82100	Diversity = 1.8814e+01	k_t
= [2, 4, 4, 3]				
Generation 830:	Best = 2.361636e+03	FES = 83100	Diversity = 1.8859e+01	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 2.361636e+03	FES = 84100	Diversity = 1.8849e+01	k_t
= [4, 2, 2, 4]				
Generation 850:	Best = 2.361636e+03	FES = 85100	Diversity = 1.8855e+01	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 2.361636e+03	FES = 86100	Diversity = 1.8694e+01	k_t
= [3, 4, 4, 4]				
Generation 870:	Best = 2.361636e+03	FES = 87100	Diversity = 1.8583e+01	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 2.360373e+03	FES = 88100	Diversity = 1.8537e+01	k_t
= [4, 3, 4, 4]				
Generation 890:	Best = 2.360373e+03	FES = 89100	Diversity = 1.8542e+01	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 2.360373e+03	FES = 90100	Diversity = 1.8526e+01	k_t
= [4, 4, 2, 2]				
Generation 910:	Best = 2.359391e+03	FES = 91100	Diversity = 1.8545e+01	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 2.359391e+03	FES = 92100	Diversity = 1.8434e+01	k_t
= [3, 3, 3, 2]				
Generation 930:	Best = 2.359391e+03	FES = 93100	Diversity = 1.8219e+01	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 2.359391e+03	FES = 94100	Diversity = 1.8017e+01	k_t
= [4, 4, 3, 3]				
Generation 950:	Best = 2.359391e+03	FES = 95100	Diversity = 1.7927e+01	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 2.359391e+03	FES = 96100	Diversity = 1.7692e+01	k_t
= [3, 2, 3, 3]				
Generation 970:	Best = 2.359238e+03	FES = 97100	Diversity = 1.7711e+01	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 2.359238e+03	FES = 98100	Diversity = 1.7711e+01	k_t
= [2, 2, 4, 3]				
Generation 990:	Best = 2.359238e+03	FES = 99100	Diversity = 1.7708e+01	k_t
= [1, 1, 1, 1]				
Generation 1000:	Best = 2.359238e+03	FES = 100100	Diversity = 1.7534e+01	k_t
= [2, 3, 3, 2]				
Generation 1010:	Best = 2.359238e+03	FES = 101100	Diversity = 1.7324e+01	k_t
= [1, 1, 1, 1]				
Generation 1020:	Best = 2.359075e+03	FES = 102100	Diversity = 1.7242e+01	k_t
= [4, 4, 2, 2]				
Generation 1030:	Best = 2.359075e+03	FES = 103100	Diversity = 1.7094e+01	k_t
= [1, 1, 1, 1]				

Generation 1040: Best = 2.359075e+03 = [4, 3, 2, 2]	FES = 104100	Diversity = 1.7090e+01	k_t
Generation 1050: Best = 2.359075e+03 = [1, 1, 1, 1]	FES = 105100	Diversity = 1.7050e+01	k_t
Generation 1060: Best = 2.357381e+03 = [4, 4, 2, 3]	FES = 106100	Diversity = 1.7039e+01	k_t
Generation 1070: Best = 2.357381e+03 = [1, 1, 1, 1]	FES = 107100	Diversity = 1.6948e+01	k_t
Generation 1080: Best = 2.357381e+03 = [4, 4, 3, 2]	FES = 108100	Diversity = 1.6950e+01	k_t
Generation 1090: Best = 2.357381e+03 = [1, 1, 1, 1]	FES = 109100	Diversity = 1.6948e+01	k_t
Generation 1100: Best = 2.357381e+03 = [3, 4, 4, 2]	FES = 110100	Diversity = 1.6862e+01	k_t
Generation 1110: Best = 2.357381e+03 = [1, 1, 1, 1]	FES = 111100	Diversity = 1.6731e+01	k_t
Generation 1120: Best = 2.357381e+03 = [4, 2, 2, 4]	FES = 112100	Diversity = 1.6609e+01	k_t
Generation 1130: Best = 2.356820e+03 = [1, 1, 1, 1]	FES = 113100	Diversity = 1.6585e+01	k_t
Generation 1140: Best = 2.356820e+03 = [3, 2, 2, 4]	FES = 114100	Diversity = 1.6433e+01	k_t
Generation 1150: Best = 2.356820e+03 = [1, 1, 1, 1]	FES = 115100	Diversity = 1.6358e+01	k_t
Generation 1160: Best = 2.356820e+03 = [2, 4, 3, 4]	FES = 116100	Diversity = 1.6161e+01	k_t
Generation 1170: Best = 2.354076e+03 = [1, 1, 1, 1]	FES = 117100	Diversity = 1.6211e+01	k_t
Generation 1180: Best = 2.350195e+03 = [3, 3, 4, 2]	FES = 118100	Diversity = 1.6189e+01	k_t
Generation 1190: Best = 2.349665e+03 = [1, 1, 1, 1]	FES = 119100	Diversity = 1.5905e+01	k_t
Generation 1200: Best = 2.347348e+03 = [2, 4, 2, 3]	FES = 120100	Diversity = 1.5713e+01	k_t
Generation 1210: Best = 2.346954e+03 = [1, 1, 1, 1]	FES = 121100	Diversity = 1.5354e+01	k_t
Generation 1220: Best = 2.344397e+03 = [3, 2, 3, 3]	FES = 122100	Diversity = 1.4956e+01	k_t
Generation 1230: Best = 2.344194e+03 = [1, 1, 1, 1]	FES = 123100	Diversity = 1.4924e+01	k_t
Generation 1240: Best = 2.341571e+03 = [4, 4, 2, 3]	FES = 124100	Diversity = 1.4888e+01	k_t
Generation 1250: Best = 2.339563e+03 = [1, 1, 1, 1]	FES = 125100	Diversity = 1.4693e+01	k_t
Generation 1260: Best = 2.339563e+03 = [2, 3, 2, 3]	FES = 126100	Diversity = 1.4563e+01	k_t
Generation 1270: Best = 2.339563e+03 = [1, 1, 1, 1]	FES = 127100	Diversity = 1.4237e+01	k_t
Generation 1280: Best = 2.336628e+03 = [4, 4, 3, 2]	FES = 128100	Diversity = 1.3923e+01	k_t
Generation 1290: Best = 2.333059e+03 = [1, 1, 1, 1]	FES = 129100	Diversity = 1.3779e+01	k_t
Generation 1300: Best = 2.332621e+03 = [2, 4, 2, 3]	FES = 130100	Diversity = 1.3049e+01	k_t

Generation 1310: Best = 2.330101e+03 = [1, 1, 1, 1]	FES = 131100	Diversity = 1.2525e+01	k_t
Generation 1320: Best = 2.329706e+03 = [3, 4, 3, 3]	FES = 132100	Diversity = 1.0152e+01	k_t
Generation 1330: Best = 2.328952e+03 = [1, 1, 1, 1]	FES = 133100	Diversity = 3.7914e+00	k_t
Generation 1340: Best = 2.327912e+03 = [4, 4, 2, 3]	FES = 134100	Diversity = 8.5483e-01	k_t
Generation 1350: Best = 2.327265e+03 = [1, 1, 1, 1]	FES = 135100	Diversity = 7.0076e-01	k_t
Generation 1360: Best = 2.326753e+03 = [2, 2, 4, 2]	FES = 136100	Diversity = 4.9938e-01	k_t
Generation 1370: Best = 2.326496e+03 = [1, 1, 1, 1]	FES = 137100	Diversity = 3.9402e-01	k_t
Generation 1380: Best = 2.326348e+03 = [4, 2, 3, 4]	FES = 138100	Diversity = 3.0160e-01	k_t
Generation 1390: Best = 2.326275e+03 = [1, 1, 1, 1]	FES = 139100	Diversity = 2.1619e-01	k_t
Generation 1400: Best = 2.326234e+03 = [3, 2, 2, 2]	FES = 140100	Diversity = 1.6680e-01	k_t
Generation 1410: Best = 2.326200e+03 = [1, 1, 1, 1]	FES = 141100	Diversity = 1.2633e-01	k_t
Generation 1420: Best = 2.326185e+03 = [3, 2, 2, 2]	FES = 142100	Diversity = 8.7915e-02	k_t
Generation 1430: Best = 2.326179e+03 = [1, 1, 1, 1]	FES = 143100	Diversity = 6.8976e-02	k_t
Generation 1440: Best = 2.326174e+03 = [3, 4, 4, 4]	FES = 144100	Diversity = 4.6408e-02	k_t
Generation 1450: Best = 2.326173e+03 = [1, 1, 1, 1]	FES = 145100	Diversity = 3.2785e-02	k_t
Generation 1460: Best = 2.326172e+03 = [2, 2, 4, 4]	FES = 146100	Diversity = 2.7079e-02	k_t
Generation 1470: Best = 2.326172e+03 = [1, 1, 1, 1]	FES = 147100	Diversity = 1.9687e-02	k_t
Generation 1480: Best = 2.326172e+03 = [3, 4, 3, 2]	FES = 148100	Diversity = 1.5235e-02	k_t
Generation 1490: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 149100	Diversity = 1.0106e-02	k_t
Generation 1500: Best = 2.326171e+03 = [4, 2, 3, 3]	FES = 150100	Diversity = 6.3496e-03	k_t
Generation 1510: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 151100	Diversity = 4.1194e-03	k_t
Generation 1520: Best = 2.326171e+03 = [4, 3, 3, 4]	FES = 152100	Diversity = 2.7282e-03	k_t
Generation 1530: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 153100	Diversity = 1.7317e-03	k_t
Generation 1540: Best = 2.326171e+03 = [2, 3, 3, 4]	FES = 154100	Diversity = 1.1672e-03	k_t
Generation 1550: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 155100	Diversity = 6.5339e-04	k_t
Generation 1560: Best = 2.326171e+03 = [2, 2, 3, 4]	FES = 156100	Diversity = 4.4917e-04	k_t
Generation 1570: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 157100	Diversity = 2.7225e-04	k_t

Generation 1580: Best = 2.326171e+03 = [3, 3, 2, 4]	FES = 158100	Diversity = 1.7789e-04	k_t
Generation 1590: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 159100	Diversity = 1.0712e-04	k_t
Generation 1600: Best = 2.326171e+03 = [4, 4, 4, 3]	FES = 160100	Diversity = 7.7265e-05	k_t
Generation 1610: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 161100	Diversity = 5.5130e-05	k_t
Generation 1620: Best = 2.326171e+03 = [3, 3, 2, 4]	FES = 162100	Diversity = 3.2279e-05	k_t
Generation 1630: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 163100	Diversity = 2.2526e-05	k_t
Generation 1640: Best = 2.326171e+03 = [2, 3, 2, 3]	FES = 164100	Diversity = 1.4451e-05	k_t
Generation 1650: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 165100	Diversity = 9.4615e-06	k_t
Generation 1660: Best = 2.326171e+03 = [2, 2, 2, 4]	FES = 166100	Diversity = 7.6769e-06	k_t
Generation 1670: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 167100	Diversity = 4.9724e-06	k_t
Generation 1680: Best = 2.326171e+03 = [2, 2, 4, 2]	FES = 168100	Diversity = 3.6564e-06	k_t
Generation 1690: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 169100	Diversity = 2.3799e-06	k_t
Generation 1700: Best = 2.326171e+03 = [4, 2, 2, 4]	FES = 170100	Diversity = 1.8241e-06	k_t
Generation 1710: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 171100	Diversity = 1.5607e-06	k_t
Generation 1720: Best = 2.326171e+03 = [2, 4, 4, 2]	FES = 172100	Diversity = 9.9746e-07	k_t
Generation 1730: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 173100	Diversity = 6.8846e-07	k_t
Generation 1740: Best = 2.326171e+03 = [4, 3, 4, 2]	FES = 174100	Diversity = 4.8520e-07	k_t
Generation 1750: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 175100	Diversity = 3.8104e-07	k_t
Generation 1760: Best = 2.326171e+03 = [3, 4, 3, 4]	FES = 176100	Diversity = 3.6206e-07	k_t
Generation 1770: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 177100	Diversity = 3.4619e-07	k_t
Generation 1780: Best = 2.326171e+03 = [3, 4, 4, 3]	FES = 178100	Diversity = 3.1910e-07	k_t
Generation 1790: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 179100	Diversity = 3.0829e-07	k_t
Generation 1800: Best = 2.326171e+03 = [4, 3, 2, 2]	FES = 180100	Diversity = 2.5844e-07	k_t
Generation 1810: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 181100	Diversity = 2.5062e-07	k_t
Generation 1820: Best = 2.326171e+03 = [2, 2, 3, 3]	FES = 182100	Diversity = 2.4455e-07	k_t
Generation 1830: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 183100	Diversity = 2.3896e-07	k_t
Generation 1840: Best = 2.326171e+03 = [2, 3, 3, 2]	FES = 184100	Diversity = 2.3589e-07	k_t

Generation 1850: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 185100	Diversity = 2.3870e-07	k_t
Generation 1860: Best = 2.326171e+03 = [4, 2, 2, 2]	FES = 186100	Diversity = 2.2557e-07	k_t
Generation 1870: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 187100	Diversity = 2.3802e-07	k_t
Generation 1880: Best = 2.326171e+03 = [3, 2, 4, 4]	FES = 188100	Diversity = 2.3827e-07	k_t
Generation 1890: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 189100	Diversity = 2.4498e-07	k_t
Generation 1900: Best = 2.326171e+03 = [3, 3, 3, 4]	FES = 190100	Diversity = 2.7605e-07	k_t
Generation 1910: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 191100	Diversity = 2.8854e-07	k_t
Generation 1920: Best = 2.326171e+03 = [4, 2, 3, 2]	FES = 192100	Diversity = 2.9866e-07	k_t
Generation 1930: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 193100	Diversity = 3.1359e-07	k_t
Generation 1940: Best = 2.326171e+03 = [3, 2, 3, 3]	FES = 194100	Diversity = 2.9314e-07	k_t
Generation 1950: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 195100	Diversity = 2.8963e-07	k_t
Generation 1960: Best = 2.326171e+03 = [3, 4, 2, 4]	FES = 196100	Diversity = 2.8958e-07	k_t
Generation 1970: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 197100	Diversity = 2.9101e-07	k_t
Generation 1980: Best = 2.326171e+03 = [4, 3, 2, 2]	FES = 198100	Diversity = 2.8449e-07	k_t
Generation 1990: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 199100	Diversity = 2.8949e-07	k_t
Generation 2000: Best = 2.326171e+03 = [4, 2, 2, 4]	FES = 200100	Diversity = 2.8554e-07	k_t
Generation 2010: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 201100	Diversity = 2.7862e-07	k_t
Generation 2020: Best = 2.326171e+03 = [2, 3, 4, 3]	FES = 202100	Diversity = 2.7010e-07	k_t
Generation 2030: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 203100	Diversity = 2.6285e-07	k_t
Generation 2040: Best = 2.326171e+03 = [2, 2, 3, 3]	FES = 204100	Diversity = 2.3922e-07	k_t
Generation 2050: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 205100	Diversity = 2.1919e-07	k_t
Generation 2060: Best = 2.326171e+03 = [2, 4, 2, 4]	FES = 206100	Diversity = 2.2810e-07	k_t
Generation 2070: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 207100	Diversity = 2.2401e-07	k_t
Generation 2080: Best = 2.326171e+03 = [3, 2, 2, 4]	FES = 208100	Diversity = 2.1581e-07	k_t
Generation 2090: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 209100	Diversity = 2.1523e-07	k_t
Generation 2100: Best = 2.326171e+03 = [2, 3, 3, 4]	FES = 210100	Diversity = 2.2231e-07	k_t
Generation 2110: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 211100	Diversity = 2.0045e-07	k_t

Generation 2120: Best = 2.326171e+03 = [2, 4, 3, 4]	FES = 212100	Diversity = 2.3527e-07	k_t
Generation 2130: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 213100	Diversity = 2.6011e-07	k_t
Generation 2140: Best = 2.326171e+03 = [4, 2, 2, 2]	FES = 214100	Diversity = 2.7460e-07	k_t
Generation 2150: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 215100	Diversity = 2.3047e-07	k_t
Generation 2160: Best = 2.326171e+03 = [3, 3, 3, 3]	FES = 216100	Diversity = 2.2607e-07	k_t
Generation 2170: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 217100	Diversity = 2.0578e-07	k_t
Generation 2180: Best = 2.326171e+03 = [2, 3, 2, 2]	FES = 218100	Diversity = 2.0708e-07	k_t
Generation 2190: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 219100	Diversity = 2.1108e-07	k_t
Generation 2200: Best = 2.326171e+03 = [3, 4, 3, 4]	FES = 220100	Diversity = 1.9689e-07	k_t
Generation 2210: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 221100	Diversity = 1.9506e-07	k_t
Generation 2220: Best = 2.326171e+03 = [4, 2, 4, 3]	FES = 222100	Diversity = 1.8578e-07	k_t
Generation 2230: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 223100	Diversity = 2.0090e-07	k_t
Generation 2240: Best = 2.326171e+03 = [3, 3, 4, 2]	FES = 224100	Diversity = 1.8861e-07	k_t
Generation 2250: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 225100	Diversity = 1.8646e-07	k_t
Generation 2260: Best = 2.326171e+03 = [4, 4, 3, 3]	FES = 226100	Diversity = 1.8622e-07	k_t
Generation 2270: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 227100	Diversity = 1.9075e-07	k_t
Generation 2280: Best = 2.326171e+03 = [2, 3, 2, 2]	FES = 228100	Diversity = 1.8189e-07	k_t
Generation 2290: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 229100	Diversity = 1.8266e-07	k_t
Generation 2300: Best = 2.326171e+03 = [3, 2, 2, 2]	FES = 230100	Diversity = 1.9896e-07	k_t
Generation 2310: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 231100	Diversity = 2.1027e-07	k_t
Generation 2320: Best = 2.326171e+03 = [3, 3, 3, 3]	FES = 232100	Diversity = 2.3646e-07	k_t
Generation 2330: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 233100	Diversity = 2.5296e-07	k_t
Generation 2340: Best = 2.326171e+03 = [4, 2, 3, 4]	FES = 234100	Diversity = 2.4643e-07	k_t
Generation 2350: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 235100	Diversity = 2.5321e-07	k_t
Generation 2360: Best = 2.326171e+03 = [4, 2, 3, 3]	FES = 236100	Diversity = 2.6503e-07	k_t
Generation 2370: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 237100	Diversity = 2.4039e-07	k_t
Generation 2380: Best = 2.326171e+03 = [4, 3, 2, 4]	FES = 238100	Diversity = 2.2217e-07	k_t

Generation 2390: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 239100	Diversity = 2.4240e-07	k_t
Generation 2400: Best = 2.326171e+03 = [2, 4, 4, 3]	FES = 240100	Diversity = 2.1270e-07	k_t
Generation 2410: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 241100	Diversity = 2.0594e-07	k_t
Generation 2420: Best = 2.326171e+03 = [2, 4, 4, 3]	FES = 242100	Diversity = 2.0215e-07	k_t
Generation 2430: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 243100	Diversity = 1.9265e-07	k_t
Generation 2440: Best = 2.326171e+03 = [4, 2, 4, 4]	FES = 244100	Diversity = 1.8577e-07	k_t
Generation 2450: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 245100	Diversity = 1.8929e-07	k_t
Generation 2460: Best = 2.326171e+03 = [2, 4, 3, 2]	FES = 246100	Diversity = 1.8584e-07	k_t
Generation 2470: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 247100	Diversity = 1.8239e-07	k_t
Generation 2480: Best = 2.326171e+03 = [3, 2, 2, 4]	FES = 248100	Diversity = 1.9381e-07	k_t
Generation 2490: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 249100	Diversity = 1.8672e-07	k_t
Generation 2500: Best = 2.326171e+03 = [4, 3, 2, 2]	FES = 250100	Diversity = 1.8317e-07	k_t
Generation 2510: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 251100	Diversity = 1.8345e-07	k_t
Generation 2520: Best = 2.326171e+03 = [2, 3, 4, 2]	FES = 252100	Diversity = 1.8836e-07	k_t
Generation 2530: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 253100	Diversity = 2.0504e-07	k_t
Generation 2540: Best = 2.326171e+03 = [3, 4, 4, 2]	FES = 254100	Diversity = 1.8274e-07	k_t
Generation 2550: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 255100	Diversity = 1.9050e-07	k_t
Generation 2560: Best = 2.326171e+03 = [4, 2, 4, 4]	FES = 256100	Diversity = 1.9350e-07	k_t
Generation 2570: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 257100	Diversity = 1.9946e-07	k_t
Generation 2580: Best = 2.326171e+03 = [3, 4, 2, 2]	FES = 258100	Diversity = 1.9388e-07	k_t
Generation 2590: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 259100	Diversity = 1.8934e-07	k_t
Generation 2600: Best = 2.326171e+03 = [3, 3, 3, 2]	FES = 260100	Diversity = 1.7597e-07	k_t
Generation 2610: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 261100	Diversity = 1.9402e-07	k_t
Generation 2620: Best = 2.326171e+03 = [2, 2, 2, 3]	FES = 262100	Diversity = 1.9129e-07	k_t
Generation 2630: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 263100	Diversity = 1.7946e-07	k_t
Generation 2640: Best = 2.326171e+03 = [2, 2, 4, 4]	FES = 264100	Diversity = 2.0108e-07	k_t
Generation 2650: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 265100	Diversity = 2.0791e-07	k_t

Generation 2660: Best = 2.326171e+03 = [4, 4, 2, 2]	FES = 266100	Diversity = 2.0566e-07	k_t
Generation 2670: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 267100	Diversity = 2.1649e-07	k_t
Generation 2680: Best = 2.326171e+03 = [3, 2, 4, 4]	FES = 268100	Diversity = 2.0072e-07	k_t
Generation 2690: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 269100	Diversity = 1.9644e-07	k_t
Generation 2700: Best = 2.326171e+03 = [3, 3, 4, 2]	FES = 270100	Diversity = 1.8699e-07	k_t
Generation 2710: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 271100	Diversity = 1.9344e-07	k_t
Generation 2720: Best = 2.326171e+03 = [3, 2, 4, 3]	FES = 272100	Diversity = 1.8322e-07	k_t
Generation 2730: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 273100	Diversity = 1.8787e-07	k_t
Generation 2740: Best = 2.326171e+03 = [2, 2, 3, 4]	FES = 274100	Diversity = 2.0163e-07	k_t
Generation 2750: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 275100	Diversity = 1.9723e-07	k_t
Generation 2760: Best = 2.326171e+03 = [4, 2, 3, 3]	FES = 276100	Diversity = 2.0658e-07	k_t
Generation 2770: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 277100	Diversity = 2.0002e-07	k_t
Generation 2780: Best = 2.326171e+03 = [4, 3, 2, 2]	FES = 278100	Diversity = 2.0209e-07	k_t
Generation 2790: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 279100	Diversity = 1.9768e-07	k_t
Generation 2800: Best = 2.326171e+03 = [3, 2, 2, 3]	FES = 280100	Diversity = 1.8890e-07	k_t
Generation 2810: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 281100	Diversity = 1.8191e-07	k_t
Generation 2820: Best = 2.326171e+03 = [3, 4, 4, 2]	FES = 282100	Diversity = 1.9610e-07	k_t
Generation 2830: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 283100	Diversity = 1.9240e-07	k_t
Generation 2840: Best = 2.326171e+03 = [2, 2, 4, 3]	FES = 284100	Diversity = 1.8824e-07	k_t
Generation 2850: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 285100	Diversity = 1.8118e-07	k_t
Generation 2860: Best = 2.326171e+03 = [3, 4, 4, 3]	FES = 286100	Diversity = 1.7243e-07	k_t
Generation 2870: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 287100	Diversity = 1.7417e-07	k_t
Generation 2880: Best = 2.326171e+03 = [3, 4, 4, 3]	FES = 288100	Diversity = 1.7419e-07	k_t
Generation 2890: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 289100	Diversity = 1.6919e-07	k_t
Generation 2900: Best = 2.326171e+03 = [3, 3, 4, 3]	FES = 290100	Diversity = 1.6764e-07	k_t
Generation 2910: Best = 2.326171e+03 = [1, 1, 1, 1]	FES = 291100	Diversity = 1.5895e-07	k_t
Generation 2920: Best = 2.326171e+03 = [3, 4, 3, 2]	FES = 292100	Diversity = 1.5260e-07	k_t

Generation 2930: Best = 2.326171e+03 FES = 293100 Diversity = 1.5000e-07 k_t
 = [1, 1, 1, 1]
 Generation 2940: Best = 2.326171e+03 FES = 294100 Diversity = 1.6319e-07 k_t
 = [4, 2, 3, 4]
 Generation 2950: Best = 2.326171e+03 FES = 295100 Diversity = 1.7956e-07 k_t
 = [1, 1, 1, 1]
 Generation 2960: Best = 2.326171e+03 FES = 296100 Diversity = 1.7815e-07 k_t
 = [2, 2, 4, 3]
 Generation 2970: Best = 2.326171e+03 FES = 297100 Diversity = 1.8467e-07 k_t
 = [1, 1, 1, 1]
 Generation 2980: Best = 2.326171e+03 FES = 298100 Diversity = 1.8247e-07 k_t
 = [3, 4, 2, 4]
 Generation 2990: Best = 2.326171e+03 FES = 299100 Diversity = 1.7045e-07 k_t
 = [1, 1, 1, 1]

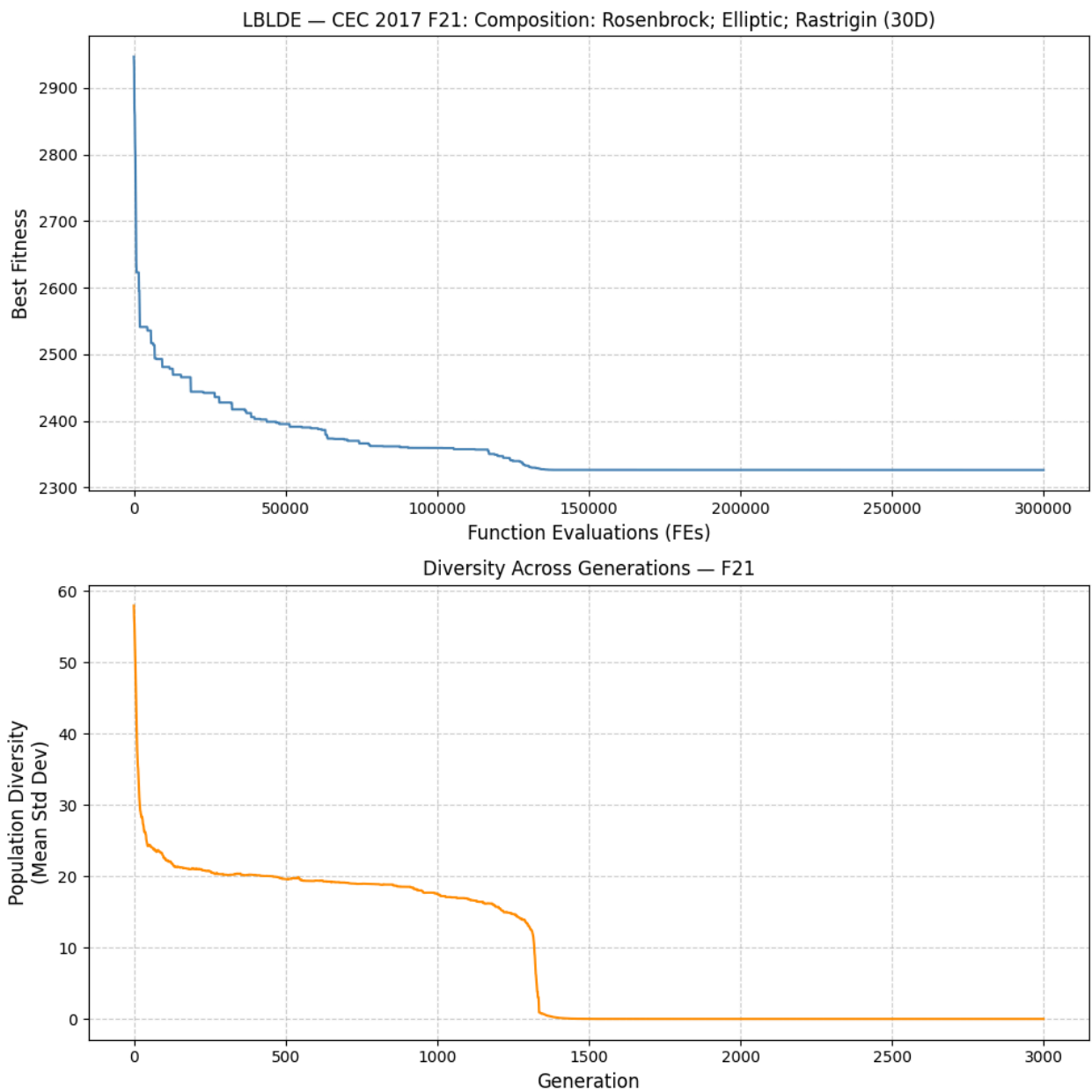
Optimization completed!

Final Best Fitness = 2.326171e+03

Total Function Evaluations = 300000

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Results:
  Best Fitness           : 2.326171e+03
  Global Optimum         : 2.100000e+03
  Error (|best-optimum|) : 2.261713e+02
  Best Solution (5 dim)  : [ 0.8161831 -29.58087244  70.41126325  21.48306088
68.54096178]
  D^(0) (init. diversity): 5.7976e+01
=====
=
  
```



Graph saved as convergence_F21.png

```
In [32]: # =====
# F22 – Composition Function 2 (Rastrigin; Griewank; Schwefel)
# =====
run_and_plot('F22', make_wrapper(composition.f22), 2200.0, 'Composition: Rastr
```

```

=====
=
LBLDE_s on CEC 2017 F22 – Single Run
=====
=
Function      : Composition: Rastrigin; Griewank; Schwefel (30D)
Bounds       : [-100, 100]
Global optimum : 2200.0
Initial D^(0) = 5.7976e+01
Generation 0: Best Fitness = 1.151639e+04 | Diversity = 5.7976e+01
Generation 10: Best = 7.391626e+03 FES = 1100 Diversity = 5.2536e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 4.325282e+03 FES = 2100 Diversity = 3.0555e+01 k_t
= [3, 3, 2, 2]
Generation 30: Best = 3.201608e+03 FES = 3100 Diversity = 1.7438e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 2.620579e+03 FES = 4100 Diversity = 9.2137e+00 k_t
= [2, 3, 2, 3]
Generation 50: Best = 2.466144e+03 FES = 5100 Diversity = 5.3536e+00 k_t
= [1, 1, 1, 1]
Generation 60: Best = 2.377082e+03 FES = 6100 Diversity = 3.1439e+00 k_t
= [2, 2, 3, 4]
Generation 70: Best = 2.346648e+03 FES = 7100 Diversity = 1.8096e+00 k_t
= [1, 1, 1, 1]
Generation 80: Best = 2.330652e+03 FES = 8100 Diversity = 1.0196e+00 k_t
= [4, 3, 4, 3]
Generation 90: Best = 2.319726e+03 FES = 9100 Diversity = 6.2725e-01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 2.314000e+03 FES = 10100 Diversity = 4.0259e-01 k_t
= [2, 4, 4, 3]
Generation 110: Best = 2.308015e+03 FES = 11100 Diversity = 3.1958e-01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 2.306093e+03 FES = 12100 Diversity = 2.6309e-01 k_t
= [3, 3, 3, 4]
Generation 130: Best = 2.304143e+03 FES = 13100 Diversity = 2.0229e-01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 2.302997e+03 FES = 14100 Diversity = 1.3439e-01 k_t
= [2, 3, 3, 3]
Generation 150: Best = 2.301556e+03 FES = 15100 Diversity = 8.6565e-02 k_t
= [1, 1, 1, 1]
Generation 160: Best = 2.301025e+03 FES = 16100 Diversity = 5.2537e-02 k_t
= [2, 4, 4, 4]
Generation 170: Best = 2.300616e+03 FES = 17100 Diversity = 3.1883e-02 k_t
= [1, 1, 1, 1]
Generation 180: Best = 2.300367e+03 FES = 18100 Diversity = 2.0474e-02 k_t
= [2, 4, 3, 2]
Generation 190: Best = 2.300213e+03 FES = 19100 Diversity = 1.1498e-02 k_t
= [1, 1, 1, 1]
Generation 200: Best = 2.300131e+03 FES = 20100 Diversity = 6.7873e-03 k_t
= [4, 4, 4, 3]
Generation 210: Best = 2.300073e+03 FES = 21100 Diversity = 4.1786e-03 k_t
= [1, 1, 1, 1]
Generation 220: Best = 2.300043e+03 FES = 22100 Diversity = 2.6789e-03 k_t
= [2, 3, 3, 4]

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Generation 230:	Best = 2.300027e+03	FES = 23100	Diversity = 1.6417e-03	k_t = [1, 1, 1, 1]
Generation 240:	Best = 2.300017e+03	FES = 24100	Diversity = 1.0439e-03	k_t = [2, 2, 3, 2]
Generation 250:	Best = 2.300011e+03	FES = 25100	Diversity = 6.6137e-04	k_t = [1, 1, 1, 1]
Generation 260:	Best = 2.300006e+03	FES = 26100	Diversity = 4.2069e-04	k_t = [4, 3, 4, 2]
Generation 270:	Best = 2.300004e+03	FES = 27100	Diversity = 2.6664e-04	k_t = [1, 1, 1, 1]
Generation 280:	Best = 2.300002e+03	FES = 28100	Diversity = 1.5736e-04	k_t = [4, 2, 2, 4]
Generation 290:	Best = 2.300001e+03	FES = 29100	Diversity = 9.6252e-05	k_t = [1, 1, 1, 1]
Generation 300:	Best = 2.300001e+03	FES = 30100	Diversity = 5.9652e-05	k_t = [4, 2, 4, 4]
Generation 310:	Best = 2.300000e+03	FES = 31100	Diversity = 3.4491e-05	k_t = [1, 1, 1, 1]
Generation 320:	Best = 2.300000e+03	FES = 32100	Diversity = 2.0147e-05	k_t = [2, 2, 3, 2]
Generation 330:	Best = 2.300000e+03	FES = 33100	Diversity = 1.1781e-05	k_t = [1, 1, 1, 1]
Generation 340:	Best = 2.300000e+03	FES = 34100	Diversity = 6.7214e-06	k_t = [4, 3, 3, 4]
Generation 350:	Best = 2.300000e+03	FES = 35100	Diversity = 3.7253e-06	k_t = [1, 1, 1, 1]
Generation 360:	Best = 2.300000e+03	FES = 36100	Diversity = 2.1830e-06	k_t = [3, 3, 3, 4]
Generation 370:	Best = 2.300000e+03	FES = 37100	Diversity = 1.2693e-06	k_t = [1, 1, 1, 1]
Generation 380:	Best = 2.300000e+03	FES = 38100	Diversity = 7.6392e-07	k_t = [4, 3, 3, 3]
Generation 390:	Best = 2.300000e+03	FES = 39100	Diversity = 4.4873e-07	k_t = [1, 1, 1, 1]
Generation 400:	Best = 2.300000e+03	FES = 40100	Diversity = 2.3657e-07	k_t = [4, 4, 2, 2]
Generation 410:	Best = 2.300000e+03	FES = 41100	Diversity = 1.2767e-07	k_t = [1, 1, 1, 1]
Generation 420:	Best = 2.300000e+03	FES = 42100	Diversity = 7.2146e-08	k_t = [3, 4, 2, 2]
Generation 430:	Best = 2.300000e+03	FES = 43100	Diversity = 3.9954e-08	k_t = [1, 1, 1, 1]
Generation 440:	Best = 2.300000e+03	FES = 44100	Diversity = 2.2631e-08	k_t = [3, 4, 4, 2]
Generation 450:	Best = 2.300000e+03	FES = 45100	Diversity = 1.2716e-08	k_t = [1, 1, 1, 1]
Generation 460:	Best = 2.300000e+03	FES = 46100	Diversity = 7.3422e-09	k_t = [2, 4, 3, 2]
Generation 470:	Best = 2.300000e+03	FES = 47100	Diversity = 4.2189e-09	k_t = [1, 1, 1, 1]
Generation 480:	Best = 2.300000e+03	FES = 48100	Diversity = 2.5396e-09	k_t = [4, 2, 3, 3]
Generation 490:	Best = 2.300000e+03	FES = 49100	Diversity = 1.3991e-09	k_t = [1, 1, 1, 1]

Generation 500:	Best = 2.300000e+03	FES = 50100	Diversity = 8.1659e-10	k_t = [4, 3, 2, 2]
Generation 510:	Best = 2.300000e+03	FES = 51100	Diversity = 4.6536e-10	k_t = [1, 1, 1, 1]
Generation 520:	Best = 2.300000e+03	FES = 52100	Diversity = 2.9124e-10	k_t = [4, 2, 3, 3]
Generation 530:	Best = 2.300000e+03	FES = 53100	Diversity = 1.7518e-10	k_t = [1, 1, 1, 1]
Generation 540:	Best = 2.300000e+03	FES = 54100	Diversity = 1.0128e-10	k_t = [2, 4, 3, 2]
Generation 550:	Best = 2.300000e+03	FES = 55100	Diversity = 6.1179e-11	k_t = [1, 1, 1, 1]
Generation 560:	Best = 2.300000e+03	FES = 56100	Diversity = 3.9195e-11	k_t = [3, 2, 4, 3]
Generation 570:	Best = 2.300000e+03	FES = 57100	Diversity = 2.5996e-11	k_t = [1, 1, 1, 1]
Generation 580:	Best = 2.300000e+03	FES = 58100	Diversity = 1.6140e-11	k_t = [4, 2, 2, 2]
Generation 590:	Best = 2.300000e+03	FES = 59100	Diversity = 1.0256e-11	k_t = [1, 1, 1, 1]
Generation 600:	Best = 2.300000e+03	FES = 60100	Diversity = 6.0908e-12	k_t = [2, 3, 3, 3]
Generation 610:	Best = 2.300000e+03	FES = 61100	Diversity = 3.5067e-12	k_t = [1, 1, 1, 1]
Generation 620:	Best = 2.300000e+03	FES = 62100	Diversity = 2.0520e-12	k_t = [4, 2, 3, 4]
Generation 630:	Best = 2.300000e+03	FES = 63100	Diversity = 1.2799e-12	k_t = [1, 1, 1, 1]
Generation 640:	Best = 2.300000e+03	FES = 64100	Diversity = 7.2636e-13	k_t = [3, 2, 3, 2]
Generation 650:	Best = 2.300000e+03	FES = 65100	Diversity = 4.1974e-13	k_t = [1, 1, 1, 1]
Generation 660:	Best = 2.300000e+03	FES = 66100	Diversity = 2.5803e-13	k_t = [3, 4, 2, 2]
Generation 670:	Best = 2.300000e+03	FES = 67100	Diversity = 1.6395e-13	k_t = [1, 1, 1, 1]
Generation 680:	Best = 2.300000e+03	FES = 68100	Diversity = 1.1112e-13	k_t = [4, 3, 3, 4]
Generation 690:	Best = 2.300000e+03	FES = 69100	Diversity = 8.9461e-14	k_t = [1, 1, 1, 1]
Generation 700:	Best = 2.300000e+03	FES = 70100	Diversity = 7.3511e-14	k_t = [4, 3, 3, 4]
Generation 710:	Best = 2.300000e+03	FES = 71100	Diversity = 6.7759e-14	k_t = [1, 1, 1, 1]
Generation 720:	Best = 2.300000e+03	FES = 72100	Diversity = 6.7944e-14	k_t = [4, 4, 4, 2]
Generation 730:	Best = 2.300000e+03	FES = 73100	Diversity = 6.8171e-14	k_t = [1, 1, 1, 1]
Generation 740:	Best = 2.300000e+03	FES = 74100	Diversity = 6.5689e-14	k_t = [4, 4, 2, 3]
Generation 750:	Best = 2.300000e+03	FES = 75100	Diversity = 6.5638e-14	k_t = [1, 1, 1, 1]
Generation 760:	Best = 2.300000e+03	FES = 76100	Diversity = 6.6532e-14	k_t = [4, 2, 4, 3]

Generation 770:	Best = 2.300000e+03	FES = 77100	Diversity = 6.5357e-14	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 2.300000e+03	FES = 78100	Diversity = 6.6683e-14	k_t
= [2, 2, 3, 3]				
Generation 790:	Best = 2.300000e+03	FES = 79100	Diversity = 6.2662e-14	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 2.300000e+03	FES = 80100	Diversity = 6.2591e-14	k_t
= [3, 2, 2, 3]				
Generation 810:	Best = 2.300000e+03	FES = 81100	Diversity = 5.8922e-14	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 2.300000e+03	FES = 82100	Diversity = 5.6480e-14	k_t
= [2, 2, 4, 4]				
Generation 830:	Best = 2.300000e+03	FES = 83100	Diversity = 5.4277e-14	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 2.300000e+03	FES = 84100	Diversity = 5.2196e-14	k_t
= [4, 4, 2, 2]				
Generation 850:	Best = 2.300000e+03	FES = 85100	Diversity = 5.1983e-14	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 2.300000e+03	FES = 86100	Diversity = 4.9882e-14	k_t
= [3, 2, 3, 4]				
Generation 870:	Best = 2.300000e+03	FES = 87100	Diversity = 4.9795e-14	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 2.300000e+03	FES = 88100	Diversity = 5.1087e-14	k_t
= [4, 4, 2, 2]				
Generation 890:	Best = 2.300000e+03	FES = 89100	Diversity = 5.0962e-14	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 2.300000e+03	FES = 90100	Diversity = 4.9023e-14	k_t
= [4, 4, 3, 3]				
Generation 910:	Best = 2.300000e+03	FES = 91100	Diversity = 5.0286e-14	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 2.300000e+03	FES = 92100	Diversity = 5.0534e-14	k_t
= [2, 2, 3, 2]				
Generation 930:	Best = 2.300000e+03	FES = 93100	Diversity = 5.0446e-14	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 2.300000e+03	FES = 94100	Diversity = 5.1695e-14	k_t
= [2, 3, 3, 4]				
Generation 950:	Best = 2.300000e+03	FES = 95100	Diversity = 5.2328e-14	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 2.300000e+03	FES = 96100	Diversity = 5.1084e-14	k_t
= [2, 4, 2, 2]				
Generation 970:	Best = 2.300000e+03	FES = 97100	Diversity = 4.9928e-14	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 2.300000e+03	FES = 98100	Diversity = 5.1133e-14	k_t
= [2, 4, 2, 2]				
Generation 990:	Best = 2.300000e+03	FES = 99100	Diversity = 5.2007e-14	k_t
= [1, 1, 1, 1]				
Generation 1000:	Best = 2.300000e+03	FES = 100100	Diversity = 5.1115e-14	k_t
= [2, 4, 3, 2]				
Generation 1010:	Best = 2.300000e+03	FES = 101100	Diversity = 5.1956e-14	k_t
= [1, 1, 1, 1]				
Generation 1020:	Best = 2.300000e+03	FES = 102100	Diversity = 5.0597e-14	k_t
= [4, 3, 2, 2]				
Generation 1030:	Best = 2.300000e+03	FES = 103100	Diversity = 4.9048e-14	k_t
= [1, 1, 1, 1]				

Generation 1040: Best = 2.300000e+03 = [3, 2, 3, 2]	FES = 104100	Diversity = 5.0133e-14	k_t
Generation 1050: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 105100	Diversity = 4.8278e-14	k_t
Generation 1060: Best = 2.300000e+03 = [4, 3, 2, 2]	FES = 106100	Diversity = 5.1048e-14	k_t
Generation 1070: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 107100	Diversity = 5.1280e-14	k_t
Generation 1080: Best = 2.300000e+03 = [2, 4, 3, 2]	FES = 108100	Diversity = 5.0005e-14	k_t
Generation 1090: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 109100	Diversity = 5.0150e-14	k_t
Generation 1100: Best = 2.300000e+03 = [2, 4, 2, 2]	FES = 110100	Diversity = 4.9959e-14	k_t
Generation 1110: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 111100	Diversity = 4.9252e-14	k_t
Generation 1120: Best = 2.300000e+03 = [2, 4, 4, 3]	FES = 112100	Diversity = 5.0193e-14	k_t
Generation 1130: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 113100	Diversity = 4.9187e-14	k_t
Generation 1140: Best = 2.300000e+03 = [2, 3, 3, 2]	FES = 114100	Diversity = 4.9804e-14	k_t
Generation 1150: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 115100	Diversity = 5.1171e-14	k_t
Generation 1160: Best = 2.300000e+03 = [3, 3, 2, 2]	FES = 116100	Diversity = 5.1586e-14	k_t
Generation 1170: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 117100	Diversity = 5.1222e-14	k_t
Generation 1180: Best = 2.300000e+03 = [4, 2, 2, 2]	FES = 118100	Diversity = 5.1619e-14	k_t
Generation 1190: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 119100	Diversity = 5.0224e-14	k_t
Generation 1200: Best = 2.300000e+03 = [4, 2, 3, 4]	FES = 120100	Diversity = 5.1431e-14	k_t
Generation 1210: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 121100	Diversity = 5.1022e-14	k_t
Generation 1220: Best = 2.300000e+03 = [3, 4, 4, 4]	FES = 122100	Diversity = 5.0942e-14	k_t
Generation 1230: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 123100	Diversity = 5.1037e-14	k_t
Generation 1240: Best = 2.300000e+03 = [4, 2, 3, 2]	FES = 124100	Diversity = 5.1770e-14	k_t
Generation 1250: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 125100	Diversity = 5.0808e-14	k_t
Generation 1260: Best = 2.300000e+03 = [2, 4, 4, 3]	FES = 126100	Diversity = 4.9908e-14	k_t
Generation 1270: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 127100	Diversity = 4.8485e-14	k_t
Generation 1280: Best = 2.300000e+03 = [3, 3, 3, 2]	FES = 128100	Diversity = 5.0448e-14	k_t
Generation 1290: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 129100	Diversity = 4.8547e-14	k_t
Generation 1300: Best = 2.300000e+03 = [3, 2, 3, 4]	FES = 130100	Diversity = 5.0705e-14	k_t

Generation 1310: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 131100	Diversity = 5.0846e-14	k_t
Generation 1320: Best = 2.300000e+03 = [4, 2, 2, 3]	FES = 132100	Diversity = 5.0384e-14	k_t
Generation 1330: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 133100	Diversity = 4.8926e-14	k_t
Generation 1340: Best = 2.300000e+03 = [4, 4, 3, 3]	FES = 134100	Diversity = 4.9092e-14	k_t
Generation 1350: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 135100	Diversity = 4.9278e-14	k_t
Generation 1360: Best = 2.300000e+03 = [3, 3, 4, 4]	FES = 136100	Diversity = 4.8778e-14	k_t
Generation 1370: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 137100	Diversity = 4.9613e-14	k_t
Generation 1380: Best = 2.300000e+03 = [3, 4, 2, 3]	FES = 138100	Diversity = 5.0643e-14	k_t
Generation 1390: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 139100	Diversity = 4.9355e-14	k_t
Generation 1400: Best = 2.300000e+03 = [3, 4, 4, 4]	FES = 140100	Diversity = 5.2248e-14	k_t
Generation 1410: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 141100	Diversity = 5.0990e-14	k_t
Generation 1420: Best = 2.300000e+03 = [3, 4, 3, 4]	FES = 142100	Diversity = 5.2658e-14	k_t
Generation 1430: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 143100	Diversity = 5.2445e-14	k_t
Generation 1440: Best = 2.300000e+03 = [4, 4, 4, 2]	FES = 144100	Diversity = 5.2618e-14	k_t
Generation 1450: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 145100	Diversity = 5.3063e-14	k_t
Generation 1460: Best = 2.300000e+03 = [2, 4, 2, 4]	FES = 146100	Diversity = 5.1070e-14	k_t
Generation 1470: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 147100	Diversity = 5.0017e-14	k_t
Generation 1480: Best = 2.300000e+03 = [4, 4, 4, 2]	FES = 148100	Diversity = 5.0401e-14	k_t
Generation 1490: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 149100	Diversity = 5.2539e-14	k_t
Generation 1500: Best = 2.300000e+03 = [4, 3, 2, 3]	FES = 150100	Diversity = 5.1645e-14	k_t
Generation 1510: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 151100	Diversity = 5.2632e-14	k_t
Generation 1520: Best = 2.300000e+03 = [4, 2, 2, 3]	FES = 152100	Diversity = 5.0741e-14	k_t
Generation 1530: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 153100	Diversity = 4.9813e-14	k_t
Generation 1540: Best = 2.300000e+03 = [2, 4, 2, 4]	FES = 154100	Diversity = 5.0900e-14	k_t
Generation 1550: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 155100	Diversity = 4.9765e-14	k_t
Generation 1560: Best = 2.300000e+03 = [4, 3, 3, 3]	FES = 156100	Diversity = 5.0173e-14	k_t
Generation 1570: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 157100	Diversity = 5.1989e-14	k_t

Generation 1580: Best = 2.300000e+03 = [3, 2, 3, 3]	FES = 158100	Diversity = 4.9979e-14	k_t
Generation 1590: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 159100	Diversity = 5.0899e-14	k_t
Generation 1600: Best = 2.300000e+03 = [2, 3, 3, 4]	FES = 160100	Diversity = 5.0618e-14	k_t
Generation 1610: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 161100	Diversity = 5.0867e-14	k_t
Generation 1620: Best = 2.300000e+03 = [3, 2, 4, 4]	FES = 162100	Diversity = 5.1401e-14	k_t
Generation 1630: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 163100	Diversity = 5.3358e-14	k_t
Generation 1640: Best = 2.300000e+03 = [3, 4, 2, 3]	FES = 164100	Diversity = 5.2030e-14	k_t
Generation 1650: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 165100	Diversity = 5.2133e-14	k_t
Generation 1660: Best = 2.300000e+03 = [3, 2, 4, 2]	FES = 166100	Diversity = 5.2608e-14	k_t
Generation 1670: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 167100	Diversity = 5.3019e-14	k_t
Generation 1680: Best = 2.300000e+03 = [2, 4, 2, 3]	FES = 168100	Diversity = 5.1272e-14	k_t
Generation 1690: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 169100	Diversity = 4.9860e-14	k_t
Generation 1700: Best = 2.300000e+03 = [2, 2, 4, 2]	FES = 170100	Diversity = 5.1213e-14	k_t
Generation 1710: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 171100	Diversity = 5.2575e-14	k_t
Generation 1720: Best = 2.300000e+03 = [4, 4, 4, 2]	FES = 172100	Diversity = 5.3276e-14	k_t
Generation 1730: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 173100	Diversity = 5.2091e-14	k_t
Generation 1740: Best = 2.300000e+03 = [4, 2, 2, 3]	FES = 174100	Diversity = 5.1951e-14	k_t
Generation 1750: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 175100	Diversity = 5.2375e-14	k_t
Generation 1760: Best = 2.300000e+03 = [4, 2, 4, 3]	FES = 176100	Diversity = 5.2484e-14	k_t
Generation 1770: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 177100	Diversity = 5.1735e-14	k_t
Generation 1780: Best = 2.300000e+03 = [3, 2, 4, 2]	FES = 178100	Diversity = 5.1428e-14	k_t
Generation 1790: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 179100	Diversity = 4.8826e-14	k_t
Generation 1800: Best = 2.300000e+03 = [2, 4, 3, 2]	FES = 180100	Diversity = 5.0014e-14	k_t
Generation 1810: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 181100	Diversity = 5.0418e-14	k_t
Generation 1820: Best = 2.300000e+03 = [4, 3, 4, 2]	FES = 182100	Diversity = 4.8666e-14	k_t
Generation 1830: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 183100	Diversity = 4.8123e-14	k_t
Generation 1840: Best = 2.300000e+03 = [2, 4, 4, 3]	FES = 184100	Diversity = 5.0018e-14	k_t

Generation 1850: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 185100	Diversity = 5.0556e-14	k_t
Generation 1860: Best = 2.300000e+03 = [4, 3, 2, 2]	FES = 186100	Diversity = 5.3123e-14	k_t
Generation 1870: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 187100	Diversity = 5.0239e-14	k_t
Generation 1880: Best = 2.300000e+03 = [2, 2, 4, 3]	FES = 188100	Diversity = 5.0026e-14	k_t
Generation 1890: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 189100	Diversity = 4.9708e-14	k_t
Generation 1900: Best = 2.300000e+03 = [3, 2, 4, 3]	FES = 190100	Diversity = 5.0945e-14	k_t
Generation 1910: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 191100	Diversity = 4.8678e-14	k_t
Generation 1920: Best = 2.300000e+03 = [4, 2, 4, 3]	FES = 192100	Diversity = 5.2177e-14	k_t
Generation 1930: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 193100	Diversity = 5.2172e-14	k_t
Generation 1940: Best = 2.300000e+03 = [4, 2, 3, 3]	FES = 194100	Diversity = 5.3352e-14	k_t
Generation 1950: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 195100	Diversity = 5.0971e-14	k_t
Generation 1960: Best = 2.300000e+03 = [3, 2, 4, 4]	FES = 196100	Diversity = 4.8751e-14	k_t
Generation 1970: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 197100	Diversity = 4.8795e-14	k_t
Generation 1980: Best = 2.300000e+03 = [3, 3, 4, 3]	FES = 198100	Diversity = 4.8465e-14	k_t
Generation 1990: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 199100	Diversity = 4.8820e-14	k_t
Generation 2000: Best = 2.300000e+03 = [2, 4, 4, 2]	FES = 200100	Diversity = 5.3559e-14	k_t
Generation 2010: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 201100	Diversity = 5.2527e-14	k_t
Generation 2020: Best = 2.300000e+03 = [2, 3, 3, 2]	FES = 202100	Diversity = 5.1566e-14	k_t
Generation 2030: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 203100	Diversity = 5.1117e-14	k_t
Generation 2040: Best = 2.300000e+03 = [3, 4, 3, 4]	FES = 204100	Diversity = 4.9477e-14	k_t
Generation 2050: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 205100	Diversity = 5.0030e-14	k_t
Generation 2060: Best = 2.300000e+03 = [3, 3, 4, 2]	FES = 206100	Diversity = 5.1887e-14	k_t
Generation 2070: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 207100	Diversity = 5.0680e-14	k_t
Generation 2080: Best = 2.300000e+03 = [4, 2, 3, 3]	FES = 208100	Diversity = 5.2796e-14	k_t
Generation 2090: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 209100	Diversity = 5.1033e-14	k_t
Generation 2100: Best = 2.300000e+03 = [3, 4, 2, 4]	FES = 210100	Diversity = 4.8640e-14	k_t
Generation 2110: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 211100	Diversity = 4.9250e-14	k_t

Generation 2120: Best = 2.300000e+03 = [4, 4, 2, 4]	FES = 212100	Diversity = 4.9313e-14	k_t
Generation 2130: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 213100	Diversity = 4.9083e-14	k_t
Generation 2140: Best = 2.300000e+03 = [3, 4, 3, 3]	FES = 214100	Diversity = 4.8459e-14	k_t
Generation 2150: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 215100	Diversity = 4.8948e-14	k_t
Generation 2160: Best = 2.300000e+03 = [3, 3, 2, 3]	FES = 216100	Diversity = 5.0865e-14	k_t
Generation 2170: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 217100	Diversity = 4.9796e-14	k_t
Generation 2180: Best = 2.300000e+03 = [3, 2, 2, 3]	FES = 218100	Diversity = 5.2197e-14	k_t
Generation 2190: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 219100	Diversity = 5.0648e-14	k_t
Generation 2200: Best = 2.300000e+03 = [4, 3, 3, 3]	FES = 220100	Diversity = 5.1467e-14	k_t
Generation 2210: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 221100	Diversity = 5.1227e-14	k_t
Generation 2220: Best = 2.300000e+03 = [4, 2, 2, 4]	FES = 222100	Diversity = 4.9969e-14	k_t
Generation 2230: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 223100	Diversity = 4.9718e-14	k_t
Generation 2240: Best = 2.300000e+03 = [3, 3, 3, 2]	FES = 224100	Diversity = 5.1845e-14	k_t
Generation 2250: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 225100	Diversity = 5.0249e-14	k_t
Generation 2260: Best = 2.300000e+03 = [3, 3, 2, 2]	FES = 226100	Diversity = 4.9005e-14	k_t
Generation 2270: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 227100	Diversity = 5.1642e-14	k_t
Generation 2280: Best = 2.300000e+03 = [3, 4, 3, 4]	FES = 228100	Diversity = 5.0783e-14	k_t
Generation 2290: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 229100	Diversity = 4.8810e-14	k_t
Generation 2300: Best = 2.300000e+03 = [4, 4, 2, 4]	FES = 230100	Diversity = 4.7380e-14	k_t
Generation 2310: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 231100	Diversity = 4.8279e-14	k_t
Generation 2320: Best = 2.300000e+03 = [4, 2, 4, 2]	FES = 232100	Diversity = 4.8461e-14	k_t
Generation 2330: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 233100	Diversity = 4.8839e-14	k_t
Generation 2340: Best = 2.300000e+03 = [3, 4, 3, 2]	FES = 234100	Diversity = 5.0252e-14	k_t
Generation 2350: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 235100	Diversity = 5.0786e-14	k_t
Generation 2360: Best = 2.300000e+03 = [2, 3, 3, 3]	FES = 236100	Diversity = 4.9649e-14	k_t
Generation 2370: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 237100	Diversity = 4.9121e-14	k_t
Generation 2380: Best = 2.300000e+03 = [2, 3, 4, 4]	FES = 238100	Diversity = 5.0312e-14	k_t

Generation 2390: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 239100	Diversity = 4.9262e-14	k_t
Generation 2400: Best = 2.300000e+03 = [3, 4, 2, 3]	FES = 240100	Diversity = 5.0721e-14	k_t
Generation 2410: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 241100	Diversity = 5.0006e-14	k_t
Generation 2420: Best = 2.300000e+03 = [4, 3, 3, 4]	FES = 242100	Diversity = 5.0899e-14	k_t
Generation 2430: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 243100	Diversity = 5.0664e-14	k_t
Generation 2440: Best = 2.300000e+03 = [3, 2, 3, 3]	FES = 244100	Diversity = 4.9620e-14	k_t
Generation 2450: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 245100	Diversity = 4.8858e-14	k_t
Generation 2460: Best = 2.300000e+03 = [3, 3, 4, 4]	FES = 246100	Diversity = 4.8811e-14	k_t
Generation 2470: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 247100	Diversity = 4.9270e-14	k_t
Generation 2480: Best = 2.300000e+03 = [2, 3, 4, 4]	FES = 248100	Diversity = 4.8777e-14	k_t
Generation 2490: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 249100	Diversity = 4.9487e-14	k_t
Generation 2500: Best = 2.300000e+03 = [3, 4, 4, 4]	FES = 250100	Diversity = 4.7649e-14	k_t
Generation 2510: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 251100	Diversity = 5.1323e-14	k_t
Generation 2520: Best = 2.300000e+03 = [2, 3, 4, 3]	FES = 252100	Diversity = 5.2176e-14	k_t
Generation 2530: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 253100	Diversity = 5.0038e-14	k_t
Generation 2540: Best = 2.300000e+03 = [2, 3, 3, 2]	FES = 254100	Diversity = 5.2587e-14	k_t
Generation 2550: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 255100	Diversity = 4.9859e-14	k_t
Generation 2560: Best = 2.300000e+03 = [2, 4, 2, 3]	FES = 256100	Diversity = 5.0122e-14	k_t
Generation 2570: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 257100	Diversity = 4.9813e-14	k_t
Generation 2580: Best = 2.300000e+03 = [3, 3, 4, 4]	FES = 258100	Diversity = 4.8852e-14	k_t
Generation 2590: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 259100	Diversity = 4.8833e-14	k_t
Generation 2600: Best = 2.300000e+03 = [3, 2, 2, 3]	FES = 260100	Diversity = 4.9565e-14	k_t
Generation 2610: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 261100	Diversity = 5.1632e-14	k_t
Generation 2620: Best = 2.300000e+03 = [4, 2, 2, 2]	FES = 262100	Diversity = 5.1184e-14	k_t
Generation 2630: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 263100	Diversity = 5.0913e-14	k_t
Generation 2640: Best = 2.300000e+03 = [2, 3, 3, 4]	FES = 264100	Diversity = 5.1966e-14	k_t
Generation 2650: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 265100	Diversity = 5.1920e-14	k_t

Generation 2660: Best = 2.300000e+03 = [3, 4, 2, 4]	FES = 266100	Diversity = 5.1580e-14	k_t
Generation 2670: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 267100	Diversity = 5.2011e-14	k_t
Generation 2680: Best = 2.300000e+03 = [3, 2, 2, 2]	FES = 268100	Diversity = 5.1718e-14	k_t
Generation 2690: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 269100	Diversity = 5.1868e-14	k_t
Generation 2700: Best = 2.300000e+03 = [2, 3, 3, 2]	FES = 270100	Diversity = 5.3386e-14	k_t
Generation 2710: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 271100	Diversity = 5.0805e-14	k_t
Generation 2720: Best = 2.300000e+03 = [4, 4, 4, 3]	FES = 272100	Diversity = 5.2123e-14	k_t
Generation 2730: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 273100	Diversity = 5.0351e-14	k_t
Generation 2740: Best = 2.300000e+03 = [2, 2, 4, 2]	FES = 274100	Diversity = 5.1220e-14	k_t
Generation 2750: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 275100	Diversity = 5.1741e-14	k_t
Generation 2760: Best = 2.300000e+03 = [4, 3, 4, 4]	FES = 276100	Diversity = 5.1113e-14	k_t
Generation 2770: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 277100	Diversity = 5.0703e-14	k_t
Generation 2780: Best = 2.300000e+03 = [2, 3, 2, 3]	FES = 278100	Diversity = 5.0984e-14	k_t
Generation 2790: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 279100	Diversity = 5.3504e-14	k_t
Generation 2800: Best = 2.300000e+03 = [2, 4, 3, 4]	FES = 280100	Diversity = 5.2951e-14	k_t
Generation 2810: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 281100	Diversity = 5.1536e-14	k_t
Generation 2820: Best = 2.300000e+03 = [3, 3, 4, 3]	FES = 282100	Diversity = 5.1196e-14	k_t
Generation 2830: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 283100	Diversity = 5.1257e-14	k_t
Generation 2840: Best = 2.300000e+03 = [2, 4, 3, 4]	FES = 284100	Diversity = 5.2500e-14	k_t
Generation 2850: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 285100	Diversity = 5.2283e-14	k_t
Generation 2860: Best = 2.300000e+03 = [2, 3, 3, 4]	FES = 286100	Diversity = 5.0730e-14	k_t
Generation 2870: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 287100	Diversity = 5.1682e-14	k_t
Generation 2880: Best = 2.300000e+03 = [4, 2, 4, 3]	FES = 288100	Diversity = 5.1041e-14	k_t
Generation 2890: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 289100	Diversity = 5.0890e-14	k_t
Generation 2900: Best = 2.300000e+03 = [4, 2, 4, 2]	FES = 290100	Diversity = 5.0210e-14	k_t
Generation 2910: Best = 2.300000e+03 = [1, 1, 1, 1]	FES = 291100	Diversity = 5.0380e-14	k_t
Generation 2920: Best = 2.300000e+03 = [2, 4, 3, 4]	FES = 292100	Diversity = 5.0836e-14	k_t

Generation 2930: Best = 2.300000e+03 FES = 293100 Diversity = 5.0341e-14 k_t
 = [1, 1, 1, 1]
 Generation 2940: Best = 2.300000e+03 FES = 294100 Diversity = 5.2634e-14 k_t
 = [3, 4, 3, 3]
 Generation 2950: Best = 2.300000e+03 FES = 295100 Diversity = 4.9375e-14 k_t
 = [1, 1, 1, 1]
 Generation 2960: Best = 2.300000e+03 FES = 296100 Diversity = 4.9721e-14 k_t
 = [3, 3, 4, 2]
 Generation 2970: Best = 2.300000e+03 FES = 297100 Diversity = 4.9038e-14 k_t
 = [1, 1, 1, 1]
 Generation 2980: Best = 2.300000e+03 FES = 298100 Diversity = 4.8993e-14 k_t
 = [2, 3, 3, 2]
 Generation 2990: Best = 2.300000e+03 FES = 299100 Diversity = 4.7348e-14 k_t
 = [1, 1, 1, 1]

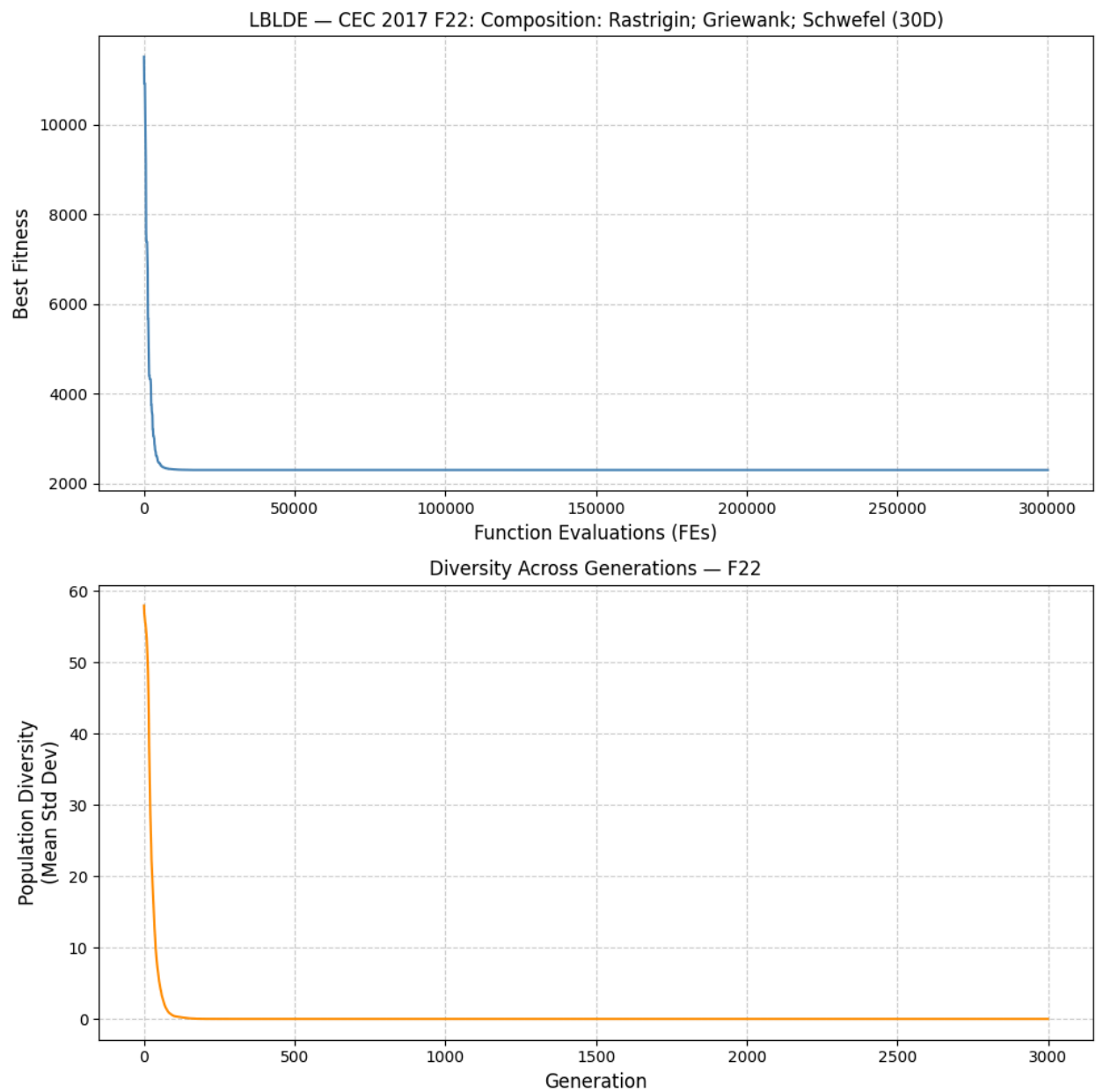
Optimization completed!

Final Best Fitness = 2.300000e+03

Total Function Evaluations = 300000

```

=====
=
Results:
  Best Fitness      : 2.300000e+03
  Global Optimum    : 2.200000e+03
  Error (|best-optimum|) : 1.000000e+02
  Best Solution (5 dim) : [-52.41266063  11.16831479  23.05456506  35.08347775
30.48935237]
  D^(0) (init. diversity): 5.7976e+01
=====
=
  
```



Graph saved as convergence_F22.png

```
In [33]: # =====
# F23 — Composition Function 3 (Rosenbrock; Ackley; Schwefel; Rastrigin)
# =====
run_and_plot('F23', make_wrapper(composition.f23), 2300.0, 'Composition: Rosen
```

```

=====
=
LBLDE_s on CEC 2017 F23 – Single Run
=====
=
Function      : Composition: Rosenbrock; Ackley; Schwefel; Rastrigin (30D)
Bounds        : [-100, 100]
Global optimum : 2300.0
Initial D^(0) = 5.7976e+01
Generation 0: Best Fitness = 4.071208e+03 | Diversity = 5.7976e+01
Generation 10: Best = 3.154118e+03 FES = 1100 Diversity = 4.6539e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 3.017265e+03 FES = 2100 Diversity = 3.1984e+01 k_t
= [3, 3, 2, 2]
Generation 30: Best = 2.912823e+03 FES = 3100 Diversity = 2.6172e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 2.912823e+03 FES = 4100 Diversity = 2.3436e+01 k_t
= [2, 4, 4, 4]
Generation 50: Best = 2.873758e+03 FES = 5100 Diversity = 2.1714e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 2.873758e+03 FES = 6100 Diversity = 2.0571e+01 k_t
= [2, 4, 2, 3]
Generation 70: Best = 2.863302e+03 FES = 7100 Diversity = 2.0008e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 2.863302e+03 FES = 8100 Diversity = 1.9350e+01 k_t
= [4, 3, 4, 3]
Generation 90: Best = 2.837713e+03 FES = 9100 Diversity = 1.8905e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 2.837713e+03 FES = 10100 Diversity = 1.8602e+01 k_t
= [4, 2, 2, 3]
Generation 110: Best = 2.837713e+03 FES = 11100 Diversity = 1.8185e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 2.812390e+03 FES = 12100 Diversity = 1.8060e+01 k_t
= [2, 4, 3, 3]
Generation 130: Best = 2.809442e+03 FES = 13100 Diversity = 1.7851e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 2.809442e+03 FES = 14100 Diversity = 1.7859e+01 k_t
= [4, 4, 2, 4]
Generation 150: Best = 2.809442e+03 FES = 15100 Diversity = 1.7714e+01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 2.809442e+03 FES = 16100 Diversity = 1.7554e+01 k_t
= [3, 3, 2, 2]
Generation 170: Best = 2.808741e+03 FES = 17100 Diversity = 1.7273e+01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 2.801962e+03 FES = 18100 Diversity = 1.7055e+01 k_t
= [3, 3, 2, 3]
Generation 190: Best = 2.801962e+03 FES = 19100 Diversity = 1.6955e+01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 2.798677e+03 FES = 20100 Diversity = 1.6881e+01 k_t
= [4, 2, 2, 4]
Generation 210: Best = 2.792047e+03 FES = 21100 Diversity = 1.6801e+01 k_t
= [1, 1, 1, 1]
Generation 220: Best = 2.792047e+03 FES = 22100 Diversity = 1.6672e+01 k_t
= [2, 3, 2, 3]

```


Generation 230:	Best = 2.782785e+03	FES = 23100	Diversity = 1.6615e+01	k_t = [1, 1, 1, 1]
Generation 240:	Best = 2.782785e+03	FES = 24100	Diversity = 1.6559e+01	k_t = [3, 3, 3, 3]
Generation 250:	Best = 2.782785e+03	FES = 25100	Diversity = 1.6461e+01	k_t = [1, 1, 1, 1]
Generation 260:	Best = 2.782785e+03	FES = 26100	Diversity = 1.6392e+01	k_t = [2, 4, 4, 2]
Generation 270:	Best = 2.782785e+03	FES = 27100	Diversity = 1.6399e+01	k_t = [1, 1, 1, 1]
Generation 280:	Best = 2.778065e+03	FES = 28100	Diversity = 1.6229e+01	k_t = [4, 3, 2, 3]
Generation 290:	Best = 2.778065e+03	FES = 29100	Diversity = 1.6184e+01	k_t = [1, 1, 1, 1]
Generation 300:	Best = 2.778001e+03	FES = 30100	Diversity = 1.6019e+01	k_t = [3, 2, 2, 4]
Generation 310:	Best = 2.768718e+03	FES = 31100	Diversity = 1.6016e+01	k_t = [1, 1, 1, 1]
Generation 320:	Best = 2.768718e+03	FES = 32100	Diversity = 1.5964e+01	k_t = [4, 2, 4, 3]
Generation 330:	Best = 2.768718e+03	FES = 33100	Diversity = 1.5932e+01	k_t = [1, 1, 1, 1]
Generation 340:	Best = 2.768718e+03	FES = 34100	Diversity = 1.5929e+01	k_t = [3, 3, 2, 3]
Generation 350:	Best = 2.766960e+03	FES = 35100	Diversity = 1.5921e+01	k_t = [1, 1, 1, 1]
Generation 360:	Best = 2.766960e+03	FES = 36100	Diversity = 1.5945e+01	k_t = [3, 3, 3, 3]
Generation 370:	Best = 2.766960e+03	FES = 37100	Diversity = 1.5895e+01	k_t = [1, 1, 1, 1]
Generation 380:	Best = 2.765594e+03	FES = 38100	Diversity = 1.5837e+01	k_t = [4, 2, 3, 4]
Generation 390:	Best = 2.763540e+03	FES = 39100	Diversity = 1.5836e+01	k_t = [1, 1, 1, 1]
Generation 400:	Best = 2.759055e+03	FES = 40100	Diversity = 1.5746e+01	k_t = [2, 3, 2, 2]
Generation 410:	Best = 2.759055e+03	FES = 41100	Diversity = 1.5650e+01	k_t = [1, 1, 1, 1]
Generation 420:	Best = 2.759055e+03	FES = 42100	Diversity = 1.5641e+01	k_t = [4, 2, 2, 4]
Generation 430:	Best = 2.759055e+03	FES = 43100	Diversity = 1.5641e+01	k_t = [1, 1, 1, 1]
Generation 440:	Best = 2.756645e+03	FES = 44100	Diversity = 1.5778e+01	k_t = [3, 4, 2, 3]
Generation 450:	Best = 2.756645e+03	FES = 45100	Diversity = 1.5755e+01	k_t = [1, 1, 1, 1]
Generation 460:	Best = 2.756645e+03	FES = 46100	Diversity = 1.5768e+01	k_t = [3, 3, 4, 4]
Generation 470:	Best = 2.755997e+03	FES = 47100	Diversity = 1.5901e+01	k_t = [1, 1, 1, 1]
Generation 480:	Best = 2.754241e+03	FES = 48100	Diversity = 1.5922e+01	k_t = [4, 3, 4, 3]
Generation 490:	Best = 2.749061e+03	FES = 49100	Diversity = 1.5787e+01	k_t = [1, 1, 1, 1]

Generation 500:	Best = 2.748788e+03	FES = 50100	Diversity = 1.5779e+01	k_t
= [2, 3, 4, 2]				
Generation 510:	Best = 2.747887e+03	FES = 51100	Diversity = 1.5769e+01	k_t
= [1, 1, 1, 1]				
Generation 520:	Best = 2.745530e+03	FES = 52100	Diversity = 1.5756e+01	k_t
= [4, 3, 3, 4]				
Generation 530:	Best = 2.744170e+03	FES = 53100	Diversity = 1.5746e+01	k_t
= [1, 1, 1, 1]				
Generation 540:	Best = 2.742892e+03	FES = 54100	Diversity = 1.5726e+01	k_t
= [2, 4, 2, 4]				
Generation 550:	Best = 2.742892e+03	FES = 55100	Diversity = 1.5718e+01	k_t
= [1, 1, 1, 1]				
Generation 560:	Best = 2.738397e+03	FES = 56100	Diversity = 1.5713e+01	k_t
= [2, 3, 4, 3]				
Generation 570:	Best = 2.738397e+03	FES = 57100	Diversity = 1.5646e+01	k_t
= [1, 1, 1, 1]				
Generation 580:	Best = 2.732564e+03	FES = 58100	Diversity = 1.5625e+01	k_t
= [3, 3, 3, 4]				
Generation 590:	Best = 2.732564e+03	FES = 59100	Diversity = 1.5611e+01	k_t
= [1, 1, 1, 1]				
Generation 600:	Best = 2.732564e+03	FES = 60100	Diversity = 1.5608e+01	k_t
= [3, 4, 3, 2]				
Generation 610:	Best = 2.732564e+03	FES = 61100	Diversity = 1.5564e+01	k_t
= [1, 1, 1, 1]				
Generation 620:	Best = 2.731584e+03	FES = 62100	Diversity = 1.5525e+01	k_t
= [2, 3, 4, 4]				
Generation 630:	Best = 2.731584e+03	FES = 63100	Diversity = 1.5501e+01	k_t
= [1, 1, 1, 1]				
Generation 640:	Best = 2.731584e+03	FES = 64100	Diversity = 1.5478e+01	k_t
= [3, 2, 2, 4]				
Generation 650:	Best = 2.730567e+03	FES = 65100	Diversity = 1.5422e+01	k_t
= [1, 1, 1, 1]				
Generation 660:	Best = 2.726587e+03	FES = 66100	Diversity = 1.5330e+01	k_t
= [4, 2, 3, 3]				
Generation 670:	Best = 2.726587e+03	FES = 67100	Diversity = 1.5283e+01	k_t
= [1, 1, 1, 1]				
Generation 680:	Best = 2.725896e+03	FES = 68100	Diversity = 1.5302e+01	k_t
= [2, 4, 4, 3]				
Generation 690:	Best = 2.725896e+03	FES = 69100	Diversity = 1.5298e+01	k_t
= [1, 1, 1, 1]				
Generation 700:	Best = 2.725896e+03	FES = 70100	Diversity = 1.5331e+01	k_t
= [4, 4, 2, 2]				
Generation 710:	Best = 2.725896e+03	FES = 71100	Diversity = 1.5267e+01	k_t
= [1, 1, 1, 1]				
Generation 720:	Best = 2.725217e+03	FES = 72100	Diversity = 1.5268e+01	k_t
= [2, 3, 4, 4]				
Generation 730:	Best = 2.725217e+03	FES = 73100	Diversity = 1.5204e+01	k_t
= [1, 1, 1, 1]				
Generation 740:	Best = 2.725217e+03	FES = 74100	Diversity = 1.5208e+01	k_t
= [2, 2, 3, 3]				
Generation 750:	Best = 2.725217e+03	FES = 75100	Diversity = 1.5208e+01	k_t
= [1, 1, 1, 1]				
Generation 760:	Best = 2.725217e+03	FES = 76100	Diversity = 1.5142e+01	k_t
= [4, 4, 2, 2]				

Generation 770:	Best = 2.725217e+03	FES = 77100	Diversity = 1.5125e+01	k_t = [1, 1, 1, 1]
Generation 780:	Best = 2.725217e+03	FES = 78100	Diversity = 1.5049e+01	k_t = [4, 3, 2, 2]
Generation 790:	Best = 2.725217e+03	FES = 79100	Diversity = 1.5069e+01	k_t = [1, 1, 1, 1]
Generation 800:	Best = 2.725217e+03	FES = 80100	Diversity = 1.5162e+01	k_t = [4, 3, 3, 3]
Generation 810:	Best = 2.723869e+03	FES = 81100	Diversity = 1.5140e+01	k_t = [1, 1, 1, 1]
Generation 820:	Best = 2.723869e+03	FES = 82100	Diversity = 1.5172e+01	k_t = [2, 2, 4, 3]
Generation 830:	Best = 2.723869e+03	FES = 83100	Diversity = 1.5193e+01	k_t = [1, 1, 1, 1]
Generation 840:	Best = 2.723869e+03	FES = 84100	Diversity = 1.5221e+01	k_t = [3, 4, 3, 3]
Generation 850:	Best = 2.723869e+03	FES = 85100	Diversity = 1.5165e+01	k_t = [1, 1, 1, 1]
Generation 860:	Best = 2.723869e+03	FES = 86100	Diversity = 1.5232e+01	k_t = [2, 4, 4, 4]
Generation 870:	Best = 2.723869e+03	FES = 87100	Diversity = 1.5067e+01	k_t = [1, 1, 1, 1]
Generation 880:	Best = 2.723336e+03	FES = 88100	Diversity = 1.5006e+01	k_t = [4, 3, 4, 4]
Generation 890:	Best = 2.723336e+03	FES = 89100	Diversity = 1.5154e+01	k_t = [1, 1, 1, 1]
Generation 900:	Best = 2.723336e+03	FES = 90100	Diversity = 1.5103e+01	k_t = [3, 4, 2, 2]
Generation 910:	Best = 2.722813e+03	FES = 91100	Diversity = 1.5162e+01	k_t = [1, 1, 1, 1]
Generation 920:	Best = 2.721532e+03	FES = 92100	Diversity = 1.5177e+01	k_t = [4, 4, 3, 2]
Generation 930:	Best = 2.721532e+03	FES = 93100	Diversity = 1.5046e+01	k_t = [1, 1, 1, 1]
Generation 940:	Best = 2.721532e+03	FES = 94100	Diversity = 1.4912e+01	k_t = [4, 2, 4, 2]
Generation 950:	Best = 2.721532e+03	FES = 95100	Diversity = 1.4885e+01	k_t = [1, 1, 1, 1]
Generation 960:	Best = 2.720881e+03	FES = 96100	Diversity = 1.4903e+01	k_t = [3, 4, 4, 3]
Generation 970:	Best = 2.720638e+03	FES = 97100	Diversity = 1.5053e+01	k_t = [1, 1, 1, 1]
Generation 980:	Best = 2.720638e+03	FES = 98100	Diversity = 1.5053e+01	k_t = [2, 2, 4, 3]
Generation 990:	Best = 2.720638e+03	FES = 99100	Diversity = 1.4945e+01	k_t = [1, 1, 1, 1]
Generation 1000:	Best = 2.720356e+03	FES = 100100	Diversity = 1.4893e+01	k_t = [4, 3, 3, 3]
Generation 1010:	Best = 2.720329e+03	FES = 101100	Diversity = 1.4815e+01	k_t = [1, 1, 1, 1]
Generation 1020:	Best = 2.719265e+03	FES = 102100	Diversity = 1.4853e+01	k_t = [2, 4, 4, 2]
Generation 1030:	Best = 2.717871e+03	FES = 103100	Diversity = 1.4846e+01	k_t = [1, 1, 1, 1]

Generation 1040: Best = 2.717871e+03 = [4, 2, 4, 2]	FES = 104100	Diversity = 1.4864e+01	k_t
Generation 1050: Best = 2.717871e+03 = [1, 1, 1, 1]	FES = 105100	Diversity = 1.4766e+01	k_t
Generation 1060: Best = 2.713160e+03 = [4, 2, 4, 4]	FES = 106100	Diversity = 1.4761e+01	k_t
Generation 1070: Best = 2.713160e+03 = [1, 1, 1, 1]	FES = 107100	Diversity = 1.4766e+01	k_t
Generation 1080: Best = 2.708111e+03 = [3, 4, 3, 3]	FES = 108100	Diversity = 1.4746e+01	k_t
Generation 1090: Best = 2.708111e+03 = [1, 1, 1, 1]	FES = 109100	Diversity = 1.4729e+01	k_t
Generation 1100: Best = 2.708111e+03 = [4, 4, 2, 3]	FES = 110100	Diversity = 1.4655e+01	k_t
Generation 1110: Best = 2.705715e+03 = [1, 1, 1, 1]	FES = 111100	Diversity = 1.4574e+01	k_t
Generation 1120: Best = 2.704509e+03 = [3, 4, 2, 2]	FES = 112100	Diversity = 1.4561e+01	k_t
Generation 1130: Best = 2.704509e+03 = [1, 1, 1, 1]	FES = 113100	Diversity = 1.4563e+01	k_t
Generation 1140: Best = 2.704509e+03 = [2, 4, 4, 3]	FES = 114100	Diversity = 1.4527e+01	k_t
Generation 1150: Best = 2.704509e+03 = [1, 1, 1, 1]	FES = 115100	Diversity = 1.4527e+01	k_t
Generation 1160: Best = 2.704509e+03 = [2, 3, 2, 3]	FES = 116100	Diversity = 1.4530e+01	k_t
Generation 1170: Best = 2.704478e+03 = [1, 1, 1, 1]	FES = 117100	Diversity = 1.4529e+01	k_t
Generation 1180: Best = 2.704478e+03 = [4, 3, 2, 4]	FES = 118100	Diversity = 1.4512e+01	k_t
Generation 1190: Best = 2.704413e+03 = [1, 1, 1, 1]	FES = 119100	Diversity = 1.4511e+01	k_t
Generation 1200: Best = 2.704413e+03 = [3, 4, 2, 4]	FES = 120100	Diversity = 1.4511e+01	k_t
Generation 1210: Best = 2.704413e+03 = [1, 1, 1, 1]	FES = 121100	Diversity = 1.4509e+01	k_t
Generation 1220: Best = 2.704413e+03 = [4, 3, 2, 3]	FES = 122100	Diversity = 1.4481e+01	k_t
Generation 1230: Best = 2.702751e+03 = [1, 1, 1, 1]	FES = 123100	Diversity = 1.4486e+01	k_t
Generation 1240: Best = 2.701922e+03 = [4, 3, 4, 3]	FES = 124100	Diversity = 1.4492e+01	k_t
Generation 1250: Best = 2.701922e+03 = [1, 1, 1, 1]	FES = 125100	Diversity = 1.4473e+01	k_t
Generation 1260: Best = 2.701453e+03 = [2, 4, 4, 2]	FES = 126100	Diversity = 1.4474e+01	k_t
Generation 1270: Best = 2.701453e+03 = [1, 1, 1, 1]	FES = 127100	Diversity = 1.4344e+01	k_t
Generation 1280: Best = 2.701453e+03 = [4, 2, 2, 3]	FES = 128100	Diversity = 1.4285e+01	k_t
Generation 1290: Best = 2.701453e+03 = [1, 1, 1, 1]	FES = 129100	Diversity = 1.4285e+01	k_t
Generation 1300: Best = 2.701453e+03 = [4, 3, 4, 2]	FES = 130100	Diversity = 1.4283e+01	k_t

Generation 1310: Best = 2.701453e+03 = [1, 1, 1, 1]	FES = 131100	Diversity = 1.4282e+01	k_t
Generation 1320: Best = 2.700166e+03 = [3, 4, 3, 2]	FES = 132100	Diversity = 1.4280e+01	k_t
Generation 1330: Best = 2.700166e+03 = [1, 1, 1, 1]	FES = 133100	Diversity = 1.4263e+01	k_t
Generation 1340: Best = 2.699553e+03 = [2, 3, 2, 4]	FES = 134100	Diversity = 1.4258e+01	k_t
Generation 1350: Best = 2.699553e+03 = [1, 1, 1, 1]	FES = 135100	Diversity = 1.4240e+01	k_t
Generation 1360: Best = 2.698397e+03 = [4, 4, 3, 4]	FES = 136100	Diversity = 1.4239e+01	k_t
Generation 1370: Best = 2.698397e+03 = [1, 1, 1, 1]	FES = 137100	Diversity = 1.4072e+01	k_t
Generation 1380: Best = 2.698397e+03 = [2, 3, 4, 4]	FES = 138100	Diversity = 1.4000e+01	k_t
Generation 1390: Best = 2.693375e+03 = [1, 1, 1, 1]	FES = 139100	Diversity = 1.3998e+01	k_t
Generation 1400: Best = 2.693375e+03 = [3, 4, 2, 2]	FES = 140100	Diversity = 1.3996e+01	k_t
Generation 1410: Best = 2.693262e+03 = [1, 1, 1, 1]	FES = 141100	Diversity = 1.3970e+01	k_t
Generation 1420: Best = 2.691948e+03 = [2, 2, 3, 3]	FES = 142100	Diversity = 1.3943e+01	k_t
Generation 1430: Best = 2.691948e+03 = [1, 1, 1, 1]	FES = 143100	Diversity = 1.3944e+01	k_t
Generation 1440: Best = 2.691948e+03 = [2, 4, 2, 4]	FES = 144100	Diversity = 1.3941e+01	k_t
Generation 1450: Best = 2.691948e+03 = [1, 1, 1, 1]	FES = 145100	Diversity = 1.3898e+01	k_t
Generation 1460: Best = 2.691948e+03 = [2, 4, 4, 4]	FES = 146100	Diversity = 1.3884e+01	k_t
Generation 1470: Best = 2.691689e+03 = [1, 1, 1, 1]	FES = 147100	Diversity = 1.3884e+01	k_t
Generation 1480: Best = 2.691084e+03 = [4, 3, 3, 4]	FES = 148100	Diversity = 1.3878e+01	k_t
Generation 1490: Best = 2.689823e+03 = [1, 1, 1, 1]	FES = 149100	Diversity = 1.3762e+01	k_t
Generation 1500: Best = 2.689823e+03 = [4, 3, 4, 2]	FES = 150100	Diversity = 1.3666e+01	k_t
Generation 1510: Best = 2.689823e+03 = [1, 1, 1, 1]	FES = 151100	Diversity = 1.3658e+01	k_t
Generation 1520: Best = 2.689412e+03 = [4, 3, 3, 2]	FES = 152100	Diversity = 1.3667e+01	k_t
Generation 1530: Best = 2.684588e+03 = [1, 1, 1, 1]	FES = 153100	Diversity = 1.3602e+01	k_t
Generation 1540: Best = 2.684468e+03 = [2, 2, 4, 3]	FES = 154100	Diversity = 1.3549e+01	k_t
Generation 1550: Best = 2.684357e+03 = [1, 1, 1, 1]	FES = 155100	Diversity = 1.3453e+01	k_t
Generation 1560: Best = 2.684051e+03 = [2, 3, 3, 2]	FES = 156100	Diversity = 1.3457e+01	k_t
Generation 1570: Best = 2.684051e+03 = [1, 1, 1, 1]	FES = 157100	Diversity = 1.3456e+01	k_t

Generation 1580: Best = 2.683380e+03 = [4, 2, 3, 2]	FES = 158100	Diversity = 1.3475e+01	k_t
Generation 1590: Best = 2.683369e+03 = [1, 1, 1, 1]	FES = 159100	Diversity = 1.3400e+01	k_t
Generation 1600: Best = 2.683369e+03 = [2, 4, 2, 4]	FES = 160100	Diversity = 1.3240e+01	k_t
Generation 1610: Best = 2.682822e+03 = [1, 1, 1, 1]	FES = 161100	Diversity = 1.3234e+01	k_t
Generation 1620: Best = 2.682822e+03 = [2, 4, 4, 4]	FES = 162100	Diversity = 1.2910e+01	k_t
Generation 1630: Best = 2.681707e+03 = [1, 1, 1, 1]	FES = 163100	Diversity = 1.2193e+01	k_t
Generation 1640: Best = 2.680838e+03 = [2, 3, 4, 2]	FES = 164100	Diversity = 1.0885e+01	k_t
Generation 1650: Best = 2.679753e+03 = [1, 1, 1, 1]	FES = 165100	Diversity = 8.7882e+00	k_t
Generation 1660: Best = 2.679654e+03 = [4, 4, 3, 3]	FES = 166100	Diversity = 8.1082e+00	k_t
Generation 1670: Best = 2.677489e+03 = [1, 1, 1, 1]	FES = 167100	Diversity = 6.9593e+00	k_t
Generation 1680: Best = 2.673413e+03 = [4, 3, 4, 4]	FES = 168100	Diversity = 6.2510e+00	k_t
Generation 1690: Best = 2.673166e+03 = [1, 1, 1, 1]	FES = 169100	Diversity = 5.3456e+00	k_t
Generation 1700: Best = 2.671841e+03 = [2, 2, 2, 2]	FES = 170100	Diversity = 4.0172e+00	k_t
Generation 1710: Best = 2.670686e+03 = [1, 1, 1, 1]	FES = 171100	Diversity = 1.4730e+00	k_t
Generation 1720: Best = 2.669995e+03 = [4, 4, 4, 3]	FES = 172100	Diversity = 1.0093e+00	k_t
Generation 1730: Best = 2.669585e+03 = [1, 1, 1, 1]	FES = 173100	Diversity = 3.5376e-01	k_t
Generation 1740: Best = 2.669004e+03 = [4, 2, 3, 3]	FES = 174100	Diversity = 2.3923e-01	k_t
Generation 1750: Best = 2.668659e+03 = [1, 1, 1, 1]	FES = 175100	Diversity = 1.7727e-01	k_t
Generation 1760: Best = 2.668539e+03 = [4, 4, 2, 2]	FES = 176100	Diversity = 1.2263e-01	k_t
Generation 1770: Best = 2.668477e+03 = [1, 1, 1, 1]	FES = 177100	Diversity = 8.8707e-02	k_t
Generation 1780: Best = 2.668421e+03 = [2, 4, 3, 3]	FES = 178100	Diversity = 6.0670e-02	k_t
Generation 1790: Best = 2.668394e+03 = [1, 1, 1, 1]	FES = 179100	Diversity = 4.3562e-02	k_t
Generation 1800: Best = 2.668389e+03 = [3, 3, 2, 4]	FES = 180100	Diversity = 2.9216e-02	k_t
Generation 1810: Best = 2.668383e+03 = [1, 1, 1, 1]	FES = 181100	Diversity = 2.0658e-02	k_t
Generation 1820: Best = 2.668380e+03 = [4, 4, 3, 2]	FES = 182100	Diversity = 1.4621e-02	k_t
Generation 1830: Best = 2.668379e+03 = [1, 1, 1, 1]	FES = 183100	Diversity = 1.0495e-02	k_t
Generation 1840: Best = 2.668378e+03 = [4, 3, 2, 4]	FES = 184100	Diversity = 7.0118e-03	k_t

Generation 1850: Best = 2.668378e+03 = [1, 1, 1, 1]	FES = 185100	Diversity = 4.8922e-03	k_t
Generation 1860: Best = 2.668378e+03 = [4, 2, 3, 2]	FES = 186100	Diversity = 3.3029e-03	k_t
Generation 1870: Best = 2.668378e+03 = [1, 1, 1, 1]	FES = 187100	Diversity = 2.2386e-03	k_t
Generation 1880: Best = 2.668377e+03 = [4, 2, 3, 3]	FES = 188100	Diversity = 1.6765e-03	k_t
Generation 1890: Best = 2.668377e+03 = [1, 1, 1, 1]	FES = 189100	Diversity = 1.2667e-03	k_t
Generation 1900: Best = 2.668377e+03 = [2, 2, 2, 4]	FES = 190100	Diversity = 9.5586e-04	k_t
Generation 1910: Best = 2.668377e+03 = [1, 1, 1, 1]	FES = 191100	Diversity = 7.3657e-04	k_t
Generation 1920: Best = 2.668377e+03 = [2, 2, 2, 4]	FES = 192100	Diversity = 5.9825e-04	k_t
Generation 1930: Best = 2.668377e+03 = [1, 1, 1, 1]	FES = 193100	Diversity = 4.7190e-04	k_t
Generation 1940: Best = 2.668377e+03 = [2, 4, 3, 4]	FES = 194100	Diversity = 3.6933e-04	k_t
Generation 1950: Best = 2.668377e+03 = [1, 1, 1, 1]	FES = 195100	Diversity = 2.9371e-04	k_t
Generation 1960: Best = 2.668377e+03 = [3, 4, 3, 4]	FES = 196100	Diversity = 2.3327e-04	k_t
Generation 1970: Best = 2.668377e+03 = [1, 1, 1, 1]	FES = 197100	Diversity = 1.9061e-04	k_t
Generation 1980: Best = 2.668377e+03 = [3, 3, 3, 2]	FES = 198100	Diversity = 1.5342e-04	k_t
Generation 1990: Best = 2.668377e+03 = [1, 1, 1, 1]	FES = 199100	Diversity = 1.2413e-04	k_t
Generation 2000: Best = 2.668377e+03 = [4, 3, 2, 4]	FES = 200100	Diversity = 9.9497e-05	k_t
Generation 2010: Best = 2.668377e+03 = [1, 1, 1, 1]	FES = 201100	Diversity = 8.6708e-05	k_t
Generation 2020: Best = 2.668377e+03 = [4, 4, 4, 4]	FES = 202100	Diversity = 7.3576e-05	k_t
Generation 2030: Best = 2.668377e+03 = [1, 1, 1, 1]	FES = 203100	Diversity = 6.1890e-05	k_t
Generation 2040: Best = 2.668377e+03 = [2, 2, 4, 4]	FES = 204100	Diversity = 5.1976e-05	k_t
Generation 2050: Best = 2.668377e+03 = [1, 1, 1, 1]	FES = 205100	Diversity = 4.5682e-05	k_t
Generation 2060: Best = 2.668377e+03 = [2, 4, 3, 4]	FES = 206100	Diversity = 4.0014e-05	k_t
Generation 2070: Best = 2.668377e+03 = [1, 1, 1, 1]	FES = 207100	Diversity = 3.4955e-05	k_t
Generation 2080: Best = 2.668377e+03 = [3, 4, 3, 4]	FES = 208100	Diversity = 3.1318e-05	k_t
Generation 2090: Best = 2.668377e+03 = [1, 1, 1, 1]	FES = 209100	Diversity = 2.6141e-05	k_t
Generation 2100: Best = 2.668377e+03 = [4, 2, 3, 3]	FES = 210100	Diversity = 2.2246e-05	k_t
Generation 2110: Best = 2.668377e+03 = [1, 1, 1, 1]	FES = 211100	Diversity = 1.8254e-05	k_t

Generation 2120: Best = 2.668377e+03 = [2, 2, 4, 2]	FES = 212100	Diversity = 1.5398e-05	k_t
Generation 2130: Best = 2.668377e+03 = [1, 1, 1, 1]	FES = 213100	Diversity = 1.3051e-05	k_t
Generation 2140: Best = 2.668377e+03 = [3, 4, 3, 2]	FES = 214100	Diversity = 1.0713e-05	k_t
Generation 2150: Best = 2.668377e+03 = [1, 1, 1, 1]	FES = 215100	Diversity = 8.8622e-06	k_t
Generation 2160: Best = 2.668377e+03 = [3, 3, 2, 2]	FES = 216100	Diversity = 7.5316e-06	k_t
Generation 2170: Best = 2.668377e+03 = [1, 1, 1, 1]	FES = 217100	Diversity = 6.5177e-06	k_t
Generation 2180: Best = 2.668377e+03 = [2, 3, 2, 4]	FES = 218100	Diversity = 5.4116e-06	k_t
Generation 2190: Best = 2.668377e+03 = [1, 1, 1, 1]	FES = 219100	Diversity = 4.5055e-06	k_t
Generation 2200: Best = 2.668377e+03 = [4, 3, 3, 2]	FES = 220100	Diversity = 3.6819e-06	k_t
Generation 2210: Best = 2.668377e+03 = [1, 1, 1, 1]	FES = 221100	Diversity = 2.9911e-06	k_t
Generation 2220: Best = 2.668377e+03 = [3, 2, 2, 4]	FES = 222100	Diversity = 2.5462e-06	k_t
Generation 2230: Best = 2.668377e+03 = [1, 1, 1, 1]	FES = 223100	Diversity = 2.1617e-06	k_t
Generation 2240: Best = 2.668377e+03 = [2, 3, 2, 3]	FES = 224100	Diversity = 1.9327e-06	k_t
Generation 2250: Best = 2.668377e+03 = [1, 1, 1, 1]	FES = 225100	Diversity = 1.6455e-06	k_t
Generation 2260: Best = 2.668377e+03 = [3, 4, 2, 3]	FES = 226100	Diversity = 1.4207e-06	k_t
Generation 2270: Best = 2.668377e+03 = [1, 1, 1, 1]	FES = 227100	Diversity = 1.2218e-06	k_t
Generation 2280: Best = 2.668377e+03 = [4, 3, 3, 4]	FES = 228100	Diversity = 1.1025e-06	k_t
Generation 2290: Best = 2.668377e+03 = [1, 1, 1, 1]	FES = 229100	Diversity = 9.4681e-07	k_t
Generation 2300: Best = 2.668377e+03 = [3, 4, 4, 3]	FES = 230100	Diversity = 8.4391e-07	k_t
Generation 2310: Best = 2.668377e+03 = [1, 1, 1, 1]	FES = 231100	Diversity = 7.8183e-07	k_t
Generation 2320: Best = 2.668377e+03 = [3, 4, 4, 4]	FES = 232100	Diversity = 7.4939e-07	k_t
Generation 2330: Best = 2.668377e+03 = [1, 1, 1, 1]	FES = 233100	Diversity = 6.5876e-07	k_t
Generation 2340: Best = 2.668377e+03 = [4, 3, 2, 2]	FES = 234100	Diversity = 6.0558e-07	k_t
Generation 2350: Best = 2.668377e+03 = [1, 1, 1, 1]	FES = 235100	Diversity = 5.5018e-07	k_t
Generation 2360: Best = 2.668377e+03 = [2, 2, 2, 2]	FES = 236100	Diversity = 4.8663e-07	k_t
Generation 2370: Best = 2.668377e+03 = [1, 1, 1, 1]	FES = 237100	Diversity = 4.4735e-07	k_t
Generation 2380: Best = 2.668377e+03 = [2, 4, 2, 3]	FES = 238100	Diversity = 4.0861e-07	k_t

Generation 2390: Best = 2.668377e+03 = [1, 1, 1, 1]	FES = 239100	Diversity = 3.7045e-07	k_t
Generation 2400: Best = 2.668377e+03 = [2, 3, 4, 4]	FES = 240100	Diversity = 3.3170e-07	k_t
Generation 2410: Best = 2.668377e+03 = [1, 1, 1, 1]	FES = 241100	Diversity = 3.1255e-07	k_t
Generation 2420: Best = 2.668377e+03 = [4, 3, 2, 3]	FES = 242100	Diversity = 3.0614e-07	k_t
Generation 2430: Best = 2.668377e+03 = [1, 1, 1, 1]	FES = 243100	Diversity = 2.9402e-07	k_t
Generation 2440: Best = 2.668377e+03 = [4, 4, 4, 4]	FES = 244100	Diversity = 2.9257e-07	k_t
Generation 2450: Best = 2.668377e+03 = [1, 1, 1, 1]	FES = 245100	Diversity = 2.7011e-07	k_t
Generation 2460: Best = 2.668377e+03 = [3, 3, 4, 3]	FES = 246100	Diversity = 2.4932e-07	k_t
Generation 2470: Best = 2.668377e+03 = [1, 1, 1, 1]	FES = 247100	Diversity = 2.3832e-07	k_t
Generation 2480: Best = 2.668377e+03 = [4, 2, 3, 4]	FES = 248100	Diversity = 2.3079e-07	k_t
Generation 2490: Best = 2.668377e+03 = [1, 1, 1, 1]	FES = 249100	Diversity = 2.2969e-07	k_t
Generation 2500: Best = 2.668377e+03 = [3, 4, 3, 2]	FES = 250100	Diversity = 2.0935e-07	k_t
Generation 2510: Best = 2.668377e+03 = [1, 1, 1, 1]	FES = 251100	Diversity = 2.0366e-07	k_t
Generation 2520: Best = 2.668377e+03 = [3, 3, 4, 3]	FES = 252100	Diversity = 1.9468e-07	k_t
Generation 2530: Best = 2.668377e+03 = [1, 1, 1, 1]	FES = 253100	Diversity = 1.9564e-07	k_t
Generation 2540: Best = 2.668377e+03 = [3, 4, 3, 3]	FES = 254100	Diversity = 1.8993e-07	k_t
Generation 2550: Best = 2.668377e+03 = [1, 1, 1, 1]	FES = 255100	Diversity = 1.8482e-07	k_t
Generation 2560: Best = 2.668377e+03 = [2, 2, 4, 4]	FES = 256100	Diversity = 1.7897e-07	k_t
Generation 2570: Best = 2.668377e+03 = [1, 1, 1, 1]	FES = 257100	Diversity = 1.7842e-07	k_t
Generation 2580: Best = 2.668377e+03 = [2, 4, 3, 4]	FES = 258100	Diversity = 1.7319e-07	k_t
Generation 2590: Best = 2.668377e+03 = [1, 1, 1, 1]	FES = 259100	Diversity = 1.7178e-07	k_t
Generation 2600: Best = 2.668377e+03 = [4, 4, 2, 3]	FES = 260100	Diversity = 1.7370e-07	k_t
Generation 2610: Best = 2.668377e+03 = [1, 1, 1, 1]	FES = 261100	Diversity = 1.7331e-07	k_t
Generation 2620: Best = 2.668377e+03 = [2, 4, 3, 3]	FES = 262100	Diversity = 1.7036e-07	k_t
Generation 2630: Best = 2.668377e+03 = [1, 1, 1, 1]	FES = 263100	Diversity = 1.6509e-07	k_t
Generation 2640: Best = 2.668377e+03 = [2, 4, 2, 4]	FES = 264100	Diversity = 1.5812e-07	k_t
Generation 2650: Best = 2.668377e+03 = [1, 1, 1, 1]	FES = 265100	Diversity = 1.5063e-07	k_t

Generation 2660: Best = 2.668377e+03 = [2, 4, 2, 4]	FES = 266100	Diversity = 1.4803e-07	k_t
Generation 2670: Best = 2.668377e+03 = [1, 1, 1, 1]	FES = 267100	Diversity = 1.4124e-07	k_t
Generation 2680: Best = 2.668377e+03 = [3, 3, 3, 3]	FES = 268100	Diversity = 1.4012e-07	k_t
Generation 2690: Best = 2.668377e+03 = [1, 1, 1, 1]	FES = 269100	Diversity = 1.3686e-07	k_t
Generation 2700: Best = 2.668377e+03 = [2, 3, 3, 2]	FES = 270100	Diversity = 1.4098e-07	k_t
Generation 2710: Best = 2.668377e+03 = [1, 1, 1, 1]	FES = 271100	Diversity = 1.4414e-07	k_t
Generation 2720: Best = 2.668377e+03 = [4, 2, 4, 2]	FES = 272100	Diversity = 1.4673e-07	k_t
Generation 2730: Best = 2.668377e+03 = [1, 1, 1, 1]	FES = 273100	Diversity = 1.4464e-07	k_t
Generation 2740: Best = 2.668377e+03 = [3, 2, 4, 3]	FES = 274100	Diversity = 1.4491e-07	k_t
Generation 2750: Best = 2.668377e+03 = [1, 1, 1, 1]	FES = 275100	Diversity = 1.4626e-07	k_t
Generation 2760: Best = 2.668377e+03 = [3, 3, 4, 4]	FES = 276100	Diversity = 1.4591e-07	k_t
Generation 2770: Best = 2.668377e+03 = [1, 1, 1, 1]	FES = 277100	Diversity = 1.4415e-07	k_t
Generation 2780: Best = 2.668377e+03 = [2, 4, 4, 4]	FES = 278100	Diversity = 1.4273e-07	k_t
Generation 2790: Best = 2.668377e+03 = [1, 1, 1, 1]	FES = 279100	Diversity = 1.4242e-07	k_t
Generation 2800: Best = 2.668377e+03 = [2, 4, 4, 3]	FES = 280100	Diversity = 1.3985e-07	k_t
Generation 2810: Best = 2.668377e+03 = [1, 1, 1, 1]	FES = 281100	Diversity = 1.4174e-07	k_t
Generation 2820: Best = 2.668377e+03 = [2, 3, 3, 2]	FES = 282100	Diversity = 1.3990e-07	k_t
Generation 2830: Best = 2.668377e+03 = [1, 1, 1, 1]	FES = 283100	Diversity = 1.4003e-07	k_t
Generation 2840: Best = 2.668377e+03 = [4, 4, 2, 3]	FES = 284100	Diversity = 1.3747e-07	k_t
Generation 2850: Best = 2.668377e+03 = [1, 1, 1, 1]	FES = 285100	Diversity = 1.3796e-07	k_t
Generation 2860: Best = 2.668377e+03 = [4, 2, 4, 3]	FES = 286100	Diversity = 1.4054e-07	k_t
Generation 2870: Best = 2.668377e+03 = [1, 1, 1, 1]	FES = 287100	Diversity = 1.4167e-07	k_t
Generation 2880: Best = 2.668377e+03 = [3, 4, 4, 2]	FES = 288100	Diversity = 1.3256e-07	k_t
Generation 2890: Best = 2.668377e+03 = [1, 1, 1, 1]	FES = 289100	Diversity = 1.2129e-07	k_t
Generation 2900: Best = 2.668377e+03 = [4, 2, 4, 2]	FES = 290100	Diversity = 1.1627e-07	k_t
Generation 2910: Best = 2.668377e+03 = [1, 1, 1, 1]	FES = 291100	Diversity = 1.1844e-07	k_t
Generation 2920: Best = 2.668377e+03 = [3, 2, 4, 4]	FES = 292100	Diversity = 1.1805e-07	k_t

Generation 2930: Best = 2.668377e+03 FES = 293100 Diversity = 1.1719e-07 k_t
 = [1, 1, 1, 1]
 Generation 2940: Best = 2.668377e+03 FES = 294100 Diversity = 1.2009e-07 k_t
 = [4, 2, 4, 4]
 Generation 2950: Best = 2.668377e+03 FES = 295100 Diversity = 1.2109e-07 k_t
 = [1, 1, 1, 1]
 Generation 2960: Best = 2.668377e+03 FES = 296100 Diversity = 1.1692e-07 k_t
 = [4, 4, 3, 3]
 Generation 2970: Best = 2.668377e+03 FES = 297100 Diversity = 1.1881e-07 k_t
 = [1, 1, 1, 1]
 Generation 2980: Best = 2.668377e+03 FES = 298100 Diversity = 1.1899e-07 k_t
 = [3, 4, 4, 2]
 Generation 2990: Best = 2.668377e+03 FES = 299100 Diversity = 1.1805e-07 k_t
 = [1, 1, 1, 1]

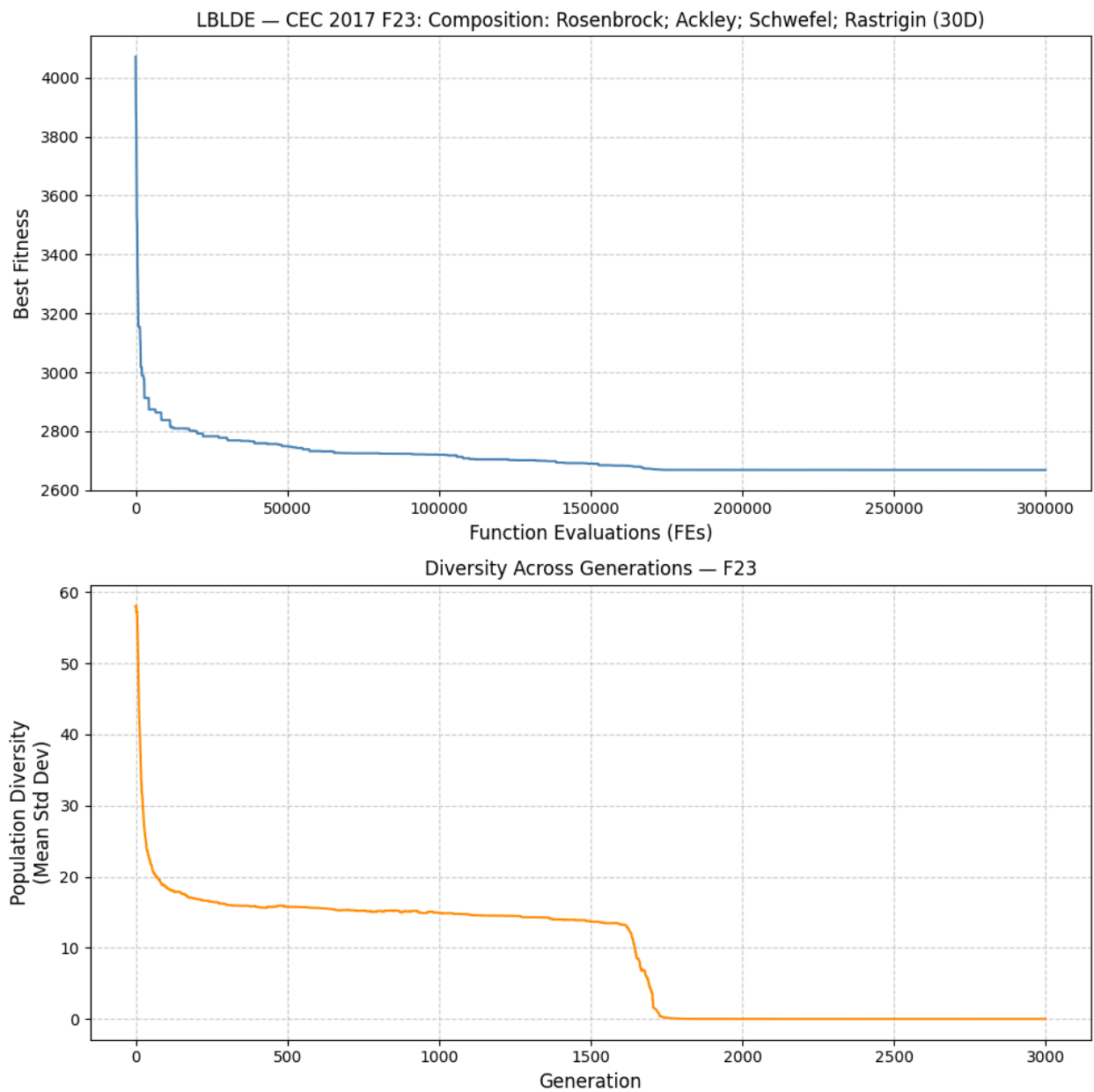
Optimization completed!

Final Best Fitness = 2.668377e+03

Total Function Evaluations = 300000

```

=====
=
Results:
  Best Fitness           : 2.668377e+03
  Global Optimum         : 2.300000e+03
  Error (|best-optimum|) : 3.683775e+02
  Best Solution (5 dim)  : [-81.50488519 -90.0718829 -77.13393      84.01059553
-72.17562064]
  D^(0) (init. diversity): 5.7976e+01
=====
=
  
```



Graph saved as convergence_F23.png

```
In [34]: # =====
# F24 — Composition Function 4 (Ackley; Elliptic; Griewank; Rastrigin)
# =====
run_and_plot('F24', make_wrapper(composition.f24), 2400.0, 'Composition: Ackle
```

```

=====
=
LBLDE_s on CEC 2017 F24 – Single Run
=====
=
Function      : Composition: Ackley; Elliptic; Griewank; Rastrigin (30D)
Bounds       : [-100, 100]
Global optimum : 2400.0
Initial D^(0) = 5.7976e+01
Generation 0: Best Fitness = 3.931320e+03 | Diversity = 5.7976e+01
Generation 10: Best = 3.330689e+03 FES = 1100 Diversity = 4.2953e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 3.165655e+03 FES = 2100 Diversity = 3.0917e+01 k_t
= [3, 3, 2, 2]
Generation 30: Best = 3.087424e+03 FES = 3100 Diversity = 2.4117e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 3.064267e+03 FES = 4100 Diversity = 2.2239e+01 k_t
= [2, 3, 2, 3]
Generation 50: Best = 3.043934e+03 FES = 5100 Diversity = 2.0001e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 3.043934e+03 FES = 6100 Diversity = 1.9079e+01 k_t
= [2, 2, 3, 4]
Generation 70: Best = 3.043934e+03 FES = 7100 Diversity = 1.8294e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 3.021890e+03 FES = 8100 Diversity = 1.7462e+01 k_t
= [3, 2, 3, 4]
Generation 90: Best = 3.010998e+03 FES = 9100 Diversity = 1.6828e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 3.010998e+03 FES = 10100 Diversity = 1.6548e+01 k_t
= [2, 3, 4, 2]
Generation 110: Best = 2.970204e+03 FES = 11100 Diversity = 1.6189e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 2.970204e+03 FES = 12100 Diversity = 1.6048e+01 k_t
= [3, 2, 2, 2]
Generation 130: Best = 2.970204e+03 FES = 13100 Diversity = 1.5937e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 2.970204e+03 FES = 14100 Diversity = 1.5695e+01 k_t
= [2, 2, 2, 4]
Generation 150: Best = 2.970204e+03 FES = 15100 Diversity = 1.5543e+01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 2.965491e+03 FES = 16100 Diversity = 1.5405e+01 k_t
= [4, 2, 2, 4]
Generation 170: Best = 2.965491e+03 FES = 17100 Diversity = 1.5328e+01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 2.964811e+03 FES = 18100 Diversity = 1.5214e+01 k_t
= [2, 4, 2, 3]
Generation 190: Best = 2.964811e+03 FES = 19100 Diversity = 1.5118e+01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 2.964811e+03 FES = 20100 Diversity = 1.5107e+01 k_t
= [3, 2, 4, 3]
Generation 210: Best = 2.963557e+03 FES = 21100 Diversity = 1.4967e+01 k_t
= [1, 1, 1, 1]
Generation 220: Best = 2.961213e+03 FES = 22100 Diversity = 1.4974e+01 k_t
= [2, 2, 2, 2]

```

Generation 230:	Best = 2.958180e+03	FES = 23100	Diversity = 1.4800e+01	k_t = [1, 1, 1, 1]
Generation 240:	Best = 2.958180e+03	FES = 24100	Diversity = 1.4703e+01	k_t = [2, 3, 4, 3]
Generation 250:	Best = 2.939228e+03	FES = 25100	Diversity = 1.4734e+01	k_t = [1, 1, 1, 1]
Generation 260:	Best = 2.939228e+03	FES = 26100	Diversity = 1.4700e+01	k_t = [3, 4, 4, 4]
Generation 270:	Best = 2.939228e+03	FES = 27100	Diversity = 1.4513e+01	k_t = [1, 1, 1, 1]
Generation 280:	Best = 2.939154e+03	FES = 28100	Diversity = 1.4527e+01	k_t = [4, 4, 4, 2]
Generation 290:	Best = 2.939154e+03	FES = 29100	Diversity = 1.4494e+01	k_t = [1, 1, 1, 1]
Generation 300:	Best = 2.929571e+03	FES = 30100	Diversity = 1.4556e+01	k_t = [4, 3, 4, 3]
Generation 310:	Best = 2.929571e+03	FES = 31100	Diversity = 1.4548e+01	k_t = [1, 1, 1, 1]
Generation 320:	Best = 2.929571e+03	FES = 32100	Diversity = 1.4444e+01	k_t = [2, 4, 3, 2]
Generation 330:	Best = 2.929571e+03	FES = 33100	Diversity = 1.4432e+01	k_t = [1, 1, 1, 1]
Generation 340:	Best = 2.929571e+03	FES = 34100	Diversity = 1.4213e+01	k_t = [4, 3, 2, 4]
Generation 350:	Best = 2.929571e+03	FES = 35100	Diversity = 1.4169e+01	k_t = [1, 1, 1, 1]
Generation 360:	Best = 2.927598e+03	FES = 36100	Diversity = 1.4123e+01	k_t = [4, 2, 2, 3]
Generation 370:	Best = 2.927598e+03	FES = 37100	Diversity = 1.4048e+01	k_t = [1, 1, 1, 1]
Generation 380:	Best = 2.927598e+03	FES = 38100	Diversity = 1.3894e+01	k_t = [4, 2, 2, 4]
Generation 390:	Best = 2.923812e+03	FES = 39100	Diversity = 1.3883e+01	k_t = [1, 1, 1, 1]
Generation 400:	Best = 2.921954e+03	FES = 40100	Diversity = 1.3736e+01	k_t = [3, 2, 4, 3]
Generation 410:	Best = 2.921163e+03	FES = 41100	Diversity = 1.3728e+01	k_t = [1, 1, 1, 1]
Generation 420:	Best = 2.915800e+03	FES = 42100	Diversity = 1.3686e+01	k_t = [2, 4, 3, 4]
Generation 430:	Best = 2.909325e+03	FES = 43100	Diversity = 1.3489e+01	k_t = [1, 1, 1, 1]
Generation 440:	Best = 2.909163e+03	FES = 44100	Diversity = 1.3384e+01	k_t = [2, 2, 2, 4]
Generation 450:	Best = 2.909163e+03	FES = 45100	Diversity = 1.3359e+01	k_t = [1, 1, 1, 1]
Generation 460:	Best = 2.908404e+03	FES = 46100	Diversity = 1.3285e+01	k_t = [2, 3, 2, 3]
Generation 470:	Best = 2.908404e+03	FES = 47100	Diversity = 1.3168e+01	k_t = [1, 1, 1, 1]
Generation 480:	Best = 2.908404e+03	FES = 48100	Diversity = 1.3152e+01	k_t = [4, 3, 4, 4]
Generation 490:	Best = 2.906744e+03	FES = 49100	Diversity = 1.3111e+01	k_t = [1, 1, 1, 1]

Generation 500:	Best = 2.901500e+03	FES = 50100	Diversity = 1.3124e+01	k_t = [2, 3, 2, 4]
Generation 510:	Best = 2.900157e+03	FES = 51100	Diversity = 1.3149e+01	k_t = [1, 1, 1, 1]
Generation 520:	Best = 2.895870e+03	FES = 52100	Diversity = 1.3033e+01	k_t = [3, 2, 2, 2]
Generation 530:	Best = 2.895071e+03	FES = 53100	Diversity = 1.3021e+01	k_t = [1, 1, 1, 1]
Generation 540:	Best = 2.894060e+03	FES = 54100	Diversity = 1.3003e+01	k_t = [3, 2, 4, 3]
Generation 550:	Best = 2.894060e+03	FES = 55100	Diversity = 1.3007e+01	k_t = [1, 1, 1, 1]
Generation 560:	Best = 2.884038e+03	FES = 56100	Diversity = 1.3033e+01	k_t = [2, 3, 3, 3]
Generation 570:	Best = 2.884038e+03	FES = 57100	Diversity = 1.2958e+01	k_t = [1, 1, 1, 1]
Generation 580:	Best = 2.884038e+03	FES = 58100	Diversity = 1.2950e+01	k_t = [4, 4, 3, 2]
Generation 590:	Best = 2.883983e+03	FES = 59100	Diversity = 1.2909e+01	k_t = [1, 1, 1, 1]
Generation 600:	Best = 2.883869e+03	FES = 60100	Diversity = 1.2919e+01	k_t = [2, 3, 3, 2]
Generation 610:	Best = 2.883511e+03	FES = 61100	Diversity = 1.2841e+01	k_t = [1, 1, 1, 1]
Generation 620:	Best = 2.883511e+03	FES = 62100	Diversity = 1.2805e+01	k_t = [4, 3, 3, 4]
Generation 630:	Best = 2.883511e+03	FES = 63100	Diversity = 1.2815e+01	k_t = [1, 1, 1, 1]
Generation 640:	Best = 2.881014e+03	FES = 64100	Diversity = 1.2815e+01	k_t = [4, 2, 4, 3]
Generation 650:	Best = 2.878975e+03	FES = 65100	Diversity = 1.2713e+01	k_t = [1, 1, 1, 1]
Generation 660:	Best = 2.878975e+03	FES = 66100	Diversity = 1.2675e+01	k_t = [3, 2, 4, 2]
Generation 670:	Best = 2.878975e+03	FES = 67100	Diversity = 1.2665e+01	k_t = [1, 1, 1, 1]
Generation 680:	Best = 2.864014e+03	FES = 68100	Diversity = 1.2623e+01	k_t = [4, 3, 2, 2]
Generation 690:	Best = 2.864014e+03	FES = 69100	Diversity = 1.2599e+01	k_t = [1, 1, 1, 1]
Generation 700:	Best = 2.863970e+03	FES = 70100	Diversity = 1.2583e+01	k_t = [2, 4, 2, 2]
Generation 710:	Best = 2.863970e+03	FES = 71100	Diversity = 1.2311e+01	k_t = [1, 1, 1, 1]
Generation 720:	Best = 2.863970e+03	FES = 72100	Diversity = 1.2097e+01	k_t = [2, 3, 4, 2]
Generation 730:	Best = 2.863970e+03	FES = 73100	Diversity = 1.2053e+01	k_t = [1, 1, 1, 1]
Generation 740:	Best = 2.863660e+03	FES = 74100	Diversity = 1.1972e+01	k_t = [2, 2, 4, 3]
Generation 750:	Best = 2.863660e+03	FES = 75100	Diversity = 1.1929e+01	k_t = [1, 1, 1, 1]
Generation 760:	Best = 2.863039e+03	FES = 76100	Diversity = 1.1787e+01	k_t = [3, 3, 4, 2]

Generation 770:	Best = 2.863039e+03	FES = 77100	Diversity = 1.1643e+01	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 2.862482e+03	FES = 78100	Diversity = 1.1593e+01	k_t
= [3, 2, 2, 2]				
Generation 790:	Best = 2.862482e+03	FES = 79100	Diversity = 1.1490e+01	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 2.861790e+03	FES = 80100	Diversity = 1.1429e+01	k_t
= [2, 4, 4, 4]				
Generation 810:	Best = 2.861133e+03	FES = 81100	Diversity = 1.1403e+01	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 2.859200e+03	FES = 82100	Diversity = 1.1079e+01	k_t
= [2, 4, 4, 2]				
Generation 830:	Best = 2.859200e+03	FES = 83100	Diversity = 1.1098e+01	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 2.858895e+03	FES = 84100	Diversity = 1.0956e+01	k_t
= [3, 4, 2, 4]				
Generation 850:	Best = 2.858895e+03	FES = 85100	Diversity = 1.0854e+01	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 2.858895e+03	FES = 86100	Diversity = 1.0792e+01	k_t
= [3, 2, 3, 4]				
Generation 870:	Best = 2.858895e+03	FES = 87100	Diversity = 1.0808e+01	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 2.858232e+03	FES = 88100	Diversity = 1.0567e+01	k_t
= [4, 3, 3, 2]				
Generation 890:	Best = 2.858232e+03	FES = 89100	Diversity = 1.0253e+01	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 2.858110e+03	FES = 90100	Diversity = 1.0117e+01	k_t
= [3, 2, 2, 2]				
Generation 910:	Best = 2.855921e+03	FES = 91100	Diversity = 1.0077e+01	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 2.854619e+03	FES = 92100	Diversity = 1.0042e+01	k_t
= [2, 2, 2, 3]				
Generation 930:	Best = 2.853233e+03	FES = 93100	Diversity = 9.9215e+00	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 2.853078e+03	FES = 94100	Diversity = 9.6266e+00	k_t
= [3, 4, 4, 3]				
Generation 950:	Best = 2.852074e+03	FES = 95100	Diversity = 8.8091e+00	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 2.851714e+03	FES = 96100	Diversity = 5.4388e+00	k_t
= [2, 4, 4, 4]				
Generation 970:	Best = 2.850771e+03	FES = 97100	Diversity = 4.9322e-01	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 2.849932e+03	FES = 98100	Diversity = 3.6101e-01	k_t
= [2, 3, 2, 3]				
Generation 990:	Best = 2.849403e+03	FES = 99100	Diversity = 2.6962e-01	k_t
= [1, 1, 1, 1]				
Generation 1000:	Best = 2.849030e+03	FES = 100100	Diversity = 1.9779e-01	k_t
= [2, 4, 4, 2]				
Generation 1010:	Best = 2.848901e+03	FES = 101100	Diversity = 1.4053e-01	k_t
= [1, 1, 1, 1]				
Generation 1020:	Best = 2.848836e+03	FES = 102100	Diversity = 9.8373e-02	k_t
= [2, 4, 4, 2]				
Generation 1030:	Best = 2.848823e+03	FES = 103100	Diversity = 7.1315e-02	k_t
= [1, 1, 1, 1]				

Generation 1040: Best = 2.848799e+03 = [4, 4, 4, 3]	FES = 104100	Diversity = 5.1018e-02	k_t
Generation 1050: Best = 2.848790e+03 = [1, 1, 1, 1]	FES = 105100	Diversity = 3.4662e-02	k_t
Generation 1060: Best = 2.848783e+03 = [2, 4, 3, 2]	FES = 106100	Diversity = 2.3002e-02	k_t
Generation 1070: Best = 2.848782e+03 = [1, 1, 1, 1]	FES = 107100	Diversity = 1.5663e-02	k_t
Generation 1080: Best = 2.848781e+03 = [2, 2, 4, 3]	FES = 108100	Diversity = 1.1227e-02	k_t
Generation 1090: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 109100	Diversity = 7.9808e-03	k_t
Generation 1100: Best = 2.848780e+03 = [2, 3, 2, 3]	FES = 110100	Diversity = 5.4778e-03	k_t
Generation 1110: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 111100	Diversity = 3.5583e-03	k_t
Generation 1120: Best = 2.848780e+03 = [3, 3, 2, 4]	FES = 112100	Diversity = 2.3625e-03	k_t
Generation 1130: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 113100	Diversity = 1.4495e-03	k_t
Generation 1140: Best = 2.848780e+03 = [4, 4, 2, 3]	FES = 114100	Diversity = 8.8331e-04	k_t
Generation 1150: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 115100	Diversity = 5.3526e-04	k_t
Generation 1160: Best = 2.848780e+03 = [2, 2, 4, 3]	FES = 116100	Diversity = 3.4172e-04	k_t
Generation 1170: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 117100	Diversity = 2.1784e-04	k_t
Generation 1180: Best = 2.848780e+03 = [4, 4, 2, 4]	FES = 118100	Diversity = 1.3504e-04	k_t
Generation 1190: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 119100	Diversity = 8.5532e-05	k_t
Generation 1200: Best = 2.848780e+03 = [2, 4, 4, 3]	FES = 120100	Diversity = 5.5777e-05	k_t
Generation 1210: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 121100	Diversity = 3.4363e-05	k_t
Generation 1220: Best = 2.848780e+03 = [4, 3, 3, 3]	FES = 122100	Diversity = 2.0006e-05	k_t
Generation 1230: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 123100	Diversity = 1.2372e-05	k_t
Generation 1240: Best = 2.848780e+03 = [3, 2, 4, 4]	FES = 124100	Diversity = 7.9191e-06	k_t
Generation 1250: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 125100	Diversity = 5.0157e-06	k_t
Generation 1260: Best = 2.848780e+03 = [2, 4, 4, 2]	FES = 126100	Diversity = 3.1149e-06	k_t
Generation 1270: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 127100	Diversity = 2.0299e-06	k_t
Generation 1280: Best = 2.848780e+03 = [3, 4, 2, 3]	FES = 128100	Diversity = 1.3360e-06	k_t
Generation 1290: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 129100	Diversity = 8.5486e-07	k_t
Generation 1300: Best = 2.848780e+03 = [4, 3, 2, 4]	FES = 130100	Diversity = 5.9073e-07	k_t

Generation 1310: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 131100	Diversity = 3.8222e-07	k_t
Generation 1320: Best = 2.848780e+03 = [4, 2, 4, 2]	FES = 132100	Diversity = 2.7398e-07	k_t
Generation 1330: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 133100	Diversity = 2.0656e-07	k_t
Generation 1340: Best = 2.848780e+03 = [2, 2, 4, 3]	FES = 134100	Diversity = 1.8768e-07	k_t
Generation 1350: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 135100	Diversity = 1.7474e-07	k_t
Generation 1360: Best = 2.848780e+03 = [2, 4, 2, 2]	FES = 136100	Diversity = 1.5515e-07	k_t
Generation 1370: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 137100	Diversity = 1.4618e-07	k_t
Generation 1380: Best = 2.848780e+03 = [2, 2, 3, 3]	FES = 138100	Diversity = 1.4122e-07	k_t
Generation 1390: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 139100	Diversity = 1.3962e-07	k_t
Generation 1400: Best = 2.848780e+03 = [3, 2, 3, 3]	FES = 140100	Diversity = 1.4247e-07	k_t
Generation 1410: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 141100	Diversity = 1.4072e-07	k_t
Generation 1420: Best = 2.848780e+03 = [2, 2, 4, 4]	FES = 142100	Diversity = 1.3988e-07	k_t
Generation 1430: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 143100	Diversity = 1.3902e-07	k_t
Generation 1440: Best = 2.848780e+03 = [4, 4, 2, 2]	FES = 144100	Diversity = 1.3975e-07	k_t
Generation 1450: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 145100	Diversity = 1.3657e-07	k_t
Generation 1460: Best = 2.848780e+03 = [3, 4, 2, 4]	FES = 146100	Diversity = 1.3438e-07	k_t
Generation 1470: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 147100	Diversity = 1.3328e-07	k_t
Generation 1480: Best = 2.848780e+03 = [3, 2, 2, 3]	FES = 148100	Diversity = 1.3290e-07	k_t
Generation 1490: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 149100	Diversity = 1.3193e-07	k_t
Generation 1500: Best = 2.848780e+03 = [3, 3, 2, 3]	FES = 150100	Diversity = 1.3248e-07	k_t
Generation 1510: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 151100	Diversity = 1.3260e-07	k_t
Generation 1520: Best = 2.848780e+03 = [3, 4, 3, 3]	FES = 152100	Diversity = 1.3279e-07	k_t
Generation 1530: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 153100	Diversity = 1.3345e-07	k_t
Generation 1540: Best = 2.848780e+03 = [2, 2, 2, 3]	FES = 154100	Diversity = 1.3242e-07	k_t
Generation 1550: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 155100	Diversity = 1.3474e-07	k_t
Generation 1560: Best = 2.848780e+03 = [3, 2, 4, 3]	FES = 156100	Diversity = 1.3966e-07	k_t
Generation 1570: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 157100	Diversity = 1.3317e-07	k_t

Generation 1580: Best = 2.848780e+03 = [3, 3, 3, 4]	FES = 158100	Diversity = 1.3801e-07	k_t
Generation 1590: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 159100	Diversity = 1.3697e-07	k_t
Generation 1600: Best = 2.848780e+03 = [4, 2, 4, 3]	FES = 160100	Diversity = 1.3888e-07	k_t
Generation 1610: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 161100	Diversity = 1.3548e-07	k_t
Generation 1620: Best = 2.848780e+03 = [4, 2, 4, 2]	FES = 162100	Diversity = 1.3390e-07	k_t
Generation 1630: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 163100	Diversity = 1.2868e-07	k_t
Generation 1640: Best = 2.848780e+03 = [4, 4, 4, 2]	FES = 164100	Diversity = 1.3126e-07	k_t
Generation 1650: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 165100	Diversity = 1.3064e-07	k_t
Generation 1660: Best = 2.848780e+03 = [2, 3, 4, 2]	FES = 166100	Diversity = 1.2762e-07	k_t
Generation 1670: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 167100	Diversity = 1.2702e-07	k_t
Generation 1680: Best = 2.848780e+03 = [4, 2, 2, 4]	FES = 168100	Diversity = 1.2480e-07	k_t
Generation 1690: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 169100	Diversity = 1.2498e-07	k_t
Generation 1700: Best = 2.848780e+03 = [2, 4, 2, 3]	FES = 170100	Diversity = 1.2354e-07	k_t
Generation 1710: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 171100	Diversity = 1.2535e-07	k_t
Generation 1720: Best = 2.848780e+03 = [2, 4, 3, 2]	FES = 172100	Diversity = 1.2224e-07	k_t
Generation 1730: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 173100	Diversity = 1.1876e-07	k_t
Generation 1740: Best = 2.848780e+03 = [3, 2, 2, 4]	FES = 174100	Diversity = 1.1948e-07	k_t
Generation 1750: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 175100	Diversity = 1.1849e-07	k_t
Generation 1760: Best = 2.848780e+03 = [3, 3, 3, 3]	FES = 176100	Diversity = 1.1899e-07	k_t
Generation 1770: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 177100	Diversity = 1.1895e-07	k_t
Generation 1780: Best = 2.848780e+03 = [3, 4, 3, 3]	FES = 178100	Diversity = 1.1626e-07	k_t
Generation 1790: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 179100	Diversity = 1.1364e-07	k_t
Generation 1800: Best = 2.848780e+03 = [3, 2, 3, 3]	FES = 180100	Diversity = 1.1541e-07	k_t
Generation 1810: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 181100	Diversity = 1.1109e-07	k_t
Generation 1820: Best = 2.848780e+03 = [3, 4, 2, 4]	FES = 182100	Diversity = 1.0982e-07	k_t
Generation 1830: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 183100	Diversity = 1.0747e-07	k_t
Generation 1840: Best = 2.848780e+03 = [3, 4, 3, 4]	FES = 184100	Diversity = 1.0209e-07	k_t

Generation 1850: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 185100	Diversity = 1.0165e-07	k_t
Generation 1860: Best = 2.848780e+03 = [4, 2, 4, 4]	FES = 186100	Diversity = 1.0197e-07	k_t
Generation 1870: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 187100	Diversity = 9.7554e-08	k_t
Generation 1880: Best = 2.848780e+03 = [2, 3, 4, 3]	FES = 188100	Diversity = 9.7242e-08	k_t
Generation 1890: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 189100	Diversity = 9.7836e-08	k_t
Generation 1900: Best = 2.848780e+03 = [2, 3, 4, 3]	FES = 190100	Diversity = 9.7104e-08	k_t
Generation 1910: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 191100	Diversity = 9.7440e-08	k_t
Generation 1920: Best = 2.848780e+03 = [2, 4, 2, 2]	FES = 192100	Diversity = 9.6741e-08	k_t
Generation 1930: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 193100	Diversity = 9.6990e-08	k_t
Generation 1940: Best = 2.848780e+03 = [4, 2, 3, 4]	FES = 194100	Diversity = 9.8665e-08	k_t
Generation 1950: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 195100	Diversity = 9.9494e-08	k_t
Generation 1960: Best = 2.848780e+03 = [2, 3, 4, 2]	FES = 196100	Diversity = 1.0028e-07	k_t
Generation 1970: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 197100	Diversity = 9.6407e-08	k_t
Generation 1980: Best = 2.848780e+03 = [4, 3, 4, 3]	FES = 198100	Diversity = 9.7464e-08	k_t
Generation 1990: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 199100	Diversity = 9.8112e-08	k_t
Generation 2000: Best = 2.848780e+03 = [3, 2, 3, 4]	FES = 200100	Diversity = 9.8346e-08	k_t
Generation 2010: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 201100	Diversity = 9.8513e-08	k_t
Generation 2020: Best = 2.848780e+03 = [3, 2, 4, 3]	FES = 202100	Diversity = 9.8228e-08	k_t
Generation 2030: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 203100	Diversity = 9.7926e-08	k_t
Generation 2040: Best = 2.848780e+03 = [4, 4, 3, 4]	FES = 204100	Diversity = 9.8689e-08	k_t
Generation 2050: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 205100	Diversity = 9.7727e-08	k_t
Generation 2060: Best = 2.848780e+03 = [3, 4, 3, 3]	FES = 206100	Diversity = 9.7500e-08	k_t
Generation 2070: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 207100	Diversity = 9.6309e-08	k_t
Generation 2080: Best = 2.848780e+03 = [3, 2, 2, 2]	FES = 208100	Diversity = 9.6402e-08	k_t
Generation 2090: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 209100	Diversity = 9.5022e-08	k_t
Generation 2100: Best = 2.848780e+03 = [2, 2, 4, 2]	FES = 210100	Diversity = 9.4553e-08	k_t
Generation 2110: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 211100	Diversity = 9.5344e-08	k_t

Generation 2120: Best = 2.848780e+03 = [2, 2, 2, 3]	FES = 212100	Diversity = 9.5887e-08	k_t
Generation 2130: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 213100	Diversity = 9.6104e-08	k_t
Generation 2140: Best = 2.848780e+03 = [4, 4, 3, 3]	FES = 214100	Diversity = 9.4298e-08	k_t
Generation 2150: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 215100	Diversity = 9.4369e-08	k_t
Generation 2160: Best = 2.848780e+03 = [2, 4, 2, 2]	FES = 216100	Diversity = 9.4140e-08	k_t
Generation 2170: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 217100	Diversity = 9.2743e-08	k_t
Generation 2180: Best = 2.848780e+03 = [3, 4, 4, 3]	FES = 218100	Diversity = 9.2655e-08	k_t
Generation 2190: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 219100	Diversity = 9.2025e-08	k_t
Generation 2200: Best = 2.848780e+03 = [4, 3, 4, 4]	FES = 220100	Diversity = 9.2807e-08	k_t
Generation 2210: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 221100	Diversity = 9.2270e-08	k_t
Generation 2220: Best = 2.848780e+03 = [2, 3, 3, 3]	FES = 222100	Diversity = 9.2131e-08	k_t
Generation 2230: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 223100	Diversity = 9.1277e-08	k_t
Generation 2240: Best = 2.848780e+03 = [4, 2, 4, 4]	FES = 224100	Diversity = 9.2139e-08	k_t
Generation 2250: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 225100	Diversity = 9.2013e-08	k_t
Generation 2260: Best = 2.848780e+03 = [3, 2, 2, 3]	FES = 226100	Diversity = 9.2216e-08	k_t
Generation 2270: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 227100	Diversity = 9.3529e-08	k_t
Generation 2280: Best = 2.848780e+03 = [3, 4, 3, 4]	FES = 228100	Diversity = 9.3033e-08	k_t
Generation 2290: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 229100	Diversity = 9.4558e-08	k_t
Generation 2300: Best = 2.848780e+03 = [2, 3, 3, 2]	FES = 230100	Diversity = 9.4896e-08	k_t
Generation 2310: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 231100	Diversity = 9.4896e-08	k_t
Generation 2320: Best = 2.848780e+03 = [3, 3, 2, 3]	FES = 232100	Diversity = 9.5428e-08	k_t
Generation 2330: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 233100	Diversity = 9.5725e-08	k_t
Generation 2340: Best = 2.848780e+03 = [4, 2, 3, 2]	FES = 234100	Diversity = 9.5383e-08	k_t
Generation 2350: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 235100	Diversity = 9.5832e-08	k_t
Generation 2360: Best = 2.848780e+03 = [4, 3, 2, 2]	FES = 236100	Diversity = 9.6147e-08	k_t
Generation 2370: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 237100	Diversity = 9.6492e-08	k_t
Generation 2380: Best = 2.848780e+03 = [3, 4, 2, 2]	FES = 238100	Diversity = 9.5410e-08	k_t

Generation 2390: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 239100	Diversity = 9.5028e-08	k_t
Generation 2400: Best = 2.848780e+03 = [2, 2, 3, 3]	FES = 240100	Diversity = 9.6617e-08	k_t
Generation 2410: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 241100	Diversity = 9.6179e-08	k_t
Generation 2420: Best = 2.848780e+03 = [3, 3, 4, 4]	FES = 242100	Diversity = 9.5531e-08	k_t
Generation 2430: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 243100	Diversity = 9.5394e-08	k_t
Generation 2440: Best = 2.848780e+03 = [4, 2, 4, 3]	FES = 244100	Diversity = 9.5177e-08	k_t
Generation 2450: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 245100	Diversity = 9.5002e-08	k_t
Generation 2460: Best = 2.848780e+03 = [4, 2, 4, 3]	FES = 246100	Diversity = 9.6064e-08	k_t
Generation 2470: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 247100	Diversity = 9.7210e-08	k_t
Generation 2480: Best = 2.848780e+03 = [4, 3, 2, 4]	FES = 248100	Diversity = 9.5844e-08	k_t
Generation 2490: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 249100	Diversity = 9.5437e-08	k_t
Generation 2500: Best = 2.848780e+03 = [4, 3, 3, 3]	FES = 250100	Diversity = 9.6581e-08	k_t
Generation 2510: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 251100	Diversity = 9.6847e-08	k_t
Generation 2520: Best = 2.848780e+03 = [4, 3, 2, 2]	FES = 252100	Diversity = 9.7164e-08	k_t
Generation 2530: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 253100	Diversity = 9.7447e-08	k_t
Generation 2540: Best = 2.848780e+03 = [3, 3, 2, 4]	FES = 254100	Diversity = 9.7440e-08	k_t
Generation 2550: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 255100	Diversity = 9.7346e-08	k_t
Generation 2560: Best = 2.848780e+03 = [4, 4, 2, 2]	FES = 256100	Diversity = 9.7041e-08	k_t
Generation 2570: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 257100	Diversity = 9.6790e-08	k_t
Generation 2580: Best = 2.848780e+03 = [3, 3, 4, 4]	FES = 258100	Diversity = 9.7098e-08	k_t
Generation 2590: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 259100	Diversity = 9.6597e-08	k_t
Generation 2600: Best = 2.848780e+03 = [3, 2, 3, 3]	FES = 260100	Diversity = 9.7547e-08	k_t
Generation 2610: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 261100	Diversity = 9.8398e-08	k_t
Generation 2620: Best = 2.848780e+03 = [4, 2, 3, 4]	FES = 262100	Diversity = 9.8329e-08	k_t
Generation 2630: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 263100	Diversity = 9.8641e-08	k_t
Generation 2640: Best = 2.848780e+03 = [2, 4, 2, 3]	FES = 264100	Diversity = 9.7830e-08	k_t
Generation 2650: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 265100	Diversity = 9.5591e-08	k_t

Generation 2660: Best = 2.848780e+03 = [2, 4, 2, 2]	FES = 266100	Diversity = 9.6730e-08	k_t
Generation 2670: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 267100	Diversity = 9.6682e-08	k_t
Generation 2680: Best = 2.848780e+03 = [3, 4, 4, 3]	FES = 268100	Diversity = 9.5711e-08	k_t
Generation 2690: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 269100	Diversity = 9.5528e-08	k_t
Generation 2700: Best = 2.848780e+03 = [3, 2, 3, 2]	FES = 270100	Diversity = 9.5899e-08	k_t
Generation 2710: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 271100	Diversity = 9.6322e-08	k_t
Generation 2720: Best = 2.848780e+03 = [2, 3, 2, 4]	FES = 272100	Diversity = 9.4715e-08	k_t
Generation 2730: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 273100	Diversity = 9.4352e-08	k_t
Generation 2740: Best = 2.848780e+03 = [3, 3, 4, 4]	FES = 274100	Diversity = 9.4535e-08	k_t
Generation 2750: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 275100	Diversity = 9.4500e-08	k_t
Generation 2760: Best = 2.848780e+03 = [3, 3, 4, 4]	FES = 276100	Diversity = 9.3722e-08	k_t
Generation 2770: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 277100	Diversity = 9.3454e-08	k_t
Generation 2780: Best = 2.848780e+03 = [3, 2, 4, 4]	FES = 278100	Diversity = 9.4395e-08	k_t
Generation 2790: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 279100	Diversity = 9.4487e-08	k_t
Generation 2800: Best = 2.848780e+03 = [2, 4, 3, 4]	FES = 280100	Diversity = 9.4196e-08	k_t
Generation 2810: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 281100	Diversity = 9.2512e-08	k_t
Generation 2820: Best = 2.848780e+03 = [3, 3, 4, 2]	FES = 282100	Diversity = 9.2558e-08	k_t
Generation 2830: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 283100	Diversity = 9.1589e-08	k_t
Generation 2840: Best = 2.848780e+03 = [3, 4, 3, 4]	FES = 284100	Diversity = 9.2018e-08	k_t
Generation 2850: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 285100	Diversity = 9.1480e-08	k_t
Generation 2860: Best = 2.848780e+03 = [4, 4, 2, 4]	FES = 286100	Diversity = 9.0886e-08	k_t
Generation 2870: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 287100	Diversity = 8.9700e-08	k_t
Generation 2880: Best = 2.848780e+03 = [4, 4, 2, 4]	FES = 288100	Diversity = 8.9292e-08	k_t
Generation 2890: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 289100	Diversity = 8.9080e-08	k_t
Generation 2900: Best = 2.848780e+03 = [2, 3, 2, 3]	FES = 290100	Diversity = 8.9560e-08	k_t
Generation 2910: Best = 2.848780e+03 = [1, 1, 1, 1]	FES = 291100	Diversity = 8.9598e-08	k_t
Generation 2920: Best = 2.848780e+03 = [4, 4, 3, 4]	FES = 292100	Diversity = 8.9109e-08	k_t

```

Generation 2930: Best = 2.848780e+03  FES = 293100  Diversity = 8.9694e-08  k_t
= [1, 1, 1, 1]
Generation 2940: Best = 2.848780e+03  FES = 294100  Diversity = 9.0676e-08  k_t
= [2, 4, 3, 2]
Generation 2950: Best = 2.848780e+03  FES = 295100  Diversity = 9.0770e-08  k_t
= [1, 1, 1, 1]
Generation 2960: Best = 2.848780e+03  FES = 296100  Diversity = 9.3087e-08  k_t
= [4, 4, 4, 4]
Generation 2970: Best = 2.848780e+03  FES = 297100  Diversity = 9.2477e-08  k_t
= [1, 1, 1, 1]
Generation 2980: Best = 2.848780e+03  FES = 298100  Diversity = 9.3117e-08  k_t
= [3, 2, 4, 4]
Generation 2990: Best = 2.848780e+03  FES = 299100  Diversity = 9.3363e-08  k_t
= [1, 1, 1, 1]

```

Optimization completed!

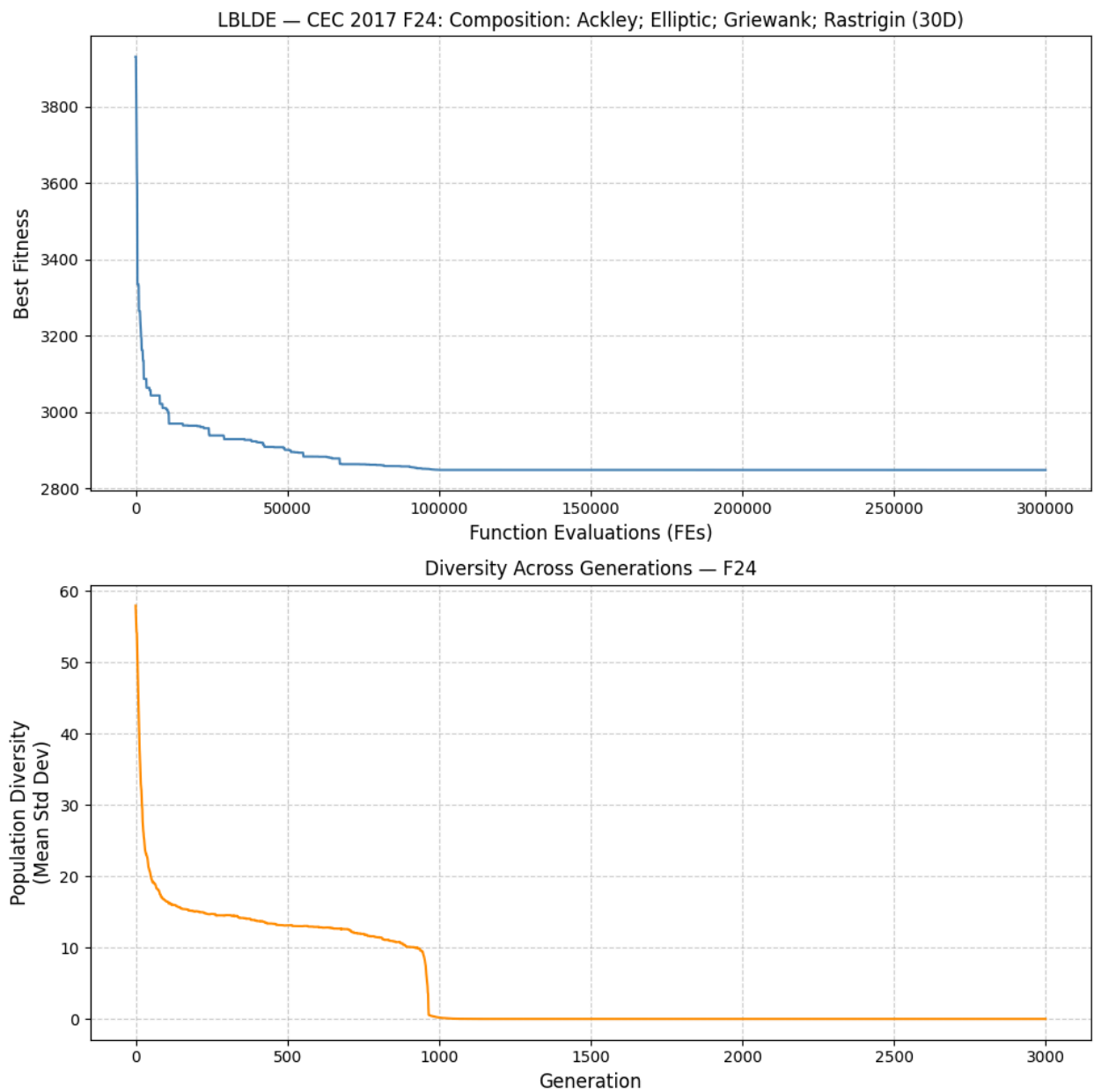
Final Best Fitness = 2.848780e+03

Total Function Evaluations = 300000

```

=====
=
Results:
  Best Fitness           : 2.848780e+03
  Global Optimum         : 2.400000e+03
  Error (|best-optimum|) : 4.487796e+02
  Best Solution (5 dim)  : [ 65.09355267  87.60919085 -50.15742971 -71.3310956
79.544047 ]
  D^(0) (init. diversity): 5.7976e+01
=====
=

```

Graph saved as convergence_F24.png

```
In [35]: # =====
# F25 — Composition Function 5 (Rastrigin; HappyCat; Ackley; Discus; Rosenbrock)
# =====
run_and_plot('F25', make_wrapper(composition.f25), 2500.0, 'Composition: Rastr
```

```

=====
=
LBLDE_s on CEC 2017 F25 – Single Run
=====
=
Function      : Composition: Rastrigin; HappyCat; Ackley; Discus; Rosenbrock
(30D)
Bounds       : [-100, 100]
Global optimum : 2500.0
Initial D^(0) = 5.7976e+01
Generation 0: Best Fitness = 2.562141e+04 | Diversity = 5.7976e+01
Generation 10: Best = 5.006037e+03 FES = 1100 Diversity = 3.5282e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 3.432786e+03 FES = 2100 Diversity = 2.2073e+01 k_t
= [3, 3, 2, 4]
Generation 30: Best = 3.111979e+03 FES = 3100 Diversity = 1.3118e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 3.040881e+03 FES = 4100 Diversity = 9.1117e+00 k_t
= [3, 4, 3, 4]
Generation 50: Best = 3.001588e+03 FES = 5100 Diversity = 6.3080e+00 k_t
= [1, 1, 1, 1]
Generation 60: Best = 2.970157e+03 FES = 6100 Diversity = 4.9307e+00 k_t
= [2, 3, 4, 3]
Generation 70: Best = 2.946347e+03 FES = 7100 Diversity = 4.0256e+00 k_t
= [1, 1, 1, 1]
Generation 80: Best = 2.933827e+03 FES = 8100 Diversity = 3.3327e+00 k_t
= [4, 3, 4, 3]
Generation 90: Best = 2.915227e+03 FES = 9100 Diversity = 2.8773e+00 k_t
= [1, 1, 1, 1]
Generation 100: Best = 2.903720e+03 FES = 10100 Diversity = 2.1955e+00 k_t
= [2, 4, 4, 3]
Generation 110: Best = 2.897496e+03 FES = 11100 Diversity = 1.8039e+00 k_t
= [1, 1, 1, 1]
Generation 120: Best = 2.892684e+03 FES = 12100 Diversity = 1.4406e+00 k_t
= [3, 3, 3, 4]
Generation 130: Best = 2.890041e+03 FES = 13100 Diversity = 1.2323e+00 k_t
= [1, 1, 1, 1]
Generation 140: Best = 2.888646e+03 FES = 14100 Diversity = 9.0482e-01 k_t
= [2, 3, 3, 3]
Generation 150: Best = 2.888082e+03 FES = 15100 Diversity = 6.1797e-01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 2.887620e+03 FES = 16100 Diversity = 4.5895e-01 k_t
= [4, 2, 3, 2]
Generation 170: Best = 2.887452e+03 FES = 17100 Diversity = 3.3126e-01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 2.887368e+03 FES = 18100 Diversity = 2.8123e-01 k_t
= [2, 4, 3, 2]
Generation 190: Best = 2.887211e+03 FES = 19100 Diversity = 2.4390e-01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 2.887156e+03 FES = 20100 Diversity = 1.9091e-01 k_t
= [4, 4, 4, 3]
Generation 210: Best = 2.887091e+03 FES = 21100 Diversity = 1.3795e-01 k_t
= [1, 1, 1, 1]
Generation 220: Best = 2.887008e+03 FES = 22100 Diversity = 1.0099e-01 k_t

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= [4, 4, 4, 3]
Generation 230: Best = 2.886938e+03 FES = 23100 Diversity = 7.2185e-02 k_t
= [1, 1, 1, 1]
Generation 240: Best = 2.886897e+03 FES = 24100 Diversity = 4.2565e-02 k_t
= [3, 3, 3, 2]
Generation 250: Best = 2.886883e+03 FES = 25100 Diversity = 2.4605e-02 k_t
= [1, 1, 1, 1]
Generation 260: Best = 2.886876e+03 FES = 26100 Diversity = 1.6263e-02 k_t
= [2, 4, 4, 3]
Generation 270: Best = 2.886871e+03 FES = 27100 Diversity = 1.0782e-02 k_t
= [1, 1, 1, 1]
Generation 280: Best = 2.886868e+03 FES = 28100 Diversity = 6.9852e-03 k_t
= [3, 4, 4, 3]
Generation 290: Best = 2.886868e+03 FES = 29100 Diversity = 4.7153e-03 k_t
= [1, 1, 1, 1]
Generation 300: Best = 2.886867e+03 FES = 30100 Diversity = 3.1599e-03 k_t
= [2, 4, 2, 3]
Generation 310: Best = 2.886867e+03 FES = 31100 Diversity = 2.1548e-03 k_t
= [1, 1, 1, 1]
Generation 320: Best = 2.886867e+03 FES = 32100 Diversity = 1.4908e-03 k_t
= [2, 2, 3, 2]
Generation 330: Best = 2.886867e+03 FES = 33100 Diversity = 8.9079e-04 k_t
= [1, 1, 1, 1]
Generation 340: Best = 2.886867e+03 FES = 34100 Diversity = 5.9157e-04 k_t
= [4, 3, 3, 4]
Generation 350: Best = 2.886867e+03 FES = 35100 Diversity = 3.9248e-04 k_t
= [1, 1, 1, 1]
Generation 360: Best = 2.886867e+03 FES = 36100 Diversity = 2.6990e-04 k_t
= [3, 3, 3, 4]
Generation 370: Best = 2.886867e+03 FES = 37100 Diversity = 1.7277e-04 k_t
= [1, 1, 1, 1]
Generation 380: Best = 2.886867e+03 FES = 38100 Diversity = 1.0934e-04 k_t
= [4, 3, 3, 3]
Generation 390: Best = 2.886867e+03 FES = 39100 Diversity = 7.5556e-05 k_t
= [1, 1, 1, 1]
Generation 400: Best = 2.886867e+03 FES = 40100 Diversity = 4.9428e-05 k_t
= [2, 3, 4, 3]
Generation 410: Best = 2.886867e+03 FES = 41100 Diversity = 3.3395e-05 k_t
= [1, 1, 1, 1]
Generation 420: Best = 2.886867e+03 FES = 42100 Diversity = 2.3323e-05 k_t
= [3, 3, 4, 3]
Generation 430: Best = 2.886867e+03 FES = 43100 Diversity = 1.6237e-05 k_t
= [1, 1, 1, 1]
Generation 440: Best = 2.886867e+03 FES = 44100 Diversity = 1.0732e-05 k_t
= [4, 3, 2, 2]
Generation 450: Best = 2.886867e+03 FES = 45100 Diversity = 7.0813e-06 k_t
= [1, 1, 1, 1]
Generation 460: Best = 2.886867e+03 FES = 46100 Diversity = 4.6750e-06 k_t
= [3, 4, 3, 3]
Generation 470: Best = 2.886867e+03 FES = 47100 Diversity = 3.0131e-06 k_t
= [1, 1, 1, 1]
Generation 480: Best = 2.886867e+03 FES = 48100 Diversity = 1.9991e-06 k_t
= [4, 4, 2, 4]
Generation 490: Best = 2.886867e+03 FES = 49100 Diversity = 1.2075e-06 k_t

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= [1, 1, 1, 1]
Generation 500: Best = 2.886867e+03 FES = 50100 Diversity = 7.8545e-07 k_t
= [2, 3, 3, 3]
Generation 510: Best = 2.886867e+03 FES = 51100 Diversity = 5.4264e-07 k_t
= [1, 1, 1, 1]
Generation 520: Best = 2.886867e+03 FES = 52100 Diversity = 3.5286e-07 k_t
= [2, 2, 2, 3]
Generation 530: Best = 2.886867e+03 FES = 53100 Diversity = 2.1124e-07 k_t
= [1, 1, 1, 1]
Generation 540: Best = 2.886867e+03 FES = 54100 Diversity = 1.4949e-07 k_t
= [4, 2, 4, 2]
Generation 550: Best = 2.886867e+03 FES = 55100 Diversity = 1.2832e-07 k_t
= [1, 1, 1, 1]
Generation 560: Best = 2.886867e+03 FES = 56100 Diversity = 1.2889e-07 k_t
= [2, 4, 4, 4]
Generation 570: Best = 2.886867e+03 FES = 57100 Diversity = 1.2931e-07 k_t
= [1, 1, 1, 1]
Generation 580: Best = 2.886867e+03 FES = 58100 Diversity = 1.2974e-07 k_t
= [2, 4, 4, 2]
Generation 590: Best = 2.886867e+03 FES = 59100 Diversity = 1.2546e-07 k_t
= [1, 1, 1, 1]
Generation 600: Best = 2.886867e+03 FES = 60100 Diversity = 1.2877e-07 k_t
= [4, 2, 2, 3]
Generation 610: Best = 2.886867e+03 FES = 61100 Diversity = 1.2253e-07 k_t
= [1, 1, 1, 1]
Generation 620: Best = 2.886867e+03 FES = 62100 Diversity = 1.2021e-07 k_t
= [3, 3, 3, 3]
Generation 630: Best = 2.886867e+03 FES = 63100 Diversity = 1.1511e-07 k_t
= [1, 1, 1, 1]
Generation 640: Best = 2.886867e+03 FES = 64100 Diversity = 1.0563e-07 k_t
= [2, 3, 3, 2]
Generation 650: Best = 2.886867e+03 FES = 65100 Diversity = 1.0347e-07 k_t
= [1, 1, 1, 1]
Generation 660: Best = 2.886867e+03 FES = 66100 Diversity = 1.0093e-07 k_t
= [2, 4, 3, 2]
Generation 670: Best = 2.886867e+03 FES = 67100 Diversity = 9.1244e-08 k_t
= [1, 1, 1, 1]
Generation 680: Best = 2.886867e+03 FES = 68100 Diversity = 8.3472e-08 k_t
= [4, 3, 2, 4]
Generation 690: Best = 2.886867e+03 FES = 69100 Diversity = 8.3571e-08 k_t
= [1, 1, 1, 1]
Generation 700: Best = 2.886867e+03 FES = 70100 Diversity = 8.3248e-08 k_t
= [4, 3, 3, 3]
Generation 710: Best = 2.886867e+03 FES = 71100 Diversity = 8.4705e-08 k_t
= [1, 1, 1, 1]
Generation 720: Best = 2.886867e+03 FES = 72100 Diversity = 8.3934e-08 k_t
= [3, 2, 2, 2]
Generation 730: Best = 2.886867e+03 FES = 73100 Diversity = 8.2034e-08 k_t
= [1, 1, 1, 1]
Generation 740: Best = 2.886867e+03 FES = 74100 Diversity = 8.3648e-08 k_t
= [3, 4, 2, 4]
Generation 750: Best = 2.886867e+03 FES = 75100 Diversity = 8.1185e-08 k_t
= [1, 1, 1, 1]
Generation 760: Best = 2.886867e+03 FES = 76100 Diversity = 8.1975e-08 k_t

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= [3, 4, 3, 2]
Generation 770: Best = 2.886867e+03 FES = 77100 Diversity = 8.2910e-08 k_t
= [1, 1, 1, 1]
Generation 780: Best = 2.886867e+03 FES = 78100 Diversity = 8.2728e-08 k_t
= [2, 4, 3, 2]
Generation 790: Best = 2.886867e+03 FES = 79100 Diversity = 8.1708e-08 k_t
= [1, 1, 1, 1]
Generation 800: Best = 2.886867e+03 FES = 80100 Diversity = 8.0972e-08 k_t
= [2, 3, 2, 3]
Generation 810: Best = 2.886867e+03 FES = 81100 Diversity = 8.0452e-08 k_t
= [1, 1, 1, 1]
Generation 820: Best = 2.886867e+03 FES = 82100 Diversity = 8.0952e-08 k_t
= [3, 3, 4, 3]
Generation 830: Best = 2.886867e+03 FES = 83100 Diversity = 7.8780e-08 k_t
= [1, 1, 1, 1]
Generation 840: Best = 2.886867e+03 FES = 84100 Diversity = 7.7896e-08 k_t
= [3, 2, 2, 3]
Generation 850: Best = 2.886867e+03 FES = 85100 Diversity = 7.9869e-08 k_t
= [1, 1, 1, 1]
Generation 860: Best = 2.886867e+03 FES = 86100 Diversity = 8.0406e-08 k_t
= [3, 3, 3, 4]
Generation 870: Best = 2.886867e+03 FES = 87100 Diversity = 7.6764e-08 k_t
= [1, 1, 1, 1]
Generation 880: Best = 2.886867e+03 FES = 88100 Diversity = 7.3140e-08 k_t
= [3, 2, 3, 3]
Generation 890: Best = 2.886867e+03 FES = 89100 Diversity = 7.2903e-08 k_t
= [1, 1, 1, 1]
Generation 900: Best = 2.886867e+03 FES = 90100 Diversity = 7.4120e-08 k_t
= [4, 4, 3, 4]
Generation 910: Best = 2.886867e+03 FES = 91100 Diversity = 7.7816e-08 k_t
= [1, 1, 1, 1]
Generation 920: Best = 2.886867e+03 FES = 92100 Diversity = 7.6835e-08 k_t
= [3, 4, 2, 2]
Generation 930: Best = 2.886867e+03 FES = 93100 Diversity = 7.9003e-08 k_t
= [1, 1, 1, 1]
Generation 940: Best = 2.886867e+03 FES = 94100 Diversity = 7.6621e-08 k_t
= [3, 4, 4, 4]
Generation 950: Best = 2.886867e+03 FES = 95100 Diversity = 7.5865e-08 k_t
= [1, 1, 1, 1]
Generation 960: Best = 2.886867e+03 FES = 96100 Diversity = 7.4968e-08 k_t
= [2, 4, 2, 4]
Generation 970: Best = 2.886867e+03 FES = 97100 Diversity = 7.4033e-08 k_t
= [1, 1, 1, 1]
Generation 980: Best = 2.886867e+03 FES = 98100 Diversity = 7.5463e-08 k_t
= [4, 3, 3, 4]
Generation 990: Best = 2.886867e+03 FES = 99100 Diversity = 7.5354e-08 k_t
= [1, 1, 1, 1]
Generation 1000: Best = 2.886867e+03 FES = 100100 Diversity = 7.5017e-08 k_t
= [4, 4, 2, 2]
Generation 1010: Best = 2.886867e+03 FES = 101100 Diversity = 7.5546e-08 k_t
= [1, 1, 1, 1]
Generation 1020: Best = 2.886867e+03 FES = 102100 Diversity = 7.1712e-08 k_t
= [4, 3, 2, 4]
Generation 1030: Best = 2.886867e+03 FES = 103100 Diversity = 7.4140e-08 k_t

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= [1, 1, 1, 1]
Generation 1040: Best = 2.886867e+03 FES = 104100 Diversity = 7.6905e-08 k_t
= [3, 2, 4, 4]
Generation 1050: Best = 2.886867e+03 FES = 105100 Diversity = 7.6545e-08 k_t
= [1, 1, 1, 1]
Generation 1060: Best = 2.886867e+03 FES = 106100 Diversity = 7.6967e-08 k_t
= [3, 3, 2, 3]
Generation 1070: Best = 2.886867e+03 FES = 107100 Diversity = 7.8059e-08 k_t
= [1, 1, 1, 1]
Generation 1080: Best = 2.886867e+03 FES = 108100 Diversity = 7.9921e-08 k_t
= [4, 4, 2, 2]
Generation 1090: Best = 2.886867e+03 FES = 109100 Diversity = 8.1311e-08 k_t
= [1, 1, 1, 1]
Generation 1100: Best = 2.886867e+03 FES = 110100 Diversity = 7.8763e-08 k_t
= [4, 3, 2, 4]
Generation 1110: Best = 2.886867e+03 FES = 111100 Diversity = 8.0401e-08 k_t
= [1, 1, 1, 1]
Generation 1120: Best = 2.886867e+03 FES = 112100 Diversity = 7.7794e-08 k_t
= [2, 2, 3, 3]
Generation 1130: Best = 2.886867e+03 FES = 113100 Diversity = 7.8523e-08 k_t
= [1, 1, 1, 1]
Generation 1140: Best = 2.886867e+03 FES = 114100 Diversity = 7.8894e-08 k_t
= [2, 2, 4, 4]
Generation 1150: Best = 2.886867e+03 FES = 115100 Diversity = 7.7885e-08 k_t
= [1, 1, 1, 1]
Generation 1160: Best = 2.886867e+03 FES = 116100 Diversity = 7.5910e-08 k_t
= [4, 2, 4, 2]
Generation 1170: Best = 2.886867e+03 FES = 117100 Diversity = 7.4344e-08 k_t
= [1, 1, 1, 1]
Generation 1180: Best = 2.886867e+03 FES = 118100 Diversity = 7.2696e-08 k_t
= [2, 2, 4, 4]
Generation 1190: Best = 2.886867e+03 FES = 119100 Diversity = 7.4546e-08 k_t
= [1, 1, 1, 1]
Generation 1200: Best = 2.886867e+03 FES = 120100 Diversity = 7.6065e-08 k_t
= [2, 2, 3, 2]
Generation 1210: Best = 2.886867e+03 FES = 121100 Diversity = 7.8657e-08 k_t
= [1, 1, 1, 1]
Generation 1220: Best = 2.886867e+03 FES = 122100 Diversity = 7.7571e-08 k_t
= [3, 4, 3, 3]
Generation 1230: Best = 2.886867e+03 FES = 123100 Diversity = 7.5509e-08 k_t
= [1, 1, 1, 1]
Generation 1240: Best = 2.886867e+03 FES = 124100 Diversity = 7.4826e-08 k_t
= [2, 2, 2, 2]
Generation 1250: Best = 2.886867e+03 FES = 125100 Diversity = 7.3989e-08 k_t
= [1, 1, 1, 1]
Generation 1260: Best = 2.886867e+03 FES = 126100 Diversity = 7.8688e-08 k_t
= [4, 3, 3, 4]
Generation 1270: Best = 2.886867e+03 FES = 127100 Diversity = 7.8967e-08 k_t
= [1, 1, 1, 1]
Generation 1280: Best = 2.886867e+03 FES = 128100 Diversity = 7.8133e-08 k_t
= [3, 3, 2, 4]
Generation 1290: Best = 2.886867e+03 FES = 129100 Diversity = 7.6439e-08 k_t
= [1, 1, 1, 1]
Generation 1300: Best = 2.886867e+03 FES = 130100 Diversity = 7.5265e-08 k_t

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= [2, 2, 4, 2]
Generation 1310: Best = 2.886867e+03 FES = 131100 Diversity = 7.5214e-08 k_t
= [1, 1, 1, 1]
Generation 1320: Best = 2.886867e+03 FES = 132100 Diversity = 7.3779e-08 k_t
= [4, 3, 2, 4]
Generation 1330: Best = 2.886867e+03 FES = 133100 Diversity = 7.4457e-08 k_t
= [1, 1, 1, 1]
Generation 1340: Best = 2.886867e+03 FES = 134100 Diversity = 7.5940e-08 k_t
= [3, 3, 4, 3]
Generation 1350: Best = 2.886867e+03 FES = 135100 Diversity = 7.7366e-08 k_t
= [1, 1, 1, 1]
Generation 1360: Best = 2.886867e+03 FES = 136100 Diversity = 7.8438e-08 k_t
= [4, 2, 3, 4]
Generation 1370: Best = 2.886867e+03 FES = 137100 Diversity = 8.0483e-08 k_t
= [1, 1, 1, 1]
Generation 1380: Best = 2.886867e+03 FES = 138100 Diversity = 7.7787e-08 k_t
= [2, 3, 3, 2]
Generation 1390: Best = 2.886867e+03 FES = 139100 Diversity = 7.6982e-08 k_t
= [1, 1, 1, 1]
Generation 1400: Best = 2.886867e+03 FES = 140100 Diversity = 7.5116e-08 k_t
= [2, 3, 4, 2]
Generation 1410: Best = 2.886867e+03 FES = 141100 Diversity = 7.1350e-08 k_t
= [1, 1, 1, 1]
Generation 1420: Best = 2.886867e+03 FES = 142100 Diversity = 7.4343e-08 k_t
= [2, 3, 4, 2]
Generation 1430: Best = 2.886867e+03 FES = 143100 Diversity = 7.7568e-08 k_t
= [1, 1, 1, 1]
Generation 1440: Best = 2.886867e+03 FES = 144100 Diversity = 7.8458e-08 k_t
= [2, 2, 2, 2]
Generation 1450: Best = 2.886867e+03 FES = 145100 Diversity = 7.9346e-08 k_t
= [1, 1, 1, 1]
Generation 1460: Best = 2.886867e+03 FES = 146100 Diversity = 7.9578e-08 k_t
= [4, 2, 4, 3]
Generation 1470: Best = 2.886867e+03 FES = 147100 Diversity = 7.7179e-08 k_t
= [1, 1, 1, 1]
Generation 1480: Best = 2.886867e+03 FES = 148100 Diversity = 7.8386e-08 k_t
= [4, 3, 3, 4]
Generation 1490: Best = 2.886867e+03 FES = 149100 Diversity = 7.6588e-08 k_t
= [1, 1, 1, 1]
Generation 1500: Best = 2.886867e+03 FES = 150100 Diversity = 7.7281e-08 k_t
= [2, 2, 3, 3]
Generation 1510: Best = 2.886867e+03 FES = 151100 Diversity = 7.5425e-08 k_t
= [1, 1, 1, 1]
Generation 1520: Best = 2.886867e+03 FES = 152100 Diversity = 7.7461e-08 k_t
= [2, 4, 3, 4]
Generation 1530: Best = 2.886867e+03 FES = 153100 Diversity = 7.6086e-08 k_t
= [1, 1, 1, 1]
Generation 1540: Best = 2.886867e+03 FES = 154100 Diversity = 7.7909e-08 k_t
= [3, 2, 4, 4]
Generation 1550: Best = 2.886867e+03 FES = 155100 Diversity = 7.8844e-08 k_t
= [1, 1, 1, 1]
Generation 1560: Best = 2.886867e+03 FES = 156100 Diversity = 7.8050e-08 k_t
= [2, 3, 4, 4]
Generation 1570: Best = 2.886867e+03 FES = 157100 Diversity = 7.4338e-08 k_t

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= [1, 1, 1, 1]
Generation 1580: Best = 2.886867e+03 FES = 158100 Diversity = 7.3511e-08 k_t
= [4, 4, 3, 4]
Generation 1590: Best = 2.886867e+03 FES = 159100 Diversity = 7.2970e-08 k_t
= [1, 1, 1, 1]
Generation 1600: Best = 2.886867e+03 FES = 160100 Diversity = 7.6950e-08 k_t
= [3, 4, 3, 4]
Generation 1610: Best = 2.886867e+03 FES = 161100 Diversity = 7.3873e-08 k_t
= [1, 1, 1, 1]
Generation 1620: Best = 2.886867e+03 FES = 162100 Diversity = 7.4721e-08 k_t
= [2, 4, 2, 3]
Generation 1630: Best = 2.886867e+03 FES = 163100 Diversity = 7.4193e-08 k_t
= [1, 1, 1, 1]
Generation 1640: Best = 2.886867e+03 FES = 164100 Diversity = 7.7626e-08 k_t
= [2, 2, 3, 3]
Generation 1650: Best = 2.886867e+03 FES = 165100 Diversity = 7.8157e-08 k_t
= [1, 1, 1, 1]
Generation 1660: Best = 2.886867e+03 FES = 166100 Diversity = 7.8372e-08 k_t
= [4, 3, 4, 2]
Generation 1670: Best = 2.886867e+03 FES = 167100 Diversity = 7.6499e-08 k_t
= [1, 1, 1, 1]
Generation 1680: Best = 2.886867e+03 FES = 168100 Diversity = 7.3671e-08 k_t
= [3, 3, 2, 2]
Generation 1690: Best = 2.886867e+03 FES = 169100 Diversity = 7.3915e-08 k_t
= [1, 1, 1, 1]
Generation 1700: Best = 2.886867e+03 FES = 170100 Diversity = 7.4466e-08 k_t
= [4, 2, 4, 4]
Generation 1710: Best = 2.886867e+03 FES = 171100 Diversity = 7.8567e-08 k_t
= [1, 1, 1, 1]
Generation 1720: Best = 2.886867e+03 FES = 172100 Diversity = 7.8846e-08 k_t
= [2, 2, 4, 2]
Generation 1730: Best = 2.886867e+03 FES = 173100 Diversity = 7.8795e-08 k_t
= [1, 1, 1, 1]
Generation 1740: Best = 2.886867e+03 FES = 174100 Diversity = 7.9725e-08 k_t
= [2, 3, 4, 3]
Generation 1750: Best = 2.886867e+03 FES = 175100 Diversity = 7.9061e-08 k_t
= [1, 1, 1, 1]
Generation 1760: Best = 2.886867e+03 FES = 176100 Diversity = 7.8731e-08 k_t
= [3, 4, 3, 4]
Generation 1770: Best = 2.886867e+03 FES = 177100 Diversity = 7.9874e-08 k_t
= [1, 1, 1, 1]
Generation 1780: Best = 2.886867e+03 FES = 178100 Diversity = 7.8053e-08 k_t
= [2, 3, 3, 2]
Generation 1790: Best = 2.886867e+03 FES = 179100 Diversity = 7.8939e-08 k_t
= [1, 1, 1, 1]
Generation 1800: Best = 2.886867e+03 FES = 180100 Diversity = 7.8378e-08 k_t
= [3, 3, 4, 3]
Generation 1810: Best = 2.886867e+03 FES = 181100 Diversity = 7.8573e-08 k_t
= [1, 1, 1, 1]
Generation 1820: Best = 2.886867e+03 FES = 182100 Diversity = 7.8076e-08 k_t
= [3, 4, 2, 3]
Generation 1830: Best = 2.886867e+03 FES = 183100 Diversity = 7.7432e-08 k_t
= [1, 1, 1, 1]
Generation 1840: Best = 2.886867e+03 FES = 184100 Diversity = 7.5556e-08 k_t

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= [4, 3, 2, 3]
Generation 1850: Best = 2.886867e+03 FES = 185100 Diversity = 7.3056e-08 k_t
= [1, 1, 1, 1]
Generation 1860: Best = 2.886867e+03 FES = 186100 Diversity = 7.4861e-08 k_t
= [2, 2, 3, 3]
Generation 1870: Best = 2.886867e+03 FES = 187100 Diversity = 7.7388e-08 k_t
= [1, 1, 1, 1]
Generation 1880: Best = 2.886867e+03 FES = 188100 Diversity = 7.6914e-08 k_t
= [4, 4, 2, 3]
Generation 1890: Best = 2.886867e+03 FES = 189100 Diversity = 7.6886e-08 k_t
= [1, 1, 1, 1]
Generation 1900: Best = 2.886867e+03 FES = 190100 Diversity = 7.7151e-08 k_t
= [2, 2, 3, 3]
Generation 1910: Best = 2.886867e+03 FES = 191100 Diversity = 7.6490e-08 k_t
= [1, 1, 1, 1]
Generation 1920: Best = 2.886867e+03 FES = 192100 Diversity = 7.5541e-08 k_t
= [2, 2, 2, 2]
Generation 1930: Best = 2.886867e+03 FES = 193100 Diversity = 7.6024e-08 k_t
= [1, 1, 1, 1]
Generation 1940: Best = 2.886867e+03 FES = 194100 Diversity = 7.4815e-08 k_t
= [4, 2, 3, 2]
Generation 1950: Best = 2.886867e+03 FES = 195100 Diversity = 7.2788e-08 k_t
= [1, 1, 1, 1]
Generation 1960: Best = 2.886867e+03 FES = 196100 Diversity = 7.4378e-08 k_t
= [3, 2, 4, 2]
Generation 1970: Best = 2.886867e+03 FES = 197100 Diversity = 7.6141e-08 k_t
= [1, 1, 1, 1]
Generation 1980: Best = 2.886867e+03 FES = 198100 Diversity = 7.5426e-08 k_t
= [2, 2, 2, 3]
Generation 1990: Best = 2.886867e+03 FES = 199100 Diversity = 7.6364e-08 k_t
= [1, 1, 1, 1]
Generation 2000: Best = 2.886867e+03 FES = 200100 Diversity = 7.7245e-08 k_t
= [3, 3, 4, 3]
Generation 2010: Best = 2.886867e+03 FES = 201100 Diversity = 7.7044e-08 k_t
= [1, 1, 1, 1]
Generation 2020: Best = 2.886867e+03 FES = 202100 Diversity = 7.7290e-08 k_t
= [3, 4, 2, 4]
Generation 2030: Best = 2.886867e+03 FES = 203100 Diversity = 7.9397e-08 k_t
= [1, 1, 1, 1]
Generation 2040: Best = 2.886867e+03 FES = 204100 Diversity = 8.0163e-08 k_t
= [2, 3, 3, 4]
Generation 2050: Best = 2.886867e+03 FES = 205100 Diversity = 8.1273e-08 k_t
= [1, 1, 1, 1]
Generation 2060: Best = 2.886867e+03 FES = 206100 Diversity = 8.1291e-08 k_t
= [3, 3, 2, 2]
Generation 2070: Best = 2.886867e+03 FES = 207100 Diversity = 8.1940e-08 k_t
= [1, 1, 1, 1]
Generation 2080: Best = 2.886867e+03 FES = 208100 Diversity = 8.2314e-08 k_t
= [3, 4, 2, 2]
Generation 2090: Best = 2.886867e+03 FES = 209100 Diversity = 8.1346e-08 k_t
= [1, 1, 1, 1]
Generation 2100: Best = 2.886867e+03 FES = 210100 Diversity = 8.3721e-08 k_t
= [4, 3, 2, 4]
Generation 2110: Best = 2.886867e+03 FES = 211100 Diversity = 8.2770e-08 k_t

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= [1, 1, 1, 1]
Generation 2120: Best = 2.886867e+03 FES = 212100 Diversity = 8.2401e-08 k_t
= [2, 3, 4, 2]
Generation 2130: Best = 2.886867e+03 FES = 213100 Diversity = 8.1148e-08 k_t
= [1, 1, 1, 1]
Generation 2140: Best = 2.886867e+03 FES = 214100 Diversity = 8.2111e-08 k_t
= [3, 3, 2, 4]
Generation 2150: Best = 2.886867e+03 FES = 215100 Diversity = 8.2488e-08 k_t
= [1, 1, 1, 1]
Generation 2160: Best = 2.886867e+03 FES = 216100 Diversity = 8.3454e-08 k_t
= [3, 3, 2, 3]
Generation 2170: Best = 2.886867e+03 FES = 217100 Diversity = 8.3966e-08 k_t
= [1, 1, 1, 1]
Generation 2180: Best = 2.886867e+03 FES = 218100 Diversity = 8.1777e-08 k_t
= [2, 4, 2, 4]
Generation 2190: Best = 2.886867e+03 FES = 219100 Diversity = 8.1776e-08 k_t
= [1, 1, 1, 1]
Generation 2200: Best = 2.886867e+03 FES = 220100 Diversity = 7.8879e-08 k_t
= [3, 2, 2, 3]
Generation 2210: Best = 2.886867e+03 FES = 221100 Diversity = 7.9208e-08 k_t
= [1, 1, 1, 1]
Generation 2220: Best = 2.886867e+03 FES = 222100 Diversity = 8.2925e-08 k_t
= [2, 2, 4, 2]
Generation 2230: Best = 2.886867e+03 FES = 223100 Diversity = 8.2531e-08 k_t
= [1, 1, 1, 1]
Generation 2240: Best = 2.886867e+03 FES = 224100 Diversity = 8.2105e-08 k_t
= [4, 3, 4, 2]
Generation 2250: Best = 2.886867e+03 FES = 225100 Diversity = 8.1705e-08 k_t
= [1, 1, 1, 1]
Generation 2260: Best = 2.886867e+03 FES = 226100 Diversity = 8.0540e-08 k_t
= [4, 3, 3, 3]
Generation 2270: Best = 2.886867e+03 FES = 227100 Diversity = 8.1910e-08 k_t
= [1, 1, 1, 1]
Generation 2280: Best = 2.886867e+03 FES = 228100 Diversity = 8.2349e-08 k_t
= [3, 3, 3, 3]
Generation 2290: Best = 2.886867e+03 FES = 229100 Diversity = 8.0728e-08 k_t
= [1, 1, 1, 1]
Generation 2300: Best = 2.886867e+03 FES = 230100 Diversity = 8.0631e-08 k_t
= [3, 2, 2, 2]
Generation 2310: Best = 2.886867e+03 FES = 231100 Diversity = 8.0451e-08 k_t
= [1, 1, 1, 1]
Generation 2320: Best = 2.886867e+03 FES = 232100 Diversity = 8.0417e-08 k_t
= [4, 4, 2, 3]
Generation 2330: Best = 2.886867e+03 FES = 233100 Diversity = 8.0417e-08 k_t
= [1, 1, 1, 1]
Generation 2340: Best = 2.886867e+03 FES = 234100 Diversity = 8.2650e-08 k_t
= [2, 3, 4, 4]
Generation 2350: Best = 2.886867e+03 FES = 235100 Diversity = 7.9470e-08 k_t
= [1, 1, 1, 1]
Generation 2360: Best = 2.886867e+03 FES = 236100 Diversity = 7.9002e-08 k_t
= [2, 4, 4, 2]
Generation 2370: Best = 2.886867e+03 FES = 237100 Diversity = 7.6449e-08 k_t
= [1, 1, 1, 1]
Generation 2380: Best = 2.886867e+03 FES = 238100 Diversity = 7.9609e-08 k_t

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= [3, 4, 4, 4]
Generation 2390: Best = 2.886867e+03  FES = 239100  Diversity = 7.9709e-08  k_t
= [1, 1, 1, 1]
Generation 2400: Best = 2.886867e+03  FES = 240100  Diversity = 7.8457e-08  k_t
= [2, 2, 4, 3]
Generation 2410: Best = 2.886867e+03  FES = 241100  Diversity = 8.0271e-08  k_t
= [1, 1, 1, 1]
Generation 2420: Best = 2.886867e+03  FES = 242100  Diversity = 7.7249e-08  k_t
= [3, 2, 2, 4]
Generation 2430: Best = 2.886867e+03  FES = 243100  Diversity = 7.6474e-08  k_t
= [1, 1, 1, 1]
Generation 2440: Best = 2.886867e+03  FES = 244100  Diversity = 7.7573e-08  k_t
= [3, 2, 2, 3]
Generation 2450: Best = 2.886867e+03  FES = 245100  Diversity = 8.1330e-08  k_t
= [1, 1, 1, 1]
Generation 2460: Best = 2.886867e+03  FES = 246100  Diversity = 8.0434e-08  k_t
= [2, 4, 2, 3]
Generation 2470: Best = 2.886867e+03  FES = 247100  Diversity = 8.0437e-08  k_t
= [1, 1, 1, 1]
Generation 2480: Best = 2.886867e+03  FES = 248100  Diversity = 7.9453e-08  k_t
= [4, 4, 4, 3]
Generation 2490: Best = 2.886867e+03  FES = 249100  Diversity = 7.8896e-08  k_t
= [1, 1, 1, 1]
Generation 2500: Best = 2.886867e+03  FES = 250100  Diversity = 7.9327e-08  k_t
= [2, 2, 2, 3]
Generation 2510: Best = 2.886867e+03  FES = 251100  Diversity = 7.5956e-08  k_t
= [1, 1, 1, 1]
Generation 2520: Best = 2.886867e+03  FES = 252100  Diversity = 7.3011e-08  k_t
= [4, 2, 4, 2]
Generation 2530: Best = 2.886867e+03  FES = 253100  Diversity = 7.5734e-08  k_t
= [1, 1, 1, 1]
Generation 2540: Best = 2.886867e+03  FES = 254100  Diversity = 7.6802e-08  k_t
= [2, 3, 4, 4]
Generation 2550: Best = 2.886867e+03  FES = 255100  Diversity = 7.9206e-08  k_t
= [1, 1, 1, 1]
Generation 2560: Best = 2.886867e+03  FES = 256100  Diversity = 7.7198e-08  k_t
= [3, 4, 2, 2]
Generation 2570: Best = 2.886867e+03  FES = 257100  Diversity = 7.7655e-08  k_t
= [1, 1, 1, 1]
Generation 2580: Best = 2.886867e+03  FES = 258100  Diversity = 7.6851e-08  k_t
= [4, 3, 4, 2]
Generation 2590: Best = 2.886867e+03  FES = 259100  Diversity = 7.6537e-08  k_t
= [1, 1, 1, 1]
Generation 2600: Best = 2.886867e+03  FES = 260100  Diversity = 7.6672e-08  k_t
= [3, 3, 3, 4]
Generation 2610: Best = 2.886867e+03  FES = 261100  Diversity = 8.0321e-08  k_t
= [1, 1, 1, 1]
Generation 2620: Best = 2.886867e+03  FES = 262100  Diversity = 7.7326e-08  k_t
= [3, 2, 2, 3]
Generation 2630: Best = 2.886867e+03  FES = 263100  Diversity = 7.8297e-08  k_t
= [1, 1, 1, 1]
Generation 2640: Best = 2.886867e+03  FES = 264100  Diversity = 7.7565e-08  k_t
= [2, 2, 4, 4]
Generation 2650: Best = 2.886867e+03  FES = 265100  Diversity = 7.6842e-08  k_t

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= [1, 1, 1, 1]
Generation 2660: Best = 2.886867e+03 FES = 266100 Diversity = 7.9530e-08 k_t
= [4, 2, 4, 4]
Generation 2670: Best = 2.886867e+03 FES = 267100 Diversity = 7.7669e-08 k_t
= [1, 1, 1, 1]
Generation 2680: Best = 2.886867e+03 FES = 268100 Diversity = 8.0084e-08 k_t
= [4, 3, 2, 3]
Generation 2690: Best = 2.886867e+03 FES = 269100 Diversity = 8.0951e-08 k_t
= [1, 1, 1, 1]
Generation 2700: Best = 2.886867e+03 FES = 270100 Diversity = 8.0206e-08 k_t
= [4, 3, 3, 2]
Generation 2710: Best = 2.886867e+03 FES = 271100 Diversity = 8.1075e-08 k_t
= [1, 1, 1, 1]
Generation 2720: Best = 2.886867e+03 FES = 272100 Diversity = 7.9263e-08 k_t
= [2, 4, 2, 4]
Generation 2730: Best = 2.886867e+03 FES = 273100 Diversity = 7.8588e-08 k_t
= [1, 1, 1, 1]
Generation 2740: Best = 2.886867e+03 FES = 274100 Diversity = 7.9405e-08 k_t
= [4, 3, 4, 4]
Generation 2750: Best = 2.886867e+03 FES = 275100 Diversity = 7.7823e-08 k_t
= [1, 1, 1, 1]
Generation 2760: Best = 2.886867e+03 FES = 276100 Diversity = 7.7914e-08 k_t
= [2, 3, 2, 4]
Generation 2770: Best = 2.886867e+03 FES = 277100 Diversity = 7.7134e-08 k_t
= [1, 1, 1, 1]
Generation 2780: Best = 2.886867e+03 FES = 278100 Diversity = 7.6847e-08 k_t
= [3, 2, 4, 2]
Generation 2790: Best = 2.886867e+03 FES = 279100 Diversity = 7.4741e-08 k_t
= [1, 1, 1, 1]
Generation 2800: Best = 2.886867e+03 FES = 280100 Diversity = 7.4509e-08 k_t
= [4, 3, 3, 4]
Generation 2810: Best = 2.886867e+03 FES = 281100 Diversity = 7.4988e-08 k_t
= [1, 1, 1, 1]
Generation 2820: Best = 2.886867e+03 FES = 282100 Diversity = 7.6485e-08 k_t
= [3, 4, 3, 2]
Generation 2830: Best = 2.886867e+03 FES = 283100 Diversity = 7.6020e-08 k_t
= [1, 1, 1, 1]
Generation 2840: Best = 2.886867e+03 FES = 284100 Diversity = 7.9473e-08 k_t
= [2, 3, 2, 4]
Generation 2850: Best = 2.886867e+03 FES = 285100 Diversity = 7.8463e-08 k_t
= [1, 1, 1, 1]
Generation 2860: Best = 2.886867e+03 FES = 286100 Diversity = 8.0290e-08 k_t
= [2, 2, 4, 3]
Generation 2870: Best = 2.886867e+03 FES = 287100 Diversity = 8.1633e-08 k_t
= [1, 1, 1, 1]
Generation 2880: Best = 2.886867e+03 FES = 288100 Diversity = 7.9873e-08 k_t
= [3, 3, 4, 4]
Generation 2890: Best = 2.886867e+03 FES = 289100 Diversity = 8.0105e-08 k_t
= [1, 1, 1, 1]
Generation 2900: Best = 2.886867e+03 FES = 290100 Diversity = 7.8899e-08 k_t
= [4, 3, 3, 2]
Generation 2910: Best = 2.886867e+03 FES = 291100 Diversity = 7.8536e-08 k_t
= [1, 1, 1, 1]
Generation 2920: Best = 2.886867e+03 FES = 292100 Diversity = 7.8614e-08 k_t

```

```

= [3, 2, 3, 4]
Generation 2930: Best = 2.886867e+03   FES = 293100   Diversity = 7.9326e-08   k_t
= [1, 1, 1, 1]
Generation 2940: Best = 2.886867e+03   FES = 294100   Diversity = 7.9648e-08   k_t
= [4, 4, 3, 2]
Generation 2950: Best = 2.886867e+03   FES = 295100   Diversity = 7.7843e-08   k_t
= [1, 1, 1, 1]
Generation 2960: Best = 2.886867e+03   FES = 296100   Diversity = 7.9090e-08   k_t
= [3, 4, 2, 4]
Generation 2970: Best = 2.886867e+03   FES = 297100   Diversity = 7.7160e-08   k_t
= [1, 1, 1, 1]
Generation 2980: Best = 2.886867e+03   FES = 298100   Diversity = 7.7148e-08   k_t
= [2, 3, 2, 2]
Generation 2990: Best = 2.886867e+03   FES = 299100   Diversity = 7.6461e-08   k_t
= [1, 1, 1, 1]

```

Optimization completed!

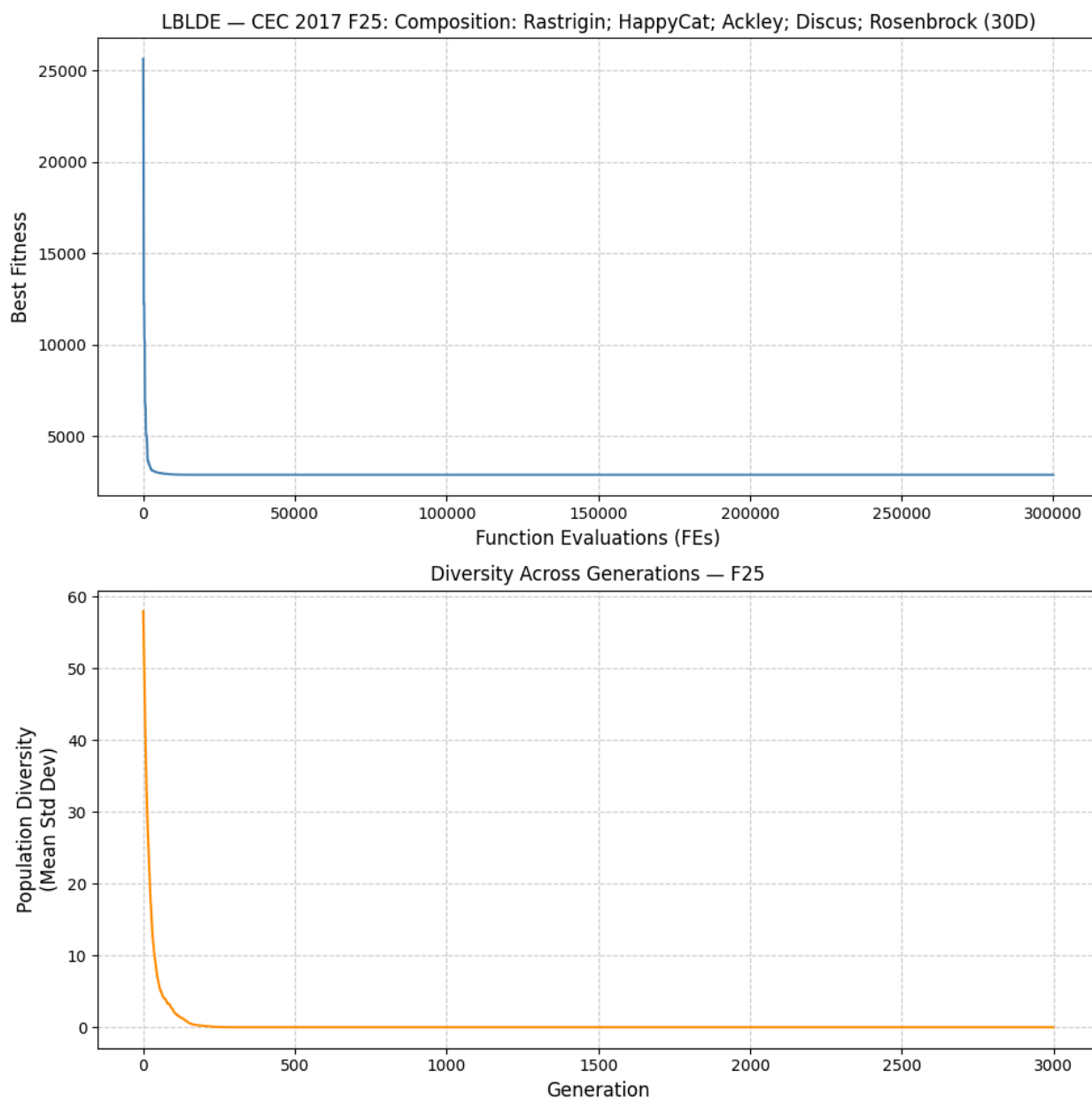
Final Best Fitness = 2.886867e+03

Total Function Evaluations = 300000

```

=====
=
Results:
  Best Fitness           : 2.886867e+03
  Global Optimum         : 2.500000e+03
  Error (|best-optimum|) : 3.868671e+02
  Best Solution (5 dim)  : [ 22.92847046  13.88133809 -17.94823897  -8.19906234
-34.56301893]
  D^(0) (init. diversity): 5.7976e+01
=====
=

```



Graph saved as convergence_F25.png

```
In [36]: # =====
# F26 – Composition Function 6 (Schaffer; Schwefel; Griewank; Rosenbrock; Rastrigin)
# =====
run_and_plot('F26', make_wrapper(composition.f26), 2600.0, 'Composition: Schaffer; Schwefel; Griewank; Rosenbrock; Rastrigin')
```

```

=====
=
LBLDE_s on CEC 2017 F26 – Single Run
=====
=
Function      : Composition: Schaffer; Schwefel; Griewank; Rosenbrock; Rastrig
in (30D)
Bounds       : [-100, 100]
Global optimum : 2600.0
Initial D^(0) = 5.7976e+01
Generation 0: Best Fitness = 1.514240e+04 | Diversity = 5.7976e+01
Generation 10: Best = 1.007613e+04 FES = 1100 Diversity = 4.4906e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 7.546877e+03 FES = 2100 Diversity = 3.2297e+01 k_t
= [3, 3, 2, 2]
Generation 30: Best = 6.968405e+03 FES = 3100 Diversity = 2.4513e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 6.393764e+03 FES = 4100 Diversity = 2.1113e+01 k_t
= [2, 3, 2, 3]
Generation 50: Best = 6.054244e+03 FES = 5100 Diversity = 1.7955e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 5.952396e+03 FES = 6100 Diversity = 1.6472e+01 k_t
= [2, 2, 3, 4]
Generation 70: Best = 5.927407e+03 FES = 7100 Diversity = 1.6019e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 5.717884e+03 FES = 8100 Diversity = 1.5292e+01 k_t
= [2, 3, 4, 2]
Generation 90: Best = 5.606634e+03 FES = 9100 Diversity = 1.5116e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 5.560767e+03 FES = 10100 Diversity = 1.4858e+01 k_t
= [4, 2, 2, 3]
Generation 110: Best = 5.422841e+03 FES = 11100 Diversity = 1.4463e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 5.397620e+03 FES = 12100 Diversity = 1.4356e+01 k_t
= [3, 3, 4, 4]
Generation 130: Best = 5.372439e+03 FES = 13100 Diversity = 1.4308e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 5.188381e+03 FES = 14100 Diversity = 1.4032e+01 k_t
= [3, 2, 2, 2]
Generation 150: Best = 5.188381e+03 FES = 15100 Diversity = 1.3926e+01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 5.188381e+03 FES = 16100 Diversity = 1.3922e+01 k_t
= [2, 3, 2, 3]
Generation 170: Best = 5.188381e+03 FES = 17100 Diversity = 1.3863e+01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 5.188381e+03 FES = 18100 Diversity = 1.3804e+01 k_t
= [4, 2, 2, 2]
Generation 190: Best = 5.059270e+03 FES = 19100 Diversity = 1.3533e+01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 5.059270e+03 FES = 20100 Diversity = 1.3470e+01 k_t
= [4, 4, 2, 2]
Generation 210: Best = 5.059270e+03 FES = 21100 Diversity = 1.3270e+01 k_t
= [1, 1, 1, 1]
Generation 220: Best = 5.041586e+03 FES = 22100 Diversity = 1.3276e+01 k_t

```

```

= [3, 2, 2, 4]
Generation 230: Best = 5.007457e+03  FES = 23100  Diversity = 1.3229e+01  k_t
= [1, 1, 1, 1]
Generation 240: Best = 4.929656e+03  FES = 24100  Diversity = 1.3239e+01  k_t
= [3, 2, 4, 3]
Generation 250: Best = 4.912261e+03  FES = 25100  Diversity = 1.3366e+01  k_t
= [1, 1, 1, 1]
Generation 260: Best = 4.876754e+03  FES = 26100  Diversity = 1.3265e+01  k_t
= [3, 3, 3, 3]
Generation 270: Best = 4.876754e+03  FES = 27100  Diversity = 1.3287e+01  k_t
= [1, 1, 1, 1]
Generation 280: Best = 4.876754e+03  FES = 28100  Diversity = 1.3266e+01  k_t
= [3, 3, 3, 2]
Generation 290: Best = 4.836632e+03  FES = 29100  Diversity = 1.3238e+01  k_t
= [1, 1, 1, 1]
Generation 300: Best = 4.826990e+03  FES = 30100  Diversity = 1.3052e+01  k_t
= [2, 3, 4, 2]
Generation 310: Best = 4.818483e+03  FES = 31100  Diversity = 1.3023e+01  k_t
= [1, 1, 1, 1]
Generation 320: Best = 4.818483e+03  FES = 32100  Diversity = 1.2997e+01  k_t
= [3, 3, 4, 2]
Generation 330: Best = 4.818483e+03  FES = 33100  Diversity = 1.3002e+01  k_t
= [1, 1, 1, 1]
Generation 340: Best = 4.706903e+03  FES = 34100  Diversity = 1.2847e+01  k_t
= [4, 4, 4, 4]
Generation 350: Best = 4.696435e+03  FES = 35100  Diversity = 1.3032e+01  k_t
= [1, 1, 1, 1]
Generation 360: Best = 4.696435e+03  FES = 36100  Diversity = 1.3116e+01  k_t
= [3, 2, 4, 4]
Generation 370: Best = 4.605940e+03  FES = 37100  Diversity = 1.2950e+01  k_t
= [1, 1, 1, 1]
Generation 380: Best = 4.605940e+03  FES = 38100  Diversity = 1.2927e+01  k_t
= [4, 3, 2, 3]
Generation 390: Best = 4.605940e+03  FES = 39100  Diversity = 1.2927e+01  k_t
= [1, 1, 1, 1]
Generation 400: Best = 4.605940e+03  FES = 40100  Diversity = 1.2843e+01  k_t
= [2, 3, 2, 3]
Generation 410: Best = 4.605940e+03  FES = 41100  Diversity = 1.2832e+01  k_t
= [1, 1, 1, 1]
Generation 420: Best = 4.605940e+03  FES = 42100  Diversity = 1.2793e+01  k_t
= [4, 4, 2, 3]
Generation 430: Best = 4.605940e+03  FES = 43100  Diversity = 1.2789e+01  k_t
= [1, 1, 1, 1]
Generation 440: Best = 4.574230e+03  FES = 44100  Diversity = 1.2741e+01  k_t
= [2, 4, 4, 2]
Generation 450: Best = 4.574230e+03  FES = 45100  Diversity = 1.2638e+01  k_t
= [1, 1, 1, 1]
Generation 460: Best = 4.561123e+03  FES = 46100  Diversity = 1.2593e+01  k_t
= [3, 2, 2, 2]
Generation 470: Best = 4.485808e+03  FES = 47100  Diversity = 1.2440e+01  k_t
= [1, 1, 1, 1]
Generation 480: Best = 4.485808e+03  FES = 48100  Diversity = 1.2399e+01  k_t
= [3, 3, 2, 2]
Generation 490: Best = 4.485808e+03  FES = 49100  Diversity = 1.2348e+01  k_t

```



```

= [1, 1, 1, 1]
Generation 500: Best = 4.484725e+03 FES = 50100 Diversity = 1.2389e+01 k_t
= [3, 2, 3, 2]
Generation 510: Best = 4.467554e+03 FES = 51100 Diversity = 1.2139e+01 k_t
= [1, 1, 1, 1]
Generation 520: Best = 4.467554e+03 FES = 52100 Diversity = 1.2118e+01 k_t
= [4, 2, 3, 3]
Generation 530: Best = 4.467554e+03 FES = 53100 Diversity = 1.2122e+01 k_t
= [1, 1, 1, 1]
Generation 540: Best = 4.466509e+03 FES = 54100 Diversity = 1.2116e+01 k_t
= [2, 3, 4, 2]
Generation 550: Best = 4.466509e+03 FES = 55100 Diversity = 1.2070e+01 k_t
= [1, 1, 1, 1]
Generation 560: Best = 4.466509e+03 FES = 56100 Diversity = 1.2066e+01 k_t
= [4, 4, 4, 2]
Generation 570: Best = 4.466509e+03 FES = 57100 Diversity = 1.2075e+01 k_t
= [1, 1, 1, 1]
Generation 580: Best = 4.466509e+03 FES = 58100 Diversity = 1.2026e+01 k_t
= [2, 2, 4, 3]
Generation 590: Best = 4.435613e+03 FES = 59100 Diversity = 1.2019e+01 k_t
= [1, 1, 1, 1]
Generation 600: Best = 4.435613e+03 FES = 60100 Diversity = 1.1938e+01 k_t
= [4, 4, 2, 3]
Generation 610: Best = 4.423645e+03 FES = 61100 Diversity = 1.1941e+01 k_t
= [1, 1, 1, 1]
Generation 620: Best = 4.423645e+03 FES = 62100 Diversity = 1.2017e+01 k_t
= [4, 2, 2, 4]
Generation 630: Best = 4.423645e+03 FES = 63100 Diversity = 1.2022e+01 k_t
= [1, 1, 1, 1]
Generation 640: Best = 4.412627e+03 FES = 64100 Diversity = 1.1969e+01 k_t
= [2, 3, 2, 2]
Generation 650: Best = 4.412627e+03 FES = 65100 Diversity = 1.1949e+01 k_t
= [1, 1, 1, 1]
Generation 660: Best = 4.412627e+03 FES = 66100 Diversity = 1.1902e+01 k_t
= [4, 2, 2, 2]
Generation 670: Best = 4.412627e+03 FES = 67100 Diversity = 1.1898e+01 k_t
= [1, 1, 1, 1]
Generation 680: Best = 4.366919e+03 FES = 68100 Diversity = 1.1904e+01 k_t
= [2, 4, 2, 2]
Generation 690: Best = 4.345912e+03 FES = 69100 Diversity = 1.1888e+01 k_t
= [1, 1, 1, 1]
Generation 700: Best = 4.345032e+03 FES = 70100 Diversity = 1.1791e+01 k_t
= [3, 3, 4, 3]
Generation 710: Best = 4.345032e+03 FES = 71100 Diversity = 1.1672e+01 k_t
= [1, 1, 1, 1]
Generation 720: Best = 4.345032e+03 FES = 72100 Diversity = 1.1635e+01 k_t
= [2, 3, 2, 2]
Generation 730: Best = 4.345032e+03 FES = 73100 Diversity = 1.1598e+01 k_t
= [1, 1, 1, 1]
Generation 740: Best = 4.345032e+03 FES = 74100 Diversity = 1.1482e+01 k_t
= [2, 2, 2, 4]
Generation 750: Best = 4.323218e+03 FES = 75100 Diversity = 1.1459e+01 k_t
= [1, 1, 1, 1]
Generation 760: Best = 4.302255e+03 FES = 76100 Diversity = 1.1449e+01 k_t

```

```

= [4, 4, 3, 3]
Generation 770: Best = 4.302255e+03 FES = 77100 Diversity = 1.1404e+01 k_t
= [1, 1, 1, 1]
Generation 780: Best = 4.298044e+03 FES = 78100 Diversity = 1.1336e+01 k_t
= [4, 2, 2, 2]
Generation 790: Best = 4.297442e+03 FES = 79100 Diversity = 1.1356e+01 k_t
= [1, 1, 1, 1]
Generation 800: Best = 4.258004e+03 FES = 80100 Diversity = 1.1235e+01 k_t
= [4, 2, 2, 2]
Generation 810: Best = 4.243925e+03 FES = 81100 Diversity = 1.1161e+01 k_t
= [1, 1, 1, 1]
Generation 820: Best = 4.243925e+03 FES = 82100 Diversity = 1.1049e+01 k_t
= [4, 2, 4, 4]
Generation 830: Best = 4.203557e+03 FES = 83100 Diversity = 1.0821e+01 k_t
= [1, 1, 1, 1]
Generation 840: Best = 4.187257e+03 FES = 84100 Diversity = 1.0764e+01 k_t
= [2, 3, 4, 4]
Generation 850: Best = 4.164226e+03 FES = 85100 Diversity = 1.0388e+01 k_t
= [1, 1, 1, 1]
Generation 860: Best = 4.114135e+03 FES = 86100 Diversity = 1.0272e+01 k_t
= [4, 4, 2, 4]
Generation 870: Best = 4.062071e+03 FES = 87100 Diversity = 1.0283e+01 k_t
= [1, 1, 1, 1]
Generation 880: Best = 4.042162e+03 FES = 88100 Diversity = 1.0121e+01 k_t
= [3, 3, 4, 3]
Generation 890: Best = 4.036698e+03 FES = 89100 Diversity = 9.8834e+00 k_t
= [1, 1, 1, 1]
Generation 900: Best = 4.036698e+03 FES = 90100 Diversity = 9.5731e+00 k_t
= [4, 3, 2, 2]
Generation 910: Best = 3.971896e+03 FES = 91100 Diversity = 9.6429e+00 k_t
= [1, 1, 1, 1]
Generation 920: Best = 3.971896e+03 FES = 92100 Diversity = 9.0978e+00 k_t
= [4, 3, 4, 3]
Generation 930: Best = 3.962244e+03 FES = 93100 Diversity = 8.3218e+00 k_t
= [1, 1, 1, 1]
Generation 940: Best = 3.949031e+03 FES = 94100 Diversity = 6.3433e+00 k_t
= [3, 3, 2, 4]
Generation 950: Best = 3.932965e+03 FES = 95100 Diversity = 3.5593e+00 k_t
= [1, 1, 1, 1]
Generation 960: Best = 3.907524e+03 FES = 96100 Diversity = 1.4233e+00 k_t
= [3, 2, 3, 4]
Generation 970: Best = 3.887566e+03 FES = 97100 Diversity = 5.4576e-01 k_t
= [1, 1, 1, 1]
Generation 980: Best = 3.877668e+03 FES = 98100 Diversity = 4.4686e-01 k_t
= [3, 3, 3, 3]
Generation 990: Best = 3.872151e+03 FES = 99100 Diversity = 3.2179e-01 k_t
= [1, 1, 1, 1]
Generation 1000: Best = 3.868730e+03 FES = 100100 Diversity = 2.3493e-01 k_t
= [4, 3, 3, 3]
Generation 1010: Best = 3.865873e+03 FES = 101100 Diversity = 1.6826e-01 k_t
= [1, 1, 1, 1]
Generation 1020: Best = 3.865111e+03 FES = 102100 Diversity = 1.1948e-01 k_t
= [3, 3, 2, 2]
Generation 1030: Best = 3.864510e+03 FES = 103100 Diversity = 8.2787e-02 k_t

```

```

= [1, 1, 1, 1]
Generation 1040: Best = 3.864242e+03 FES = 104100 Diversity = 5.8205e-02 k_t
= [3, 2, 3, 4]
Generation 1050: Best = 3.864116e+03 FES = 105100 Diversity = 3.5670e-02 k_t
= [1, 1, 1, 1]
Generation 1060: Best = 3.864082e+03 FES = 106100 Diversity = 2.2712e-02 k_t
= [4, 2, 3, 4]
Generation 1070: Best = 3.864062e+03 FES = 107100 Diversity = 1.4807e-02 k_t
= [1, 1, 1, 1]
Generation 1080: Best = 3.864055e+03 FES = 108100 Diversity = 9.3320e-03 k_t
= [4, 3, 3, 4]
Generation 1090: Best = 3.864050e+03 FES = 109100 Diversity = 5.7817e-03 k_t
= [1, 1, 1, 1]
Generation 1100: Best = 3.864049e+03 FES = 110100 Diversity = 3.9361e-03 k_t
= [2, 3, 3, 3]
Generation 1110: Best = 3.864048e+03 FES = 111100 Diversity = 2.3800e-03 k_t
= [1, 1, 1, 1]
Generation 1120: Best = 3.864048e+03 FES = 112100 Diversity = 1.5825e-03 k_t
= [3, 4, 2, 3]
Generation 1130: Best = 3.864048e+03 FES = 113100 Diversity = 1.0527e-03 k_t
= [1, 1, 1, 1]
Generation 1140: Best = 3.864048e+03 FES = 114100 Diversity = 7.0934e-04 k_t
= [4, 4, 2, 2]
Generation 1150: Best = 3.864048e+03 FES = 115100 Diversity = 4.7447e-04 k_t
= [1, 1, 1, 1]
Generation 1160: Best = 3.864048e+03 FES = 116100 Diversity = 3.1704e-04 k_t
= [4, 3, 4, 2]
Generation 1170: Best = 3.864048e+03 FES = 117100 Diversity = 1.9521e-04 k_t
= [1, 1, 1, 1]
Generation 1180: Best = 3.864048e+03 FES = 118100 Diversity = 1.1728e-04 k_t
= [2, 2, 4, 3]
Generation 1190: Best = 3.864048e+03 FES = 119100 Diversity = 7.8181e-05 k_t
= [1, 1, 1, 1]
Generation 1200: Best = 3.864048e+03 FES = 120100 Diversity = 4.6379e-05 k_t
= [4, 2, 4, 3]
Generation 1210: Best = 3.864048e+03 FES = 121100 Diversity = 2.6000e-05 k_t
= [1, 1, 1, 1]
Generation 1220: Best = 3.864048e+03 FES = 122100 Diversity = 1.7282e-05 k_t
= [4, 2, 3, 4]
Generation 1230: Best = 3.864048e+03 FES = 123100 Diversity = 1.1365e-05 k_t
= [1, 1, 1, 1]
Generation 1240: Best = 3.864048e+03 FES = 124100 Diversity = 7.0123e-06 k_t
= [4, 4, 4, 2]
Generation 1250: Best = 3.864048e+03 FES = 125100 Diversity = 4.4155e-06 k_t
= [1, 1, 1, 1]
Generation 1260: Best = 3.864048e+03 FES = 126100 Diversity = 2.8407e-06 k_t
= [2, 2, 2, 2]
Generation 1270: Best = 3.864048e+03 FES = 127100 Diversity = 1.8940e-06 k_t
= [1, 1, 1, 1]
Generation 1280: Best = 3.864048e+03 FES = 128100 Diversity = 1.3396e-06 k_t
= [4, 2, 4, 2]
Generation 1290: Best = 3.864048e+03 FES = 129100 Diversity = 8.1568e-07 k_t
= [1, 1, 1, 1]
Generation 1300: Best = 3.864048e+03 FES = 130100 Diversity = 5.3492e-07 k_t

```

```

= [2, 4, 3, 3]
Generation 1310: Best = 3.864048e+03 FES = 131100 Diversity = 3.3224e-07 k_t
= [1, 1, 1, 1]
Generation 1320: Best = 3.864048e+03 FES = 132100 Diversity = 2.1683e-07 k_t
= [4, 2, 3, 3]
Generation 1330: Best = 3.864048e+03 FES = 133100 Diversity = 1.4077e-07 k_t
= [1, 1, 1, 1]
Generation 1340: Best = 3.864048e+03 FES = 134100 Diversity = 1.0346e-07 k_t
= [2, 2, 2, 3]
Generation 1350: Best = 3.864048e+03 FES = 135100 Diversity = 8.6804e-08 k_t
= [1, 1, 1, 1]
Generation 1360: Best = 3.864048e+03 FES = 136100 Diversity = 7.7279e-08 k_t
= [3, 3, 2, 2]
Generation 1370: Best = 3.864048e+03 FES = 137100 Diversity = 7.4837e-08 k_t
= [1, 1, 1, 1]
Generation 1380: Best = 3.864048e+03 FES = 138100 Diversity = 7.2392e-08 k_t
= [4, 2, 3, 2]
Generation 1390: Best = 3.864048e+03 FES = 139100 Diversity = 6.9328e-08 k_t
= [1, 1, 1, 1]
Generation 1400: Best = 3.864048e+03 FES = 140100 Diversity = 6.7551e-08 k_t
= [2, 4, 2, 4]
Generation 1410: Best = 3.864048e+03 FES = 141100 Diversity = 6.6623e-08 k_t
= [1, 1, 1, 1]
Generation 1420: Best = 3.864048e+03 FES = 142100 Diversity = 6.5013e-08 k_t
= [4, 2, 2, 2]
Generation 1430: Best = 3.864048e+03 FES = 143100 Diversity = 6.3581e-08 k_t
= [1, 1, 1, 1]
Generation 1440: Best = 3.864048e+03 FES = 144100 Diversity = 6.2860e-08 k_t
= [4, 3, 4, 3]
Generation 1450: Best = 3.864048e+03 FES = 145100 Diversity = 6.2298e-08 k_t
= [1, 1, 1, 1]
Generation 1460: Best = 3.864048e+03 FES = 146100 Diversity = 6.2135e-08 k_t
= [3, 3, 2, 3]
Generation 1470: Best = 3.864048e+03 FES = 147100 Diversity = 6.2365e-08 k_t
= [1, 1, 1, 1]
Generation 1480: Best = 3.864048e+03 FES = 148100 Diversity = 6.1164e-08 k_t
= [4, 2, 3, 3]
Generation 1490: Best = 3.864048e+03 FES = 149100 Diversity = 6.0292e-08 k_t
= [1, 1, 1, 1]
Generation 1500: Best = 3.864048e+03 FES = 150100 Diversity = 6.1461e-08 k_t
= [2, 2, 3, 4]
Generation 1510: Best = 3.864048e+03 FES = 151100 Diversity = 6.0745e-08 k_t
= [1, 1, 1, 1]
Generation 1520: Best = 3.864048e+03 FES = 152100 Diversity = 6.0688e-08 k_t
= [4, 2, 4, 3]
Generation 1530: Best = 3.864048e+03 FES = 153100 Diversity = 5.9732e-08 k_t
= [1, 1, 1, 1]
Generation 1540: Best = 3.864048e+03 FES = 154100 Diversity = 5.9828e-08 k_t
= [2, 3, 3, 3]
Generation 1550: Best = 3.864048e+03 FES = 155100 Diversity = 6.0033e-08 k_t
= [1, 1, 1, 1]
Generation 1560: Best = 3.864048e+03 FES = 156100 Diversity = 6.0441e-08 k_t
= [3, 4, 3, 3]
Generation 1570: Best = 3.864048e+03 FES = 157100 Diversity = 6.0814e-08 k_t

```

```

= [1, 1, 1, 1]
Generation 1580: Best = 3.864048e+03 FES = 158100 Diversity = 6.0612e-08 k_t
= [4, 2, 4, 2]
Generation 1590: Best = 3.864048e+03 FES = 159100 Diversity = 6.0694e-08 k_t
= [1, 1, 1, 1]
Generation 1600: Best = 3.864048e+03 FES = 160100 Diversity = 6.0608e-08 k_t
= [3, 3, 4, 2]
Generation 1610: Best = 3.864048e+03 FES = 161100 Diversity = 6.0247e-08 k_t
= [1, 1, 1, 1]
Generation 1620: Best = 3.864048e+03 FES = 162100 Diversity = 5.8916e-08 k_t
= [4, 4, 4, 3]
Generation 1630: Best = 3.864048e+03 FES = 163100 Diversity = 5.9019e-08 k_t
= [1, 1, 1, 1]
Generation 1640: Best = 3.864048e+03 FES = 164100 Diversity = 5.9181e-08 k_t
= [4, 4, 2, 4]
Generation 1650: Best = 3.864048e+03 FES = 165100 Diversity = 5.9015e-08 k_t
= [1, 1, 1, 1]
Generation 1660: Best = 3.864048e+03 FES = 166100 Diversity = 5.8659e-08 k_t
= [2, 2, 2, 4]
Generation 1670: Best = 3.864048e+03 FES = 167100 Diversity = 5.7891e-08 k_t
= [1, 1, 1, 1]
Generation 1680: Best = 3.864048e+03 FES = 168100 Diversity = 5.7397e-08 k_t
= [4, 4, 3, 4]
Generation 1690: Best = 3.864048e+03 FES = 169100 Diversity = 5.6608e-08 k_t
= [1, 1, 1, 1]
Generation 1700: Best = 3.864048e+03 FES = 170100 Diversity = 5.7217e-08 k_t
= [3, 2, 3, 3]
Generation 1710: Best = 3.864048e+03 FES = 171100 Diversity = 5.6901e-08 k_t
= [1, 1, 1, 1]
Generation 1720: Best = 3.864048e+03 FES = 172100 Diversity = 5.7130e-08 k_t
= [4, 3, 2, 3]
Generation 1730: Best = 3.864048e+03 FES = 173100 Diversity = 5.6551e-08 k_t
= [1, 1, 1, 1]
Generation 1740: Best = 3.864048e+03 FES = 174100 Diversity = 5.7183e-08 k_t
= [2, 4, 4, 2]
Generation 1750: Best = 3.864048e+03 FES = 175100 Diversity = 5.7606e-08 k_t
= [1, 1, 1, 1]
Generation 1760: Best = 3.864048e+03 FES = 176100 Diversity = 5.6717e-08 k_t
= [2, 3, 3, 2]
Generation 1770: Best = 3.864048e+03 FES = 177100 Diversity = 5.6109e-08 k_t
= [1, 1, 1, 1]
Generation 1780: Best = 3.864048e+03 FES = 178100 Diversity = 5.6878e-08 k_t
= [3, 3, 4, 3]
Generation 1790: Best = 3.864048e+03 FES = 179100 Diversity = 5.6383e-08 k_t
= [1, 1, 1, 1]
Generation 1800: Best = 3.864048e+03 FES = 180100 Diversity = 5.6159e-08 k_t
= [3, 2, 3, 4]
Generation 1810: Best = 3.864048e+03 FES = 181100 Diversity = 5.5824e-08 k_t
= [1, 1, 1, 1]
Generation 1820: Best = 3.864048e+03 FES = 182100 Diversity = 5.4760e-08 k_t
= [2, 2, 4, 2]
Generation 1830: Best = 3.864048e+03 FES = 183100 Diversity = 5.4898e-08 k_t
= [1, 1, 1, 1]
Generation 1840: Best = 3.864048e+03 FES = 184100 Diversity = 5.5366e-08 k_t

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= [3, 3, 2, 3]
Generation 1850: Best = 3.864048e+03 FES = 185100 Diversity = 5.5675e-08 k_t
= [1, 1, 1, 1]
Generation 1860: Best = 3.864048e+03 FES = 186100 Diversity = 5.5562e-08 k_t
= [2, 2, 4, 2]
Generation 1870: Best = 3.864048e+03 FES = 187100 Diversity = 5.4537e-08 k_t
= [1, 1, 1, 1]
Generation 1880: Best = 3.864048e+03 FES = 188100 Diversity = 5.4418e-08 k_t
= [2, 3, 4, 3]
Generation 1890: Best = 3.864048e+03 FES = 189100 Diversity = 5.5064e-08 k_t
= [1, 1, 1, 1]
Generation 1900: Best = 3.864048e+03 FES = 190100 Diversity = 5.5486e-08 k_t
= [2, 2, 4, 4]
Generation 1910: Best = 3.864048e+03 FES = 191100 Diversity = 5.6516e-08 k_t
= [1, 1, 1, 1]
Generation 1920: Best = 3.864048e+03 FES = 192100 Diversity = 5.5994e-08 k_t
= [4, 4, 2, 4]
Generation 1930: Best = 3.864048e+03 FES = 193100 Diversity = 5.7238e-08 k_t
= [1, 1, 1, 1]
Generation 1940: Best = 3.864048e+03 FES = 194100 Diversity = 5.7120e-08 k_t
= [3, 3, 4, 2]
Generation 1950: Best = 3.864048e+03 FES = 195100 Diversity = 5.7739e-08 k_t
= [1, 1, 1, 1]
Generation 1960: Best = 3.864048e+03 FES = 196100 Diversity = 5.7193e-08 k_t
= [3, 2, 2, 2]
Generation 1970: Best = 3.864048e+03 FES = 197100 Diversity = 5.6755e-08 k_t
= [1, 1, 1, 1]
Generation 1980: Best = 3.864048e+03 FES = 198100 Diversity = 5.7176e-08 k_t
= [4, 3, 4, 2]
Generation 1990: Best = 3.864048e+03 FES = 199100 Diversity = 5.7103e-08 k_t
= [1, 1, 1, 1]
Generation 2000: Best = 3.864048e+03 FES = 200100 Diversity = 5.7311e-08 k_t
= [2, 4, 4, 3]
Generation 2010: Best = 3.864048e+03 FES = 201100 Diversity = 5.6783e-08 k_t
= [1, 1, 1, 1]
Generation 2020: Best = 3.864048e+03 FES = 202100 Diversity = 5.6371e-08 k_t
= [3, 4, 2, 3]
Generation 2030: Best = 3.864048e+03 FES = 203100 Diversity = 5.5874e-08 k_t
= [1, 1, 1, 1]
Generation 2040: Best = 3.864048e+03 FES = 204100 Diversity = 5.5639e-08 k_t
= [3, 3, 2, 2]
Generation 2050: Best = 3.864048e+03 FES = 205100 Diversity = 5.5865e-08 k_t
= [1, 1, 1, 1]
Generation 2060: Best = 3.864048e+03 FES = 206100 Diversity = 5.5231e-08 k_t
= [2, 2, 2, 3]
Generation 2070: Best = 3.864048e+03 FES = 207100 Diversity = 5.5260e-08 k_t
= [1, 1, 1, 1]
Generation 2080: Best = 3.864048e+03 FES = 208100 Diversity = 5.5035e-08 k_t
= [2, 2, 4, 4]
Generation 2090: Best = 3.864048e+03 FES = 209100 Diversity = 5.4178e-08 k_t
= [1, 1, 1, 1]
Generation 2100: Best = 3.864048e+03 FES = 210100 Diversity = 5.3754e-08 k_t
= [2, 4, 4, 3]
Generation 2110: Best = 3.864048e+03 FES = 211100 Diversity = 5.4311e-08 k_t

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= [1, 1, 1, 1]
Generation 2120: Best = 3.864048e+03 FES = 212100 Diversity = 5.4915e-08 k_t
= [3, 2, 4, 4]
Generation 2130: Best = 3.864048e+03 FES = 213100 Diversity = 5.4880e-08 k_t
= [1, 1, 1, 1]
Generation 2140: Best = 3.864048e+03 FES = 214100 Diversity = 5.4765e-08 k_t
= [2, 2, 3, 2]
Generation 2150: Best = 3.864048e+03 FES = 215100 Diversity = 5.3054e-08 k_t
= [1, 1, 1, 1]
Generation 2160: Best = 3.864048e+03 FES = 216100 Diversity = 5.3876e-08 k_t
= [2, 3, 2, 3]
Generation 2170: Best = 3.864048e+03 FES = 217100 Diversity = 5.4078e-08 k_t
= [1, 1, 1, 1]
Generation 2180: Best = 3.864048e+03 FES = 218100 Diversity = 5.4123e-08 k_t
= [4, 3, 2, 4]
Generation 2190: Best = 3.864048e+03 FES = 219100 Diversity = 5.3383e-08 k_t
= [1, 1, 1, 1]
Generation 2200: Best = 3.864048e+03 FES = 220100 Diversity = 5.4046e-08 k_t
= [3, 3, 4, 2]
Generation 2210: Best = 3.864048e+03 FES = 221100 Diversity = 5.3946e-08 k_t
= [1, 1, 1, 1]
Generation 2220: Best = 3.864048e+03 FES = 222100 Diversity = 5.3021e-08 k_t
= [3, 4, 2, 3]
Generation 2230: Best = 3.864048e+03 FES = 223100 Diversity = 5.2894e-08 k_t
= [1, 1, 1, 1]
Generation 2240: Best = 3.864048e+03 FES = 224100 Diversity = 5.2917e-08 k_t
= [3, 3, 2, 3]
Generation 2250: Best = 3.864048e+03 FES = 225100 Diversity = 5.2225e-08 k_t
= [1, 1, 1, 1]
Generation 2260: Best = 3.864048e+03 FES = 226100 Diversity = 5.3076e-08 k_t
= [3, 3, 3, 3]
Generation 2270: Best = 3.864048e+03 FES = 227100 Diversity = 5.3320e-08 k_t
= [1, 1, 1, 1]
Generation 2280: Best = 3.864048e+03 FES = 228100 Diversity = 5.2572e-08 k_t
= [3, 4, 3, 2]
Generation 2290: Best = 3.864048e+03 FES = 229100 Diversity = 5.2844e-08 k_t
= [1, 1, 1, 1]
Generation 2300: Best = 3.864048e+03 FES = 230100 Diversity = 5.2479e-08 k_t
= [4, 2, 2, 4]
Generation 2310: Best = 3.864048e+03 FES = 231100 Diversity = 5.2738e-08 k_t
= [1, 1, 1, 1]
Generation 2320: Best = 3.864048e+03 FES = 232100 Diversity = 5.2714e-08 k_t
= [3, 3, 3, 4]
Generation 2330: Best = 3.864048e+03 FES = 233100 Diversity = 5.2088e-08 k_t
= [1, 1, 1, 1]
Generation 2340: Best = 3.864048e+03 FES = 234100 Diversity = 5.1248e-08 k_t
= [4, 4, 3, 2]
Generation 2350: Best = 3.864048e+03 FES = 235100 Diversity = 5.0870e-08 k_t
= [1, 1, 1, 1]
Generation 2360: Best = 3.864048e+03 FES = 236100 Diversity = 5.0749e-08 k_t
= [3, 2, 3, 3]
Generation 2370: Best = 3.864048e+03 FES = 237100 Diversity = 5.0352e-08 k_t
= [1, 1, 1, 1]
Generation 2380: Best = 3.864048e+03 FES = 238100 Diversity = 5.0213e-08 k_t

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= [4, 2, 4, 4]
Generation 2390: Best = 3.864048e+03 FES = 239100 Diversity = 5.0182e-08 k_t
= [1, 1, 1, 1]
Generation 2400: Best = 3.864048e+03 FES = 240100 Diversity = 5.0734e-08 k_t
= [2, 3, 2, 3]
Generation 2410: Best = 3.864048e+03 FES = 241100 Diversity = 5.0517e-08 k_t
= [1, 1, 1, 1]
Generation 2420: Best = 3.864048e+03 FES = 242100 Diversity = 5.0879e-08 k_t
= [2, 4, 3, 2]
Generation 2430: Best = 3.864048e+03 FES = 243100 Diversity = 5.1254e-08 k_t
= [1, 1, 1, 1]
Generation 2440: Best = 3.864048e+03 FES = 244100 Diversity = 5.0919e-08 k_t
= [3, 4, 2, 2]
Generation 2450: Best = 3.864048e+03 FES = 245100 Diversity = 5.0730e-08 k_t
= [1, 1, 1, 1]
Generation 2460: Best = 3.864048e+03 FES = 246100 Diversity = 5.0466e-08 k_t
= [3, 3, 4, 4]
Generation 2470: Best = 3.864048e+03 FES = 247100 Diversity = 5.0630e-08 k_t
= [1, 1, 1, 1]
Generation 2480: Best = 3.864048e+03 FES = 248100 Diversity = 5.0414e-08 k_t
= [3, 3, 2, 3]
Generation 2490: Best = 3.864048e+03 FES = 249100 Diversity = 5.0577e-08 k_t
= [1, 1, 1, 1]
Generation 2500: Best = 3.864048e+03 FES = 250100 Diversity = 5.0290e-08 k_t
= [2, 4, 2, 2]
Generation 2510: Best = 3.864048e+03 FES = 251100 Diversity = 5.0301e-08 k_t
= [1, 1, 1, 1]
Generation 2520: Best = 3.864048e+03 FES = 252100 Diversity = 4.9958e-08 k_t
= [2, 3, 2, 3]
Generation 2530: Best = 3.864048e+03 FES = 253100 Diversity = 4.9006e-08 k_t
= [1, 1, 1, 1]
Generation 2540: Best = 3.864048e+03 FES = 254100 Diversity = 4.8789e-08 k_t
= [2, 2, 4, 4]
Generation 2550: Best = 3.864048e+03 FES = 255100 Diversity = 4.9083e-08 k_t
= [1, 1, 1, 1]
Generation 2560: Best = 3.864048e+03 FES = 256100 Diversity = 4.9068e-08 k_t
= [4, 2, 4, 4]
Generation 2570: Best = 3.864048e+03 FES = 257100 Diversity = 4.8973e-08 k_t
= [1, 1, 1, 1]
Generation 2580: Best = 3.864048e+03 FES = 258100 Diversity = 4.9035e-08 k_t
= [3, 4, 4, 4]
Generation 2590: Best = 3.864048e+03 FES = 259100 Diversity = 4.8714e-08 k_t
= [1, 1, 1, 1]
Generation 2600: Best = 3.864048e+03 FES = 260100 Diversity = 4.8412e-08 k_t
= [3, 2, 3, 4]
Generation 2610: Best = 3.864048e+03 FES = 261100 Diversity = 4.8641e-08 k_t
= [1, 1, 1, 1]
Generation 2620: Best = 3.864048e+03 FES = 262100 Diversity = 4.8849e-08 k_t
= [3, 3, 4, 4]
Generation 2630: Best = 3.864048e+03 FES = 263100 Diversity = 4.8283e-08 k_t
= [1, 1, 1, 1]
Generation 2640: Best = 3.864048e+03 FES = 264100 Diversity = 4.8290e-08 k_t
= [2, 3, 4, 3]
Generation 2650: Best = 3.864048e+03 FES = 265100 Diversity = 4.8212e-08 k_t

```



```

= [1, 1, 1, 1]
Generation 2660: Best = 3.864048e+03  FES = 266100  Diversity = 4.7772e-08  k_t
= [3, 4, 4, 4]
Generation 2670: Best = 3.864048e+03  FES = 267100  Diversity = 4.7056e-08  k_t
= [1, 1, 1, 1]
Generation 2680: Best = 3.864048e+03  FES = 268100  Diversity = 4.7141e-08  k_t
= [4, 4, 4, 4]
Generation 2690: Best = 3.864048e+03  FES = 269100  Diversity = 4.6821e-08  k_t
= [1, 1, 1, 1]
Generation 2700: Best = 3.864048e+03  FES = 270100  Diversity = 4.6332e-08  k_t
= [4, 2, 3, 4]
Generation 2710: Best = 3.864048e+03  FES = 271100  Diversity = 4.6943e-08  k_t
= [1, 1, 1, 1]
Generation 2720: Best = 3.864048e+03  FES = 272100  Diversity = 4.6884e-08  k_t
= [4, 2, 3, 3]
Generation 2730: Best = 3.864048e+03  FES = 273100  Diversity = 4.6536e-08  k_t
= [1, 1, 1, 1]
Generation 2740: Best = 3.864048e+03  FES = 274100  Diversity = 4.5993e-08  k_t
= [3, 4, 3, 3]
Generation 2750: Best = 3.864048e+03  FES = 275100  Diversity = 4.6331e-08  k_t
= [1, 1, 1, 1]
Generation 2760: Best = 3.864048e+03  FES = 276100  Diversity = 4.6062e-08  k_t
= [3, 3, 4, 3]
Generation 2770: Best = 3.864048e+03  FES = 277100  Diversity = 4.5817e-08  k_t
= [1, 1, 1, 1]
Generation 2780: Best = 3.864048e+03  FES = 278100  Diversity = 4.5569e-08  k_t
= [4, 4, 3, 2]
Generation 2790: Best = 3.864048e+03  FES = 279100  Diversity = 4.4994e-08  k_t
= [1, 1, 1, 1]
Generation 2800: Best = 3.864048e+03  FES = 280100  Diversity = 4.4309e-08  k_t
= [3, 3, 4, 3]
Generation 2810: Best = 3.864048e+03  FES = 281100  Diversity = 4.4341e-08  k_t
= [1, 1, 1, 1]
Generation 2820: Best = 3.864048e+03  FES = 282100  Diversity = 4.4479e-08  k_t
= [4, 4, 2, 2]
Generation 2830: Best = 3.864048e+03  FES = 283100  Diversity = 4.4819e-08  k_t
= [1, 1, 1, 1]
Generation 2840: Best = 3.864048e+03  FES = 284100  Diversity = 4.4816e-08  k_t
= [4, 2, 4, 4]
Generation 2850: Best = 3.864048e+03  FES = 285100  Diversity = 4.4234e-08  k_t
= [1, 1, 1, 1]
Generation 2860: Best = 3.864048e+03  FES = 286100  Diversity = 4.4902e-08  k_t
= [3, 4, 2, 2]
Generation 2870: Best = 3.864048e+03  FES = 287100  Diversity = 4.4198e-08  k_t
= [1, 1, 1, 1]
Generation 2880: Best = 3.864048e+03  FES = 288100  Diversity = 4.4177e-08  k_t
= [4, 2, 3, 2]
Generation 2890: Best = 3.864048e+03  FES = 289100  Diversity = 4.4009e-08  k_t
= [1, 1, 1, 1]
Generation 2900: Best = 3.864048e+03  FES = 290100  Diversity = 4.3340e-08  k_t
= [4, 2, 3, 3]
Generation 2910: Best = 3.864048e+03  FES = 291100  Diversity = 4.3649e-08  k_t
= [1, 1, 1, 1]
Generation 2920: Best = 3.864048e+03  FES = 292100  Diversity = 4.4558e-08  k_t

```

```

= [3, 4, 2, 3]
Generation 2930: Best = 3.864048e+03  FES = 293100  Diversity = 4.3645e-08  k_t
= [1, 1, 1, 1]
Generation 2940: Best = 3.864048e+03  FES = 294100  Diversity = 4.3240e-08  k_t
= [3, 4, 2, 2]
Generation 2950: Best = 3.864048e+03  FES = 295100  Diversity = 4.2969e-08  k_t
= [1, 1, 1, 1]
Generation 2960: Best = 3.864048e+03  FES = 296100  Diversity = 4.3395e-08  k_t
= [4, 3, 2, 2]
Generation 2970: Best = 3.864048e+03  FES = 297100  Diversity = 4.2421e-08  k_t
= [1, 1, 1, 1]
Generation 2980: Best = 3.864048e+03  FES = 298100  Diversity = 4.2511e-08  k_t
= [3, 3, 4, 4]
Generation 2990: Best = 3.864048e+03  FES = 299100  Diversity = 4.2764e-08  k_t
= [1, 1, 1, 1]

```

Optimization completed!

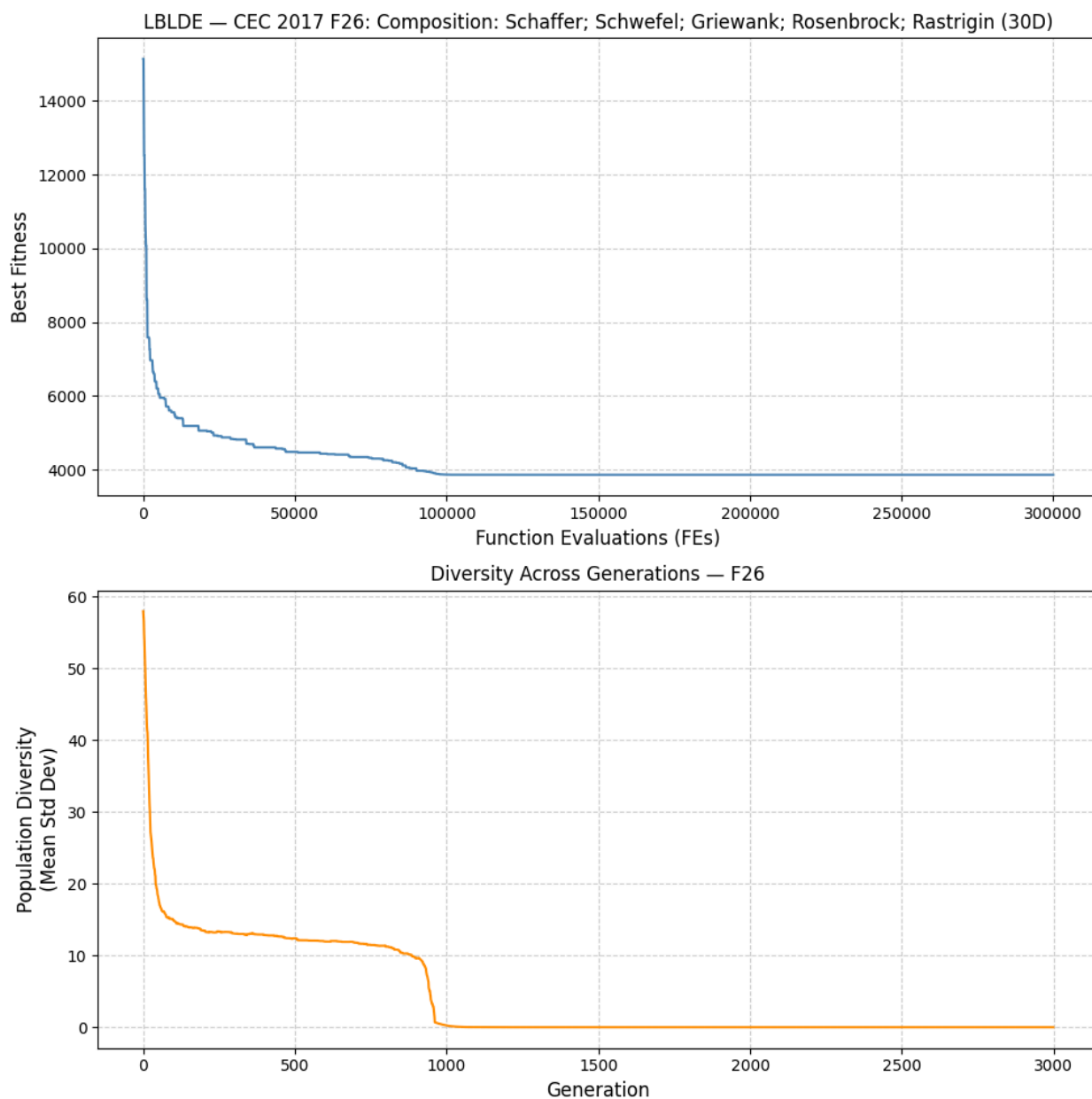
Final Best Fitness = 3.864048e+03

Total Function Evaluations = 300000

```

=====
=
Results:
  Best Fitness          : 3.864048e+03
  Global Optimum        : 2.600000e+03
  Error (|best-optimum|) : 1.264048e+03
  Best Solution (5 dim)  : [ 10.4811935  -65.77033991 -14.69155508 -27.35456489
-37.41367751]
  D^(0) (init. diversity): 5.7976e+01
=====
=

```



Graph saved as convergence_F26.png

```
In [37]: # =====
# F27 — Composition Function 7 (HGBat; Rastrigin; Schwefel; Bent Cigar; Elliptic)
# =====
run_and_plot('F27', make_wrapper(composition.f27), 2700.0, 'Composition: HGBat')
```

```

=====
=
LBLDE_s on CEC 2017 F27 – Single Run
=====
=
Function      : Composition: HGBat; Rastrigin; Schwefel; Bent Cigar; Elliptic;
Schaffer (30D)
Bounds        : [-100, 100]
Global optimum : 2700.0
Initial D^(0) = 5.7976e+01
Generation 0: Best Fitness = 4.906147e+03 | Diversity = 5.7976e+01
Generation 10: Best = 3.857412e+03 FES = 1100 Diversity = 4.9509e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 3.405959e+03 FES = 2100 Diversity = 4.1848e+01 k_t
= [3, 3, 2, 2]
Generation 30: Best = 3.364499e+03 FES = 3100 Diversity = 3.7505e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 3.344802e+03 FES = 4100 Diversity = 3.3259e+01 k_t
= [2, 4, 4, 4]
Generation 50: Best = 3.291793e+03 FES = 5100 Diversity = 3.3132e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 3.291793e+03 FES = 6100 Diversity = 3.2414e+01 k_t
= [2, 3, 3, 4]
Generation 70: Best = 3.287134e+03 FES = 7100 Diversity = 3.2376e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 3.269408e+03 FES = 8100 Diversity = 3.0987e+01 k_t
= [4, 3, 4, 3]
Generation 90: Best = 3.266306e+03 FES = 9100 Diversity = 2.9159e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 3.263581e+03 FES = 10100 Diversity = 2.7242e+01 k_t
= [4, 2, 2, 3]
Generation 110: Best = 3.254577e+03 FES = 11100 Diversity = 2.6112e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 3.250609e+03 FES = 12100 Diversity = 2.5906e+01 k_t
= [3, 4, 2, 3]
Generation 130: Best = 3.243483e+03 FES = 13100 Diversity = 2.5147e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 3.239772e+03 FES = 14100 Diversity = 2.4394e+01 k_t
= [3, 2, 4, 3]
Generation 150: Best = 3.238586e+03 FES = 15100 Diversity = 2.3662e+01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 3.236349e+03 FES = 16100 Diversity = 2.3868e+01 k_t
= [3, 3, 4, 4]
Generation 170: Best = 3.234856e+03 FES = 17100 Diversity = 2.2901e+01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 3.232219e+03 FES = 18100 Diversity = 2.1709e+01 k_t
= [2, 4, 4, 4]
Generation 190: Best = 3.232210e+03 FES = 19100 Diversity = 2.0828e+01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 3.228905e+03 FES = 20100 Diversity = 1.8532e+01 k_t
= [4, 2, 2, 2]
Generation 210: Best = 3.227790e+03 FES = 21100 Diversity = 1.6885e+01 k_t
= [1, 1, 1, 1]
Generation 220: Best = 3.225825e+03 FES = 22100 Diversity = 1.6247e+01 k_t

```

```

= [2, 3, 3, 3]
Generation 230: Best = 3.224653e+03 FES = 23100 Diversity = 1.4481e+01 k_t
= [1, 1, 1, 1]
Generation 240: Best = 3.224653e+03 FES = 24100 Diversity = 1.3744e+01 k_t
= [4, 4, 4, 4]
Generation 250: Best = 3.223638e+03 FES = 25100 Diversity = 1.2671e+01 k_t
= [1, 1, 1, 1]
Generation 260: Best = 3.221453e+03 FES = 26100 Diversity = 1.2385e+01 k_t
= [4, 4, 2, 3]
Generation 270: Best = 3.220554e+03 FES = 27100 Diversity = 1.2257e+01 k_t
= [1, 1, 1, 1]
Generation 280: Best = 3.219854e+03 FES = 28100 Diversity = 1.1046e+01 k_t
= [2, 3, 2, 2]
Generation 290: Best = 3.219094e+03 FES = 29100 Diversity = 9.9577e+00 k_t
= [1, 1, 1, 1]
Generation 300: Best = 3.216731e+03 FES = 30100 Diversity = 9.7308e+00 k_t
= [2, 2, 2, 3]
Generation 310: Best = 3.216458e+03 FES = 31100 Diversity = 7.4055e+00 k_t
= [1, 1, 1, 1]
Generation 320: Best = 3.215599e+03 FES = 32100 Diversity = 7.2210e+00 k_t
= [3, 3, 2, 2]
Generation 330: Best = 3.215155e+03 FES = 33100 Diversity = 7.0771e+00 k_t
= [1, 1, 1, 1]
Generation 340: Best = 3.214615e+03 FES = 34100 Diversity = 6.6813e+00 k_t
= [3, 2, 2, 4]
Generation 350: Best = 3.212846e+03 FES = 35100 Diversity = 6.3416e+00 k_t
= [1, 1, 1, 1]
Generation 360: Best = 3.212759e+03 FES = 36100 Diversity = 6.1620e+00 k_t
= [3, 2, 2, 3]
Generation 370: Best = 3.209801e+03 FES = 37100 Diversity = 5.9093e+00 k_t
= [1, 1, 1, 1]
Generation 380: Best = 3.209654e+03 FES = 38100 Diversity = 5.3298e+00 k_t
= [3, 3, 3, 2]
Generation 390: Best = 3.209654e+03 FES = 39100 Diversity = 4.8796e+00 k_t
= [1, 1, 1, 1]
Generation 400: Best = 3.209626e+03 FES = 40100 Diversity = 4.6077e+00 k_t
= [4, 2, 2, 4]
Generation 410: Best = 3.209078e+03 FES = 41100 Diversity = 4.0777e+00 k_t
= [1, 1, 1, 1]
Generation 420: Best = 3.208734e+03 FES = 42100 Diversity = 3.6927e+00 k_t
= [2, 3, 2, 3]
Generation 430: Best = 3.208709e+03 FES = 43100 Diversity = 3.1318e+00 k_t
= [1, 1, 1, 1]
Generation 440: Best = 3.208641e+03 FES = 44100 Diversity = 2.5958e+00 k_t
= [4, 4, 3, 2]
Generation 450: Best = 3.207914e+03 FES = 45100 Diversity = 1.9287e+00 k_t
= [1, 1, 1, 1]
Generation 460: Best = 3.207690e+03 FES = 46100 Diversity = 1.5667e+00 k_t
= [3, 4, 3, 2]
Generation 470: Best = 3.206807e+03 FES = 47100 Diversity = 1.1910e+00 k_t
= [1, 1, 1, 1]
Generation 480: Best = 3.206705e+03 FES = 48100 Diversity = 1.0109e+00 k_t
= [2, 3, 3, 4]
Generation 490: Best = 3.206488e+03 FES = 49100 Diversity = 8.4838e-01 k_t

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= [1, 1, 1, 1]
Generation 500: Best = 3.206327e+03 FES = 50100 Diversity = 7.5608e-01 k_t
= [2, 2, 2, 4]
Generation 510: Best = 3.206108e+03 FES = 51100 Diversity = 6.2429e-01 k_t
= [1, 1, 1, 1]
Generation 520: Best = 3.205941e+03 FES = 52100 Diversity = 4.8204e-01 k_t
= [4, 2, 2, 4]
Generation 530: Best = 3.205851e+03 FES = 53100 Diversity = 3.9746e-01 k_t
= [1, 1, 1, 1]
Generation 540: Best = 3.205642e+03 FES = 54100 Diversity = 3.3179e-01 k_t
= [4, 3, 2, 4]
Generation 550: Best = 3.205527e+03 FES = 55100 Diversity = 2.7495e-01 k_t
= [1, 1, 1, 1]
Generation 560: Best = 3.205483e+03 FES = 56100 Diversity = 2.2052e-01 k_t
= [3, 3, 2, 3]
Generation 570: Best = 3.205463e+03 FES = 57100 Diversity = 1.9252e-01 k_t
= [1, 1, 1, 1]
Generation 580: Best = 3.205418e+03 FES = 58100 Diversity = 1.6143e-01 k_t
= [4, 2, 3, 4]
Generation 590: Best = 3.205373e+03 FES = 59100 Diversity = 1.3923e-01 k_t
= [1, 1, 1, 1]
Generation 600: Best = 3.205358e+03 FES = 60100 Diversity = 1.1644e-01 k_t
= [4, 4, 3, 4]
Generation 610: Best = 3.205333e+03 FES = 61100 Diversity = 9.8085e-02 k_t
= [1, 1, 1, 1]
Generation 620: Best = 3.205328e+03 FES = 62100 Diversity = 8.6824e-02 k_t
= [4, 4, 3, 2]
Generation 630: Best = 3.205318e+03 FES = 63100 Diversity = 7.5941e-02 k_t
= [1, 1, 1, 1]
Generation 640: Best = 3.205314e+03 FES = 64100 Diversity = 6.1832e-02 k_t
= [2, 2, 4, 3]
Generation 650: Best = 3.205306e+03 FES = 65100 Diversity = 5.7925e-02 k_t
= [1, 1, 1, 1]
Generation 660: Best = 3.205302e+03 FES = 66100 Diversity = 5.1422e-02 k_t
= [4, 3, 4, 3]
Generation 670: Best = 3.205299e+03 FES = 67100 Diversity = 4.4565e-02 k_t
= [1, 1, 1, 1]
Generation 680: Best = 3.205296e+03 FES = 68100 Diversity = 4.0888e-02 k_t
= [4, 2, 2, 2]
Generation 690: Best = 3.205292e+03 FES = 69100 Diversity = 3.8279e-02 k_t
= [1, 1, 1, 1]
Generation 700: Best = 3.205289e+03 FES = 70100 Diversity = 3.4513e-02 k_t
= [3, 4, 2, 4]
Generation 710: Best = 3.205287e+03 FES = 71100 Diversity = 3.1210e-02 k_t
= [1, 1, 1, 1]
Generation 720: Best = 3.205285e+03 FES = 72100 Diversity = 3.0034e-02 k_t
= [3, 4, 3, 4]
Generation 730: Best = 3.205283e+03 FES = 73100 Diversity = 2.7989e-02 k_t
= [1, 1, 1, 1]
Generation 740: Best = 3.205283e+03 FES = 74100 Diversity = 2.6752e-02 k_t
= [3, 4, 2, 2]
Generation 750: Best = 3.205281e+03 FES = 75100 Diversity = 2.3098e-02 k_t
= [1, 1, 1, 1]
Generation 760: Best = 3.205281e+03 FES = 76100 Diversity = 2.1673e-02 k_t

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= [3, 3, 3, 4]
Generation 770: Best = 3.205279e+03 FES = 77100 Diversity = 1.9599e-02 k_t
= [1, 1, 1, 1]
Generation 780: Best = 3.205279e+03 FES = 78100 Diversity = 1.9274e-02 k_t
= [2, 2, 4, 4]
Generation 790: Best = 3.205278e+03 FES = 79100 Diversity = 1.8523e-02 k_t
= [1, 1, 1, 1]
Generation 800: Best = 3.205277e+03 FES = 80100 Diversity = 1.8926e-02 k_t
= [3, 2, 4, 3]
Generation 810: Best = 3.205277e+03 FES = 81100 Diversity = 1.7458e-02 k_t
= [1, 1, 1, 1]
Generation 820: Best = 3.205276e+03 FES = 82100 Diversity = 1.6107e-02 k_t
= [2, 4, 2, 2]
Generation 830: Best = 3.205276e+03 FES = 83100 Diversity = 1.5177e-02 k_t
= [1, 1, 1, 1]
Generation 840: Best = 3.205275e+03 FES = 84100 Diversity = 1.4630e-02 k_t
= [3, 3, 2, 2]
Generation 850: Best = 3.205275e+03 FES = 85100 Diversity = 1.3952e-02 k_t
= [1, 1, 1, 1]
Generation 860: Best = 3.205275e+03 FES = 86100 Diversity = 1.2413e-02 k_t
= [4, 3, 2, 4]
Generation 870: Best = 3.205274e+03 FES = 87100 Diversity = 1.2111e-02 k_t
= [1, 1, 1, 1]
Generation 880: Best = 3.205274e+03 FES = 88100 Diversity = 1.1797e-02 k_t
= [4, 3, 4, 4]
Generation 890: Best = 3.205274e+03 FES = 89100 Diversity = 1.1685e-02 k_t
= [1, 1, 1, 1]
Generation 900: Best = 3.205274e+03 FES = 90100 Diversity = 1.1647e-02 k_t
= [2, 2, 3, 3]
Generation 910: Best = 3.205273e+03 FES = 91100 Diversity = 1.1001e-02 k_t
= [1, 1, 1, 1]
Generation 920: Best = 3.205273e+03 FES = 92100 Diversity = 1.0083e-02 k_t
= [3, 4, 3, 4]
Generation 930: Best = 3.205273e+03 FES = 93100 Diversity = 1.0152e-02 k_t
= [1, 1, 1, 1]
Generation 940: Best = 3.205273e+03 FES = 94100 Diversity = 1.0019e-02 k_t
= [3, 2, 2, 3]
Generation 950: Best = 3.205273e+03 FES = 95100 Diversity = 9.4120e-03 k_t
= [1, 1, 1, 1]
Generation 960: Best = 3.205273e+03 FES = 96100 Diversity = 8.6490e-03 k_t
= [4, 3, 4, 3]
Generation 970: Best = 3.205272e+03 FES = 97100 Diversity = 8.3581e-03 k_t
= [1, 1, 1, 1]
Generation 980: Best = 3.205272e+03 FES = 98100 Diversity = 7.9738e-03 k_t
= [4, 4, 3, 4]
Generation 990: Best = 3.205272e+03 FES = 99100 Diversity = 7.3796e-03 k_t
= [1, 1, 1, 1]
Generation 1000: Best = 3.205272e+03 FES = 100100 Diversity = 6.8470e-03 k_t
= [4, 2, 4, 2]
Generation 1010: Best = 3.205272e+03 FES = 101100 Diversity = 6.4530e-03 k_t
= [1, 1, 1, 1]
Generation 1020: Best = 3.205272e+03 FES = 102100 Diversity = 6.1207e-03 k_t
= [4, 4, 2, 3]
Generation 1030: Best = 3.205272e+03 FES = 103100 Diversity = 6.1138e-03 k_t

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= [1, 1, 1, 1]
Generation 1040: Best = 3.205272e+03 FES = 104100 Diversity = 6.1343e-03 k_t
= [4, 4, 4, 4]
Generation 1050: Best = 3.205272e+03 FES = 105100 Diversity = 5.9543e-03 k_t
= [1, 1, 1, 1]
Generation 1060: Best = 3.205272e+03 FES = 106100 Diversity = 5.4073e-03 k_t
= [2, 3, 3, 3]
Generation 1070: Best = 3.205272e+03 FES = 107100 Diversity = 5.2891e-03 k_t
= [1, 1, 1, 1]
Generation 1080: Best = 3.205272e+03 FES = 108100 Diversity = 4.9551e-03 k_t
= [3, 2, 4, 3]
Generation 1090: Best = 3.205272e+03 FES = 109100 Diversity = 4.9213e-03 k_t
= [1, 1, 1, 1]
Generation 1100: Best = 3.205272e+03 FES = 110100 Diversity = 4.5165e-03 k_t
= [2, 4, 3, 3]
Generation 1110: Best = 3.205272e+03 FES = 111100 Diversity = 4.2551e-03 k_t
= [1, 1, 1, 1]
Generation 1120: Best = 3.205272e+03 FES = 112100 Diversity = 4.0196e-03 k_t
= [3, 2, 4, 4]
Generation 1130: Best = 3.205272e+03 FES = 113100 Diversity = 3.7920e-03 k_t
= [1, 1, 1, 1]
Generation 1140: Best = 3.205272e+03 FES = 114100 Diversity = 3.3805e-03 k_t
= [2, 4, 4, 2]
Generation 1150: Best = 3.205272e+03 FES = 115100 Diversity = 3.4030e-03 k_t
= [1, 1, 1, 1]
Generation 1160: Best = 3.205272e+03 FES = 116100 Diversity = 3.1003e-03 k_t
= [4, 3, 4, 3]
Generation 1170: Best = 3.205272e+03 FES = 117100 Diversity = 2.9468e-03 k_t
= [1, 1, 1, 1]
Generation 1180: Best = 3.205272e+03 FES = 118100 Diversity = 2.8556e-03 k_t
= [4, 2, 2, 3]
Generation 1190: Best = 3.205272e+03 FES = 119100 Diversity = 2.8736e-03 k_t
= [1, 1, 1, 1]
Generation 1200: Best = 3.205271e+03 FES = 120100 Diversity = 2.8570e-03 k_t
= [3, 2, 4, 4]
Generation 1210: Best = 3.205271e+03 FES = 121100 Diversity = 2.9886e-03 k_t
= [1, 1, 1, 1]
Generation 1220: Best = 3.205271e+03 FES = 122100 Diversity = 2.8744e-03 k_t
= [3, 2, 4, 3]
Generation 1230: Best = 3.205271e+03 FES = 123100 Diversity = 2.7146e-03 k_t
= [1, 1, 1, 1]
Generation 1240: Best = 3.205271e+03 FES = 124100 Diversity = 2.7273e-03 k_t
= [2, 2, 4, 4]
Generation 1250: Best = 3.205271e+03 FES = 125100 Diversity = 2.5428e-03 k_t
= [1, 1, 1, 1]
Generation 1260: Best = 3.205271e+03 FES = 126100 Diversity = 2.4149e-03 k_t
= [2, 3, 2, 3]
Generation 1270: Best = 3.205271e+03 FES = 127100 Diversity = 2.2732e-03 k_t
= [1, 1, 1, 1]
Generation 1280: Best = 3.205271e+03 FES = 128100 Diversity = 2.1787e-03 k_t
= [4, 4, 3, 2]
Generation 1290: Best = 3.205271e+03 FES = 129100 Diversity = 2.1205e-03 k_t
= [1, 1, 1, 1]
Generation 1300: Best = 3.205271e+03 FES = 130100 Diversity = 2.0067e-03 k_t

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= [4, 2, 2, 4]
Generation 1310: Best = 3.205271e+03 FES = 131100 Diversity = 1.8786e-03 k_t
= [1, 1, 1, 1]
Generation 1320: Best = 3.205271e+03 FES = 132100 Diversity = 1.8961e-03 k_t
= [3, 4, 3, 2]
Generation 1330: Best = 3.205271e+03 FES = 133100 Diversity = 1.8715e-03 k_t
= [1, 1, 1, 1]
Generation 1340: Best = 3.205271e+03 FES = 134100 Diversity = 1.7684e-03 k_t
= [4, 3, 4, 4]
Generation 1350: Best = 3.205271e+03 FES = 135100 Diversity = 1.7307e-03 k_t
= [1, 1, 1, 1]
Generation 1360: Best = 3.205271e+03 FES = 136100 Diversity = 1.6772e-03 k_t
= [3, 3, 3, 3]
Generation 1370: Best = 3.205271e+03 FES = 137100 Diversity = 1.5861e-03 k_t
= [1, 1, 1, 1]
Generation 1380: Best = 3.205271e+03 FES = 138100 Diversity = 1.5814e-03 k_t
= [2, 2, 3, 4]
Generation 1390: Best = 3.205271e+03 FES = 139100 Diversity = 1.4985e-03 k_t
= [1, 1, 1, 1]
Generation 1400: Best = 3.205271e+03 FES = 140100 Diversity = 1.4193e-03 k_t
= [2, 4, 2, 4]
Generation 1410: Best = 3.205271e+03 FES = 141100 Diversity = 1.4432e-03 k_t
= [1, 1, 1, 1]
Generation 1420: Best = 3.205271e+03 FES = 142100 Diversity = 1.4402e-03 k_t
= [3, 2, 3, 2]
Generation 1430: Best = 3.205271e+03 FES = 143100 Diversity = 1.5166e-03 k_t
= [1, 1, 1, 1]
Generation 1440: Best = 3.205271e+03 FES = 144100 Diversity = 1.4593e-03 k_t
= [2, 2, 4, 2]
Generation 1450: Best = 3.205271e+03 FES = 145100 Diversity = 1.5139e-03 k_t
= [1, 1, 1, 1]
Generation 1460: Best = 3.205271e+03 FES = 146100 Diversity = 1.5441e-03 k_t
= [2, 4, 3, 3]
Generation 1470: Best = 3.205271e+03 FES = 147100 Diversity = 1.4910e-03 k_t
= [1, 1, 1, 1]
Generation 1480: Best = 3.205271e+03 FES = 148100 Diversity = 1.5103e-03 k_t
= [2, 2, 2, 3]
Generation 1490: Best = 3.205271e+03 FES = 149100 Diversity = 1.5126e-03 k_t
= [1, 1, 1, 1]
Generation 1500: Best = 3.205271e+03 FES = 150100 Diversity = 1.5487e-03 k_t
= [4, 3, 2, 3]
Generation 1510: Best = 3.205271e+03 FES = 151100 Diversity = 1.4764e-03 k_t
= [1, 1, 1, 1]
Generation 1520: Best = 3.205271e+03 FES = 152100 Diversity = 1.4596e-03 k_t
= [4, 3, 3, 4]
Generation 1530: Best = 3.205271e+03 FES = 153100 Diversity = 1.3935e-03 k_t
= [1, 1, 1, 1]
Generation 1540: Best = 3.205271e+03 FES = 154100 Diversity = 1.3332e-03 k_t
= [2, 4, 3, 4]
Generation 1550: Best = 3.205271e+03 FES = 155100 Diversity = 1.3021e-03 k_t
= [1, 1, 1, 1]
Generation 1560: Best = 3.205271e+03 FES = 156100 Diversity = 1.2341e-03 k_t
= [3, 4, 4, 4]
Generation 1570: Best = 3.205271e+03 FES = 157100 Diversity = 1.0886e-03 k_t

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= [1, 1, 1, 1]
Generation 1580: Best = 3.205271e+03 FES = 158100 Diversity = 9.9188e-04 k_t
= [2, 3, 2, 2]
Generation 1590: Best = 3.205271e+03 FES = 159100 Diversity = 9.3061e-04 k_t
= [1, 1, 1, 1]
Generation 1600: Best = 3.205271e+03 FES = 160100 Diversity = 9.1743e-04 k_t
= [4, 4, 2, 4]
Generation 1610: Best = 3.205271e+03 FES = 161100 Diversity = 9.3443e-04 k_t
= [1, 1, 1, 1]
Generation 1620: Best = 3.205271e+03 FES = 162100 Diversity = 9.3536e-04 k_t
= [4, 2, 3, 2]
Generation 1630: Best = 3.205271e+03 FES = 163100 Diversity = 8.6128e-04 k_t
= [1, 1, 1, 1]
Generation 1640: Best = 3.205271e+03 FES = 164100 Diversity = 8.8104e-04 k_t
= [2, 2, 3, 3]
Generation 1650: Best = 3.205271e+03 FES = 165100 Diversity = 8.3144e-04 k_t
= [1, 1, 1, 1]
Generation 1660: Best = 3.205271e+03 FES = 166100 Diversity = 8.2523e-04 k_t
= [4, 3, 3, 2]
Generation 1670: Best = 3.205271e+03 FES = 167100 Diversity = 8.0977e-04 k_t
= [1, 1, 1, 1]
Generation 1680: Best = 3.205271e+03 FES = 168100 Diversity = 7.7566e-04 k_t
= [4, 4, 3, 4]
Generation 1690: Best = 3.205271e+03 FES = 169100 Diversity = 7.5633e-04 k_t
= [1, 1, 1, 1]
Generation 1700: Best = 3.205271e+03 FES = 170100 Diversity = 7.4734e-04 k_t
= [2, 3, 4, 4]
Generation 1710: Best = 3.205271e+03 FES = 171100 Diversity = 7.5040e-04 k_t
= [1, 1, 1, 1]
Generation 1720: Best = 3.205271e+03 FES = 172100 Diversity = 7.4508e-04 k_t
= [3, 4, 4, 3]
Generation 1730: Best = 3.205271e+03 FES = 173100 Diversity = 7.2710e-04 k_t
= [1, 1, 1, 1]
Generation 1740: Best = 3.205271e+03 FES = 174100 Diversity = 7.1091e-04 k_t
= [3, 4, 4, 2]
Generation 1750: Best = 3.205271e+03 FES = 175100 Diversity = 7.3463e-04 k_t
= [1, 1, 1, 1]
Generation 1760: Best = 3.205271e+03 FES = 176100 Diversity = 6.8141e-04 k_t
= [2, 4, 3, 2]
Generation 1770: Best = 3.205271e+03 FES = 177100 Diversity = 7.3368e-04 k_t
= [1, 1, 1, 1]
Generation 1780: Best = 3.205271e+03 FES = 178100 Diversity = 7.8476e-04 k_t
= [4, 3, 2, 3]
Generation 1790: Best = 3.205271e+03 FES = 179100 Diversity = 7.5136e-04 k_t
= [1, 1, 1, 1]
Generation 1800: Best = 3.205271e+03 FES = 180100 Diversity = 7.6428e-04 k_t
= [4, 2, 4, 3]
Generation 1810: Best = 3.205271e+03 FES = 181100 Diversity = 7.5612e-04 k_t
= [1, 1, 1, 1]
Generation 1820: Best = 3.205271e+03 FES = 182100 Diversity = 7.0679e-04 k_t
= [4, 4, 3, 2]
Generation 1830: Best = 3.205271e+03 FES = 183100 Diversity = 6.8471e-04 k_t
= [1, 1, 1, 1]
Generation 1840: Best = 3.205271e+03 FES = 184100 Diversity = 6.4176e-04 k_t

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= [3, 3, 2, 4]
Generation 1850: Best = 3.205271e+03 FES = 185100 Diversity = 6.0426e-04 k_t
= [1, 1, 1, 1]
Generation 1860: Best = 3.205271e+03 FES = 186100 Diversity = 5.6424e-04 k_t
= [3, 4, 4, 3]
Generation 1870: Best = 3.205271e+03 FES = 187100 Diversity = 5.2381e-04 k_t
= [1, 1, 1, 1]
Generation 1880: Best = 3.205271e+03 FES = 188100 Diversity = 5.1220e-04 k_t
= [2, 4, 3, 2]
Generation 1890: Best = 3.205271e+03 FES = 189100 Diversity = 5.1632e-04 k_t
= [1, 1, 1, 1]
Generation 1900: Best = 3.205271e+03 FES = 190100 Diversity = 5.4238e-04 k_t
= [3, 3, 2, 4]
Generation 1910: Best = 3.205271e+03 FES = 191100 Diversity = 5.3434e-04 k_t
= [1, 1, 1, 1]
Generation 1920: Best = 3.205271e+03 FES = 192100 Diversity = 5.1284e-04 k_t
= [3, 3, 2, 3]
Generation 1930: Best = 3.205271e+03 FES = 193100 Diversity = 4.9956e-04 k_t
= [1, 1, 1, 1]
Generation 1940: Best = 3.205271e+03 FES = 194100 Diversity = 5.0795e-04 k_t
= [4, 3, 3, 2]
Generation 1950: Best = 3.205271e+03 FES = 195100 Diversity = 4.9122e-04 k_t
= [1, 1, 1, 1]
Generation 1960: Best = 3.205271e+03 FES = 196100 Diversity = 5.4172e-04 k_t
= [2, 2, 4, 2]
Generation 1970: Best = 3.205271e+03 FES = 197100 Diversity = 5.4585e-04 k_t
= [1, 1, 1, 1]
Generation 1980: Best = 3.205271e+03 FES = 198100 Diversity = 4.9382e-04 k_t
= [4, 4, 2, 2]
Generation 1990: Best = 3.205271e+03 FES = 199100 Diversity = 4.8589e-04 k_t
= [1, 1, 1, 1]
Generation 2000: Best = 3.205271e+03 FES = 200100 Diversity = 4.5512e-04 k_t
= [4, 3, 2, 3]
Generation 2010: Best = 3.205271e+03 FES = 201100 Diversity = 4.7869e-04 k_t
= [1, 1, 1, 1]
Generation 2020: Best = 3.205271e+03 FES = 202100 Diversity = 4.7209e-04 k_t
= [2, 3, 2, 2]
Generation 2030: Best = 3.205271e+03 FES = 203100 Diversity = 4.3804e-04 k_t
= [1, 1, 1, 1]
Generation 2040: Best = 3.205271e+03 FES = 204100 Diversity = 4.0847e-04 k_t
= [4, 4, 3, 3]
Generation 2050: Best = 3.205271e+03 FES = 205100 Diversity = 3.9216e-04 k_t
= [1, 1, 1, 1]
Generation 2060: Best = 3.205271e+03 FES = 206100 Diversity = 3.7816e-04 k_t
= [4, 4, 4, 3]
Generation 2070: Best = 3.205271e+03 FES = 207100 Diversity = 3.8722e-04 k_t
= [1, 1, 1, 1]
Generation 2080: Best = 3.205271e+03 FES = 208100 Diversity = 4.1340e-04 k_t
= [3, 3, 4, 4]
Generation 2090: Best = 3.205271e+03 FES = 209100 Diversity = 4.2846e-04 k_t
= [1, 1, 1, 1]
Generation 2100: Best = 3.205271e+03 FES = 210100 Diversity = 4.3756e-04 k_t
= [3, 3, 2, 3]
Generation 2110: Best = 3.205271e+03 FES = 211100 Diversity = 3.9940e-04 k_t

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= [1, 1, 1, 1]
Generation 2120: Best = 3.205271e+03 FES = 212100 Diversity = 4.1842e-04 k_t
= [2, 3, 2, 2]
Generation 2130: Best = 3.205271e+03 FES = 213100 Diversity = 4.8259e-04 k_t
= [1, 1, 1, 1]
Generation 2140: Best = 3.205271e+03 FES = 214100 Diversity = 4.5777e-04 k_t
= [4, 3, 4, 3]
Generation 2150: Best = 3.205271e+03 FES = 215100 Diversity = 4.6759e-04 k_t
= [1, 1, 1, 1]
Generation 2160: Best = 3.205271e+03 FES = 216100 Diversity = 5.1900e-04 k_t
= [3, 3, 3, 3]
Generation 2170: Best = 3.205271e+03 FES = 217100 Diversity = 5.5679e-04 k_t
= [1, 1, 1, 1]
Generation 2180: Best = 3.205271e+03 FES = 218100 Diversity = 6.4055e-04 k_t
= [4, 3, 3, 2]
Generation 2190: Best = 3.205271e+03 FES = 219100 Diversity = 6.4686e-04 k_t
= [1, 1, 1, 1]
Generation 2200: Best = 3.205271e+03 FES = 220100 Diversity = 5.9573e-04 k_t
= [2, 3, 2, 2]
Generation 2210: Best = 3.205271e+03 FES = 221100 Diversity = 6.0711e-04 k_t
= [1, 1, 1, 1]
Generation 2220: Best = 3.205271e+03 FES = 222100 Diversity = 5.3776e-04 k_t
= [4, 3, 4, 4]
Generation 2230: Best = 3.205271e+03 FES = 223100 Diversity = 5.4131e-04 k_t
= [1, 1, 1, 1]
Generation 2240: Best = 3.205271e+03 FES = 224100 Diversity = 5.4538e-04 k_t
= [3, 2, 3, 4]
Generation 2250: Best = 3.205271e+03 FES = 225100 Diversity = 5.7860e-04 k_t
= [1, 1, 1, 1]
Generation 2260: Best = 3.205271e+03 FES = 226100 Diversity = 5.7115e-04 k_t
= [2, 2, 4, 4]
Generation 2270: Best = 3.205271e+03 FES = 227100 Diversity = 5.7004e-04 k_t
= [1, 1, 1, 1]
Generation 2280: Best = 3.205271e+03 FES = 228100 Diversity = 5.3340e-04 k_t
= [4, 4, 3, 4]
Generation 2290: Best = 3.205271e+03 FES = 229100 Diversity = 5.1832e-04 k_t
= [1, 1, 1, 1]
Generation 2300: Best = 3.205271e+03 FES = 230100 Diversity = 4.8879e-04 k_t
= [2, 3, 3, 3]
Generation 2310: Best = 3.205271e+03 FES = 231100 Diversity = 4.6218e-04 k_t
= [1, 1, 1, 1]
Generation 2320: Best = 3.205271e+03 FES = 232100 Diversity = 4.6578e-04 k_t
= [2, 3, 3, 4]
Generation 2330: Best = 3.205271e+03 FES = 233100 Diversity = 4.5410e-04 k_t
= [1, 1, 1, 1]
Generation 2340: Best = 3.205271e+03 FES = 234100 Diversity = 4.4105e-04 k_t
= [3, 4, 3, 3]
Generation 2350: Best = 3.205271e+03 FES = 235100 Diversity = 4.1015e-04 k_t
= [1, 1, 1, 1]
Generation 2360: Best = 3.205271e+03 FES = 236100 Diversity = 3.9854e-04 k_t
= [2, 4, 4, 3]
Generation 2370: Best = 3.205271e+03 FES = 237100 Diversity = 4.0185e-04 k_t
= [1, 1, 1, 1]
Generation 2380: Best = 3.205271e+03 FES = 238100 Diversity = 4.1185e-04 k_t

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= [4, 4, 3, 4]
Generation 2390: Best = 3.205271e+03 FES = 239100 Diversity = 4.1703e-04 k_t
= [1, 1, 1, 1]
Generation 2400: Best = 3.205271e+03 FES = 240100 Diversity = 3.9012e-04 k_t
= [2, 2, 3, 3]
Generation 2410: Best = 3.205271e+03 FES = 241100 Diversity = 3.5409e-04 k_t
= [1, 1, 1, 1]
Generation 2420: Best = 3.205271e+03 FES = 242100 Diversity = 3.5526e-04 k_t
= [2, 4, 2, 3]
Generation 2430: Best = 3.205271e+03 FES = 243100 Diversity = 3.5071e-04 k_t
= [1, 1, 1, 1]
Generation 2440: Best = 3.205271e+03 FES = 244100 Diversity = 3.1992e-04 k_t
= [2, 4, 2, 2]
Generation 2450: Best = 3.205271e+03 FES = 245100 Diversity = 3.2598e-04 k_t
= [1, 1, 1, 1]
Generation 2460: Best = 3.205271e+03 FES = 246100 Diversity = 3.2448e-04 k_t
= [2, 4, 3, 2]
Generation 2470: Best = 3.205271e+03 FES = 247100 Diversity = 3.0119e-04 k_t
= [1, 1, 1, 1]
Generation 2480: Best = 3.205271e+03 FES = 248100 Diversity = 2.9251e-04 k_t
= [4, 3, 4, 4]
Generation 2490: Best = 3.205271e+03 FES = 249100 Diversity = 2.8825e-04 k_t
= [1, 1, 1, 1]
Generation 2500: Best = 3.205271e+03 FES = 250100 Diversity = 2.9319e-04 k_t
= [4, 4, 3, 2]
Generation 2510: Best = 3.205271e+03 FES = 251100 Diversity = 2.7294e-04 k_t
= [1, 1, 1, 1]
Generation 2520: Best = 3.205271e+03 FES = 252100 Diversity = 2.8189e-04 k_t
= [2, 2, 2, 3]
Generation 2530: Best = 3.205271e+03 FES = 253100 Diversity = 2.9807e-04 k_t
= [1, 1, 1, 1]
Generation 2540: Best = 3.205271e+03 FES = 254100 Diversity = 3.2224e-04 k_t
= [3, 3, 4, 3]
Generation 2550: Best = 3.205271e+03 FES = 255100 Diversity = 2.8299e-04 k_t
= [1, 1, 1, 1]
Generation 2560: Best = 3.205271e+03 FES = 256100 Diversity = 2.6111e-04 k_t
= [3, 4, 4, 3]
Generation 2570: Best = 3.205271e+03 FES = 257100 Diversity = 2.2769e-04 k_t
= [1, 1, 1, 1]
Generation 2580: Best = 3.205271e+03 FES = 258100 Diversity = 2.1267e-04 k_t
= [3, 2, 4, 3]
Generation 2590: Best = 3.205271e+03 FES = 259100 Diversity = 2.0483e-04 k_t
= [1, 1, 1, 1]
Generation 2600: Best = 3.205271e+03 FES = 260100 Diversity = 1.9150e-04 k_t
= [2, 2, 2, 4]
Generation 2610: Best = 3.205271e+03 FES = 261100 Diversity = 1.7893e-04 k_t
= [1, 1, 1, 1]
Generation 2620: Best = 3.205271e+03 FES = 262100 Diversity = 1.5199e-04 k_t
= [3, 4, 3, 3]
Generation 2630: Best = 3.205271e+03 FES = 263100 Diversity = 1.8125e-04 k_t
= [1, 1, 1, 1]
Generation 2640: Best = 3.205271e+03 FES = 264100 Diversity = 1.8248e-04 k_t
= [4, 3, 2, 3]
Generation 2650: Best = 3.205271e+03 FES = 265100 Diversity = 1.8084e-04 k_t

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= [1, 1, 1, 1]
Generation 2660: Best = 3.205271e+03 FES = 266100 Diversity = 1.6978e-04 k_t
= [4, 2, 4, 2]
Generation 2670: Best = 3.205271e+03 FES = 267100 Diversity = 1.5964e-04 k_t
= [1, 1, 1, 1]
Generation 2680: Best = 3.205271e+03 FES = 268100 Diversity = 1.6125e-04 k_t
= [3, 4, 3, 2]
Generation 2690: Best = 3.205271e+03 FES = 269100 Diversity = 1.5607e-04 k_t
= [1, 1, 1, 1]
Generation 2700: Best = 3.205271e+03 FES = 270100 Diversity = 1.5027e-04 k_t
= [4, 4, 3, 3]
Generation 2710: Best = 3.205271e+03 FES = 271100 Diversity = 1.4677e-04 k_t
= [1, 1, 1, 1]
Generation 2720: Best = 3.205271e+03 FES = 272100 Diversity = 1.3834e-04 k_t
= [3, 4, 2, 2]
Generation 2730: Best = 3.205271e+03 FES = 273100 Diversity = 1.3811e-04 k_t
= [1, 1, 1, 1]
Generation 2740: Best = 3.205271e+03 FES = 274100 Diversity = 1.4262e-04 k_t
= [4, 3, 4, 3]
Generation 2750: Best = 3.205271e+03 FES = 275100 Diversity = 1.5183e-04 k_t
= [1, 1, 1, 1]
Generation 2760: Best = 3.205271e+03 FES = 276100 Diversity = 1.9750e-04 k_t
= [3, 4, 3, 3]
Generation 2770: Best = 3.205271e+03 FES = 277100 Diversity = 2.0638e-04 k_t
= [1, 1, 1, 1]
Generation 2780: Best = 3.205271e+03 FES = 278100 Diversity = 2.3679e-04 k_t
= [3, 3, 4, 4]
Generation 2790: Best = 3.205271e+03 FES = 279100 Diversity = 2.5754e-04 k_t
= [1, 1, 1, 1]
Generation 2800: Best = 3.205271e+03 FES = 280100 Diversity = 2.7359e-04 k_t
= [4, 2, 4, 4]
Generation 2810: Best = 3.205271e+03 FES = 281100 Diversity = 2.8302e-04 k_t
= [1, 1, 1, 1]
Generation 2820: Best = 3.205271e+03 FES = 282100 Diversity = 3.2447e-04 k_t
= [2, 3, 4, 3]
Generation 2830: Best = 3.205271e+03 FES = 283100 Diversity = 3.7396e-04 k_t
= [1, 1, 1, 1]
Generation 2840: Best = 3.205271e+03 FES = 284100 Diversity = 3.6344e-04 k_t
= [2, 2, 4, 2]
Generation 2850: Best = 3.205271e+03 FES = 285100 Diversity = 3.4210e-04 k_t
= [1, 1, 1, 1]
Generation 2860: Best = 3.205271e+03 FES = 286100 Diversity = 3.2636e-04 k_t
= [4, 2, 2, 4]
Generation 2870: Best = 3.205271e+03 FES = 287100 Diversity = 3.0504e-04 k_t
= [1, 1, 1, 1]
Generation 2880: Best = 3.205271e+03 FES = 288100 Diversity = 2.8067e-04 k_t
= [2, 2, 4, 2]
Generation 2890: Best = 3.205271e+03 FES = 289100 Diversity = 2.5688e-04 k_t
= [1, 1, 1, 1]
Generation 2900: Best = 3.205271e+03 FES = 290100 Diversity = 2.3753e-04 k_t
= [4, 2, 2, 4]
Generation 2910: Best = 3.205271e+03 FES = 291100 Diversity = 2.1776e-04 k_t
= [1, 1, 1, 1]
Generation 2920: Best = 3.205271e+03 FES = 292100 Diversity = 1.9621e-04 k_t

```

```

= [3, 3, 3, 2]
Generation 2930: Best = 3.205271e+03   FES = 293100   Diversity = 1.8991e-04   k_t
= [1, 1, 1, 1]
Generation 2940: Best = 3.205271e+03   FES = 294100   Diversity = 1.9350e-04   k_t
= [3, 2, 3, 2]
Generation 2950: Best = 3.205271e+03   FES = 295100   Diversity = 1.8458e-04   k_t
= [1, 1, 1, 1]
Generation 2960: Best = 3.205271e+03   FES = 296100   Diversity = 1.8181e-04   k_t
= [4, 4, 3, 4]
Generation 2970: Best = 3.205271e+03   FES = 297100   Diversity = 1.8139e-04   k_t
= [1, 1, 1, 1]
Generation 2980: Best = 3.205271e+03   FES = 298100   Diversity = 1.5985e-04   k_t
= [3, 4, 3, 2]
Generation 2990: Best = 3.205271e+03   FES = 299100   Diversity = 1.5241e-04   k_t
= [1, 1, 1, 1]

```

Optimization completed!

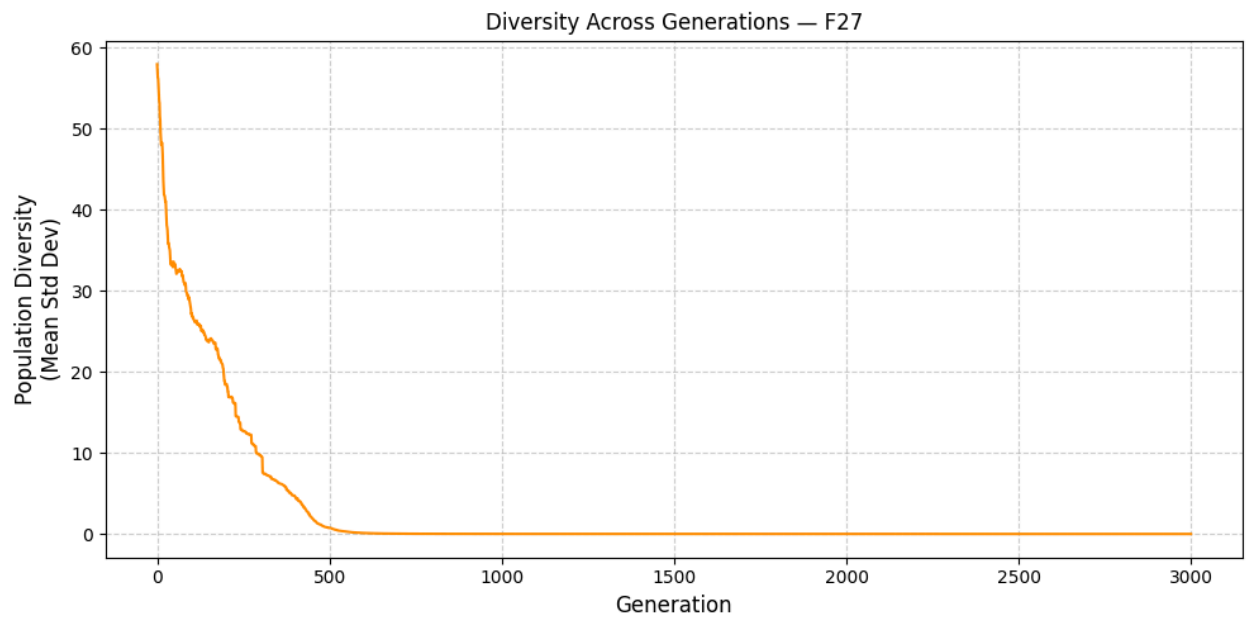
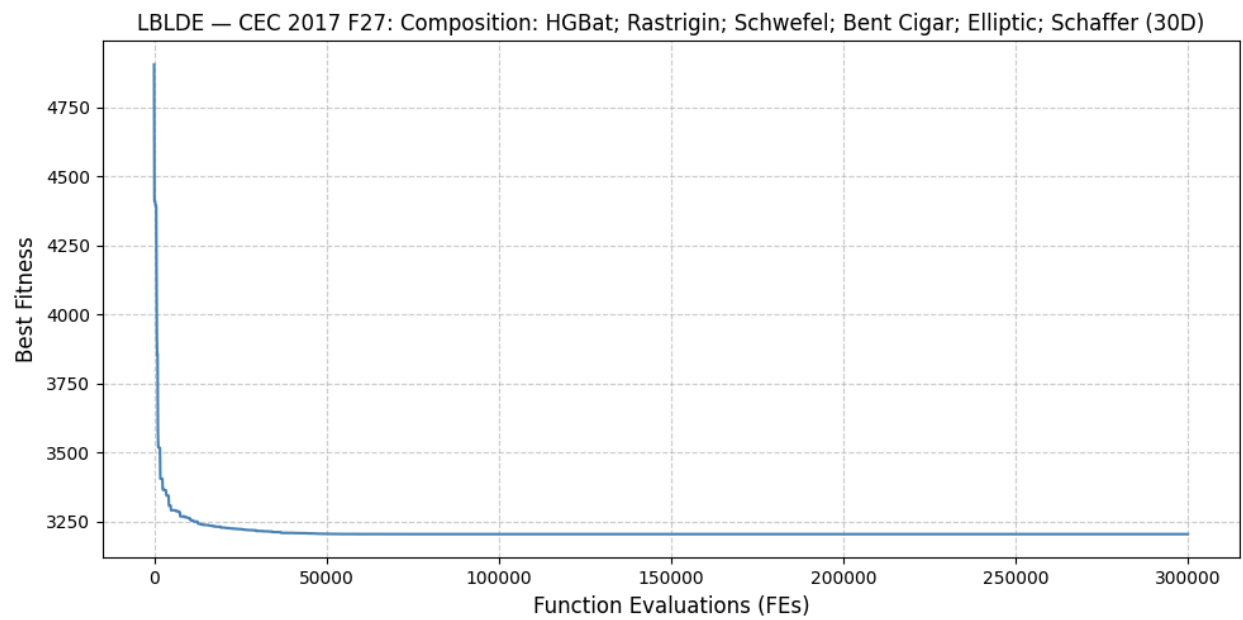
Final Best Fitness = 3.205271e+03

Total Function Evaluations = 300000

```

=====
=
Results:
  Best Fitness           : 3.205271e+03
  Global Optimum         : 2.700000e+03
  Error (|best-optimum|) : 5.052711e+02
  Best Solution (5 dim)  : [-93.04706256 -99.99999866 -99.99999994 -95.57678878
-33.69137897]
  D^(0) (init. diversity): 5.7976e+01
=====
=

```



Graph saved as convergence_F27.png

```
In [38]: # =====
# F28 — Composition Function 8 (Ackley; Griewank; Discus; Rosenbrock; HappyCat)
# =====
run_and_plot('F28', make_wrapper(composition.f28), 2800.0, 'Composition: Ackle
```



```

=====
=
LBLDE_s on CEC 2017 F28 – Single Run
=====
=
Function      : Composition: Ackley; Griewank; Discus; Rosenbrock; HappyCat; S
chaffer (30D)
Bounds       : [-100, 100]
Global optimum : 2800.0
Initial D^(0) = 5.7976e+01
Generation 0: Best Fitness = 1.208847e+04 | Diversity = 5.7976e+01
Generation 10: Best = 5.382141e+03 FES = 1100 Diversity = 3.8920e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 4.378384e+03 FES = 2100 Diversity = 2.5997e+01 k_t
= [3, 3, 2, 2]
Generation 30: Best = 3.832182e+03 FES = 3100 Diversity = 1.9962e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 3.545754e+03 FES = 4100 Diversity = 1.3957e+01 k_t
= [3, 4, 3, 4]
Generation 50: Best = 3.414864e+03 FES = 5100 Diversity = 1.1692e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 3.383938e+03 FES = 6100 Diversity = 9.2669e+00 k_t
= [2, 3, 4, 3]
Generation 70: Best = 3.336609e+03 FES = 7100 Diversity = 7.0519e+00 k_t
= [1, 1, 1, 1]
Generation 80: Best = 3.317977e+03 FES = 8100 Diversity = 4.9887e+00 k_t
= [4, 3, 4, 3]
Generation 90: Best = 3.303917e+03 FES = 9100 Diversity = 3.8200e+00 k_t
= [1, 1, 1, 1]
Generation 100: Best = 3.294658e+03 FES = 10100 Diversity = 3.2501e+00 k_t
= [2, 4, 4, 3]
Generation 110: Best = 3.288627e+03 FES = 11100 Diversity = 2.3001e+00 k_t
= [1, 1, 1, 1]
Generation 120: Best = 3.282569e+03 FES = 12100 Diversity = 1.7514e+00 k_t
= [3, 3, 3, 4]
Generation 130: Best = 3.277372e+03 FES = 13100 Diversity = 1.6537e+00 k_t
= [1, 1, 1, 1]
Generation 140: Best = 3.273863e+03 FES = 14100 Diversity = 1.3937e+00 k_t
= [2, 3, 3, 3]
Generation 150: Best = 3.270476e+03 FES = 15100 Diversity = 1.0492e+00 k_t
= [1, 1, 1, 1]
Generation 160: Best = 3.268747e+03 FES = 16100 Diversity = 9.4042e-01 k_t
= [4, 2, 3, 2]
Generation 170: Best = 3.267340e+03 FES = 17100 Diversity = 8.0153e-01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 3.266454e+03 FES = 18100 Diversity = 6.6961e-01 k_t
= [4, 4, 2, 2]
Generation 190: Best = 3.265389e+03 FES = 19100 Diversity = 6.1842e-01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 3.264608e+03 FES = 20100 Diversity = 5.5059e-01 k_t
= [4, 4, 4, 3]
Generation 210: Best = 3.264284e+03 FES = 21100 Diversity = 4.5024e-01 k_t
= [1, 1, 1, 1]
Generation 220: Best = 3.263766e+03 FES = 22100 Diversity = 3.9405e-01 k_t

```

```

= [2, 3, 3, 4]
Generation 230: Best = 3.263623e+03 FES = 23100 Diversity = 2.9062e-01 k_t
= [1, 1, 1, 1]
Generation 240: Best = 3.263394e+03 FES = 24100 Diversity = 2.6711e-01 k_t
= [4, 4, 4, 3]
Generation 250: Best = 3.263351e+03 FES = 25100 Diversity = 2.1351e-01 k_t
= [1, 1, 1, 1]
Generation 260: Best = 3.263278e+03 FES = 26100 Diversity = 1.8722e-01 k_t
= [4, 3, 3, 3]
Generation 270: Best = 3.263200e+03 FES = 27100 Diversity = 1.7505e-01 k_t
= [1, 1, 1, 1]
Generation 280: Best = 3.263165e+03 FES = 28100 Diversity = 1.4682e-01 k_t
= [3, 3, 4, 4]
Generation 290: Best = 3.263115e+03 FES = 29100 Diversity = 1.3754e-01 k_t
= [1, 1, 1, 1]
Generation 300: Best = 3.263076e+03 FES = 30100 Diversity = 1.2499e-01 k_t
= [2, 3, 2, 3]
Generation 310: Best = 3.263042e+03 FES = 31100 Diversity = 1.1268e-01 k_t
= [1, 1, 1, 1]
Generation 320: Best = 3.263001e+03 FES = 32100 Diversity = 1.0641e-01 k_t
= [2, 2, 3, 2]
Generation 330: Best = 3.262970e+03 FES = 33100 Diversity = 9.9255e-02 k_t
= [1, 1, 1, 1]
Generation 340: Best = 3.262940e+03 FES = 34100 Diversity = 1.0101e-01 k_t
= [2, 3, 4, 4]
Generation 350: Best = 3.262904e+03 FES = 35100 Diversity = 8.9871e-02 k_t
= [1, 1, 1, 1]
Generation 360: Best = 3.262888e+03 FES = 36100 Diversity = 8.5082e-02 k_t
= [3, 3, 2, 3]
Generation 370: Best = 3.262862e+03 FES = 37100 Diversity = 8.1515e-02 k_t
= [1, 1, 1, 1]
Generation 380: Best = 3.262843e+03 FES = 38100 Diversity = 7.8063e-02 k_t
= [2, 4, 2, 3]
Generation 390: Best = 3.262819e+03 FES = 39100 Diversity = 7.5105e-02 k_t
= [1, 1, 1, 1]
Generation 400: Best = 3.262798e+03 FES = 40100 Diversity = 7.9300e-02 k_t
= [2, 3, 2, 2]
Generation 410: Best = 3.262781e+03 FES = 41100 Diversity = 7.6616e-02 k_t
= [1, 1, 1, 1]
Generation 420: Best = 3.262762e+03 FES = 42100 Diversity = 7.4408e-02 k_t
= [3, 3, 4, 3]
Generation 430: Best = 3.262748e+03 FES = 43100 Diversity = 6.7100e-02 k_t
= [1, 1, 1, 1]
Generation 440: Best = 3.262726e+03 FES = 44100 Diversity = 6.9273e-02 k_t
= [3, 4, 4, 2]
Generation 450: Best = 3.262707e+03 FES = 45100 Diversity = 8.4237e-02 k_t
= [1, 1, 1, 1]
Generation 460: Best = 3.262681e+03 FES = 46100 Diversity = 8.5952e-02 k_t
= [2, 4, 3, 2]
Generation 470: Best = 3.262658e+03 FES = 47100 Diversity = 8.9584e-02 k_t
= [1, 1, 1, 1]
Generation 480: Best = 3.262642e+03 FES = 48100 Diversity = 8.2900e-02 k_t
= [4, 2, 3, 3]
Generation 490: Best = 3.262622e+03 FES = 49100 Diversity = 8.9965e-02 k_t

```

```

= [1, 1, 1, 1]
Generation 500: Best = 3.262598e+03 FES = 50100 Diversity = 8.0384e-02 k_t
= [3, 2, 2, 4]
Generation 510: Best = 3.262572e+03 FES = 51100 Diversity = 8.7808e-02 k_t
= [1, 1, 1, 1]
Generation 520: Best = 3.262548e+03 FES = 52100 Diversity = 8.3119e-02 k_t
= [3, 3, 4, 3]
Generation 530: Best = 3.262530e+03 FES = 53100 Diversity = 7.1179e-02 k_t
= [1, 1, 1, 1]
Generation 540: Best = 3.262512e+03 FES = 54100 Diversity = 7.2487e-02 k_t
= [4, 2, 4, 2]
Generation 550: Best = 3.262497e+03 FES = 55100 Diversity = 7.0643e-02 k_t
= [1, 1, 1, 1]
Generation 560: Best = 3.262482e+03 FES = 56100 Diversity = 7.4184e-02 k_t
= [2, 4, 4, 4]
Generation 570: Best = 3.262455e+03 FES = 57100 Diversity = 7.9068e-02 k_t
= [1, 1, 1, 1]
Generation 580: Best = 3.262443e+03 FES = 58100 Diversity = 7.6829e-02 k_t
= [4, 2, 3, 4]
Generation 590: Best = 3.262427e+03 FES = 59100 Diversity = 6.8151e-02 k_t
= [1, 1, 1, 1]
Generation 600: Best = 3.262409e+03 FES = 60100 Diversity = 6.6555e-02 k_t
= [3, 2, 2, 3]
Generation 610: Best = 3.262397e+03 FES = 61100 Diversity = 5.4709e-02 k_t
= [1, 1, 1, 1]
Generation 620: Best = 3.262382e+03 FES = 62100 Diversity = 5.6346e-02 k_t
= [3, 3, 3, 3]
Generation 630: Best = 3.262371e+03 FES = 63100 Diversity = 5.5795e-02 k_t
= [1, 1, 1, 1]
Generation 640: Best = 3.262359e+03 FES = 64100 Diversity = 5.1824e-02 k_t
= [4, 2, 3, 2]
Generation 650: Best = 3.262342e+03 FES = 65100 Diversity = 6.5961e-02 k_t
= [1, 1, 1, 1]
Generation 660: Best = 3.262324e+03 FES = 66100 Diversity = 6.7755e-02 k_t
= [2, 4, 4, 2]
Generation 670: Best = 3.262306e+03 FES = 67100 Diversity = 6.8142e-02 k_t
= [1, 1, 1, 1]
Generation 680: Best = 3.262292e+03 FES = 68100 Diversity = 7.1955e-02 k_t
= [4, 2, 3, 2]
Generation 690: Best = 3.262266e+03 FES = 69100 Diversity = 7.9147e-02 k_t
= [1, 1, 1, 1]
Generation 700: Best = 3.262243e+03 FES = 70100 Diversity = 8.0401e-02 k_t
= [3, 2, 4, 3]
Generation 710: Best = 3.262221e+03 FES = 71100 Diversity = 7.2265e-02 k_t
= [1, 1, 1, 1]
Generation 720: Best = 3.262204e+03 FES = 72100 Diversity = 6.4361e-02 k_t
= [4, 4, 4, 3]
Generation 730: Best = 3.262192e+03 FES = 73100 Diversity = 6.0586e-02 k_t
= [1, 1, 1, 1]
Generation 740: Best = 3.262182e+03 FES = 74100 Diversity = 5.7111e-02 k_t
= [2, 4, 3, 2]
Generation 750: Best = 3.262167e+03 FES = 75100 Diversity = 5.7545e-02 k_t
= [1, 1, 1, 1]
Generation 760: Best = 3.262155e+03 FES = 76100 Diversity = 5.8518e-02 k_t

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= [3, 4, 3, 2]
Generation 770: Best = 3.262142e+03 FES = 77100 Diversity = 5.7396e-02 k_t
= [1, 1, 1, 1]
Generation 780: Best = 3.262131e+03 FES = 78100 Diversity = 5.7072e-02 k_t
= [3, 2, 2, 2]
Generation 790: Best = 3.262123e+03 FES = 79100 Diversity = 5.4420e-02 k_t
= [1, 1, 1, 1]
Generation 800: Best = 3.262112e+03 FES = 80100 Diversity = 5.1303e-02 k_t
= [4, 2, 3, 4]
Generation 810: Best = 3.262104e+03 FES = 81100 Diversity = 5.0419e-02 k_t
= [1, 1, 1, 1]
Generation 820: Best = 3.262095e+03 FES = 82100 Diversity = 4.6078e-02 k_t
= [3, 2, 3, 4]
Generation 830: Best = 3.262087e+03 FES = 83100 Diversity = 4.7335e-02 k_t
= [1, 1, 1, 1]
Generation 840: Best = 3.262076e+03 FES = 84100 Diversity = 4.8062e-02 k_t
= [4, 4, 2, 2]
Generation 850: Best = 3.262068e+03 FES = 85100 Diversity = 5.1219e-02 k_t
= [1, 1, 1, 1]
Generation 860: Best = 3.262060e+03 FES = 86100 Diversity = 4.9338e-02 k_t
= [4, 2, 2, 2]
Generation 870: Best = 3.262051e+03 FES = 87100 Diversity = 5.0395e-02 k_t
= [1, 1, 1, 1]
Generation 880: Best = 3.262044e+03 FES = 88100 Diversity = 4.7085e-02 k_t
= [3, 4, 2, 3]
Generation 890: Best = 3.262035e+03 FES = 89100 Diversity = 4.0285e-02 k_t
= [1, 1, 1, 1]
Generation 900: Best = 3.262030e+03 FES = 90100 Diversity = 3.5765e-02 k_t
= [2, 2, 3, 2]
Generation 910: Best = 3.262027e+03 FES = 91100 Diversity = 3.4212e-02 k_t
= [1, 1, 1, 1]
Generation 920: Best = 3.262021e+03 FES = 92100 Diversity = 3.4697e-02 k_t
= [3, 3, 4, 3]
Generation 930: Best = 3.262016e+03 FES = 93100 Diversity = 3.3153e-02 k_t
= [1, 1, 1, 1]
Generation 940: Best = 3.262012e+03 FES = 94100 Diversity = 2.9585e-02 k_t
= [3, 4, 2, 4]
Generation 950: Best = 3.262006e+03 FES = 95100 Diversity = 2.9438e-02 k_t
= [1, 1, 1, 1]
Generation 960: Best = 3.262000e+03 FES = 96100 Diversity = 3.4059e-02 k_t
= [2, 3, 4, 4]
Generation 970: Best = 3.261993e+03 FES = 97100 Diversity = 3.9083e-02 k_t
= [1, 1, 1, 1]
Generation 980: Best = 3.261984e+03 FES = 98100 Diversity = 4.5031e-02 k_t
= [2, 2, 2, 2]
Generation 990: Best = 3.261977e+03 FES = 99100 Diversity = 4.7330e-02 k_t
= [1, 1, 1, 1]
Generation 1000: Best = 3.261971e+03 FES = 100100 Diversity = 4.4798e-02 k_t
= [3, 3, 3, 2]
Generation 1010: Best = 3.261965e+03 FES = 101100 Diversity = 4.0390e-02 k_t
= [1, 1, 1, 1]
Generation 1020: Best = 3.261959e+03 FES = 102100 Diversity = 3.5440e-02 k_t
= [2, 2, 4, 4]
Generation 1030: Best = 3.261956e+03 FES = 103100 Diversity = 2.9066e-02 k_t

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= [1, 1, 1, 1]
Generation 1040: Best = 3.261953e+03 FES = 104100 Diversity = 2.3815e-02 k_t
= [2, 2, 2, 3]
Generation 1050: Best = 3.261950e+03 FES = 105100 Diversity = 2.4188e-02 k_t
= [1, 1, 1, 1]
Generation 1060: Best = 3.261948e+03 FES = 106100 Diversity = 2.3015e-02 k_t
= [4, 2, 2, 3]
Generation 1070: Best = 3.261945e+03 FES = 107100 Diversity = 2.3558e-02 k_t
= [1, 1, 1, 1]
Generation 1080: Best = 3.261941e+03 FES = 108100 Diversity = 2.6914e-02 k_t
= [3, 3, 2, 2]
Generation 1090: Best = 3.261938e+03 FES = 109100 Diversity = 2.4338e-02 k_t
= [1, 1, 1, 1]
Generation 1100: Best = 3.261935e+03 FES = 110100 Diversity = 2.4776e-02 k_t
= [2, 2, 2, 3]
Generation 1110: Best = 3.261932e+03 FES = 111100 Diversity = 2.2879e-02 k_t
= [1, 1, 1, 1]
Generation 1120: Best = 3.261929e+03 FES = 112100 Diversity = 2.4047e-02 k_t
= [4, 4, 2, 4]
Generation 1130: Best = 3.261926e+03 FES = 113100 Diversity = 2.8009e-02 k_t
= [1, 1, 1, 1]
Generation 1140: Best = 3.261923e+03 FES = 114100 Diversity = 2.8080e-02 k_t
= [4, 4, 2, 2]
Generation 1150: Best = 3.261920e+03 FES = 115100 Diversity = 2.6910e-02 k_t
= [1, 1, 1, 1]
Generation 1160: Best = 3.261918e+03 FES = 116100 Diversity = 2.1923e-02 k_t
= [4, 3, 2, 3]
Generation 1170: Best = 3.261916e+03 FES = 117100 Diversity = 2.1298e-02 k_t
= [1, 1, 1, 1]
Generation 1180: Best = 3.261913e+03 FES = 118100 Diversity = 2.1308e-02 k_t
= [3, 4, 4, 3]
Generation 1190: Best = 3.261910e+03 FES = 119100 Diversity = 2.1754e-02 k_t
= [1, 1, 1, 1]
Generation 1200: Best = 3.261907e+03 FES = 120100 Diversity = 2.3637e-02 k_t
= [3, 3, 4, 4]
Generation 1210: Best = 3.261905e+03 FES = 121100 Diversity = 2.1914e-02 k_t
= [1, 1, 1, 1]
Generation 1220: Best = 3.261902e+03 FES = 122100 Diversity = 2.4300e-02 k_t
= [2, 3, 4, 4]
Generation 1230: Best = 3.261900e+03 FES = 123100 Diversity = 2.1041e-02 k_t
= [1, 1, 1, 1]
Generation 1240: Best = 3.261897e+03 FES = 124100 Diversity = 2.4606e-02 k_t
= [3, 4, 2, 3]
Generation 1250: Best = 3.261894e+03 FES = 125100 Diversity = 2.0895e-02 k_t
= [1, 1, 1, 1]
Generation 1260: Best = 3.261893e+03 FES = 126100 Diversity = 1.9265e-02 k_t
= [2, 3, 2, 3]
Generation 1270: Best = 3.261891e+03 FES = 127100 Diversity = 1.7117e-02 k_t
= [1, 1, 1, 1]
Generation 1280: Best = 3.261890e+03 FES = 128100 Diversity = 1.6575e-02 k_t
= [4, 3, 2, 2]
Generation 1290: Best = 3.261888e+03 FES = 129100 Diversity = 1.3447e-02 k_t
= [1, 1, 1, 1]
Generation 1300: Best = 3.261887e+03 FES = 130100 Diversity = 1.4156e-02 k_t

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= [3, 3, 3, 3]
Generation 1310: Best = 3.261886e+03 FES = 131100 Diversity = 1.1414e-02 k_t
= [1, 1, 1, 1]
Generation 1320: Best = 3.261886e+03 FES = 132100 Diversity = 9.8888e-03 k_t
= [3, 3, 3, 2]
Generation 1330: Best = 3.261885e+03 FES = 133100 Diversity = 8.7841e-03 k_t
= [1, 1, 1, 1]
Generation 1340: Best = 3.261885e+03 FES = 134100 Diversity = 7.9882e-03 k_t
= [2, 4, 2, 4]
Generation 1350: Best = 3.261884e+03 FES = 135100 Diversity = 1.0111e-02 k_t
= [1, 1, 1, 1]
Generation 1360: Best = 3.261884e+03 FES = 136100 Diversity = 1.0495e-02 k_t
= [2, 3, 4, 4]
Generation 1370: Best = 3.261883e+03 FES = 137100 Diversity = 9.9960e-03 k_t
= [1, 1, 1, 1]
Generation 1380: Best = 3.261882e+03 FES = 138100 Diversity = 9.2203e-03 k_t
= [4, 3, 3, 2]
Generation 1390: Best = 3.261882e+03 FES = 139100 Diversity = 1.1003e-02 k_t
= [1, 1, 1, 1]
Generation 1400: Best = 3.261881e+03 FES = 140100 Diversity = 1.1628e-02 k_t
= [3, 4, 2, 2]
Generation 1410: Best = 3.261880e+03 FES = 141100 Diversity = 1.2058e-02 k_t
= [1, 1, 1, 1]
Generation 1420: Best = 3.261878e+03 FES = 142100 Diversity = 1.4224e-02 k_t
= [2, 2, 4, 3]
Generation 1430: Best = 3.261877e+03 FES = 143100 Diversity = 1.7149e-02 k_t
= [1, 1, 1, 1]
Generation 1440: Best = 3.261876e+03 FES = 144100 Diversity = 1.2725e-02 k_t
= [2, 2, 2, 4]
Generation 1450: Best = 3.261875e+03 FES = 145100 Diversity = 1.3449e-02 k_t
= [1, 1, 1, 1]
Generation 1460: Best = 3.261874e+03 FES = 146100 Diversity = 1.3440e-02 k_t
= [3, 3, 3, 2]
Generation 1470: Best = 3.261873e+03 FES = 147100 Diversity = 1.1084e-02 k_t
= [1, 1, 1, 1]
Generation 1480: Best = 3.261873e+03 FES = 148100 Diversity = 1.1231e-02 k_t
= [4, 4, 4, 4]
Generation 1490: Best = 3.261872e+03 FES = 149100 Diversity = 1.1252e-02 k_t
= [1, 1, 1, 1]
Generation 1500: Best = 3.261871e+03 FES = 150100 Diversity = 9.0840e-03 k_t
= [3, 3, 3, 2]
Generation 1510: Best = 3.261871e+03 FES = 151100 Diversity = 9.8800e-03 k_t
= [1, 1, 1, 1]
Generation 1520: Best = 3.261870e+03 FES = 152100 Diversity = 8.5902e-03 k_t
= [3, 4, 4, 4]
Generation 1530: Best = 3.261869e+03 FES = 153100 Diversity = 1.1675e-02 k_t
= [1, 1, 1, 1]
Generation 1540: Best = 3.261868e+03 FES = 154100 Diversity = 1.2919e-02 k_t
= [3, 3, 3, 3]
Generation 1550: Best = 3.261867e+03 FES = 155100 Diversity = 1.3296e-02 k_t
= [1, 1, 1, 1]
Generation 1560: Best = 3.261866e+03 FES = 156100 Diversity = 1.2792e-02 k_t
= [2, 2, 2, 3]
Generation 1570: Best = 3.261865e+03 FES = 157100 Diversity = 1.0402e-02 k_t

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= [1, 1, 1, 1]
Generation 1580: Best = 3.261865e+03 FES = 158100 Diversity = 9.8763e-03 k_t
= [3, 3, 4, 2]
Generation 1590: Best = 3.261864e+03 FES = 159100 Diversity = 1.2021e-02 k_t
= [1, 1, 1, 1]
Generation 1600: Best = 3.261863e+03 FES = 160100 Diversity = 1.0315e-02 k_t
= [4, 2, 3, 2]
Generation 1610: Best = 3.261863e+03 FES = 161100 Diversity = 7.0278e-03 k_t
= [1, 1, 1, 1]
Generation 1620: Best = 3.261863e+03 FES = 162100 Diversity = 5.2868e-03 k_t
= [2, 4, 3, 3]
Generation 1630: Best = 3.261862e+03 FES = 163100 Diversity = 5.5969e-03 k_t
= [1, 1, 1, 1]
Generation 1640: Best = 3.261862e+03 FES = 164100 Diversity = 5.7984e-03 k_t
= [4, 2, 3, 3]
Generation 1650: Best = 3.261862e+03 FES = 165100 Diversity = 6.3042e-03 k_t
= [1, 1, 1, 1]
Generation 1660: Best = 3.261861e+03 FES = 166100 Diversity = 5.0634e-03 k_t
= [4, 2, 2, 2]
Generation 1670: Best = 3.261861e+03 FES = 167100 Diversity = 4.6489e-03 k_t
= [1, 1, 1, 1]
Generation 1680: Best = 3.261861e+03 FES = 168100 Diversity = 4.2592e-03 k_t
= [3, 3, 4, 4]
Generation 1690: Best = 3.261861e+03 FES = 169100 Diversity = 4.1078e-03 k_t
= [1, 1, 1, 1]
Generation 1700: Best = 3.261861e+03 FES = 170100 Diversity = 4.8356e-03 k_t
= [4, 3, 4, 2]
Generation 1710: Best = 3.261860e+03 FES = 171100 Diversity = 5.2359e-03 k_t
= [1, 1, 1, 1]
Generation 1720: Best = 3.261860e+03 FES = 172100 Diversity = 4.7471e-03 k_t
= [4, 2, 2, 2]
Generation 1730: Best = 3.261860e+03 FES = 173100 Diversity = 4.0973e-03 k_t
= [1, 1, 1, 1]
Generation 1740: Best = 3.261860e+03 FES = 174100 Diversity = 5.0153e-03 k_t
= [2, 3, 4, 2]
Generation 1750: Best = 3.261860e+03 FES = 175100 Diversity = 6.4445e-03 k_t
= [1, 1, 1, 1]
Generation 1760: Best = 3.261859e+03 FES = 176100 Diversity = 5.8682e-03 k_t
= [3, 3, 3, 4]
Generation 1770: Best = 3.261859e+03 FES = 177100 Diversity = 5.1739e-03 k_t
= [1, 1, 1, 1]
Generation 1780: Best = 3.261859e+03 FES = 178100 Diversity = 4.7197e-03 k_t
= [2, 4, 4, 2]
Generation 1790: Best = 3.261859e+03 FES = 179100 Diversity = 3.6307e-03 k_t
= [1, 1, 1, 1]
Generation 1800: Best = 3.261859e+03 FES = 180100 Diversity = 2.4609e-03 k_t
= [2, 2, 3, 2]
Generation 1810: Best = 3.261859e+03 FES = 181100 Diversity = 3.0528e-03 k_t
= [1, 1, 1, 1]
Generation 1820: Best = 3.261859e+03 FES = 182100 Diversity = 2.9581e-03 k_t
= [4, 2, 2, 2]
Generation 1830: Best = 3.261859e+03 FES = 183100 Diversity = 4.0750e-03 k_t
= [1, 1, 1, 1]
Generation 1840: Best = 3.261858e+03 FES = 184100 Diversity = 4.1557e-03 k_t

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= [4, 4, 4, 2]
Generation 1850: Best = 3.261858e+03 FES = 185100 Diversity = 5.5773e-03 k_t
= [1, 1, 1, 1]
Generation 1860: Best = 3.261858e+03 FES = 186100 Diversity = 6.6184e-03 k_t
= [2, 3, 2, 4]
Generation 1870: Best = 3.261857e+03 FES = 187100 Diversity = 7.0582e-03 k_t
= [1, 1, 1, 1]
Generation 1880: Best = 3.261857e+03 FES = 188100 Diversity = 8.6043e-03 k_t
= [4, 3, 4, 4]
Generation 1890: Best = 3.261856e+03 FES = 189100 Diversity = 9.1251e-03 k_t
= [1, 1, 1, 1]
Generation 1900: Best = 3.261856e+03 FES = 190100 Diversity = 7.4234e-03 k_t
= [2, 2, 4, 3]
Generation 1910: Best = 3.261856e+03 FES = 191100 Diversity = 6.2756e-03 k_t
= [1, 1, 1, 1]
Generation 1920: Best = 3.261855e+03 FES = 192100 Diversity = 6.0754e-03 k_t
= [4, 2, 4, 4]
Generation 1930: Best = 3.261855e+03 FES = 193100 Diversity = 6.1596e-03 k_t
= [1, 1, 1, 1]
Generation 1940: Best = 3.261855e+03 FES = 194100 Diversity = 5.9421e-03 k_t
= [3, 2, 2, 2]
Generation 1950: Best = 3.261855e+03 FES = 195100 Diversity = 4.5172e-03 k_t
= [1, 1, 1, 1]
Generation 1960: Best = 3.261855e+03 FES = 196100 Diversity = 4.9704e-03 k_t
= [3, 3, 4, 4]
Generation 1970: Best = 3.261854e+03 FES = 197100 Diversity = 4.5156e-03 k_t
= [1, 1, 1, 1]
Generation 1980: Best = 3.261854e+03 FES = 198100 Diversity = 3.6262e-03 k_t
= [3, 3, 4, 2]
Generation 1990: Best = 3.261854e+03 FES = 199100 Diversity = 3.6449e-03 k_t
= [1, 1, 1, 1]
Generation 2000: Best = 3.261854e+03 FES = 200100 Diversity = 4.4378e-03 k_t
= [4, 2, 4, 3]
Generation 2010: Best = 3.261854e+03 FES = 201100 Diversity = 3.7026e-03 k_t
= [1, 1, 1, 1]
Generation 2020: Best = 3.261854e+03 FES = 202100 Diversity = 3.7294e-03 k_t
= [3, 2, 3, 2]
Generation 2030: Best = 3.261854e+03 FES = 203100 Diversity = 4.2833e-03 k_t
= [1, 1, 1, 1]
Generation 2040: Best = 3.261853e+03 FES = 204100 Diversity = 4.5638e-03 k_t
= [4, 4, 2, 2]
Generation 2050: Best = 3.261853e+03 FES = 205100 Diversity = 3.5663e-03 k_t
= [1, 1, 1, 1]
Generation 2060: Best = 3.261853e+03 FES = 206100 Diversity = 3.5394e-03 k_t
= [4, 4, 4, 2]
Generation 2070: Best = 3.261853e+03 FES = 207100 Diversity = 4.3210e-03 k_t
= [1, 1, 1, 1]
Generation 2080: Best = 3.261853e+03 FES = 208100 Diversity = 5.4408e-03 k_t
= [3, 3, 4, 3]
Generation 2090: Best = 3.261853e+03 FES = 209100 Diversity = 5.4219e-03 k_t
= [1, 1, 1, 1]
Generation 2100: Best = 3.261853e+03 FES = 210100 Diversity = 4.6615e-03 k_t
= [4, 4, 2, 3]
Generation 2110: Best = 3.261852e+03 FES = 211100 Diversity = 4.1575e-03 k_t

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= [1, 1, 1, 1]
Generation 2120: Best = 3.261852e+03 FES = 212100 Diversity = 3.6813e-03 k_t
= [3, 2, 4, 3]
Generation 2130: Best = 3.261852e+03 FES = 213100 Diversity = 4.4168e-03 k_t
= [1, 1, 1, 1]
Generation 2140: Best = 3.261852e+03 FES = 214100 Diversity = 3.9740e-03 k_t
= [3, 2, 2, 4]
Generation 2150: Best = 3.261852e+03 FES = 215100 Diversity = 4.3055e-03 k_t
= [1, 1, 1, 1]
Generation 2160: Best = 3.261852e+03 FES = 216100 Diversity = 5.1320e-03 k_t
= [4, 3, 4, 3]
Generation 2170: Best = 3.261852e+03 FES = 217100 Diversity = 4.8409e-03 k_t
= [1, 1, 1, 1]
Generation 2180: Best = 3.261851e+03 FES = 218100 Diversity = 4.2044e-03 k_t
= [4, 3, 4, 4]
Generation 2190: Best = 3.261851e+03 FES = 219100 Diversity = 4.6744e-03 k_t
= [1, 1, 1, 1]
Generation 2200: Best = 3.261851e+03 FES = 220100 Diversity = 3.9008e-03 k_t
= [4, 4, 3, 4]
Generation 2210: Best = 3.261851e+03 FES = 221100 Diversity = 4.3431e-03 k_t
= [1, 1, 1, 1]
Generation 2220: Best = 3.261851e+03 FES = 222100 Diversity = 3.8018e-03 k_t
= [3, 3, 4, 3]
Generation 2230: Best = 3.261851e+03 FES = 223100 Diversity = 3.1843e-03 k_t
= [1, 1, 1, 1]
Generation 2240: Best = 3.261851e+03 FES = 224100 Diversity = 3.5325e-03 k_t
= [2, 2, 2, 3]
Generation 2250: Best = 3.261850e+03 FES = 225100 Diversity = 4.1190e-03 k_t
= [1, 1, 1, 1]
Generation 2260: Best = 3.261850e+03 FES = 226100 Diversity = 3.8854e-03 k_t
= [3, 4, 4, 2]
Generation 2270: Best = 3.261850e+03 FES = 227100 Diversity = 3.4355e-03 k_t
= [1, 1, 1, 1]
Generation 2280: Best = 3.261850e+03 FES = 228100 Diversity = 3.8712e-03 k_t
= [3, 2, 3, 2]
Generation 2290: Best = 3.261850e+03 FES = 229100 Diversity = 3.4875e-03 k_t
= [1, 1, 1, 1]
Generation 2300: Best = 3.261850e+03 FES = 230100 Diversity = 4.1793e-03 k_t
= [3, 2, 3, 3]
Generation 2310: Best = 3.261850e+03 FES = 231100 Diversity = 4.4226e-03 k_t
= [1, 1, 1, 1]
Generation 2320: Best = 3.261850e+03 FES = 232100 Diversity = 3.4572e-03 k_t
= [2, 4, 3, 2]
Generation 2330: Best = 3.261850e+03 FES = 233100 Diversity = 3.5375e-03 k_t
= [1, 1, 1, 1]
Generation 2340: Best = 3.261850e+03 FES = 234100 Diversity = 2.8889e-03 k_t
= [4, 4, 4, 3]
Generation 2350: Best = 3.261850e+03 FES = 235100 Diversity = 2.5888e-03 k_t
= [1, 1, 1, 1]
Generation 2360: Best = 3.261849e+03 FES = 236100 Diversity = 2.3165e-03 k_t
= [3, 2, 3, 4]
Generation 2370: Best = 3.261849e+03 FES = 237100 Diversity = 2.4431e-03 k_t
= [1, 1, 1, 1]
Generation 2380: Best = 3.261849e+03 FES = 238100 Diversity = 2.4781e-03 k_t

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= [3, 3, 2, 3]
Generation 2390: Best = 3.261849e+03  FES = 239100  Diversity = 2.0612e-03  k_t
= [1, 1, 1, 1]
Generation 2400: Best = 3.261849e+03  FES = 240100  Diversity = 2.6259e-03  k_t
= [2, 4, 3, 2]
Generation 2410: Best = 3.261849e+03  FES = 241100  Diversity = 3.1011e-03  k_t
= [1, 1, 1, 1]
Generation 2420: Best = 3.261849e+03  FES = 242100  Diversity = 2.9359e-03  k_t
= [3, 2, 4, 3]
Generation 2430: Best = 3.261849e+03  FES = 243100  Diversity = 2.4685e-03  k_t
= [1, 1, 1, 1]
Generation 2440: Best = 3.261849e+03  FES = 244100  Diversity = 2.1338e-03  k_t
= [4, 4, 2, 3]
Generation 2450: Best = 3.261849e+03  FES = 245100  Diversity = 2.4306e-03  k_t
= [1, 1, 1, 1]
Generation 2460: Best = 3.261849e+03  FES = 246100  Diversity = 2.8067e-03  k_t
= [3, 3, 4, 3]
Generation 2470: Best = 3.261849e+03  FES = 247100  Diversity = 3.1117e-03  k_t
= [1, 1, 1, 1]
Generation 2480: Best = 3.261849e+03  FES = 248100  Diversity = 2.7165e-03  k_t
= [4, 3, 3, 3]
Generation 2490: Best = 3.261849e+03  FES = 249100  Diversity = 2.3085e-03  k_t
= [1, 1, 1, 1]
Generation 2500: Best = 3.261849e+03  FES = 250100  Diversity = 2.2846e-03  k_t
= [4, 4, 4, 2]
Generation 2510: Best = 3.261849e+03  FES = 251100  Diversity = 2.8658e-03  k_t
= [1, 1, 1, 1]
Generation 2520: Best = 3.261849e+03  FES = 252100  Diversity = 2.3363e-03  k_t
= [4, 4, 4, 2]
Generation 2530: Best = 3.261849e+03  FES = 253100  Diversity = 2.4219e-03  k_t
= [1, 1, 1, 1]
Generation 2540: Best = 3.261849e+03  FES = 254100  Diversity = 3.2707e-03  k_t
= [2, 4, 3, 3]
Generation 2550: Best = 3.261848e+03  FES = 255100  Diversity = 3.3617e-03  k_t
= [1, 1, 1, 1]
Generation 2560: Best = 3.261848e+03  FES = 256100  Diversity = 3.3060e-03  k_t
= [2, 4, 2, 2]
Generation 2570: Best = 3.261848e+03  FES = 257100  Diversity = 2.8303e-03  k_t
= [1, 1, 1, 1]
Generation 2580: Best = 3.261848e+03  FES = 258100  Diversity = 2.4568e-03  k_t
= [4, 3, 3, 2]
Generation 2590: Best = 3.261848e+03  FES = 259100  Diversity = 2.2222e-03  k_t
= [1, 1, 1, 1]
Generation 2600: Best = 3.261848e+03  FES = 260100  Diversity = 2.4042e-03  k_t
= [4, 3, 3, 4]
Generation 2610: Best = 3.261848e+03  FES = 261100  Diversity = 2.6986e-03  k_t
= [1, 1, 1, 1]
Generation 2620: Best = 3.261848e+03  FES = 262100  Diversity = 2.1853e-03  k_t
= [3, 3, 4, 3]
Generation 2630: Best = 3.261848e+03  FES = 263100  Diversity = 2.4183e-03  k_t
= [1, 1, 1, 1]
Generation 2640: Best = 3.261848e+03  FES = 264100  Diversity = 2.4995e-03  k_t
= [3, 3, 4, 3]
Generation 2650: Best = 3.261848e+03  FES = 265100  Diversity = 2.7634e-03  k_t

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= [1, 1, 1, 1]
Generation 2660: Best = 3.261848e+03 FES = 266100 Diversity = 2.2010e-03 k_t
= [4, 3, 2, 2]
Generation 2670: Best = 3.261848e+03 FES = 267100 Diversity = 2.1074e-03 k_t
= [1, 1, 1, 1]
Generation 2680: Best = 3.261848e+03 FES = 268100 Diversity = 2.3250e-03 k_t
= [3, 2, 3, 2]
Generation 2690: Best = 3.261848e+03 FES = 269100 Diversity = 1.7499e-03 k_t
= [1, 1, 1, 1]
Generation 2700: Best = 3.261848e+03 FES = 270100 Diversity = 1.9808e-03 k_t
= [4, 3, 3, 3]
Generation 2710: Best = 3.261848e+03 FES = 271100 Diversity = 2.1876e-03 k_t
= [1, 1, 1, 1]
Generation 2720: Best = 3.261848e+03 FES = 272100 Diversity = 2.1601e-03 k_t
= [4, 3, 2, 2]
Generation 2730: Best = 3.261848e+03 FES = 273100 Diversity = 2.0048e-03 k_t
= [1, 1, 1, 1]
Generation 2740: Best = 3.261848e+03 FES = 274100 Diversity = 1.9788e-03 k_t
= [3, 4, 2, 3]
Generation 2750: Best = 3.261848e+03 FES = 275100 Diversity = 2.2861e-03 k_t
= [1, 1, 1, 1]
Generation 2760: Best = 3.261848e+03 FES = 276100 Diversity = 2.0454e-03 k_t
= [2, 2, 3, 4]
Generation 2770: Best = 3.261847e+03 FES = 277100 Diversity = 2.0368e-03 k_t
= [1, 1, 1, 1]
Generation 2780: Best = 3.261847e+03 FES = 278100 Diversity = 2.1708e-03 k_t
= [2, 4, 4, 2]
Generation 2790: Best = 3.261847e+03 FES = 279100 Diversity = 2.5972e-03 k_t
= [1, 1, 1, 1]
Generation 2800: Best = 3.261847e+03 FES = 280100 Diversity = 3.4163e-03 k_t
= [2, 3, 3, 2]
Generation 2810: Best = 3.261847e+03 FES = 281100 Diversity = 3.8120e-03 k_t
= [1, 1, 1, 1]
Generation 2820: Best = 3.261847e+03 FES = 282100 Diversity = 2.9201e-03 k_t
= [2, 3, 4, 2]
Generation 2830: Best = 3.261847e+03 FES = 283100 Diversity = 2.5332e-03 k_t
= [1, 1, 1, 1]
Generation 2840: Best = 3.261847e+03 FES = 284100 Diversity = 2.2386e-03 k_t
= [2, 4, 2, 4]
Generation 2850: Best = 3.261847e+03 FES = 285100 Diversity = 2.0292e-03 k_t
= [1, 1, 1, 1]
Generation 2860: Best = 3.261847e+03 FES = 286100 Diversity = 2.1155e-03 k_t
= [4, 3, 3, 3]
Generation 2870: Best = 3.261847e+03 FES = 287100 Diversity = 2.2778e-03 k_t
= [1, 1, 1, 1]
Generation 2880: Best = 3.261847e+03 FES = 288100 Diversity = 2.4985e-03 k_t
= [4, 4, 3, 2]
Generation 2890: Best = 3.261847e+03 FES = 289100 Diversity = 2.5906e-03 k_t
= [1, 1, 1, 1]
Generation 2900: Best = 3.261847e+03 FES = 290100 Diversity = 2.7964e-03 k_t
= [2, 2, 2, 4]
Generation 2910: Best = 3.261847e+03 FES = 291100 Diversity = 2.5631e-03 k_t
= [1, 1, 1, 1]
Generation 2920: Best = 3.261847e+03 FES = 292100 Diversity = 2.5529e-03 k_t

```

```

= [2, 2, 2, 2]
Generation 2930: Best = 3.261847e+03  FES = 293100  Diversity = 2.3179e-03  k_t
= [1, 1, 1, 1]
Generation 2940: Best = 3.261847e+03  FES = 294100  Diversity = 1.6546e-03  k_t
= [2, 3, 4, 4]
Generation 2950: Best = 3.261847e+03  FES = 295100  Diversity = 2.1545e-03  k_t
= [1, 1, 1, 1]
Generation 2960: Best = 3.261847e+03  FES = 296100  Diversity = 2.1315e-03  k_t
= [2, 3, 4, 3]
Generation 2970: Best = 3.261847e+03  FES = 297100  Diversity = 2.1371e-03  k_t
= [1, 1, 1, 1]
Generation 2980: Best = 3.261847e+03  FES = 298100  Diversity = 1.9293e-03  k_t
= [3, 3, 3, 4]
Generation 2990: Best = 3.261847e+03  FES = 299100  Diversity = 2.2226e-03  k_t
= [1, 1, 1, 1]

```

Optimization completed!

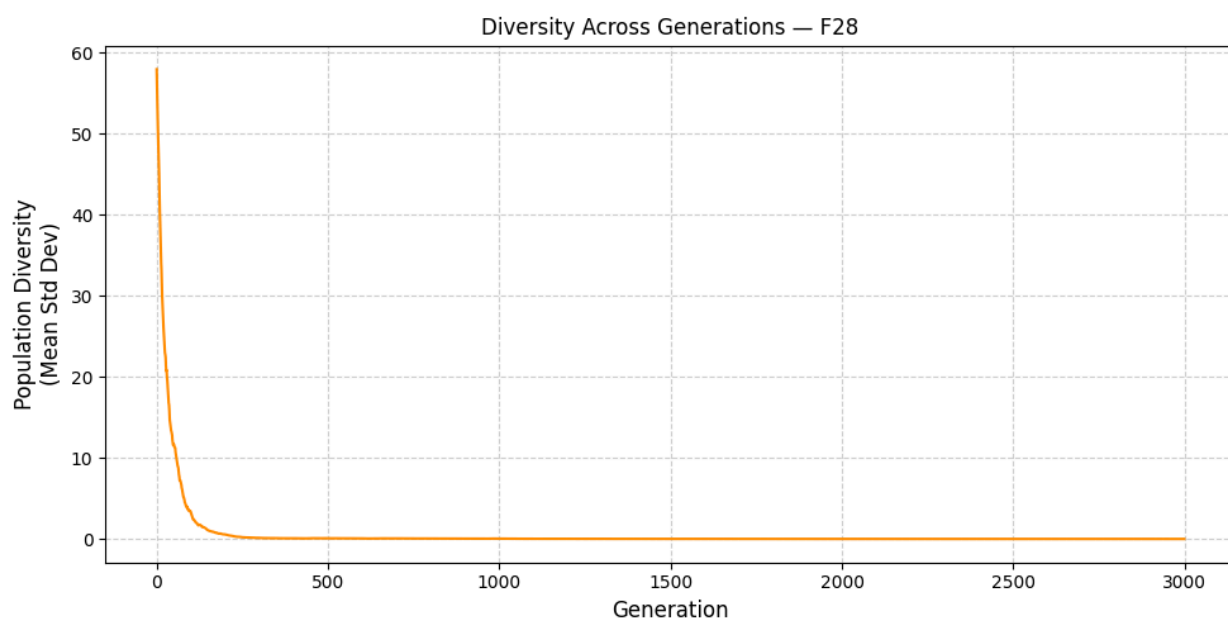
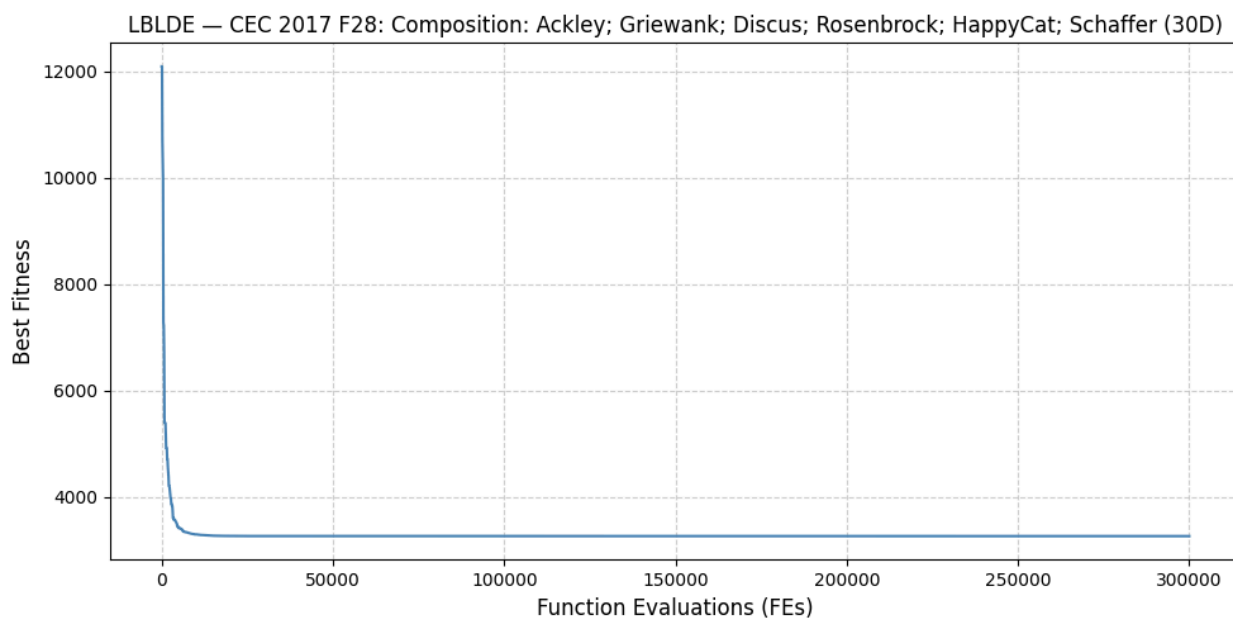
Final Best Fitness = 3.261847e+03

Total Function Evaluations = 300000

```

=====
=
Results:
  Best Fitness          : 3.261847e+03
  Global Optimum        : 2.800000e+03
  Error (|best-optimum|) : 4.618465e+02
  Best Solution (5 dim)  : [-33.81401232  60.51250749 -44.71020645  76.16963619
-54.64912157]
  D^(0) (init. diversity): 5.7976e+01
=====
=

```



Graph saved as convergence_F28.png

```
In [39]: # =====
# F29 — Composition Function 9 (F15; F16; F17)
# =====
run_and_plot('F29', make_wrapper(composition.f29), 2900.0, 'Composition: F15;
```

```

=====
=
LBLDE_s on CEC 2017 F29 – Single Run
=====
=
Function      : Composition: F15; F16; F17 (30D)
Bounds       : [-100, 100]
Global optimum : 2900.0
Initial D^(0) = 5.7976e+01
Generation 0: Best Fitness = 8.977257e+03 | Diversity = 5.7976e+01
Generation 10: Best = 6.025172e+03 FES = 1100 Diversity = 4.6932e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 5.105831e+03 FES = 2100 Diversity = 4.1024e+01 k_t
= [3, 3, 2, 2]
Generation 30: Best = 4.690996e+03 FES = 3100 Diversity = 4.0635e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 4.690996e+03 FES = 4100 Diversity = 3.8753e+01 k_t
= [3, 4, 3, 4]
Generation 50: Best = 4.679200e+03 FES = 5100 Diversity = 3.8061e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 4.532762e+03 FES = 6100 Diversity = 3.7523e+01 k_t
= [2, 3, 4, 3]
Generation 70: Best = 4.495210e+03 FES = 7100 Diversity = 3.7775e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 4.241919e+03 FES = 8100 Diversity = 3.7497e+01 k_t
= [3, 4, 4, 4]
Generation 90: Best = 4.190990e+03 FES = 9100 Diversity = 3.7412e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 4.190990e+03 FES = 10100 Diversity = 3.7287e+01 k_t
= [4, 3, 2, 2]
Generation 110: Best = 4.088641e+03 FES = 11100 Diversity = 3.6606e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 4.088641e+03 FES = 12100 Diversity = 3.6231e+01 k_t
= [2, 3, 3, 2]
Generation 130: Best = 4.088641e+03 FES = 13100 Diversity = 3.5825e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 3.990185e+03 FES = 14100 Diversity = 3.5811e+01 k_t
= [4, 4, 4, 3]
Generation 150: Best = 3.990185e+03 FES = 15100 Diversity = 3.5345e+01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 3.990185e+03 FES = 16100 Diversity = 3.4439e+01 k_t
= [4, 3, 2, 2]
Generation 170: Best = 3.990185e+03 FES = 17100 Diversity = 3.4644e+01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 3.968131e+03 FES = 18100 Diversity = 3.4797e+01 k_t
= [4, 2, 4, 3]
Generation 190: Best = 3.952447e+03 FES = 19100 Diversity = 3.4640e+01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 3.944782e+03 FES = 20100 Diversity = 3.4518e+01 k_t
= [2, 4, 4, 4]
Generation 210: Best = 3.944782e+03 FES = 21100 Diversity = 3.4527e+01 k_t
= [1, 1, 1, 1]
Generation 220: Best = 3.869122e+03 FES = 22100 Diversity = 3.4390e+01 k_t
= [4, 2, 2, 2]

```

Generation 230:	Best = 3.869122e+03	FES = 23100	Diversity = 3.3942e+01	k_t
= [1, 1, 1, 1]				
Generation 240:	Best = 3.869122e+03	FES = 24100	Diversity = 3.4204e+01	k_t
= [3, 4, 2, 4]				
Generation 250:	Best = 3.868978e+03	FES = 25100	Diversity = 3.4354e+01	k_t
= [1, 1, 1, 1]				
Generation 260:	Best = 3.794780e+03	FES = 26100	Diversity = 3.4129e+01	k_t
= [2, 3, 2, 2]				
Generation 270:	Best = 3.794780e+03	FES = 27100	Diversity = 3.3828e+01	k_t
= [1, 1, 1, 1]				
Generation 280:	Best = 3.794780e+03	FES = 28100	Diversity = 3.3535e+01	k_t
= [4, 3, 3, 4]				
Generation 290:	Best = 3.794780e+03	FES = 29100	Diversity = 3.3370e+01	k_t
= [1, 1, 1, 1]				
Generation 300:	Best = 3.757988e+03	FES = 30100	Diversity = 3.3501e+01	k_t
= [3, 2, 4, 4]				
Generation 310:	Best = 3.694957e+03	FES = 31100	Diversity = 3.3778e+01	k_t
= [1, 1, 1, 1]				
Generation 320:	Best = 3.674431e+03	FES = 32100	Diversity = 3.3493e+01	k_t
= [3, 2, 4, 3]				
Generation 330:	Best = 3.674431e+03	FES = 33100	Diversity = 3.3270e+01	k_t
= [1, 1, 1, 1]				
Generation 340:	Best = 3.674431e+03	FES = 34100	Diversity = 3.3142e+01	k_t
= [4, 4, 3, 3]				
Generation 350:	Best = 3.674431e+03	FES = 35100	Diversity = 3.2815e+01	k_t
= [1, 1, 1, 1]				
Generation 360:	Best = 3.622596e+03	FES = 36100	Diversity = 3.2327e+01	k_t
= [4, 3, 4, 3]				
Generation 370:	Best = 3.622596e+03	FES = 37100	Diversity = 3.1637e+01	k_t
= [1, 1, 1, 1]				
Generation 380:	Best = 3.622596e+03	FES = 38100	Diversity = 3.1304e+01	k_t
= [4, 2, 4, 3]				
Generation 390:	Best = 3.622596e+03	FES = 39100	Diversity = 3.1056e+01	k_t
= [1, 1, 1, 1]				
Generation 400:	Best = 3.622596e+03	FES = 40100	Diversity = 3.0756e+01	k_t
= [4, 2, 2, 4]				
Generation 410:	Best = 3.610374e+03	FES = 41100	Diversity = 3.0588e+01	k_t
= [1, 1, 1, 1]				
Generation 420:	Best = 3.610374e+03	FES = 42100	Diversity = 3.0169e+01	k_t
= [4, 2, 3, 3]				
Generation 430:	Best = 3.610374e+03	FES = 43100	Diversity = 3.0182e+01	k_t
= [1, 1, 1, 1]				
Generation 440:	Best = 3.608107e+03	FES = 44100	Diversity = 2.9802e+01	k_t
= [2, 4, 4, 2]				
Generation 450:	Best = 3.608107e+03	FES = 45100	Diversity = 2.9706e+01	k_t
= [1, 1, 1, 1]				
Generation 460:	Best = 3.607498e+03	FES = 46100	Diversity = 2.9555e+01	k_t
= [2, 3, 4, 3]				
Generation 470:	Best = 3.607498e+03	FES = 47100	Diversity = 2.8958e+01	k_t
= [1, 1, 1, 1]				
Generation 480:	Best = 3.559403e+03	FES = 48100	Diversity = 2.8979e+01	k_t
= [4, 3, 2, 4]				
Generation 490:	Best = 3.559403e+03	FES = 49100	Diversity = 2.8741e+01	k_t
= [1, 1, 1, 1]				

Generation 500:	Best = 3.558461e+03	FES = 50100	Diversity = 2.8795e+01	k_t
= [3, 3, 2, 3]				
Generation 510:	Best = 3.557327e+03	FES = 51100	Diversity = 2.8664e+01	k_t
= [1, 1, 1, 1]				
Generation 520:	Best = 3.551764e+03	FES = 52100	Diversity = 2.8712e+01	k_t
= [2, 2, 3, 3]				
Generation 530:	Best = 3.544750e+03	FES = 53100	Diversity = 2.8485e+01	k_t
= [1, 1, 1, 1]				
Generation 540:	Best = 3.544750e+03	FES = 54100	Diversity = 2.8453e+01	k_t
= [3, 2, 2, 4]				
Generation 550:	Best = 3.537657e+03	FES = 55100	Diversity = 2.8203e+01	k_t
= [1, 1, 1, 1]				
Generation 560:	Best = 3.525406e+03	FES = 56100	Diversity = 2.7819e+01	k_t
= [2, 4, 3, 3]				
Generation 570:	Best = 3.508309e+03	FES = 57100	Diversity = 2.7567e+01	k_t
= [1, 1, 1, 1]				
Generation 580:	Best = 3.508309e+03	FES = 58100	Diversity = 2.7268e+01	k_t
= [4, 4, 4, 4]				
Generation 590:	Best = 3.507004e+03	FES = 59100	Diversity = 2.6618e+01	k_t
= [1, 1, 1, 1]				
Generation 600:	Best = 3.494607e+03	FES = 60100	Diversity = 2.6057e+01	k_t
= [3, 2, 2, 3]				
Generation 610:	Best = 3.482167e+03	FES = 61100	Diversity = 2.5630e+01	k_t
= [1, 1, 1, 1]				
Generation 620:	Best = 3.481543e+03	FES = 62100	Diversity = 2.5194e+01	k_t
= [2, 2, 4, 3]				
Generation 630:	Best = 3.474919e+03	FES = 63100	Diversity = 2.4081e+01	k_t
= [1, 1, 1, 1]				
Generation 640:	Best = 3.471875e+03	FES = 64100	Diversity = 2.2609e+01	k_t
= [3, 2, 3, 3]				
Generation 650:	Best = 3.458717e+03	FES = 65100	Diversity = 2.1879e+01	k_t
= [1, 1, 1, 1]				
Generation 660:	Best = 3.458412e+03	FES = 66100	Diversity = 2.1106e+01	k_t
= [3, 4, 3, 4]				
Generation 670:	Best = 3.455579e+03	FES = 67100	Diversity = 1.9182e+01	k_t
= [1, 1, 1, 1]				
Generation 680:	Best = 3.454567e+03	FES = 68100	Diversity = 1.8094e+01	k_t
= [2, 3, 2, 3]				
Generation 690:	Best = 3.451833e+03	FES = 69100	Diversity = 1.7528e+01	k_t
= [1, 1, 1, 1]				
Generation 700:	Best = 3.447086e+03	FES = 70100	Diversity = 1.6729e+01	k_t
= [4, 3, 2, 4]				
Generation 710:	Best = 3.436641e+03	FES = 71100	Diversity = 1.5031e+01	k_t
= [1, 1, 1, 1]				
Generation 720:	Best = 3.436641e+03	FES = 72100	Diversity = 1.4322e+01	k_t
= [3, 2, 3, 4]				
Generation 730:	Best = 3.436641e+03	FES = 73100	Diversity = 1.3680e+01	k_t
= [1, 1, 1, 1]				
Generation 740:	Best = 3.423925e+03	FES = 74100	Diversity = 1.3518e+01	k_t
= [2, 4, 4, 3]				
Generation 750:	Best = 3.418852e+03	FES = 75100	Diversity = 1.3345e+01	k_t
= [1, 1, 1, 1]				
Generation 760:	Best = 3.418852e+03	FES = 76100	Diversity = 1.3083e+01	k_t
= [2, 3, 3, 4]				

Generation 770:	Best = 3.413022e+03	FES = 77100	Diversity = 1.2175e+01	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 3.404068e+03	FES = 78100	Diversity = 1.2277e+01	k_t
= [4, 2, 3, 4]				
Generation 790:	Best = 3.404068e+03	FES = 79100	Diversity = 1.2229e+01	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 3.404068e+03	FES = 80100	Diversity = 1.2017e+01	k_t
= [3, 2, 2, 2]				
Generation 810:	Best = 3.404068e+03	FES = 81100	Diversity = 1.1956e+01	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 3.399932e+03	FES = 82100	Diversity = 1.1580e+01	k_t
= [4, 3, 3, 2]				
Generation 830:	Best = 3.399932e+03	FES = 83100	Diversity = 1.1440e+01	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 3.397708e+03	FES = 84100	Diversity = 1.1329e+01	k_t
= [3, 3, 2, 3]				
Generation 850:	Best = 3.397708e+03	FES = 85100	Diversity = 1.1066e+01	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 3.394682e+03	FES = 86100	Diversity = 1.0774e+01	k_t
= [3, 4, 3, 2]				
Generation 870:	Best = 3.394682e+03	FES = 87100	Diversity = 1.0628e+01	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 3.394682e+03	FES = 88100	Diversity = 1.0338e+01	k_t
= [4, 2, 4, 2]				
Generation 890:	Best = 3.394682e+03	FES = 89100	Diversity = 1.0169e+01	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 3.382693e+03	FES = 90100	Diversity = 9.9916e+00	k_t
= [2, 3, 3, 3]				
Generation 910:	Best = 3.382693e+03	FES = 91100	Diversity = 9.6575e+00	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 3.379739e+03	FES = 92100	Diversity = 9.3783e+00	k_t
= [3, 2, 3, 2]				
Generation 930:	Best = 3.379739e+03	FES = 93100	Diversity = 9.3863e+00	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 3.379068e+03	FES = 94100	Diversity = 9.3392e+00	k_t
= [2, 3, 3, 4]				
Generation 950:	Best = 3.379021e+03	FES = 95100	Diversity = 9.1164e+00	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 3.379021e+03	FES = 96100	Diversity = 9.2047e+00	k_t
= [3, 3, 3, 4]				
Generation 970:	Best = 3.379021e+03	FES = 97100	Diversity = 9.1587e+00	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 3.376704e+03	FES = 98100	Diversity = 9.0425e+00	k_t
= [3, 2, 2, 4]				
Generation 990:	Best = 3.376704e+03	FES = 99100	Diversity = 8.9907e+00	k_t
= [1, 1, 1, 1]				
Generation 1000:	Best = 3.376704e+03	FES = 100100	Diversity = 8.8073e+00	k_t
= [3, 2, 3, 2]				
Generation 1010:	Best = 3.376704e+03	FES = 101100	Diversity = 8.5943e+00	k_t
= [1, 1, 1, 1]				
Generation 1020:	Best = 3.376704e+03	FES = 102100	Diversity = 8.4967e+00	k_t
= [3, 4, 3, 3]				
Generation 1030:	Best = 3.376632e+03	FES = 103100	Diversity = 8.4192e+00	k_t
= [1, 1, 1, 1]				

Generation 1040: Best = 3.376632e+03 = [4, 3, 3, 3]	FES = 104100	Diversity = 8.5511e+00	k_t
Generation 1050: Best = 3.376632e+03 = [1, 1, 1, 1]	FES = 105100	Diversity = 8.4993e+00	k_t
Generation 1060: Best = 3.374176e+03 = [3, 4, 2, 2]	FES = 106100	Diversity = 8.1779e+00	k_t
Generation 1070: Best = 3.371852e+03 = [1, 1, 1, 1]	FES = 107100	Diversity = 8.1790e+00	k_t
Generation 1080: Best = 3.371852e+03 = [2, 2, 4, 2]	FES = 108100	Diversity = 8.1402e+00	k_t
Generation 1090: Best = 3.369572e+03 = [1, 1, 1, 1]	FES = 109100	Diversity = 8.1014e+00	k_t
Generation 1100: Best = 3.369572e+03 = [2, 4, 2, 4]	FES = 110100	Diversity = 8.0418e+00	k_t
Generation 1110: Best = 3.369572e+03 = [1, 1, 1, 1]	FES = 111100	Diversity = 8.0206e+00	k_t
Generation 1120: Best = 3.369572e+03 = [2, 3, 3, 4]	FES = 112100	Diversity = 8.0335e+00	k_t
Generation 1130: Best = 3.369572e+03 = [1, 1, 1, 1]	FES = 113100	Diversity = 8.0505e+00	k_t
Generation 1140: Best = 3.369447e+03 = [4, 4, 3, 4]	FES = 114100	Diversity = 7.9387e+00	k_t
Generation 1150: Best = 3.369447e+03 = [1, 1, 1, 1]	FES = 115100	Diversity = 7.9238e+00	k_t
Generation 1160: Best = 3.369447e+03 = [4, 2, 4, 4]	FES = 116100	Diversity = 8.0098e+00	k_t
Generation 1170: Best = 3.362613e+03 = [1, 1, 1, 1]	FES = 117100	Diversity = 8.0135e+00	k_t
Generation 1180: Best = 3.362613e+03 = [3, 3, 3, 2]	FES = 118100	Diversity = 7.9955e+00	k_t
Generation 1190: Best = 3.362613e+03 = [1, 1, 1, 1]	FES = 119100	Diversity = 7.9611e+00	k_t
Generation 1200: Best = 3.360691e+03 = [4, 4, 3, 2]	FES = 120100	Diversity = 7.9185e+00	k_t
Generation 1210: Best = 3.360653e+03 = [1, 1, 1, 1]	FES = 121100	Diversity = 7.8692e+00	k_t
Generation 1220: Best = 3.360653e+03 = [3, 2, 3, 4]	FES = 122100	Diversity = 7.7627e+00	k_t
Generation 1230: Best = 3.360653e+03 = [1, 1, 1, 1]	FES = 123100	Diversity = 7.6050e+00	k_t
Generation 1240: Best = 3.360653e+03 = [2, 3, 4, 4]	FES = 124100	Diversity = 7.5256e+00	k_t
Generation 1250: Best = 3.356546e+03 = [1, 1, 1, 1]	FES = 125100	Diversity = 7.4926e+00	k_t
Generation 1260: Best = 3.353876e+03 = [4, 2, 4, 2]	FES = 126100	Diversity = 7.4079e+00	k_t
Generation 1270: Best = 3.353876e+03 = [1, 1, 1, 1]	FES = 127100	Diversity = 7.3835e+00	k_t
Generation 1280: Best = 3.353876e+03 = [4, 4, 3, 4]	FES = 128100	Diversity = 7.2800e+00	k_t
Generation 1290: Best = 3.353876e+03 = [1, 1, 1, 1]	FES = 129100	Diversity = 7.1615e+00	k_t
Generation 1300: Best = 3.353876e+03 = [3, 4, 4, 2]	FES = 130100	Diversity = 7.0584e+00	k_t

Generation 1310: Best = 3.353876e+03 = [1, 1, 1, 1]	FES = 131100	Diversity = 7.0474e+00	k_t
Generation 1320: Best = 3.353876e+03 = [4, 3, 3, 2]	FES = 132100	Diversity = 7.0435e+00	k_t
Generation 1330: Best = 3.353876e+03 = [1, 1, 1, 1]	FES = 133100	Diversity = 7.0511e+00	k_t
Generation 1340: Best = 3.353876e+03 = [2, 4, 2, 2]	FES = 134100	Diversity = 6.9633e+00	k_t
Generation 1350: Best = 3.353876e+03 = [1, 1, 1, 1]	FES = 135100	Diversity = 6.7883e+00	k_t
Generation 1360: Best = 3.353876e+03 = [4, 4, 4, 2]	FES = 136100	Diversity = 6.7848e+00	k_t
Generation 1370: Best = 3.353876e+03 = [1, 1, 1, 1]	FES = 137100	Diversity = 6.5307e+00	k_t
Generation 1380: Best = 3.353876e+03 = [4, 4, 3, 3]	FES = 138100	Diversity = 6.4473e+00	k_t
Generation 1390: Best = 3.353876e+03 = [1, 1, 1, 1]	FES = 139100	Diversity = 6.2922e+00	k_t
Generation 1400: Best = 3.353611e+03 = [4, 4, 3, 2]	FES = 140100	Diversity = 6.2472e+00	k_t
Generation 1410: Best = 3.353611e+03 = [1, 1, 1, 1]	FES = 141100	Diversity = 6.1494e+00	k_t
Generation 1420: Best = 3.353611e+03 = [3, 2, 4, 2]	FES = 142100	Diversity = 5.9748e+00	k_t
Generation 1430: Best = 3.353611e+03 = [1, 1, 1, 1]	FES = 143100	Diversity = 5.7335e+00	k_t
Generation 1440: Best = 3.351952e+03 = [2, 3, 4, 2]	FES = 144100	Diversity = 5.6726e+00	k_t
Generation 1450: Best = 3.351952e+03 = [1, 1, 1, 1]	FES = 145100	Diversity = 5.5638e+00	k_t
Generation 1460: Best = 3.351952e+03 = [2, 2, 3, 4]	FES = 146100	Diversity = 5.4167e+00	k_t
Generation 1470: Best = 3.348799e+03 = [1, 1, 1, 1]	FES = 147100	Diversity = 5.2637e+00	k_t
Generation 1480: Best = 3.344298e+03 = [2, 4, 2, 4]	FES = 148100	Diversity = 5.1300e+00	k_t
Generation 1490: Best = 3.342669e+03 = [1, 1, 1, 1]	FES = 149100	Diversity = 5.0363e+00	k_t
Generation 1500: Best = 3.342621e+03 = [2, 4, 2, 2]	FES = 150100	Diversity = 4.9204e+00	k_t
Generation 1510: Best = 3.342106e+03 = [1, 1, 1, 1]	FES = 151100	Diversity = 4.8598e+00	k_t
Generation 1520: Best = 3.341153e+03 = [2, 3, 3, 2]	FES = 152100	Diversity = 4.6636e+00	k_t
Generation 1530: Best = 3.337676e+03 = [1, 1, 1, 1]	FES = 153100	Diversity = 4.6322e+00	k_t
Generation 1540: Best = 3.335897e+03 = [3, 3, 3, 3]	FES = 154100	Diversity = 4.5834e+00	k_t
Generation 1550: Best = 3.335056e+03 = [1, 1, 1, 1]	FES = 155100	Diversity = 4.5437e+00	k_t
Generation 1560: Best = 3.333729e+03 = [2, 4, 3, 2]	FES = 156100	Diversity = 4.3911e+00	k_t
Generation 1570: Best = 3.333728e+03 = [1, 1, 1, 1]	FES = 157100	Diversity = 4.0610e+00	k_t

Generation 1580: Best = 3.333156e+03 = [3, 2, 3, 4]	FES = 158100	Diversity = 3.8162e+00	k_t
Generation 1590: Best = 3.332371e+03 = [1, 1, 1, 1]	FES = 159100	Diversity = 3.2598e+00	k_t
Generation 1600: Best = 3.330275e+03 = [2, 3, 4, 3]	FES = 160100	Diversity = 2.7526e+00	k_t
Generation 1610: Best = 3.330275e+03 = [1, 1, 1, 1]	FES = 161100	Diversity = 1.9581e+00	k_t
Generation 1620: Best = 3.330275e+03 = [2, 2, 4, 2]	FES = 162100	Diversity = 1.5180e+00	k_t
Generation 1630: Best = 3.328145e+03 = [1, 1, 1, 1]	FES = 163100	Diversity = 1.1754e+00	k_t
Generation 1640: Best = 3.328145e+03 = [2, 2, 4, 2]	FES = 164100	Diversity = 1.0814e+00	k_t
Generation 1650: Best = 3.328145e+03 = [1, 1, 1, 1]	FES = 165100	Diversity = 1.0323e+00	k_t
Generation 1660: Best = 3.327782e+03 = [2, 4, 3, 3]	FES = 166100	Diversity = 9.3486e-01	k_t
Generation 1670: Best = 3.327182e+03 = [1, 1, 1, 1]	FES = 167100	Diversity = 9.0734e-01	k_t
Generation 1680: Best = 3.326527e+03 = [3, 4, 2, 4]	FES = 168100	Diversity = 9.4763e-01	k_t
Generation 1690: Best = 3.326043e+03 = [1, 1, 1, 1]	FES = 169100	Diversity = 7.8566e-01	k_t
Generation 1700: Best = 3.326043e+03 = [2, 2, 3, 3]	FES = 170100	Diversity = 7.6998e-01	k_t
Generation 1710: Best = 3.325953e+03 = [1, 1, 1, 1]	FES = 171100	Diversity = 7.4528e-01	k_t
Generation 1720: Best = 3.325876e+03 = [4, 3, 2, 3]	FES = 172100	Diversity = 7.4823e-01	k_t
Generation 1730: Best = 3.325641e+03 = [1, 1, 1, 1]	FES = 173100	Diversity = 7.5271e-01	k_t
Generation 1740: Best = 3.325242e+03 = [2, 3, 3, 2]	FES = 174100	Diversity = 7.4371e-01	k_t
Generation 1750: Best = 3.324866e+03 = [1, 1, 1, 1]	FES = 175100	Diversity = 7.2681e-01	k_t
Generation 1760: Best = 3.324706e+03 = [2, 2, 4, 2]	FES = 176100	Diversity = 7.1520e-01	k_t
Generation 1770: Best = 3.324601e+03 = [1, 1, 1, 1]	FES = 177100	Diversity = 7.1508e-01	k_t
Generation 1780: Best = 3.324529e+03 = [2, 4, 4, 2]	FES = 178100	Diversity = 7.0769e-01	k_t
Generation 1790: Best = 3.324486e+03 = [1, 1, 1, 1]	FES = 179100	Diversity = 6.9176e-01	k_t
Generation 1800: Best = 3.324486e+03 = [4, 2, 2, 2]	FES = 180100	Diversity = 6.7382e-01	k_t
Generation 1810: Best = 3.324406e+03 = [1, 1, 1, 1]	FES = 181100	Diversity = 6.5989e-01	k_t
Generation 1820: Best = 3.324113e+03 = [4, 3, 4, 3]	FES = 182100	Diversity = 6.5352e-01	k_t
Generation 1830: Best = 3.324045e+03 = [1, 1, 1, 1]	FES = 183100	Diversity = 6.4841e-01	k_t
Generation 1840: Best = 3.323916e+03 = [3, 2, 3, 4]	FES = 184100	Diversity = 6.5302e-01	k_t

Generation 1850: Best = 3.323763e+03 = [1, 1, 1, 1]	FES = 185100	Diversity = 6.3526e-01	k_t
Generation 1860: Best = 3.323157e+03 = [3, 4, 4, 3]	FES = 186100	Diversity = 6.2596e-01	k_t
Generation 1870: Best = 3.322816e+03 = [1, 1, 1, 1]	FES = 187100	Diversity = 6.3602e-01	k_t
Generation 1880: Best = 3.322816e+03 = [2, 4, 3, 4]	FES = 188100	Diversity = 6.1785e-01	k_t
Generation 1890: Best = 3.322816e+03 = [1, 1, 1, 1]	FES = 189100	Diversity = 5.9284e-01	k_t
Generation 1900: Best = 3.322816e+03 = [2, 3, 4, 4]	FES = 190100	Diversity = 5.7285e-01	k_t
Generation 1910: Best = 3.322555e+03 = [1, 1, 1, 1]	FES = 191100	Diversity = 5.6993e-01	k_t
Generation 1920: Best = 3.322555e+03 = [3, 2, 2, 2]	FES = 192100	Diversity = 5.6550e-01	k_t
Generation 1930: Best = 3.322324e+03 = [1, 1, 1, 1]	FES = 193100	Diversity = 5.6300e-01	k_t
Generation 1940: Best = 3.322267e+03 = [4, 4, 2, 4]	FES = 194100	Diversity = 5.6181e-01	k_t
Generation 1950: Best = 3.322267e+03 = [1, 1, 1, 1]	FES = 195100	Diversity = 5.4660e-01	k_t
Generation 1960: Best = 3.322267e+03 = [3, 4, 4, 2]	FES = 196100	Diversity = 5.4018e-01	k_t
Generation 1970: Best = 3.322267e+03 = [1, 1, 1, 1]	FES = 197100	Diversity = 5.4060e-01	k_t
Generation 1980: Best = 3.322246e+03 = [3, 3, 3, 2]	FES = 198100	Diversity = 5.2904e-01	k_t
Generation 1990: Best = 3.322020e+03 = [1, 1, 1, 1]	FES = 199100	Diversity = 5.3037e-01	k_t
Generation 2000: Best = 3.322020e+03 = [4, 4, 3, 4]	FES = 200100	Diversity = 5.3258e-01	k_t
Generation 2010: Best = 3.321966e+03 = [1, 1, 1, 1]	FES = 201100	Diversity = 5.2874e-01	k_t
Generation 2020: Best = 3.321733e+03 = [2, 4, 3, 3]	FES = 202100	Diversity = 5.2534e-01	k_t
Generation 2030: Best = 3.321686e+03 = [1, 1, 1, 1]	FES = 203100	Diversity = 5.2537e-01	k_t
Generation 2040: Best = 3.321686e+03 = [3, 2, 3, 3]	FES = 204100	Diversity = 5.1619e-01	k_t
Generation 2050: Best = 3.321578e+03 = [1, 1, 1, 1]	FES = 205100	Diversity = 5.1184e-01	k_t
Generation 2060: Best = 3.321554e+03 = [2, 2, 2, 3]	FES = 206100	Diversity = 5.0612e-01	k_t
Generation 2070: Best = 3.321554e+03 = [1, 1, 1, 1]	FES = 207100	Diversity = 4.9897e-01	k_t
Generation 2080: Best = 3.321554e+03 = [2, 3, 3, 3]	FES = 208100	Diversity = 4.9128e-01	k_t
Generation 2090: Best = 3.321489e+03 = [1, 1, 1, 1]	FES = 209100	Diversity = 5.0299e-01	k_t
Generation 2100: Best = 3.321484e+03 = [3, 3, 4, 2]	FES = 210100	Diversity = 5.0340e-01	k_t
Generation 2110: Best = 3.321364e+03 = [1, 1, 1, 1]	FES = 211100	Diversity = 4.9927e-01	k_t

Generation 2120: Best = 3.321364e+03 = [2, 3, 4, 4]	FES = 212100	Diversity = 4.9948e-01	k_t
Generation 2130: Best = 3.321213e+03 = [1, 1, 1, 1]	FES = 213100	Diversity = 4.9013e-01	k_t
Generation 2140: Best = 3.321168e+03 = [2, 2, 4, 4]	FES = 214100	Diversity = 4.8615e-01	k_t
Generation 2150: Best = 3.321153e+03 = [1, 1, 1, 1]	FES = 215100	Diversity = 4.6524e-01	k_t
Generation 2160: Best = 3.321069e+03 = [4, 3, 4, 4]	FES = 216100	Diversity = 4.6405e-01	k_t
Generation 2170: Best = 3.321023e+03 = [1, 1, 1, 1]	FES = 217100	Diversity = 4.6445e-01	k_t
Generation 2180: Best = 3.321005e+03 = [2, 4, 2, 3]	FES = 218100	Diversity = 4.6645e-01	k_t
Generation 2190: Best = 3.320576e+03 = [1, 1, 1, 1]	FES = 219100	Diversity = 4.4471e-01	k_t
Generation 2200: Best = 3.320568e+03 = [4, 4, 2, 3]	FES = 220100	Diversity = 4.3848e-01	k_t
Generation 2210: Best = 3.320568e+03 = [1, 1, 1, 1]	FES = 221100	Diversity = 4.3562e-01	k_t
Generation 2220: Best = 3.320420e+03 = [2, 2, 3, 2]	FES = 222100	Diversity = 4.3348e-01	k_t
Generation 2230: Best = 3.320419e+03 = [1, 1, 1, 1]	FES = 223100	Diversity = 4.2845e-01	k_t
Generation 2240: Best = 3.320354e+03 = [2, 4, 3, 4]	FES = 224100	Diversity = 4.2162e-01	k_t
Generation 2250: Best = 3.320353e+03 = [1, 1, 1, 1]	FES = 225100	Diversity = 4.1880e-01	k_t
Generation 2260: Best = 3.320282e+03 = [4, 3, 4, 4]	FES = 226100	Diversity = 4.1153e-01	k_t
Generation 2270: Best = 3.320079e+03 = [1, 1, 1, 1]	FES = 227100	Diversity = 4.1069e-01	k_t
Generation 2280: Best = 3.320065e+03 = [2, 4, 3, 3]	FES = 228100	Diversity = 3.9986e-01	k_t
Generation 2290: Best = 3.319970e+03 = [1, 1, 1, 1]	FES = 229100	Diversity = 3.5948e-01	k_t
Generation 2300: Best = 3.319937e+03 = [4, 3, 2, 2]	FES = 230100	Diversity = 3.1920e-01	k_t
Generation 2310: Best = 3.319833e+03 = [1, 1, 1, 1]	FES = 231100	Diversity = 2.7087e-01	k_t
Generation 2320: Best = 3.319695e+03 = [4, 4, 2, 3]	FES = 232100	Diversity = 1.8380e-01	k_t
Generation 2330: Best = 3.319558e+03 = [1, 1, 1, 1]	FES = 233100	Diversity = 1.5012e-01	k_t
Generation 2340: Best = 3.319485e+03 = [3, 2, 2, 2]	FES = 234100	Diversity = 1.1986e-01	k_t
Generation 2350: Best = 3.319289e+03 = [1, 1, 1, 1]	FES = 235100	Diversity = 8.9746e-02	k_t
Generation 2360: Best = 3.319251e+03 = [4, 2, 3, 3]	FES = 236100	Diversity = 7.4143e-02	k_t
Generation 2370: Best = 3.319149e+03 = [1, 1, 1, 1]	FES = 237100	Diversity = 6.9159e-02	k_t
Generation 2380: Best = 3.319107e+03 = [3, 4, 2, 2]	FES = 238100	Diversity = 6.2236e-02	k_t

Generation 2390: Best = 3.319050e+03 = [1, 1, 1, 1]	FES = 239100	Diversity = 5.4283e-02	k_t
Generation 2400: Best = 3.319025e+03 = [2, 4, 2, 3]	FES = 240100	Diversity = 4.9065e-02	k_t
Generation 2410: Best = 3.318976e+03 = [1, 1, 1, 1]	FES = 241100	Diversity = 4.2823e-02	k_t
Generation 2420: Best = 3.318959e+03 = [3, 3, 4, 2]	FES = 242100	Diversity = 3.8533e-02	k_t
Generation 2430: Best = 3.318942e+03 = [1, 1, 1, 1]	FES = 243100	Diversity = 3.1875e-02	k_t
Generation 2440: Best = 3.318930e+03 = [3, 4, 4, 3]	FES = 244100	Diversity = 2.3216e-02	k_t
Generation 2450: Best = 3.318923e+03 = [1, 1, 1, 1]	FES = 245100	Diversity = 2.0130e-02	k_t
Generation 2460: Best = 3.318910e+03 = [3, 3, 3, 4]	FES = 246100	Diversity = 1.7361e-02	k_t
Generation 2470: Best = 3.318905e+03 = [1, 1, 1, 1]	FES = 247100	Diversity = 1.5227e-02	k_t
Generation 2480: Best = 3.318901e+03 = [3, 2, 3, 2]	FES = 248100	Diversity = 1.3397e-02	k_t
Generation 2490: Best = 3.318899e+03 = [1, 1, 1, 1]	FES = 249100	Diversity = 1.2596e-02	k_t
Generation 2500: Best = 3.318895e+03 = [4, 2, 2, 2]	FES = 250100	Diversity = 1.1619e-02	k_t
Generation 2510: Best = 3.318892e+03 = [1, 1, 1, 1]	FES = 251100	Diversity = 1.0714e-02	k_t
Generation 2520: Best = 3.318891e+03 = [3, 4, 3, 4]	FES = 252100	Diversity = 9.2957e-03	k_t
Generation 2530: Best = 3.318889e+03 = [1, 1, 1, 1]	FES = 253100	Diversity = 7.8054e-03	k_t
Generation 2540: Best = 3.318888e+03 = [3, 4, 3, 2]	FES = 254100	Diversity = 7.4791e-03	k_t
Generation 2550: Best = 3.318887e+03 = [1, 1, 1, 1]	FES = 255100	Diversity = 6.3828e-03	k_t
Generation 2560: Best = 3.318885e+03 = [3, 4, 3, 4]	FES = 256100	Diversity = 5.7824e-03	k_t
Generation 2570: Best = 3.318884e+03 = [1, 1, 1, 1]	FES = 257100	Diversity = 5.3375e-03	k_t
Generation 2580: Best = 3.318883e+03 = [3, 3, 3, 3]	FES = 258100	Diversity = 5.7377e-03	k_t
Generation 2590: Best = 3.318883e+03 = [1, 1, 1, 1]	FES = 259100	Diversity = 6.3691e-03	k_t
Generation 2600: Best = 3.318882e+03 = [2, 4, 4, 3]	FES = 260100	Diversity = 5.4870e-03	k_t
Generation 2610: Best = 3.318882e+03 = [1, 1, 1, 1]	FES = 261100	Diversity = 5.4694e-03	k_t
Generation 2620: Best = 3.318881e+03 = [4, 3, 4, 3]	FES = 262100	Diversity = 4.9509e-03	k_t
Generation 2630: Best = 3.318880e+03 = [1, 1, 1, 1]	FES = 263100	Diversity = 4.3075e-03	k_t
Generation 2640: Best = 3.318880e+03 = [3, 2, 2, 3]	FES = 264100	Diversity = 4.0443e-03	k_t
Generation 2650: Best = 3.318879e+03 = [1, 1, 1, 1]	FES = 265100	Diversity = 4.3512e-03	k_t

Generation 2660: Best = 3.318879e+03 = [3, 3, 3, 4]	FES = 266100	Diversity = 4.4547e-03	k_t
Generation 2670: Best = 3.318878e+03 = [1, 1, 1, 1]	FES = 267100	Diversity = 4.1826e-03	k_t
Generation 2680: Best = 3.318878e+03 = [3, 4, 3, 4]	FES = 268100	Diversity = 4.1302e-03	k_t
Generation 2690: Best = 3.318877e+03 = [1, 1, 1, 1]	FES = 269100	Diversity = 3.8242e-03	k_t
Generation 2700: Best = 3.318877e+03 = [2, 4, 4, 3]	FES = 270100	Diversity = 4.1363e-03	k_t
Generation 2710: Best = 3.318876e+03 = [1, 1, 1, 1]	FES = 271100	Diversity = 3.7787e-03	k_t
Generation 2720: Best = 3.318876e+03 = [2, 3, 4, 3]	FES = 272100	Diversity = 3.6116e-03	k_t
Generation 2730: Best = 3.318876e+03 = [1, 1, 1, 1]	FES = 273100	Diversity = 3.3727e-03	k_t
Generation 2740: Best = 3.318876e+03 = [4, 3, 4, 2]	FES = 274100	Diversity = 3.3972e-03	k_t
Generation 2750: Best = 3.318875e+03 = [1, 1, 1, 1]	FES = 275100	Diversity = 3.1115e-03	k_t
Generation 2760: Best = 3.318875e+03 = [3, 4, 2, 2]	FES = 276100	Diversity = 2.8740e-03	k_t
Generation 2770: Best = 3.318875e+03 = [1, 1, 1, 1]	FES = 277100	Diversity = 3.0064e-03	k_t
Generation 2780: Best = 3.318874e+03 = [4, 3, 2, 2]	FES = 278100	Diversity = 2.7516e-03	k_t
Generation 2790: Best = 3.318874e+03 = [1, 1, 1, 1]	FES = 279100	Diversity = 2.6531e-03	k_t
Generation 2800: Best = 3.318874e+03 = [3, 4, 4, 2]	FES = 280100	Diversity = 3.2806e-03	k_t
Generation 2810: Best = 3.318873e+03 = [1, 1, 1, 1]	FES = 281100	Diversity = 3.2437e-03	k_t
Generation 2820: Best = 3.318873e+03 = [2, 3, 4, 2]	FES = 282100	Diversity = 3.2123e-03	k_t
Generation 2830: Best = 3.318873e+03 = [1, 1, 1, 1]	FES = 283100	Diversity = 3.0369e-03	k_t
Generation 2840: Best = 3.318873e+03 = [4, 2, 4, 4]	FES = 284100	Diversity = 3.2571e-03	k_t
Generation 2850: Best = 3.318872e+03 = [1, 1, 1, 1]	FES = 285100	Diversity = 3.3225e-03	k_t
Generation 2860: Best = 3.318872e+03 = [3, 2, 3, 2]	FES = 286100	Diversity = 3.0082e-03	k_t
Generation 2870: Best = 3.318872e+03 = [1, 1, 1, 1]	FES = 287100	Diversity = 3.2034e-03	k_t
Generation 2880: Best = 3.318872e+03 = [4, 2, 3, 4]	FES = 288100	Diversity = 2.7976e-03	k_t
Generation 2890: Best = 3.318871e+03 = [1, 1, 1, 1]	FES = 289100	Diversity = 2.6251e-03	k_t
Generation 2900: Best = 3.318871e+03 = [3, 4, 4, 2]	FES = 290100	Diversity = 2.4573e-03	k_t
Generation 2910: Best = 3.318871e+03 = [1, 1, 1, 1]	FES = 291100	Diversity = 2.2308e-03	k_t
Generation 2920: Best = 3.318871e+03 = [3, 4, 3, 4]	FES = 292100	Diversity = 2.2342e-03	k_t

Generation 2930: Best = 3.318871e+03 FES = 293100 Diversity = 2.1207e-03 k_t
 = [1, 1, 1, 1]
 Generation 2940: Best = 3.318871e+03 FES = 294100 Diversity = 2.2151e-03 k_t
 = [2, 3, 2, 2]
 Generation 2950: Best = 3.318871e+03 FES = 295100 Diversity = 2.2282e-03 k_t
 = [1, 1, 1, 1]
 Generation 2960: Best = 3.318870e+03 FES = 296100 Diversity = 2.1292e-03 k_t
 = [4, 2, 2, 4]
 Generation 2970: Best = 3.318870e+03 FES = 297100 Diversity = 2.1971e-03 k_t
 = [1, 1, 1, 1]
 Generation 2980: Best = 3.318870e+03 FES = 298100 Diversity = 2.0018e-03 k_t
 = [4, 2, 4, 3]
 Generation 2990: Best = 3.318870e+03 FES = 299100 Diversity = 2.0675e-03 k_t
 = [1, 1, 1, 1]

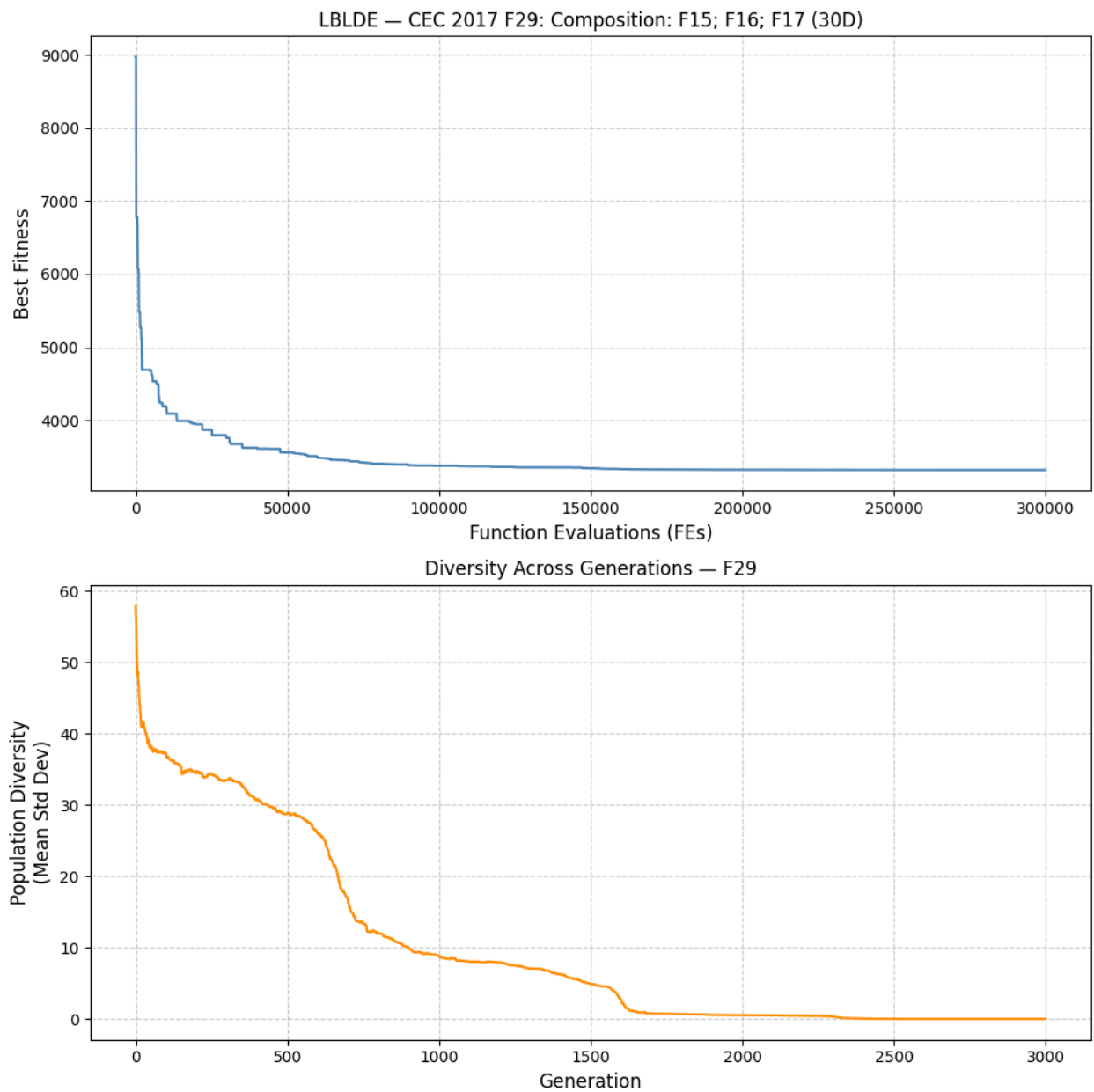
Optimization completed!

Final Best Fitness = 3.318870e+03

Total Function Evaluations = 300000

```

=====
=
Results:
  Best Fitness           : 3.318870e+03
  Global Optimum         : 2.900000e+03
  Error (|best-optimum|) : 4.188696e+02
  Best Solution (5 dim)  : [-22.17208885 -95.83435705  89.75294063  78.33546217
99.40746292]
  D^(0) (init. diversity): 5.7976e+01
=====
=
  
```



Graph saved as convergence_F29.png

```
In [40]: # =====
# F30 — Composition Function 10 (F15; F18; F19)
# =====
run_and_plot('F30', make_wrapper(composition.f30), 3000.0, 'Composition: F15;
```

```

=====
=
LBLDE_s on CEC 2017 F30 – Single Run
=====
=
Function      : Composition: F15; F18; F19 (30D)
Bounds       : [-100, 100]
Global optimum : 3000.0
Initial D^(0) = 5.7976e+01
Generation 0: Best Fitness = 3.426498e+10 | Diversity = 5.7976e+01
Generation 10: Best = 9.581993e+08 FES = 1100 Diversity = 4.5220e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 1.074427e+08 FES = 2100 Diversity = 4.0635e+01 k_t
= [3, 3, 2, 2]
Generation 30: Best = 2.507155e+07 FES = 3100 Diversity = 4.2452e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 1.682511e+07 FES = 4100 Diversity = 3.9980e+01 k_t
= [2, 2, 2, 3]
Generation 50: Best = 8.149917e+06 FES = 5100 Diversity = 3.7433e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 5.547742e+06 FES = 6100 Diversity = 3.7749e+01 k_t
= [2, 2, 4, 3]
Generation 70: Best = 2.759102e+06 FES = 7100 Diversity = 3.6633e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 1.703143e+06 FES = 8100 Diversity = 3.6441e+01 k_t
= [3, 2, 4, 2]
Generation 90: Best = 1.043932e+06 FES = 9100 Diversity = 3.5319e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 9.095971e+05 FES = 10100 Diversity = 3.2639e+01 k_t
= [3, 2, 4, 4]
Generation 110: Best = 3.908822e+05 FES = 11100 Diversity = 3.0195e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 2.235297e+05 FES = 12100 Diversity = 2.7328e+01 k_t
= [4, 2, 4, 3]
Generation 130: Best = 1.462376e+05 FES = 13100 Diversity = 2.6321e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 1.145377e+05 FES = 14100 Diversity = 2.4997e+01 k_t
= [4, 4, 4, 3]
Generation 150: Best = 7.413146e+04 FES = 15100 Diversity = 2.1919e+01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 5.187693e+04 FES = 16100 Diversity = 2.0750e+01 k_t
= [2, 2, 2, 4]
Generation 170: Best = 3.745285e+04 FES = 17100 Diversity = 1.9204e+01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 3.016043e+04 FES = 18100 Diversity = 1.8218e+01 k_t
= [4, 4, 2, 3]
Generation 190: Best = 2.520676e+04 FES = 19100 Diversity = 1.7117e+01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 1.740927e+04 FES = 20100 Diversity = 1.5927e+01 k_t
= [4, 2, 3, 2]
Generation 210: Best = 1.381336e+04 FES = 21100 Diversity = 1.4128e+01 k_t
= [1, 1, 1, 1]
Generation 220: Best = 1.343635e+04 FES = 22100 Diversity = 1.3154e+01 k_t
= [3, 2, 2, 3]

```

Generation 230:	Best = 1.247537e+04	FES = 23100	Diversity = 1.2365e+01	k_t
= [1, 1, 1, 1]				
Generation 240:	Best = 1.082428e+04	FES = 24100	Diversity = 1.0917e+01	k_t
= [2, 4, 3, 4]				
Generation 250:	Best = 1.036413e+04	FES = 25100	Diversity = 9.3501e+00	k_t
= [1, 1, 1, 1]				
Generation 260:	Best = 9.900334e+03	FES = 26100	Diversity = 9.0090e+00	k_t
= [4, 3, 3, 2]				
Generation 270:	Best = 9.264003e+03	FES = 27100	Diversity = 8.1464e+00	k_t
= [1, 1, 1, 1]				
Generation 280:	Best = 9.264003e+03	FES = 28100	Diversity = 7.5778e+00	k_t
= [3, 4, 2, 3]				
Generation 290:	Best = 8.874882e+03	FES = 29100	Diversity = 6.9784e+00	k_t
= [1, 1, 1, 1]				
Generation 300:	Best = 8.629044e+03	FES = 30100	Diversity = 6.9218e+00	k_t
= [4, 4, 4, 4]				
Generation 310:	Best = 8.341640e+03	FES = 31100	Diversity = 6.9336e+00	k_t
= [1, 1, 1, 1]				
Generation 320:	Best = 8.232545e+03	FES = 32100	Diversity = 6.7150e+00	k_t
= [4, 4, 3, 4]				
Generation 330:	Best = 7.965538e+03	FES = 33100	Diversity = 6.6362e+00	k_t
= [1, 1, 1, 1]				
Generation 340:	Best = 7.844151e+03	FES = 34100	Diversity = 6.5978e+00	k_t
= [3, 3, 2, 2]				
Generation 350:	Best = 7.646555e+03	FES = 35100	Diversity = 6.2864e+00	k_t
= [1, 1, 1, 1]				
Generation 360:	Best = 7.601649e+03	FES = 36100	Diversity = 6.2486e+00	k_t
= [3, 2, 4, 4]				
Generation 370:	Best = 7.449324e+03	FES = 37100	Diversity = 5.8545e+00	k_t
= [1, 1, 1, 1]				
Generation 380:	Best = 7.290244e+03	FES = 38100	Diversity = 5.2888e+00	k_t
= [2, 3, 3, 3]				
Generation 390:	Best = 7.254940e+03	FES = 39100	Diversity = 5.0444e+00	k_t
= [1, 1, 1, 1]				
Generation 400:	Best = 7.202369e+03	FES = 40100	Diversity = 4.4869e+00	k_t
= [4, 4, 4, 4]				
Generation 410:	Best = 7.047186e+03	FES = 41100	Diversity = 3.8380e+00	k_t
= [1, 1, 1, 1]				
Generation 420:	Best = 7.016817e+03	FES = 42100	Diversity = 3.8203e+00	k_t
= [2, 2, 2, 2]				
Generation 430:	Best = 6.928064e+03	FES = 43100	Diversity = 3.4703e+00	k_t
= [1, 1, 1, 1]				
Generation 440:	Best = 6.858842e+03	FES = 44100	Diversity = 2.9122e+00	k_t
= [2, 3, 4, 2]				
Generation 450:	Best = 6.796715e+03	FES = 45100	Diversity = 3.1682e+00	k_t
= [1, 1, 1, 1]				
Generation 460:	Best = 6.722141e+03	FES = 46100	Diversity = 3.1278e+00	k_t
= [2, 3, 3, 4]				
Generation 470:	Best = 6.713804e+03	FES = 47100	Diversity = 3.4281e+00	k_t
= [1, 1, 1, 1]				
Generation 480:	Best = 6.672927e+03	FES = 48100	Diversity = 3.1703e+00	k_t
= [4, 4, 4, 2]				
Generation 490:	Best = 6.632583e+03	FES = 49100	Diversity = 3.3646e+00	k_t
= [1, 1, 1, 1]				

Generation 500:	Best = 6.616431e+03	FES = 50100	Diversity = 3.2587e+00	k_t
= [2, 2, 2, 2]				
Generation 510:	Best = 6.582219e+03	FES = 51100	Diversity = 3.3397e+00	k_t
= [1, 1, 1, 1]				
Generation 520:	Best = 6.531190e+03	FES = 52100	Diversity = 3.3831e+00	k_t
= [3, 3, 4, 3]				
Generation 530:	Best = 6.485624e+03	FES = 53100	Diversity = 3.4593e+00	k_t
= [1, 1, 1, 1]				
Generation 540:	Best = 6.376626e+03	FES = 54100	Diversity = 3.2766e+00	k_t
= [2, 4, 2, 3]				
Generation 550:	Best = 6.345899e+03	FES = 55100	Diversity = 3.2031e+00	k_t
= [1, 1, 1, 1]				
Generation 560:	Best = 6.302470e+03	FES = 56100	Diversity = 3.8856e+00	k_t
= [2, 2, 4, 2]				
Generation 570:	Best = 6.269424e+03	FES = 57100	Diversity = 4.7193e+00	k_t
= [1, 1, 1, 1]				
Generation 580:	Best = 6.233212e+03	FES = 58100	Diversity = 5.2005e+00	k_t
= [3, 4, 2, 3]				
Generation 590:	Best = 6.199630e+03	FES = 59100	Diversity = 5.2604e+00	k_t
= [1, 1, 1, 1]				
Generation 600:	Best = 6.154990e+03	FES = 60100	Diversity = 5.0421e+00	k_t
= [3, 2, 2, 4]				
Generation 610:	Best = 6.110013e+03	FES = 61100	Diversity = 4.9050e+00	k_t
= [1, 1, 1, 1]				
Generation 620:	Best = 6.042675e+03	FES = 62100	Diversity = 4.5993e+00	k_t
= [3, 4, 2, 4]				
Generation 630:	Best = 6.008832e+03	FES = 63100	Diversity = 4.4586e+00	k_t
= [1, 1, 1, 1]				
Generation 640:	Best = 5.977422e+03	FES = 64100	Diversity = 4.8898e+00	k_t
= [3, 2, 2, 3]				
Generation 650:	Best = 5.919012e+03	FES = 65100	Diversity = 4.9692e+00	k_t
= [1, 1, 1, 1]				
Generation 660:	Best = 5.919012e+03	FES = 66100	Diversity = 4.9003e+00	k_t
= [2, 4, 3, 4]				
Generation 670:	Best = 5.824796e+03	FES = 67100	Diversity = 4.4052e+00	k_t
= [1, 1, 1, 1]				
Generation 680:	Best = 5.814193e+03	FES = 68100	Diversity = 4.5486e+00	k_t
= [4, 3, 4, 2]				
Generation 690:	Best = 5.749501e+03	FES = 69100	Diversity = 4.6795e+00	k_t
= [1, 1, 1, 1]				
Generation 700:	Best = 5.693422e+03	FES = 70100	Diversity = 4.5196e+00	k_t
= [2, 2, 3, 2]				
Generation 710:	Best = 5.669493e+03	FES = 71100	Diversity = 4.3056e+00	k_t
= [1, 1, 1, 1]				
Generation 720:	Best = 5.610027e+03	FES = 72100	Diversity = 3.8377e+00	k_t
= [3, 4, 3, 2]				
Generation 730:	Best = 5.597390e+03	FES = 73100	Diversity = 3.5315e+00	k_t
= [1, 1, 1, 1]				
Generation 740:	Best = 5.549599e+03	FES = 74100	Diversity = 3.3972e+00	k_t
= [4, 4, 4, 3]				
Generation 750:	Best = 5.549599e+03	FES = 75100	Diversity = 3.1799e+00	k_t
= [1, 1, 1, 1]				
Generation 760:	Best = 5.527343e+03	FES = 76100	Diversity = 3.0244e+00	k_t
= [2, 3, 3, 2]				

Generation 770:	Best = 5.495186e+03	FES = 77100	Diversity = 2.8648e+00	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 5.460597e+03	FES = 78100	Diversity = 2.6742e+00	k_t
= [2, 3, 4, 3]				
Generation 790:	Best = 5.443061e+03	FES = 79100	Diversity = 2.5712e+00	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 5.416852e+03	FES = 80100	Diversity = 2.4491e+00	k_t
= [4, 3, 2, 2]				
Generation 810:	Best = 5.356341e+03	FES = 81100	Diversity = 2.5043e+00	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 5.356341e+03	FES = 82100	Diversity = 2.3682e+00	k_t
= [2, 2, 3, 2]				
Generation 830:	Best = 5.350467e+03	FES = 83100	Diversity = 2.3149e+00	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 5.335152e+03	FES = 84100	Diversity = 2.1735e+00	k_t
= [4, 4, 2, 2]				
Generation 850:	Best = 5.323433e+03	FES = 85100	Diversity = 2.1460e+00	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 5.314227e+03	FES = 86100	Diversity = 2.0996e+00	k_t
= [4, 3, 2, 2]				
Generation 870:	Best = 5.285326e+03	FES = 87100	Diversity = 2.0277e+00	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 5.284670e+03	FES = 88100	Diversity = 1.9782e+00	k_t
= [3, 2, 4, 4]				
Generation 890:	Best = 5.275282e+03	FES = 89100	Diversity = 1.8178e+00	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 5.258671e+03	FES = 90100	Diversity = 1.6830e+00	k_t
= [2, 3, 3, 3]				
Generation 910:	Best = 5.258671e+03	FES = 91100	Diversity = 1.6373e+00	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 5.240001e+03	FES = 92100	Diversity = 1.5389e+00	k_t
= [4, 3, 4, 4]				
Generation 930:	Best = 5.231166e+03	FES = 93100	Diversity = 1.6735e+00	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 5.220060e+03	FES = 94100	Diversity = 1.6012e+00	k_t
= [2, 3, 3, 3]				
Generation 950:	Best = 5.212106e+03	FES = 95100	Diversity = 1.5192e+00	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 5.212106e+03	FES = 96100	Diversity = 1.3915e+00	k_t
= [2, 4, 2, 2]				
Generation 970:	Best = 5.208947e+03	FES = 97100	Diversity = 1.2378e+00	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 5.203838e+03	FES = 98100	Diversity = 1.0604e+00	k_t
= [3, 3, 4, 4]				
Generation 990:	Best = 5.197247e+03	FES = 99100	Diversity = 9.6934e-01	k_t
= [1, 1, 1, 1]				
Generation 1000:	Best = 5.192370e+03	FES = 100100	Diversity = 8.9428e-01	k_t
= [4, 2, 4, 2]				
Generation 1010:	Best = 5.190613e+03	FES = 101100	Diversity = 8.5329e-01	k_t
= [1, 1, 1, 1]				
Generation 1020:	Best = 5.188222e+03	FES = 102100	Diversity = 8.3594e-01	k_t
= [4, 3, 2, 3]				
Generation 1030:	Best = 5.182882e+03	FES = 103100	Diversity = 8.1243e-01	k_t
= [1, 1, 1, 1]				

Generation 1040: Best = 5.181045e+03 = [2, 3, 2, 4]	FES = 104100	Diversity = 8.3763e-01	k_t
Generation 1050: Best = 5.179534e+03 = [1, 1, 1, 1]	FES = 105100	Diversity = 7.9579e-01	k_t
Generation 1060: Best = 5.175539e+03 = [3, 2, 4, 2]	FES = 106100	Diversity = 8.6059e-01	k_t
Generation 1070: Best = 5.175460e+03 = [1, 1, 1, 1]	FES = 107100	Diversity = 8.1215e-01	k_t
Generation 1080: Best = 5.169570e+03 = [2, 3, 3, 3]	FES = 108100	Diversity = 8.0185e-01	k_t
Generation 1090: Best = 5.168790e+03 = [1, 1, 1, 1]	FES = 109100	Diversity = 7.8856e-01	k_t
Generation 1100: Best = 5.168790e+03 = [3, 2, 2, 3]	FES = 110100	Diversity = 7.5142e-01	k_t
Generation 1110: Best = 5.167279e+03 = [1, 1, 1, 1]	FES = 111100	Diversity = 6.8469e-01	k_t
Generation 1120: Best = 5.165478e+03 = [2, 2, 2, 3]	FES = 112100	Diversity = 6.6640e-01	k_t
Generation 1130: Best = 5.164855e+03 = [1, 1, 1, 1]	FES = 113100	Diversity = 6.5610e-01	k_t
Generation 1140: Best = 5.164855e+03 = [3, 4, 3, 3]	FES = 114100	Diversity = 6.6170e-01	k_t
Generation 1150: Best = 5.161890e+03 = [1, 1, 1, 1]	FES = 115100	Diversity = 6.6421e-01	k_t
Generation 1160: Best = 5.161890e+03 = [2, 2, 4, 3]	FES = 116100	Diversity = 6.5413e-01	k_t
Generation 1170: Best = 5.161890e+03 = [1, 1, 1, 1]	FES = 117100	Diversity = 6.1849e-01	k_t
Generation 1180: Best = 5.159542e+03 = [2, 3, 2, 2]	FES = 118100	Diversity = 6.1764e-01	k_t
Generation 1190: Best = 5.159542e+03 = [1, 1, 1, 1]	FES = 119100	Diversity = 6.1869e-01	k_t
Generation 1200: Best = 5.157417e+03 = [3, 3, 4, 4]	FES = 120100	Diversity = 6.1408e-01	k_t
Generation 1210: Best = 5.156265e+03 = [1, 1, 1, 1]	FES = 121100	Diversity = 5.9674e-01	k_t
Generation 1220: Best = 5.156265e+03 = [2, 3, 2, 4]	FES = 122100	Diversity = 6.0405e-01	k_t
Generation 1230: Best = 5.150263e+03 = [1, 1, 1, 1]	FES = 123100	Diversity = 6.9034e-01	k_t
Generation 1240: Best = 5.148244e+03 = [2, 4, 4, 3]	FES = 124100	Diversity = 7.8600e-01	k_t
Generation 1250: Best = 5.147253e+03 = [1, 1, 1, 1]	FES = 125100	Diversity = 9.7285e-01	k_t
Generation 1260: Best = 5.135997e+03 = [3, 4, 4, 3]	FES = 126100	Diversity = 9.8547e-01	k_t
Generation 1270: Best = 5.135142e+03 = [1, 1, 1, 1]	FES = 127100	Diversity = 8.5872e-01	k_t
Generation 1280: Best = 5.135024e+03 = [3, 3, 3, 3]	FES = 128100	Diversity = 8.3649e-01	k_t
Generation 1290: Best = 5.129785e+03 = [1, 1, 1, 1]	FES = 129100	Diversity = 7.6400e-01	k_t
Generation 1300: Best = 5.127887e+03 = [2, 2, 4, 4]	FES = 130100	Diversity = 7.1480e-01	k_t

Generation 1310: Best = 5.126082e+03 = [1, 1, 1, 1]	FES = 131100	Diversity = 7.1634e-01	k_t
Generation 1320: Best = 5.125305e+03 = [3, 2, 3, 3]	FES = 132100	Diversity = 6.2316e-01	k_t
Generation 1330: Best = 5.120127e+03 = [1, 1, 1, 1]	FES = 133100	Diversity = 5.8836e-01	k_t
Generation 1340: Best = 5.118777e+03 = [3, 2, 2, 2]	FES = 134100	Diversity = 6.0128e-01	k_t
Generation 1350: Best = 5.115819e+03 = [1, 1, 1, 1]	FES = 135100	Diversity = 5.6589e-01	k_t
Generation 1360: Best = 5.112500e+03 = [2, 4, 3, 4]	FES = 136100	Diversity = 5.8951e-01	k_t
Generation 1370: Best = 5.111769e+03 = [1, 1, 1, 1]	FES = 137100	Diversity = 5.6857e-01	k_t
Generation 1380: Best = 5.111769e+03 = [2, 3, 3, 4]	FES = 138100	Diversity = 5.6822e-01	k_t
Generation 1390: Best = 5.107620e+03 = [1, 1, 1, 1]	FES = 139100	Diversity = 5.2992e-01	k_t
Generation 1400: Best = 5.107620e+03 = [4, 4, 3, 2]	FES = 140100	Diversity = 5.1851e-01	k_t
Generation 1410: Best = 5.107620e+03 = [1, 1, 1, 1]	FES = 141100	Diversity = 4.9298e-01	k_t
Generation 1420: Best = 5.105095e+03 = [3, 3, 4, 4]	FES = 142100	Diversity = 4.8908e-01	k_t
Generation 1430: Best = 5.105095e+03 = [1, 1, 1, 1]	FES = 143100	Diversity = 4.5660e-01	k_t
Generation 1440: Best = 5.105077e+03 = [4, 4, 4, 4]	FES = 144100	Diversity = 4.3264e-01	k_t
Generation 1450: Best = 5.103833e+03 = [1, 1, 1, 1]	FES = 145100	Diversity = 4.4837e-01	k_t
Generation 1460: Best = 5.101392e+03 = [4, 4, 3, 3]	FES = 146100	Diversity = 4.5155e-01	k_t
Generation 1470: Best = 5.100077e+03 = [1, 1, 1, 1]	FES = 147100	Diversity = 4.2889e-01	k_t
Generation 1480: Best = 5.097812e+03 = [3, 4, 4, 4]	FES = 148100	Diversity = 4.1624e-01	k_t
Generation 1490: Best = 5.097609e+03 = [1, 1, 1, 1]	FES = 149100	Diversity = 4.2353e-01	k_t
Generation 1500: Best = 5.097593e+03 = [2, 2, 3, 4]	FES = 150100	Diversity = 4.5744e-01	k_t
Generation 1510: Best = 5.097298e+03 = [1, 1, 1, 1]	FES = 151100	Diversity = 4.8235e-01	k_t
Generation 1520: Best = 5.094522e+03 = [3, 3, 3, 2]	FES = 152100	Diversity = 4.6642e-01	k_t
Generation 1530: Best = 5.094522e+03 = [1, 1, 1, 1]	FES = 153100	Diversity = 4.5612e-01	k_t
Generation 1540: Best = 5.093962e+03 = [4, 4, 3, 3]	FES = 154100	Diversity = 4.3719e-01	k_t
Generation 1550: Best = 5.092213e+03 = [1, 1, 1, 1]	FES = 155100	Diversity = 4.3206e-01	k_t
Generation 1560: Best = 5.092213e+03 = [2, 4, 4, 2]	FES = 156100	Diversity = 3.9749e-01	k_t
Generation 1570: Best = 5.092213e+03 = [1, 1, 1, 1]	FES = 157100	Diversity = 3.9125e-01	k_t

Generation 1580: Best = 5.092001e+03 = [4, 4, 2, 2]	FES = 158100	Diversity = 3.8917e-01	k_t
Generation 1590: Best = 5.089972e+03 = [1, 1, 1, 1]	FES = 159100	Diversity = 3.8046e-01	k_t
Generation 1600: Best = 5.088811e+03 = [3, 3, 3, 2]	FES = 160100	Diversity = 3.9226e-01	k_t
Generation 1610: Best = 5.088512e+03 = [1, 1, 1, 1]	FES = 161100	Diversity = 3.9149e-01	k_t
Generation 1620: Best = 5.088512e+03 = [2, 3, 3, 2]	FES = 162100	Diversity = 4.0207e-01	k_t
Generation 1630: Best = 5.088512e+03 = [1, 1, 1, 1]	FES = 163100	Diversity = 4.0254e-01	k_t
Generation 1640: Best = 5.087632e+03 = [3, 2, 4, 4]	FES = 164100	Diversity = 4.0847e-01	k_t
Generation 1650: Best = 5.086044e+03 = [1, 1, 1, 1]	FES = 165100	Diversity = 4.0740e-01	k_t
Generation 1660: Best = 5.086044e+03 = [3, 4, 3, 3]	FES = 166100	Diversity = 4.3881e-01	k_t
Generation 1670: Best = 5.085811e+03 = [1, 1, 1, 1]	FES = 167100	Diversity = 4.1922e-01	k_t
Generation 1680: Best = 5.085527e+03 = [2, 3, 2, 2]	FES = 168100	Diversity = 4.1752e-01	k_t
Generation 1690: Best = 5.084837e+03 = [1, 1, 1, 1]	FES = 169100	Diversity = 4.1130e-01	k_t
Generation 1700: Best = 5.084443e+03 = [4, 2, 3, 2]	FES = 170100	Diversity = 4.0427e-01	k_t
Generation 1710: Best = 5.084315e+03 = [1, 1, 1, 1]	FES = 171100	Diversity = 3.8860e-01	k_t
Generation 1720: Best = 5.083889e+03 = [3, 3, 4, 3]	FES = 172100	Diversity = 3.9397e-01	k_t
Generation 1730: Best = 5.083889e+03 = [1, 1, 1, 1]	FES = 173100	Diversity = 4.1510e-01	k_t
Generation 1740: Best = 5.083857e+03 = [3, 2, 4, 3]	FES = 174100	Diversity = 4.1785e-01	k_t
Generation 1750: Best = 5.082983e+03 = [1, 1, 1, 1]	FES = 175100	Diversity = 4.1042e-01	k_t
Generation 1760: Best = 5.082983e+03 = [2, 4, 3, 4]	FES = 176100	Diversity = 4.3597e-01	k_t
Generation 1770: Best = 5.082983e+03 = [1, 1, 1, 1]	FES = 177100	Diversity = 4.5750e-01	k_t
Generation 1780: Best = 5.082007e+03 = [2, 2, 4, 4]	FES = 178100	Diversity = 4.4751e-01	k_t
Generation 1790: Best = 5.079142e+03 = [1, 1, 1, 1]	FES = 179100	Diversity = 4.4977e-01	k_t
Generation 1800: Best = 5.079142e+03 = [3, 2, 4, 2]	FES = 180100	Diversity = 4.9388e-01	k_t
Generation 1810: Best = 5.079074e+03 = [1, 1, 1, 1]	FES = 181100	Diversity = 5.4540e-01	k_t
Generation 1820: Best = 5.078335e+03 = [4, 3, 4, 4]	FES = 182100	Diversity = 5.2840e-01	k_t
Generation 1830: Best = 5.077364e+03 = [1, 1, 1, 1]	FES = 183100	Diversity = 5.1082e-01	k_t
Generation 1840: Best = 5.077090e+03 = [4, 4, 2, 3]	FES = 184100	Diversity = 4.6500e-01	k_t

Generation 1850: Best = 5.076173e+03 = [1, 1, 1, 1]	FES = 185100	Diversity = 4.6331e-01	k_t
Generation 1860: Best = 5.074864e+03 = [2, 2, 4, 3]	FES = 186100	Diversity = 4.5556e-01	k_t
Generation 1870: Best = 5.074428e+03 = [1, 1, 1, 1]	FES = 187100	Diversity = 4.4573e-01	k_t
Generation 1880: Best = 5.072294e+03 = [4, 4, 4, 3]	FES = 188100	Diversity = 4.3315e-01	k_t
Generation 1890: Best = 5.071634e+03 = [1, 1, 1, 1]	FES = 189100	Diversity = 4.3716e-01	k_t
Generation 1900: Best = 5.070837e+03 = [4, 3, 4, 2]	FES = 190100	Diversity = 4.3273e-01	k_t
Generation 1910: Best = 5.069882e+03 = [1, 1, 1, 1]	FES = 191100	Diversity = 4.3939e-01	k_t
Generation 1920: Best = 5.068229e+03 = [3, 4, 2, 4]	FES = 192100	Diversity = 4.6466e-01	k_t
Generation 1930: Best = 5.066890e+03 = [1, 1, 1, 1]	FES = 193100	Diversity = 4.8464e-01	k_t
Generation 1940: Best = 5.065874e+03 = [4, 4, 4, 2]	FES = 194100	Diversity = 5.0197e-01	k_t
Generation 1950: Best = 5.065439e+03 = [1, 1, 1, 1]	FES = 195100	Diversity = 5.0287e-01	k_t
Generation 1960: Best = 5.064009e+03 = [2, 2, 4, 4]	FES = 196100	Diversity = 4.9552e-01	k_t
Generation 1970: Best = 5.049773e+03 = [1, 1, 1, 1]	FES = 197100	Diversity = 5.5929e-01	k_t
Generation 1980: Best = 5.049773e+03 = [4, 4, 3, 4]	FES = 198100	Diversity = 8.9251e-01	k_t
Generation 1990: Best = 5.046239e+03 = [1, 1, 1, 1]	FES = 199100	Diversity = 8.5452e-01	k_t
Generation 2000: Best = 5.043764e+03 = [2, 4, 4, 2]	FES = 200100	Diversity = 7.0859e-01	k_t
Generation 2010: Best = 5.043764e+03 = [1, 1, 1, 1]	FES = 201100	Diversity = 6.2218e-01	k_t
Generation 2020: Best = 5.039695e+03 = [4, 4, 3, 3]	FES = 202100	Diversity = 5.7482e-01	k_t
Generation 2030: Best = 5.039395e+03 = [1, 1, 1, 1]	FES = 203100	Diversity = 5.2309e-01	k_t
Generation 2040: Best = 5.039039e+03 = [3, 4, 2, 4]	FES = 204100	Diversity = 5.0068e-01	k_t
Generation 2050: Best = 5.037019e+03 = [1, 1, 1, 1]	FES = 205100	Diversity = 4.7356e-01	k_t
Generation 2060: Best = 5.036987e+03 = [2, 2, 4, 2]	FES = 206100	Diversity = 4.3739e-01	k_t
Generation 2070: Best = 5.035902e+03 = [1, 1, 1, 1]	FES = 207100	Diversity = 4.0103e-01	k_t
Generation 2080: Best = 5.035902e+03 = [2, 2, 2, 4]	FES = 208100	Diversity = 3.8450e-01	k_t
Generation 2090: Best = 5.035656e+03 = [1, 1, 1, 1]	FES = 209100	Diversity = 3.6187e-01	k_t
Generation 2100: Best = 5.035656e+03 = [3, 4, 4, 3]	FES = 210100	Diversity = 3.6004e-01	k_t
Generation 2110: Best = 5.035656e+03 = [1, 1, 1, 1]	FES = 211100	Diversity = 3.5992e-01	k_t

Generation 2120: Best = 5.034478e+03 = [4, 4, 3, 3]	FES = 212100	Diversity = 3.5245e-01	k_t
Generation 2130: Best = 5.034478e+03 = [1, 1, 1, 1]	FES = 213100	Diversity = 3.4196e-01	k_t
Generation 2140: Best = 5.034413e+03 = [3, 4, 2, 4]	FES = 214100	Diversity = 3.1648e-01	k_t
Generation 2150: Best = 5.034413e+03 = [1, 1, 1, 1]	FES = 215100	Diversity = 3.1362e-01	k_t
Generation 2160: Best = 5.033612e+03 = [2, 2, 2, 3]	FES = 216100	Diversity = 3.0844e-01	k_t
Generation 2170: Best = 5.033273e+03 = [1, 1, 1, 1]	FES = 217100	Diversity = 3.0278e-01	k_t
Generation 2180: Best = 5.033273e+03 = [4, 2, 3, 2]	FES = 218100	Diversity = 2.8954e-01	k_t
Generation 2190: Best = 5.033273e+03 = [1, 1, 1, 1]	FES = 219100	Diversity = 2.7950e-01	k_t
Generation 2200: Best = 5.033273e+03 = [3, 4, 3, 3]	FES = 220100	Diversity = 2.8251e-01	k_t
Generation 2210: Best = 5.033273e+03 = [1, 1, 1, 1]	FES = 221100	Diversity = 2.8019e-01	k_t
Generation 2220: Best = 5.032070e+03 = [3, 2, 4, 3]	FES = 222100	Diversity = 2.7683e-01	k_t
Generation 2230: Best = 5.032070e+03 = [1, 1, 1, 1]	FES = 223100	Diversity = 2.7092e-01	k_t
Generation 2240: Best = 5.031827e+03 = [4, 2, 2, 2]	FES = 224100	Diversity = 2.6266e-01	k_t
Generation 2250: Best = 5.030104e+03 = [1, 1, 1, 1]	FES = 225100	Diversity = 2.6254e-01	k_t
Generation 2260: Best = 5.030102e+03 = [3, 2, 4, 3]	FES = 226100	Diversity = 2.5925e-01	k_t
Generation 2270: Best = 5.030025e+03 = [1, 1, 1, 1]	FES = 227100	Diversity = 2.5784e-01	k_t
Generation 2280: Best = 5.030025e+03 = [2, 4, 3, 3]	FES = 228100	Diversity = 2.5668e-01	k_t
Generation 2290: Best = 5.029667e+03 = [1, 1, 1, 1]	FES = 229100	Diversity = 2.4741e-01	k_t
Generation 2300: Best = 5.028889e+03 = [4, 2, 2, 3]	FES = 230100	Diversity = 2.4982e-01	k_t
Generation 2310: Best = 5.028656e+03 = [1, 1, 1, 1]	FES = 231100	Diversity = 2.4630e-01	k_t
Generation 2320: Best = 5.028184e+03 = [3, 4, 3, 4]	FES = 232100	Diversity = 2.5098e-01	k_t
Generation 2330: Best = 5.028184e+03 = [1, 1, 1, 1]	FES = 233100	Diversity = 2.4450e-01	k_t
Generation 2340: Best = 5.028184e+03 = [4, 2, 2, 3]	FES = 234100	Diversity = 2.4059e-01	k_t
Generation 2350: Best = 5.028178e+03 = [1, 1, 1, 1]	FES = 235100	Diversity = 2.3734e-01	k_t
Generation 2360: Best = 5.028178e+03 = [3, 4, 2, 4]	FES = 236100	Diversity = 2.4192e-01	k_t
Generation 2370: Best = 5.028178e+03 = [1, 1, 1, 1]	FES = 237100	Diversity = 2.4416e-01	k_t
Generation 2380: Best = 5.028124e+03 = [3, 2, 4, 2]	FES = 238100	Diversity = 2.4817e-01	k_t

Generation 2390: Best = 5.028049e+03 = [1, 1, 1, 1]	FES = 239100	Diversity = 2.4779e-01	k_t
Generation 2400: Best = 5.026628e+03 = [3, 3, 2, 3]	FES = 240100	Diversity = 2.4837e-01	k_t
Generation 2410: Best = 5.026628e+03 = [1, 1, 1, 1]	FES = 241100	Diversity = 2.5596e-01	k_t
Generation 2420: Best = 5.026628e+03 = [4, 4, 4, 3]	FES = 242100	Diversity = 2.5646e-01	k_t
Generation 2430: Best = 5.026628e+03 = [1, 1, 1, 1]	FES = 243100	Diversity = 2.5311e-01	k_t
Generation 2440: Best = 5.026628e+03 = [4, 3, 2, 3]	FES = 244100	Diversity = 2.5154e-01	k_t
Generation 2450: Best = 5.026628e+03 = [1, 1, 1, 1]	FES = 245100	Diversity = 2.4390e-01	k_t
Generation 2460: Best = 5.026628e+03 = [4, 3, 2, 2]	FES = 246100	Diversity = 2.4183e-01	k_t
Generation 2470: Best = 5.026628e+03 = [1, 1, 1, 1]	FES = 247100	Diversity = 2.3198e-01	k_t
Generation 2480: Best = 5.026628e+03 = [3, 2, 4, 2]	FES = 248100	Diversity = 2.3251e-01	k_t
Generation 2490: Best = 5.026628e+03 = [1, 1, 1, 1]	FES = 249100	Diversity = 2.2334e-01	k_t
Generation 2500: Best = 5.026349e+03 = [4, 2, 2, 2]	FES = 250100	Diversity = 2.1834e-01	k_t
Generation 2510: Best = 5.026349e+03 = [1, 1, 1, 1]	FES = 251100	Diversity = 2.1706e-01	k_t
Generation 2520: Best = 5.026349e+03 = [3, 4, 4, 4]	FES = 252100	Diversity = 2.1420e-01	k_t
Generation 2530: Best = 5.026349e+03 = [1, 1, 1, 1]	FES = 253100	Diversity = 2.0156e-01	k_t
Generation 2540: Best = 5.026349e+03 = [4, 3, 4, 2]	FES = 254100	Diversity = 1.9632e-01	k_t
Generation 2550: Best = 5.026142e+03 = [1, 1, 1, 1]	FES = 255100	Diversity = 1.9032e-01	k_t
Generation 2560: Best = 5.026124e+03 = [3, 4, 3, 2]	FES = 256100	Diversity = 1.8801e-01	k_t
Generation 2570: Best = 5.026124e+03 = [1, 1, 1, 1]	FES = 257100	Diversity = 1.7772e-01	k_t
Generation 2580: Best = 5.026124e+03 = [4, 2, 2, 3]	FES = 258100	Diversity = 1.8081e-01	k_t
Generation 2590: Best = 5.026124e+03 = [1, 1, 1, 1]	FES = 259100	Diversity = 1.7703e-01	k_t
Generation 2600: Best = 5.026048e+03 = [3, 2, 4, 2]	FES = 260100	Diversity = 1.7480e-01	k_t
Generation 2610: Best = 5.026048e+03 = [1, 1, 1, 1]	FES = 261100	Diversity = 1.7453e-01	k_t
Generation 2620: Best = 5.026048e+03 = [4, 2, 4, 2]	FES = 262100	Diversity = 1.7388e-01	k_t
Generation 2630: Best = 5.026008e+03 = [1, 1, 1, 1]	FES = 263100	Diversity = 1.6771e-01	k_t
Generation 2640: Best = 5.025634e+03 = [4, 3, 4, 2]	FES = 264100	Diversity = 1.6075e-01	k_t
Generation 2650: Best = 5.025634e+03 = [1, 1, 1, 1]	FES = 265100	Diversity = 1.5190e-01	k_t

Generation 2660: Best = 5.025634e+03 = [4, 2, 2, 4]	FES = 266100	Diversity = 1.5348e-01	k_t
Generation 2670: Best = 5.025137e+03 = [1, 1, 1, 1]	FES = 267100	Diversity = 1.5297e-01	k_t
Generation 2680: Best = 5.025082e+03 = [2, 4, 4, 4]	FES = 268100	Diversity = 1.5142e-01	k_t
Generation 2690: Best = 5.024917e+03 = [1, 1, 1, 1]	FES = 269100	Diversity = 1.4494e-01	k_t
Generation 2700: Best = 5.024917e+03 = [4, 2, 2, 3]	FES = 270100	Diversity = 1.3838e-01	k_t
Generation 2710: Best = 5.024885e+03 = [1, 1, 1, 1]	FES = 271100	Diversity = 1.3441e-01	k_t
Generation 2720: Best = 5.024885e+03 = [3, 2, 2, 2]	FES = 272100	Diversity = 1.3299e-01	k_t
Generation 2730: Best = 5.024751e+03 = [1, 1, 1, 1]	FES = 273100	Diversity = 1.3056e-01	k_t
Generation 2740: Best = 5.024689e+03 = [3, 2, 4, 2]	FES = 274100	Diversity = 1.3112e-01	k_t
Generation 2750: Best = 5.024689e+03 = [1, 1, 1, 1]	FES = 275100	Diversity = 1.3078e-01	k_t
Generation 2760: Best = 5.024689e+03 = [3, 2, 4, 2]	FES = 276100	Diversity = 1.2954e-01	k_t
Generation 2770: Best = 5.024533e+03 = [1, 1, 1, 1]	FES = 277100	Diversity = 1.2888e-01	k_t
Generation 2780: Best = 5.024522e+03 = [3, 2, 2, 2]	FES = 278100	Diversity = 1.2600e-01	k_t
Generation 2790: Best = 5.024522e+03 = [1, 1, 1, 1]	FES = 279100	Diversity = 1.1758e-01	k_t
Generation 2800: Best = 5.024507e+03 = [2, 2, 4, 2]	FES = 280100	Diversity = 1.1192e-01	k_t
Generation 2810: Best = 5.024497e+03 = [1, 1, 1, 1]	FES = 281100	Diversity = 1.0424e-01	k_t
Generation 2820: Best = 5.024497e+03 = [3, 2, 2, 3]	FES = 282100	Diversity = 9.6272e-02	k_t
Generation 2830: Best = 5.024460e+03 = [1, 1, 1, 1]	FES = 283100	Diversity = 8.0008e-02	k_t
Generation 2840: Best = 5.024267e+03 = [3, 4, 4, 3]	FES = 284100	Diversity = 7.9963e-02	k_t
Generation 2850: Best = 5.024215e+03 = [1, 1, 1, 1]	FES = 285100	Diversity = 6.3408e-02	k_t
Generation 2860: Best = 5.023953e+03 = [3, 3, 3, 4]	FES = 286100	Diversity = 6.2428e-02	k_t
Generation 2870: Best = 5.023950e+03 = [1, 1, 1, 1]	FES = 287100	Diversity = 6.1608e-02	k_t
Generation 2880: Best = 5.023817e+03 = [3, 2, 2, 2]	FES = 288100	Diversity = 5.7402e-02	k_t
Generation 2890: Best = 5.023758e+03 = [1, 1, 1, 1]	FES = 289100	Diversity = 5.3518e-02	k_t
Generation 2900: Best = 5.023673e+03 = [2, 4, 4, 4]	FES = 290100	Diversity = 5.0642e-02	k_t
Generation 2910: Best = 5.023482e+03 = [1, 1, 1, 1]	FES = 291100	Diversity = 4.5101e-02	k_t
Generation 2920: Best = 5.023305e+03 = [2, 3, 4, 4]	FES = 292100	Diversity = 4.4788e-02	k_t

Generation 2930: Best = 5.023159e+03 FES = 293100 Diversity = 4.3638e-02 k_t
 = [1, 1, 1, 1]
 Generation 2940: Best = 5.023145e+03 FES = 294100 Diversity = 4.3393e-02 k_t
 = [3, 2, 4, 4]
 Generation 2950: Best = 5.023113e+03 FES = 295100 Diversity = 4.2098e-02 k_t
 = [1, 1, 1, 1]
 Generation 2960: Best = 5.022807e+03 FES = 296100 Diversity = 4.1029e-02 k_t
 = [3, 2, 4, 2]
 Generation 2970: Best = 5.022798e+03 FES = 297100 Diversity = 4.0676e-02 k_t
 = [1, 1, 1, 1]
 Generation 2980: Best = 5.022798e+03 FES = 298100 Diversity = 3.9108e-02 k_t
 = [3, 2, 3, 4]
 Generation 2990: Best = 5.022798e+03 FES = 299100 Diversity = 3.7857e-02 k_t
 = [1, 1, 1, 1]

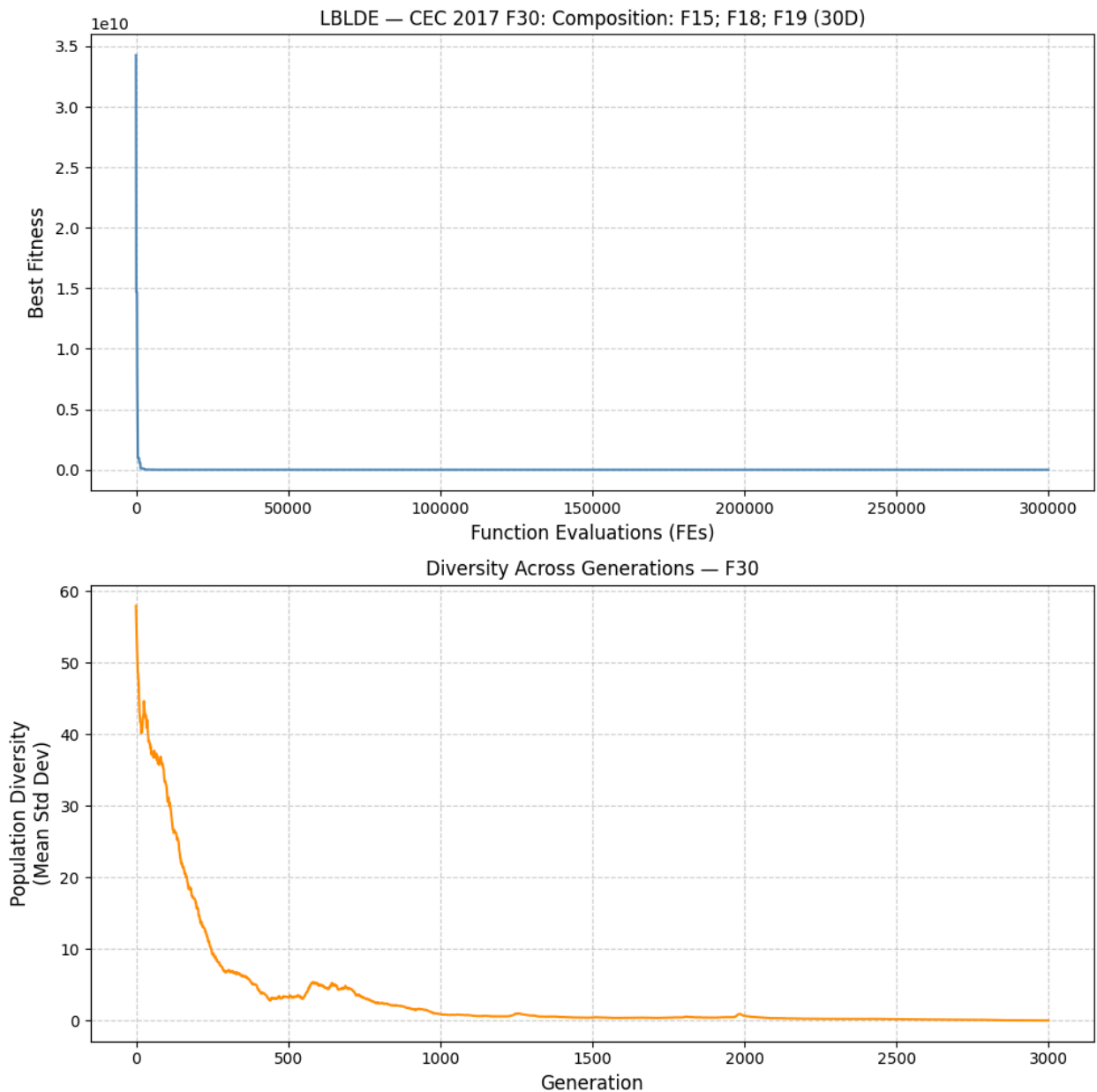
Optimization completed!

Final Best Fitness = 5.022720e+03

Total Function Evaluations = 300000

```

=====
=
Results:
  Best Fitness           : 5.022720e+03
  Global Optimum         : 3.000000e+03
  Error (|best-optimum|) : 2.022720e+03
  Best Solution (5 dim)  : [-21.54658761 -13.11327205  99.99180969  61.46376541
-23.77535202]
  D^(0) (init. diversity): 5.7976e+01
=====
=
  
```



Graph saved as convergence_F30.png

Section 7. Compiled Results

Print out a summary of all results in a tabular format

```
In [41]: print('=' * 80)
print('SUMMARY — LBLDE_s on CEC 2017 (30D), Single Run, Seed=42')
print('=' * 80)
print(f'{"Function":<10} {"Best Fitness":>18} {"Global Optimum":>18} {"Error |
print('-' * 70)
for fn in [f'F{i}' for i in range(1, 31)]:
```

```

if fn not in all_results:
    continue
r = all_results[fn]
if r['best_fit'] is None:
    print(f'{fn:<10} {"N/A (deprecated)":>18} {r["f_global"]:>18.2f} {"N/A"')
else:
    print(f'{fn:<10} {r["best_fit"]:>18.6e} {r["f_global"]:>18.2f} {r["err
print('=' * 80)

```

```

=====
=
SUMMARY – LBLDE_s on CEC 2017 (30D), Single Run, Seed=42
=====
=

```

Function	Best Fitness	Global Optimum	Error best-opt
F1	3.937546e+03	100.00	3.837546e+03
F2	1.233162e+11	200.00	1.233162e+11
F3	3.000000e+02	300.00	2.667123e-05
F4	4.042891e+02	400.00	4.289085e+00
F5	5.348235e+02	500.00	3.482354e+01
F6	6.000000e+02	600.00	3.819354e-05
F7	7.562905e+02	700.00	5.629047e+01
F8	8.278588e+02	800.00	2.785884e+01
F9	9.004543e+02	900.00	4.543240e-01
F10	2.636273e+03	1000.00	1.636273e+03
F11	1.114726e+03	1100.00	1.472576e+01
F12	3.164549e+03	1200.00	1.964549e+03
F13	1.312004e+03	1300.00	1.200392e+01
F14	1.428482e+03	1400.00	2.848173e+01
F15	1.507010e+03	1500.00	7.009543e+00
F16	1.910254e+03	1600.00	3.102544e+02
F17	1.734966e+03	1700.00	3.496593e+01
F18	1.822323e+03	1800.00	2.232350e+01
F19	1.906146e+03	1900.00	6.146145e+00
F20	2.023239e+03	2000.00	2.323937e+01
F21	2.326171e+03	2100.00	2.261713e+02
F22	2.300000e+03	2200.00	1.000000e+02
F23	2.668377e+03	2300.00	3.683775e+02
F24	2.848780e+03	2400.00	4.487796e+02
F25	2.886867e+03	2500.00	3.868671e+02
F26	3.864048e+03	2600.00	1.264048e+03
F27	3.205271e+03	2700.00	5.052711e+02
F28	3.261847e+03	2800.00	4.618465e+02
F29	3.318870e+03	2900.00	4.188696e+02
F30	5.022720e+03	3000.00	2.022720e+03

```

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```

```
In [42]: !zip plots.zip convergence_F{1..30}.png
```


adding: convergence_F1.png (deflated 23%)
adding: convergence_F2.png (deflated 21%)
adding: convergence_F3.png (deflated 21%)
adding: convergence_F4.png (deflated 23%)
adding: convergence_F5.png (deflated 20%)
adding: convergence_F6.png (deflated 21%)
adding: convergence_F7.png (deflated 22%)
adding: convergence_F8.png (deflated 20%)
adding: convergence_F9.png (deflated 23%)
adding: convergence_F10.png (deflated 20%)
adding: convergence_F11.png (deflated 21%)
adding: convergence_F12.png (deflated 19%)
adding: convergence_F13.png (deflated 21%)
adding: convergence_F14.png (deflated 20%)
adding: convergence_F15.png (deflated 20%)
adding: convergence_F16.png (deflated 19%)
adding: convergence_F17.png (deflated 18%)
adding: convergence_F18.png (deflated 21%)
adding: convergence_F19.png (deflated 18%)
adding: convergence_F20.png (deflated 17%)
adding: convergence_F21.png (deflated 20%)
adding: convergence_F22.png (deflated 23%)
adding: convergence_F23.png (deflated 20%)
adding: convergence_F24.png (deflated 20%)
adding: convergence_F25.png (deflated 22%)
adding: convergence_F26.png (deflated 20%)
adding: convergence_F27.png (deflated 20%)
adding: convergence_F28.png (deflated 21%)
adding: convergence_F29.png (deflated 19%)
adding: convergence_F30.png (deflated 21%)